

SPECIFICATIONS

**Replace Roofs of Buildings 1, 2, 3/16, and 31
Department of Veterans Affairs
Mann-Grandstaff VA Medical Center
4815 North Assembly St.
Spokane, Washington 99205**

Project No. 668-24-103

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Department of Veterans Affairs

Network Contracting Office 20

35% SCHEMATIC DESIGN – 04.28.2025

Department of Veterans Affairs
VA Spokane Healthcare System
Project No.: 668-24-103
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Sections in blue will be edited at 95% submission.

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SECTION 00 01 15
LIST OF DRAWINGS

The drawings listed below accompanying this specification form a part of the contract.

GENERAL

GI-000	Cover Sheet
GI-001	Sheet Index, Applicable Standards & General Notes
GI-002	Phasing Plan
GI-100	Bldg 1: Existing Roof Plan
GI-101	Bldg 1: Existing Roof Photos
GI-102	Bldg 1: Existing Roof Photos
GI-200	Bldg 2: Existing Roof Plan
GI-201	Bldg 2: Existing Roof Photos
GI-202	Bldg 2: Existing Roof Photos
GI-300	Bldg 3 & 16: Existing Roof Plan
GI-301	Bldg 3 & 16: Existing Roof Photos
GI-310	Bldg 31: Existing Roof Plan
GI-311	Bldg 31: Existing Roof Photos

ARCHITECTURAL

AD-100	Bldg 1: Roof Demolition Plan
AD-200	Bldg 2: Roof Demolition Plan
AD-300	Bldg 3 & 16: Roof Demolition Plan
AD-310	Bldg 31: Roof Demolition Plan
AE-100	Bldg 1: Roof Renovation Plan
AE-200	Bldg 2: Roof Renovation Plan
AE-300	Bldg 3 & 16: Roof Renovation Plan
AE-310	Bldg 31: Roof Renovation Plan
AE-900	Roof Details
AE-901	Roof Details

- - - E N D - - -

SECTION 01 00 00
GENERAL REQUIREMENTS

~~SPEC WRITER NOTES:~~

- ~~1. Delete between // --- //if not applicable to project. Include following Notice on 8a contracts only.~~
- ~~2. //NOTICE: The provisions of this Section, GENERAL REQUIREMENTS, pertain only to the contract between the Small Business Administration and its selected subcontractor(s).//~~
- ~~3. Verify identification requirements for all construction workers required by the medical center. ADD a statement concerning Photo ID requirements.~~
- ~~4. Provide general description of the project in 1.2.A.~~
- ~~5. "COR" is the term typically utilized in most paragraphs. "COR" is utilized on major and CFM-managed projects.~~

GENERAL

1.1 SAFETY REQUIREMENTS

- A. Refer to section 01 35 26, SAFETY REQUIREMENTS for safety and infection control requirements.

1.2 GENERAL INTENTION

- A. Contractor shall completely prepare site for building operations, including demolition and removal of existing structures, and furnish labor and materials and perform work for the roofing replacement of Buildings 1, 2, 3/16 & 31, at the Mann-Grandstaff VA Medical Center, 4815 North Assembly Street, Spokane, WA 99205, as required by drawings and specifications.

~~SPEC WRITER NOTE: Following paragraph is applicable for construction at existing hospitals.~~

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- B. Visits to the site by Bidders may be made only by appointment with the ~~//Medical Center Engineering Officer.// Cemetery Director~~ //Contracting Officer's Representative (COR).

~~SPEC WRITER NOTE: Include following paragraph only on A/E projects.~~

- C. Offices of MA Architects, as Architect-Engineers, will render certain technical services during construction. Such services shall be considered as advisory to the Government and shall not be construed as expressing or implying a contractual act of the Government without affirmations by Contracting Officer (CO) or his duly authorized representative (COR).

~~SPEC WRITER NOTE: The following paragraph is intended for use on Central Office projects and A/E projects where certain tests are to be performed by a VA retained Testing Laboratory.~~

- D. Before placement and installation of work subject to tests by testing laboratory retained by Department of Veterans Affairs, the Contractor shall notify the COR in sufficient time to enable testing laboratory personnel to be present at the site in time for proper taking and testing of specimens and field inspection. Such prior notice shall be not less than three workdays unless otherwise designated by the COR.
- E. All employees of general contractor and subcontractors shall comply with VA security management program and obtain permission of the VA police, be identified by project and employer, and restricted from unauthorized access.

~~SPEC WRITER NOTES:~~

- ~~1. In paragraph below change title, "GENERAL CONSTRUCTION" as necessary to reflect project title, such as A/C, Sewers, etc.~~
- ~~2. Unit price items will only be used when their use is directed by the Project Manager. When using unit price items insert FAR clause~~

~~52.215-26 in the contract
specifications.~~

1.3 STATEMENT OF BID ITEM(S)

- A. ITEM I, GENERAL CONSTRUCTION BASE BID: Work includes general construction, alterations, ~~roads, walks, grading, drainage,~~
~~//————//~~ necessary removal of existing structures, re-roofing
of areas shown on the construction documents ~~and construction~~ and certain other items.

~~SPEC WRITER NOTE:~~

- ~~1. AE should define the specific scope of the project in the bid items to provide a project overview. A breakdown of additional items should be added to cover the work as required.~~
- ~~2. If prices are needed for Alternate Bid Items, describe Alternate Items below and show items on SOLICITATION, OFFER AND AWARD (SF-1442). Coordinate with Project Manager. Alternates should be deductive, ordered from largest to smallest, and be cumulative in nature to address the requirements of 10-20% deduct requirements based on project type. Multiple small deducts are not desirable and only degrade project quality.~~

~~B. ALTERNATE NO.1: BASE BID except~~

~~delete~~_____

~~C. ALTERNATE NO.1: ALTERNATE 1 except delete~~

1.4 SPECIFICATIONS AND DRAWINGS FOR CONTRACTOR

- A. Drawings and contract documents may be obtained from the website where the solicitation is posted. Additional copies will be at Contractor's expense.

~~SPEC WRITER NOTE: Modify following
article to suit the project.
Coordinate with Medical Center.~~

1.5 CONSTRUCTION SECURITY REQUIREMENTS

A. Security Plan:

1. The security plan defines both physical and administrative security procedures that will remain effective for the entire duration of the project.
2. The General Contractor is responsible for assuring that all sub-contractors working on the project and their employees also comply with these regulations.

B. Security Procedures:

1. General Contractor's employees shall not enter the project site without appropriate badge. They may also be subject to inspection of their personal effects when entering or leaving the project site.
2. Before starting work the General Contractor shall give one week's notice to the COR so that security ~~//escort//~~ arrangements~~//~~ can be provided for the employees. This notice is separate from any notices required for utility shutdown described later in this section.
3. No photography of VA premises is allowed without written permission of the COR. Patients and staff are not to be photographed at any time.
4. VA reserves the right to close down or shut down the project site and order General Contractor's employees off the premises in the event of a national emergency. The General Contractor may return to the site only with the written approval of the COR.

~~C. Guards:~~

- ~~1. The General Contractor shall provide unarmed guards at the project site // 24 hours a days, 7 days a week// // after construction hours//.~~

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~~2. The Contractor shall provide the guards and VA police with communication devices as directed.~~

~~3. The general Contractor shall install equipment for recording guard rounds to ensure systematic checking of the premises.~~

~~D.C.~~ Key Control:

~~SPEC WRITER NOTE: Modify following article to suit the project. Coordinate with Medical Center.~~

1. The General Contractor shall provide duplicate keys and lock combinations to the COR for the purpose of security inspections of every area of project including tool boxes and parked machines and take any emergency action.

~~2. The General Contractor shall //install all permanent cores at completion of the work// turn over all permanent lock cylinders to the VA locksmith for permanent installation //. See Section 08 71 00, DOOR HARDWARE and coordinate.~~

~~E.D.~~ Document Control:

1. Before starting any work, the General Contractor/Sub Contractors shall submit an electronic security memorandum describing the approach to following goals and maintaining confidentiality of "sensitive information".
2. The General Contractor is responsible for safekeeping of all drawings, project manual and other project information. This information shall be shared only with those with a specific need to accomplish the project.
3. Certain documents, sketches, videos or photographs and drawings may be marked "Law Enforcement Sensitive" or "Sensitive Unclassified". Secure such information in separate containers and limit the access to only those who will need it for the project. Return the information to the COR upon request.
4. These security documents shall not be removed or transmitted from the project site without the written approval of the COR.

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5. All paper waste or electronic media such as CD's and diskettes shall be shredded and destroyed in a manner acceptable to the VA.
6. Notify the COR and Site Security Officer immediately when there is a loss or compromise of "sensitive information".
7. All electronic information shall be stored in specified location following VA standards and procedures using an Engineering Document Management Software (EDMS).
 - a) Security, access and maintenance of all project drawings, both scanned and electronic shall be performed and tracked through the EDMS system.
 - b) "Sensitive information" including drawings and other documents may be attached to e-mail provided all VA encryption procedures are followed.

F.E. Motor Vehicle Restrictions

1. Vehicle authorization request shall be required for any vehicle entering the site and such request shall be submitted 24 hours before the date and time of access. Access shall be restricted to picking up and dropping off materials and supplies.
2. ~~//An unlimited//~~A limited~~//~~ number of ~~/(2 to 5)//~~ permits shall be issued for General Contractor and its employees for parking in designated areas only. Contractor to coordinate with VA Medical Center Facility Manager.

~~SPEC WRITER NOTE: Coordinate editing with facility Safety Manager/Officer at VA medical facilities. Edit subparagraphs C, E, G, H, M, P and Q carefully as they directly relate to interim life safety measures required in or adjacent to construction affecting occupied buildings by the Joint Commission on Accreditation of Healthcare Organizations. At other sites, edit for project and delete // and facility Safety // Manager // Officer // provisions.~~

1.6 OPERATIONS AND STORAGE AREAS (FAR 52.236-10)

- A. The Contractor shall confine all operations (including storage of materials) on Government premises to areas authorized or approved by the COR. The Contractor shall hold and save the Government, its officers and agents, free and harmless from liability of any nature occasioned by the Contractor's performance.
- B. Temporary buildings (e.g., storage sheds, shops, offices) and utilities may be erected by the Contractor only with the approval of the COR and shall be built with labor and materials furnished by the Contractor without expense to the Government. The temporary buildings and utilities shall remain the property of the Contractor and shall be removed by the Contractor at its expense upon completion of the work. With the written consent of the COR, the buildings and utilities may be abandoned and need not be removed.
- C. The Contractor shall, under regulations prescribed by the COR, use only established roadways, or use temporary roadways constructed by the Contractor when and as authorized by the COR. When materials are transported in prosecuting the work, vehicles shall not be loaded beyond the loading capacity recommended by the manufacturer of the vehicle or prescribed by any Federal, State, or local law or regulation. When it is necessary to cross curbs or sidewalks, the Contractor shall protect them from damage. The Contractor shall repair or pay for the repair of any damaged curbs, sidewalks, or roads.

~~SPEC WRITER NOTES:~~

- ~~1. Use Paragraphs D through O for projects at existing sites and Paragraphs E* through F* for projects at new stations.~~
- ~~2. If construction fences are required, and if there are any requirements or limitations on workers' parking and access by VA or Contractor, they should be so stated in this article.~~

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- ~~3. Check with requirements of Network Analysis System when that Section is included on project, because the Project Analysis Group generally composes the following subparagraphs G through H' to supplement the NAS section.~~
- ~~4. On large projects, the drawings should indicate the space available to the Contractor. On small projects the COR may determine what space may be made available.~~

- D. Working space and space available for storing materials shall be shown on the drawings, as determined by the COR.
- E. Workers are subject to rules of conduct of the Medical Center-
~~// Cemetery // applicable to their conduct.~~
- F. ~~//~~Execute work in such a manner as to interfere as little as possible with work being done by others. Keep roads clear of construction materials, debris, standing construction equipment and vehicles at all times.~~//~~
- G. Execute work so as to interfere as little as possible with normal functioning of the Medical Center ~~// Cemetery //~~ as a whole, including operations of utility services, fire protection systems and any existing equipment, and with work being done by others.
~~//~~Use of equipment and tools that transmit vibrations and noises through the building structure, are not permitted in buildings that are occupied, during construction, jointly by patients or medical personnel, and Contractor's personnel, except as permitted by the COR where required by limited working space.~~//~~
1. Do not store materials and equipment in other than assigned areas.
 2. ~~//~~Schedule delivery of materials and equipment to immediate construction working areas within buildings in use by Department of Veterans Affairs in quantities sufficient for not more than two work days. ~~//~~Provide unobstructed access to ~~//~~Medical Center ~~// Cemetery //~~ areas required to remain in operation.

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3. Where access by ~~//Medical Center~~ ~~//Cemetery~~ personnel to vacated portions of buildings is not required, storage of Contractor's materials and equipment will be permitted subject to fire and safety requirements.

H. ~~//Utilities Services~~: Where necessary to cut existing pipes, electrical wires, conduits, cables, etc., of utility services, or of fire protection systems or communications systems (except telephone), they shall be cut and capped at suitable places where shown; or, in absence of such indication, where directed by the COR. All such actions shall be coordinated with the COR or Utility Company involved:

1. Whenever it is required that a connection fee be paid to a public utility provider for new permanent service to the construction project, for such items as water, sewer, electricity, gas or steam, payment of such fee shall be the responsibility of the Government and not the Contractor.

I. Phasing:

1. The Medical Center must maintain its operation 24 hours a day 7 days a week. Therefore, any interruption in service must be scheduled and coordinated with the COR to ensure that no lapses in operation occur. It is the CONTRACTOR'S responsibility to develop a work plan and schedule detailing, at a minimum, the procedures to be employed, the equipment and materials to be used, the interim life safety measure to be used during the work, and a schedule defining the duration of the work with milestone subtasks. The work to be outlined shall include, but not be limited to:

~~2.~~ To ensure such executions, Contractor shall furnish the COR with a schedule of approximate phasing dates on which the Contractor intends to accomplish work in each specific area of site, building or portion thereof. In addition, Contractor shall notify the COR two weeks in advance of the proposed date of starting work in each specific area of site, building or

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portion thereof. Arrange such phasing dates to ensure accomplishment of this work in successive phases mutually agreeable to the Medical Center Director, COR and Contractor. ~~as follows:~~

~~SPEC WRITER NOTE: Set up phasing by buildings, wings, floors, or areas in accordance with information received from Medical Center through Project Director.~~

~~Phase I:~~

~~Phase II:~~

~~SPEC WRITER NOTE: Following paragraphs J and K shall be used where entire buildings are vacated. Coordinate with preceding phasing paragraph. Edit and use paragraphs J and K as indicated by project requirements. If the project includes NAS, the Project Analysis Group should furnish information about which buildings or areas that cannot be vacated during construction.~~

~~J. //Building (s) No. (s) // Part of Bldg.// _____ will be vacated by Government in accordance with above phasing beginning immediately after date of receipt of Notice to Proceed and turned over to Contractor.~~

~~SPEC WRITER NOTE: Use the first sentence of Paragraph K, including material enclosed within " //, " only if all areas will be vacated by VA during the periods of alterations. Delete material within " // " from first sentence and include subparagraph 1 and 2 if one or more areas will be occupied by VA during the periods of alterations.~~

~~K-J. //Building(s) No.(s) _____ All buildings will be occupied during performance of work. // . // ; but immediate areas of alterations will be vacated. //~~

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~~1. //Certain areas of Building(s) No. (s) _____ will be occupied by Medical Center personnel for various periods as listed below:—~~

AREA	PERIOD
(a)	_____
(b)	_____
(c)	_____
(d)	_____

~~2.1.~~ Contractor shall take all measures and provide all material necessary for protecting existing equipment and property in affected areas of construction against dust and debris, so that equipment and affected areas to be used in the Medical Centers operations will not be hindered. Contractor shall permit access to Department of Veterans Affairs personnel and patients through other construction areas which serve as routes of access to such affected areas and equipment. These routes whether access or egress shall be isolated from the construction area by temporary partitions and have walking surfaces, lighting etc. to facilitate patient and staff access. Coordinate alteration work in areas occupied by Department of Veterans Affairs so that Medical Center operations will continue during the construction period.

~~3. Immediate areas of alterations not mentioned in preceding Subparagraph 1 will be temporarily vacated while alterations are performed. //~~

~~L.K.~~ //Construction Fence: Before construction operations begin, Contractor shall provide a chain link construction fence, 2.1m (seven feet) minimum height, around the construction area indicated on the drawings. Provide gates as required for access with necessary hardware, including hasps and padlocks. Fasten fence fabric to terminal posts with tension bands and to line posts and top and bottom rails with tie wires spaced at maximum

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375mm (15 inches). Bottom of fences shall extend to 25mm (one inch) above grade. Remove the fence when directed by the COR.

M-L. When a building and/or construction site is turned over to Contractor, Contractor shall accept entire responsibility including upkeep and maintenance therefore:

1. Contractor shall maintain a minimum temperature of 4 degrees C (40 degrees F) at all times, except as otherwise specified.
2. Contractor shall maintain in operating condition existing fire protection and alarm equipment. In connection with fire alarm equipment, Contractor shall make arrangements for pre-inspection of site with Fire Department or Company (Department of Veterans Affairs or municipal) whichever will be required to respond to an alarm from Contractor's employee or watchman.

~~SPEC WRITER NOTE: If anticipated work involves a serious disruption of services, the permitted down time and other limitations should be mentioned in the specification.~~

N-M. Utilities Services: Maintain existing utility services for the ~~// Medical Center // Cemetery //~~ at all times. Provide temporary facilities, labor, materials, equipment, connections, and utilities to assure uninterrupted services. Where necessary to cut existing water, steam, gases, sewer or air pipes, or conduits, wires, cables, etc. of utility services or of fire protection systems and communications systems (including telephone), they shall be cut and capped at suitable places where shown; or, in absence of such indication, where directed by the COR.

1. No utility service such as water, gas, steam, sewers or electricity, or fire protection systems and communications systems may be interrupted without prior approval of the ~~// COR- // [Chief Engineer] [Chief of Facilities Management]~~.
Electrical work shall be accomplished with all affected

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- circuits or equipment de-energized. When an electrical outage cannot be accomplished, work on any energized circuits or equipment shall not commence without a detailed work plan, and the CORMedical Center Director's prior knowledge and written approval. Refer to specification Sections 26-05-11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, 27-05-11- REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS and 28-05-00, COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY for additional requirements.
2. Contractor shall submit a request to interrupt any such services to the COR, in writing, 7 days in advance of proposed interruption. Request shall state reason, date, exact time of, and approximate duration of such interruption.
 3. Contractor will be advised (in writing) of approval of request, or of which other date and/or time such interruption will cause least inconvenience to operations of the Medical Center // Cemetery //. Interruption time approved by the CORMedical Center may occur at other than Contractor's normal working hours.
 4. Major interruptions of any system must be requested, in writing, at least 15 calendar days prior to the desired time and shall be performed as directed by the COR.
 5. In case of a contract construction emergency, service will be interrupted on approval of the COR. Such approval will be confirmed in writing as soon as practical.
 6. Whenever it is required that a connection fee be paid to a public utility provider for new permanent service to the construction project, for such items as water, sewer, electricity, gas or steam, payment of such fee shall be the responsibility of the Government and not the Contractor.
- 0-N. Abandoned Lines: All service lines such as wires, cables, conduits, ducts, pipes and the like, and their hangers or supports, shall be removed back to their source. Those which are

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indicated to be abandoned but are not required to be entirely removed, shall be sealed, capped or plugged at the main, branch or panel they originate from. The lines shall not be capped in finished areas, but shall be removed and sealed, capped or plugged in ceilings, within furred spaces, in unfinished areas, or within walls or partitions; so that they are completely behind the finished surfaces.

~~P.O.~~ To minimize interference of construction activities with flow of Medical Center traffic, comply with the following:

1. Keep roads, walks and entrances to grounds, to parking and to occupied areas of buildings clear of construction materials, debris and standing construction equipment and vehicles. ~~//~~

~~Wherever excavation for new utility lines cross existing roads, at least one lane must be open to traffic at all times with approval. //~~

- ~~2. Method and scheduling of required cutting, altering and removal of existing roads, walks and entrances must be approved by the COR.~~

~~Q.P.~~ Coordinate the work for this contract with other construction operations as directed by COR. This includes the scheduling of traffic and the use of roadways, ~~as specified in Article, USE OF ROADWAYS.~~

~~SPEC WRITER NOTE: Use following Subparagraph O on existing cemetery projects only.~~

~~R. Coordination of Construction with Cemetery Director: The burial activities at a National Cemetery shall take precedence over construction activities. The Contractor must cooperate and coordinate with the Cemetery Director, through the COR, in arranging construction schedule to cause the least possible interference with cemetery activities in actual burial areas. Construction noise during the interment services shall not disturb the service. Trucks and workers shall not pass through the service area during this period:~~

- ~~1. The Contractor is required to discontinue his work sufficiently in advance of Easter Sunday, Mother's Day, Father's Day, Memorial Day, Veteran's Day and/or Federal holidays, to permit him to clean up all areas of operation adjacent to existing burial plots before these dates.~~
- ~~2. Cleaning up shall include the removal of all equipment, tools, materials and debris and leaving the areas in a clean, neat condition.~~

~~SPEC WRITER NOTE: Include the following references to Supply Representative only on STATION LEVEL PROJECTS.~~

1.7 ALTERATIONS

- A. Survey: Before any work is started, the Contractor shall make a thorough survey with the COR and a representative of VA Supply Service, of ~~// buildings //~~ areas of buildings ~~//~~ in which alterations occur and areas which are anticipated routes of access, and furnish a report, signed by ~~// both, //~~ all three, ~~//~~ to the COR. This report shall list by ~~rooms and spaces~~ roof areas:
 1. Existing condition and types of ~~resilient flooring, doors, windows~~ roofing, walls and other surfaces not required to be altered throughout ~~//~~ affected areas of the ~~// building. //~~ buildings. ~~//~~
 2. Existence and conditions of items such as plumbing fixtures and accessories, electrical fixtures, mechanical equipment, ~~venetian blinds, shades,~~ etc., required by drawings to be either reused or relocated, or both.
 3. Shall note any discrepancies between drawings and existing conditions at site.
 4. Shall designate areas for working space, materials storage and routes of access to areas within buildings where alterations occur and which have been agreed upon by Contractor and COR.
- B. Any items required by drawings to be either reused or relocated or both, found during this survey to be nonexistent, or in

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opinion of ~~//the~~ COR ~~// and/or Supply Representative //~~, to be in such condition that their use is impossible or impractical, shall be furnished and/or replaced by Contractor with new items in accordance with specifications which will be furnished by Government. Provided the contract work is changed by reason of this subparagraph B, the contract will be modified accordingly, under provisions of clause entitled "DIFFERING SITE CONDITIONS" (FAR 52.236-2) and "CHANGES" (FAR 52.243-4).

C. Re-Survey: Thirty days before expected partial or final inspection date, the Contractor and COR together shall make a thorough re-survey of the areas of buildings involved. They shall furnish a report on conditions then existing, of ~~resilient flooring, doors, windows~~roofing, walls and other surfaces as compared with conditions of same as noted in first condition survey report:

1. Re-survey report shall also list any damage caused by Contractor to such ~~flooring and other~~items and surfaces, despite protection measures; and, will form basis for determining extent of repair work required of Contractor to restore damage caused by Contractor's workers in executing work of this contract.

D. Protection: Provide the following protective measures:

1. Wherever existing roof surfaces are disturbed they shall be protected against water infiltration. In case of leaks, they shall be repaired immediately upon discovery.
2. Temporary protection against damage for portions of existing structures and grounds where work is to be done, materials handled and equipment moved and/or relocated.
3. Protection of interior of existing structures at all times, from damage, dust and weather inclemency. Wherever work is performed, floor surfaces that are to remain in place shall be adequately protected prior to starting work, and this

protection shall be maintained intact until all work in the area is completed.

1.8 DISPOSAL AND RETENTION

A. Materials and equipment accruing from work removed and from demolition of buildings or structures, or parts thereof, shall be disposed of as follows:

1. Reserved items which are to remain property of the Government are ~~//~~identified by attached tags ~~//~~or ~~//~~noted on drawings or in specifications~~//~~ as items to be stored. Items that remain property of the Government shall be removed or dislodged from present locations in such a manner as to prevent damage which would be detrimental to re-installation and reuse. Store such items where directed by COR.
2. Items not reserved shall become property of the Contractor and be removed by Contractor from the~~//~~Medical Center~~// Cemetery~~~~//~~.
3. Items of portable equipment and furnishings located in rooms and spaces in which work is to be done under this contract shall remain the property of the Government. When rooms and spaces are vacated by the Department of Veterans Affairs during the alteration period, such items which are NOT required by drawings and specifications to be either relocated or reused will be removed by the Government in advance of work to avoid interfering with Contractor's operation.

~~SPEC WRITER NOTE: Use the following subparagraph only when there are existing PCB transformers or capacitors to be removed and disposed of by the Contractor. Verify the need for this subparagraph with the Team Electrical Engineer.~~

- ~~4. //PCB Transformers // and Capacitors // : The Contractor shall be responsible for disposal of the Polychlorinated Biphenyl (PCB) transformers // and capacitors //. The transformers // and capacitors // shall be taken out of service and handled in~~

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~~accordance with the procedures of the Environmental Protection Agency (EPA) and the Department of Transportation (DOT) as outlined in Code of Federal Regulation (CFR), Titled 40 and 49 respectively. The EPA's Toxic Substance Control Act (TSCA) Compliance Program Policy Nos. 6-PCB-6 and 6-PCB-7 also apply. Upon removal of PCB transformers // and capacitors // for disposal, the "originator" copy of the Uniform Hazardous Waste Manifest (EPA Form 8700-22), along with the Uniform Hazardous Waste Manifest Continuation Sheet (EPA Form 8700-22A) shall be returned to the COR who will annotate the contract file and transmit the Manifest to the Medical Center's // Cemetery's // Chief.~~

~~a) Copies of the following listed CFR titles may be obtained from the Government Printing Office:~~

~~40 CFR 261.....Identification and Listing of Hazardous Waste~~

~~40 CFR 262.....Standards Applicable to Generators of Hazardous Waste~~

~~40 CFR 263.....Standards Applicable to Transporters of Hazardous Waste~~

~~40 CFR 761.....PCB Manufacturing, Processing, Distribution in Commerce, and use Prohibitions~~

~~49 CFR 172.....Hazardous Material tables and Hazardous Material Communications Regulations~~

~~49 CFR 173.....Shippers - General Requirements for Shipments and Packaging~~

~~49 CFR 173.....Subpart A General~~

~~49 CFR 173.....Subpart B Preparation of Hazardous Material for Transportation~~

~~49 CFR 173.....Subpart J Other Regulated Material; Definitions and Preparation~~

~~TSCA.....Compliance Program Policy Nos. 6-PCB-6 and 6-PCB-7//~~

~~SPEC WRITER NOTE: Tailor or delete the following Article and paragraphs if the scope of work encompasses only interior work.~~

1.9 ~~//~~PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS (FAR 52.236-9)

- A. The Contractor shall preserve and protect all structures, equipment, and vegetation (such as trees, shrubs, and grass) on or adjacent to the work site, which are not to be removed and which do not unreasonably interfere with the work required under this contract. The Contractor shall only remove trees when specifically authorized to do so, and shall avoid damaging vegetation that will remain in place. If any limbs or branches of trees are broken during contract performance, or by the careless operation of equipment, or by workers, the Contractor shall trim those limbs or branches with a clean cut and paint the cut with a tree-pruning compound as directed by the COR.
- B. The Contractor shall protect from damage all existing improvements and utilities at or near the work site and on adjacent property of a third party, the locations of which are made known to or should be known by the Contractor. The Contractor shall repair any damage to those facilities, including those that are the property of a third party, resulting from failure to comply with the requirements of this contract or failure to exercise reasonable care in performing the work. If the Contractor fails or refuses to repair the damage promptly, the COR may have the necessary work performed and charge the cost to the Contractor.

~~C. Refer to Section 01 57 19, TEMPORARY ENVIRONMENTAL CONTROLS, for additional requirements on protecting vegetation, soils and the environment. Refer to Articles, "Alterations", "Restoration", and "Operations and Storage Areas" for additional instructions concerning repair of damage to structures and site improvements.~~

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~~SPEC WRITER NOTE: Use following paragraph when the disturbed area on the site exceeds one acre. Confirm with the Project Manager.~~

D.C. Refer to FAR clause 52.236-7, "Permits and Responsibilities," which is included in General Conditions. A National Pollutant Discharge Elimination System (NPDES) permit is required for this project. The Contractor is considered an "operator" under the permit and has extensive responsibility for compliance with permit requirements. VA will make the permit application available at the (appropriate medical center) office. The apparent low bidder, contractor and affected subcontractors shall furnish all information and certifications that are required to comply with the permit process and permit requirements. Many of the permit requirements will be satisfied by completing construction as shown and specified. Some requirements involve the Contractor's method of operations and operations planning and the Contractor is responsible for employing best management practices. The affected activities often include, but are not limited to the following:

1. Designating areas for equipment maintenance and repair;
2. Providing waste receptacles at convenient locations and provide regular collection of wastes;
3. Locating equipment wash down areas on site, and provide appropriate control of wash-waters;
4. Providing protected storage areas for chemicals, paints, solvents, fertilizers, and other potentially toxic materials; and
5. Providing adequately maintained sanitary facilities. ~~++~~

1.10 RESTORATION

- A. Remove, cut, alter, replace, patch and repair existing work as necessary to install new work. Except as otherwise shown or specified, do not cut, alter or remove any structural work, and do not disturb any ducts, plumbing, steam, gas, or electric work

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without approval of the COR. Existing work to be altered or extended and that is found to be defective in any way, shall be reported to the COR before it is disturbed. Materials and workmanship used in restoring work, shall conform in type and quality to that of original existing construction, except as otherwise shown or specified.

- B. Upon completion of contract, deliver work complete and undamaged. Existing work (walls, ceilings, partitions, floors, mechanical and electrical work, lawns, paving, roads, walks, etc.) disturbed or removed as a result of performing required new work, shall be patched, repaired, reinstalled, or replaced with new work, and refinished and left in as good condition as existed before commencing work.
- C. At Contractor's own expense, Contractor shall immediately restore to service and repair any damage caused by Contractor's workers to existing piping and conduits, wires, cables, etc., of utility services or of fire protection systems and communications systems (including telephone) which are not scheduled for discontinuance or abandonment.
- D. Expense of repairs to such utilities and systems not shown on drawings or locations of which are unknown will be covered by adjustment to contract time and price in accordance with clause entitled "CHANGES" (FAR 52.243-4) and "DIFFERING SITE CONDITIONS" (FAR 52.236-2).

~~SPEC WRITER NOTE: Check with Project Director to determine if soil report is available at the Medical Center. Delete if not exterior work.~~

~~1.11 // PHYSICAL DATA - SOIL CONDITIONS~~

- ~~A. Data and information furnished or referred to below is for the Contractor's information. The Government shall not be responsible for any interpretation of or conclusion drawn from the data or information by the Contractor.~~

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~~SPEC WRITER NOTE: Insert name and address of testing facility performing the soils investigation work.~~

~~1. The indications of physical conditions on the drawings and in the specifications are the result of site investigations by _____.~~

~~{FAR 52.236-4}~~

~~B. Subsurface conditions have been developed by core borings and test pits. Logs of subsurface exploration are shown diagrammatically on drawings.~~

~~C. A copy of the soil report will be made available for inspection by bidders upon request to the Engineering Officer at the VA Medical Center, _____ and shall be considered part of the contract documents.~~

~~D. Government does not guarantee that other materials will not be encountered nor that proportions, conditions or character of several materials will not vary from those indicated by explorations. Bidders are expected to examine site of work and logs of borings; and, after investigation, decide for themselves character of materials and make their bids accordingly. Upon proper application to Department of Veterans Affairs, bidders will be permitted to make subsurface explorations of their own at site. //~~

~~1.12 //PROFESSIONAL SURVEYING SERVICES~~

~~A. A registered professional land surveyor or registered civil engineer whose services are retained and paid for by the Contractor shall perform services specified herein and in other specification sections. The Contractor shall certify that the land surveyor or civil engineer is not one who is a regular employee of the Contractor, and that the land surveyor or civil engineer has no financial interest in this contract. //~~

~~SPEC WRITER NOTE: Delete inapplicable portions of above and following article. Do not include for small additions to buildings, such as~~

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~~stairways, etc. For one story construction omit the words enclosed in paragraph 1.14, D, by //-----//.~~

~~1.13 //LAYOUT OF WORK~~

~~A. The Contractor shall lay out the work from Government established base lines and bench marks, indicated on the drawings, and shall be responsible for all measurements in connection with the layout. The Contractor shall furnish, at Contractor's own expense, all stakes, templates, platforms, equipment, tools, materials, and labor required to lay out any part of the work. The Contractor shall be responsible for executing the work to the lines and grades that may be established or indicated by the Contracting Officer. The Contractor shall also be responsible for maintaining and preserving all stakes and other marks established by the Contracting Officer until authorized to remove them. If such marks are destroyed by the Contractor or through Contractor's negligence before their removal is authorized, the Contracting Officer may replace them and deduct the expense of the replacement from any amounts due or to become due to the Contractor.~~

~~(FAR 52.236-17)~~

~~B. Establish and plainly mark // center lines for each building and corner of column lines and/or addition to each existing building, // lines for each gravesite control monument, // and such other lines and grades that are reasonably necessary to properly assure that location, orientation, and elevations established for // each such structure and/or addition, // roads, // parking lots, // gravesite control monuments, // are in accordance with lines and elevations shown on contract drawings.~~

~~C. Following completion of general mass excavation and before any other permanent work is performed, establish and plainly mark (through use of appropriate batter boards or other means) sufficient additional survey control points or system of points as may be necessary to assure proper alignment, orientation, and~~

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~~grade of all major features of work. Survey shall include, but not be limited to, location of lines and grades of footings, exterior walls, center lines of columns in both directions, major utilities and elevations of floor slabs:~~

~~1. Such additional survey control points or system of points thus established shall be checked and certified by a registered land surveyor or registered civil engineer. Furnish such certification to the COR before any work (such as footings, floor slabs, columns, walls, utilities and other major controlling features) is placed.~~

~~D. //During progress of work, and particularly as work progresses from floor to floor, Contractor shall have line grades and plumbness of all major form work checked and certified by a registered land surveyor or registered civil engineer as meeting requirements of contract drawings. Furnish such certification to the COR before any major items of concrete work are placed. In addition, // Contractor shall // also // furnish to the COR certificates from a registered land surveyor or registered civil engineer that the following work is complete in every respect as required by contract drawings.~~

~~1. Lines of each building and/or addition.~~

~~2. Elevations of bottoms of footings and tops of floors of each building and/or addition.~~

~~3. Lines and elevations of sewers and of all outside distribution systems.~~

~~SPEC WRITER NOTE: Use following for cemetery projects as required.~~

~~4. //Lines of grave plot documentation. //~~

~~5. Lines of elevations of all swales and interment areas.~~

~~6. Lines and elevations of roads, streets // and parking lots. //~~

~~SPEC WRITER NOTE: Use following Paragraph E on non cemetery projects and use Paragraph E* on cemetery projects.~~

- ~~E. Whenever changes from contract drawings are made in line or grading requiring certificates, record such changes on a reproducible drawing bearing the registered land surveyor or registered civil engineer seal, and forward these drawings upon completion of work to COR.~~
- ~~F. Upon completion of the work, the Contractor shall furnish the COR one electronic copy and reproducible drawings at the scale of the contract drawings, showing the finished grade on the grid developed for constructing the work, including burial monuments and fifty foot stationing along new road centerlines. These drawings shall bear the seal of the registered land surveyor or registered civil engineer.~~
- ~~G. The Contractor shall perform the surveying and layout work of this and other articles and specifications in accordance with the provisions of Article "Professional Surveying Services". //~~

~~1.141.11~~ -AS-BUILT DRAWINGS

- A. The contractor shall maintain two full size sets of as-built drawings which will be kept current during construction of the project, to include all contract changes, modifications and clarifications.
- B. All variations shall be shown in the same general detail as used in the contract drawings. To ensure compliance, as-built drawings shall be made available for the COR review, as often as requested.
- C. Contractor shall deliver two approved completed sets of as-built drawings in the electronic version (scanned PDF) to the ~~//COR //~~ ~~[Chief Engineer][Chief of Facilities Management]~~ within 15 calendar days after each completed phase and after the acceptance of the project by the COR.
- D. Paragraphs A, B, & C shall also apply to all shop drawings.

1.151.12 **WARRANTY MANAGEMENT**

A. Warranty Management Plan: Develop a warranty management plan which contains information relevant to FAR 52.246-21 Warranty of Construction in ~~//~~~~---~~~~//~~ at least 30 days before the planned pre-warranty conference, submit ~~//one set//~~ ~~//~~~~---~~~~//~~ **four sets**~~//~~ of the warranty management plan. Include within the warranty management plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan must be in narrative form and contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesman, or of engineering background, not necessarily familiar with this contract. The term "status" as indicated below must include due date and whether item has been submitted or was approved. Warranty information made available during the construction phase must be submitted to the COR for approval prior to each monthly invoice for payment. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period will begin on the date of the project acceptance and continue for the product warranty period. A joint 4 month and 9 month warranty inspection will be conducted, measured from time of acceptance, by the Contactor and the COR. Include in the warranty management plan, but not limited to, the following:

1. Roles and responsibilities of all personnel associated with the warranty process, including points of contact and telephone numbers within the company of the Contractor, subcontractors, manufacturers or suppliers involved.
2. Furnish with each warranty the name, address and telephone number of each of the guarantor's representatives nearest project location.
3. Listing and status of delivery of all Certificates of Warranty for extended warranty items, to include roofs, HVAC balancing, pumps, motors, transformers and for all commissioned systems

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such as fire protection and alarm systems, sprinkler systems and lightning protection systems, etc.

4. A list for each warranted equipment item, feature of construction or system indicating:
 - a. Name of item.
 - b. Model and serial numbers.
 - c. Location where installed.
 - d. Name and phone numbers of manufacturers and suppliers.
 - e. Name and phone numbers of manufacturers or suppliers.
 - f. Names, addresses and phone numbers of sources of spare parts.
 - g. Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have extended warranties must be indicated with separate warranty expiration dates.
 - h. Starting point and duration of warranty period.
 - i. Summary of maintenance procedures required to continue the warranty in force.
 - j. Cross-reference to specific pertinent Operation and Maintenance manuals.
 - k. Organizations, names and phone numbers of persons to call for warranty service.
 - l. Typical response time and repair time expected for various warranted equipment.
5. The plans for attendance at the 4 and 9-month post construction warranty inspections conducted by the government.
6. Procedure and status of tagging of all equipment covered by extended warranties.
7. Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty and/or safety reasons.

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B. Performance & Payment Bonds: The Performance & Payment Bonds must remain effective throughout the construction period.

1. In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the COR will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.
2. In the event sufficient funds are not available to cover the construction warranty work performed by the Government at the contractor's expenses, the COR will have the right to recoup expenses from the bonding company.
3. Following oral or written notification of required construction warranty repair work, the Contractor shall respond in a timely manner. Written verification will follow oral instructions. Failure to respond will be cause for the COR to proceed against the Contractor.

C. Pre-Warranty Conference: Prior to contract completion, and at a time designated by the COR, the Contractor shall meet with the COR to develop a mutual understanding with respect to the requirements of this section. Communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the COR for the execution of the construction warranty will be established/ reviewed at this meeting. In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contract will be located within the local service area of the warranted construction, be continuously available and be

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responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in conjunction with other portions of this provision.

D. Contractor's Response to Construction Warranty Service Requirements:

E. Following oral or written notification by the Contracting Officer, the Contractor shall respond to construction warranty service requirements in accordance with the "Construction Warranty Service Priority List" and the three categories of priorities listed below. Submit a report on any warranty item that has been repaired during the warranty period. Include within the report the cause of the problem, date reported, corrective action taken, and when the repair was completed. If the Contractor does not perform the construction warranty within the timeframe specified, the Government will perform the work and back charge the construction warranty payment item established.

1. First Priority Code 1. Perform onsite inspection to evaluate situation, and determine course of action within 4 hours, initiate work within 6 hours and work continuously to completion or relief.
2. Second Priority Code 2. Perform onsite inspection to evaluate situation, and determine course of action within 8 hours, initiate work within 24 hours and work continuously to completion or relief.
3. Third Priority Code 3. All other work to be initiated within 3 work days and work continuously to completion or relief.
4. The "Construction Warranty Service Priority List" is as follows:

~~SPEC WRITER NOTE: List below is not intended to be all inclusive of areas requiring attention for warranty services. List shall serve as a template only. Spec writer shall edit and alter list as needed to have~~

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~~Contractor to address warranty
services per facility needs.~~

- a) Code 1-Life Safety Systems
 - 1) Fire suppression systems.
 - 2) Fire alarm system(s).
- b) Code 1-Air Conditioning Systems
 - 1) Air conditioning leak in part of the building, if causing damage.
 - 2) Air conditioning system not cooling properly.
- c) Code 1 Doors
 - 1) Overhead doors not operational, causing a security, fire or safety problem.
 - 1) Interior, exterior personnel doors or hardware, not functioning properly, causing security, fire or safety problem.
- d) Code 3-Doors
 - 1) Overhead doors not operational.
 - 2) Interior/exterior personnel doors or hardware not functioning properly.
- e) Code 1-Electrical
 - 1) Power failure (entire area or any building operational after 1600 hours).
 - 2) Security lights.
 - 3) Smoke detectors.
- f) Code 2-Electrical
 - 1) Power failure (no power to a room or part of building).
Receptacle and lights not operational (in a room or part of building).
- g) Code 3-Electrical
 - 1) Exterior lights not operational.
- h) Code 1-Gas
 - 1) Leaks and pipeline breaks.
- i) Code 1-Heat

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1) Power failure affecting heat.

j) Code 1-Plumbing

1) Hot water heater failure.

2) Leaking water supply pipes

k) Code 2-Plumbing

1) Flush valves not operating properly

2) Fixture drain, supply line or any water pipe leaking.

3) Toilet leaking at base.

l) Code 3- Plumbing

1) Leaky faucets.

m) Code 3-Interior

1) Floors damaged.

2) Paint chipping or peeling.

3) Casework damaged.

n) Code 1-Roof Leaks

1) Damage to property is occurring.

o) Code 2-Water (Exterior)

1) No water to facility.

p) Code 2-Water (Hot)

1) No hot water in portion of building listed.

q) Code 3

1) All work not listed above.

F. Warranty Tags: At the time of installation, tag each warranted item with a durable, oil and water-resistant tag approved by the COR. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also submit ~~two~~ record copies of the warranty tags showing the layout and design. The date of acceptance and the QC signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

Warranty Tags
Type of product/material

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Warranty Tags
Model number
Serial number
Contract number
Warranty period from/to
Inspector's signature
Construction Contractor
Address
Telephone number
Warranty Contact
Address
Telephone number
Warranty response time priority code

~~SPEC WRITER NOTE: Determine necessary paragraphs from below and delete non-applicable portions.~~

1-161.13 USE OF ROADWAYS

A. For hauling, use only established public roads and roads on ~~//~~ Medical Center ~~// Cemetery //~~ property and, when authorized by the COR, such temporary roads which are necessary in the performance of contract work. Temporary roads shall be constructed and restoration performed by the Contractor at Contractor's expense. When necessary to cross curbing, sidewalks, or similar construction, they must be protected by well-constructed bridges.

~~B. //When new permanent roads are to be a part of this contract, Contractor may construct them immediately for use to facilitate building operations. These roads may be used by all who have business thereon within zone of building operations. //~~

~~C. //When certain buildings (or parts of certain buildings) are required to be completed in advance of general date of completion, all roads leading thereto must be completed and~~

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~~available for use at time set for completion of such buildings or parts thereof. //~~

~~SPEC WRITER NOTE: Before including COR Field Office, check with Project Director to determine if Medical Center can provide office space for the COR in an existing building. Note these facilities are typically only required for major projects or others as authorized by CFM.~~

~~1.17 // RESIDENT ENGINEER'S FIELD OFFICE~~

- ~~A. The Contractor shall, within fifteen (15) days after receipt of Notice to Proceed, provide where shown on the drawings a temporary field office, furniture, and two inch deep gravel surfaced area for use of the COR. Office and furniture shall be new.~~
- ~~B. The field office shall provide not less than // 67 square meters (720 gross square feet) // 134 square meters (1440 gross square feet) // of floor area in one unit. Installation of the office shall meet all local codes.~~
- ~~C. Provide office with two, 900 mm (three foot) wide exterior doors, including hardware and OSHA approved platform and stairs leading to grade.~~
- ~~D. Enclose the entire perimeter of the office from the floor to the ground and finish to match exterior. Provide R7 insulation and seal tight to ground with a painted 19 mm (3/4 inch) exterior grade plywood skirt.~~
- ~~E. Exterior finishes shall be manufacturer's standards.~~
- ~~F. Provide floor, wall, and roof with not less than R5 insulation.~~
- ~~G. Interior finishes shall consist of resilient flooring, plywood paneling or painted wallboard on walls, and acoustical tile ceilings. Interior doors may be either painted or stained.~~
- ~~H. Interior shall be subdivided with full height partitions to provide // one office, // two offices, // one sample room, // one toilet // two separate toilets //. Provide each space with 900 mm~~

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~~(three foot) wide door with master keyed locks. Section off an area with a low partition and counter for the secretary's desk // s //.~~

- ~~I. Provide 750 mm (2-1/2 feet) wide by 900 mm (3 feet) high operable windows; two in each room (none required in sample room), except provide only one 600 mm (2 foot) high window in toilet room(s). Window openings shall be fitted with security bars to prevent any forced entry. The door//s// of field office shall have a hasp and padlock and also deadbolts keyed from both sides.~~
- ~~J. Provide sufficient fluorescent lighting in each room to deliver 750 lux (70 foot-candles) of light at desk top height without the aid of daylight. Provide one light switch in each room.~~
- ~~K. Provide one duplex receptacle in each wall of each room. If a wall is 3.0 m (10 feet) long or more, provide two receptacles for each 3.0 m (10 feet), or portion thereof, of wall. Provide two duplex receptacles in low partition at secretary's desk.~~
- ~~L. The Contractor shall provide the following:~~
 - ~~1. Electricity, hot and cold water, and necessary utility services (except telephone).~~
 - ~~2. All necessary piping, power circuits network cabling, cat 5e or better cabling for phones and computers, electrical fixtures, lighting, and other items necessary to provide a habitable structure for the purpose intended. The number of network and electrical receptacles will be as per attached drawing of the field office.~~
 - ~~3. Thermostatically controlled, centralized heating and air conditioning system designed to maintain the temperature between 21 and 27 degrees C (70 and 80 degrees F) with 50 percent relative humidity maintained during the air conditioning season.~~
 - ~~4. One water closet, lavatory, mirror, toilet paper dispenser, paper towel dispenser, soap dispenser, towel bar, and two-prong coat hooks for each toilet room.~~

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~~SPEC WRITER NOTE: Verify with Project Manager.~~

~~5. //The contractor to install a suitable alarm system for the field office //.~~

~~M. Contractor shall, for the duration of the COR occupancy, provide the following:~~

~~1. Satisfactory conditions in and around the field office and parking area.~~

~~2. Maintenance of gravel surfaced area, including the area for parking, in an acceptable condition for vehicle and foot traffic at all times.~~

~~SPEC WRITER NOTE: Confirm with Project Manager as to who pays for utilities.~~

~~3. Maintenance of utility services.~~

~~SPEC WRITER NOTE: Delete following subparagraph 4 unless specifically instructed by the Project Manager to leave it in.~~

~~4. //Daily janitorial services and supplies (toilet paper, soap, etc.). //~~

~~5. Potable water, fuel and electric power for normal office uses, including lights, heating and air conditioning.~~

~~N. The Contractor shall provide the following new items:~~

~~SPEC WRITER NOTE: Use first list of furniture 67 square meters for 720 square foot office. Second list for 134 square meters (1440 square foot) office.~~

~~QUANTITY REQUIRED~~

~~//1 workstation with adjustable keying desk or drawer 738 mm H x 1.5 m W x 760 mm D (size 29 1/2" H x 60" W x 30" D)~~

~~1 Printer stand 663 mm H x 1.5 m W x 750 mm D (size 26 1/2" H x 60" W x 30" D)~~

~~3 Office desks, double pedestal~~

~~1 Conference table 900 mm x 1.8 m (size 3' x 6')~~

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~~1 Plan table 1.2 m x 2.1 m (4' x 7')~~
~~3 Work tables 750 mm x 1.8 m (folding 30" x 72")~~
~~1 Office chair~~
~~4 Swivel chairs with arms~~
~~6 Conference chairs (armless & folding)~~
~~2 Arm Chairs~~
~~4 Lockable 5 drawer file cabinets, letter size~~
~~1 Drawing rack, with 12 750 mm (12 30 inch) "Plan Hold" drawing holders, freestanding~~
~~1 Shelves for sample room, 7 adjustable Shelves, 305 mm W x 900 mm L (12" W x 3' L)~~
~~3 Bookcases~~
~~1 Electric water cooler~~
~~1 Metal storage cabinet, 900 mm x 450 mm x 1.8 m (36" x 18" x 72") with six shelves~~
~~2 workstations with adjustable keying desk or drawer 738 mm H x 1.5 m W 750 mm D (size 29 1/2" H x 60" W x 30" D)~~
~~2 Printer stands 738 mm H x 1500 mm W x 750 mm D (size 29 1/2" H x 60" H x 30" D)~~
~~7 Office desks, double pedestal~~
~~2 Conference tables 900 mm x 1800 mm (size 3' x 6')~~
~~1 Plan table 1200 mm x 6 meters (4' x 20')~~
~~7 Work tables 750 mm x 1800 mm (folding 30" x 72")~~
~~2 Office chairs~~
~~7 Swivel chairs with arms~~
~~12 Conference chairs (armless and folding)~~
~~7 Arm chairs~~
~~8 Lockable 5 drawer file cabinets, letter size~~
~~2 Drawing racks, each with 12 750 mm (12 30 inch) "Plan Hold" drawing holders, freestanding~~
~~7 Bookcases~~
~~1 Electric water cooler~~

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~~4 Shelves for sample 900 mm x 450 mm x 1.8 m (36" x 18" x 72")
high, 7 adjustable shelves//~~

~~SPEC WRITER NOTE: Delete following
paragraph 0 if COR field office will
not require relocating during the
tenure of this contract.~~

~~0. COR field office and facilities shall be relocated once after its
initial installation at the Contractor's expense. Relocation
consists of moving the field office and facilities to a location
within the VA site designated by the COR together with providing
and maintaining utilities, parking area, sanitary facilities and
janitorial service in new location until completion and final
acceptance of project.~~

~~SPEC WRITER NOTE: Check with the
Project Manager to determine if field
office (trailer) is to become property
of the Government. This condition will
usually occur on phased projects
(projects involving more than one
construction contract).~~

~~P. At the completion of all work, including the punch list, the COR
field office and facilities shall become the property of the //
Contractor and Contractor shall remove same, including utility
connections, from the // Medical Center // Cemetery//. The site
shall be restored to original condition and finished in
accordance with contract requirements. // Government and be left
intact, including utility connections, for future use by
Department of Veterans Affairs. // All 5 drawer file cabinets
provided shall become the property of the Government.~~

~~Q. The Contractor shall furnish floor plans for approval by the COR
prior to furnishing the field office. //~~

1.181.14 TEMPORARY USE OF MECHANICAL AND ELECTRICAL EQUIPMENT

~~A. Use of new installed mechanical and electrical equipment to
provide heat, ventilation, plumbing, light and power will be
permitted subject to written approval and compliance with the
following provisions:~~

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- ~~1. Permission to use each unit or system must be given by COR in writing. If the equipment is not installed and maintained in accordance with the written agreement and following provisions, the COR will withdraw permission for use of the equipment.~~
- ~~2. Electrical installations used by the equipment shall be completed in accordance with the drawings and specifications to prevent damage to the equipment and the electrical systems, i.e. transformers, relays, circuit breakers, fuses, conductors, motor controllers and their overload elements shall be properly sized, coordinated and adjusted. Installation of temporary electrical equipment or devices shall be in accordance with NFPA 70, National Electrical Code, (2014 Edition), Article 590, Temporary Installations. Voltage supplied to each item of equipment shall be verified to be correct and it shall be determined that motors are not overloaded. The electrical equipment shall be thoroughly cleaned before using it and again immediately before final inspection including vacuum cleaning and wiping clean interior and exterior surfaces.~~
- ~~3. Units shall be properly lubricated, balanced, and aligned. Vibrations must be eliminated.~~
- ~~4. Automatic temperature control systems for preheat coils shall function properly and all safety controls shall function to prevent coil freeze-up damage.~~
- ~~5. The air filtering system utilized shall be that which is designed for the system when complete, and all filter elements shall be replaced at completion of construction and prior to testing and balancing of system.~~
- ~~6. All components of heat production and distribution system, metering equipment, condensate returns, and other auxiliary facilities used in temporary service shall be cleaned prior to use; maintained to prevent corrosion internally and externally~~

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~~during use; and cleaned, maintained and inspected prior to acceptance by the Government. // Boilers, pumps, feedwater heaters and auxiliary equipment must be operated as a complete system and be fully maintained by operating personnel. Boiler water must be given complete and continuous chemical treatment. //~~

~~B. Prior to final inspection, the equipment or parts used which show wear and tear beyond normal, shall be replaced with identical replacements, at no additional cost to the Government.~~

~~C. This paragraph shall not reduce the requirements of the mechanical and electrical specifications sections.~~

~~D. Any damage to the equipment or excessive wear due to prolonged use will be repaired replaced by the contractor at the contractor's expense.~~

~~SPEC WRITER NOTE: Use following Article whenever Contractor may or may not use existing elevators. Information must be secured from Medical Center as to whether elevator is for exclusive or temporary use of Contractor and between what hours it may be used by Contractor.~~

1.191.15 // TEMPORARY USE OF EXISTING ELEVATORS

~~A. Contractor will not be allowed the use of existing elevators. Outside type hoist shall be used by Contractor for transporting materials and equipment.~~

~~B.A. // Use of existing // elevator // elevators // for handling building materials and Contractor's personnel will be permitted subject to following provisions:~~

- ~~1. Contractor makes all arrangements with the COR for use of elevators. The COR will ascertain that elevators are in proper condition. Contractor may use elevators Nos. _____ in Building Nos. _____ // for exclusive use // for daily use between the hours of _____. // and for special nonrecurring time intervals when permission is granted. Personnel for~~

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- operating elevators will not be provided by the Department of Veterans Affairs.
2. Contractor covers and provides maximum protection of following elevator components:
- a) Entrance jambs, heads soffits and threshold plates.
 - b) Entrance columns, canopy, return panels and inside surfaces of car enclosure walls.
 - c) Finish flooring.
3. Government will accept hoisting ropes of elevator and rope of each speed governor if they are worn under normal operation. However, if these ropes are damaged by action of foreign matter such as sand, lime, grit, stones, etc., during temporary use, they shall be removed and replaced by new hoisting ropes at the contractors expense.
4. If brake lining of elevators are excessively worn or damaged during temporary use, they shall be removed and replaced by new brake lining at the contractors expense.
5. All parts of main controller, starter, relay panel, selector, etc., worn or damaged during temporary use shall be removed and replaced with new parts at the contractors expense, if recommended by elevator inspector after elevator is released by Contractor.
6. Place elevator in condition equal, less normal wear, to that existing at time it was placed in service of Contractor as approved by the COR.

~~SPEC WRITER NOTE: Omit Paragraphs 1.18, A*, 4, 5, and 6 unless elevator is for exclusive use of Contractor and for not less than thirty days duration.~~

~~SPEC WRITER NOTE: Following Article 1.19 should be used when Contractor is permitted to use new elevators.~~

~~1.20 // TEMPORARY USE OF NEW ELEVATORS~~

~~A. The Contractor and his personnel shall be permitted use of new elevator(s) subject to the following provisions:~~

- ~~1. Contractor shall make arrangements with the COR for use of elevator(s). Contractor may obtain elevator(s) for exclusive use.~~
- ~~2. Prior to the use of elevator(s), the Contractor shall have the elevator(s) inspected and accepted by an ASME accredited, certified elevator safety inspector. The acceptance report shall be submitted to the COR.~~
- ~~3. Submit to the COR the schedule and procedures for maintaining equipment. Indicate the day or days of the week and total hours required for maintenance. A report shall be submitted to the COR monthly indicating the type of maintenance conducted, hours used, and any repairs made to the elevator(s).~~
- ~~4. The Contractor shall be responsible for enforcing the maintenance procedures as per VA and manufacturers recommendations and requirements.~~
- ~~5. During temporary use of elevator(s) all repairs, equipment replacement and cost of maintenance shall be the responsibility of the Contractor.~~
- ~~6. Personnel for operating elevator(s) shall not be provided by the Department of Veterans Affairs.~~
- ~~7. Contractor shall cover and provide maximum protection of the entire elevator(s) installation.~~
- ~~8. The Contractor shall arrange for the elevator company to perform operation of the elevator(s) so that an ASME accredited, certified elevator safety inspector can evaluate the equipment. The Contractor shall be responsible for any costs of the elevator company.~~
- ~~9. All elevator(s) parts worn or damaged during temporary use shall be removed and replaced with new parts at the contractor's expense. This shall be determined by an ASME~~

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~~accredited certified elevator safety inspector after temporary use and before acceptance by the Government. Submit report to the COR for approval.~~

~~10. Elevator shall be tested as required by the testing section of the elevator(s) specifications before acceptance by the Department of Veterans Affairs. The Contractor shall be responsible for all cost associated with testing and inspection. //~~

1.211.16 TEMPORARY TOILETS

~~SPEC WRITER NOTE: Check with the Project Manager to verify if any existing toilets can be used by contractor's workers.~~

A. Provide where directed, (for use of all Contractor's workers) ample temporary sanitary toilet accommodations with suitable sewer and water connections; or, when approved by COR, provide suitable dry closets where directed. Keep such places clean and free from flies and all connections and appliances connected therewith are to be removed prior to completion of contract, and premises left perfectly clean.

~~1. *Contractor may have for use of Contractor's workers, such toilet accommodations as may be assigned to Contractor by // Medical Center // Cemetery //. Contractor shall keep such places clean and be responsible for any damage done thereto by Contractor's workers. Failure to maintain satisfactory condition in toilets will deprive Contractor of the privilege to use such toilets.~~

~~SPEC WRITER NOTE: Consult Project Manager for use of paragraph 1.21. Contractor shall furnish utilities from commercial sources for construction on new sites. When practical, Contractor shall furnish utilities from commercial sources for new buildings and large additions on existing sites. Otherwise Medical Center will furnish utilities to~~

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~~Contractor at no cost for alterations
and remodeling. Check with Project
Manager concerning practical use of
meters.~~

1.221.17 - AVAILABILITY AND USE OF UTILITY SERVICES

- A. The Government shall make all reasonably required amounts of utilities available to the Contractor from existing outlets and supplies, as specified in the contract. The amount to be paid by the Contractor for chargeable electrical services shall be the prevailing rates charged to the Government. The Contractor shall carefully conserve any utilities furnished without charge.
- B. The Contractor, at Contractor's expense and in a workmanlike manner, in compliance with code and as satisfactory to the COR, shall install and maintain all necessary temporary connections and distribution lines, and all meters required to measure the amount of electricity used for the purpose of determining charges. Before final acceptance of the work by the Government, the Contractor shall remove all the temporary connections, distribution lines, meters, and associated paraphernalia and repair restore the infrastructure as required.
- C. Contractor shall install meters at Contractor's expense and furnish the ~~//~~Medical Center ~~// Cemetery //~~ a monthly record of the Contractor's usage of electricity as hereinafter specified.
- D. Heat: Furnish temporary heat necessary to prevent injury to work and materials through dampness and cold. Use of open salamanders or any temporary heating devices which may be fire hazards or may smoke and damage finished work, will not be permitted. Maintain minimum temperatures as specified for various materials:
 - 1. Obtain heat by connecting to the Medical Center heating distribution system.

~~SPEC WRITER NOTE: Coordinate with
Project Manager to confirm the
following:~~

- a) ~~//~~Steam is available at no cost to Contractor.~~//~~

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~~SPEC WRITER NOTE: Use sub-paragraph 1 in connection with existing projects only. Omit inappropriate sub-paragraphs.~~

E. Electricity (for Construction and Testing): Furnish all temporary electric services.

1. Obtain electricity by connecting to the Medical Center electrical distribution system. The Contractor shall meter and pay for electricity required for electric cranes and hoisting devices, electrical welding devices and any electrical heating devices providing temporary heat. Electricity for all other uses is available at no cost to the Contractor.

F. Water (for Construction and Testing): Furnish temporary water service.

1. Obtain water by connecting to the Medical Center water distribution system. Provide reduced pressure backflow preventer at each connection as per code. Water is available at no cost to the Contractor.
2. Maintain connections, pipe, fittings and fixtures and conserve water-use so none is wasted. Failure to stop leakage or other wastes will be cause for revocation (at COR discretion) of use of water from the Medical Center's system.

G. Fuel: Natural and LP gas and burner fuel oil required for boiler cleaning, normal initial boiler-burner setup and adjusting, and for performing the specified boiler tests will be furnished by the Government. Fuel required for prolonged boiler-burner setup, adjustments, or modifications due to improper design or operation of boiler, burner, or control devices shall be furnished and paid by the Contractor at Contractor's expense.

~~1.23 //NEW TELEPHONE EQUIPMENT~~

- ~~A. The contractor shall coordinate with the work of installation of telephone equipment by others. This work shall be completed before the building is turned over to VA. //~~

~~SPEC WRITER NOTE: Delete TESTS when not applicable to project. Check with Team Mechanical and Electrical Engineers.~~

~~1.24 //TESTS~~

- ~~A. As per specification section 23 05 93 the contractor shall provide a written testing //and commissioning// plan complete with component level, equipment level, sub-system level and system level breakdowns. The plan will provide a schedule and a written sequence of what will be tested, how and what the expected outcome will be. This document will be submitted for approval prior to commencing work. The contractor shall document the results of the approved plan and submit for approval with the as built documentation.~~
- ~~B. Pre-test mechanical and electrical equipment and systems and make corrections required for proper operation of such systems before requesting final tests. Final test will not be conducted unless pre-tested.~~
- ~~C. Conduct final tests required in various sections of specifications in presence of an authorized representative of the Contracting Officer. Contractor shall furnish all labor, materials, equipment, instruments, and forms, to conduct and record such tests.~~
- ~~D. Mechanical and electrical systems shall be balanced, controlled and coordinated. A system is defined as the entire system which must be coordinated to work together during normal operation to produce results for which the system is designed. For example, air conditioning supply air is only one part of entire system which provides comfort conditions for a building. Other related components are return air, exhaust air, steam, chilled water, refrigerant, hot water, controls and electricity, etc. Another example of a system which involves several components of different disciplines is a boiler installation. Efficient and acceptable boiler operation depends upon the coordination and~~

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~~proper operation of fuel, combustion air, controls, steam, feedwater, condensate and other related components.~~

~~E. All related components as defined above shall be functioning when any system component is tested. Tests shall be completed within a reasonable period of time during which operating and environmental conditions remain reasonably constant and are typical of the design conditions.~~

~~F. Individual test result of any component, where required, will only be accepted when submitted with the test results of related components and of the entire system. //~~

1.251.18 - INSTRUCTIONS

- A. Contractor shall furnish Maintenance and Operating manuals (hard copies and electronic) and verbal instructions when required by the various sections of the specifications and as hereinafter specified.
- B. Manuals: Maintenance and operating manuals and one compact disc (four hard copies and one electronic copy each) for each separate piece of equipment shall be delivered to the COR coincidental with the delivery of the equipment to the job site. Manuals shall be complete, detailed guides for the maintenance and operation of equipment. They shall include complete information necessary for starting, adjusting, maintaining in continuous operation for long periods of time and dismantling and reassembling of the complete units and sub-assembly components. Manuals shall include an index covering all component parts clearly cross-referenced to diagrams and illustrations. Illustrations shall include "exploded" views showing and identifying each separate item. Emphasis shall be placed on the use of special tools and instruments. The function of each piece of equipment, component, accessory and control shall be clearly and thoroughly explained. All necessary precautions for the operation of the equipment and the reason for each precaution shall be clearly set forth. Manuals must reference the exact model, style and size of the piece of

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equipment and system being furnished. Manuals referencing equipment similar to but of a different model, style, and size than that furnished will not be accepted.

- C. Instructions: Contractor shall provide qualified, factory-trained manufacturers' representatives to give detailed training to assigned Department of Veterans Affairs personnel in the operation and complete maintenance for each piece of equipment. All such training will be at the job site. These requirements are more specifically detailed in the various technical sections. Instructions for different items of equipment that are component parts of a complete system, shall be given in an integrated, progressive manner. All instructors for every piece of component equipment in a system shall be available until instructions for all items included in the system have been completed. This is to assure proper instruction in the operation of inter-related systems. All instruction periods shall be at such times as scheduled by the COR and shall be considered concluded only when the COR is satisfied in regard to complete and thorough coverage. The contractor shall submit a course outline with associated material to the COR for review and approval prior to scheduling training to ensure the subject matter covers the expectations of the VA and the contractual requirements. The Department of Veterans Affairs reserves the right to request the removal of, and substitution for, any instructor who, in the opinion of the COR, does not demonstrate sufficient qualifications in accordance with requirements for instructors above.

~~1.26 GOVERNMENT FURNISHED PROPERTY~~

- ~~A. The Government shall deliver to the Contractor, the Government furnished property shown on the // Schedule // drawings //.~~

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~~B. Equipment furnished by Government to be installed by Contractor will be furnished to Contractor at the // Medical Center // Cemetery //.~~

~~C. //Contractor shall be prepared to receive this equipment from Government and store or place such equipment not less than 90 days before Completion Date of project. //~~

~~1. //Storage space for equipment will be provided by the Government and the Contractor shall be prepared to unload and store such equipment therein upon its receipt at the // Medical Center // Cemetery. //~~

~~D. Notify Contracting Officer in writing, 60 days in advance, of date on which Contractor will be prepared to receive equipment furnished by Government. Arrangements will then be made by the Government for delivery of equipment.~~

~~1. Immediately upon delivery of equipment, Contractor shall arrange for a joint inspection thereof with a representative of the Government. At such time the Contractor shall acknowledge receipt of equipment described, make notations, and immediately furnish the Government representative with a written statement as to its condition or shortages.~~

~~2. Contractor thereafter is responsible for such equipment until such time as acceptance of contract work is made by the Government.~~

~~E. Equipment furnished by the Government will be delivered in a partially assembled (knock down) condition in accordance with existing standard commercial practices, complete with all fittings, fastenings, and appliances necessary for connections to respective services installed under contract. All fittings and appliances (i.e., couplings, ells, tees, nipples, piping, conduits, cables, and the like) necessary to make the connection between the Government furnished equipment item and the utility stub up shall be furnished and installed by the contractor at no additional cost to the Government.~~

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~~F. Completely assemble and install the Government furnished equipment in place ready for proper operation in accordance with specifications and drawings.~~

~~G. Furnish supervision of installation of equipment at construction site by qualified factory trained technicians regularly employed by the equipment manufacturer.~~

~~SPEC WRITER NOTE: Check with Equipment Specification Writer concerning what equipment is scheduled to be relocated before using or omitting this article.~~

~~1.271.19~~ **RELOCATED // EQUIPMENT // AND ITEMS //**

A. Contractor shall disconnect, dismantle as necessary, remove and reinstall in new location, all existing equipment //and items // indicated //by symbol "R"// or otherwise shown to be relocated by the Contractor.

B. Perform relocation of such equipment or items at such times and in such a manner as directed by the COR.

C. Suitably cap existing service lines, such as steam, condensate return, water, drain, gas, air, vacuum and/or electrical, at the main whenever such lines are disconnected from equipment to be relocated. Remove abandoned lines in finished areas and cap as specified herein before under paragraph "Abandoned Lines".

D. Provide all mechanical and electrical service connections, fittings, fastenings and any other materials necessary for assembly and installation of relocated equipment; and leave such equipment in proper operating condition.

E. //Contractor shall employ services of an installation engineer, who is an authorized representative of the manufacturer of this equipment to supervise assembly and installation of existing // ~~remote dictating machine, // X-ray, // dental // and // laundry~~ //equipment, required to be relocated. //

F. All service lines such as noted above for relocated equipment shall be in place at point of relocation ready for use before any existing equipment is disconnected. Make relocated existing

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equipment ready for operation or use immediately after
reinstallation.

~~SPEC WRITER NOTE: Following paragraph
is to be used only in connection with
bed producing projects. Allow 630
square meters (7,000 square feet up to
300 beds; 930 square meters (10,000
square feet) over 300 beds and up to
500 beds; 1860 square meters (20,000
square feet) over 500 beds.~~

~~1.28 STORAGE SPACE FOR DEPARTMENT OF VETERANS AFFAIRS EQUIPMENT~~

~~A. Contractor shall complete approximately _____ square meters
_____ (square feet) of space in building accessible from
ground level without use of elevators for storage of certain
materials and equipment by Department of Veterans Affairs.~~

~~SPEC WRITER NOTE: Omit subparagraph 4
below when utilities are furnished by
Government.~~

- ~~1. Provide such space with adequate light, ventilation and heat
in season and lock for adequate security. Contractor shall
also install and connect portion of nearest specified fire
protection system including all apparatus for instant use to
provide water for adequate fire protection of storage space.~~
- ~~2. Storage space shall be turned over to Contracting Officer
ninety days prior to Completion Date of the buildings
involved.~~
- ~~3. Forward two sets of drawings to Contracting Officer through
the COR 120 days prior to Completion Date of building;
drawings shall indicate those areas which will be made
available to Department of Veterans Affairs for temporary
storage.~~
- ~~4. All cost for utility services for such storage space shall be
borne by Contractor until entire building is turned over for
occupancy.~~

~~B. "Completion Date" shall mean that date as established by Contracting Officer upon which Contractor will turn over entire project or portions thereof to the Government.~~

~~SPEC WRITER NOTE: Use construction sign when construction cost estimate is and over \$2,000,000. Note construction sign is not required for interior projects.~~

1.291.20 // CONSTRUCTION SIGN

- A. Provide a Construction Sign where directed by the COR. All wood members shall be of framing lumber. Cover sign frame with 0.7 mm (24 gage) galvanized sheet steel nailed securely around edges and on all bearings. Provide three 100 by 100 mm (4 inch by 4 inch) posts (or equivalent round posts) set 1200 mm (four feet) into ground. Set bottom of sign level at 900 mm (three feet) above ground and secure to posts with through bolts. Make posts full height of sign. Brace posts with 50 x 100 mm (two by four inch) material as directed.
- B. Paint all surfaces of sign and posts two coats of white gloss paint. Border and letters shall be of black gloss paint, except project title which shall be blue gloss paint.
- C. Maintain sign and remove it when directed by the COR.
- D. Detail Drawing of construction sign shall show required legend and other characteristics of sign. ~~is // attached hereto and made a part of this specification. // shown on the drawings. //~~

~~SPEC WRITER NOTE: Specify safety sign for all projects when construction cost estimate is over \$2,000,000, except where the work is such that a sign cannot be observed by the Contractor's personnel sometime during the working day. //~~

1.301.21 SAFETY SIGN

- A. Provide a Safety Sign where directed by the COR. Face of sign shall be 19 mm (3/4 inch) thick exterior grade plywood. Provide

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two 100 mm by 100 mm (four by four inch) posts extending full height of sign and 900 mm (three feet) into ground. Set bottom of sign level at 1200 mm (four feet) above ground.

- B. Paint all surfaces of Safety Sign and posts with one prime coat and two coats of white gloss paint. Letters and design shall be painted with gloss paint of colors noted.
- C. Maintain sign and remove it when directed by the COR.
- D. Drawing details in VA Signage Design Manual, Section 11 Specialty Signs (found on VA TIL) show required legend and other characteristics of sign. ~~and are // attached hereto and is made a part of this specification. // shown on the drawings. //~~
- E. Post the number of accident free days on a daily basis.

~~SPEC WRITER NOTE: Photographs are not required for Station Level projects. Insert total number of photos in second line below. Number of photographs required shall be within limits included in following table:~~

Number of photographs	
Estimated Cost	No. of Photographs
Up to \$250,000	50 to 100
Up to \$500,000	100 to 150
Up to \$1,000,000	150 to 200
Up to \$2,000,000	200 to 250
Up to \$5,000,000	250 to 300
Up to \$10,000,000	300 to 400
More than \$10,000,000	400 to 500

~~SPEC WRITER NOTE: Use the following paragraph for new buildings and major building additions only. Edit quantities and types as applicable. Indexing system is also optional.~~

1.311.22 //PHOTOGRAPHIC DOCUMENTATION

- A. During the construction period through completion, provide photographic documentation of construction progress and at selected milestones. ~~including electronic indexing, navigation,~~

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~~storage and remote access to the documentation, as per these specifications. The commercial photographer or the subcontractor used for this work shall meet the following qualifications:~~

- ~~1. Demonstrable minimum experience of three (3) years in operation providing documentation and advanced indexing/navigation systems including a representative portfolio of construction projects of similar type, size, duration and complexity as the Project.~~
- ~~2. Demonstrable ability to service projects throughout North America, which shall be demonstrated by a representative portfolio of active projects of similar type, size, duration and complexity as the Project.~~

B. Photographic documentation elements:

1. Each digital image shall be taken with a professional grade camera with minimum size of 6 megapixels (MP) capable of producing 200x250mm (8 x 10 inch) prints with a minimum of 2272 x 1704 pixels and 400x500mm (16 x 20 inch) prints with a minimum 2592 x 1944 pixels.
2. ~~Indexing and navigation system shall utilize actual AUTOCAD construction drawings, making such drawings interactive on an on-line interface. For all documentation referenced herein, indexing and navigation must be organized by both time (date-stamped) and location throughout the project.~~Photos shall document all phases of construction and shall be updated weekly until the project is completed. A minimum of five (5) phtos shall be taken during each workday. Photos shall be submitted each month along with the project invoice for monthly payment.
3. ~~Documentation shall combine indexing and navigation system with inspection-grade digital photography designed to capture actual conditions throughout construction and at critical milestones. Documentation shall be accessible on-line through~~

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- ~~use of an internet connection. Documentation shall allow for secure multiple-user access, simultaneously, on-line.~~
- ~~4. Before construction, the building pad, adjacent streets, roadways, parkways, driveways, curbs, sidewalks, landscaping, adjacent utilities and adjacent structures surrounding the building pad and site shall be documented. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive architectural drawings. If site work or pad preparation is extensive, this documentation may be required immediately before construction and at several pre-determined intervals before building work commences.~~
- ~~5. Construction progress for all trades shall be tracked at pre-determined intervals, but not less than once every thirty (30) calendar days ("Progressions"). Progression documentation shall track both the exterior and interior construction of the building. Exterior Progressions shall track 360 degrees around the site and each building. Interior Progressions shall track interior improvements beginning when stud work commences and continuing until Project completion.~~
- ~~6. As-built condition of pre-foundation utilities and site utilities shall be documented prior to pouring footers, placing concrete and/or backfilling. This process shall include all underground and in-slab utilities within the building(s) envelope(s) and utility runs in the immediate vicinity of the building(s) envelope(s). This may also include utilities enclosed in slab-on-deck in multi-story buildings. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive site utility plans.~~
- ~~7. As-built conditions of mechanical, electrical, plumbing and all other systems shall be documented post-inspection and pre-insulation, sheet rock or dry wall installation. This process~~

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- ~~shall include all finished systems located in the walls and ceilings of all buildings at the Project. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive architectural drawings.~~
- ~~8. As built conditions of exterior skin and elevations shall be documented with an increased concentration of digital photographs as directed by the COR in order to capture pre-determined focal points, such as waterproofing, window flashing, radiused steel work, architectural or Exterior Insulation and Finish Systems (EIFS) detailing. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive elevations or elevation details.~~
- ~~9. As built finished conditions of the interior of each building including floors, ceilings and walls shall be documented at certificate of occupancy or equivalent, or just prior to occupancy, or both, as directed by the COR. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive architectural drawings.~~
- ~~10. Miscellaneous events that occur during any Contractor site visit, or events captured by the Department of Veterans Affairs independently, shall be dated, labeled and inserted into a Section in the navigation structure entitled "Slideshows," allowing this information to be stored in the same "place" as the formal scope.~~
- ~~11. Customizable project specific digital photographic documentation of other details or milestones. Indexing and navigation accomplished through interactive architectural plans.~~
- ~~12. Monthly (29 max) exterior progressions (360 degrees around the project) and slideshows (all elevations and building~~

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- envelope). ~~The slideshows allow for the inclusion of Department of Veterans Affairs pictures, aerial photographs, and timely images which do not fit into any regular monthly photopath.~~
- ~~13. Weekly (21 Max) Site Progressions — Photographic documentation capturing the project at different stages of construction. These progressions shall capture underground utilities, excavation, grading, backfill, landscaping and road construction throughout the duration of the project.~~
- ~~14. Regular (8 max) interior progressions of all walls of the entire project to begin at time of substantial framed or as directed by the COR through to completion.~~
- ~~15. Detailed Exact-Built of all Slabs for all project slab pours just prior to placing concrete or as directed by the COR.~~
- ~~16. Detailed Interior exact built overlapping photos of the entire building to include documentation of all mechanical, electrical and plumbing systems in every wall and ceiling, to be conducted after rough-ins are complete, just prior to insulation and or drywall, or as directed by COR.~~
- ~~17. Finished detailed Interior exact built overlapping photos of all walls, ceilings, and floors to be scheduled by COR prior to occupancy.~~
- ~~18. In event a greater or lesser number of images than specified above are required by the COR, adjustment in contract price will be made in accordance with clause entitled "CHANGES" (FAR 52.243-4).~~
- ~~C. Images shall be taken by a commercial photographer and must show distinctly, at as large a scale as possible, all parts of work embraced in the picture.~~
- ~~D. Coordination of photo shoots is accomplished through COR. Contractor shall also attend construction team meetings as necessary. Contractor's operations team shall provide regular updates regarding the status of the documentation, including~~

- ~~photo shoots concluded, the availability of new Progressions or Exact-Builts viewable on-line and anticipated future shoot dates.~~
- ~~E. Contractor shall provide all on-line domain/web hosting, security measures, and redundant server back-up of the documentation.~~
- ~~F. Contractor shall provide technical support related to using the system or service.~~
- ~~G. Upon completion of the project, final copies of the documentation (the "Permanent Record") with the indexing and navigation system embedded (and active) shall be provided in an electronic media format, typically a DVD or external hard drive. Permanent Record shall have Building Information Modeling (BIM) interface capabilities. On-line access terminates upon delivery of the Permanent Record.~~

1.321.23 - ~~FINAL ELEVATION DIGITAL IMAGES~~

- ~~A. A minimum of four (4) images of each elevation shall be taken with a minimum 6 MP camera, by a professional photographer with different settings to allow the COR to select the image to be printed. All images are provided to the RE on a CD.~~
- ~~B. Photographs shall be taken upon completion, including landscaping. They shall be taken on a clear sunny day to obtain sufficient detail to show depth and to provide clear, sharp pictures. Pictures shall be 400 mm x 500 mm (16 by 20 inches), printed on regular weight paper, matte finish archival grade photographic paper and produced by a RA4 process from the digital image with a minimum 300 PPI. Identifying data shall be carried on label affixed to back of photograph without damage to photograph and shall be similar to that provided for final construction photographs.~~
- ~~C. Furnish six (6) 400 mm x 500 mm (16 by 20 inch) color prints of the following buildings constructed under this project (elevations as selected by the RE from the images taken above). Photographs shall be artistically composed showing full front elevations. All images shall become property of the Government.~~

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~~Each of the selected six prints shall be place in a frame with a minimum of 2 inches of appropriate matting as a border. Provide a selection of a minimum of 3 different frames from which the SRE will select one style to frame all six prints. Photographs with frames shall be delivered to the COR in boxes suitable for shipping.~~

- ~~1. Hospital Building No. ____.~~
- ~~2. Clinical Building No. ____.~~
- ~~3. Nursing Home Care Building No. ____.~~
- ~~4. Chapel Building No. ____.~~
- ~~5. Boiler Plant Building No. ____.~~

~~1.33 // HISTORIC PRESERVATION~~

- ~~A. Where the Contractor or any of the Contractor's employees, prior to, or during the construction work, are advised of or discover any possible archeological, historical and/or cultural resources, the Contractor shall immediately notify the COR verbally, and then with a written follow up. //~~

~~1.34 // VA TRIRICA CPMS~~

~~SPEC WRITER NOTE: Delete or include and edit the following article after consulting with the Project Manager. This requirement, at present, is only for major projects. The term "major medical facility project" means a project for the construction, alteration, or acquisition of a medical facility involving a total expenditure of more than \$20,000,000, but such term does not include an acquisition by exchange."~~

- ~~A. VA contractors, selected by award to perform work, are required to get access to the VA TRIRICA CPMS. The TRIRICA CPMS is the management and collaborative environment that the VA uses for all Major, Minor and Non-Recurring Maintenance (NRM) projects within the Office of Construction & Facilities Management (CFM), Veterans Health Administration (VHA), National Cemetery~~

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~~Administration (NCA), and the Veterans Benefits Administration (VBA).~~

~~B. The contractor is solely responsible for acquiring access to the VA TRIRICA CPMS.~~

~~C. To gain access to the VA TRIRICA CPMS the contractor is encouraged to follow the licensing process outline as specified below:~~

~~D. Requirement: TRIRICA is the management and collaborative environment that VA uses for all construction projects. VA requires its contractors to procure TRIRICA access as part of the cost of performance for a VA construction related contract.~~

~~E. Access Request and Payment can be made through the following URL <https://valicensing.onefi.com/>~~

~~F. Inquiries or to request additional services, contact the following:~~

~~—Craig Alsheimer, Federal Account Manager~~

~~—Computerized Facility Integrations, LLC~~

~~—18000 West Nine Mile Road~~

~~—Suite 700~~

~~—Southfield, MI 48075~~

~~—Email: calsheimer@goefi.com~~

~~—Phone: 248-557-4234 Extension 6010; 410-292-7006~~

~~G. Process:~~

~~1. Once the contractor has been notified by VA of the award and a unique contract number, the contractor can enter a request for access to TRIRICA at URL <https://valicensing.onefi.com/>~~

~~2. CFI will process the request for access and payment. CFI will create the USER ID and a password. Security provisions required to align the contractor to the Contract Number will be entered and an email will be generated and submitted to the requestor.~~

~~3. CFI will also provide standard terms and conditions related to the transaction and use agreement. //~~

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SECTION 01 32 16.15
PROJECT SCHEDULES
(SMALL PROJECTS - DESIGN/BID/BUILD)

PART 1- GENERAL

1.1 DESCRIPTION:

- A. The Contractor shall develop a Critical Path Method (CPM) plan and schedule demonstrating fulfillment of the contract requirements (Project Schedule), and shall keep the Project Schedule up-to-date in accordance with the requirements of this section and shall utilize the plan for scheduling, coordinating and monitoring work under this contract (including all activities of subcontractors, equipment vendors and suppliers). Conventional Critical Path Method (CPM) technique shall be utilized to satisfy both time and cost applications.

1.2 CONTRACTOR'S REPRESENTATIVE:

- A. The Contractor shall designate an authorized representative responsible for the Project Schedule including preparation, review and progress reporting with and to the Contracting Officer's Representative (COR).
- B. The Contractor's representative shall have direct project control and complete authority to act on behalf of the Contractor in fulfilling the requirements of this specification section.
- C. The Contractor's representative shall have the option of developing the project schedule within their organization or to engage the services of an outside consultant. If an outside scheduling consultant is utilized, Section 1.3 of this specification will apply.

1.3 CONTRACTOR'S CONSULTANT:

- A. The Contractor shall submit a qualification proposal to the COR, within 10 days of bid acceptance. The qualification proposal shall include:
1. The name and address of the proposed consultant.
 2. Information to show that the proposed consultant has the qualifications to meet the requirements specified in the preceding paragraph.
 3. A representative sample of prior construction projects, which the proposed consultant has performed complete project scheduling services. These representative samples shall be of similar size and scope.

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- B. The COR has the right to approve or disapprove the proposed consultant, and will notify the Contractor of the VA decision within seven calendar days from receipt of the qualification proposal. In case of disapproval, the Contractor shall resubmit another consultant within 10 calendar days for renewed consideration. The Contractor shall have their scheduling consultant approved prior to submitting any schedule for approval.

1.4 COMPUTER PRODUCED SCHEDULES

- A. The contractor shall provide monthly, to the Department of Veterans Affairs (VA), all computer-produced time/cost schedules and reports generated from monthly project updates. This monthly computer service will include: three copies of up to five different reports (inclusive of all pages) available within the user defined reports of the scheduling software approved by the COR; a hard copy listing of all project schedule changes, and associated data, made at the update and an electronic file of this data; and the resulting monthly updated schedule in PDM format. These must be submitted with and substantively support the contractor's monthly payment request and the signed look ahead report. The COR shall identify the five different report formats that the contractor shall provide.
- B. The contractor shall be responsible for the correctness and timeliness of the computer-produced reports. The Contractor shall also responsible for the accurate and timely submittal of the updated project schedule and all CPM data necessary to produce the computer reports and payment request that is specified.
- C. The VA will report errors in computer-produced reports to the Contractor's representative within ten calendar days from receipt of reports. The Contractor shall reprocess the computer-produced reports and associated diskette(s), when requested by the COR, to correct errors which affect the payment and schedule for the project.

1.5 THE COMPLETE PROJECT SCHEDULE SUBMITTAL

- A. Within 45 calendar days after receipt of Notice to Proceed, the Contractor shall submit for the COR's review; three blue line copies of the interim schedule on sheets of paper 765 x 1070 mm (30 x 42 inches) and an electronic file in the previously approved CPM schedule program. The submittal shall also include three copies of a computer-produced activity/event ID schedule showing project duration; phase completion

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dates; and other data, including event cost. Each activity/event on the computer-produced schedule shall contain as a minimum, but not limited to, activity/event ID, activity/event description, duration, budget amount, early start date, early finish date, late start date, late finish date and total float. Work activity/event relationships shall be restricted to finish-to-start or start-to-start without lead or lag constraints. Activity/event date constraints, not required by the contract, will not be accepted unless submitted to and approved by the COR. The contractor shall make a separate written detailed request to the COR identifying these date constraints and secure the COR's written approval before incorporating them into the network diagram. The COR's separate approval of the Project Schedule shall not excuse the contractor of this requirement. Logic events (non-work) will be permitted where necessary to reflect proper logic among work events, but must have zero duration. The complete working schedule shall reflect the Contractor's approach to scheduling the complete project.

The final Project Schedule in its original form shall contain no contract changes or delays which may have been incurred during the final network diagram development period and shall reflect the entire contract duration as defined in the bid documents. These changes/delays shall be entered at the first update after the final Project Schedule has been approved. The Contractor should provide their requests for time and supporting time extension analysis for contract time as a result of contract changes/delays, after this update, and in accordance with Article, ADJUSTMENT OF CONTRACT COMPLETION.

- B. Within 30 calendar days after receipt of the complete project interim Project Schedule and the complete final Project Schedule, the COR, will do one or both of the following:
1. Notify the Contractor concerning his actions, opinions, and objections.
 2. A meeting with the Contractor at or near the job site for joint review, correction or adjustment of the proposed plan will be scheduled if required. Within 14 calendar days after the joint review, the Contractor shall revise and shall submit three blue line copies of the revised Project Schedule, three copies of the revised computer-produced activity/event ID schedule and a revised electronic file as specified by the COR. The revised submission will

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- be reviewed by the COR and, if found to be as previously agreed upon, will be approved.
- C. The approved baseline schedule and the computer-produced schedule(s) generated there from shall constitute the approved baseline schedule until subsequently revised in accordance with the requirements of this section.
- ~~D. The Complete Project Schedule shall contain approximately _____ work activities/events.~~

1.6 WORK ACTIVITY/EVENT COST DATA

- A. The Contractor shall cost load all work activities/events except procurement activities. The cumulative amount of all cost loaded work activities/events (including alternates) shall equal the total contract price. Prorate overhead, profit and general conditions on all work activities/events for the entire project length. The contractor shall generate from this information cash flow curves indicating graphically the total percentage of work activity/event dollar value scheduled to be in place on early finish, late finish. These cash flow curves will be used by the COR to assist him in determining approval or disapproval of the cost loading. Negative work activity/event cost data will not be acceptable, except on VA issued contract changes.
- B. The Contractor shall cost load work activities/events for guarantee period services, test, balance and adjust various systems in accordance with the provisions in Article, FAR 52.232 - 5 (PAYMENT UNDER FIXED-PRICE CONSTRUCTION CONTRACTS) and VAAR 852.232 ~~--//Article 70-~~
~~Without NAS-CPM//_Article 71 Including NAS-CPM//_for~~ (PAYMENTS UNDER FIXED PRICE CONSTRUCTION).
- C. In accordance with FAR 52.236 - 1 (PERFORMANCE OF WORK BY THE CONTRACTOR), the Contractor shall submit, simultaneously with the cost per work activity/event of the construction schedule required by this Section, a responsibility code for all activities/events of the project for which the Contractor's forces will perform the work.
- D. The Contractor shall cost load work activities/events for all BID ITEMS including ASBESTOS ABATEMENT. The sum of each BID ITEM work shall equal the value of the bid item in the Contractors' bid.

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1.7 PROJECT SCHEDULE REQUIREMENTS

- A. Show on the project schedule the sequence of work activities/events required for complete performance of all items of work. The Contractor Shall:
1. Show activities/events as:
 - a. Contractor's time required for submittal of shop drawings, templates, fabrication, delivery and similar pre-construction work.
 - b. COR's and Architect-Engineer's review and approval of shop drawings, equipment schedules, samples, template, or similar items.
 - c. Interruption of VA Facilities utilities, delivery of Government furnished equipment, and rough-in drawings, project phasing and any other specification requirements.
 - d. Test, balance and adjust various systems and pieces of equipment, maintenance and operation manuals, instructions and preventive maintenance tasks.
 - e. VA inspection and acceptance activity/event with a minimum duration of five work days at the end of each phase and immediately preceding any VA move activity/event required by the contract phasing for that phase.
 2. Show not only the activities/events for actual construction work for each trade category of the project, but also trade relationships to indicate the movement of trades from one area, floor, or building, to another area, floor, or building, for at least five trades who are performing major work under this contract.
 3. Break up the work into activities/events of a duration no longer than 20 work days each or one reporting period, except as to non-construction activities/events (i.e., procurement of materials, delivery of equipment, concrete and asphalt curing) and any other activities/events for which the COR may approve the showing of a longer duration. The duration for VA approval of any required submittal, shop drawing, or other submittals will not be less than 20 work days.
 4. Describe work activities/events clearly, so the work is readily identifiable for assessment of completion. Activities/events labeled

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- "start," "continue," or "completion," are not specific and will not be allowed. Lead and lag time activities will not be acceptable.
5. The schedule shall be generally numbered in such a way to reflect either discipline, phase or location of the work.
- B. The Contractor shall submit the following supporting data in addition to the project schedule:
1. The appropriate project calendar including working days and holidays.
 2. The planned number of shifts per day.
 3. The number of hours per shift.
- Failure of the Contractor to include this data shall delay the review of the submittal until the COR is in receipt of the missing data.
- C. To the extent that the Project Schedule or any revised Project Schedule shows anything not jointly agreed upon, it shall not be deemed to have been approved by the COR. Failure to include any element of work required for the performance of this contract shall not excuse the Contractor from completing all work required within any applicable completion date of each phase regardless of the COR's approval of the Project Schedule.
- D. Compact Disk Requirements and CPM Activity/Event Record Specifications: Submit to the VA an electronic file(s) containing one file of the data required to produce a schedule, reflecting all the activities/events of the complete project schedule being submitted.

1.8 PAYMENT TO THE CONTRACTOR:

- A. Monthly, the contractor shall submit an application and certificate for payment using ~~//VA Form 10-6001a // or //the AIA application and certificate for payment documents G702 & G703 //~~ reflecting updated schedule activities and cost data in accordance with the provisions of the following Article, PAYMENT AND PROGRESS REPORTING, as the basis upon which progress payments will be made pursuant to Article, FAR 52.232 - 5 (PAYMENT UNDER FIXED-PRICE CONSTRUCTION CONTRACTS) and VAAR 852.232 - ~~//_Article 70 Without NAS CPM//~~ //Article 71 Including NAS-CPM//for (PAYMENTS UNDER FIXED PRICE CONSTRUCTION). The Contractor shall be entitled to a monthly progress payment upon approval of estimates as determined from the currently approved updated project schedule. Monthly payment requests shall include: a listing of all

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agreed upon project schedule changes and associated data; and an electronic file (s) of the resulting monthly updated schedule.

- B. Approval of the Contractor's monthly Application for Payment shall be contingent, among other factors, on the submittal of a satisfactory monthly update of the project schedule.

1.9 PAYMENT AND PROGRESS REPORTING

- A. Monthly schedule update meetings will be held on dates mutually agreed to by the COR and the Contractor. Contractor and their CPM consultant (if applicable) shall attend all monthly schedule update meetings. The Contractor shall accurately update the Project Schedule and all other data required and provide this information to the COR three work days in advance of the schedule update meeting. Job progress will be reviewed to verify:
1. Actual start and/or finish dates for updated/completed activities/events.
 2. Remaining duration for each activity/event started, or scheduled to start, but not completed.
 3. Logic, time and cost data for change orders, and supplemental agreements that are to be incorporated into the Project Schedule.
 4. Changes in activity/event sequence and/or duration which have been made, pursuant to the provisions of following Article, ADJUSTMENT OF CONTRACT COMPLETION.
 5. Completion percentage for all completed and partially completed activities/events.
 6. Logic and duration revisions required by this section of the specifications.
 7. Activity/event duration and percent complete shall be updated independently.
- B. After completion of the joint review, the contractor shall generate an updated computer-produced calendar-dated schedule and supply the COR with reports in accordance with the Article, COMPUTER PRODUCED SCHEDULES, specified.
- C. After completing the monthly schedule update, the contractor's representative or scheduling consultant shall rerun all current period contract change(s) against the prior approved monthly project schedule. The analysis shall only include original workday durations and schedule logic agreed upon by the contractor and COR for the contract change(s).

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When there is a disagreement on logic and/or durations, the Contractor shall use the schedule logic and/or durations provided and approved by the COR. After each rerun update, the resulting electronic project schedule data file shall be appropriately identified and submitted to the VA in accordance to the requirements listed in articles 1.4 and 1.7. This electronic submission is separate from the regular monthly project schedule update requirements and shall be submitted to the COR within fourteen (14) calendar days of completing the regular schedule update. Before inserting the contract changes durations, care must be taken to ensure that only the original durations will be used for the analysis, not the reported durations after progress. In addition, once the final network diagram is approved, the contractor must recreate all manual progress payment updates on this approved network diagram and associated reruns for contract changes in each of these update periods as outlined above for regular update periods. This will require detailed record keeping for each of the manual progress payment updates.

- D. Following approval of the CPM schedule, the VA, the General Contractor, its approved CPM Consultant, RE office representatives, and all subcontractors needed, as determined by the SRE, shall meet to discuss the monthly updated schedule. The main emphasis shall be to address work activities to avoid slippage of project schedule and to identify any necessary actions required to maintain project schedule during the reporting period. The Government representatives and the Contractor should conclude the meeting with a clear understanding of those work and administrative actions necessary to maintain project schedule status during the reporting period. This schedule coordination meeting will occur after each monthly project schedule update meeting utilizing the resulting schedule reports from that schedule update. If the project is behind schedule, discussions should include ways to prevent further slippage as well as ways to improve the project schedule status, when appropriate.

1.10 RESPONSIBILITY FOR COMPLETION

- A. If it becomes apparent from the current revised monthly progress schedule that phasing or contract completion dates will not be met, the Contractor shall execute some or all of the following remedial actions:

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1. Increase construction manpower in such quantities and crafts as necessary to eliminate the backlog of work.
 2. Increase the number of working hours per shift, shifts per working day, working days per week, the amount of construction equipment, or any combination of the foregoing to eliminate the backlog of work.
 3. Reschedule the work in conformance with the specification requirements.
- B. Prior to proceeding with any of the above actions, the Contractor shall notify and obtain approval from the COR for the proposed schedule changes. If such actions are approved, the representative schedule revisions shall be incorporated by the Contractor into the Project Schedule before the next update, at no additional cost to the Government.

1.11 CHANGES TO THE SCHEDULE

- A. Within 30 calendar days after VA acceptance and approval of any updated project schedule, the Contractor shall submit a revised electronic file (s) and a list of any activity/event changes including predecessors and successors for any of the following reasons:
1. Delay in completion of any activity/event or group of activities/events, which may be involved with contract changes, strikes, unusual weather, and other delays will not relieve the Contractor from the requirements specified unless the conditions are shown on the CPM as the direct cause for delaying the project beyond the acceptable limits.
 2. Delays in submittals, or deliveries, or work stoppage are encountered which make rescheduling of the work necessary.
 3. The schedule does not represent the actual prosecution and progress of the project.
 4. When there is, or has been, a substantial revision to the activity/event costs regardless of the cause for these revisions.
- B. CPM revisions made under this paragraph which affect the previously approved computer-produced schedules for Government furnished equipment, vacating of areas by the VA Facility, contract phase(s) and sub phase(s), utilities furnished by the Government to the Contractor, or any other previously contracted item, shall be furnished in writing to the COR for approval.

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- C. COR's approval for the revised project schedule and all relevant data is contingent upon compliance with all other paragraphs of this section and any other previous agreements by the COR.
- D. The cost of revisions to the project schedule resulting from contract changes will be included in the proposal for changes in work as specified in FAR 52.243 - 4 (Changes), and will be based on the complexity of the revision or contract change, man hours expended in analyzing the change, and the total cost of the change.
- E. The cost of revisions to the Project Schedule not resulting from contract changes is the responsibility of the Contractor.

1.12 ADJUSTMENT OF CONTRACT COMPLETION

- A. The contract completion time will be adjusted only for causes specified in this contract. Request for an extension of the contract completion date by the Contractor shall be supported with a justification, CPM data and supporting evidence as the COR may deem necessary for determination as to whether or not the Contractor is entitled to an extension of time under the provisions of the contract. Submission of proof based on revised activity/event logic, durations (in work days) and costs is obligatory to any approvals. The schedule must clearly display that the Contractor has used, in full, all the float time available for the work involved in this request. The COR's determination as to the total number of days of contract extension will be based upon the current computer-produced calendar-dated schedule for the time period in question and all other relevant information.
- B. Actual delays in activities/events which, according to the computer-produced calendar-dated schedule, do not affect the extended and predicted contract completion dates shown by the critical path in the network, will not be the basis for a change to the contract completion date. The COR will within a reasonable time after receipt of such justification and supporting evidence, review the facts and advise the Contractor in writing of the COR's decision.
- C. The Contractor shall submit each request for a change in the contract completion date to the COR in accordance with the provisions specified under FAR 52.243 - 4 (Changes). The Contractor shall include, as a part of each change order proposal, a sketch showing all CPM logic revisions, duration (in work days) changes, and cost changes, for work

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in question and its relationship to other activities on the approved network diagram.

- D. All delays due to non-work activities/events such as RFI's, WEATHER, STRIKES, and similar non-work activities/events shall be analyzed on a month by month basis.

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SECTION 01 33 23

SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This specification defines the general requirements and procedures for submittals. A submittal is information submitted for VA review to establish compliance with the contract documents.
- B. Detailed submittal requirements are found in the technical sections of the contract specifications. The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective technical specifications at no additional cost to the government.
- C. VA approval of a submittal does not relieve the Contractor of the responsibility for any error which may exist. The Contractor is responsible for fully complying with all contract requirements and the satisfactory construction of all work, including the need to check, confirm, and coordinate the work of all subcontractors for the project. Non-compliant material incorporated in the work will be removed and replaced at the Contractor's expense.

1.2 DEFINITIONS

- A. Preconstruction Submittals: Submittals which are required prior to issuing contract notice to proceed or starting construction. For example, Certificates of insurance; Surety bonds; Site-specific safety plan; Construction progress schedule; Schedule of values; Submittal register; List of proposed subcontractors; etc.
- B. Shop Drawings: Drawings, diagrams, and schedules specifically prepared by or for the Contractor to illustrate some portion of the work. Coordination Shop Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be integrated and coordinated.
- C. Product Data: Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions, and brochures, which describe and illustrate size, physical appearance, and other characteristics of materials, systems, or equipment for some portion of the work. Samples of warranty language when the contract requires extended product warranties.
- D. Samples: Physical examples of materials, equipment, or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged. Color samples from the manufacturer's standard line (or custom color samples if specified) to be

used in selecting or approving colors for the project. Field samples and mock-ups constructed to establish standards by which the ensuing work can be judged.

- E. Design Data: Calculations, mix designs, analyses, or other data pertaining to a part of work.
- F. Test Reports: Report which includes findings of a test required to be performed by the Contractor on an actual portion of the work. Report which includes findings of a test made at the job site or on a sample taken from the job site, on a portion of work during or after installation.
- G. Certificates: Document required of Contractor, or of a manufacturer, supplier, installer, or subcontractor through Contractor. The purpose is to document procedures, acceptability of methods, or personnel qualifications for a portion of the work.
- H. Manufacturer's Instructions: Pre-printed material describing installation of a product, system, or material, including special notices and Material Safety Data Sheets (MSDS) concerning impedances, hazards, and safety precautions.
- I. Manufacturer's Field Reports: Documentation of the testing and verification actions taken by manufacturer's representative at the job site on a portion of the work, during or after installation, to confirm compliance with manufacturer's standards or instructions. The documentation must indicate whether the material, product, or system has passed or failed the test.
- J. Operation and Maintenance Data: Manufacturer data that is required to operate, maintain, troubleshoot, and repair equipment, including manufacturer's help, parts list, and product line documentation. This data shall be incorporated in an operations and maintenance manual.
- K. Closeout Submittals: Documentation necessary to properly close out a construction contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a phase of construction on a multi-phase contract.

L. Request for Information (RFI's): A document submitted by the Contractor requesting clarification of a portion of the Contract Documents shall be hereinafter referred to as an RFI. A properly prepared Request for Information shall include a detailed written statement that indicates the specific Drawing or Specification in need of clarification and the nature of the clarification requested.

- a. Drawings shall be identified by the drawing number and location on the drawing sheet.

b. Specifications shall be identified by Section number and paragraph.

a-c. RFI shall include a cross-reference to earlier similar RFS's, as applicable.

1.3 SUBMITTAL REGISTER

- A. The submittal register will list items of equipment and materials for which submittals are required by the specifications. This list may not be all inclusive and additional submittals may be required by the specifications. The Contractor is not relieved from supplying submittals required by the contract ~~documents~~ documents, but which have been omitted from the submittal register.
- B. The submittal register will serve as a scheduling document for submittals and will be used to control submittal actions throughout the contract period (Refer to Specification Section 01 32 16.15-1.7).
- C. The VA will provide the initial submittal register in electronic format is attached to the end of this section. Thereafter, the Contractor shall track all submittals by maintaining a complete list, including completion of all data columns, including dates on which submittals are received and returned by the VA.
- D. The Contractor shall update the submittal register as submittal actions occur and maintain the submittal register at the project site until final acceptance of all work by Contracting Officer.
- E. The Contractor shall submit formal monthly updates to the submittal register in electronic format. Each monthly update shall document actual submission and approval dates for each submittal.

1.4 SUBMITTAL SCHEDULING

- A. Submittals are to be scheduled, submitted, reviewed, and approved prior to the acquisition of the material or equipment.
- B. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. Allow time for potential resubmittal.
- C. No delay costs or time extensions will be allowed for time lost in late submittals or resubmittals.
- D. All submittals are required to be approved by the VA Contracting Officer or their Representative (COR) prior to the start of the specified work activity.

1.5 SUBMITTAL PREPARATION

- A. Each submittal is to be complete and in sufficient detail to allow ready determination of compliance with contract requirements.
- B. Collect required data for each specific material, product, unit of work, or system into a single submittal. Prominently mark choices, options, and portions applicable to the submittal. Partial submittals will not be accepted for expedition of construction effort. Submittal will be returned without review if incomplete.
- C. If available product data is incomplete, provide Contractor-prepared documentation to supplement product data and satisfy submittal requirements.
- D. All irrelevant or unnecessary data shall be removed from the submittal to facilitate accuracy and timely processing. Submittals that contain an excessive amount of irrelevant or unnecessary data will be returned with review.
- E. Provide a transmittal form for each submittal with the following information:
 - 1. Project title, location and Project number.
 - 2. Construction contract number.
 - 3. Date of the drawings and revisions.
 - 4. Name, address, and telephone number of subcontractor, supplier, manufacturer, and any other subcontractor associated with the submittal.
 - 5. List paragraph number of the specification section and sheet number of the contract drawings by which the submittal is required.
 - 6. When a resubmission, add alphabetic suffix on submittal description. For example, submittal 18 would become 18A, to indicate resubmission.
 - 7. Product identification and location in project.
- F. The Contractor is responsible for reviewing and certifying that all submittals are in compliance with contract requirements before submitting for VA review. Proposed deviations from the contract requirements are to be clearly identified. All deviations submitted must include a ~~side-by-side~~ side-by-side comparison of item being proposed against item specified. Failure to point out deviations will result in the VA requiring removal and replacement of such work at the Contractor's expense.
- G. The Contractor shall ~~S~~stamp, sign, and date each submittal transmittal form indicating action taken. Submittals not bearing the Contractor's review and approval will be returned without review for corrective action.
- H. Stamp used by the Contractor on the submittal transmittal form to certify that the submittal meets contract requirements is to be similar to the following:

CONTRACTOR
(Firm Name)
_____ Approved
_____ Approved with corrections as noted on submittal data and/or attached sheets(s)
SIGNATURE: _____
TITLE: _____
DATE: _____

1.6 SUBMITTAL FORMAT AND TRANSMISSION

- A. Provide submittals in electronic format, with the exception of material samples. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer.
- B. Compile the electronic submittal file as a single, complete document. Name the electronic submittal file specifically according to its contents (Project, Specification Section and Submittal number).
- C. Electronic files must be of sufficient quality that all information is legible. Generate PDF files from original documents so that the text included in the PDF file is both searchable and can be copied. If documents are scanned, Optical Character Resolution (OCR) routines are required.

~~D. E-mail electronic submittal documents smaller than 5MB in size to e-mail addresses as directed by the Contracting Officer.~~

~~DE. The Contractor is responsible for setting up, providing, and maintaining web-based Project Management software site for the duration of the construction contract period of performance for the distribution of submittals and RFI's. The file sharing system platform must be accessible from the VA computer network. Provide electronic documents over 15MB through an electronic File Transfer Protocol (FTP) file sharing system. Confirm that the electronic FTP file sharing system can be accessed from the VA computer network. The Contractor is responsible for setting up, providing, and maintaining the electronic FTP file sharing system for the construction contract period of performance.~~

EF. Provide hard copies of submittals when requested by the Contracting Officer or their Representative (COR). Up to 3 additional hard copies of any submittal may be requested at the discretion of the Contracting Officer, at no additional cost to the VA.

1.7 SHOP DRAWINGS

A. Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents.

1. Preparation: Fully illustrate requirements in the Contract Documents. Include the following information, as applicable:

- a. Identification of products.
- b. Schedules.
- c. Compliance with specified standards.
- d. Notation of coordination requirements.
- e. Notation of dimensions established by field measurement.
- f. Relationship and attachment to adjoining construction clearly indicated.
- g. Seal and signature of professional engineer if specified.

2. BIM Incorporation: Develop and incorporate Shop Drawing files into BIM established for Project.

1.87 SAMPLES

- A. Submit two sets of physical samples showing range of variation, for each required item.
- B. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified.

- C. When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.
- D. Before submitting samples, the Contractor is to ensure that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.
- E. The VA reserves the right to disapprove any material or equipment which previously has proven unsatisfactory in service.
- F. Physical samples supplied may be requested back for use in the project after reviewed and approved.

1.98 OPERATION AND MAINTENANCE (O&M) DATA

- A. Submit data specified for a given item within 30 calendar days after the item is delivered to the contract site.
- B. In the event the Contractor fails to deliver O&M Data within the time limits specified, the Contracting Officer may withhold from progress payments 50 percent of the price of the item for which such O&M Data is applicable.

1.109 TEST REPORTS

- A. COR may require specific test after work has been installed or completed which could require contractor to repair test area at no additional cost to contract.
- B. Field Test Reports: Submit written reports indicating and interpreting results of field tests performed, either during installation of product(s) or after product(s) are install in their final location, for compliance with requirements in the Contract Documents.
- C. Material Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in the Contract Documents.

1.11 DELEGATED DESIGN SERVICES

- A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated.
- B. Delegated Design Services Certification: In addition to Shop Drawings, Product Data, and other required submittals, submit digitally signed PDF file paper copies of certificate, signed and sealed by the responsible design

professional, for each product and system specifically assigned to Contractor to be designed or certified by a design professional.

1.12 CONTRACTOR'S REQUEST FOR INFORMATION

- A. When the Contractor is unable to determine from the Contract Documents the materials, or system to be installed; or when an item of construction are required to occupy the same space (interference); or when an item of work is described differently at two places in the contract documents (contradicting direction); the Contractor shall request clarification of the indeterminate or interfering item. The RFI procedure herein shall be the method of requesting clarification.
- B. Contractor shall endeavor to keep the number of RFIs to a minimum by thoroughly examining all Contract Drawings, Specifications, Change Orders, answered RFIs, Submittals, and prior project-related communications to insure that the requested information does not already exist. RFIs that request information available in the Contract Documents or previously-issued communications will be declared a "non-conforming RFI" and returned by the VA.
- C. RFI's shall be submitted through the online Construction Management Platform established by the General Contractor.
- D. RFI's from subcontractors or material suppliers shall be submitted through and reviewed by the Contractor prior to submittal to the VA. The Contractor shall utilize a standardized RFI form acceptable to the VA, completely filled out, and shall not just state "see attached subcontractor RFI."
- E. RFI's shall not be used for the following purposes:
 - 1. To request approval of submittals
 - 2. To request approval of substitutions
 - 3. To request approval of means & methods of installation of work nor coordination of interference(s) for which the Contractor is responsible to coordinate between trades.

1.12~~9~~ VA REVIEW OF SUBMITTALS AND REQUESTS FOR INFORMATION (RFI)

- A. The VA will review all submittals and RFI's for compliance with the technical requirements of the contract documents. The Architect-Engineer for this project will assist the VA in reviewing all submittals and determining contractual compliance. Review will be only for conformance with the design intent of the contract documents.

- B. Period of review for submittals and RFI's -begins when the VA COR receives submittal from the Contractor prior to 3pm of the Federal working day it is submitted in the time period of the facility for which work is being performed. Submittals or RFI's submitted after 3pm will be considered received the following Federal Working Day.-
- C. Period of review for each resubmittal is the same as for initial submittal.
- D. VA review period is 15 Federal working days for submittals.
- E. VA review period is 10 Federal working days for RFI.
- F. The VA will return submittals to the Contractor with the following notations:
 - 1. "Reviewed": authorizes the Contractor to proceed with the work covered.
 - 2. "Reviewed ~~with changes~~ as -as noted": authorizes the Contractor to proceed with the work covered provided the Contractor incorporates the noted comments and makes the noted corrections.
 - 3. "Revise and ~~R~~esubmit": indicates noncompliance with the contract requirements or that submittal is incomplete. Resubmit with appropriate changes and corrections. No work shall proceed for this item until resubmittal is approved.
 - 4. "Not ~~R~~esubmitted": indicates submittal does not have evidence of being reviewed and approved by Contractor or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals after taking appropriate action.

1.113 APPROVED SUBMITTALS

- A. The VA ~~approval-review~~ of submittals is not to be construed as a complete check, and indicates only that the general method of construction, materials, detailing, and other information are satisfactory.
- B. VA ~~reviewapproval~~ of a submittal does not relieve the Contractor of the responsibility for any error which may exist. The Contractor is responsible for fully complying with all contract requirements and the satisfactory construction of all work, including the need to check, confirm, and coordinate the work of all subcontractors for the project. Non-compliant material incorporated in the work will be removed and replaced at the Contractor's expense.
- C. After submittals have been ~~approved~~reviewed, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.
- D. Retain a copy of all ~~approved-reviewed~~ submittals at project site, including approved samples readily available upon request by the COR.

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1.1⁴² WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained.

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**SECTION 01 35 26
SAFETY REQUIREMENTS**

1.1 APPLICABLE PUBLICATIONS:

A. Latest publications listed below form part of this Article to extent referenced. Publications are referenced in text by basic designations only.

B. American Society of Safety Professionals (ASSP):

A10.1-2011.....Pre-Project & Pre-Task Safety and Health
Planning

A10.34-2012.....Protection of the Public on or Adjacent to
Construction Sites

A10.38-2013.....Basic Elements of an Employer's Program to
Provide a Safe and Healthful Work Environment
American National Standard Construction and
Demolition Operations

Z359.0-2012.....Definitions and Nomenclature Used for Fall
Protection and Fall Arrest

Z359.1-2016.....The Fall Protection Code

C. American Society for Testing and Materials (ASTM):

E84-2013.....Surface Burning Characteristics of Building
Materials

D. The Facilities Guidelines Institute (FGI):

FGI Guidelines-2010Guidelines for Design and Construction of
Healthcare Facilities

E. National Fire Protection Association (NFPA):

10-2018.....Standard for Portable Fire Extinguishers

30-2018.....Flammable and Combustible Liquids Code

51B-2019.....Standard for Fire Prevention During Welding,
Cutting and Other Hot Work

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70-2020.....National Electrical Code

70B-2019.....Recommended Practice for Electrical Equipment
Maintenance

70E-2018Standard for Electrical Safety in the Workplace

241-2019.....Standard for Safeguarding Construction,
Alteration, and Demolition Operations

703-2024.....Standard for Fire Retardant-Treated Wood and
Fire-Retardant Coatings for Building Materials

F. The Joint Commission (TJC)

TJC ManualComprehensive Accreditation and Certification
Manual

G. U.S. Nuclear Regulatory Commission

10 CFR 20Standards for Protection Against Radiation

H. U.S. Occupational Safety and Health Administration (OSHA):

29 CFR 1904Reporting and Recording Injuries & Illnesses

29 CFR 1910Safety and Health Regulations for General
Industry

29 CFR 1926Safety and Health Regulations for Construction
Industry

CPL 2-0.124.....Multi-Employer Citation Policy

I. US Army Corps of Engineers Safety and Occupational Health (SOH)
Requirements Publication No. EM 385-1-1, 2024 edition

1.2 DEFINITIONS:

A. Critical Lift. A lift with the hoisted load exceeding 75% of the crane's maximum capacity; lifts made out of the view of the operator (blind picks); lifts involving two or more cranes; personnel being hoisted; and special hazards such as lifts over occupied facilities, loads lifted close to power-lines, and lifts in high winds or where other adverse environmental conditions exist; and any lift which the crane operator believes is critical.

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B. OSHA "Competent Person" (CP). One who is capable of identifying existing and predictable hazards in the surroundings and working conditions which are unsanitary, hazardous or dangerous to employees, and who has the authorization to take prompt corrective measures to eliminate them (see 29 CFR 1926.32(f)).

C. Competent Person, Confined Space

The CP, Confined Space, is a person meeting the competent person requirements as defined EM 385-1-1 Appendix Q, with thorough knowledge of OSHA's Confined Space Standard, 29 CFR 1910.146, and designated in writing to be responsible for the immediate supervision, implementation and monitoring of the confined space program, who through training, knowledge and experience in confined space entry is capable of identifying, evaluating and addressing existing and potential confined space hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

D. Competent Person, Cranes and Rigging.

The CP, Cranes and Rigging, as defined in EM 385-1-1 Appendix Q, is a person meeting the competent person, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the Crane and Rigging Program, who through training, knowledge and experience in crane and rigging is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures regarding such hazards.

E. Competent Person, Excavation/Trenching.

A CP, Excavation/Trenching, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and 29 CFR 1926, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the excavation/trenching program, who through training, knowledge and experience in excavation/trenching is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

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F. Competent Person, Fall Protection

The CP, Fall Protection, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and in accordance with ASSE/SAFE Z359.0, who has been designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the fall protection program, who through training, knowledge and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating and addressing existing and potential fall hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

G. Competent Person, Scaffolding

The CP, Scaffolding is a person meeting the competent person requirements in EM 385-1-1 Appendix Q and designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the scaffolding program. The CP for Scaffolding has enough training, knowledge and experience in scaffolding to correctly identify, evaluate and address existing and potential hazards and has the authority to take prompt corrective measures with regard to these hazards. CP qualifications must be documented and include experience on the specific scaffolding systems/types being used, assessment of the base material that the scaffold will be erected upon, load calculations for materials and personnel, and erection and dismantling. The CP for scaffolding must have a documented, minimum of 8-hours of scaffold training to include training on the specific type of scaffold being used (e.g. mast-climbing, adjustable, tubular frame), in accordance with EM 385-1-1 Section 22.B.02.

H. Competent Person (CP) Trainer

A competent person trainer as defined in EM 385-1-1 Appendix Q, who is qualified in the material presented, and who possesses a working knowledge of applicable technical regulations, standards, equipment and systems related to the subject matter on which they are training Competent Persons. A competent person trainer must be familiar with the typical hazards and the equipment used in the industry they are instructing. The training provided by the competent person trainer must be appropriate to that specific industry. The competent person

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trainer must evaluate the knowledge and skills of the competent persons as part of the training process.

I. High Risk Activities

High Risk Activities are activities that involve work at heights, crane and rigging, excavations and trenching, scaffolding, electrical work, and confined space entry.

J. "Qualified Person" means one who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training and experience, has successfully demonstrated his ability to solve or resolve problems relating to the subject matter, the work, or the project.

K. Qualified Person, Fall Protection (QP for FP)

A QP for FP is a person meeting the requirements of EM 385-1-1 Appendix Q, and ASSE/SAFE Z359.0, with a recognized degree or professional certificate and with extensive knowledge, training and experience in the fall protection and rescue field who is capable of designing, analyzing, and evaluating and specifying fall protection and rescue systems.

L. USACE Property and Equipment

Interpret "USACE" property and equipment specified in USACE EM 385-1-1 as Government property and equipment.

M. High Visibility Accident. Any mishap which may generate publicity or high visibility.

N. Mishap: Mishap in this specification is defined according to the EM 385-1-1. A mishap is any unplanned, undesired event that occurs during the course of work being performed. This includes accidents, incidents, and near misses.

O. Mishap Criticality Categories

1. No impact/Near-Miss - near miss incidents that shall be investigated and reported to the VA within 24 hours.
2. Minor incident/impact - incidents that require first aid or result in minor equipment damage (less than \$5000). These

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incidents must be investigated and reported to the VA within 24 hours.

3. Moderate incident/impact - Any work-related injury or illness that results in any of the following. These incidents must be investigated and are required to be reported to the VA within 2 hours.
 - a. Days away from work (any time lost after day of injury/illness onset);
 - b. Restricted work;
 - c. Transfer to another job;
 - d. Medical treatment beyond first aid;
 - e. Loss of consciousness;
 - f. A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (1) through (5) above or,
 - g. any incident that leads to major equipment damage (greater than \$5000).
 4. Major incident/impact - Any mishap that leads to fatalities, hospitalizations, amputations, and losses of an eye as a result of contractors' activities. Or any incident which leads to major property damage (greater than \$20,000) and/or may generate publicity or high visibility. These incidents must be investigated and are required to be reported to the VA as soon as practical, but not later than 2 hours after the incident.
- P. Load Handling Equipment (LHE)
- LHE is a term used to describe cranes, hoists and all other hoisting equipment (hoisting equipment means equipment, including crane, derricks, hoists and power operated equipment used with rigging to raise, lower or horizontally move a load).
- Q. Medical Treatment. Treatment administered by a physician or by registered professional personnel under the standing orders of a

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physician. Medical treatment does not include first aid treatment even through provided by physician or registered personnel.

1.3 SUBMITTAL REQUIRMENTS:

A. Government approval or concurrence is required for submittals with a "G" designation; submittals not having a "G" designation are submitted for information only.

B. Preconstruction Submittals

1. Accident Prevention Plan (APP); G

C. Reports

1. Monthly Contractor Health Safety and Environmental (HS&E) Performance Report; G
2. Notifications and Reports;
3. Mishap Reports;
4. Near-Miss Reports;
5. LHE Inspection Reports
6. Monthly Exposure Hour Reports;

D. Work Plans

1. Standard Lift Plan; G
2. Critical Lift Plan; G
3. Activity Hazard Analysis (AHA); G
4. Confined Space Entry Permit
5. Hot Work Permit
6. Radiography Operation Planning Work Sheet; G
7. Portable Gauge Operations Planning Worksheet; G

E. Certificates

1. Contractor Safety Self-Evaluation Checklist
2. Crane Operators/Riggers Certifications; G

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3. Certificate of Compliance
4. Mobile Cranes Inspection Certificate
5. License Certificates
6. Machinery & Mechanized Equipment Certification Form

1.4 REGULATORY REQUIREMENTS:

- A. In addition to the detailed requirements included in the provisions of this contract, comply with the most recent edition of USACE EM 385-1-1, comply with 29 CFR 1926, comply with 29 CFR 1910 as incorporated by reference within 29 CFR 1926, comply with ASSP A10.34, and all applicable federal, state, and local laws, ordinances, criteria, rules and regulations. Submit matters of interpretation of standards for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern except with specific approval and acceptance by the Contracting Officer's Representative (COR).

~~SPEC WRITER NOTE: VHA Directive 7715-17 requires inclusion of FAR Clause 52.236-13, Accident Prevention in all construction contracts. Paragraph (f) of the requisite clause, which requires the contractor to develop an Accident Prevention Plan (APP) and Activity Hazard Analyses (AHAs), should be routinely included with the clause as most construction is sufficiently hazardous to warrant inclusion.~~

1.5 ACCIDENT PREVENTION PLAN (APP) :

- A. The APP (aka Construction Safety & Health Plan) shall interface with the Contractor's overall safety and health program. Include any portions of the Contractor's overall safety and health program referenced in the APP in the applicable APP element and ensure it is site-specific. The Government considers the Prime Contractor to be the "controlling authority" for all worksite safety and health of each subcontractor(s). Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the contract and the penalties for noncompliance, coordinating the work to prevent

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one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that Mishap prevention responsibilities are being carried out.

B. The APP shall be prepared as follows:

1. Written in English by a qualified person who is employed by the Prime Contractor articulating the specific work and hazards pertaining to the contract. Specifically articulating the safety requirements found within these VA contract safety specifications and the latest version of the United States Army Corps of Engineers - Safety and Occupational Health (SOH) Requirements - EM 385-1-1 Manual. Model language and format can be found in Appendix A of the EM 385-1-1 Manual
2. Address both the Prime Contractors and the subcontractors work operations.
3. State measures to be taken to control hazards associated with materials, services, or equipment provided by suppliers.
4. Address all the elements/sub-elements and in order as follows:
 - a. **SIGNATURE SHEET.** Title, signature, and phone number of the following:
 - 1) Plan preparer (Qualified Person such as corporate safety staff person or contracted Certified Safety Professional with construction safety experience);
 - 2) Plan approver (company/corporate officers authorized to obligate the company);
 - 3) Plan concurrence (e.g., Chief of Operations, Corporate Chief of Safety, Corporate Industrial Hygienist, project manager or superintendent, project safety professional). Provide concurrence of other applicable corporate and project personnel (Contractor).
 - b. **BACKGROUND INFORMATION.** List the following:
 - 1) Contractor;

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- 2) Contract number;
 - 3) Project name;
 - 4) Brief project description, description of work to be performed, and location; phases of work anticipated (these will require an AHA).
- c. **STATEMENT OF SAFETY AND HEALTH POLICY.** Provide a copy of current corporate/company Safety and Health Policy Statement, detailing commitment to providing a safe and healthful workplace for all employees. The Contractor's written safety program goals, objectives, and accident experience goals for this contract should be provided. The Statement of Safety and Health Policy must be signed by a company executive.
- d. **RESPONSIBILITIES AND LINES OF AUTHORITIES.** Provide the following:
- 1) A statement of the employer's ultimate responsibility for the implementation of his Safety and Occupational Health (SOH) program;
 - 2) Identification and accountability of personnel responsible for safety at both corporate and project level. Contracts specifically requiring safety or industrial hygiene personnel shall include a copy of their resumes.
 - 3) The names of Competent and/or Qualified Person(s) and proof of competency/qualification to meet specific OSHA or EM 385-1-1 Competent/Qualified Person(s) requirements must be attached.;
 - 4) Requirements that no work shall be performed unless a designated competent person is present on the job site;
 - 5) Requirements for pre-task Activity Hazard Analysis (AHAs);
 - 6) Lines of authority;
 - 7) Policies and procedures regarding noncompliance with safety requirements (to include disciplinary actions for violation of safety requirements) should be identified;

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e. **SUBCONTRACTORS AND SUPPLIERS.** If applicable, provide procedures for coordinating SOH activities with other employers on the job site:

- 1) Identification of subcontractors and suppliers (if known);
- 2) Safety responsibilities of subcontractors and suppliers.

f. **TRAINING.**

- 1) Site-specific SOH orientation training at the time of initial hire or assignment to the project for every employee before working on the project site is required.
- 2) Mandatory training and certifications that are applicable to this project (e.g., explosive actuated tools, crane operator, rigger, crane signal person, fall protection, electrical lockout/NFPA 70E, machine/equipment lockout, confined space, etc.) and any requirements for periodic retraining/recertification area required.
- 3) Procedures for ongoing safety and health training for supervisors and employees shall be established to address changes in site hazards/conditions.
- 4) OSHA 10-hour Construction Outreach training within the past five years is required for all workers on site and the OSHA 30-hour Construction Outreach training within the past five years is required for Trade Competent Persons (CPs)

g. **SAFETY AND HEALTH INSPECTIONS.**

- 1) Specific assignment of responsibilities for a minimum daily job site safety and health inspection during periods of work activity: Who will conduct (e.g., "Site Safety and Health CP"), proof of inspector's training/qualifications, when inspections will be conducted, procedures for documentation, deficiency tracking system, and follow-up procedures.
- 2) Any external inspections/certifications that may be required (e.g., contracted CSP or CSHT)

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h. MISHAP INVESTIGATION & REPORTING.

1) The APP shall include identify person(s) responsible to provide the following to the COR:

1. Monthly HS&E Performance Report (Exposure Hours Data);
2. Mishap investigation reports;
3. Project site injury and illness logs;
4. Near-Miss reports;

i. PLANS (PROGRAMS, PROCEDURES) REQUIRED. Based on a risk assessment of contracted activities and on mandatory OSHA and/or EM 385-1-1 compliance programs, the Contractor shall address all applicable occupational, patient, and public safety risks in site-specific compliance and accident prevention plans. These Plans shall include but are not limited to procedures for addressing the risks associated with the following:

- 1) Emergency response;
- 2) Contingency for severe weather;
- 3) Fire Prevention;
- 4) Medical Support;
- 5) Posting of emergency telephone numbers;
- 6) Prevention of alcohol and drug abuse;
- 7) Site sanitation (housekeeping, drinking water, toilets);
- 8) Night operations and lighting;
- 9) Hazard communication program;
- 10) Welding/Cutting "Hot" work;
- 11) Electrical Safe Work Practices (Electrical LOTO/NFPA 70E);
- 12) General Electrical Safety;
- 13) Hazardous energy control (LOTO);

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- 14) Site-Specific Fall Protection & Prevention;
- 15) Excavation/trenching;
- 16) Asbestos abatement;
- 17) Lead abatement;
- 18) Crane Critical lift;
- 19) Respiratory protection;
- 20) Health hazard control program;
- 21) Radiation Safety Program;
- 22) Abrasive blasting;
- 23) Heat/Cold Stress Monitoring;
- 24) Crystalline Silica Monitoring (Assessment);
- 25) Demolition plan (to include engineering survey);
- 26) Formwork and shoring erection and removal;
- 27) Pre-Cast Concrete;
- 28) Public (Mandatory compliance with ANSI/ASSP A10.34-2012);
- 29) Sub-contractor or cross-trade coordination.

- C. Submit the APP to the COR for review for compliance with contract requirements in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES 15 calendar days prior to the date of the preconstruction safety conference for acceptance. Work cannot proceed without an accepted APP.
- D. Once accepted by the COR, the APP and attachments will be enforced as part of the contract. Disregarding the provisions of this contract or the accepted APP will be cause for stopping of work, at the discretion of the COR in accordance with FAR Clause 52.236-13, *Accident Prevention*, until the matter has been rectified.
- E. Once work begins, changes to the accepted APP shall be made with the knowledge and concurrence of the COR. Should any severe hazard

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exposure, i.e. imminent danger, become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the COR within 2 hours of discovery.

Eliminate/remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions to safeguard onsite personnel, visitors, the public and the environment.

1.6 ACTIVITY HAZARD ANALYSES (AHAS) :

- A. AHAs are also known as Job Hazard Analyses, Job Safety Analyses, and Activity Safety Analyses. Before beginning each work activity involving a type of work presenting hazards not experienced in previous project operations or where a new work crew or sub-contractor is to perform the work, the Contractor(s) performing that work activity shall prepare an AHA (Example electronic AHA forms can be found on the US Army Corps of Engineers web site)
- B. AHAs shall define the activities being performed and identify the work sequences, the specific anticipated hazards, site conditions, equipment, materials, and the control measures to be implemented to eliminate or reduce each hazard to an acceptable level of risk.
- C. Work shall not begin until the AHA for the work activity has been accepted by the COR and discussed with all engaged in the activity, including the Contractor, subcontractor(s), and Government on-site representatives at preparatory and initial control phase meetings.
 - 1. The names of the Competent/Qualified Person(s) required for a particular activity (for example, excavations, scaffolding, fall protection, other activities as specified by OSHA, EM 385-1-1, or other State and Local agencies) shall be identified and included in the AHA. Certification of their competency/qualification shall be submitted to the Government Designated Authority (GDA) for acceptance prior to the start of that work activity.
 - 2. The AHA shall be reviewed and modified as necessary to address changing site conditions, operations, or change of competent/qualified person(s).
 - a. If more than one Competent/Qualified Person is used on the AHA activity, a list of names shall be submitted as an attachment to

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the AHA. Those listed must be Competent/Qualified for the type of work involved in the AHA and familiar with current site safety issues.

- b. If a new Competent/Qualified Person (not on the original list) is added, the list shall be updated (an administrative action not requiring an updated AHA). The new person shall acknowledge in writing that he or she has reviewed the AHA and is familiar with current site safety issues.
3. Submit AHAs to the COR for review for compliance with contract requirements in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES for review at least 15 calendar days prior to the start of each phase. Subsequent AHAs as shall be formatted as amendments to the APP. The analysis should be used during daily inspections to ensure the implementation and effectiveness of the activity's safety and health controls.
4. The AHA list will be reviewed periodically (at least weekly) at the Contractor supervisory safety meeting and updated as necessary when procedures, scheduling, or hazards change.
5. Develop the activity hazard analyses using the project schedule as the basis for the activities performed. All activities listed on the project schedule will require an AHA. The AHAs will be developed by the contractor, supplier, or subcontractor and provided to the prime contractor for review and approval and then submitted to the COR for review for compliance.

~~SPEC WRITER NOTE: Include FAR Clause
52.236-26, Preconstruction Safety
Conference to ensure that it takes place.~~

1.7 PRECONSTRUCTION SAFETY CONFERENCE:

- A. Contractor representatives who have a responsibility or significant role in implementation of the accident prevention program, as required by 29 CFR 1926.20(b)(1), on the project shall attend the pre-construction safety conference to gain a mutual understanding of its implementation. This includes the project superintendent, subcontractor

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superintendents, and any other assigned safety and health professionals.

- B. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the contract. This list of proposed AHAs will be reviewed at the conference and an agreement will be reached between the Contractor and the COR as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, review, and acceptance of AHAs to preclude project delays.

1.8 SAFETY MEETINGS:

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors on the project location. The SSHO, supervisors, foremen, or CDSOs must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the COR on request. Notify the COR of all scheduled meetings 7 calendar days in advance.

~~SPEC WRITER NOTE: Projects over \$30M or executed by the Office of Construction and Facilities Management (CFM) are required to have a dedicated SSHO on-site; Projects under \$30M should consider a full time dedicated SSHO as part of the Pre-Construction Risk Assessment (PCRA) process. Facilities that want to remove the requirement should include documentation as to why it was removed as part of the PCRA.~~

1.9 "SITE SAFETY AND HEALTH OFFICER" (SSHO) AND "COMPETENT PERSON" (CP):

- A. The Prime Contractor shall provide a Safety oversight team that includes a minimum of one (1) person at each project site, for each shift, to function as the Site Safety and Health Officer (SSHO). The

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Prime Contractor shall provide a minimum of one "Full-Time" SSHO at each project site, for each shift (with no other duties) that holds as current, a professional safety certification with at least 3 years of dedicated construction safety related experience. The SSHO shall ensure that the requirements of the VA and of 29 CFR 1926.16 are met for the project. The SSHO must be at the work site at all times, during construction activities, to implement and administer the Contractor's safety program and government-accepted Accident Prevention Plan. If the SSHO is off-site for a period longer than 8 hours, or one shift, and not exceeding 40 hours, or 5 shifts, a pre-approved Alternate Safety Officer shall be provided and shall fulfill the same roles and responsibilities as the primary SSHO. The Alternate Safety Officer shall have the required training, experience, and qualifications in accordance with EM 385-1-1 Section 01.A.17, and all associated subparagraphs. When the SSHO is temporarily (not to exceed 8 hours) off-site, a Designated Representative (DR) from the Prime Contractors' staff, as identified in the AHA may be used in lieu of an Alternate Safety Officer and shall be on the project site, at all times, when work is being performed. Note that the DR is a collateral duty safety position, with safety duties in addition to their full-time occupation. Each subcontractor shall designate a minimum of one CP in compliance with 29 CFR 1926.20 (b) (2) that will be identified as a CP to administer their individual safety programs.

- B. Further, all specialized Competent Persons for the work crews will be supplied by the respective contractor as required by 29 CFR 1926 (i.e. Asbestos, Electrical, Cranes, & Derricks, Demolition, Fall Protection, Fire Safety/Life Safety, Ladder, Rigging, Scaffolds, and Trenches/Excavations).
- C. These Competent Persons can have collateral duties as the subcontractor's superintendent and/or work crew lead persons as well as fill more than one specialized CP role (i.e. Asbestos, Electrical, Cranes, & Derricks, Demolition, Fall Protection, Fire Safety/Life Safety, Ladder, Rigging, Scaffolds, and Trenches/Excavations). However, the SSHO has be a separate qualified individual from the Prime Contractor's Superintendent and/or Quality Control Manager with duties only as the SSHO

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- D. The SSHO or an equally qualified Designated Representative/alternate will maintain a presence on the site during construction operations in accordance with FAR Clause 52.236-6: *Superintendence by the Contractor*. CPs will maintain presence during their construction activities in accordance with above mentioned clause. A listing of the designated SSHO and all known CPs shall be submitted prior to the start of work as part of the APP with the training documentation and/or AHA as listed in Section 1.8 below.
- E. The repeated presence of uncontrolled hazards during a contractor's work operations will result in the designated CP as being deemed incompetent and result in the required removal of the employee in accordance with FAR Clause 52.236-5: Material and Workmanship, Paragraph (c).

1.10 TRAINING:

- A. The designated Prime Contractor SSHO must meet the requirements of all applicable OSHA standards and be capable (through training, experience, and qualifications) of ensuring that the requirements of 29 CFR 1926.16 and other appropriate Federal, State and local requirements are met for the project.
- B. As a minimum the SSHO must have completed the OSHA 30-hour Construction Safety Outreach class within the past five (5) years and:
1. Seven (7) years of construction industry safety related experience.
 2. OR have a safety and health degree from an accredited university or college and five (5) years of construction industry safety related experience
 3. OR hold as current, a Certified Safety Professional (CSP) or a Construction Health and Safety Technician (CHST) certification and five (3) years of construction industry safety related experience.
- C. The Alternate Safety Officer shall meet, at a minimum, the requirements of EM 385-1-1 Section 1 and have five (5) years of construction industry safety experience. If the SSHO does not have a current certification, certification must be obtained within 90 days, maximum, of contract award.

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- D. All designated CPs shall have completed the OSHA 30-hour Construction Safety course and/or EM 385-1-1 40-hour training within the past 5 years. In addition, all CPs with high hazard work operations (such as operations involving asbestos, electrical, cranes, demolition, work at heights/fall protection, fire safety/life safety, ladder, rigging, scaffolds, and trenches/excavations) shall have a specialized formal course in the hazard recognition and control associated with those high hazard work operations. Documented "repeat" deficiencies in the execution of safety requirements will require retaking the requisite formal course.
- E. All other construction workers shall have the OSHA 10-hour Construction Safety Outreach course within the past 5 years and any necessary safety training to be able to identify hazards within their work environment.
- F. Submit training records associated with the above training requirements to the COR for review for compliance with contract requirements in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES 15 calendar days prior to the date of the preconstruction safety conference for acceptance.
- G. Prior to any worker for the contractor or subcontractors beginning work, they shall undergo a safety briefing provided by the SSHO or his/her designated representative. As a minimum, this briefing shall include information on the site-specific hazards, construction limits, VAMC safety guidelines, means of egress, break areas, work hours, locations of restrooms, use of VAMC equipment, emergency procedures, accident reporting etc. Documentation shall be provided to the COR that individuals have undergone contractor's safety briefing.
- H. Ongoing safety training will be accomplished in the form of weekly documented safety meeting.

1.11 INSPECTIONS:

- A. The SSHO shall conduct frequent and regular safety inspections (daily) of the site and each of the subcontractors CPs shall conduct frequent and regular safety inspections (daily) of their work operations as required by 29 CFR 1926.20(b)(2). Each week, the SSHO shall conduct a formal documented inspection with a written report of the entire

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construction areas with the subcontractors' "Trade Safety and Health CPs" present in their work areas. Coordinate with, and report findings and corrective actions weekly to the COR.

B. A Certified Safety Professional (CSP) with specialized knowledge in construction safety or a certified Construction Safety and Health Technician (CSHT) shall randomly conduct a monthly site safety inspection. The CSP or CSHT shall be one that is not a part of the immediate site project team. The individual can be a corporate safety professional or independently contracted who is not an immediate member of the construction project site team. The CSP or CSHT will provide their certificate number on the required report for verification as necessary.

1. Results of the inspection will be documented with tracking of the identified hazards to abatement.
2. The COR will be notified immediately prior to start of the inspection and invited to accompany the inspection.
3. Identified hazard and controls will be discussed to come to a mutual understanding to ensure abatement and prevent future reoccurrence.
4. A report of the inspection findings with status of abatement will be provided to the COR within one week of the onsite inspection.

1.12 MISHAPS, OSHA 300 LOGS, AND EXPOSURE-HOURS:

A. The prime contractor shall establish and maintain a Near-Miss and Mishap reporting, recordkeeping, and analysis system to track and analyze all injuries and illnesses, high visibility incidents, and accidental property damage (both government and contractor) that occur on site. Notify and provide an initial report to the COR as soon as practical, but no more than two hours after any Moderate or Major Mishap, High Visibility Incidents, or any weight handling and hoisting equipment mishap. No Impact/Near-Miss and Minor Mishaps shall be reported within 24 hours or as soon as practical. Within the notification the sender shall include contractor name; contract title; type of contract; name of activity, installation or location where mishap occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and

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brief description of mishap (to include type of construction equipment used, PPE used, etc.). Preserve the conditions and evidence on the mishap site until the COR determines whether a government investigation will be conducted.

- B. Conduct a mishap investigation for all Mishaps including mishaps resulting in at least \$20,000 in damages without injury, to establish the root cause(s) of the mishap. The Mishap investigation shall include images, 5 whys, the injured person's firsthand account, any witness accounts, methods of procedures, related AHA to the task, and corrective action plan signed by the president, vice president, or appropriate corporate-level leadership identified in the company's org chart for the project. Additionally, complete the VA Form 2162 (or equivalent) and provide the report to the COR within 7 calendar days of the accident. The COR will provide copies of any required or special forms.
- C. A summation of all exposure-hours worked by the contractor and associated sub-contractors for each month will be reported to the COR no later than the 10th day of the following month.
- D. A summation of all Minor, Moderate, and Major incidents experienced on site by the contractor and associated sub-contractors for each month will be provided to the COR monthly. The contractor and associated sub-contractors' OSHA 300 logs will be made available to the COR upon request.

1.13 PERSONAL PROTECTIVE EQUIPMENT (PPE):

- A. PPE is governed in all areas by the nature of the work the employee is performing. For example, specific PPE required for performing work on electrical equipment is identified in NFPA 70E, Standard for Electrical Safety in the Workplace.
- B. Any PPE above and beyond the Mandatory PPE identified below and determined by the COR to be necessary in the performance of the Governments duties relative to this contract, shall be provided to the Government upon written notification by the COR and at no additional expense to the Government. All PPE provided to the Government shall

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also be accompanied by any relevant or required training necessary to ensure its proper use.

C. Mandatory Minimum PPE includes:

1. Hard Hats - unless written authorization is given by the COR in circumstances of work operations that have limited potential for falling object hazards such as during finishing work or minor remodeling. With authorization to relax the requirement of hard hats, if a worker becomes exposed to an overhead falling object hazard, then hard hats would be required in accordance with the OSHA and/or EM 385-1-1 regulations.
2. Safety glasses - unless written authorization is given by the COR in circumstances of no eye hazards, appropriate safety glasses meeting the ANSI Z.87.1 standard must be worn by each person on site.
3. Appropriate Safety Shoes - based on the hazards present, safety shoes meeting the requirements of ASTM F2413-11 shall be worn by each person on site unless written authorization is given by the COR in circumstances of no foot hazards.
4. Hearing protection - Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks.
5. High-visibility Safety Apparel - appropriate brightly colored clothing that feature some type of reflective material on them, such as safety vests, shall be worn by personnel or visitors on construction sites.

~~SPEC WRITER NOTE: VA Medical Facility Construction Safety Committee Chair designates who will complete VHA Pre-Construction Risk Assessments (PCRA) for each construction, renovation or maintenance project. Project or construction activity scope and complexity determine the need for multiple VHA PCRA's. VHA PCRA's are to be completed and documented in accordance with Directive 7715. The VHA PCRA form must be used to assess and document all construction-associated hazards that affect VA medical facilities, their~~

~~occupants, services and mission-essential functions and capabilities. VHA-PCRA's are intended to eliminate or minimize construction-associated risks. The VHA-PCRA form, use instructions and other guidance are available at <http://vaww.hcfp.va.gov/resources/vha-pre-construction-risk-assessment-pera>. NOTE: This is an internal VA website that is not available to the public.~~

1.14 PRE-CONSTRUCTION RISK ASSESSMENT

- A. Control of all construction-associated hazards that affect VA medical facilities, their occupants, services and mission-essential functions and capabilities is critical in all medical center facilities. VHA Pre-Construction Risk Assessments (PCRA's) for construction, renovation and maintenance projects are included with this contract solicitation with required mitigations of identified hazards. VHA-PCRA's will be re-validated and updated as needed based on but not limited to changes from original designs, affected individuals, areas/locations, scope, contractor means and methods, safety requirements, phasing, contractor competencies and capabilities.

~~SPEC WRITER NOTE: Based upon construction activities where the initial infection control review within the VHA-PCRA's identifies potential risks of infectious disease transmission affecting the care, treatment or services of patients or residents include this paragraph. A detailed analysis must be conducted using the VHA Infection Control Risk Assessment for Construction, Renovation and Maintenance form to document in-depth infection risk assessments and identification of mitigation actions/activities. The VHA-ICRA form use instructions and other guidance are available at: <http://vaww.hcfp.va.gov/resources/vha-infection-control-risk-assessment-icra>. NOTE: This is an internal VA website that is not available to the public.~~

- B. Infection Prevention and Control is critical in all medical center facilities. Interior construction activities causing disturbance of existing dust, or creating new dust, must be conducted within ventilation-controlled areas that minimize the flow of airborne

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particles into patient areas. A detailed analysis of potential risks for infectious disease transmission affecting the care, treatment or services of patients or residents has been conducted. VHA Infection Control Risk Assessments (ICRAs) are included with this contract solicitation with required mitigation actions/activities. VHA-ICRAs will be re-validated and updated as needed based on changes in original designs, affected individuals, area(s) or location(s), scope, contractor means and methods, infection prevention and control requirements, differing site conditions, phasing, contractor competencies and capabilities, and infectious disease outbreaks.

C. For work occurring at a VA medical facility, coordinate with the facility Safety Manager and COR, as several aspects of this section directly relate to infection control risk assessments required in or adjacent to construction affecting occupied buildings accredited by The Joint Commission.

D. Products and Materials:

1. Sheet Plastic: Fire retardant polyethylene, 6-mil thickness meeting local fire codes.
2. Barrier Doors: Self Closing ~~// One hour // Two hour //~~ fire-rated ~~//~~ solid core wood in steel frame, painted.
3. Dust proof ~~// one-hour // two-hour //~~ fire-rated ~~//~~ drywall.
4. High Efficiency Particulate Air-Equipped filtration machine rated at 95% capture of 0.3 microns including pollen, mold spores and dust particles. HEPA filters should have ASHRAE 85 or other prefilter to extend the useful life of the HEPA. Provide both primary and secondary filtrations units. Maintenance of equipment and replacement of the HEPA filters and other filters will be in accordance with manufacturer's instructions.
5. Exhaust Hoses: Heavy duty, flexible steel reinforced; Ventilation Blower Hose.
6. Adhesive Walk-off Mats: Provide minimum size mats of 24 inches x 36 inches.

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7. Disinfectant: EPA-registered, Hospital-approved disinfectant or equivalent product.
8. Portable Ceiling Access Module.

~~SPEC WRITER NOTES:~~

~~1. VAMC's Infection Control Risk Assessment (ICRA) Team shall prepare an Infection Control plan and continue oversight during design, planning and construction on a regular basis. (VHA Directive 7715-17).~~

~~2. ICRA Team may provide a separate document to be included as part of the contract documents or may modify requirements included in the following article.~~

1.15 DUST CONTROL

- A. Contractor shall verify that dust will not be introduced into the facility through intake vents or building openings. HEPA filtration on intake vents is required where dust may be introduced.
- B. Dust created from disturbance of soil such as from vehicle movement will be wetted with use of a water truck as necessary.
- C. All cutting, drilling, grinding, sanding, or disturbance of materials shall be accomplished with tools equipped with either local exhaust ventilation (i.e. vacuum systems) or wet suppression controls.

~~SPEC WRITER NOTE: Perform a TB pre-construction risk assessment for the transmission of Tuberculosis (TB) to the contracted construction workers based upon the construction site location, patient population, hospital layout, and the defined risk as outlined in the "CDC Guidelines for preventing the transmission of Mycobacterium Tuberculosis in Health-Care Setting, 2005". A pre placement tuberculin screening is only required when/if contracted construction worker(s) have been determined to be at risk for transmission of TB to them based upon this TB pre construction risk assessment. Include the following section only as applicable.~~

1.16 TUBERCULOSIS SCREENING

- A. Contractor shall provide written certification that all contract employees assigned to the work site have had a pre-placement tuberculin screening within 90 days prior to assignment to the worksite and been found have negative TB screening reactions. Contractors shall be required to show documentation of negative TB screening reactions for any additional workers who are added after the 90-day requirement before they will be allowed to work on the work site. NOTE: This can be the Center for Disease Control (CDC) and Prevention and two-step skin testing or a Food and Drug Administration (FDA)-approved blood test.
1. Contract employees manifesting positive screening reactions to the tuberculin shall be examined according to current CDC guidelines prior to working on VHA property.
 2. Subsequently, if the employee is found without evidence of active (infectious) pulmonary TB, a statement documenting examination by a physician shall be on file with the employer (construction contractor), noting that the employee with a positive tuberculin screening test is without evidence of active (infectious) pulmonary TB.
 3. If the employee is found with evidence of active (infectious) pulmonary TB, the employee shall require treatment with a subsequent statement to the fact on file with the employer before being allowed to return to work on VHA property.

~~SPEC WRITER NOTE: Coordinate editing with facility Safety Manager/Officer at VA medical facilities. Edit subparagraphs D, F, H, I, N, Q and R carefully as they directly relate to interim life safety measures required in or adjacent to construction affecting occupied buildings by the Joint Commission on Accreditation of Healthcare Organizations. At other sites, edit for project and delete // and facility Safety // Manager // Officer// provisions.~~

1.17 FIRE SAFETY

- A. Fire Safety Plan: Establish and maintain a site-specific fire protection program in accordance with 29 CFR 1926. Prior to start of work, prepare a plan detailing project-specific fire safety measures, including periodic status reports, and submit to the COR for review for compliance with contract requirements in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES. This plan may be an element of the Accident Prevention Plan.
- B. Site and Building Access: Maintain free and unobstructed access to facility emergency services and for fire, police and other emergency response forces in accordance with NFPA 241.
- C. For work occurring at a VA medical facility, coordinate with the facility Safety Manager and COR, as several aspects of this section directly relate to interim life safety measures required in or adjacent to construction affecting occupied buildings accredited by The Joint Commission.
- D. Separate temporary facilities, such as trailers, storage sheds, and dumpsters, from existing buildings and new construction by distances in accordance with NFPA 241. For small facilities with less than 6 m (20 feet) exposing overall length, separate by 3m (10 feet).
- E. Temporary Construction Partitions:

~~SPEC WRITER NOTE: Where phasing drawings are used, show locations and hourly fire ratings of anticipated temporary construction partitions and hourly fire ratings of nearby existing construction on phasing drawings. Detail unusual conditions.~~

- 1. Install and maintain temporary construction partitions to provide smoke-tight separations between ~~// construction areas // the areas that are described in phasing requirements //~~ and adjoining areas. Construct partitions of gypsum board or fire-retardant treated plywood (fire-retardant treated in accordance with NFPA 703) on both sides of fire-retardant treated wood framing or metal steel studs. Extend the partitions through suspended ceilings to floor slab deck

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- or roof. Seal joints and penetrations. At door openings, install Class C, ¾ hour fire/smoke rated doors with self-closing devices.
2. Install ~~// one-hour // two-hour // fire-rated //~~ temporary construction partitions as shown on drawings to maintain integrity of existing exit stair enclosures, exit passageways, fire-rated enclosures of hazardous areas, horizontal exits, smoke barriers, vertical shafts and openings enclosures.
 3. Close openings in smoke barriers and fire-rated construction to maintain fire ratings. Seal penetrations with listed through-penetration firestop materials ~~in accordance with Section 07-84-00, FIRESTOPPING.~~
- F. Temporary Heating and Electrical: Install, use and maintain installations in accordance with 29 CFR 1926, NFPA 241 and NFPA 70.
- G. Means of Egress: Do not block exiting for occupied buildings, including paths from exits to roads. Minimize disruptions and coordinate with the COR.
- H. Egress Routes for Construction Workers: Maintain free and unobstructed egress. Inspect daily. Report findings and corrective actions weekly to the COR.
- I. Fire Extinguishers: Provide and maintain extinguishers in construction areas and temporary storage areas in accordance with 29 CFR 1926, NFPA 241 and NFPA 10.
- J. Flammable and Combustible Liquids: Store, dispense and use liquids in accordance with 29 CFR 1926, NFPA 241 and NFPA 30.
- K. Standpipes: Install and extend standpipes up with each floor in accordance with 29 CFR 1926 and NFPA 241. Do not charge wet standpipes subject to freezing until weather protected.
- ~~SPEC WRITER NOTE: Modify to suit design.
Coordinate with phasing.~~
- L. Sprinklers: Install, test and activate new automatic sprinklers prior to removing existing sprinklers.

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- M. Existing Fire Protection: Do not impair automatic sprinklers, smoke and heat detection, and fire alarm systems, except for portions immediately under construction, and temporarily for connections. Provide fire watch for impairments more than 4 hours in a 24-hour period. Request interruptions in accordance with Article, OPERATIONS AND STORAGE AREAS, and coordinate with the COR. All existing or temporary fire protection systems (fire alarms, sprinklers) located in construction areas shall be tested as coordinated with the medical center. Parameters for the testing and results of any tests performed shall be recorded by the medical center and copies provided to the COR.
- N. Smoke Detectors: Prevent accidental operation. Remove temporary covers at end of work operations each day. Coordinate with the COR.

~~SPEC WRITER NOTE: Use facility permit process at existing VA medical facilities. For other sites, use contractor's process.~~

- O. Hot Work: Perform and safeguard hot work operations in accordance with NFPA 241 and NFPA 51B. Coordinate with the COR at least 24 hours in advance. Designate contractor's responsible project-site fire prevention program manager to permit hot work.
- P. Fire Hazard Prevention and Safety Inspections: Inspect entire construction areas weekly. Coordinate with, and report findings and corrective actions weekly to the COR.
- Q. Smoking: Smoking is prohibited in and adjacent to construction areas inside existing buildings and additions under construction. In separate and detached buildings under construction, smoking is prohibited except in designated smoking rest areas.
- R. Dispose of waste and debris in accordance with NFPA 241. Remove from buildings daily.

~~SPEC WRITER NOTE: If it is anticipated that work will be performed in compartmentalized areas, add the following subparagraph.~~

- S. If required, submit documentation to the COR that personnel have been trained in the fire safety aspects of working in areas with impaired structural or compartmentalization features.

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1.18 ELECTRICAL

- A. All electrical work shall comply with NFPA 70 (NEC), NFPA 70B, NFPA 70E, 29 CFR Part 1910 Subpart J - General Environmental Controls, 29 CFR Part 1910 Subpart S - Electrical, and 29 CFR 1926 Subpart K in addition to other references required by contract.
- B. All qualified persons performing electrical work under this contract shall be licensed journeyman or master electricians. All apprentice electricians performing under this contract shall be deemed unqualified persons unless they are working under the immediate supervision of a licensed electrician or master electrician.
- C. All electrical work will be accomplished de-energized and in the Electrically Safe Work Condition (refer to NFPA 70E for Work Involving Electrical Hazards, including Exemptions to Work Permit). Any Contractor, subcontractor or temporary worker who fails to fully comply with this requirement is subject to immediate termination in accordance with FAR clause 52.236-5(c). Only in rare circumstance when achieving an electrically safe work condition prior to beginning work would increase or cause additional hazards or is infeasible due to equipment design or operational limitations is energized work permitted. The COR with approval of the Medical Center Director will make the determination if the circumstances would meet the exception outlined above. An AHA and permit specific to energized work activities will be developed, reviewed, and accepted by the VA prior to the start of that activity.
 - 1. Development of a Hazardous Electrical Energy Control Procedure is required prior to de-energization. A single Simple Lockout/Tagout Procedure for multiple work operations can only be used for work involving qualified person(s) de-energizing one set of conductors or circuit part source. Task specific Complex Lockout/Tagout Procedures are required at all other times.
 - 2. Verification of the absence of voltage after de-energization and lockout/tagout is considered "energized electrical work" (live work) under NFPA 70E, and shall only be performed by qualified persons wearing appropriate shock protective (voltage rated) gloves and arc rate personal protective clothing and equipment, using Underwriters

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Laboratories (UL) tested and appropriately rated contact electrical testing instruments or equipment appropriate for the environment in which they will be used.

3. Personal Protective Equipment (PPE) and electrical testing instruments will be readily available for inspection by the COR.
- D. Before beginning any electrical work, an Activity Hazard Analysis (AHA) will be conducted to include Shock Hazard and Arc Flash Hazard analyses (NFPA Tables can be used only as a last alternative and it is strongly suggested a full Arc Flash Hazard Analyses be conducted). Work shall not begin until the AHA for the work activity and permit for energized work has been reviewed and accepted by the COR and discussed with all engaged in the activity, including the Contractor, subcontractor(s), and Government on-site representatives at preparatory and initial control phase meetings.
- E. Ground-fault circuit interrupters. GFCI protection shall be provided where an employee is operating or using cord- and plug-connected tools related to construction activity supplied by 125-volt, 15-, 20-, or 30-ampere circuits. Where employees operate or use equipment supplied by greater than 125-volt, 15-, 20-, or 30- ampere circuits, GFCI protection or an assured equipment grounding conductor program shall be implemented in accordance with NFPA 70E - 2015, Chapter 1, Article 110.4(C) (2) .

1.19 FALL PROTECTION

- A. The fall protection (FP) threshold height requirement is 4ft for ALL WORK in government occupied sites, unless the OSHA 29 CFR 1926 or EM 385-1-1 requirements are more stringent, to include steel erection activities, systems-engineered activities (prefabricated) metal buildings, residential (wood) construction and scaffolding work. For work occurring at new "greenfield" sites or sites with 24/7 Contractor controlled 100% restricted access, the fall protection (FP) threshold height may be increased to 6ft unless the OSHA 29 CFR 1926 or EM 385-1-1 requirements are more stringent.
1. The use of a Safety Monitoring System (SMS) as a fall protection method is prohibited.

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2. The use of Controlled Access Zone (CAZ) as a fall protection method is prohibited.
3. A Warning Line System (WLS) may ONLY be used on floors or flat or low-sloped roofs (between 0 - 18.4 degrees or 4:12 slope) and shall be erected around all sides of the work area (See 29 CFR 1926.502(f) for construction of WLS requirements). Working within the WLS does not require FP. No worker shall be allowed in the area between the roof or floor edge and the WLS without FP. FP is required when working outside the WLS.
4. Fall protection while using a ladder will be governed by the more stringent of OSHA and EM 385-1-1 requirements.

1.20 SCAFFOLDS AND OTHER WORK PLATFORMS

- A. All scaffolds and other work platforms construction activities shall comply with 29 CFR 1926 Subpart L.
- B. The fall protection (FP) threshold height requirement is 4 ft as stated in Section 1.16.
- C. The following hierarchy and prohibitions shall be followed in selecting appropriate work platforms.
 1. Scaffolds, platforms, or temporary floors shall be provided for all work except that can be performed safely from the ground or similar footing.
 2. Ladders less than 20 feet may be used as work platforms only when use of small hand tools or handling of light material is involved.
 3. Ladder jacks, lean-to, and prop-scaffolds are prohibited.
 4. Emergency descent devices shall not be used as working platforms.
- D. Contractors shall use a scaffold tagging system in which all scaffolds are tagged by the Competent Person. Tags shall be color-coded: green indicates the scaffold has been inspected and is safe to use; red indicates the scaffold is unsafe to use. Tags shall be readily visible, made of materials that will withstand the environment in which they are used, be legible and shall include:
 1. The Competent Person's name and signature;

2. Dates of initial and last inspections.

E. Mast Climbing work platforms: When access ladders, including masts designed as ladders, exceed 20 ft (6 m) in height, positive fall protection shall be used.

~~1.21 EXCAVATION AND TRENCHES~~

~~A. All excavation and trenching work shall comply with 29 CFR 1926 Subpart P. Excavations less than 5 feet in depth require evaluation by the contractor's "Competent Person" (CP) for determination of the necessity of an excavation protective system where kneeling, laying in, or stooping within the excavation is required.~~

~~B. All excavations and trenches 24 inches in depth or greater shall require a written trenching and excavation permit (NOTE — some States and other local jurisdictions require separate state/jurisdiction-issued excavation permits). The permit shall have two sections, one section will be completed prior to digging or drilling and the other will be completed prior to personnel entering the excavations greater than 5 feet in depth. Each section of the permit shall be provided to the COR prior to proceeding with digging or drilling and prior to proceeding with entering the excavation. After completion of the work and prior to opening a new section of an excavation, the permit shall be closed out and provided to the COR. The permit shall be maintained onsite, and the first section of the permit shall include the following:~~

- ~~1. Estimated start time & stop time~~
- ~~2. Specific location and nature of the work.~~
- ~~3. Indication of the contractor's "Competent Person" (CP) in excavation safety with qualifications and signature. Formal course in excavation safety is required by the contractor's CP.~~
- ~~4. Indication of whether soil or concrete removal to an offsite location is necessary.~~
- ~~5. Indication of whether soil samples are required to determine soil contamination.~~

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- ~~6. Indication of coordination with local authority (i.e. "One Call") or contractor's effort to determine utility location with search and survey equipment.~~
- ~~7. Indication of review of site drawings for proximity of utilities to digging/drilling.~~
- ~~C. The second section of the permit for excavations greater than five feet in depth shall include the following:~~
 - ~~1. Determination of OSHA classification of soil. Soil samples will be from freshly dug soil with samples taken from different soil type layers as necessary and placed at a safe distance from the excavation by the excavating equipment. A pocket penetrometer will be utilized in determination of the unconfined compression strength of the soil for comparison against OSHA table (Less than 0.5 Tons/FT2 — Type C, 0.5 Tons/FT2 to 1.5 Tons/FT2 — Type B, greater than 1.5 Tons/FT2 — Type A without condition to reduce to Type B).~~
 - ~~2. Indication of selected protective system (sloping/benching, shoring, shielding). When soil classification is identified as "Type A" or "Solid Rock", only shoring or shielding or Professional Engineer designed systems can be used for protection. A Sloping/Benching system may only be used when classifying the soil as Type B or Type C. Refer to Appendix B of 29 CFR 1926, Subpart P for further information on protective systems designs.~~
 - ~~3. Indication of the spoil pile being stored at least 2 feet from the edge of the excavation and safe access being provided within 25 feet of the workers.~~
 - ~~4. Indication of assessment for a potential toxic, explosive, or oxygen deficient atmosphere where oxygen deficiency (atmospheres containing less than 19.5 percent oxygen) or a hazardous atmosphere exists or could reasonably be expected to exist. Internal combustion engine equipment is not allowed in an excavation without providing force-air ventilation to lower the concentration to below OSHA PELs, providing sufficient oxygen levels, and atmospheric testing as necessary to ensure safe levels are maintained.~~

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~~C. As required by OSHA 29 CFR 1926.651(b)(1), the estimated location of utility installations, such as sewer, telephone, fuel, electric, water lines, or any other underground installations that reasonably may be expected to be encountered during excavation work, shall be determined prior to opening an excavation.~~

~~1. The planned dig site will be outlined/marked in white prior to locating the utilities.~~

~~2. Used of the American Public Works Association Uniform Color Code is required for the marking of the proposed excavation and located utilities.~~

~~3. 811 will be called two business days before digging on all local or State lands and public Right of Ways.~~

~~4. Digging will not commence until all known utilities are marked.~~

~~5. Utility markings will be maintained~~

~~D. Excavations will be hand dug or excavated by other similar safe and acceptable means as excavation operations approach within 3 to 5 feet of identified underground utilities. Exploratory bar or other detection equipment will be utilized as necessary to further identify the location of underground utilities.~~

~~E. Excavations greater than 20 feet in depth require a Professional Engineer designed excavation protective system.~~

1.22 CRANES

A. All crane work shall comply with 29 CFR 1926 Subpart CC and EM 385-1-1.

B. Prior to operating a crane, the operator must be licensed, qualified or certified to operate the crane. Thus, all the provisions contained with Subpart CC are effective and there is no "Phase In" date.

C. A detailed lift plan for all lifts shall be submitted to the COR 14 days prior to the scheduled lift complete with route for truck carrying load, crane load analysis, siting of crane and path of swing and all other elements of a critical lift plan where the lift meets the definition of a critical lift. Critical lifts require a more comprehensive lift plan to minimize the potential of crane failure

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and/or catastrophic loss. The plan must be reviewed and accepted by the General Contractor before being submitted to the VA for review. The lift will not be allowed to proceed without prior acceptance of this document by the COR.

D. Crane operators shall not carry loads

1. over the general public or VAMC personnel
2. over any occupied building unless
 - a. The top two or more floors are vacated under the lift affected area AND a Contractor provided engineering analysis, performed by a Professional Engineer, demonstrates that overhead protection provided by the roof and vacated floors is sufficient to protect against the potential loss of a load being lifted or a minimum of 300 psf of protection is provided, whichever is greater
 - b. OR Overhead protection is designed by a Professional Engineer and installed over the roof by the Contractor to protect the occupants below. The installed system shall be sufficient to protect against the potential loss of a load being lifted or a minimum of 300 psf of protection is provided, whichever is greater.
 - c. OR A formal request may be submitted to the COR for documentation, if available, indicating the roof of the building has been previously designed and constructed using a reinforced concrete roof slab or other means to provide overhead protection from construction activities specifically associated with the vertical or horizontal expansion of the same building AND a Contractor provided engineering analysis, performed by a Professional Engineer, demonstrates that overhead protection provided by the roof is sufficient to protect against the potential loss of a load being lifted or a minimum of 300 psf of protection is provided, whichever is greater.

1.23 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

- A. All installation, maintenance, and servicing of equipment or machinery shall comply with 29 CFR 1910.147 except for specifically referenced

operations in 29 CFR 1926 such as concrete & masonry equipment [1926.702(j)], heavy machinery & equipment [1926.600(a)(3)(i)], and process safety management of highly hazardous chemicals (1926.64). Control of hazardous electrical energy during the installation, maintenance, or servicing of electrical equipment shall comply with Section 1.15 to include NFPA 70E and other VA specific requirements discussed in the section.

~~1.24 CONFINED SPACE ENTRY~~

- ~~A. All confined space entry shall comply with 29 CFR 1926, Subpart AA except for specifically referenced operations in 29 CFR 1926 such as excavations/trenches [1926.651(g)].~~
- ~~B. A site specific Confined Space Entry Plan (including permitting process) shall be developed and submitted to the COR.~~

1.25 WELDING AND CUTTING

As specified in section 1.14, Hot Work: Perform and safeguard hot work operations in accordance with NFPA 241 and NFPA 51B. Coordinate with the COR. Obtain permits from the COR at least 24 hours in advance. Designate contractor's responsible project-site fire prevention program manager to permit hot work.

1.26 LADDERS

- A. All Ladder use shall comply with 29 CFR 1926 Subpart X and EM 385-1-1.
 - B. All portable ladders shall be of sufficient length and shall be placed so that workers will not stretch or assume a hazardous position.
 - C. Manufacturer safety labels shall be in place on ladders
 - D. Step Ladders shall not be used in the closed position
 - E. Top steps or cap of step ladders shall not be used as a step
 - F. Portable ladders, used as temporary access, shall extend at least 3 ft (0.9 m) above the upper landing surface.
1. When a 3 ft (0.9-m) extension is not possible, a grasping device (such as a grab rail) shall be provided to assist workers in mounting and dismounting the ladder.

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2. In no case shall the length of the ladder be such that ladder deflection under a load would, by itself, cause the ladder to slip from its support.

G. Ladders shall be inspected for visible defects on a daily basis and after any occurrence that could affect their safe use. Broken or damaged ladders shall be immediately tagged "DO NOT USE," or with similar wording, and withdrawn from service until restored to a condition meeting their original design.

1.27 FLOOR & WALL OPENINGS

A. All floor and wall openings shall comply with 29 CFR 1926 Subpart M and EM 385-1-1.

B. Floor and roof holes/openings are any that measure over 2 in (51 mm) in any direction of a walking/working surface which persons may trip or fall into or where objects may fall to the level below. Skylights located in floors or roofs are considered floor or roof hole/openings.

C. All floor, roof openings or hole into which a person can accidentally walk or fall through shall be guarded either by a railing system with toe boards along all exposed sides or a load-bearing cover. When the cover is not in place, the opening or hole shall be protected by a removable guardrail system or shall be attended when the guarding system has been removed, or other fall protection system.

1. Covers shall be capable of supporting, without failure, at least twice the weight of the worker, equipment and material combined.

2. Covers shall be secured when installed, clearly marked with the word "HOLE", "COVER" or "Danger, Roof Opening-Do Not Remove" or color-coded or equivalent methods (e.g., red or orange "X"). Workers must be made aware of the meaning for color coding and equivalent methods.

3. Roofing material, such as roofing membrane, insulation or felts, covering or partly covering openings or holes, shall be immediately cut out. No hole or opening shall be left unattended unless covered.

4. Non-load-bearing skylights shall be guarded by a load-bearing skylight screen, cover, or railing system along all exposed sides.

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5. Workers are prohibited from standing/walking on skylights.

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SECTION 01 42 19
REFERENCE STANDARDS

PART 1 - GENERAL

1.1 DESCRIPTION

This section specifies the availability and source of references and standards specified in the project manual under paragraphs APPLICABLE PUBLICATIONS and/or shown on the drawings.

1.2 AVAILABILITY OF SPECIFICATIONS LISTED IN THE GSA INDEX OF FEDERAL SPECIFICATIONS, STANDARDS AND COMMERCIAL ITEM DESCRIPTIONS FPMR PART 101-29 (FAR 52.211-1) (AUG 1998)

- A. The GSA Index of Federal Specifications, Standards and Commercial Item Descriptions, FPMR Part 101-29 and copies of specifications, standards, and commercial item descriptions cited in the solicitation may be obtained for a fee by submitting a request to - GSA Federal Supply Service, Specifications Section, Suite 8100, 470 East L'Enfant Plaza, SW, Washington, DC 20407, Telephone (202) 619-8925, Facsimile (202) 619-8978.
- B. If the General Services Administration, Department of Agriculture, or Department of Veterans Affairs issued this solicitation, a single copy of specifications, standards, and commercial item descriptions cited in this solicitation may be obtained free of charge by submitting a request to the addressee in paragraph (a) of this provision. Additional copies will be issued for a fee.

1.3 AVAILABILITY FOR EXAMINATION OF SPECIFICATIONS NOT LISTED IN THE GSA INDEX OF FEDERAL SPECIFICATIONS, STANDARDS AND COMMERCIAL ITEM DESCRIPTIONS (FAR 52.211-4) (JUN 1988)

The specifications and standards cited in this solicitation can be examined at the following location:

DEPARTMENT OF VETERANS AFFAIRS
Office of Construction & Facilities Management
Facilities Quality Service (00CFM1A)
425 Eye Street N.W, (sixth floor)
Washington, DC 20001
Telephone Numbers: (202) 632-5249 or (202) 632-5178
Between 9:00 AM - 3:00 PM

1.4 AVAILABILITY OF SPECIFICATIONS NOT LISTED IN THE GSA INDEX OF FEDERAL SPECIFICATIONS, STANDARDS AND COMMERCIAL ITEM DESCRIPTIONS (FAR 52.211-3) (JUN 1988)

The specifications cited in this solicitation may be obtained from the associations or organizations listed below.

AA	Aluminum Association Inc. http://www.aluminum.org
AABC	Associated Air Balance Council https://www.aabc.com
AAMA	American Architectural Manufacturer's Association http://www.aamanet.org
AASHTO	American Association of State Highway and Transportation Officials http://www.aashto.org
AATCC	American Association of Textile Chemists and Colorists http://www.aatcc.org
ACGIH	American Conference of Governmental Industrial Hygienists http://www.acgih.org
ACI	American Concrete Institute http://www.aci-int.net
ACPA	American Concrete Pipe Association http://www.concrete-pipe.org
ACPPA	American Concrete Pressure Pipe Association http://www.acppa.org
ADC	Air Diffusion Council http://flexibleduct.org
AGA	American Gas Association http://www.aga.org
AGC	Associated General Contractors of America http://www.agc.org

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AGMA	American Gear Manufacturers Association, Inc. http://www.agma.org
AH	American Hort https://www.americanhort.org
AHAM	Association of Home Appliance Manufacturers http://www.aham.org
AIA	American Institute of Architects http://www.aia.org
AISC	American Institute of Steel Construction http://www.aisc.org
AISI	American Iron and Steel Institute http://www.steel.org
AITC	American Institute of Timber Construction https://aitc-glulam.org
AMCA	Air Movement and Control Association, Inc. http://www.amca.org
ANSI	American National Standards Institute, Inc. http://www.ansi.org
APA	The Engineered Wood Association http://www.apawood.org
ARI	Air-Conditioning and Refrigeration Institute http://www.ari.org
ARPM	Association for Rubber Product Manufacturers https://arpm.com
ASABE	American Society of Agricultural and Biological Engineers https://www.asabe.org
ASCE	American Society of Civil Engineers http://www.asce.org

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ASHRAE	American Society of Heating, Refrigerating, and Air-Conditioning Engineers http://www.ashrae.org
ASME	American Society of Mechanical Engineers http://www.asme.org
ASSE	American Society of Sanitary Engineering International http://www.asse-plumbing.org
ASTM	American Society for Testing and Materials International http://www.astm.org
AWI	Architectural Woodwork Institute https://www.awinet.org
AWS	American Welding Society https://www.aws.org
AWWA	American Water Works Association https://www.awwa.org
BHMA	Builders Hardware Manufacturers Association https://www.buildershardware.com
BIA	The Brick Industry Association http://www.gobrick.com
CAGI	Compressed Air and Gas Institute https://www.cagi.org
CGA	Compressed Gas Association, Inc. https://www.cganet.com
CI	The Chlorine Institute, Inc. https://www.chlorineinstitute.org
CISCA	Ceilings and Interior Systems Construction Association https://www.cisca.org
CISPI	Cast Iron Soil Pipe Institute https://www.cispi.org

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CLFMI	Chain Link Fence Manufacturers Institute https://www.chainlinkinfo.org
CPA	Composite Panel Association https://www.compositepanel.org
CPMB	Concrete Plant Manufacturers Bureau https://www.cpmc.org
CRA	California Redwood Association http://www.calredwood.org
CRSI	Concrete Reinforcing Steel Institute https://www.crsi.org
CTI	Cooling Technology Institute https://www.cti.org
DHA	Decorative Hardwoods Association https://www.decorativehardwoods.org
DHI	Door and Hardware Institute https://www.dhi.org
EGSA	Electrical Generating Systems Association http://www.egsa.org
EEI	Edison Electric Institute https://www.eei.org
EPA	United States Environmental Protection Agency https://www.epa.gov
ETL	ETL Testing Services http://www.intertek.com
FAA	Federal Aviation Administration https://www.faa.gov
FCC	Federal Communications Commission https://www.fcc.gov

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FPS	Forest Products Society http://www.forestprod.org
GANA	Glass Association of North America http://www.glasswebsite.com
FM	Factory Mutual Global Insurance https://www.fmglobal.com
GA	Gypsum Association https://gypsum.org
GSA	General Services Administration https://www.gsa.gov
HI	Hydraulic Institute http://www.pumps.org
ICC	International Code Council https://shop.iccsafe.org
ICEA	Insulated Cable Engineers Association https://www.icea.net
ICAC	Institute of Clean Air Companies http://www.icac.com
IEEE	Institute of Electrical and Electronics Engineers https://www.ieee.org/
IGMA	Insulating Glass Manufacturers Alliance https://www.igmaonline.org
IMSA	International Municipal Signal Association http://www.imsasafety.org
MBMA	Metal Building Manufacturers Association https://www.mbma.com
MSS	Manufacturers Standardization Society of the Valve and Fittings Industry http://msshq.org

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NAAMM National Association of Architectural Metal Manufacturers
<https://www.naamm.org>

PHCC Plumbing-Heating-Cooling Contractors Association
<https://www.phccweb.org>

NBS National Bureau of Standards
See - NIST

NBBI The National Board of Boiler and Pressure Vessel Inspectors
<https://www.nationalboard.org>

NEC National Electric Code
See - NFPA National Fire Protection Association

NEMA National Electrical Manufacturers Association
<https://www.nema.org>

NFPA National Fire Protection Association
<https://www.nfpa.org>

NHLA National Hardwood Lumber Association
<https://www.nhla.com>

NIH National Institute of Health
<https://www.nih.gov>

NIST National Institute of Standards and Technology
<https://www.nist.gov>

NELMA Northeastern Lumber Manufacturers Association, Inc.
<http://www.nelma.org>

NPA National Particleboard Association
(See CPA, Composite Panel Association)

NSF National Sanitation Foundation
<http://www.nsf.org>

OSHA Occupational Safety and Health Administration
Department of Labor
<https://www.osha.gov>

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PCA	Portland Cement Association https://www.cement.org
PCI	Precast Prestressed Concrete Institute https://www.pci.org
PPI	Plastics Pipe Institute https://www.plasticpipe.org
PEI	Porcelain Enamel Institute http://www.porcelainenamel.com
PTI	Post-Tensioning Institute http://www.post-tensioning.org
RFCI	Resilient Floor Covering Institute https://www.rfci.com
RIS	Redwood Inspection Service (See Western Wood Products Association) https://www.wwpa.org
SCMA	Southern Cypress Manufacturers Association http://www.cypressinfo.org
SDI	Steel Door Institute http://www.steeldoor.org
SJI	Steel Joist Institute https://www.steeljoist.org
SMACNA	Sheet Metal & Air-Conditioning Contractors' National Association https://www.smacna.org
SSPC	The Society for Protective Coatings https://www.sspc.org
STI	Steel Tank Institute https://www.steeltank.com
SWI	Steel Window Institute https://www.steelwindows.com

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TCNA Tile Council of North America

<https://www.tcnatile.com>

TEMA Tubular Exchanger Manufacturers Association

<http://www.tema.org>

TPI Truss Plate Institute

<https://www.tpinst.org>

UBC The Uniform Building Code
(See ICC)

UL Underwriters' Laboratories Incorporated

<https://www.ul.com>

ULC Underwriters' Laboratories of Canada

<https://www.ulc.ca>

WCLB West Coast Lumber Inspection Bureau

<http://www.wclib.org>

WDMA Window and Door Manufacturers Association

<https://www.wdma.com>

WRCLA Western Red Cedar Lumber Association

<https://www.realcedar.com>

WWPA Western Wood Products Association

<http://www.wwpa.org>

- - - E N D - - -

SECTION 01 45 00
QUALITY CONTROL

PART 1 - GENERAL

1.1 DESCRIPTION

This section specifies requirements for Contractor Quality Control (CQC) for Design-Bid-Build (DBB) or Design-Build (DB) construction projects. This section can be used for both project types.

1.2 APPLICABLE PUBLICATIONS

- A. The publication listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.
- B. ASTM International (ASTM)
 - 1. D3740 - (2012a) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
 - 2. E329 - (2014a) Standard Specification for Agencies Engaged in the Testing and/or Inspection of Materials Used in Construction

1.3 SUBMITTALS

Government approval is required for all submittals. CQC inspection reports shall be submitted under this Specification section and follow the [Applicable CQC Control Phase (Preparatory, Initial, or Follow-Up)]: [Applicable Specification section] naming convention.

- 1. Preconstruction Submittals
 - a. Interim CQC Plan
 - b. CQC Plan
 - c. Additional Requirements for Design Quality Control (DQC) Plan
- 2. Design Data
 - a. Discipline-Specific Checklists
 - b. Design Quality Control
- 3. Test Reports
 - a. Verification Statement

PART 2 PRODUCTS - NOT USED

PART 3 - EXECUTION

3.1 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system. that complies with the FAR Clause 52.246.12 titled "Inspection of Construction". QC consists of plans, procedures, and organization necessary to produce an end product which complies with the Contract requirements. The QC system covers all design and construction operations, both onsite and offsite, and be keyed to the proposed design and construction sequence. The project superintendent will be held responsible for the quality of work and is subject to removal by the Contracting Officer's Representative (COR) for non-compliance with the quality requirements specified in the Contract. In this context the highest level manager responsible for the overall construction activities at the site, including quality and production is the project superintendent. The project superintendent maintains a physical presence at the site at all times and is responsible for all construction and related activities at the site, except as otherwise acceptable to the COR.

3.2 CQC PLAN:

- A. Submit the CQC Plan no later than ~~// CO or Designee to determine during Constructability review - 15/30/60~~ //15 days after receipt of Notice to Proceed (NTP), , proposed to implement the requirements of the FAR Clause 52.246.12 titled "Inspection of Construction". The Government will consider an Interim CQC Plan for the first ~~//30 days~~ to match timeline established immediately above ~~//15~~ days of operation, which must be accepted within ~~// 5, 10, or 20~~ //30 business days of NTP. Design and/or construction will be permitted to begin only after acceptance of the CQC Plan or acceptance of an Interim plan applicable to the particular feature of work to be started. Work outside of the accepted Interim CQC Plan will not be permitted to begin until acceptance of a CQC Plan or another Interim CQC Plan containing the additional work scope is accepted.
- B. Content of the CQC Plan: Include, as a minimum, the following to cover all design and construction operations, both onsite and offsite,

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including work by subcontractors, designers of record consultants, architects/engineers (A/E), fabricators, suppliers, and purchasing agents:

1. A description of the QC organization, including a chart showing lines of authority and acknowledgement that the CQC staff will implement the three phase control system for all aspects of the work specified. Include a CQC System Manager that reports to the project superintendent.
2. The name, qualifications (in resume format) duties, responsibilities, and authorities of each person assigned a CQC function.
3. A copy of the letter to the CQC System Manager signed by an authorized official of the firm which describes the responsibilities and delegates sufficient authorities to adequately perform the functions of the CQC System Manager, including authority to stop work which is not in compliance with the Contract. Letters of direction to all other various quality control representatives outlining duties, authorities, and responsibilities will be issued by the CQC System Manager. Furnish copies of these letters to the COR.
4. Procedures for scheduling, reviewing, certifying, and managing submittals including those of subcontractors, designers of record, consultants, A/E's offsite fabricators, suppliers and purchasing agents. These procedures must be in accordance with Section 01 33 23 Shop Drawings, Product Data, and Samples.
5. Control, verification, and acceptance of testing procedures for each specific test to include the test name, specification paragraph requiring test, feature of work to be tested, test frequency, and person responsible for each test (Laboratory facilities approved by the COR are required to be used).
6. Procedures for tracking Preparatory, Initial, and Follow-Up control phases and control, verification, and acceptance tests including documentation.
7. Procedures for tracking design and construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected.

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8. Reporting procedures, including proposed reporting formats.
 9. A list of the definable features of work. A definable feature of work is a task which is separate and distinct from other tasks has separate control requirements, and is identified by different trades or disciplines, or it is work by the same trade in a different environment. Although each section of specifications can generally be considered as a definable feature of work, there are frequently more than one definable feature under a particular section. This list will be agreed upon during the Coordination meeting.
 10. Coordinate schedule work with Special Inspections—~~required by Section 01-45-35 Special Inspections~~, the Statement of Special Inspections and Schedule of Special Inspections. Where the applicable Code issue by the International Code Council (ICC) calls for inspections by the Building Official, the Contractor must include the inspections in the CQC Plan and must perform the inspections required by the applicable ICC. The Contractor must perform these inspections using independent qualified inspectors. Include the Special Inspection Plan requirements in the CQC Plan.
- C. Additional Requirements for Design Quality Control (DQC) Plan: The following additional requirements apply to the DQC Plan for DB projects only and not DBB projects:
1. Submit and maintain a DQC Plan as an effective QC program which assures that all services required by this contract are performed and provided in a manner that meets professional architectural and engineering quality standards. As a minimum, all documents must be technically reviewed by competent, independent reviewers identified in the DQC Plan. The same element that produced the product may not perform the independent technical review (ITR). Correct errors and deficiencies in the design documents prior to submitting them to the Government.
 2. Include the design schedule in the master project schedule, showing the sequence of events involved in carrying out the project design tasks within the specific Contract period. This should be at a detailed level of scheduling sufficient to identify all major design tasks, including those that control the flow of work. Include review and correction periods associated with each item. This should be a forward planning as well as a project monitoring tool. The schedule

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reflects calendar days and not dates for each activity. If the schedule is changed, submit a revised schedule reflecting the change within 7 calendar days. Include in the DQC Plan the discipline-specific checklists to be used during the design and quality control of each submittal. Submit at each design phase as part of the project documentation these completed discipline-specific checklists.

3. Implement the DQC Plan by a DQC Manager who has the responsibility of being cognizant of and assuring that all documents on the project have been coordinated. This individual must be a person who has verifiable engineering or architectural design experience and is a Professional Engineer or Registered Architect within the state of Construction location. Notify the COR, in writing, of the name of the individual, and the name of an alternate person assigned to the position.

~~SPEC WRITER NOTE: The Contracting Officer or Authorized designee will notify the Contractor in writing of the acceptance of the DQC Plan. After acceptance, any changes proposed by the Contractor are subject to the acceptance of the Government prior to implementation by the Contractor.~~

- D. Acceptance of Plan: Acceptance of the Contractor's plan is required prior to the start of design and construction. Acceptance is conditional and will be predicated on satisfactory performance during the design and construction. The Government reserves the right to require the Contractor to make changes in the CQC Plan and operations including removal of personnel as necessary, to obtain the quality specified.
- E. Notification of Changes: After acceptance of the CQC Plan, notify the COR in writing of any proposed change. Proposed changes are subject to acceptance by the Government prior to implementation by the Contractor.

3.3 COORDINATION MEETING:

After the Preconstruction Conference Post-award Conference before start of design or construction, and prior to acceptance by the Government of the CQC Plan, meet with the COR to discuss the Contractor's quality control system. Submit the CQC Plan a minimum of ~~// 2 or 5 //~~ business

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days prior to the Coordination Meeting. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CC operations, design activities (if applicable), control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the **Government**, signed by both the Contractor and the COR will become a part of the contract file. There can be occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

3.4 QUALITY CONTROL ORGANIZATION:

- A. Personnel Requirements: The requirements for the CQC organization are a Safety and Health Manager, CQC System Manager, a Design Quality Manager (if applicable), and sufficient number of additional qualified personnel to ensure safety and Contract compliance. The Safety and Health Manager shall satisfy the requirements of Specification 01 35 26 Safety Requirements and reports directly to a senior project (or corporate) official independent from the CQC System Manager. The Safety and Health Manager will also serve as a member of the CQC Staff. Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly will also be included as part of the CQC organization. The Contractor's CQC staff maintains a presence at the site at all times during progress of the work and have complete authority and responsibility to take any action necessary to ensure Contract compliance. The CQC staff will be subject to acceptance by the COR. Provide adequate office space, filing systems, and other resources as necessary to maintain an effective and fully functional CQC organization. Promptly complete and furnish all letters, material submittals, shop drawings submittals, schedules and all other project documentation to the CQC organization. The CQC organization is responsible to maintain these documents and records at the site at all times, except as otherwise acceptable to the Government.

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- B. CQC System Manager: Identify as CQC System Manager an individual within the onsite work organization that is responsible for overall management of CQC and has the authority to act in all CQC matters for the Contractor. The CQC system Manager is required to be a ~~//graduate~~ engineer, graduate architect, or a graduate of construction management, with a minimum of ~~// PM or SRE to determine qualifications based on project complexity at construction review //~~5 years construction experience on construction similar to the scope of this Contract. ~~//construction person with a minimum of // PM or SRE to determine qualifications based on project complexity at construction review //~~ years in related work. ~~//~~ This CQC System manager is on the site at all times during construction and is employed by the General Contractor. ~~//~~ The CQC System Manager is assigned as CQC System Manager but has duties as project superintendent in addition to quality control. ~~//~~ Identify in the plan an alternate to serve in the event of the CDQC System Manager's absence. The requirements for the alternate are the same as the CQC System Manager.
- C. CQC Personnel: In addition to CQC personnel specified elsewhere in the contract, provide as part of the CQC organization specialized personnel to assist in the CQC System Manager for the following areas, as applicable: electrical, mechanical, civil, structural, environmental, architectural, materials technician submittals clerk, Commissioning Agent/LEED specialist, and low voltage systems. These individuals or specified technical companies ~~// are directly employed by the General Contractor and cannot be employed by a supplier or subcontractor on this project //~~ are employees of the prime or subcontractor ~~//~~; be responsible to the CQC System Manager; be physically present at the construction site during work on the specialized personnel's areas of responsibility; have the necessary education or experience in accordance with the Experience Matrix listed herein. These individuals ~~// have no other duties other than quality control. //~~ can perform other duties but need to be allowed sufficient time to perform the specialized personnel's assigned quality controls duties as described in the CQC Plan. ~~//~~ A single person can cover more than one area provided that the single person is qualified to perform QC activities in each designated and that workload allows. ~~//~~

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EXPERIENCE MATRIX

Area	Qualifications
Civil	Graduate Civil Engineer or Construction Manager with 2 years experience in the type of work being performed on this project or technician with 5 years related experience.
Mechanical	Graduate Mechanical Engineer with 2 years experience or construction professional with 5 years of experience supervising mechanical features of work in the field with a construction company.
Electrical	Graduate Electrical Engineer with 2 years related experience or construction professional with 5 years of experience supervising electrical features of work in the field with a construction company.
Structural	Graduate Civil Engineer (with Structural Track or Focus), Structural Engineer, or Construction Manager with 2 years experience or construction professional with 5 years experience supervising structural features of work in the field with a construction company.
Architectural	Graduate Architect with 2 years experience or construction professional with 5 years of related experience.
Environmental	Graduate Environmental Engineer with 3 years experience.
Submittals	Submittal Clerk with 1 year experience.
Concrete, Pavement, and Soils	Materials Technician with 2 years experience for the appropriate area.
Testing, Adjusting, and Balancing (TAB)	Specialist must be a member of AABC or an experienced technician of the firm certified by the NEBB.
Design Quality Control Manager	Registered Architect or Professional Engineer

~~SPEC WRITE NOTE: The CQM Training certificate expires after 5 years. If the CQC System Manager's certificate has~~

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~~expired, retake the course to remain
current.~~

- D. Additional Requirements: In addition to the above experience and education requirements, the CQC System Manager and Alternate CQC System Manager are required to have completed the Construction Quality Management (CQM) for Construction course. If the CQC System Manager does not have a current specification, obtain the CQM for Contractors course identification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Command and the Army Corps of Engineers. Contact the COR for information on the next scheduled class.
- E. Organizational Changes: Maintain the CQC staff at full strength at all times. When it is necessary to make changes to the CQC staff, revise the CQC Plan to reflect the changes and submit the changes to the COR.

3.5 **SUBMITTALS AND DELIVERABLES:** Submittals have to comply with the requirements in Section 01 33 23 Shop Drawings, Product Data, and Samples. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the contract requirements. When Section 01 91 00 General Commissioning Requirements is included in the contract, the submittals required by the section have to be coordinated with the Section 01 33 23 Shop Drawings, Product Data, and Samples to ensure adequate time is allowed for each type of submittal required.

3.6 CONTROL:

- A. CQC is the means by which the Contractor ensures that the construction, to include that of subcontractors and suppliers, complies with the requirements of the contract. At least three phases of control are required to be conducted by the CQC System Manager for each definable feature of the construction work as follows:
1. Preparatory Phase: This phase is performed prior to beginning work on each definable feature of work after all required plans/documents/materials are approved/accepted, and after copies are at the work site. This phase includes:
 - a. A review of each paragraph of applicable specifications, references codes, and standards. Make available during the preparatory inspection a copy of those sections of referenced

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codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.

- b. Review of the Contract drawings.
- c. Check to assure that all materials and equipment have been tested, submitted, and approved.
- d. Review of provisions that have been made to provide required control inspection and testing.
- e. Review Special Inspections required by Section 01 45 35 Special Inspections, that Statement of Special Inspections and the Schedule of Specials Inspections.
- f. Examination of the work area to assure that all required preliminary work has been completed and is in compliance with the Contract.
- g. Examination of required materials, equipment, and sample work to assure that they are on hand conform to approved shop drawings or submitted data, and are properly stored.
- h. Review of the appropriate Activity Hazard Analysis (AHA) to assure safety requirements are met.
- i. Discussion of procedures for controlling quality of the work including repetitive deficiencies. Document construction tolerances and workmanship standards - contract defined or industry standard if not contract defined - for that feature of work.
- j. Check to ensure that the portion of the plan for the work to be performed has been accepted by the COR.
- k. Discussion of the initial control phase.
- l. The Government needs to be notified at least 48 hours or 2 business days in advance of beginning the Preparatory control phase. Include a meeting conducted by the CQC System Manager and attended by the superintendent, other CQC personnel (as applicable), and the foreman responsible for the definable feature. Document the results of the Preparatory phase actions by separate minutes prepared by the CQC System Manager and attach to the daily CQC report. Instruct applicable workers as

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to the acceptable level of workmanship required in order to meet contract specifications.

- B. Initial Phase: This phase is accomplished at the beginning of a definable feature of work. Accomplish the following:
1. Check work to ensure that it is in full compliance with contract requirements. Review minutes of the Preparatory meeting.
 2. Verify adequacy of controls to ensure full contract compliance. Verify the required control inspection and testing is in compliance with the contract.
 3. Establish level of workmanship and verify that it meets minimum acceptable workmanship standards. Compare with required sample panels as appropriate.
 4. Resolve all differences.
 5. Check safety to include compliance with an upgrading of the safety plan and activity hazard analysis. Review the activity analysis with each worker.
 6. The Government needs to be notified at least 48 hours or 2 business days in advance of beginning the initial phase for definable features of work. Prepare separate minutes of this phase by the CQC System Manager and attach to the daily CQC report. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with Follow-Up phases.
 7. The initial phase for each definable feature of work is repeated for each new crew to work onsite, or any time acceptable specified quality standards are not being met.
 8. Coordinate scheduled work with Special Inspections required by Section 01 45 35 Special Inspections, the Statement of Special Inspections, and the Schedule of Special Inspections.
- C. Follow-Up Phase: Perform daily checks to assure control activities, including control testing, are providing continued compliance with contract requirements until the completion of the particular feature of work. Record the checks in the CQC documentation. Conduct final Follow-Up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work. Do not build upon nor conceal non-conforming work. Coordinate scheduled work with Special Inspections required by Section 01 45 35 Special

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Inspections, the Statement of Special Inspections, and the Schedule of Special Inspections

- D. Additional Preparatory and Initial Phases on the same definable features of work if: the quality ongoing work is unacceptable; if there are changes in the applicable CQC staff, onsite production supervision or work crew; if work on a definable feature is resumed after a substantial period of inactivity, or if other problems develop.

3.7 TESTS

- A. Testing Procedure: Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance test when specified. Procure the services of a Department of Veteran Affairs approved testing laboratory or establish an approved testing laboratory at the project site. Perform the following activities and record and provide the following data:
1. Verify that testing procedures comply with contract requirements.
 2. Verify that facilities and testing equipment are available and comply with testing standards.
 3. Check test instrument calibration data against certified standards.
 4. Verify that recording forms and test identification control number system, including all of the test documentation requirements, have been prepared.
 5. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the unique sequential control number identifying the test. If approved by the COR actual test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the COR. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

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- B. Testing Laboratories: All testing laboratories must be validated through the procedures contained in Specification ~~section 01 45 29~~ Testing Laboratory Services.
1. Capability Check: The Government reserves the right to check laboratory equipment in the proposed laboratory for compliance with the standards set forth in the contract specifications and to check the laboratory technician's testing procedures and techniques. Laboratories utilized for testing soils, concrete, asphalt and steel is required to meet criteria detailed in ASTM D3740 and ASTM E329.
 2. Capability Recheck: If the selected laboratory fails the capability check, the Contractor will be assessed a charge equal to value of recheck to reimburse the Government for each succeeding recheck of the laboratory or the checking of a subsequently selected laboratory. Such costs will be deducted from the Contract amount due the Contractor.
- C. Onsite Laboratory: The Government reserves the right to utilize the Contractor's control testing laboratory and equipment to make assurance tests, and to check the Contractor's testing procedures, techniques, and test results at no additional cost to the Government.

3.8 COMPLETION INSPECTION

- A. Punch-Out Inspection: Conduct an inspection of the work by the CQC system Manager near the end of the work, or any increment of the work established by the specifications. Prepare and include in the CQC documentation a punch list of items which do not conform to the approved drawings and specifications. Include within the list of deficiencies the estimated date by which the deficiencies will be corrected. Make a second inspection the CQC System Manager or staff to ascertain that all deficiencies have been corrected. Once this is accomplished, notify the Government that the facility is ready for the Government Pre-Final Inspection.
- B. Pre-Final Inspection: The Government will perform the Pre-Final Inspection to verify that the facility is complete and ready to be occupied. A Government Pre-Final Punch List may be developed as a result of this inspection. Ensure that all items on this list have been corrected before notifying the Government, so that a Final Acceptance Inspection with the customer can be scheduled. Correct any items noted

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on the Pre-Final Inspection in a timely manner. These inspections and any deficiency corrections required by this paragraph need to be accomplished within the time slated for completion of the entire work or any particular increment of the work if the project is divided into increments by separate construction completion dates.

- C. Final Acceptance Inspection: The Contractor's QC Inspection personnel, plus the superintendent or other primary management person, and the COR is required to be in attendance at the Final Acceptance Inspection. Additional Government personnel can also be in attendance. The Final Acceptance Inspection will be formally scheduled by the COR based upon results of the Pre-Final Inspection. Notify the COR at least 14 days prior to the Final Acceptance Inspection and include the Contractor's assurance that all specific items previously identified to the Contractor as being unacceptable, along with all remaining work performed under the contract, will be complete and acceptable by the date schedule for the Final Acceptance Inspection. Failure of the Contractor to have all contract work acceptably complete for this inspection will be cause for the COR to bill the Contractor for the Government's additional inspection cost in accordance with FAR Clause 52.246-12 titled "Inspection of Construction".

3.9 DOCUMENTATION

- A. Quality Control Activities: Maintain current records providing factual evidence that required QC activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:
1. The name and area of responsibility of the Contractor/Subcontractor
 2. Operating plant/equipment with hours worked, idle, or down for repair.
 3. Work performed each day, giving location, description, and by whom. When Network Analysis (NAS) is used, identify each phase of work performed each day by NAS activity number.
 4. Test and control activities performed with results and references to specification/drawing requirements. Identify the Control Phase (Preparatory, Initial, and/or Follow-Up). List deficiencies noted, along with corrective action.

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5. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specification/drawing requirements.
 6. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.
 7. Offsite surveillance activities, including actions taken.
 8. Job safety evaluations stating what was checked, results, and instructions or corrective actions.
 9. Instructions given/received and conflicts in plans and specifications.
 10. Provide documentation of design quality control activities. For independent design reviews, provide, as a minimum, identification of the Independent Technical Reviewer (ITR) team, the ITR review comments, responses, and the record of resolution of the comments.
- B. Verification Statement: Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract. Furnish the original and one copy of these records in report form to the Government daily with 1 week after the date covered by the report, except that reports need not be submitted for days on which no work is performed. As a minimum, prepare and submit on report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the contract. The first report following a day of no work will be for that day only. Reports need to be signed and dated by the CQC System Manager. Include copies of test reports and copies of reports prepared by all subordinate QC personnel within the CQC System Manager Report.

3.10 SAMPLE FORMS

Templates of various quality control reports can be found on the Whole Building Design Guide website at https://www.wbdg.org/FFC/NAVGRAPH/01%2045%2000.00%2020_quality_control_reports.pdf

- 3.11 **NOTIFICATION OF NONCOMPLIANCE:** The COR will notify the Contractor of any detected noncompliance with the foregoing requirements. The Contractor should take immediate corrective action after receipt of

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such notice. Such notice, when delivered to the Contractor at the work site will be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the COR can issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of claim for extension of time or for excess costs or damages by the Contractor.

--- End of Section ---

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SECTION 01 45 35
SPECIAL INSPECTIONS

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This guide specification will be applicable to both new buildings and existing building rehabilitations/renovations. In addition to the Special Inspection and testing specified requirements, a registered design professional must perform structural observations during construction. All observed deficiencies will be immediately reported to the Contracting Officer. The registered design professional performing these observations will be a representative of the Designer of Record (DOR) for the building being constructed.
- B. Structural observations are required for the following project conditions per IBC Chapter 17:
 - 1. Seismic Design Category D, E or F; and assigned to Risk Cat III, IV or V.
 - 2. Seismic Design Category D, E or F; and with a height greater than 22860 mm 75 ft.
 - 3. Seismic Design Category E, assigned to Risk Category I or II and the building is greater than two stories above grade plane.
 - 4. Nominal design wind speed in excess of 49 m/sec 110 mph; and assigned to Risk Cat III, IV or V.
 - 5. Nominal design wind speed in excess of 49 m/sec 110 mph; and with a height greater than 23 m 75 ft.

1.2 APPLICABLE PUBLICATIONS

- A. The publication listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.
- B. American Society of Civil Engineers (ASCE)
 - 1. ASCE 7 - (2010; Errata 2011; Supp 2 2013) Minimum Design Loads for Buildings and Other Structures
- C. International Code Council (ICC)
 - 2. ICC IBC - (2015) International Building Code

1.3 GENERAL REQUIREMENTS

- A. Perform Special Inspections in accordance with the Statement of Special Inspections, Schedule of Special Inspections and Chapter 17 of ICC IBC. The Statement of Special Inspections and Schedule of Special Inspections are included as an attachment to this specification. Special Inspections are to be performed by an independent third party and are intended to ensure that the work of the prime contractor is in accordance with the Contract Documents and applicable building codes. Special inspections do not take the place of the three phases of control inspections performed by the Contractor's QC Manager or any testing and inspections required by other sections of the specifications.
- B. Structural observations will be performed by the Government. The contractor must provide notification to the Contracting Officer 14 days prior to the following points of construction that structural observations need to occur:

SPEC. WRITE NOTE: Define the points in construction that structural observations need to occur in paragraph below - delete if unused

//1. The Government shall perform structural observations during construction on the following dates //____//

//2. The Government shall perform structural observations during construction on the following dates //____//

1.4 DEFINITIONS

- A. Continuous Special Inspections - The constant monitoring of specific tasks by a special inspector. These inspections must be carried out continuously over the duration of the particular tasks.
- B. Periodic Special Inspections - Special Inspections by the special inspector who is intermittently present where the work to be inspected has been or is being performed. Specific time interval on a specific Special Inspection should be indicated on the Schedule of Special Inspections.
- C. Perform - Perform these Special Inspections tasks for each welded joint or member.
- D. Observe - Observe these Special Inspections items on a random daily basis. Operations need not be delayed pending these inspections.

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- E. Special Inspector (SI) - A qualified person retained by the contractor and approved by the Contracting Officer as having the competence necessary to inspect a particular type of construction requiring Special Inspections. The SI must be an independent third party hired directly by the Prime Contractor.
- F. Associate Special Inspector (ASI) - A qualified person who assists the SI in performing Special Inspections but must perform inspection under the direct supervision of the SI and cannot perform inspections without the SI on site.
- G. Third Party - A third party inspector must not be company employee of the Contractor or any Sub-Contractor performing the work to be inspected.
- H. Special Inspector of Record (SIOR) - SIOR must be an independent third party hired directly by the Prime Contractor and is required for the following project conditions:
 - 1. Seismic Design Category D, E, or F; and assigned to Risk Category III, IV, or V.
 - 2. Seismic Design Category D, E, or F; and with a height greater than 22860mm 75 ft.
 - 3. Seismic Design Category E, assigned to Risk Category I or II and the building is greater than two (2) stories above grade plane.
 - 4. Nominal design wind speed in excess of 49 m/sec 100 mph; and assigned to Risk Category III, IV, or V.
 - 5. Nominal design wind speed in excess of 49 mm/sec 100mph; and with a height greater than 23m 75ft.
 - 6. In addition to these conditions, the DOR is encouraged to consider using an SIOR on large magnitude or critical projects where this additional level of quality control is affordable.
- I. Contracting Officer - The Government official having overall authority for administrative contracting actions. Certain contracting actions may be delegated to the Contracting Officer's Representative (COR).
- J. Contractor's Quality Control (QC) Manager - An individual retained by the prime contractor and qualified in accordance with the Section 01 45 00.00 10 QUALITY CONTROL having the overall responsibility for the contractor's QC organization.
- K. Designer of Record (DOR) - A registered design professional is contracted by the Government as an A/E responsible for the overall

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design and review of submittal documents prepared by others. The DOR is registered or licensed to practice their respective design profession as defined by the statutory requirements of the professional registration laws in state in which the design professional works. The DOR is also referred to as the Engineer of Record (EOR) in design code documents.

- L. Statement of Special Inspections (SSI) - A document developed by the DOR identifying the material, systems, components and work required to have Special Inspections and covering the following:
1. List of the Architectural Designated Seismic Systems - these components are in or attached to a Risk Category IV or V structure and are needed for continued operation of the facility or their failure could impair the continued operation of the facility.
 2. List of the Mechanical Designated Seismic Systems
 - a. For Seismic Design Category C or Risk V, list the following:
 - 1) Heating, ventilation, and air-conditioning (HVAC) ductwork containing hazardous materials and anchorage of such ductwork
 - 2) Piping systems and mechanical units containing flammable, combustible, or highly toxic materials.
 - b. For Seismic Design Category D, E, or F or Risk Category V list mechanical system that meet one of the following:
 - 1) Life safety component required to function after an earthquake
 - 2) Component that contains hazardous content,
 - 3) All components in an essential facility needed for continued operation after an earthquake.
 3. List of the Electrical Designated Systems
 - a. For Seismic Design Category C or Risk V, list the anchorage of electrical equipment used for emergency or standby power systems.
 - b. For Seismic Design Category D, E or F list electrical system that meet one of the following:
 - 1) Life safety component required to function after an earthquake
 - 2) Component that contains hazardous content,
 - 3) All components in an essential facility needed for continued operation after an earthquake.

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4. List of elements that are part of the progressive collapse resistance system.
- a. Provide a description of the following as they apply:
- 1) Elements of the tie force system consisting of internal longitudinal and transverse, vertical, and peripheral tie.
 - 2) Elements of the alternate path system.
 - 3) Elements having enhanced local resistance. The Statement of Special Inspections and the Schedule of Special Inspections will be included as an attachment to this specification
 - a) Schedule of Special Inspections - A schedule which lists each of the required Special Inspections, the extent to which each Special Inspections is to be performed, and the required frequency for each in accordance with ICC IBC Chapter 17. Template found here:



SECTION 01 45 35
Schedule of SI Temp



SECTION 01 45 35
Statement of SI Tem

- b) Designated Seismic System - Those nonstructural components that require design in accordance with ASCE 7 Chapter 13 and for which the component importance factor, I_p , is greater than 1.0. This designation applies to systems that are required to be operational following the Design Earthquake for RC I - IV structures and following the MCER for RC V structures. All systems in RC V facilities designated as MC-1 in accordance with UFC 3-310-04 are considered part of the Designated Seismic Systems. // Designated Seismic Systems will be identified by Owner and will have an Importance Factor $I_p = 1.5$ //
- M. Submittals: Government approval is required for all submittals. CQC Special Inspection reports shall be submitted under this Specification section and follow the [Special Inspection]: [Applicable Specification section or description] naming convention. Submit the following:
1. SD-01 Preconstruction Submittals;
 2. SIOR Letter of Acceptance;
 3. Special Inspections Project Manual;
 4. Special Inspections Agency's Written Practices

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5. NDT Procedures and Equipment' Calibration Records;
 6. SD-06 Test Reports;
 7. Special Inspections
 8. Daily Reports;
 9. Special Inspections; Biweekly Reports;
 10. SD-07 Certificates;
 11. Fabrication Plant
 12. Steel Truss Plant;
 13. Wood Truss Plant;
 14. AC472 Accreditation;
 15. Steel Joist Institute Membership;
 16. Precast Concrete Institute (PCI) Certified Plant;
 17. Certificate of Compliance;
 18. Special Inspector of Record Qualifications;
 19. Special Inspector Qualifications;
 20. Qualification Records for NDT technicians;
 21. SD-11 Closeout Submittals;
 22. Interim Final Report of Special Inspections;
 23. Comprehensive Final Report of Special Inspections;
- N. Special Inspector Qualifications: Submit qualifications for each SI, ASI, and the SIOR from the following certifying associations:
Associated Air Balance Council (AABC); American Concrete Institute (ACI); Association of the Wall and Ceiling Industry (AWCI); American Welding Society (AWS); Factory Mutual (FM); International Code Council (ICC); Nondestructive Testing (NDT); National Institute for Certification in Engineering Technologies (NICET); Precast/Prestressed Concrete Institute (PCI); Post-Tensioning Institute (PTI); Underwriters Laboratories (UL). Qualifications should be in accordance with the following minimums // ; PM or SRE can restrict qualifications to the higher standards shown if multiple options are shown for a role based on complexity of project //.

QUALIFICATIONS

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Area	Special Inspector	Associated Special Inspector	SIOR
Steel Construction and High Strength Bolting	ICC Structural Steel and Bolting Special Inspector certificate with on year of related experience, or Registered Professional Engineer with related experience.	Engineer-In-Training with one year of related experience.	
Welding Structural Steel (For highly complex steel use only AWS Certified Welding Inspectors)	ICC Welding Special Inspector certificate with one year of related experience or AWS Certified Welding Inspector	AWS Certified Associate Welding Inspector	
Nondestructive Testing of Welds	NDT Level II Certificate	NDT Level II Certificate plus one year of related experience	
Cold Formed Steel Framing	ICC Structural Steel and Bolting Special Inspector certificate with on year of related experience, or ICC Commercial Building Inspector with one year of experience; or Registered Professional Engineer with related experience.	Engineer-In-Training with one year of related experience.	

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Area	Special Inspector	Associated Special Inspector	SIOR
Concrete Construction	ICC Reinforced Concrete Special Inspector Certificate with one year of related experience, or ACI Concrete Construction Special Inspector, or NICET Concrete Technician Level III Certificate in Construction Materials Testing, or, Registered Professional Engineer with related experience	ACI Concrete Construction Special Inspector in Training, or Engineer-In-Training with one year of related experience	
Prestressed Concrete Construction	ICC Pre-stressed Special Inspector Certificate with one year of related experience, or PCI Quality Control Technician/ Inspector Level II Certificate with one year of related experience, or Registered Professional Engineer with related experience.	PCI Quality Control Technician/ Inspector Level I Certificate with one year of related experience, or Engineer-In-Training with one year of related experience	
Post-Tensioned Concrete Construction	PTI Level 2 Unbonded PT Inspector Certificate, or Registered Professional Engineer with related experience	PTI Level 1 Unbonded PT Inspector Certificate with one year of related experience, or Engineer-In-Training with one year of related experience	
Masonry Construction	ICC Structural Masonry Special Inspector Certificate with one year of related experience, or Registered Professional Engineer with related experience	Engineer-In-Training with one year of related experience	

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Area	Special Inspector	Associated Special Inspector	SIOR
Wood	ICC Commercial Building Inspector Certificate with one year of related experience, or ICC Residential Building Inspector with one year of experience, or Registered Professional Engineer with related experience	Engineer-In-Training with one year of related experience	
Verification of Site Soil Condition, Fill Placement, and Load-Bearing Requirements	ICC Soils Special Inspector Certificate with one year of related experience, or NICET Soils Technician Level II Certificate in Construction Material Testing, or NICET Geotechnical Engineering Technician Level II Construction or Generalist Certificate, or Geologist-In-Training with one year of related experience, or Registered Professional Engineer with related experience	NICET Soils Technician Level I Certificate in Construction Material Testing with one year of related experience, or NICET Geotechnical Engineering Technician Level I Construction, or Generalist Certificate with one year of related experience, or Engineer-In-Training with one year of related experience	

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Area	Special Inspector	Associated Special Inspector	SIOR
Deep Foundations	NICET Soils Technician Level II Certificate in Construction Material Testing, or NICET Geotechnical Engineering Technician Level II Construction or Generalist Certificate, or Geologist-In-Training with one year of related experience, or Registered Professional Engineer with related experience	NICET Soils Technician Level I Certificate in Construction Material Testing with one year of related experience, or NICET Geotechnical Engineering Technician Level I Construction or Generalist Certificate with one year of related experience, or Engineer-In-Training with one year of related experience	
Sprayed Fire Resistant Manual	ICC Spray-applied Fireproofing Special Inspector Certificate, or ICC Fire Inspector I Certificate with one year of related experience, or Registered Professional Engineer with related experience	Engineer-In-Training with one year of related experience	
Mastic and Intumescent Fire Resistant Coatings	ICC Spray-applied Fireproofing Special Inspector Certificate, or ICC Fire Inspector I Certificate with one year of related experience, or Registered Professional Engineer with related experience	Engineer-In-Training with one year of related experience.	

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Area	Special Inspector	Associated Special Inspector	SIOR
Exterior Insulation and Finish Systems (EIFS)	AWCI EIFS Inspector Certificate, or Exterior Design Institute Certificate, or Registered Professional Engineer with related experience	Engineer-In-Training with one year of related experience	
Fire-Resistant Penetrations and Joints	Passed the UL Firestop Exam with one year of related experience, or Passed the FM Firestop Exam with one year of related experience, or Registered Professional Engineer with related experience	Engineer-In-Training with one year of related experience.	
Smoke Control	AABC Technician Certification with one year of related experience, or Registered Professional Engineer with related experience	Engineer-In-Training with one year of related experience.	
SIOR			Registered Professional Engineer

PART 2 - PRODUCTS

2.1 FABRICATORS SPECIAL INSPECTION

- A. Special Inspections of fabricator's work performed in the fabricator's shop is required to be inspected in accordance with the Statement of Special Inspections and the Schedule of Special Inspections unless the fabricator is certified by the approved agency to perform such work without Special Inspections. Submit the applicable certification(s) from the following list to the Contracting Officer for information to

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allow work performed in the fabricator's shop to not be subjected to Special Inspections.

- B. The following certifications meet the requirements for fabricator approval in accordance with paragraph 1704.2.5.2 of IBC:
1. American Institute of Steel Construction (AISC) Certified Fabrication Plant, Category STD.
 2. Truss Plate Institute (TPI) steel truss plate quality assurance program certification.
 3. Truss Plate Institute (TPI) wood truss plate quality assurance program certification.
 4. International Accreditation Service, AC472 Accreditation Steel Joist Institute Membership
 5. Precast Concrete Institute (PCI) Certified Plant, Group C
- C. At the completion of fabrication, submit a certificate of compliance, to be included with the comprehensive final report of Special **Inspections, stating that the materials supplied and work performed by** the fabricator are in accordance the construction documents.

PART 3 - EXECUTION

3.1 RESPONSIBILITIES MATRIX

Inspector	Responsibility	Condition
SIOR	a. Supervise all Special Inspectors required by the contract documents and the IBC. b. Submit a SIOR Letter of Acceptance to the Contracting Officer attesting to acceptance of the duties of SIOR, signed and sealed by the SIOR. c. Verify the qualifications of all of the Special Inspectors. d. Verify the qualifications of fabricators.	Applicable when SIOR is required
	e. Submit Special Inspections agency's written practices for the monitoring and control of the agency's operations to include the following: 1. The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualifications and certification of inspection personnel.	Applicable when SIOR is required and when the structural design is required to follow AISC341 for seismic

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Inspector	Responsibility	Condition
	2. The agency's inspection procedures, including general inspection, material controls, and visual welding inspection. f. Submit qualification records for nondestructive testing (NDT) technicians designated for the project. Submit NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.	design of steel structures
	g. Prepare a Special Inspections Project Manual, which will cover the following: 1. Roles and responsibilities of the following individuals during Special Inspections: SIOR, SI, General Contractor, Subcontractors, QC Manager, and DOR. 2. Organizational chart and/or communication plan, indicating lines of communication 3. Contractor's internal plan for scheduling inspections. Address items such as timeliness of inspection requests, who to contact for inspection requests, and availability of alternate inspectors. Contractor's internal plan for scheduling inspections. Address items such as timeliness of inspection requests, who to contact for inspection requests, and availability of alternate inspectors. 4. Indicate the government reporting procedures. 5. Propose forms or templates to be used by SI and SIOR to document inspections. 6. Indicate procedures for tracking nonconforming work and verification that corrective work is complete. 7. Indicate how the SIOR and/or SI will participate in weekly QC meetings. 8. Indicate how Special Inspections of shop fabricated items will be handled when the fabricator's shop is not certified per paragraph FABRICATOR SPECIAL INSPECTIONS. 9. Include a section in the manual that covers each specific item requiring Special Inspections that is indicated on the Schedule of Special Inspections. Provide names and qualifications of each special inspector who will be performing the Special Inspections for each specific item. Provide detail on how the Special Inspections are to be carried out for each item so that the expectations are	Applicable when SIOR is required

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Inspector	Responsibility	Condition
	clear for the General Contractor and the Subcontractor performing the work. Make a copy of the Special Inspections Project Manual available on the job site during construction. Submit a copy of the Special Inspections Project Manual for approval. h. Attend coordination and mutual understanding meeting where the information in the Special Inspections Project Manual will be reviewed to verify that all parties have a clear understanding of the Special Inspections provisions and the individual duties and responsibilities of each party. i. Maintain a 3- ring binder for the Special Inspector's daily and biweekly reports and the Special Inspections Project Manual. This file must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the DOR. j. Submit a copy of the Special Inspector's daily reports to the QC Manager. k. Discrepancies that are observed during Special Inspections must be reported to the QC Manager for correction. If discrepancies are not corrected before the special inspector leaves the site the observed discrepancies must be documented in the daily report. l. Submit a biweekly Special Inspections report until all work requiring Special Inspections is complete. A report is required for each biweekly period in which Special Inspections activity occurs, and must include the following: 1. A brief summary of the work performed during the reporting time frame. 2. Changes and/or discrepancies with the drawings, specifications, and mechanical or electrical component certification if they require seismic systems, that were observed during the reporting period. 3. Discrepancies which were resolved or corrected. 4. A list of nonconforming items requiring resolution. 5. All applicable test results including nondestructive testing reports. //m. For large, complex projects, at the completion of each Definable Feature of Work (DFOW) requiring Special Inspections, submit an	

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Inspector	Responsibility	Condition
	interim final report of Special Inspections that documents the Special Inspections completed for that DFOW and corrections of all discrepancies noted in the daily reports. The interim final report of Special Inspections must be signed, dated and bear the seal of the SIOR.]. //	
QC Manager	a. If there is no SIOR, QC Manager must Supervise all Special Inspectors required by the contract documents and the IBC; Verify the qualifications of all of the Special Inspectors; Verify the qualifications of fabricators; Maintain a 3-ring binder for the Special Inspector's daily and biweekly reports. This file must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the DOR.	Applicable when SIOR is not required
	b. Maintain a rework items list that includes discrepancies noted on the Special Inspectors daily report.	n/a
Special Inspectors	a. Inspect all elements of the project for which the special inspector is qualified to inspect and are identified in the Schedule of Special Inspections. b. Attend preparatory phase meetings related to the Definable Feature of Work (DFOW) for which the special inspector is qualified to inspect.	
	c. Submit Special Inspections agency's written practices for the monitoring and control of the agency's operations to include the following: 1. The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualifications and certification of inspection personnel. 2. The agency's inspection procedures, including general inspection, material controls, and visual welding inspection. d. Submit qualification records for nondestructive testing (NDT) technicians designated for the project. e. Submit NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.]	Applicable when SIOR is NOT required and when the structural design is required to follow AISC 341 for seismic design of steel structures

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Inspector	Responsibility	Condition
	f. Submit a copy of the daily reports to the QC Manager. g. Discrepancies that are observed during Special Inspections must be reported to the QC Manager for correction. If discrepancies are not corrected before the special inspector leaves the site the observed discrepancies must be documented in the daily report. h. Submit a biweekly Special Inspection Report until all inspections are complete. A report is required for each biweekly period in which Special Inspections activity occurs, and must include the following: 1. A brief summary of the work performed during the reporting time frame 2. Changes and/or discrepancies with the drawings, specifications, and mechanical or electrical component certification if they require seismic systems that were observed during the reporting period. 3. Discrepancies which were resolved or corrected. 4. A list of nonconforming items requiring resolution. 5. All applicable test result including nondestructive testing reports. //i. For large, complex projects, at the completion of each Definable Feature of Work (DFOW) requiring Special Inspections, submit an interim final report of Special Inspections that documents the Special Inspections completed for that DFOW and corrections of all discrepancies noted in the daily reports. The interim final report of Special Inspections must be signed, dated and bear the seal of the SIOR.]. // j. At the completion of the project submit a comprehensive final report of Special Inspections that documents the Special Inspections completed for the project and corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed, dated and indicate the certification of the special inspector qualifying them to conduct the inspection.	Applicable when SIOR is not required

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Inspector	Responsibility	Condition
	k. Submit daily reports to the SIOR	Applicable when SIOR is required

3.2 DEFECTIVE WORK

Check work as it progresses, but failure to detect any defective work or materials must in no way prevent later rejection if defective work or materials are discovered, nor obligate the Government to accept such work.

-- End of Section -

SECTION 01 57 19
TEMPORARY ENVIRONMENTAL CONTROLS

~~SPEC WRITER NOTE:~~

- ~~1. Refer to and edit this Section per the environmental protection actions required and identified in the specific project mitigation memorandum on file with the Project Director. Note that certain states like FL and NC have stringent requirements that the facility may recognize and incorporate. Delete or add information between // and any other items applicable to project. Renumber the paragraphs as applicable.~~
- ~~2. "COR" is the term typically utilized in most paragraphs. "COR" is utilized on major and CFM-managed projects.~~

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This section specifies the control of environmental pollution and damage that the Contractor must consider for air, water, and land resources. It includes management of visual aesthetics, noise, solid waste, radiant energy, and radioactive materials, as well as other pollutants and resources encountered or generated by the Contractor. The Contractor is obligated to consider specified control measures with the costs included within the various contract items of work.
- B. Environmental pollution and damage is defined as the presence of chemical, physical, or biological elements or agents which:
 1. Adversely effect human health or welfare,
 2. Unfavorably alter ecological balances of importance to human life,
 3. Effect other species of importance to humankind, or;
 4. Degrade the utility of the environment for aesthetic, cultural, and historical purposes.
- C. Definitions of Pollutants:
 1. Chemical Waste: Petroleum products, bituminous materials, salts, acids, alkalis, herbicides, pesticides, organic chemicals, and inorganic wastes.
 2. Debris: Combustible and noncombustible wastes, such as leaves, tree trimmings, ashes, and waste materials resulting from construction or maintenance and repair work.

3. Sediment: Soil and other debris that has been eroded and transported by runoff water.
4. Solid Waste: Rubbish, debris, garbage, and other discarded solid materials resulting from industrial, commercial, and agricultural operations and from community activities.
5. Surface Discharge: The term "Surface Discharge" implies that the water is discharged with possible sheeting action and subsequent soil erosion may occur. Waters that are surface discharged may terminate in drainage ditches, storm sewers, creeks, and/or "water of the United States" and would require a permit to discharge water from the governing agency.
6. Rubbish: Combustible and noncombustible wastes such as paper, boxes, glass and crockery, metal and lumber scrap, tin cans, and bones.
7. Sanitary Wastes:
 - a. Sewage: Domestic sanitary sewage and human and animal waste.
 - b. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

1.2 QUALITY CONTROL

- A. Establish and maintain quality control for the environmental protection of all items set forth herein.
- B. Record on daily reports any problems in complying with laws, regulations, and ordinances. Note any corrective action taken.

1.3 REFERENCES

- A. The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.
- B. U.S. National Archives and Records Administration (NARA):
33 CFR 328.....Definitions

1.4 SUBMITTALS

- A. In accordance with Section, 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES, furnish the following:
 1. Environmental Protection Plan: After the contract is awarded and prior to the commencement of the work, the Contractor shall meet with the Contracting Officer's Representative (COR) to discuss the proposed Environmental Protection Plan and to develop mutual understanding relative to details of environmental protection. Not more than 20 days after the meeting, the Contractor shall prepare

and submit to the ~~Contracting Officer~~ //COR for approval, a written and/or graphic Environmental Protection Plan including, but not limited to, the following:

- a. Name(s) of person(s) within the Contractor's organization who is (are) responsible for ensuring adherence to the Environmental Protection Plan.
- b. Name(s) and qualifications of person(s) responsible for manifesting hazardous waste to be removed from the site.
- c. Name(s) and qualifications of person(s) responsible for training the Contractor's environmental protection personnel.
- d. Description of the Contractor's environmental protection personnel training program.
- e. A list of Federal, State, and local laws, regulations, and permits concerning environmental protection, pollution control, noise control and abatement that are applicable to the Contractor's proposed operations and the requirements imposed by those laws, regulations, and permits.
- f. Methods for protection of features to be preserved within authorized work areas including trees, shrubs, vines, grasses, ground cover, landscape features, air and water quality, fish and wildlife, soil, historical, and archeological and cultural resources.
- g. Procedures to provide the environmental protection that comply with the applicable laws and regulations. Describe the procedures to correct pollution of the environment due to accident, natural causes, or failure to follow the procedures as described in the Environmental Protection Plan.
- h. Permits, licenses, and the location of the solid waste disposal area.
- i. Drawings showing locations of any ~~proposed~~ //temporary excavations or embankments for haul roads, // stream crossings, // material storage areas, structures, sanitary facilities, // and stockpiles of excess or spoil materials. Include as part of an Erosion Control Plan approved by the District Office of the U.S. Soil Conservation Service and the Department of Veterans Affairs.
- j. Environmental Monitoring Plans for the job site including land, water, air, and noise.

k. Work Area Plan showing the proposed activity in each portion of the area and identifying the areas of limited use or nonuse. Plan should include measures for marking the limits of use areas. This plan may be incorporated within the Erosion Control Plan.

l. Inclusion of "best management practices" and methodologies.

B. Approval of the Contractor's Environmental Protection Plan will not relieve the Contractor of responsibility for adequate and continued control of pollutants and other environmental protection measures.

1.5 PROTECTION OF ENVIRONMENTAL RESOURCES

A. Protect environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire period of this contract. Confine activities to areas defined by the specifications and drawings.

B. Protection of Land Resources: Prior to construction, identify all land resources to be preserved within the work area. Do not remove, cut, deface, injure, or destroy land resources including trees, shrubs, vines, grasses, topsoil, and landforms without permission from the COR. Do not fasten or attach ropes, cables, or guys to trees for anchorage unless specifically authorized, or where special emergency use is permitted. Provide erosion control plans, in phases where required.

1. Work Area Limits: Prior to any construction, mark the areas that require work to be performed under this contract. Mark or fence isolated areas within the general work area that are to be saved and protected. Protect monuments, works of art, and markers before construction operations begin. Convey to all personnel the purpose of marking and protecting all necessary objects.

2. Protection of Landscape: Protect trees, shrubs, vines, grasses, land forms, and other landscape features shown on the drawings to be preserved by marking, fencing, or using any other approved techniques.

a. Box and protect from damage existing trees and shrubs to remain on the construction site.

b. Immediately repair all damage to existing trees and shrubs by trimming, cleaning, and painting with antiseptic tree paint.

c. Do not store building materials or perform construction activities closer to existing trees or shrubs than the farthest extension of their limbs.

3. Reduction of Exposure of Unprotected Erodible Soils: Plan and conduct earthwork to minimize the duration of exposure of unprotected soils. Clear areas in reasonably sized increments only as needed to use. Form earthwork to final grade as shown. Immediately protect side slopes and back slopes upon completion of rough grading.
4. Temporary Protection of Disturbed Areas: Construct diversion ditches, benches, and berms to retard and divert runoff from the construction site to protected drainage areas approved under paragraph 208 of the Clean Water Act.

~~SPEC WRITER NOTE: The design year storm is determined by the downstream environment to be protected. Implement appropriate protection based on the estimate of damage to the downstream environment versus the design year storm that will cause damage. If permanent sediment basins are necessary for the particular project, include these permanent facilities in the project design and the contract documents. If permanent basins are not required, delete reference thereto.~~

- a. Sediment Basins: Trap sediment from construction areas in temporary or permanent sediment basins that accommodate the runoff of a local ~~//_____// (design year)~~ storm. After each storm, pump the basins dry and remove the accumulated sediment. Control overflow/drainage with paved weirs or by vertical overflow pipes, draining from the surface.
 - b. Reuse or conserve the collected topsoil sediment as directed by the COR. ~~Topsoil use and requirements are specified in Section 31 20 00, EARTH MOVING.~~
 - c. Institute effluent quality monitoring programs as required by Federal, State, and local environmental agencies.
5. Erosion and Sedimentation Control Devices: The erosion and sediment controls selected and maintained by the Contractor shall be such that water quality standards are not violated as a result of the Contractor's activities. Construct or install all temporary and permanent erosion and sedimentation control features ~~//shown.//~~ on the Environmental Protection Plan. ~~//Maintain temporary erosion and~~

sediment control measures such as berms, dikes, drains, sedimentation basins, grassing, and mulching, until permanent drainage and erosion control facilities are completed and operative.

6. Manage borrow areas on ~~//~~ and off ~~//~~ Government property to minimize erosion and to prevent sediment from entering nearby water courses or lakes.
 7. Manage and control spoil areas on ~~//~~ and off ~~//~~ Government property to limit spoil to areas ~~//~~ shown ~~//~~ on the Environmental Protection Plan ~~//~~ and prevent erosion of soil or sediment from entering nearby water courses or lakes.
 8. Protect adjacent areas from despoilment by temporary excavations and embankments.
 9. Handle and dispose of solid wastes in such a manner that will prevent contamination of the environment. Place solid wastes (excluding clearing debris) in containers that are emptied on a regular schedule. Transport all solid waste off Government property and dispose of waste in compliance with Federal, State, and local requirements.
 10. Store chemical waste away from the work areas in corrosion resistant containers and dispose of waste in accordance with Federal, State, and local regulations.
 11. Handle discarded materials other than those included in the solid waste category as directed by the COR.
- C. Protection of Water Resources: Keep construction activities under surveillance, management, and control to avoid pollution of surface and ground waters and sewer systems. Implement management techniques to control water pollution by the listed construction activities that are included in this contract.
1. Washing and Curing Water: Do not allow wastewater directly derived from construction activities to enter water areas. Collect and place wastewater in retention ponds allowing the suspended material to settle, the pollutants to separate, or the water to evaporate.
 2. Control movement of materials and equipment at stream crossings during construction to prevent violation of water pollution control standards of the Federal, State, or local government.

~~SPEC WRITER NOTE: Specify additional operations unique to this contract.~~

3. Monitor water areas affected by construction.
- D. Protection of Fish and Wildlife Resources: Keep construction activities under surveillance, management, and control to minimize interference with, disturbance of, or damage to fish and wildlife. Prior to beginning construction operations, list species that require specific attention along with measures for their protection.
- E. Protection of Air Resources: Keep construction activities under surveillance, management, and control to minimize pollution of air resources. Burning is not permitted on the job site. Keep activities, equipment, processes, and work operated or performed, in strict accordance with the State ~~of // insert Name of State and title of State Air Pollution Statute, Rule, or Regulation //~~ and Federal emission and performance laws and standards. Maintain ambient air quality standards set by the Environmental Protection Agency, for those construction operations and activities specified.
1. Particulates: Control dust particles, aerosols, and gaseous by-products from all construction activities, processing, and preparation of materials (such as from asphaltic batch plants) at all times, including weekends, holidays, and hours when work is not in progress.
 2. Particulates Control: Maintain all excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and all other work areas within or outside the project boundaries free from particulates which would cause a hazard or a nuisance. Sprinklering, chemical treatment of an approved type, light bituminous treatment, baghouse, scrubbers, electrostatic precipitators, or other methods are permitted to control particulates in the work area.
 3. Hydrocarbons and Carbon Monoxide: Control monoxide emissions from equipment to Federal and State allowable limits.
 4. Odors: Control odors of construction activities and prevent obnoxious odors from occurring.
- F. Reduction of Noise: Minimize noise using every action possible. Perform noise-producing work in less sensitive hours of the day or week as directed by the COR. Maintain noise-produced work at or below the decibel levels and within the time periods specified.

1. Perform construction activities involving repetitive, high-level impact noise only between 8:00 ~~//==//~~a.m. and 6:00 ~~//==//~~p.m unless otherwise permitted by local ordinance or the COR. Repetitive impact noise on the property shall not exceed the following dB limitations:

Time Duration of Impact Noise	Sound Level in dB
More than 12 minutes in any hour	70
Less than 30 seconds of any hour	85
Less than three minutes of any hour	80
Less than 12 minutes of any hour	75

~~SPEC WRITER NOTE: Insert additional information as needed when unique to a particular VA Medical Center site.~~

2. Provide sound-deadening devices on equipment and take noise abatement measures that are necessary to comply with the requirements of this contract, consisting of, but not limited to, the following:
- a. Maintain maximum permissible construction equipment noise levels at 15 meter (50 feet) (dBA):

EARTHMOVING		MATERIALS HANDLING	
FRONT LOADERS	75	CONCRETE MIXERS	75
BACKHOES	75	CONCRETE PUMPS	75
DOZERS	75	CRANES	75
TRACTORS	75	DERRICKS IMPACT	75
SCAPERS	80	PILE DRIVERS	95
GRADERS	75	JACK HAMMERS	75
TRUCKS	75	ROCK DRILLS	80
PAVERS, STATIONARY	80	PNEUMATIC TOOLS	80
PUMPS	75	BLASTING	//==//
GENERATORS	75	SAWS	75
COMPRESSORS	75	VIBRATORS	75

- b. Use shields or other physical barriers to restrict noise transmission.

- c. Provide soundproof housings or enclosures for noise-producing machinery.
 - d. Use efficient silencers on equipment air intakes.
 - e. Use efficient intake and exhaust mufflers on internal combustion engines that are maintained so equipment performs below noise levels specified.
 - f. Line hoppers and storage bins with sound deadening material.
 - g. Conduct truck loading, unloading, and hauling operations so that noise is kept to a minimum.
3. Measure sound level for noise exposure due to the construction at least once every five successive working days while work is being performed above 55 ~~dB~~ dB(A) noise level. Measure noise exposure at the property line or 15 m (50 feet) from the noise source, whichever is greater. Measure the sound levels on the A weighing network of a General Purpose sound level meter at slow response. To minimize the effect of reflective sound waves at buildings, take measurements at 900 to 1800 mm (three to six feet) in front of any building face. Submit the recorded information to the COR noting any problems and the alternatives for mitigating actions.
- G. Restoration of Damaged Property: If any direct or indirect damage is done to public or private property resulting from any act, omission, neglect, or misconduct, the Contractor shall restore the damaged property to a condition equal to that existing before the damage at no additional cost to the Government. Repair, rebuild, or restore property as directed or make good such damage in an acceptable manner.
- H. Final Clean-up: On completion of project and after removal of all debris, rubbish, and temporary construction, Contractor shall leave the construction area in a clean condition satisfactory to the COR. Cleaning shall include off the station disposal of all items and materials not required to be salvaged, as well as all debris and rubbish resulting from demolition and new work operations.

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SECTION 01 74 19
CONSTRUCTION WASTE MANAGEMENT

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This section specifies the requirements for the management of non-hazardous building construction and demolition waste.
- B. Waste disposal in landfills shall be minimized to the greatest extent possible. Of the inevitable waste that is generated, as much of the waste material as economically feasible shall be salvaged, recycled or reused.
- C. Contractor shall use all reasonable means to divert construction and demolition waste from landfills and incinerators, and facilitate their salvage and recycle not limited to the following:
- D. Waste Management Plan development and implementation.
- E. Techniques to minimize waste generation.
- F. Sorting and separating of waste materials.
- G. Salvage of existing materials and items for reuse or resale.
- H. Recycling of materials that cannot be reused or sold.
- I. At a minimum the following waste categories shall be diverted from landfills:
 - 1. Soil.
 - 2. Inerts (eg, concrete, masonry and asphalt).
 - 3. Clean dimensional wood and palette wood.
 - 4. Green waste (biodegradable landscaping materials).
 - 5. Engineered wood products (plywood, particle board and I-joists, etc).
 - 6. Metal products (eg, steel, wire, beverage containers, copper, etc).
 - 7. Sheathings.
 - 8. Cardboard, paper and packaging.
 - 9. Bitumen roofing materials.
 - 10. Plastics (eg. ABS, PVC).
 - 11. Carpet and/or pad.
 - 12. Gypsum board.
 - 13. Insulation.
 - 14. Paint.
 - 15. Flourescent lamps.

1.2 RELATED WORK

- A. Section 02 41 00, DEMOLITION.
- B. Section 01 00 00, GENERAL REQUIREMENTS.
 - ~~1. //C. Lead Paint: Section 02 83 33.13, LEAD BASED PAINT REMOVAL AND DISPOSAL.//~~
 - ~~2. //D. Division 1 Sustainability specifications//~~

1.3 QUALITY ASSURANCE

- A. Contractor shall practice efficient waste management when sizing, cutting and installing building products. Processes shall be employed to ensure the generation of as little waste as possible. Construction /Demolition waste includes products of the following:
 - 1. Excess or unusable construction materials.
 - 2. Packaging used for construction products.
 - 3. Poor planning and/or layout.
 - 4. Construction error.
 - 5. Over ordering.
 - 6. Weather damage.
 - 7. Contamination.
 - 8. Mishandling.
 - 9. Breakage.
- B. Establish and maintain the management of non-hazardous building construction and demolition waste set forth herein. Conduct a site assessment to estimate the types of materials that will be generated by demolition and construction.
- C. Contractor shall develop and implement procedures to recycle construction and demolition waste to a minimum of ~~++50++//++//~~ percent.
- D. Contractor shall be responsible for implementation of any special programs involving rebates or similar incentives related to recycling. Any revenues or savings obtained from salvage or recycling shall accrue to the contractor.
- E. Contractor shall provide all demolition, removal and legal disposal of materials. Contractor shall ensure that facilities used for recycling, reuse and disposal shall be permitted for the intended use to the extent required by local, state, federal regulations.
- F. Contractor shall assign a specific area to facilitate separation of materials for reuse, salvage, recycling, and return. Such areas are to

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be kept neat and clean and clearly marked in order to avoid contamination or mixing of materials.

- G. Contractor shall provide on-site instructions and supervision of separation, handling, salvaging, recycling, reuse and return methods to be used by all parties during waste generating stages.
- H. Record on daily reports any problems in complying with laws, regulations and ordinances with corrective action taken.

1.4 TERMINOLOGY

- A. Class III Landfill: A landfill that accepts non-hazardous resources such as household, commercial and industrial waste resulting from construction, remodeling, repair and demolition operations.
- B. Clean: Untreated and unpainted; uncontaminated with adhesives, oils, solvents, mastics and like products.
- C. Construction and Demolition Waste: Includes all non-hazardous resources resulting from construction, remodeling, alterations, repair and demolition operations.
- D. Dismantle: The process of parting out a building in such a way as to preserve the usefulness of its materials and components.
- E. Disposal: Acceptance of solid wastes at a legally operating facility for the purpose of land filling (includes Class III landfills and inert fills).
- F. Inert Backfill Site: A location, other than inert fill or other disposal facility, to which inert materials are taken for the purpose of filling an excavation, shoring or other soil engineering operation.
- G. Inert Fill: A facility that can legally accept inert waste, such as asphalt and concrete exclusively for the purpose of disposal.
- H. Inert Solids/Inert Waste: Non-liquid solid resources including, but not limited to, soil and concrete that does not contain hazardous waste or soluble pollutants at concentrations in excess of water-quality objectives established by a regional water board, and does not contain significant quantities of decomposable solid resources.
- I. Mixed Debris: Loads that include commingled recyclable and non-recyclable materials generated at the construction site.
- J. Mixed Debris Recycling Facility: A solid resource processing facility that accepts loads of mixed construction and demolition debris for the purpose of recovering re-usable and recyclable materials and disposing non-recyclable materials.

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- K. Permitted Waste Hauler: A company that holds a valid permit to collect and transport solid wastes from individuals or businesses for the purpose of recycling or disposal.
- L. Recycling: The process of sorting, cleansing, treating, and reconstituting materials for the purpose of using the altered form in the manufacture of a new product. Recycling does not include burning, incinerating or thermally destroying solid waste.
1. On-site Recycling - Materials that are sorted and processed on site for use in an altered state in the work, i.e. concrete crushed for use as a sub-base in paving.
 2. Off-site Recycling - Materials hauled to a location and used in an altered form in the manufacture of new products.
- M. Recycling Facility: An operation that can legally accept materials for the purpose of processing the materials into an altered form for the manufacture of new products. Depending on the types of materials accepted and operating procedures, a recycling facility may or may not be required to have a solid waste facilities permit or be regulated by the local enforcement agency.
- N. Reuse: Materials that are recovered for use in the same form, on-site or off-site.
- O. Return: To give back reusable items or unused products to vendors for credit.
- P. Salvage: To remove waste materials from the site for resale or re-use by a third party.
- Q. Source-Separated Materials: Materials that are sorted by type at the site for the purpose of reuse and recycling.
- R. Solid Waste: Materials that have been designated as non-recyclable and are discarded for the purposes of disposal.
- S. Transfer Station: A facility that can legally accept solid waste for the purpose of temporarily storing the materials for re-loading onto other trucks and transporting them to a landfill for disposal, or recovering some materials for re-use or recycling.

1.5 SUBMITTALS

- A. In accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, and SAMPLES, furnish the following:
- B. Prepare and submit to the //Contracting Officer's Representative (COR) ~~//Resident Engineer//~~ a written demolition debris management

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plan. The plan shall include, but not be limited to, the following information:

1. Procedures to be used for debris management.
2. Techniques to be used to minimize waste generation.
3. Analysis of the estimated job site waste to be generated:
 - a. List of each material and quantity to be salvaged, reused, recycled.
 - b. List of each material and quantity proposed to be taken to a landfill.
4. Detailed description of the Means/Methods to be used for material handling.
 - a. On site: Material separation, storage, protection where applicable.
 - b. Off site: Transportation means and destination. Include list of materials.
 - 1) Description of materials to be site-separated and self-hauled to designated facilities.
 - 2) Description of mixed materials to be collected by designated waste haulers and removed from the site.
 - a) The names and locations of mixed debris reuse and recycling facilities or sites.
 - b) The names and locations of trash disposal landfill facilities or sites.
 - c) Documentation that the facilities or sites are approved to receive the materials.
- C. Designated Manager responsible for instructing personnel, supervising, documenting and administer over meetings relevant to the Waste Management Plan.
- D. Monthly summary of construction and demolition debris diversion and disposal, quantifying all materials generated at the work site and disposed of or diverted from disposal through recycling.
- E. Target waste diversion rate by material and an overall diversion rate.
- F. Final report documenting the results of implementation of the preconstruction waste management plan.

1.6 APPLICABLE PUBLICATIONS

- A. Publications listed below form a part of this specification to the extent referenced. Publications are referenced by the basic designation

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only. In the event that criteria requirements conflict, the most stringent requirements shall be met.

B. U.S. Green Building Council (USGBC): LEED Green Building Rating System for New Construction

1. Green Building Initiative (GBI): Green Globes for New Construction 2019

1.7 RECORDS

A. Maintain records to document the quantity of waste generated; the quantity of waste diverted through sale, reuse, or recycling; and the quantity of waste disposed by landfill or incineration. Records shall be kept in accordance with the ~~//LEED Reference Guide and LEED Template//~~ ~~// Green Globes for New Construction 2019 Technical Reference Manual.//~~ Contractor to maintain records of waste removal throughout the duration of the project.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. List of each material and quantity to be salvaged, recycled, reused.
- B. List of each material and quantity proposed to be taken to a landfill.
- C. Material tracking data: Receiving parties, dates removed, transportation costs, weight tickets, tipping fees, manifests, invoices, net total costs or savings.

PART 3 - EXECUTION

3.1 COLLECTION

- A. Provide all necessary containers, bins and storage areas to facilitate effective waste management.
- B. Clearly identify containers, bins and storage areas so that recyclable materials are separated from trash and can be transported to respective recycling facility for processing.
- C. Hazardous wastes shall be separated, stored, disposed of according to local, state, federal regulations.

3.2 DISPOSAL

- A. Contractor shall be responsible for transporting and disposing of materials that cannot be delivered to a source-separated or mixed materials recycling facility to a transfer station or disposal facility

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that can accept the materials in accordance with state and federal regulations.

- B. Construction or demolition materials with no practical reuse or that cannot be salvaged or recycled shall be disposed of at a landfill or incinerator.

3.3 REPORT

- A. With each application for progress payment, submit a summary of construction and demolition debris diversion and disposal including beginning and ending dates of period covered.
- B. Quantify all materials diverted from landfill disposal through salvage or recycling during the period with the receiving parties, dates removed, transportation costs, weight tickets, manifests, invoices. Include the net total costs or savings for each salvaged or recycled material.
- C. Quantify all materials disposed of during the period with the receiving parties, dates removed, transportation costs, weight tickets, tipping fees, manifests, invoices. Include the net total costs for each disposal.

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SECTION 01 81 13
SUSTAINABLE CONSTRUCTION REQUIREMENTS

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This Section describes general requirements and procedures to comply with federal mandates and U.S. Department of Veterans Affairs (VA) policies for sustainable construction.
- B. The Design Professional has selected materials and utilized integrated design processes that achieve the Government's objectives. Contractor is responsible to maintain and support these objectives in developing means and methods for performing work and in proposing product substitutions or changes to specified processes. Obtain approval from Contracting Officer's Representative (COR) for all changes and substitutions to materials or processes. Proposed changes must meet, or exceed, materials or processes specified.

1.2 RELATED WORK

- A. Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.
- B. Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT.

~~SPEC WRITER NOTE:~~

- ~~1. Retain one of the following to reflect the appropriate rating system being used for certification or verification.~~
- ~~2. The following sustainability certification requirements apply only to projects for which design began prior to August 2nd, 2017.~~

~~C. // Section 01 81 13.01 SUSTAINABILITY CERTIFICATION REQUIREMENTS - LEED NC v3. //~~

~~D. // Section 01 81 13.02 SUSTAINABILITY CERTIFICATION REQUIREMENTS - LEED NC v4. //~~

~~E. // Section 01 81 13.11 SUSTAINABILITY CERTIFICATION REQUIREMENTS - LEED FOR HEALTHCARE v3. //~~

~~F. // Section 01 81 13.21 SUSTAINABILITY CERTIFICATION REQUIREMENTS - GREEN GLOBES. //~~

~~G.C.~~ Section 01 91 00 GENERAL COMMISSIONING REQUIREMENTS.

1.3 DEFINITIONS

- A. Recycled Content: Recycled content of materials is defined according to Federal Trade Commission Guides for the Use of Environmental Marketing

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Claims (16 CFR Part 260). Recycled content value of a material assembly is determined by weight. Recycled fraction of assembly is multiplied by cost of assembly to determine recycled content value.

1. "Post-Consumer" material is defined as waste material generated by households or by commercial, industrial, and institutional facilities in their role as end users of the product, which can no longer be used for its intended purpose.
 2. "Pre-Consumer" material is defined as material diverted from waste stream during the manufacturing process. Excluded is reutilization of materials such as rework, regrind, or scrap generated in a process and capable of being reclaimed within the same process that generated it.
- B. Biobased Products: Biobased products are derived from plants and other renewable agricultural, marine, and forestry materials and provide an alternative to conventional petroleum derived products. Biobased products include diverse categories such as lubricants, cleaning products, inks, fertilizers, and bioplastics.
- C. Low Pollutant-Emitting Materials: Materials and products which are minimally odorous, irritating, or harmful to comfort and well-being of installers and occupants.
- D. Volatile Organic Compounds (VOC): Chemicals that are emitted as gases from certain solids or liquids. VOCs include a variety of chemicals, some of which may have short- and long-term adverse health effects.

1.4 REFERENCE STANDARDS

- A. Carpet and Rug Institute Green Label Plus program.
- B. U.S. Department of Agriculture BioPreferred program (USDA BioPreferred).
- C. U.S. Environmental Protection Agency Comprehensive Procurement Guidelines (CPG).
- D. U.S. Environmental Protection Agency WaterSense Program (WaterSense).
- E. U.S. Environmental Protection Agency ENERGY STAR Program (ENERGY STAR).
- F. U. S. Department of Energy Federal Energy Management Program (FEMP).
- G. Green Electronic Council EPEAT Program (EPEAT).

1.5 SUBMITTALS

- A. All submittals to be provided by contractor to COR.
- B. Sustainability Action Plan:

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1. Submit documentation as required by this section; provide additional copies of typical submittals required under technical sections when sustainable construction requires copies of record submittals.
2. Within 30 days after Preconstruction Meeting provide a narrative plan for complying with requirements stipulated within this section.
3. Sustainability Action Plan must:
 - a. Make reference to sustainable construction submittals defined by this section.
 - b. Address all items listed under PERFORMANCE CRITERIA.
 - c. Indicate individual(s) responsible for implementing the plan.
- C. Low Pollutant-Emitting Materials Tracking Spreadsheet: Within 30 days after Preconstruction Meeting provide a preliminary Low Pollutant-Emitting Materials Tracking Spreadsheet. The Low Pollutant-Emitting Materials Tracking Spreadsheet must be an electronic file and include all materials on Project in categories described under Low Pollutant-Emitting Materials in 01 81 13.
- D. Construction Indoor Air Quality (IAQ) Management Plan:
 1. Not more than 30 days after Preconstruction Meeting provide a Construction IAQ Management Plan as an electronic file including descriptions of the following:
 - a. Instruction procedures for meeting or exceeding minimum requirements of ANSI/SMACNA 008-2008, Chapter 3, including procedures for HVAC Protection, Source Control, Pathway Interruption, Housekeeping, and Scheduling.
 - b. Instruction procedures for protecting absorptive materials stored on-site or installed from moisture damage.
 - c. Schedule of submission of photographs of on-site construction IAQ management measures such as protection of ducts and on-site stored oil installed absorptive materials.
 - d. Instruction procedures if air handlers must be used during construction, including a description of filtration media to be used at each return air grille.
 - e. Instruction procedure for replacing all air-filtration media immediately prior to occupancy after completion of construction, including a description of filtration media to be used at each air handling or air supply unit.

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f. Instruction procedures and schedule for implementing building flush-out.

E. Product Submittals:

1. Recycled Content: Submit product data from manufacturer indicating percentages by weight of post-consumer and pre-consumer recycled content for products having recycled content (excluding MEP systems equipment and components).
2. Biobased Content: Submit product data for products to be installed or used which are included in any of the USDA BioPreferred program's product categories. Data to include percentage of biobased content and source of biobased material.
3. Low Pollutant-Emitting Materials: Submit product data confirming compliance with relevant requirements for all materials on Project in categories described under Low Pollutant-Emitting Materials in 01 81 13.
4. For applicable products and equipment, submit product documentation confirming ENERGY STAR label, FEMP certification, WaterSense, and/or EPEAT certification.

F. Sustainable Construction Progress Reports: Concurrent with each Application for Payment, submit a Sustainable Construction Progress Report to confirm adherence with Sustainability Action Plan.

1. Include narratives of revised strategies for bringing work progress into compliance with plan and product submittal data.
2. Include updated and current Low Pollutant-Emitting Materials Tracking Spreadsheet.
3. Include construction waste tracking, in tons or cubic yards, including waste description, whether diverted or landfilled, hauler, and percent diverted for comingled quantities; and excluding land-clearing debris and soil. Provide haul receipts and documentation of diverted percentages for comingled wastes.

G. Closeout Submittals: Within 14 days after Substantial Completion provide the following:

1. Final version of Low Pollutant-Emitting Materials Tracking Spreadsheet.
2. Manufacturer's cut sheets and product data highlighting the Minimum Efficiency Reporting Value (MERV) for filtration media installed at

return air grilles during construction if permanently installed air handling units are used during construction.

3. Manufacturer's cut sheets and product data highlighting the Minimum Efficiency Reporting Value (MERV) for final filtration media in air handling units.
4. Minimum 18 construction photographs including six photographs taken on three different occasions during construction of ANSI/SMACNA 008-2008, Chapter 3 approaches employed, along with a brief description of each approach, documenting implementation of IAQ management measures, such as protection of ducts and on-site stored or installed absorptive materials.
5. Flush-out Documentation:
 - a. Product data for filtration media used during flush-out.
 - b. Product data for filtration media installed immediately prior to occupancy.
 - c. Signed statement describing building air flush-out procedures including dates when flush-out was begun and completed and statement that filtration media was replaced after flush-out.

1.6 QUALITY ASSURANCE

- A. Preconstruction Meeting: After award of Contract and prior to commencement of Work, schedule and conduct meeting with COR/Resident Engineer and Architect to discuss the Project Sustainable Action Plan content as it applies to submittals, project delivery, required Construction Indoor Air Quality (IAQ) Management Plan, and other Sustainable Construction Requirements. The purpose of this meeting is to develop a mutual understanding of the Sustainable Construction Requirements and coordination of contractor's management of these requirements with the COR Contracting Officer and the Construction Quality Manager.
- B. Construction Job Conferences: Status of compliance with Sustainable Construction Requirements of these specifications will be an agenda item at regular job meetings conducted during the course of work at the site.

1.7 APPLICABLE PUBLICATIONS

- A. Publications listed below form a part of this specification to extent referenced. Publications are referenced in text by basic designation

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only. Comply with applicable provisions and recommendations of the following, except as otherwise shown or specified.

~~SPEC WRITER NOTES:~~

- ~~1. Remove reference citations that do not remain in Part 2 or Part 3 of edited specification.~~
- ~~2. Verify and make dates indicated for remaining citations the most current at date of submittal; determine changes from date indicated on the TIL download of the section and modify requirements impacted by the changes.~~

- B. Green Seal Standard GS-11, Paints, 1st Edition, May 20, 1993.
- C. Green Seal Standard GC-03, Anti-Corrosive Paints, 2nd Edition, January 7, 1997.
- D. Green Seal Standard GC-36, Commercial Adhesives, October 19, 2000.
- E. South Coast Air Quality Management District (SCAQMD) Rule 1113, Architectural Coatings, rules in effect on January 1, 2004.
- F. South Coast Air Quality Management District (SCAQMD) Rule 1168, July 1, 2005 and rule amendment date of January 7, 2005.
- G. Sheet Metal and Air Conditioning National Contractors' Association (SMACNA) IAQ Guidelines for Occupied Buildings under Construction, 2nd Edition (ANSI/SMACNA 008-2008), Chapter 3.
- H. California Department of Public Health Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers, Version 1.1, Emission Testing method for California Specification 01350 (CDPH Standard Method V1.1-2010).
- I. Federal Trade Commission Guides for the Use of Environmental Marketing Claims (16 CFR Part 260).
- J. ASHRAE Standard 52.2-2007.

PART 2 - PRODUCTS

2.1 PERFORMANCE CRITERIA

- A. Construction waste diversion from landfill disposal must comprise at least 50 percent of total construction waste, excluding land clearing debris and soil. Alternative daily cover (ADC) does not qualify as material diverted from disposal.
- B. Low Pollutant-Emitting Materials:

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1. Adhesives, sealants and sealant primers applied on site within the weatherproofing membrane must comply with VOC limits of SCAQMD Rule 1168:

a. Flooring Adhesives and Sealants:

- 1) Indoor carpet adhesives: 50 g/L.
- 2) Wood Flooring Adhesive: 100 g/L.
- 3) Rubber Floor Adhesives: 60 g/L.
- 4) Subfloor Adhesives: 50 g/L.
- 5) Ceramic Tile Adhesives and Grout: 65 g/L.
- 6) Cove Base Adhesives: 50 g/L.
- 7) Multipurpose Construction Adhesives: 70 g/L.
- 8) Porous Material (Except Wood) Substrate: 50 g/L.
- 9) Wood Substrate: 30 g/L.
- 10) Architectural Non-Porous Sealant Primer: 250 g/L.
- 11) Architectural Porous Sealant Primer: 775 g/L.
- 12) Other Sealant Primer: 750 g/L.
- 13) Structural Wood Member Adhesive: 140 g/L.
- 14) Sheet-Applied Rubber Lining Operations: 850 g/L.
- 15) Top and Trim Adhesive: 250 g/L.
- 16) Architectural Sealant: 250 g/L.
- 17) Other Sealant: 420 g/L.

b. Non-Flooring Adhesives and Sealants:

- 1) Drywall and Panel Adhesives: 50 g/L.
- 2) Multipurpose Construction Adhesives: 70 g/L.
- 3) Structural Glazing Adhesives: 100 g/L.
- 4) Metal-to-Metal Substrate Adhesives: 30 g/L.
- 5) Plastic Foam Substrate Adhesive: 50 g/L.
- 6) Porous Material (Except Wood) Substrate Adhesive: 50 g/L.
- 7) Wood Substrate Adhesive: 30 g/L.
- 8) Fiberglass Substrate Adhesive: 80 g/L.
- 9) Architectural Non-Porous Sealant Primer: 250 g/L.
- 10) Architectural Porous Sealant Primer: 775 g/L.
- 11) Other Sealant Primer: 750 g/L.
- 12) PVC Welding Adhesives: 510 g/L.
- 13) CPVC Welding Adhesives: 490 g/L.
- 14) ABS Welding Adhesives: 325 g/L.
- 15) Plastic Cement Welding Adhesives: 250 g/L.

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- 16) Adhesive Primer for Plastic: 550 g/L.
 - 17) Contact Adhesive: 80 g/L.
 - 18) Special Purpose Contact Adhesive: 250 g/L.
 - 19) Structural Wood Member Adhesive: 140 g/L.
 - 20) Sheet Applied Rubber Lining Operations: 850 g/L.
 - 21) Top and Trim Adhesive: 250 g/L.
 - 22) Architectural Sealants: 250 g/L.
 - 23) Other Sealants: 420 g/L.
2. Aerosol adhesives applied on site within the weatherproofing membrane must comply with the following Green Seal GS-36.
- a. Aerosol Adhesive, General-Purpose Mist Spray: 65 percent VOCs by weight.
 - b. Aerosol Adhesive, General-Purpose Web Spray: 55 percent VOCs by weight.
 - c. Special-Purpose Aerosol Adhesive (All Types): 70 percent VOCs by weight.
3. Paints and coatings applied on site within the weatherproofing membrane must comply with the following criteria:
- a. VOC content limits for paints and coatings established in Green Seal Standard GS-11.
 - b. VOC content limit for anti-corrosive and anti-rust paints applied to interior ferrous metal substrates of 250 g/L established in Green Seal GC-03.
 - c. Clear wood finishes, floor coatings, stains, primers, sealers, and shellacs applied to interior elements must not exceed VOC content limits established in SCAQMD Rule 1113.
 - d. Comply with the following VOC content limits:
 - 1) Anti-Corrosive/Antirust Paints: 250 g/L.
 - 2) Clear Wood Finish, Lacquer: 550 g/L.
 - 3) Clear Wood Finish, Sanding Sealer: 350 g/L.
 - 4) Clear Wood Finish, Varnish: 350 g/L.
 - 5) Floor Coating: 100 g/L.
 - 6) Interior Flat Paint, Coating or Primer: 50 g/L.
 - 7) Interior Non-Flat Paint, Coating or Primer: 150 g/L.
 - 8) Sealers and Undercoaters: 200 g/L.
 - 9) Shellac, Clear: 730 g/L.
 - 10) Shellac, Pigmented: 550 g/L.

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- 11) Stain: 250 g/L.
- 12) Clear Brushing Lacquer: 680 g/L.
- 13) Concrete Curing Compounds: 350 g/L.
- 14) Japans/Faux Finishing Coatings: 350 g/L.
- 15) Magnesite Cement Coatings: 450 g/L.
- 16) Pigmented Lacquer: 550 g/L.
- 17) Waterproofing Sealers: 250 g/L.
- 18) Wood Preservatives: 350 g/L.
- 19) Low-Solids Coatings: 120 g/L.

~~4. Carpet installed in building interior must comply with one of the following:~~

- ~~a. Meet testing and product requirements of the Carpet and Rug Institute Green Label Plus program.~~
- ~~b. Maximum VOC concentrations specified in CDPH Standard Method V1.1-2010, using office scenario at the 14 day time point.~~

~~5. Each non-carpet flooring element installed in building interior which is not inherently non emitting (stone, ceramic, powder coated metals, plated or anodized metal, glass, concrete, clay brick, and unfinished or untreated solid wood flooring) must comply with one of the following:~~

- ~~a. Meet requirements of the FloorScore standard as shown with testing by an independent third party.~~
- ~~b. Maximum VOC concentrations specified in CDPH Standard Method V1.1-2010, using office scenario at 14 day time point.~~

~~6.4.~~ Composite wood and agrifiber products used within the weatherproofing membrane must contain no added urea-formaldehyde resins.

~~7.5.~~ Laminating adhesives used to fabricate on-site and shop-applied composite wood and agrifiber assemblies must not contain added urea-formaldehyde.

C. Recycled Content:

- 1. Any products being installed or used that are listed on EPA Comprehensive Procurement Guidelines designated product list must meet or exceed the EPA's recycled content recommendations. The EPA Comprehensive Procurement Guidelines categories include:

- a. Building insulation.
- ~~b. Cement and concrete.~~

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~~e.b.~~ _____ Consolidated and reprocessed latex paint.

~~d. Floor tiles.~~

~~e. Flowable fill.~~

~~f. Laminated paperboard.~~

~~g. Modular threshold ramps.~~

~~h. Nonpressure pipe.~~

~~i. Patio blocks.~~

~~j. Railroad grade crossing surfaces.~~

~~k.c.~~ _____ Roofing materials.

~~l. Shower and restroom dividers/partitions.~~

~~m.d.~~ _____ Structural fiberboard.

~~n. Nylon carpet and nylon carpet backing.~~

~~o. Compost and fertilizer made from recovered organic materials.~~

~~p. Hydraulic mulch.~~

~~q. Lawn and garden edging.~~

~~r. Plastic lumber landscaping timbers and posts.~~

~~s. Park benches and picnic tables.~~

~~t. Plastic fencing.~~

~~u. Playground equipment.~~

~~v. Playground surfaces.~~

~~w. Bike racks.~~

D. Biobased Content:

1. Materials and equipment being installed or used that are listed on the USDA BioPreferred program product category list must meet or exceed USDA's minimum biobased content threshold. Refer to individual specification sections for detailed requirements applicable to that section.

- a. USDA BioPreferred program categories include:

~~SPEC WRITER NOTE:~~

- ~~1. Edit following list appropriately for project.~~

1) Adhesive and Mastic Removers.

~~2) Carpets.~~

~~3)2)~~ _____ Cleaners.

~~4) Composite Panels.~~

~~5)3)~~ _____ Corrosion Preventatives.

~~6) Erosion Control Materials.~~

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~~7)4)~~ 4) Dust Suppressants.

~~8)~~ Fertilizers.

~~9)~~ Floor Cleaners and Protectors.

~~10)~~ Floor Coverings (Non-Carpet).

~~11)~~ Glass Cleaners.

~~12)~~ Hydraulic Fluids.

~~13)5)~~ Industrial Cleaners.

~~14)~~ Interior Paints and Coatings.

~~15)~~ Mulch and Compost Materials.

~~16)6)~~ Multipurpose Cleaners.

~~17)~~ Multipurpose Lubricants.

~~18)~~ Packaging Films.

~~19)7)~~ Paint Removers.

~~20)~~ Plastic Insulating Foam.

~~21)~~ Pneumatic Equipment Lubricants.

~~22)8)~~ Roof Coatings.

~~23)~~ Wastewater Systems Coatings.

~~24)~~ Water Tank Coatings.

~~25)~~ Wood and Concrete Sealers.

~~26)~~ Wood and Concrete Stains.

~~E. Materials, products, and equipment being installed which fall into a category covered by the WaterSense program must be WaterSense-labeled or meet or exceed WaterSense program performance requirements, unless disallowed for infection control reasons.~~

~~1. WaterSense categories include:~~

- ~~a. Bathroom Faucets~~
- ~~b. Commercial Toilets~~
- ~~c. Irrigation Controllers~~
- ~~d. Pre-Rinse Spray Valves~~
- ~~e. Residential Toilets~~
- ~~f. Showerheads~~
- ~~g. Spray Sprinkler Bodies~~
- ~~h. Urinals~~

~~F.E.~~ F.E. Materials, products, and equipment being installed which fall into any of the following product categories must be Energy Star-labeled.

1. Applicable Energy Star product categories as of 09/14/2017 include:

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- a. ~~Appliances+Roof Products.~~
 - 1) ~~Air Purifiers and Cleaners.~~
 - 2) ~~Clothes Dryers (Residential).~~
 - 3) ~~Clothes Washers (Commercial & Residential).~~
 - 4) ~~Dehumidifiers.~~
 - 5) ~~Dishwashers (Residential).~~
 - 6) ~~Freezers (Residential).~~
 - 7) ~~Refrigerators (Residential).~~
- b. ~~Electronics and Information Technology:~~
 - 1) ~~Audio/Video Equipment.~~
 - 2) ~~Computers.~~
 - 3) ~~Data Center Storage.~~
 - 4) ~~Digital Media Player.~~
 - 5) ~~Enterprise Servers.~~
 - 6) ~~Imaging Equipment.~~
 - 7) ~~Monitors.~~
 - 8) ~~Professional Displays.~~
 - 9) ~~Set Top and Cable Boxes.~~
 - 10) ~~Telephones.~~
 - 11) ~~Televisions.~~
 - 12) ~~Uninterruptible Power Supplies.~~
 - 13) ~~Voice over Internet Protocol (VoIP) Phones.~~
- c. ~~Food Service Equipment (Commercial):~~
 - 1) ~~Dishwashers.~~
 - 2) ~~Fryers.~~
 - 3) ~~Griddles.~~
 - 4) ~~Hot Food Holding Cabinets.~~
 - 5) ~~Ice Makers.~~
 - 6) ~~Ovens.~~
 - 7) ~~Refrigerators and Freezers.~~
 - 8) ~~Steam Cookers.~~
 - 9) ~~Vending Machines.~~
- d. ~~Heating and Cooling Equipment:~~
 - 1) ~~Air Source Heat Pumps (Residential).~~
 - 2) ~~Boilers.~~
 - 3) ~~Ceiling Fans (Residential).~~
 - 4) ~~Central Air Conditioners (Residential).~~

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- ~~5) Ductless Heating and Cooling (Residential).~~
- ~~6) Furnaces (Residential).~~
- ~~7) Water Heaters.~~
- ~~8) Geothermal Heat Pumps (Residential).~~
- ~~9) Light Commercial Heating and Cooling Equipment.~~
- ~~10) Room Air Conditioners (Residential).~~
- ~~11) Ventilation Fans (Residential).~~

~~e. Other:~~

- ~~1) Decorative Light Strings.~~
- ~~2) Electric Vehicle Supply Equipment.~~
- ~~3) Laboratory Grade Refrigerators and Freezers.~~
- ~~4) Light Bulbs.~~
- ~~5) Light Fixtures.~~
- ~~6) Pool Pumps.~~
- ~~7) Roof Products.~~
- ~~8) Water Coolers.~~
- ~~9) Windows, Doors, and Skylights.~~

~~G. Materials, products, and equipment being installed which fall into any of the following categories must be FEMP-designated. FEMP-designated product categories as of 09/14/2017 include:~~

- ~~1. Boilers (Commercial).~~
- ~~2. Dishwashers (Commercial).~~
- ~~3. Electric Chillers, Air-Cooled (Commercial).~~
- ~~4. Electric Chillers, Water-Cooled (Commercial).~~
- ~~5. Exterior Lighting.~~
- ~~6. Fluorescent Ballasts.~~
- ~~7. Fluorescent Lamps, General Service.~~
- ~~8. Ice Machines, Water-Cooled.~~
- ~~9. Industrial Lighting (High/Low Bay).~~
- ~~10. Light Emitting Diode (LED) Luminaires.~~

~~H. Electronic products and equipment being installed which fall into any of the following categories shall be EPEAT registered. Electronic products and equipment covered by EPEAT program as of 09/14/2017 include:~~

- ~~1. Computers.~~
- ~~2. Displays.~~
- ~~3. Imaging Equipment.~~

~~4. Televisions.~~

PART 3 - EXECUTION

3.1 FIELD QUALITY CONTROL

A. Construction Indoor Air Quality Management:

1. During construction, meet or exceed recommended control measures of ANSI/SMACNA 008-2008, Chapter 3.
2. Protect stored on-site and installed absorptive materials from moisture damage.
3. If permanently installed air handlers are used during construction, filtration media with a minimum efficiency reporting value (MERV) of 8 must be used at each return air grille, as determined by ASHRAE Standard 52.2-1999 (with errata but without addenda). Replace all filtration media immediately prior to occupancy.

~~4. Perform building flush-out as follows:~~

- ~~a. After construction ends, prior to occupancy and with interior finishes installed, perform a building flush-out by supplying a total volume of 14000 cu. ft. of outdoor air per sq. ft. of floor area while maintaining an internal temperature of at least 60 degrees Fahrenheit and a relative humidity no higher than 60 percent. OR~~
- ~~b. If occupancy is desired prior to flush-out completion, the space may be occupied following delivery of a minimum of 3500 cu. ft. of outdoor air per sq. ft. of floor area to the space. Once a space is occupied, it must be ventilated at a minimum rate of 0.30 cfm per sq. ft. of outside air or design minimum outside air rate determined until a total of 14000 cu. ft./sq. ft. of outside air has been delivered to the space. During each day of flush-out period, ventilation must begin a minimum of three hours prior to occupancy and continue during occupancy.~~

~~SPEC WRITER NOTE:~~

- ~~1. The following is a requirement for the State of California; although, it could be applied to any project.~~

~~5.4. //~~ Provide construction dust control to comply with SCAQMD Rule 403. ~~//~~

-----END-----

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**SECTION 01 91 00
GENERAL COMMISSIONING REQUIREMENTS**

PART 1 - GENERAL

1.1 COMMISSIONING DESCRIPTION

- A. This Section 01 91 00 GENERAL COMMISSIONING REQUIREMENTS shall form the basis of the construction phase commissioning process and procedures. The Commissioning Agent shall add, modify, and refine the commissioning procedures, as approved by the Department of Veterans Affairs (VA), to suit field conditions and actual manufacturer's equipment, incorporate test data and procedure results, and provide detailed scheduling for all commissioning tasks.
- B. Various sections of the project specifications require equipment startup, testing, and adjusting services. Requirements for startup, testing, and adjusting services specified in the Division 7, Division 21, Division 22, Division 23, Division 26, Division 27, Division 28, and Division 31 series sections of these specifications are intended to be provided in coordination with the commissioning services and are not intended to duplicate services. The Contractor shall coordinate the work required by individual specification sections with the commissioning services requirements specified herein.
- C. Where individual testing, adjusting, or related services are required in the project specifications and not specifically required by this commissioning requirements specification, the specified services shall be provided and copies of documentation, as required by those specifications shall be submitted to the VA and the Commissioning Agent to be indexed for future reference.
- D. Where training or educational services for VA are required and specified in other sections of the specifications, including but not limited to Division 7, Division 8, Division 21, Division 22, Division 23, Division 26, Division 27, Division 28, and Division 31 series sections of the specification, these services are intended to be provided in addition to the training and educational services specified herein.
- E. Commissioning is a systematic process of verifying that the building systems perform interactively according to the construction documents and the VA's operational needs. The commissioning process shall encompass and coordinate the system documentation, equipment startup,

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control system calibration, testing and balancing, performance testing and training. Commissioning during the construction and post-occupancy phases is intended to achieve the following specific objectives according to the contract documents:

1. Verify that the applicable equipment and systems are installed in accordance with the contract documents and according to the manufacturer's recommendations.
 2. Verify and document proper integrated performance of equipment and systems.
 3. Verify that Operations & Maintenance documentation is complete.
 4. Verify that all components requiring servicing can be accessed, serviced and removed without disturbing nearby components including ducts, piping, cabling or wiring.
 5. Verify that the VA's operating personnel are adequately trained to enable them to operate, monitor, adjust, maintain, and repair building systems in an effective and energy-efficient manner.
 6. Document the successful achievement of the commissioning objectives listed above.
- F. The commissioning process does not take away from or reduce the responsibility of the Contractor to provide a finished and fully functioning product.

1.2 CONTRACTUAL RELATIONSHIPS

- A. For this construction project, the Department of Veterans Affairs contracts with a Contractor to provide construction services. The contracts are administered by the VA Contracting Officer and the Resident Engineer as the designated representative of the Contracting Officer. On this project, the authority to modify the contract in any way is strictly limited to the authority of the Contracting Officer.
- B. In this project, only two contract parties are recognized and communications on contractual issues are strictly limited to VA Resident Engineer and the Contractor. It is the practice of the VA to require that communications between other parties to the contracts (Subcontractors and Vendors) be conducted through the Resident Engineer and Contractor. It is also the practice of the VA that communications between other parties of the project (Commissioning Agent and Architect/Engineer) be conducted through the Resident Engineer.

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- C. Whole Building Commissioning is a process that relies upon frequent and direct communications, as well as collaboration between all parties to the construction process. By its nature, a high level of communication and cooperation between the Commissioning Agent and all other parties (Architects, Engineers, Subcontractors, Vendors, third party testing agencies, etc.) is essential to the success of the Commissioning effort.
- D. With these fundamental practices in mind, the commissioning process described herein has been developed to recognize that, in the execution of the Commissioning Process, the Commissioning Agent must develop effective methods to communicate with every member of the construction team involved in delivering commissioned systems while simultaneously respecting the exclusive contract authority of the Contracting Officer and Resident Engineer. Thus, the procedures outlined in this specification must be executed within the following limitations:
1. No communications (verbal or written) from the Commissioning Agent shall be deemed to constitute direction that modifies the terms of any contract between the Department of Veterans Affairs and the Contractor.
 2. Commissioning Issues identified by the Commissioning Agent will be delivered to the Resident Engineer and copied to the designated Commissioning Representatives for the Contractor and subcontractors on the Commissioning Team for information only in order to expedite the communication process. These issues must be understood as the professional opinion of the Commissioning Agent and as suggestions for resolution.
 3. In the event that any Commissioning Issues and suggested resolutions are deemed by the Resident Engineer to require either an official interpretation of the construction documents or require a modification of the contract documents, the Contracting Officer or Resident Engineer will issue an official directive to this effect.
 4. All parties to the Commissioning Process shall be individually responsible for alerting the Resident Engineer of any issues that they deem to constitute a potential contract change prior to acting on these issues.
 5. Authority for resolution or modification of design and construction issues rests solely with the Contracting Officer or Resident

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Engineer, with appropriate technical guidance from the
Architect/Engineer and/or Commissioning Agent.

~~Spec Writer Note: Edit the following paragraph to include only those related work sections that are included in the contract. Add related work references as necessary and delete any related work references not included in the contract.~~

1.3 RELATED WORK

- A. Section 01 00 00 GENERAL REQUIREMENTS.
- ~~B. Section 01 32 16.01 ARCHITECTURAL AND ENGINEERING CPM SCHEDULES~~
- C. Section 01 32.16.15 PROJECT SCHEDULES (SMALL PROJECTS -
DESIGN/BID/BUILD)
- ~~D. Section 01 32.16.16 NETWORK ANALYSIS SCHEDULES (DESIGN-BUILD ONLY)~~
- ~~E. Section 01 32.16.17 NETWORK ANALYSIS SCHEDULES-MAJOR DESIGN/BUILD
PROJECTS~~
- F. Section 01 33 23 SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES
- G. Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS
- H. Section 07 08 00 FACILITY EXTERIOR CLOSURE COMMISSIONING.
- ~~I. Section 21 08 00 COMMISSIONING OF FIRE PROTECTION SYSTEMS.~~
- ~~J. Section 22 08 00 COMMISSIONING OF PLUMBING SYSTEMS.~~
- ~~K. Section 23 08 00 COMMISSIONING OF HVAC SYSTEMS.~~
- ~~L. Section 26 08 00 COMMISSIONING OF ELECTRICAL SYSTEMS.~~
- ~~M. Section 27 08 00 COMMISSIONING OF COMMUNICATIONS SYSTEMS.~~
- ~~N. Section 28 08 00 COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY
SYSTEMS.~~
- ~~O. Section 33 08 00 COMMISSIONING OF SITE UTILITIES.~~

1.4 SUMMARY

- A. This Section includes general requirements that apply to implementation of commissioning without regard to systems, subsystems, and equipment being commissioned.
- B. The commissioning activities have been developed to support the VA requirements to meet guidelines for Federal Leadership in Environmental, Energy, and Economic Performance.

~~SPEC WRITER NOTE: Paragraph C is for LEED projects. Paragraphs D is for Green Globes Projects. Retain either Paragraph C or Paragraph D as appropriate. Delete paragraph C.2 if enhanced commissioning for LEED is not included in the project.~~

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~~Delete paragraph C.3 if the M&V credit is not included in the project.~~

~~C. The commissioning activities have been developed to support the United States Green Building Council's (USGBC) LEED™ rating program and to support delivery of project performance in accordance with the VA requirements developed for the project to support the following credits:~~

- ~~1. Commissioning activities and documentation for the LEED™ section on "Energy and Atmosphere" and the prerequisite of "Fundamental Building Systems Commissioning."~~
- ~~2. Commissioning activities and documentation for the LEED™ section on "Energy and Atmosphere" requirements for the "Enhanced Building System Commissioning" credit.~~
- ~~3. Activities and documentation for the LEED™ section on "Measurement and Verification" requirements for the Measurement and Verification credit.~~

~~D. The commissioning activities have been developed to support the Green Buildings Initiative's Green Globes rating program and to support delivery of project performance in accordance with the VA requirements developed for the project.~~

1.5 ACRONYMS

List of Acronyms	
Acronym	Meaning
A/E	Architect / Engineer Design Team
AHJ	Authority Having Jurisdiction
ASHRAE	Association Society for Heating Air Condition and Refrigeration Engineers
BOD	Basis of Design
BSC	Building Systems Commissioning
CCTV	Closed Circuit Television
CD	Construction Documents
CMMS	Computerized Maintenance Management System
CO	Contracting Officer (VA)
COR	Contracting Officer's Representative (see also VA-RE)
COBie	Construction Operations Building Information Exchange
CPC	Construction Phase Commissioning
Cx	Commissioning

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List of Acronyms	
Acronym	Meaning
CxA	Commissioning Agent
CxM	Commissioning Manager
CxR	Commissioning Representative
DPC	Design Phase Commissioning
FPT	Functional Performance Test
GBI-GG	Green Building Initiative - Green Globes
HVAC	Heating, Ventilation, and Air Conditioning
LEED	Leadership in Energy and Environmental Design
NC	Department of Veterans Affairs National Cemetery
NCA	Department of Veterans Affairs National Cemetery Administration
NEBB	National Environmental Balancing Bureau
O&M	Operations & Maintenance
OPR	Owner's Project Requirements
PFC	Pre-Functional Checklist
PFT	Pre-Functional Test
SD	Schematic Design
SO	Site Observation
TAB	Test Adjust and Balance
VA	Department of Veterans Affairs
VAMC	VA Medical Center
VA CFM	VA Office of Construction and Facilities Management
VACO	VA Central Office
VA PM	VA Project Manager
VA-RE	VA Resident Engineer
USGBC	United States Green Building Council

1.6 DEFINITIONS

Acceptance Phase Commissioning: Commissioning tasks executed after most construction has been completed, most Site Observations and Static Tests have been completed and Pre-Functional Testing has been completed and accepted. The main commissioning activities performed during this phase are verification that the installed systems are functional by conducting Systems Functional Performance tests and Owner Training.

Accuracy: The capability of an instrument to indicate the true value of a measured quantity.

Back Check: A back check is a verification that an agreed upon solution to a design comment has been adequately addressed in a subsequent design review

Basis of Design (BOD): The Engineer's Basis of Design is comprised of two components: the Design Criteria and the Design Narrative, these documents record the concepts, calculations, decisions, and product selections used to meet the Owner's Project Requirements (OPR) and to satisfy applicable regulatory requirements, standards, and guidelines.

Benchmarks: Benchmarks are the comparison of a building's energy usage to other similar buildings and to the building itself. For example, ENERGY STAR Portfolio Manager is a frequently used and nationally recognized building energy benchmarking tool.

Building Information Modeling (BIM): Building Information Modeling is a parametric database which allows a building to be designed and constructed virtually in 3D, and provides reports both in 2D views and as schedules. This electronic information can be extracted and reused for pre-populating facility management CMMS systems. Building Systems Commissioning (BSC): NEBB acronym used to designate its commissioning program.

Calibrate: The act of comparing an instrument of unknown accuracy with a standard of known accuracy to detect, correlate, report, or eliminate by adjustment any variation in the accuracy of the tested instrument.

CCTV: Closed circuit Television. Normally used for security surveillance and alarm detections as part of a special electrical security system.

COBie: Construction Operations Building Information Exchange (COBie) is an electronic industry data format used to transfer information developed during design, construction, and commissioning into the Computer Maintenance Management Systems (CMMS) used to operate facilities. See the Whole Building Design Guide website for further information (<http://www.wbdg.org/resources/cobie.php>)

Commissionability: Defines a design component or construction process that has the necessary elements that will allow a system or component to be effectively measured, tested, operated and commissioned

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Commissioning Agent (CxA): The qualified Commissioning Professional who administers the Cx process by managing the Cx team and overseeing the Commissioning Process. Where CxA is used in this specification it means the Commissioning Agent, members of his staff or appointed members of the commissioning team. Note that LEED uses the term Commissioning Authority in lieu of Commissioning Agent.

Commissioning Checklists: Lists of data or inspections to be verified to ensure proper system or component installation, operation, and function. Verification checklists are developed and used during all phases of the commissioning process to verify that the Owner's Project Requirements (OPR) is being achieved.

Commissioning Design Review: The commissioning design review is a collaborative review of the design professionals design documents for items pertaining to the following: owner's project requirements; basis of design; operability and maintainability (O&M) including documentation; functionality; training; energy efficiency, control systems' sequence of operations including building automation system features; commissioning specifications and the ability to functionally test the systems.

Commissioning Issue: A condition identified by the Commissioning Agent or other member of the Commissioning Team that adversely affects the commissionability, operability, maintainability, or functionality of a system, equipment, or component. A condition that is in conflict with the Contract Documents and/or performance requirements of the installed systems and components. (See also - Commissioning Observation).

Commissioning Manager (CxM): A qualified individual appointed by the Contractor to manage the commissioning process on behalf of the Contractor.

Commissioning Observation: An issue identified by the Commissioning Agent or other member of the Commissioning Team that does not conform to the project OPR, contract documents or standard industry best practices. (See also Commissioning Issue)

Commissioning Plan: A document that outlines the commissioning process, commissioning scope and defines responsibilities, processes, schedules, and the documentation requirements of the Commissioning Process.

Commissioning Process: A quality focused process for enhancing the delivery of a project. The process focuses upon verifying and

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documenting that the facility and all of its systems, components, and assemblies are planned, designed, installed, tested, can be operated, and maintained to meet the Owner's Project Requirements.

Commissioning Report: The final commissioning document which presents the commissioning process results for the project. Cx reports include an executive summary, the commissioning plan, issue log, correspondence, and all appropriate check sheets and test forms.

Commissioning Representative (CxR): An individual appointed by a sub-contractor to manage the commissioning process on behalf of the sub-contractor.

Commissioning Specifications: The contract documents that detail the objective, scope and implementation of the commissioning process as developed in the Commissioning Plan.

Commissioning Team: Individual team members whose coordinated actions are responsible for implementing the Commissioning Process.

Construction Phase Commissioning: All commissioning efforts executed during the construction process after the design phase and prior to the Acceptance Phase Commissioning.

Contract Documents (CD): Contract documents include design and construction contracts, price agreements and procedure agreements. Contract Documents also include all final and complete drawings, specifications and all applicable contract modifications or supplements.

Construction Phase Commissioning (CPC): All commissioning efforts executed during the construction process after the design phase and prior to the Acceptance Phase Commissioning.

Coordination Drawings: Drawings showing the work of all trades that are used to illustrate that equipment can be installed in the space allocated without compromising equipment function or access for maintenance and replacement. These drawings graphically illustrate and dimension manufacturers' recommended maintenance clearances. On mechanical projects, coordination drawings include structural steel, ductwork, major piping and electrical conduit and show the elevations and locations of the above components.

Data Logging: The monitoring and recording of temperature, flow, current, status, pressure, etc. of equipment using stand-alone data recorders.

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Deferred System Test: Tests that cannot be completed at the end of the acceptance phase due to ambient conditions, schedule issues or other conditions preventing testing during the normal acceptance testing period.

Deficiency: See "Commissioning Issue".

Design Criteria: A listing of the VA Design Criteria outlining the project design requirements, including its source. These are used during the design process to show the design elements meet the OPR.

Design Intent: The overall term that includes the OPR and the BOD. It is a detailed explanation of the ideas, concepts, and criteria that are defined by the owner to be important. The design intent documents are utilized to provide a written record of these ideas, concepts and criteria.

Design Narrative: A written description of the proposed design solutions that satisfy the requirements of the OPR.

Design Phase Commissioning (DPC): All commissioning tasks executed during the design phase of the project.

Environmental Systems: Systems that use a combination of mechanical equipment, airflow, water flow and electrical energy to provide heating, ventilating, air conditioning, humidification, and dehumidification for the purpose of human comfort or process control of temperature and humidity.

Executive Summary: A section of the Commissioning report that reviews the general outcome of the project. It also includes any unresolved issues, recommendations for the resolution of unresolved issues and all deferred testing requirements.

Functionality: This defines a design component or construction process which will allow a system or component to operate or be constructed in a manner that will produce the required outcome of the OPR.

Functional Test Procedure (FTP): A written protocol that defines methods, steps, personnel, and acceptance criteria for tests conducted on components, equipment, assemblies, systems, and interfaces among systems.

Industry Accepted Best Practice: A design component or construction process that has achieved industry consensus for quality performance and functionality. Refer to the current edition of the NEBB Design Phase Commissioning Handbook for examples.

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Installation Verification: Observations or inspections that confirm the system or component has been installed in accordance with the contract documents and to industry accepted best practices.

Integrated System Testing: Integrated Systems Testing procedures entail testing of multiple integrated systems performance to verify proper functional interface between systems. Typical Integrated Systems Testing includes verifying that building systems respond properly to loss of utility, transfer to emergency power sources, re-transfer from emergency power source to normal utility source; interface between HVAC controls and Fire Alarm systems for equipment shutdown, interface between Fire Alarm system and elevator control systems for elevator recall and shutdown; interface between Fire Alarm System and Security Access Control Systems to control access to spaces during fire alarm conditions; and other similar tests as determined for each specific project.

Issues Log: A formal and ongoing record of problems or concerns - and their resolution - that have been raised by members of the Commissioning Team during the course of the Commissioning Process.

Lessons Learned Workshop: A workshop conducted to discuss and document project successes and identify opportunities for improvements for future projects.

Maintainability: A design component or construction process that will allow a system or component to be effectively maintained. This includes adequate room for access to adjust and repair the equipment. Maintainability also includes components that have readily obtainable repair parts or service.

Manual Test: Testing using hand-held instruments, immediate control system readouts or direct observation to verify performance (contrasted to analyzing monitored data taken over time to make the 'observation').

Owner's Project Requirements (OPR): A written document that details the project requirements and the expectations of how the building and its systems will be used and operated. These include project goals, measurable performance criteria, cost considerations, benchmarks, success criteria, and supporting information.

Peer Review: A formal in-depth review separate from the commissioning review processes. The level of effort and intensity is much greater than a typical commissioning facilitation or extended commissioning

review. The VA usually hires an independent third-party (called the IDIQ A/E) to conduct peer reviews.

Precision: The ability of an instrument to produce repeatable readings of the same quantity under the same conditions. The precision of an instrument refers to its ability to produce a tightly grouped set of values around the mean value of the measured quantity.

Pre-Design Phase Commissioning: Commissioning tasks performed prior to the commencement of design activities that includes project programming and the development of the commissioning process for the project

Pre-Functional Checklist (PFC): A form used by the contractor to verify that appropriate components are onsite, correctly installed, set up, calibrated, functional and ready for functional testing.

Pre-Functional Test (PFT): An inspection or test that is done before functional testing. PFT's include installation verification and system and component start up tests.

Procedure or Protocol: A defined approach that outlines the execution of a sequence of work or operations. Procedures are used to produce repeatable and defined results.

Range: The upper and lower limits of an instrument's ability to measure the value of a quantity for which the instrument is calibrated.

Resolution: This word has two meanings in the Cx Process. The first refers to the smallest change in a measured variable that an instrument can detect. The second refers to the implementation of actions that correct a tested or observed deficiency.

Site Observation Visit: On-site inspections and observations made by the Commissioning Agent for the purpose of verifying component, equipment, and system installation, to observe contractor testing, equipment start-up procedures, or other purposes.

Site Observation Reports (SO): Reports of site inspections and observations made by the Commissioning Agent. Observation reports are intended to provide early indication of an installation issue which will need correction or analysis.

Special System Inspections: Inspections required by a local code authority prior to occupancy and are not normally a part of the commissioning process.

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Static Tests: Tests or inspections that validate a specified static condition such as pressure testing. Static tests may be specification or code initiated.

Start Up Tests: Tests that validate the component or system is ready for automatic operation in accordance with the manufactures requirements.

Systems Manual: A system-focused composite document that includes all information required for the owners operators to operate the systems.

Test Procedure: A written protocol that defines methods, personnel, and expectations for tests conducted on components, equipment, assemblies, systems, and interfaces among systems.

Testing: The use of specialized and calibrated instruments to measure parameters such as: temperature, pressure, vapor flow, air flow, fluid flow, rotational speed, electrical characteristics, velocity, and other data in order to determine performance, operation, or function.

Testing, Adjusting, and Balancing (TAB): A systematic process or service applied to heating, ventilating and air-conditioning (HVAC) systems and other environmental systems to achieve and document air and hydronic flow rates. The standards and procedures for providing these services are referred to as "Testing, Adjusting, and Balancing" and are described in the Procedural Standards for the Testing, Adjusting and Balancing of Environmental Systems, published by NEBB or AABC.

Thermal Scans: Thermographic pictures taken with an Infrared Thermographic Camera. Thermographic pictures show the relative temperatures of objects and surfaces and are used to identify leaks, thermal bridging, thermal intrusion, electrical overload conditions, moisture containment, and insulation failure.

Training Plan: A written document that details, in outline form the expectations of the operator training. Training agendas should include instruction on how to obtain service, operate, startup, shutdown and maintain all systems and components of the project.

Trending: Monitoring over a period of time with the building automation system.

Unresolved Commissioning Issue: Any Commissioning Issue that, at the time that the Final Report or the Amended Final Report is issued that has not been either resolved by the construction team or accepted by

the VA. Validation: The process by which work is verified as complete and operating correctly:

1. First party validation occurs when a firm or individual verifying the task is the same firm or individual performing the task.
2. Second party validation occurs when the firm or individual verifying the task is under the control of the firm performing the task or has other possibilities of financial conflicts of interest in the resolution (Architects, Designers, General Contractors and Third Tier Subcontractors or Vendors).
3. Third party validation occurs when the firm verifying the task is not associated with or under control of the firm performing or designing the task.

Verification: The process by which specific documents, components, equipment, assemblies, systems, and interfaces among systems are confirmed to comply with the criteria described in the Owner's Project Requirements.

Warranty Phase Commissioning: Commissioning efforts executed after a project has been completed and accepted by the Owner. Warranty Phase Commissioning includes follow-up on verification of system performance, measurement and verification tasks and assistance in identifying warranty issues and enforcing warranty provisions of the construction contract.

Warranty Visit: A commissioning meeting and site review where all outstanding warranty issues and deferred testing is reviewed and discussed.

Whole Building Commissioning: Commissioning of building systems such as Building Envelope, HVAC, Electrical, Special Electrical (Fire Alarm, Security & Communications), Plumbing and Fire Protection as described in this specification.

1.7 SYSTEMS TO BE COMMISSIONED

- A. Commissioning of a system or systems specified for this project is part of the construction process. Documentation and testing of these systems, as well as training of the VA's Operation and Maintenance personnel, is required in cooperation with the VA and the Commissioning Agent.

~~SPEC WRITER NOTE: Paragraph B should list the specific systems that will be~~

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~~commissioned. Edit the list as necessary for specific projects. The list below should match the list included in Sections XX 08 00 COMMISSIONING OF "XX" SYSTEMS included in the various Divisions' Technical Specifications. It is recommended that the list included in Sections XX 08 00 be developed first and then copied/pasted into the list below. Both lists should be identical to prevent confusion.~~

B. The following systems will be commissioned as part of this project:

Systems To Be Commissioned	
System	Description
Building Exterior Closure	
Foundations (excluding structural)	Standard, special, slab-on-grade, vapor barriers, air barriers
Basements	Basement walls, crawl spaces, waterproofing, drainage
Superstructure	Floor construction, roof construction, sunshades, connections to adjacent structures
Exterior Closure	Exterior walls, exterior windows, exterior doors, louvers, grilles and sunscreens,
Roofing	Roof system (including parapet), roof openings (skylights, pipe chases, ducts, equipment curbs, etc.)
Note:	The emphasis on commissioning the above building envelope systems is on control of air flow, heat flow, noise, infrared, ultraviolet, rain penetration, moisture, durability, security, reliability, constructability, maintainability, and sustainability.
Specialties	
Patient Bed Service Walls	Medical gas certification and cross check, electrical connections
Equipment	

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Systems To Be Commissioned	
System	Description
Parking Control Equipment	Barriers
Laboratory Fume Hoods	Fume Hood Certification
Biological Safety Cabinets	Cabinet Certification
Packaged Incinerators	Combustion Testing, Cycle Certification
Conveying Equipment	
Electric Dumbwaiters	Interface with other systems (Fire Alarm, etc.) [ASTM testing and certification by others]
Elevators	Interface with other systems (fire alarm, etc.) [ASTM testing and certification by others]
Escalators	Interface with other systems (fire alarm, etc.) [ASTM testing and certification by others]
Material Delivery Systems	Interface with other systems (fire alarm, elevators, etc.)
Pneumatic Tube Systems	Interface with other systems (fire alarm, etc.)
Fire Suppression	
Fire Pump	Fire Pump, jockey pump, fire pump controller/ATS
Fire Sprinkler Systems	Wet pipe system, dry pipe system, pre-action system, special agent systems
Plumbing	
Domestic Water Distribution	Booster pumps, backflow preventers, water softeners, potable water storage tanks
Domestic Hot Water Systems	Water heaters**, heat exchangers, circulation pumps, point-of-use water heaters*
Sewerage Pump Systems	Sewage ejectors
Wastewater Pump Systems	Sump pumps

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Systems To Be Commissioned	
System	Description
Sanitary Waste Interceptors	Grease interceptors, acid neutralizers
General Service Air Systems	Packaged compressor systems, air dryers, filtration
Medical Air Systems	Packaged medical air compressor units. Outlet certification, cross-connection verification
Medical Vacuum Systems	Packaged medical vacuum units, outlet certification, cross-connection verification
Dental Air Systems	Packaged dental air compressor units, outlet certification, cross-connect verification
Dental Evacuation and Vacuum Systems	Packaged Dental Evacuation units, packaged dental vacuum units, outlet certification, cross-connection verification
Waste Anesthesia Gas Systems	Packaged Waste Anesthesia Gas units, outlet certification, cross-connection verification
Medical Gas Systems (other than Medical Air Systems)	Medical gas (oxygen, nitrogen, nitrous oxide, etc.) tank/manifold systems, outlet certification, cross-connection verification
Chemical Waste Systems	Chemical storage tanks, neutralization systems, ventilation, process control
Reverse Osmosis Systems	Packaged Reverse Osmosis systems
Water De-Alkalizing Systems	Package Water De-Alkalizing systems
HVAC	
Noise and Vibration Control	Noise and vibration levels for critical equipment such as Air Handlers, Chillers, Cooling Towers, Boilers, Generators, etc. will be commissioned as part of the system commissioning

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Systems To Be Commissioned	
System	Description
Direct Digital Control System**	Operator Interface Computer, Operator Work Station (including graphics, point mapping, trends, alarms), Network Communications Modules and Wiring, Integration Panels. [DDC Control panels will be commissioned with the systems controlled by the panel]
Chilled Water System**	Chillers (centrifugal, rotary screw, air-cooled), pumps (primary, secondary, variable primary), VFDs associated with chilled water system components, DDC Control Panels (including integration with Building Control System)
Condenser Water System**	Cooling Towers, Fluid Coolers, heat exchangers/economizers, pumps, VFDs associated with condenser water system components, DDC control panels.
Steam/Heating Hot Water System**	Boilers, boiler feed water system, economizers/heat recovery equipment, condensate recovery, water treatment, boiler fuel system, controls, interface with facility DDC system.
HVAC Air Handling Systems**	Air handling Units, packaged rooftop AHU, Outdoor Air conditioning units, humidifiers, DDC control panels
HVAC Ventilation/Exhaust Systems	General exhaust, toilet exhaust, laboratory exhaust, isolation exhaust, room pressurization control systems
HVAC Energy Recovery Systems**	Heat Wheels, Heat Recovery Loops, AHU Integrated Heat Recovery
HVAC Terminal Unit Systems**	VAV Terminal Units, CAV terminal units, fan coil units, fin-tube radiation, unit heaters
Decentralized Unitary HVAC Systems*	Split-system HVAC systems, controls, interface with facility DDC

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Systems To Be Commissioned	
System	Description
Unitary Heat Pump Systems**	Water source heat pumps, controls, interface with facility DDC
Humidity Control Systems	Humidifiers, de-humidifiers, controls, interface with facility DDC
Hydronic Distribution Systems	Pumps, DDC control panels, heat exchangers,
Facility Fuel Systems	Boiler fuel system, generator fuel system
Geothermal Energy Direct Use Heating **	Geothermal well, ground heat exchanger, geothermal pumps, heat exchanger, valves, instrumentation
Solar Energy Heating Systems **	Solar collectors, heat exchangers, storage tanks, solar boosted domestic hot water heater, pumps, valves, instrumentation
Facility Fuel Gas Systems	Witness Natural gas piping pressure testing, natural gas compressors and storage, propane storage
Smoke Evacuation System	Atrium smoke evacuation, other smoke evacuation and smoke management systems, controls, interface with other systems (fire alarm), emergency operation.
Electrical	
Medium Voltage Electrical Distribution Systems	Medium Voltage Switchgear, Medium Voltage Switches, Underground ductbank and distribution, Pad Mount Transformers, Medium Voltage Load Interrupter Switches,
Grounding & Bonding Systems	Witness 3rd party testing, review reports
Electric Power Monitoring Systems	Metering, sub-metering, power monitoring systems, PLC control systems
Electrical System Protective Device Study	Review reports, verify field settings consistent with Study

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Systems To Be Commissioned	
System	Description
Secondary Unit Substations	Medium voltage components, transformers, low voltage distribution, verify breaker testing results (injection current, etc)
Low Voltage Distribution System	Normal power distribution system, Life safety power distribution system, critical power distribution system, equipment power distribution system, switchboards, distribution panels, panelboards, verify breaker testing results (injection current, etc)
Emergency Power Generation Systems	Generators, Generator paralleling switchgear, automatic transfer switches, PLC and other control systems
Lighting & Lighting Control** Systems	Emergency lighting, occupancy sensors, lighting control systems, architectural dimming systems, theatrical dimming systems, exterior lighting and controls
Cathodic Protection Systems	Review 3rd party testing results.
Lightning Protection System	Witness 3rd party testing, review reports
Communications	
Grounding & Bonding System	Witness 3rd party testing, review reports
Structured Cabling System	Witness 3rd party testing, review reports
Master Antenna Television System	Witness 3rd party testing, review reports
Public Address & Mass Notification Systems	Witness 3rd party testing, review reports
Intercom & Program Systems	Witness 3rd party testing, review reports

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Systems To Be Commissioned	
System	Description
Nurse Call & Code Blue Systems	Witness 3rd party testing, review reports
Security Emergency Call Systems	Witness 3rd party testing, review reports
Duress Alarm Systems	Witness 3rd party testing, review reports
Electronic Safety and Security	
Grounding & Bonding	Witness 3rd party testing, review reports
Physical Access Control Systems	Witness 3rd party testing, review reports
Access Control Systems	Witness 3rd party testing, review reports
Security Access Detection Systems	Witness 3rd party testing, review reports
Video Surveillance System	Witness 3rd party testing, review reports
Electronic Personal Protection System	Witness 3rd party testing, review reports
Fire Detection and Alarm System	100% device acceptance testing, battery draw-down test, verify system monitoring, verify interface with other systems.
Renewable Energy Sources	
Geothermal Energy Electrical Generation Systems **	Geothermal well, DC-AC Inverters, storage batteries, turbine generator modules, switchgear, combiner boxes, instrumentation, monitoring and control systems
Solar Energy Electrical Power Generation Systems **	Solar collector modules, DC-AC inverter, storage batteries, combiners, Switchgear, instrumentation, monitoring and control systems
Wind Energy Electrical Power Generation Systems **	Wind Turbines, DC-AC inverter, storage batteries, combiners, switchgear, instrumentation, monitoring and control systems

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Systems To Be Commissioned	
System	Description
Site Utilities	
Water Utilities	City Water Service Entrance, Backflow Prevention, Pressure Control, Booster Pumps, Irrigation Systems
Sanitary Sewerage Utilities	City Sanitary Connection, Waste Treatment Systems
Storm Drainage Utilities	City Storm Water Connection, Site Storm Water Distribution
Energy Distribution Utilities	Connection to Third Party Energy (Steam, High Temp Hot Water, Chilled Water) Supply Systems, Metering, Pressure Control
Transportation	
Active Traffic Barrier Systems	Witness 3rd party testing
Integrated Systems Tests	
Loss of Power Response	Loss of power to building, loss of power to campus, restoration of power to building, restoration of power to campus.
Fire Alarm Response	Integrated System Response to Fire Alarm Condition and Return to Normal
Table Notes	
** Denotes systems that LEED requires to be commissioned to comply with the LEED Fundamental Commissioning pre-requisite.	

1.8 COMMISSIONING TEAM

- A. The commissioning team shall consist of, but not be limited to, representatives of Contractor, including Project Superintendent and subcontractors, installers, schedulers, suppliers, and specialists deemed appropriate by the Department of Veterans Affairs (VA) and Commissioning Agent.
- B. Members Appointed by Contractor:

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1. Contractor' Commissioning Manager: The designated person, company, or entity that plans, schedules and coordinates the commissioning activities for the construction team.
2. Contractor's Commissioning Representative(s): Individual(s), each having authority to act on behalf of the entity he or she represents, explicitly organized to implement the commissioning process through coordinated actions.

C. Members Appointed by VA:

1. Commissioning Agent: The designated person, company, or entity that plans, schedules, and coordinates the commissioning team to implement the commissioning process. The VA will engage the CxA under a separate contract.
2. User: Representatives of the facility user and operation and maintenance personnel.
3. A/E: Representative of the Architect and engineering design professionals.

1.9 VA'S COMMISSIONING RESPONSIBILITIES

- A. Appoint an individual, company or firm to act as the Commissioning Agent.
- B. Assign operation and maintenance personnel and schedule them to participate in commissioning team activities including, but not limited to, the following:
 1. Coordination meetings.
 2. Training in operation and maintenance of systems, subsystems, and equipment.
 3. Testing meetings.
 4. Witness and assist in Systems Functional Performance Testing.
 5. Demonstration of operation of systems, subsystems, and equipment.
- C. Provide the Construction Documents, prepared by Architect and approved by VA, to the Commissioning Agent and for use in managing the commissioning process, developing the commissioning plan, systems manuals, and reviewing the operation and maintenance training plan.

1.10 CONTRACTOR'S COMMISSIONING RESPONSIBILITIES

- A. The Contractor shall assign a Commissioning Manager to manage commissioning activities of the Contractor, and subcontractors.
- B. The Contractor shall ensure that the commissioning responsibilities outlined in these specifications are included in all subcontracts and

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that subcontractors comply with the requirements of these specifications.

- C. The Contractor shall ensure that each installing subcontractor shall assign representatives with expertise and authority to act on behalf of the subcontractor and schedule them to participate in and perform commissioning team activities including, but not limited to, the following:

1. Participate in commissioning coordination meetings.
2. Conduct operation and maintenance training sessions in accordance with approved training plans.
3. Verify that Work is complete and systems are operational according to the Contract Documents, including calibration of instrumentation and controls.
4. Evaluate commissioning issues and commissioning observations identified in the Commissioning Issues Log, field reports, test reports or other commissioning documents. In collaboration with entity responsible for system and equipment installation, recommend corrective action.
5. Review and comment on commissioning documentation.
6. Participate in meetings to coordinate Systems Functional Performance Testing.
7. Provide schedule for operation and maintenance data submittals, equipment startup, and testing to Commissioning Agent for incorporation into the commissioning plan.
8. Provide information to the Commissioning Agent for developing commissioning plan.
9. Participate in training sessions for VA's operation and maintenance personnel.
10. Provide technicians who are familiar with the construction and operation of installed systems and who shall develop specific test procedures to conduct Systems Functional Performance Testing of installed systems.

1.11 COMMISSIONING AGENT'S RESPONSIBILITIES

- A. Organize and lead the commissioning team.
- B. Prepare the commissioning plan. See Paragraph 1.11-A of this specification Section for further information.

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- C. Review and comment on selected submittals from the Contractor for general conformance with the Construction Documents. Review and comment on the ability to test and operate the system and/or equipment, including providing gages, controls and other components required to operate, maintain, and test the system. Review and comment on performance expectations of systems and equipment and interfaces between systems relating to the Construction Documents.
- D. At the beginning of the construction phase, conduct an initial construction phase coordination meeting for the purpose of reviewing the commissioning activities and establishing tentative schedules for operation and maintenance submittals; operation and maintenance training sessions; TAB Work; Pre-Functional Checklists, Systems Functional Performance Testing; and project completion.
- E. Convene commissioning team meetings for the purpose of coordination, communication, and conflict resolution; discuss status of the commissioning processes. Responsibilities include arranging for facilities, preparing agenda and attendance lists, and notifying participants. The Commissioning Agent shall prepare and distribute minutes to commissioning team members and attendees within five workdays of the commissioning meeting.
- F. Observe construction and report progress, observations and issues. Observe systems and equipment installation for adequate accessibility for maintenance and component replacement or repair, and for general conformance with the Construction Documents.
- G. Prepare Project specific Pre-Functional Checklists and Systems Functional Performance Test procedures.
- H. Coordinate Systems Functional Performance Testing schedule with the Contractor.
- I. Witness selected systems startups.
- J. Verify selected Pre-Functional Checklists completed and submitted by the Contractor.
- K. Witness and document Systems Functional Performance Testing.
- L. Compile test data, inspection reports, and certificates and include them in the systems manual and commissioning report.

~~Spec Writer's Note: Coordinate the specification reference in Para J below with the specific project spec section dealing with O&M Data requirements.~~

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- M. Review and comment on operation and maintenance (O&M) documentation and systems manual outline for compliance with the Contract Documents. Operation and maintenance documentation requirements are specified in Paragraph 1.25, Section 01 00 00 GENERAL REQUIREMENTS.
- N. Review operation and maintenance training program developed by the Contractor. Verify training plans provide qualified instructors to conduct operation and maintenance training.
- O. Prepare commissioning Field Observation Reports.
- P. Prepare the Final Commissioning Report.
- Q. Return to the site at 10 months into the 12 month warranty period and review with facility staff the current building operation and the condition of outstanding issues related to the original and seasonal Systems Functional Performance Testing. Also interview facility staff and identify problems or concerns they have operating the building as originally intended. Make suggestions for improvements and for recording these changes in the O&M manuals. Identify areas that may come under warranty or under the original construction contract. Assist facility staff in developing reports, documents and requests for services to remedy outstanding problems.
- R. Assemble the final commissioning documentation, including the Final Commissioning Report and Addendum to the Final Commissioning Report.

1.12 COMMISSIONING DOCUMENTATION

- A. Commissioning Plan: A document, prepared by Commissioning Agent, that outlines the schedule, allocation of resources, and documentation requirements of the commissioning process, and shall include, but is not limited, to the following:
 - 1. Plan for delivery and review of submittals, systems manuals, and other documents and reports. Identification of the relationship of these documents to other functions and a detailed description of submittals that are required to support the commissioning processes. Submittal dates shall include the latest date approved submittals must be received without adversely affecting commissioning plan.
 - 2. Description of the organization, layout, and content of commissioning documentation (including systems manual) and a

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- detailed description of documents to be provided along with identification of responsible parties.
3. Identification of systems and equipment to be commissioned.
 4. Schedule of Commissioning Coordination meetings.
 5. Identification of items that must be completed before the next operation can proceed.
 6. Description of responsibilities of commissioning team members.
 7. Description of observations to be made.
 8. Description of requirements for operation and maintenance training.
 9. Schedule for commissioning activities with dates coordinated with overall construction schedule.
 10. Process and schedule for documenting changes on a continuous basis to appear in Project Record Documents.
 11. Process and schedule for completing prestart and startup checklists for systems, subsystems, and equipment to be verified and tested.
 12. Preliminary Systems Functional Performance Test procedures.
- B. Systems Functional Performance Test Procedures: The Commissioning Agent will develop Systems Functional Performance Test Procedures for each system to be commissioned, including subsystems, or equipment and interfaces or interlocks with other systems. Systems Functional Performance Test Procedures will include a separate entry, with space for comments, for each item to be tested. Preliminary Systems Functional Performance Test Procedures will be provided to the VA, Architect/Engineer, and Contractor for review and comment. The Systems Performance Test Procedure will include test procedures for each mode of operation and provide space to indicate whether the mode under test responded as required. Each System Functional Performance Test procedure, regardless of system, subsystem, or equipment being tested, shall include, but not be limited to, the following:
1. Name and identification code of tested system.
 2. Test number.
 3. Time and date of test.
 4. Indication of whether the record is for a first test or retest following correction of a problem or issue.
 5. Dated signatures of the person performing test and of the witness, if applicable.
 6. Individuals present for test.

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7. Observations and Issues.
8. Issue number, if any, generated as the result of test.
- C. Pre-Functional Checklists: The Commissioning Agent will prepare Pre-Functional Checklists. Pre-Functional Checklists shall be completed and signed by the Contractor, verifying that systems, subsystems, equipment, and associated controls are ready for testing. The Commissioning Agent will spot check Pre-Functional Checklists to verify accuracy and readiness for testing. Inaccurate or incomplete Pre-Functional Checklists shall be returned to the Contractor for correction and resubmission.
- D. Test and Inspection Reports: The Commissioning Agent will record test data, observations, and measurements on Systems Functional Performance Test Procedure. The report will also include recommendation for system acceptance or non-acceptance. Photographs, forms, and other means appropriate for the application shall be included with data. Commissioning Agent Will compile test and inspection reports and test and inspection certificates and include them in systems manual and commissioning report.

~~Spec Writer Note: Modify Para E below to include a reference to other paragraph(s) that may be included to require that the Contractor is liable for any costs incurred by the VA for retesting. These costs may include additional fees to the Commissioning Agent and/or A/E.~~

- E. Corrective Action Documents: The Commissioning Agent will document corrective action taken for systems and equipment that fail tests. The documentation will include any required modifications to systems and equipment and/or revisions to test procedures, if any. The Commissioning Agent will witness and document any retesting of systems and/or equipment requiring corrective action and document retest results. Costs for corrective work and/or retesting shall be borne by the Contractor, and at no additional cost to the Owner.
- F. Commissioning Issues Log: The Commissioning Agent will prepare and maintain Commissioning Issues Log that describes Commissioning Issues and Commissioning Observations that are identified during the Commissioning process. These observations and issues include, but are not limited to, those that are at variance with the Contract Documents.

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The Commissioning Issues Log will identify and track issues as they are encountered, the party responsible for resolution, progress toward resolution, and document how the issue was resolved. The Master Commissioning Issues Log will also track the status of unresolved issues.

1. Creating a Commissioning Issues Log Entry:

- a. Identify the issue with unique numeric or alphanumeric identifier by which the issue may be tracked.
- b. Assign a descriptive title for the issue.
- c. Identify date and time of the issue.
- d. Identify test number of test being performed at the time of the observation, if applicable, for cross reference.
- e. Identify system, subsystem, and equipment to which the issue applies.
- f. Identify location of system, subsystem, and equipment.
- g. Include information that may be helpful in diagnosing or evaluating the issue.
- h. Note recommended corrective action.
- i. Identify commissioning team member responsible for corrective action.
- j. Identify expected date of correction.
- k. Identify person that identified the issue.

2. Documenting Issue Resolution:

- a. Log date correction is completed or the issue is resolved.
- b. Describe corrective action or resolution taken. Include description of diagnostic steps taken to determine root cause of the issue, if any.
- c. Identify changes to the Contract Documents that may require action.
- d. State that correction was completed and system, subsystem, and equipment are ready for retest, if applicable.
- e. Identify person(s) who corrected or resolved the issue.
- f. Identify person(s) verifying the issue resolution.

G. Final Commissioning Report: The Commissioning Agent will document results of the commissioning process, including unresolved issues, and performance of systems, subsystems, and equipment. The Commissioning Report will indicate whether systems, subsystems, and equipment have

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been properly installed and are performing according to the Contract Documents. This report will be used by the Department of Veterans Affairs when determining that systems will be accepted. This report will be used to evaluate systems, subsystems, and equipment and will serve as a future reference document during VA occupancy and operation. It shall describe components and performance that exceed requirements of the Contract Documents and those that do not meet requirements of the Contract Documents. The commissioning report will include, but is not limited to, the following:

1. Lists and explanations of substitutions; compromises; variances with the Contract Documents; record of conditions; and, if appropriate, recommendations for resolution. Design Narrative documentation maintained by the Commissioning Agent.
2. Commissioning plan.
3. Pre-Functional Checklists completed by the Contractor, with annotation of the Commissioning Agent review and spot check.
4. Systems Functional Performance Test Procedures, with annotation of test results and test completion.
5. Commissioning Issues Log.
6. Listing of deferred and off-season test(s) not performed, including the schedule for their completion.

H. Addendum to Final Commissioning Report: The Commissioning Agent will prepare an Addendum to the Final Commissioning Report near the end of the Warranty Period. The Addendum will indicate whether systems, subsystems, and equipment are complete and continue to perform according to the Contract Documents. The Addendum to the Final Commissioning Report shall include, but is not limited to, the following:

1. Documentation of deferred and off season test(s) results.
2. Completed Systems Functional Performance Test Procedures for off season test(s).
3. Documentation that unresolved system performance issues have been resolved.
4. Updated Commissioning Issues Log, including status of unresolved issues.
5. Identification of potential Warranty Claims to be corrected by the Contractor.

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- ~~I. Systems Manual: The Commissioning Agent will gather required information and compile the Systems Manual. The Systems Manual will include, but is not limited to, the following:~~
- ~~1. Design Narrative, including system narratives, schematics, single-line diagrams, flow diagrams, equipment schedules, and changes made throughout the Project.~~
 - ~~2. Reference to Final Commissioning Plan.~~
 - ~~3. Reference to Final Commissioning Report.~~
 - ~~4. Approved Operation and Maintenance Data as submitted by the Contractor.~~

1.13 SUBMITTALS

A. Preliminary Commissioning Plan Submittal: The Commissioning Agent has prepared a Preliminary Commissioning Plan based on the final Construction Documents. The Preliminary Commissioning Plan is included as an Appendix to this specification section. The Preliminary Commissioning Plan is provided for information only. It contains preliminary information about the following commissioning activities:

1. The Commissioning Team: A list of commissioning team members by organization.
2. Systems to be commissioned. A detailed list of systems to be commissioned for the project. This list also provides preliminary information on systems/equipment submittals to be reviewed by the Commissioning Agent; preliminary information on Pre-Functional Checklists that are to be completed; preliminary information on Systems Performance Testing, including information on testing sample size (where authorized by the VA).
3. Commissioning Team Roles and Responsibilities: Preliminary roles and responsibilities for each Commissioning Team member.
4. Commissioning Documents: A preliminary list of commissioning-related documents, include identification of the parties responsible for preparation, review, approval, and action on each document.
5. Commissioning Activities Schedule: Identification of Commissioning Activities, including Systems Functional Testing, the expected duration and predecessors for the activity.
6. Pre-Functional Checklists: Preliminary Pre-Functional Checklists for equipment, components, subsystems, and systems to be commissioned. These Preliminary Pre-Functional Checklists provide

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guidance on the level of detailed information the Contractor shall include on the final submission.

7. Systems Functional Performance Test Procedures: Preliminary step-by-step System Functional Performance Test Procedures to be used during Systems Functional Performance Testing. These Preliminary Systems Functional Performance procedures provide information on the level of testing rigor, and the level of Contractor support required during performance of system's testing.
- B. Final Commissioning Plan Submittal: Based on the Final Construction Documents and the Contractor's project team, the Commissioning Agent will prepare the Final Commissioning Plan as described in this section. The Commissioning Agent will submit three hard copies and three sets of electronic files of Final Commissioning Plan. The Contractor shall review the Commissioning Plan and provide any comments to the VA. The Commissioning Agent will incorporate review comments into the Final Commissioning Plan as directed by the VA.
- C. Systems Functional Performance Test Procedure: The Commissioning Agent will submit preliminary Systems Functional Performance Test Procedures to the Contractor, and the VA for review and comment. The Contractor shall return review comments to the VA and the Commissioning Agent. The VA will also return review comments to the Commissioning Agent. The Commissioning Agent will incorporate review comments into the Final Systems Functional Test Procedures to be used in Systems Functional Performance Testing.
- D. Pre-Functional Checklists: The Commissioning Agent will submit Pre-Functional Checklists to be completed by the Contractor.
- E. Test and Inspection Reports: The Commissioning Agent will submit test and inspection reports to the VA with copies to the Contractor and the Architect/Engineer.
- F. Corrective Action Documents: The Commissioning Agent will submit corrective action documents to the VA Resident Engineer with copies to the Contractor and Architect.
- G. Preliminary Commissioning Report Submittal: The Commissioning Agent will submit three electronic copies of the preliminary commissioning report. One electronic copy, with review comments, will be returned to the Commissioning Agent for preparation of the final submittal.

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H. Final Commissioning Report Submittal: The Commissioning Agent will submit four sets of electronically formatted information of the final commissioning report to the VA. The final submittal will incorporate comments as directed by the VA.

I. Data for Commissioning:

1. The Commissioning Agent will request in writing from the Contractor specific information needed about each piece of commissioned equipment or system to fulfill requirements of the Commissioning Plan.
2. The Commissioning Agent may request further documentation as is necessary for the commissioning process or to support other VA data collection requirements, including Construction Operations Building Information Exchange (COBIE), Building Information Modeling (BIM), etc.

1.14 COMMISSIONING PROCESS

A. The Commissioning Agent will be responsible for the overall management of the commissioning process as well as coordinating scheduling of commissioning tasks with the VA and the Contractor. As directed by the VA, the Contractor shall incorporate Commissioning tasks, including, but not limited to, Systems Functional Performance Testing (including predecessors) with the Master Construction Schedule.

Spec writer's note: Coordinate the number of days listed in the following paragraphs with the VA Resident Engineer.

B. Within //XX// days of contract award, the Contractor shall designate a specific individual as the Commissioning Manager (CxM) to manage and lead the commissioning effort on behalf of the Contractor. The Commissioning Manager shall be the single point of contact and communications for all commissioning related services by the Contractor.

C. Within //XX// days of contract award, the Contractor shall ensure that each subcontractor designates specific individuals as Commissioning Representatives (CXR) to be responsible for commissioning related tasks. The Contractor shall ensure the designated Commissioning Representatives participate in the commissioning process as team members providing commissioning testing services, equipment operation, adjustments, and corrections if necessary. The Contractor shall ensure

that all Commissioning Representatives shall have sufficient authority to direct their respective staff to provide the services required, and to speak on behalf of their organizations in all commissioning related contractual matters.

1.15 QUALITY ASSURANCE

- A. Instructor Qualifications: Factory authorized service representatives shall be experienced in training, operation, and maintenance procedures for installed systems, subsystems, and equipment.
- B. Test Equipment Calibration: The Contractor shall comply with test equipment manufacturer's calibration procedures and intervals. Recalibrate test instruments immediately whenever instruments have been repaired following damage or dropping. Affix calibration tags to test instruments. Instruments shall have been calibrated within six months prior to use.

1.16 COORDINATION

- A. Management: The Commissioning Agent will coordinate the commissioning activities with the VA and Contractor. The Commissioning Agent will submit commissioning documents and information to the VA. All commissioning team members shall work together to fulfill their contracted responsibilities and meet the objectives of the contract documents.
- B. Scheduling: The Contractor shall work with the Commissioning Agent and the VA to incorporate the commissioning activities into the construction schedule. The Commissioning Agent will provide sufficient information (including, but not limited to, tasks, durations and predecessors) on commissioning activities to allow the Contractor and the VA to schedule commissioning activities. All parties shall address scheduling issues and make necessary notifications in a timely manner in order to expedite the project and the commissioning process. The Contractor shall update the Master Construction as directed by the VA.
- C. Initial Schedule of Commissioning Events: The Commissioning Agent will provide the initial schedule of primary commissioning events in the Commissioning Plan and at the commissioning coordination meetings. The Commissioning Plan will provide a format for this schedule. As construction progresses, more detailed schedules will be developed by the Contractor with information from the Commissioning Agent.

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- D. Commissioning Coordinating Meetings: The Commissioning Agent will conduct periodic Commissioning Coordination Meetings of the commissioning team to review status of commissioning activities, to discuss scheduling conflicts, and to discuss upcoming commissioning process activities.
- E. Pretesting Meetings: The Commissioning Agent will conduct pretest meetings of the commissioning team to review startup reports, Pre-Functional Checklist results, Systems Functional Performance Testing procedures, testing personnel and instrumentation requirements.
- F. Systems Functional Performance Testing Coordination: The Contractor shall coordinate testing activities to accommodate required quality assurance and control services with a minimum of delay and to avoid necessity of removing and replacing construction to accommodate testing and inspecting. The Contractor shall coordinate the schedule times for tests, inspections, obtaining samples, and similar activities.

PART 2 - PRODUCTS

2.1 TEST EQUIPMENT

- A. The Contractor shall provide all standard and specialized testing equipment required to perform Systems Functional Performance Testing. Test equipment required for Systems Functional Performance Testing will be identified in the detailed System Functional Performance Test Procedure prepared by the Commissioning Agent.
- B. Data logging equipment and software required to test equipment shall be provided by the Contractor.
- C. All testing equipment shall be of sufficient quality and accuracy to test and/or measure system performance with the tolerances specified in the Specifications. If not otherwise noted, the following minimum requirements apply: Temperature sensors and digital thermometers shall have a certified calibration within the past year to an accuracy of 0.5 °C (1.0 °F) and a resolution of + or - 0.1 °C (0.2 °F). Pressure sensors shall have an accuracy of + or - 2.0% of the value range being measured (not full range of meter) and have been calibrated within the last year. All equipment shall be calibrated according to the manufacturer's recommended intervals and following any repairs to the equipment. Calibration tags shall be affixed or certificates readily available.

PART 3 - EXECUTION

3.1 COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES

A. The following table outlines the roles and responsibilities for the Commissioning Team members during the Construction Phase:

Spec Writer's Notes: Edit the following tables to describe the roles and responsibilities for each commissioning team member for each of the commissioning tasks as appropriate for the project.

Construction Phase		CxA = Commissioning Agent RE = Resident Engineer A/E = Design Arch/Engineer PC = Prime Contractor O&M = Gov't Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commissioning Roles & Responsibilities							
Category	Task Description	CxA	RE	A/E	PC	O&M	Notes
Meetings	Construction Commissioning Kick Off meeting	L	A	P	P	O	
	Commissioning Meetings	L	A	P	P	O	
	Project Progress Meetings	P	A	P	L	O	
	Controls Meeting	L	A	P	P	O	
Coordination	Coordinate with [OGC's, AHJ, Vendors, etc.] to ensure that Cx interacts properly with other systems as needed to support the OPR and BOD.	L	A	P	P	N/A	
Cx Plan & Spec	Final Commissioning Plan	L	A	R	R	O	

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Construction Phase		CxA = Commissioning Agent RE = Resident Engineer A/E = Design Arch/Engineer PC = Prime Contractor O&M = Gov't Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commissioning Roles & Responsibilities							
Category	Task Description	CxA	RE	A/E	PC	O&M	Notes
Schedules	Duration Schedule for Commissioning Activities	L	A	R	R	N/A	
OPR and BOD	Maintain OPR on behalf of Owner	L	A	R	R	O	
	Maintain BOD/DID on behalf of Owner	L	A	R	R	O	
Document Reviews	TAB Plan Review	L	A	R	R	O	
	Submittal and Shop Drawing Review	R	A	R	L	O	
	Review Contractor Equipment Startup Checklists	L	A	R	R	N/A	
	Review Change Orders, ASI, and RFI	L	A	R	R	N/A	
Site Observations	Witness Factory Testing	P	A	P	L	O	
	Construction Observation Site Visits	L	A	R	R	O	
Functional Test Protocols	Final Pre-Functional Checklists	L	A	R	R	O	
	Final Functional Performance Test Protocols	L	A	R	R	O	

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Construction Phase		CxA = Commissioning Agent RE = Resident Engineer A/E = Design Arch/Engineer PC = Prime Contractor O&M = Gov't Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commissioning Roles & Responsibilities							
Category	Task Description	CxA	RE	A/E	PC	O&M	Notes
Technical Activities	Issues Resolution Meetings	P	A	P	L	O	
Reports and Logs	Status Reports	L	A	R	R	O	
	Maintain Commissioning Issues Log	L	A	R	R	O	

B. The following table outlines the roles and responsibilities for the Commissioning Team members during the Acceptance Phase:

Acceptance Phase		CxA = Commissioning Agent RE = Resident Engineer A/E = Design Arch/Engineer PC = Prime Contractor O&M = Gov't Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commissioning Roles & Responsibilities							
Category	Task Description	CxA	RE	A/E	PC	O&M	Notes
Meetings	Commissioning Meetings	L	A	P	P	O	
	Project Progress Meetings	P	A	P	L	O	
	Pre-Test Coordination Meeting	L	A	P	P	O	
	Lessons Learned and Commissioning Report Review Meeting	L	A	P	P	O	

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Acceptance Phase		CxA = Commissioning Agent RE = Resident Engineer A/E = Design Arch/Engineer PC = Prime Contractor O&M = Gov't Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commissioning Roles & Responsibilities							
Category	Task Description	CxA	RE	A/E	PC	O&M	Notes
Coordination	Coordinate with [OGC's, AHJ, Vendors, etc.] to ensure that Cx interacts properly with other systems as needed to support OPR and BOD	L	P	P	P	O	
Cx Plan & Spec	Maintain/Update Commissioning Plan	L	A	R	R	O	
Schedules	Prepare Functional Test Schedule	L	A	R	R	O	
OPR and BOD	Maintain OPR on behalf of Owner	L	A	R	R	O	
	Maintain BOD/DID on behalf of Owner	L	A	R	R	O	
Document Reviews	Review Completed Pre-Functional Checklists	L	A	R	R	O	
	Pre-Functional Checklist Verification	L	A	R	R	O	
	Review Operations & Maintenance Manuals	L	A	R	R	R	
	Training Plan Review	L	A	R	R	R	
	Warranty Review	L	A	R	R	O	
	Review TAB Report	L	A	R	R	O	
Site Observations	Construction Observation Site Visits	L	A	R	R	O	
	Witness Selected Equipment Startup	L	A	R	R	O	

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Acceptance Phase		CxA = Commissioning Agent RE = Resident Engineer A/E = Design Arch/Engineer PC = Prime Contractor O&M = Gov't Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Category	Task Description	CxA	RE	A/E	PC	O&M	Notes
Functional Test Protocols	TAB Verification	L	A	R	R	O	
	Systems Functional Performance Testing	L	A	P	P	P	
	Retesting	L	A	P	P	P	
Technical Activities	Issues Resolution Meetings	P	A	P	L	O	
	Systems Training	L	S	R	P	P	
Reports and Logs	Status Reports	L	A	R	R	O	
	Maintain Commissioning Issues Log	L	A	R	R	O	
	Final Commissioning Report	L	A	R	R	R	
	Prepare Systems Manuals	L	A	R	R	R	

C. The following table outlines the roles and responsibilities for the Commissioning Team members during the Warranty Phase:

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Warranty Phase		CxA = Commissioning Agent					L = Lead
Commissioning Roles & Responsibilities		RE = Resident Engineer A/E = Design Arch/Engineer PC = Prime Contractor O&M = Gov't Facility O&M					P = Participate A = Approve R = Review O = Optional
Category	Task Description	CxA	RE	A/E	PC	O&M	Notes
Meetings	Post-Occupancy User Review Meeting	L	A	O	P	P	
Site Observations	Periodic Site Visits	L	A	O	O	P	
Functional Test Protocols	Deferred and/or seasonal Testing	L	A	O	P	P	
Technical Activities	Issues Resolution Meetings	L	S	O	O	P	
	Post-Occupancy Warranty Checkup and review of Significant Outstanding Issues	L	A		R	P	
Reports and Logs	Final Commissioning Report Amendment	L	A		R	R	
	Status Reports	L	A		R	R	

3.2 STARTUP, INITIAL CHECKOUT, AND PRE-FUNCTIONAL CHECKLISTS

A. The following procedures shall apply to all equipment and systems to be commissioned, according to Part 1, Systems to Be Commissioned.

1. Pre-Functional Checklists are important to ensure that the equipment and systems are hooked up and operational. These ensure that Systems Functional Performance Testing may proceed without unnecessary delays. Each system to be commissioned shall have a full Pre-Functional Checklist completed by the Contractor prior to Systems Functional Performance Testing. No sampling strategies are used.

a. The Pre-Functional Checklist will identify the trades responsible for completing the checklist. The Contractor shall ensure the appropriate trades complete the checklists.

b. The Commissioning Agent will review completed Pre-Functional Checklists and field-verify the accuracy of the completed checklist using sampling techniques.

~~2. Startup and Initial Checkout Plan: The Contractor shall develop detailed startup plans for all equipment. The primary role of the Contractor in this process is to ensure that there is written documentation that each of the manufacturer recommended procedures have been completed. Parties responsible for startup shall be identified in the Startup Plan and in the checklist forms.~~

~~a. The Contractor shall develop the full startup plan by combining (or adding to) the checklists with the manufacturer's detailed startup and checkout procedures from the O&M manual data and the field checkout sheets normally used by the Contractor. The plan shall include checklists and procedures with specific boxes or lines for recording and documenting the checking and inspections of each procedure and a summary statement with a signature block at the end of the plan.~~

~~b. The full startup plan shall at a minimum consist of the following items:~~

~~1) The Pre-Functional Checklists.~~

~~2) The manufacturer's standard written startup procedures copied from the installation manuals with check boxes by each procedure and a signature block added by hand at the end.~~

~~3) The manufacturer's normally used field checkout sheets.~~

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- ~~e. The Commissioning Agent will submit the full startup plan to the VA and Contractor for review. Final approval will be by the VA.~~
- ~~d. The Contractor shall review and evaluate the procedures and the format for documenting them, noting any procedures that need to be revised or added.~~
- ~~3. Sensor and Actuator Calibration~~
 - ~~a. All field installed temperature, relative humidity, CO2 and pressure sensors and gages, and all actuators (dampers and valves) on all equipment shall be calibrated using the methods described in Division 21, Division 22, Division 23, Division 26, Division 27, and Division 28 specifications.~~
 - ~~b. All procedures used shall be fully documented on the Pre-Functional Checklists or other suitable forms, clearly referencing the procedures followed and written documentation of initial, intermediate and final results.~~
- ~~4. Execution of Equipment Startup~~

~~Spec Writer Note: Coordinate the number of week's lead time with the Resident Engineer.~~

- ~~a. //Four// //insert number// weeks prior to equipment startup, the Contractor shall schedule startup and checkout with the VA and Commissioning Agent. The performance of the startup and checkout shall be directed and executed by the Contractor.~~
 - ~~b. The Commissioning Agent will observe the startup procedures for selected pieces of primary equipment.~~
 - ~~c. The Contractor shall execute startup and provide the VA and Commissioning Agent with a signed and dated copy of the completed startup checklists, and contractor tests.~~
 - ~~d. Only individuals that have direct knowledge and witnessed that a line item task on the Startup Checklist was actually performed shall initial or check that item off. It is not acceptable for witnessing supervisors to fill out these forms.~~
- 3.3 DEFICIENCIES, NONCONFORMANCE, AND APPROVAL IN CHECKLISTS AND STARTUP**
- A. The Contractor shall clearly list any outstanding items of the initial startup and Pre-Functional Checklist procedures that were not completed successfully, at the bottom of the procedures form or on an attached sheet. The procedures form and any outstanding deficiencies shall be

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provided to the VA and the Commissioning Agent within two days of completion.

- B. The Commissioning Agent will review the report and submit comments to the VA. The Commissioning Agent will work with the Contractor to correct and verify deficiencies or uncompleted items. The Commissioning Agent will involve the VA and others as necessary. The Contractor shall correct all areas that are noncompliant or incomplete in the checklists in a timely manner, and shall notify the VA and Commissioning Agent as soon as outstanding items have been corrected. The Contractor shall submit an updated startup report and a Statement of Correction on the original noncompliance report. When satisfactorily completed, the Commissioning Agent will recommend approval of the checklists and startup of each system to the VA.
- C. The Contractor shall be responsible for resolution of deficiencies as directed the VA.

Spec Writer Note: The following paragraph on Phased Commissioning is for projects with phased delivery of buildings or parts of buildings. It is intended to notify the Contractor that some Commissioning Activities, such as startup, functional testing, etc. will be conducted as the various phases of the project near completion and will require multiple mobilizations of the startup and functional testing teams to support the phased commissioning. Delete the following paragraph when phased delivery is not part of the project.

3.4 PHASED COMMISSIONING

- A. The project may require startup and initial checkout to be executed in phases. This phasing shall be planned and scheduled in a coordination meeting of the VA, Commissioning Agent, and the Contractor. Results will be added to the master construction schedule and the commissioning schedule.

~~3.5 DDC SYSTEM TRENDING FOR COMMISSIONING~~

- ~~A. Trending is a method of testing as a standalone method or to augment manual testing. The Contractor shall trend any and all points of the system or systems at intervals specified below.~~

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- ~~B. Alarms are a means to notify the system operator that abnormal conditions are present in the system. Alarms shall be structured into three tiers - Critical, Priority, and Maintenance.~~
- ~~1. Critical alarms are intended to be alarms that require the immediate attention of and action by the Operator. These alarms shall be displayed on the Operator Workstation in a popup style window that is graphically linked to the associated unit's graphical display. The popup style window shall be displayed on top of any active window within the screen, including non DDC system software.~~
 - ~~2. Priority level alarms are to be printed to a printer which is connected to the Operator's Work Station located within the engineer's office. Additionally Priority level alarms shall be able to be monitored and viewed through an active alarm application. Priority level alarms are alarms which shall require reaction from the operator or maintenance personnel within a normal work shift, and not immediate action.~~
 - ~~3. Maintenance alarms are intended to be minor issues which would require examination by maintenance personnel within the following shift. These alarms shall be generated in a scheduled report automatically by the DDC system at the start of each shift. The generated maintenance report will be printed to a printer located within the engineer's office.~~
- ~~C. The Contractor shall provide a wireless internet network in the building for use during controls programming, checkout, and commissioning. This network will allow project team members to more effectively program, view, manipulate and test control devices while being in the same room as the controlled device.~~
- ~~D. The Contractor shall provide graphical trending through the DDC control system of systems being commissioned. Trending requirements are indicated below and included with the Systems Functional Performance Test Procedures. Trending shall occur before, during and after Systems Functional Performance Testing. The Contractor shall be responsible for producing graphical representations of the trended DDC points that show each system operating properly during steady state conditions as well as during the System Functional Testing. These graphical reports shall be submitted to the Resident Engineer and Commissioning Agent for review and analysis before, during dynamic operation, and after Systems~~

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~~Functional Performance Testing. The Contractor shall provide, but not limited to, the following trend requirements and trend submissions:~~

- ~~1. Pre-testing, Testing, and Post-testing – Trend reports of trend logs and graphical trend plots are required as defined by the Commissioning Agent. The trend log points, sampling rate, graphical plot configuration, and duration will be dictated by the Commissioning Agent. At any time during the Commissioning Process the Commissioning Agent may recommend changes to aspects of trending as deemed necessary for proper system analysis. The Contractor shall implement any changes as directed by the Resident Engineer. Any pre-test trend analysis comments generated by the Commissioning Team should be addressed and resolved by the Contractor, as directed by the Resident Engineer, prior to the execution of Systems Functional Performance Testing.~~
- ~~2. Dynamic plotting – The Contractor shall also provide dynamic plotting during Systems Functional Performance testing at frequent intervals for points determined by the Systems Functional Performance Test Procedure. The graphical plots will be formatted and plotted at durations listed in the Systems Functional Performance Test Procedure.~~
- ~~3. Graphical plotting – The graphical plots shall be provided with a dual y-axis allowing 15 or more trend points (series) plotted simultaneously on the graph with each series in distinct color. The plots will further require title, axis naming, legend etc. all described by the Systems Functional Performance Test Procedure. If this cannot be sufficiently accomplished directly in the Direct Digital Control System then it is the responsibility of the Contractor to plot these trend logs in Microsoft Excel.~~
- ~~4. The following tables indicate the points to be trended and alarmed by system. The Operational Trend Duration column indicates the trend duration for normal operations. The Testing Trend Duration column indicates the trend duration prior to Systems Functional Performance Testing and again after Systems Functional Performance Testing. The Type column indicates point type: AI = Analog Input, AO = Analog Output, DI = Digital Input, DO = Digital Output, Calc = Calculated Point. In the Trend Interval Column, COV = Change of Value. The Alarm Type indicates the alarm priority; C = Critical, P~~

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~~= Priority, and M = Maintenance. The Alarm Range column indicates when the point is considered in the alarm state. The Alarm Delay column indicates the length of time the point must remain in an alarm state before the alarm is recorded in the DDC. The intent is to allow minor, short duration events to be corrected by the DDC system prior to recording an alarm.~~

~~Spec Writer Note: The following tables provide guidelines for system trends and alarms. Coordinate the types of systems and point names with the construction documents. Verify alarm priorities, ranges and delay. The Design Engineer may elect to include trending and alarm information on the DDC Control Schematics and Sequences of Operations in the Construction Drawing set or in the DDC Control Specifications. Verify the control drawings and/or DDC specifications have included reference to this section of 01 91 00. If adequately included in the drawings or specifications, the following tables should be deleted to prevent duplication and possible conflicts.~~

Dual-Path Air Handling Unit Trending and Alarms

Point	Type	Trend Interval	Operationa l Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
OA Temperature	AI	15 Min	24 hours	3 days	N/A		
RA Temperature	AI	15 Min	24 hours	3 days	N/A		
RA Humidity	AI	15 Min	24 hours	3 days	P	>60% RH	10 min
Mixed Air Temp	AI	None	None	None	N/A		
SA Temp	AI	15 Min	24 hours	3 days	C	+5°F from SP	10 min
Supply Fan Speed	AI	15 Min	24 hours	3 days	N/A		
Return Fan Speed	AI	15 Min	24 hours	3 days	N/A		
RA Pre-Filter Status	AI	None	None	None	N/A		
OA Pre-Filter Status	AI	None	None	None	N/A		
After Filter Status	AI	None	None	None	N/A		

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Dual Path Air Handling Unit Trending and Alarms							
Point	Type	Trend Interval	Operations 1-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
SA Flow	AI	15 Min	24 hours	3 days	E	±10% from SP	10 min
OA Supply Temp	AI	15 Min	24 hours	3 days	P	±5°F from SP	10 min
RA Supply Temp	AI	15 Min	24 hours	3 days	N/A		
RA CHW Valve Position	AI	15 Min	24 hours	3 days	N/A		
OA CHW Valve Position	AI	15 Min	24 hours	3 days	N/A		
OA HW Valve Position	AI	15 Min	24 hours	3 days	N/A		
OA Flow	AI	15 Min	24 hours	3 days	P	±10% from SP	5 min
RA Flow	AI	15 Min	24 hours	3 days	P	±10% from SP	5 min
Initial UVC Intensity (%)	AI	None	None	None	N/A		
Duct Pressure	AI	15 Min	24 hours	3 days	E	±25% from SP	6 min
CO2 Level	AI	15 Min	24 hours	3 days	P	±10% from SP	10 min
Supply Fan Status	DI	COV	24 hours	3 days	E	Status ↔ Command	10 min
Return Fan Status	DI	COV	24 hours	3 days	E	Status ↔ Command	10 Min
High Static Status	DI	COV	24 hours	3 days	P	True	1 min
Fire Alarm Status	DI	COV	24 hours	3 days	E	True	5 min
Freeze Stat Level 1	DI	COV	24 hours	3 days	E	True	10 min
Freeze Stat Level 2	DI	COV	24 hours	3 days	E	True	5 min
Freeze Stat Level 3	DI	COV	24 hours	3 days	P	True	1 min
Fire/Smoke Damper Status	DI	COV	24 hours	3 days	P	Closed	1 min
Emergency AHU Shutdown	DI	COV	24 hours	3 days	P	True	1 min
Exhaust Fan #1 Status	DI	COV	24 hours	3 days	E	Status ↔ Command	10 min

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Dual Path Air Handling Unit Trending and Alarms							
Point	Type	Trend Interval	Operations 1-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Exhaust Fan #2 Status	DI	COV	24 hours	3 days	C	Status ↔ Command	10 min
Exhaust Fan #3 Status	DI	COV	24 hours	3 days	C	Status ↔ Command	10 min
OA Alarm	DI	COV	24 hours	3 days	C	True	10 min
High Static Alarm	DI	COV	24 hours	3 days	C	True	10 min
UVC Emitter Alarm	DI	COV	24 hours	3 days	P	True	10 min
CO2 Alarm	DI	COV	24 hours	3 days	P	True	10 min
Power Failure	DI	COV	24 hours	3 days	P	True	1 min
Supply Fan Speed	AO	15 Min	24 hours	3 days	N/A		
Return Fan Speed	AO	15 Min	24 hours	3 days	N/A		
RA CHW Valve Position	AO	15 Min	24 hours	3 days	N/A		
OA CHW Valve Position	AO	15 Min	24 hours	3 days	N/A		
OA HW Valve Position	AO	15 Min	24 hours	3 days	N/A		
Supply Fan S/S	DO	COV	24 hours	3 days	N/A		
Return Fan S/S	DO	COV	24 hours	3 days	N/A		
Fire/Smoke Dampers	DO	COV	24 hours	3 days	N/A		
Exhaust Fan S/S	DO	COV	24 hours	3 days	N/A		
Exhaust Fan S/S	DO	COV	24 hours	3 days	N/A		
Exhaust Fan S/S	DO	COV	24 hours	3 days	N/A		
AHU Energy	Cale	1 Hour	30 day	N/A	N/A		

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Terminal Unit (VAV, CAV, etc.) Trending and Alarms							
Point	Type	Trend Interval	Operationa l Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Space Temperature	AI	15 Min	12 hours	3 days	P	±5°F from SP	10 min
Air Flow	AI	15 Min	12 hours	3 days	P	±5°F from SP	10 min
SA Temperature	AI	15 Min	12 hours	3 days	P	±5°F from SP	10 min
Local Setpoint	AI	15 Min	12 hours	3 days	M	±10°F from SP	60 min
Space Humidity	AI	15 Min	12 hours	3 days	P	> 60% RH	5 min
Unoccupied Override	DI	COV	12 hours	3 days	M	N/A	12 Hours
Refrigerator Alarm	DI	COV	12 hours	3 days	C	N/A	10 min
Damper Position	AO	15 Minutes	12 hours	3 days	N/A		
Heating coil Valve Position	AO	15 Minutes	12 hours	3 days	N/A		

4-Pipe Fan Coil Trending and Alarms							
Point	Type	Trend Interval	Operationa l Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Space Temperature	AI	15 Minutes	12 hours	3 days	P	±5°F from SP	10 min
SA Temperature	AI	15 Minutes	12 hours	3 days	P	±5°F from SP	10 min
Pre-Filter Status	AI	None	None	None	M	> SP	1 hour
Water Sensor	DI	COV	12 hours	3 days	M	N/A	30 Min
Cooling Coil Valve Position	AO	15 Minutes	12 hours	3 days	N/A		
Heating coil Valve Position	AO	15 Minutes	12 hours	3 days	N/A		

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4-Pipe Fan Coil Trending and Alarms

Point	Type	Trend Interval	Operationa l Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Fan Coil ON/OFF	DO	COV	12 hours	3 days	M	Status ↔ Command	30 min

2-Pipe Fan Coil Unit Trending and Alarms

Point	Type	Trend Interval	Operationa l Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Space Temperature	AI	15 Minutes	12 hours	3 days	P	±5°F from SP	10 min
SA Temperature	AI	15 Minutes	12 hours	3 days	P	±5°F from SP	10 min
Pre-Filter Status	AI	None	None	None	M	> SP	1 hour
Water Sensor	DI	COV	12 hours	3 days	M	N/A	30 Min
Cooling Coil Valve Position	AO	15 Minutes	12 hours	3 days	N/A		
Fan Coil ON/OFF	DO	COV	12 hours	3 days	M	Status ↔ Command	30 min

Unit Heater Trending and Alarms

Point	Type	Trend Interval	Operationa l Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Space Temperature	AI	15 Minutes	12 hours	3 days	P	±5°F from SP	10 min
Heating Valve Position	AO	15 Minutes	12 hours	3 days	N/A		

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Unit Heater Trending and Alarms							
Point	Type	Trend Interval	Operationa l-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Unit Heater ON/OFF	DO	COV	12 hours	3 days	M	Status ↔ Command	30 min

Steam and Condensate Pumps Trending and Alarms							
Point	Type	Trend Interval	Operationa l-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Steam Flow (LB/HR)	AI	15 Minutes	12 hours	3 days	N/A		
Condensate Pump Run Hours	AI	15 Minutes	12 hours	3 days	N/A		
Water Meter (GPM)	AI	15 Minutes	12 hours	3 days	N/A		
Electric Meter (KW/H)	AI	15 Minutes	12 hours	3 days	N/A		
Irrigation Meter (GPM)	AI	15 Minutes	12 hours	3 days	N/A		
Chilled Water Flow (TONS)	AI	15 Minutes	12 hours	3 days	N/A		
Condensate Flow (GPM)	AI	15 Minutes	12 hours	3 days	N/A		
High Water Level Alarm	DI	COV	12 hours	3 days	C	True	5 Min
Condensate Pump Start/Stop	DO	COV	12 hours	3 days	P	Status ↔ Command	10 min

Domestic Hot Water Trending and Alarms							
Point	Type	Trend Interval	Operationa l-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Domestic HW Setpoint WH-1	AI	15 Minute	12 Hours	3 days	N/A		
Domestic HW Setpoint WH-2	AI	15 Minute	12 Hours	3 days	N/A		

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Domestic Hot Water Trending and Alarms							
Point	Type	Trend Interval	Operationa l Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Domestic HW Temperature	AI	15 Minute	12 Hours	3 days	C	> 135 °F	10 Min
Domestic HW Temperature	AI	15 Minute	12 Hours	3 days	P	±5°F from SP	10 Min
Dom. Circ. Pump #1 Status	DI	COV	12 Hours	3 days	M	Status <=> Command	30 min
Dom. Circ. Pump #2 Status	DI	COV	12 Hours	3 days	M	Status <=> Command	30 min
Dom. Circ. Pump #1 Start/Stop	DO	COV	12 Hours	3 days	N/A		
Dom. Circ. Pump #2 Start/Stop	DO	COV	12 Hours	3 days	N/A		
Domestic HW Start/Stop	DO	COV	12 Hours	3 days	N/A		

Hydronic Hot Water Trending and Alarms							
Point	Type	Trend Interval	Operationa l Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
System HWS Temperature	AI	15 min	12 hours	3 days	C	±5°F from SP	10 Min
System HWR Temperature	AI	15 min	12 hours	3 days	M	±15°F from SP	300 Min
HX-1 Entering Temperature	AI	15 min	12 hours	3 days	P	±5°F from SP	10 Min
HX-2 Entering Temperature	AI	15 min	12 hours	3 days	P	±5°F from SP	10 Min
HX-2 Leaving Temperature	AI	15 min	12 hours	3 days	P	±5°F from SP	10 Min
System Flow (GPM)	AI	15 min	12 hours	3 days	N/A		
System Differential Pressure	AI	15 min	12 hours	3 days	P	±10% from SP	8 Min
				3 days			

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Hydronic Hot Water Trending and Alarms							
Point	Type	Trend Interval	Operations 1-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
HW Pump 1 Status	DI	COV	12 Hours	3 days	E	Status ↔ Command	30 min
HW Pump 2 Status	DI	COV	12 Hours	3 days	E	Status ↔ Command	30 min
HW Pump 1 VFD Speed	AO	15 Min	12 Hours	3 days	N/A		
HW Pump 2 VFD Speed	AO	15 Min	12 Hours	3 days	N/A		
Steam Station #1 1/3 Control Valve Position	AO	15 Min	12 Hours	3 days	N/A		
Steam Station #1 2/3 Control Valve Position	AO	15 Min	12 Hours	3 days	N/A		
Steam Station #2 1/3 Control Valve Position	AO	15 Min	12 Hours	3 days	N/A		
Steam Station #2 2/3 Control Valve Position	AO	15 Min	12 Hours	3 days	N/A		
Steam Station Bypass Valve Position	AO	15 Min	12 Hours	3 days	N/A		
HW Pump 1 Start/Stop	DO	COV	12 Hours	3 days	N/A		
HW Pump 2 Start/Stop	DO	COV	12 Hours	3 days	N/A		
HWR #1 Valve	DO	COV	12 Hours	3 days	N/A		
HWR #2 Valve	DO	COV	12 Hours	3 days	N/A		

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Chilled Water System Trending and Alarms							
Point	Type	Trend Interval	Operations 1-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Chiller 1 Entering Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 1 Leaving Temperature	AI	15 Minutes	12 Hours	3 days	P	±5°F from SP	10 Min
Chiller 1 Flow	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 1 Percent Load	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 1 KW Consumption	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 1 Tonnage	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 2 Entering Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 2 Leaving Temperature	AI	15 Minutes	12 Hours	3 days	P	±5°F from SP	10 Min
Chiller 2 Flow	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 2 Percent Load	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 2 KW Consumption	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 2 Tonnage	AI	15 Minutes	12 Hours	3 days	N/A		
Primary Loop Decoupler Flow	AI	15 Minutes	12 Hours	3 days	N/A		
Primary Loop Flow	AI	15 Minutes	12 Hours	3 days	N/A		
Primary Loop Supply Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Secondary Loop Differential Pressure	AI	15 Minutes	12 Hours	3 days	P	±5% from SP	10 Min
Secondary Loop Flow	AI	15 Minutes	12 Hours	3 days	N/A		
Secondary Loop Supply Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Secondary Loop Return Temperature	AI	15 Minutes	12 Hours	3 days	N/A		

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Point	Type	Trend Interval	Operations 1-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Secondary Loop Tonnage	AI	15 Minutes	12 Hours	3 days	N/A		
Primary Loop Pump 1 Status	DI	COV	12 Hours	3 days	E	Status ↔ Command	30 min
Primary Loop Pump 2 Status	DI	COV	12 Hours	3 days	E	Status ↔ Command	30 min
Secondary Loop Pump 1 Status	DI	COV	12 Hours	3 days	E	Status ↔ Command	30 min
Secondary Loop Pump 2 Status	DI	COV	12 Hours	3 days	E	Status ↔ Command	30 min
Chiller 1 Status	DI	COV	12 Hours	3 days	E	Status ↔ Command	30 min
Chiller 1 Evaporator Iso-Valve	DI	COV	12 Hours	3 days	N/A		
Chiller 1 Evaporator Flow Switch	DI	COV	12 Hours	3 days	N/A		
Chiller 1 Unit Alarm	DI	COV	12 Hours	3 days	E	True	10 Min
Chiller 2 Status	DI	COV	12 Hours	3 days	E	Status ↔ Command	30 min
Chiller 2 Evaporator Iso-Valve	DI	COV	12 Hours	3 days	N/A		
Chiller 2 Evaporator Flow Switch	DI	COV	12 Hours	3 days	N/A		
Chiller 2 Unit Alarm	DI	COV	12 Hours	3 days	E	True	10 Min
Refrigerant Detector	DI	COV	12 Hours	3 days	E	True	10 Min
Refrigerant Exhaust Fan Status	DI	COV	12 Hours	3 days	M	Status ↔ Command	30 min
Emergency Shutdown	DI	COV	12 Hours	3 days	P	True	1 Min

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Chilled Water System Trending and Alarms							
Point	Type	Trend Interval	Operations 1-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Primary Loop Pump 1 VFD Speed	AO	15 Minutes	12 Hours	3 days	N/A		
Primary Loop Pump 2 VFD Speed	AO	15 Minutes	12 Hours	3 days	N/A		
Secondary Loop Pump 1 VFD Speed	AO	15 Minutes	12 Hours	3 days	N/A		
Secondary Loop Pump 2 VFD Speed	AO	15 Minutes	12 Hours	3 days	N/A		
Primary Pump 1 Start / Stop	DO	COV	12 Hours	3 days	N/A		
Primary Pump 2 Start / Stop	DO	COV	12 Hours	3 days	N/A		
Secondary Pump 1 Start / Stop	DO	COV	12 Hours	3 days	N/A		
Secondary Pump 2 Start / Stop	DO	COV	12 Hours	3 days	N/A		
Chiller 1 Enable	DO	COV	12 Hours	3 days	N/A		
Chiller 1 Iso Valve Command	DO	COV	12 Hours	3 days	N/A		
Chiller 2 Enable	DO	COV	12 Hours	3 days	N/A		
Chiller 2 Iso Valve Command	DO	COV	12 Hours	3 days	N/A		
Refrigerant Exhaust Fan Start / Stop	DO	COV	12 Hours	3 days	N/A		

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Condenser Water System Trending and Alarms							
Point	Type	Trend Interval	Operations 1-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Chiller 1 Condenser Entering Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 1 Condenser Leaving Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 2 Condenser Entering Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller 2 Condenser Leaving Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower 1 Supply Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower 1 Return Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower 1 Basin Temp	AI	15 Minutes	12 Hours	3 days	P	< 45 oF	10 Min
Cooling Tower 2 Supply Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower 2 Return Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower 2 Basin Temp	AI	15 Minutes	12 Hours	3 days	P	< 45 oF	10 Min
Condenser Water Supply Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Condenser Water Return Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Outdoor Air Wet Bulb	AI	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower 1 Fan Status	DI	COV	12 Hours	3 days	P	Status <> Command	1 min
Cooling Tower 1 Basin Heat	DI	COV	12 Hours	3 days	N/A		
Cooling Tower 1 Heat Trace	DI	COV	12 Hours	3 days	N/A		
Cooling Tower 2 Fan Status	DI	COV	12 Hours	3 days	P	Status <> Command	1 min
Cooling Tower 2 Basin Heat	DI	COV	12 Hours	3 days	N/A		
Cooling Tower 2 Heat Trace	DI	COV	12 Hours	3 days	N/A		

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Condenser Water System Trending and Alarms							
Point	Type	Trend Interval	Operations 1-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Chiller 1 Isolation Valve	DI	COV	12 Hours	3 days	P	Status ↔ Command	1 min
Chiller 2 Isolation Valve	DI	COV	12 Hours	3 days	P	Status ↔ Command	1 min
Condenser Water Pump 1 Status	DI	COV	12 Hours	3 days	P	Status ↔ Command	1 min
Condenser Water Pump 2 Status	DI	COV	12 Hours	3 days	P	Status ↔ Command	1 min
Chiller 1 Condenser Bypass Valve	AO	15 Minutes	12 Hours	3 days	N/A		
Chiller 2 Condenser Bypass Valve	AO	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower 1 Bypass Valve	AO	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower 1 Fan Speed	AO	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower 2 Bypass Valve	AO	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower 2 Fan Speed	AO	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower 1 Fan Start / Stop	DO	COV	12 Hours	3 days	N/A		
Cooling Tower 2 Fan Start / Stop	DO	COV	12 Hours	3 days	N/A		
Condenser Water Pump 1 Start / Stop	DO	COV	12 Hours	3 days	N/A		
Condenser Water Pump 2 Start / Stop	DO	COV	12 Hours	3 days	N/A		

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Steam Boiler System Trending and Alarms							
Point	Type	Trend Interval	Operations 1 Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Boiler 1 Steam Pressure	AI	15 Minutes	12 Hours	3 days	P	±5% from SP	10 Min
Boiler 1 Steam Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Boiler 1 Fire Signal	AI	15 Minutes	12 Hours	3 days	N/A		
Boiler 2 Steam Pressure	AI	15 Minutes	12 Hours	3 days	P	±5% from SP	10 Min
Boiler 2 Steam Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Boiler 2 Fire Signal	AI	15 Minutes	12 Hours	3 days	N/A		
System Steam Pressure	AI	15 Minutes	12 Hours	3 days	P	±5% from SP	10 Min
Boiler 1 Enable	DI	COV	12 Hours	3 days	N/A		
Boiler 1 Status	DI	COV	12 Hours	3 days	P	Status ↔ Command	10 min
Boiler 1 Alarm	DI	COV	12 Hours	3 days	C	True	1 Min
Boiler 1 on Fuel Oil	DI	COV	12 Hours	3 days	N/A		
Boiler 1 Low Water Alarm	DI	COV	12 Hours	3 days	C	True	5 Min
Boiler 1 High Water Alarm	DI	COV	12 Hours	3 days	C	True	5 Min
Boiler 1 Feed Pump	DI	COV	12 Hours	3 days	N/A		
Boiler 2 Enable	DI	COV	12 Hours	3 days	N/A		
Boiler 2 Status	DI	COV	12 Hours	3 days	P	Status ↔ Command	10 min
Boiler 2 Alarm	DI	COV	12 Hours	3 days	C	True	1 Min
Boiler 2 on Fuel Oil	DI	COV	12 Hours	3 days	N/A		
Boiler 2 Low Water Alarm	DI	COV	12 Hours	3 days	C	True	5 Min
Boiler 2 High Water Alarm	DI	COV	12 Hours	3 days	C	True	5 Min

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Steam Boiler System Trending and Alarms							
Point	Type	Trend Interval	Operational Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Boiler 2 Feed Pump	DI	COV	12 Hours	3 days	N/A		
Combustion Damper Status	DI	COV	12 Hours	3 days	P	Status ↔ Command	5 min
Condensate Recovery Pump Status	DI	COV	12 Hours	3 days	P	Status ↔ Command	5 min
Boiler 1 Feed Pump Start / Stop	DO	COV	12 Hours	3 days	N/A		
Boiler 2 Start / Stop	DO	COV	12 Hours	3 days	N/A		
Combustion Damper Command	DO	COV	12 Hours	3 days	N/A		
Condensate Recovery Pump Start / Stop	DO	COV	12 Hours	3 days	N/A		

Hot Water Boiler System Trending and Alarms							
Point	Type	Trend Interval	Operational Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Outside Air Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Boiler 1 Fire Signal	AI	15 Minutes	12 Hours	3 days	N/A		
Boiler 1 Entering Water Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Boiler 1 Leaving Water Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Boiler 2 Fire Signal	AI	15 Minutes	12 Hours	3 days	N/A		

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Hot Water Boiler System Trending and Alarms							
Point	Type	Trend Interval	Operations 1 Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Boiler 2 Entering Water Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Boiler 2 Leaving Water Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Hot Water Supply Temperature	AI	15 Minutes	12 Hours	3 days	P	±5 °F from SP	10 Min
Hot Water Return Temperature	AI	15 Minutes	12 Hours	3 days	N/A		
Secondary Loop Differential Pressure	AI	15 Minutes	12 Hours	3 days	C	±5% from SP	10 Min
Lead Boiler	AI	15 Minutes	12 Hours	3 days	N/A		
Boiler 1 Enable	DI	COV	12 Hours	3 days	N/A		
Boiler 1 Status	DI	COV	12 Hours	3 days	P	Status ↔ Command	10 min
Boiler 1 Isolation Valve	DI	COV	12 Hours	3 days	N/A		
Boiler 1 on Fuel Oil	DI	COV	12 Hours	3 days	N/A		
Boiler 1 Alarm	DI	COV	12 Hours	3 days	C	True	1 Min
Boiler 2 Enable	DI	COV	12 Hours	3 days	N/A		
Boiler 2 Status	DI	COV	12 Hours	3 days	P	Status ↔ Command	10 min
Boiler 2 Isolation Valve	DI	COV	12 Hours	3 days	N/A		
Boiler 2 on Fuel Oil	DI	COV	12 Hours	3 days	N/A		
Boiler 2 Alarm	DI	COV	12 Hours	3 days	C	True	1 Min
Combustion Dampers Open	DI	COV	12 Hours	3 days	P	Status ↔ Command	10 min

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Hot Water Boiler System Trending and Alarms							
Point	Type	Trend Interval	Operations 1-Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Primary Pump 1-Status	DI	COV	12-Hours	3-days	P	Status ↔ Command	10 min
Primary Pump 2-Status	DI	COV	12-Hours	3-days	P	Status ↔ Command	10 min
Secondary Pump 1-Status	DI	COV	12-Hours	3-days	P	Status ↔ Command	10 min
Secondary Pump 2-Status	DI	COV	12-Hours	3-days	P	Status ↔ Command	10 min
Primary Pump 1-VFD-Speed	AO	COV	12-Hours	3-days	N/A		
Primary Pump 2-VFD-Speed	AO	COV	12-Hours	3-days	N/A		
Secondary Pump 1-VFD Speed	AO	COV	12-Hours	3-days	N/A		
Secondary Pump 2-VFD Speed	AO	COV	12-Hours	3-days	N/A		
Hot Water System-Enable	DO	COV	12-Hours	3-days	N/A		
Combustion Dampers Command	DO	COV	12-Hours	3-days	N/A		
Primary Pump 1-Start / Stop	DO	COV	12-Hours	3-days	N/A		
Primary Pump 2-Start / Stop	DO	COV	12-Hours	3-days	N/A		
Secondary Pump 1-Start / Stop	DO	COV	12-Hours	3-days	N/A		
Secondary Pump 2-Start / Stop	DO	COV	12-Hours	3-days	N/A		

E. The Contractor shall provide the following information prior to Systems Functional Performance Testing. Any documentation that is modified

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~~after submission shall be recorded and resubmitted to the Resident Engineer and Commissioning Agent.~~

~~1. Point to Point checkout documentation;~~

~~2. Sensor field calibration documentation including system name, sensor/point name, measured value, DDC value, and Correction Factor.~~

~~3. A sensor calibration table listing the referencing the location of procedures to following in the O&M manuals, and the frequency at which calibration should be performed for all sensors, separated by system, subsystem, and type. The calibration requirements shall be submitted both in the O&M manuals and separately in a standalone document containing all sensors for inclusion in the commissioning documentation. The following table is a sample that can be used as a template for submission.~~

SYSTEM		
Sensor	Calibration Frequency	O&M Calibration Procedure Reference
Discharge air temperature	Once a year	Volume I Section D.3.aa
Discharge static pressure	Every 6 months	Volume II Section A.1.e

~~4. Loop tuning documentation and constants for each loop of the building systems. The documentation shall be submitted in outline or table separated by system, control type (e.g. heating valve temperature control); proportional, integral and derivative constants, interval (and bias if used) for each loop. The following table is a sample that can be used as a template for submission.~~

AIR HANDLING UNIT AHU-1				
Control Reference	Proportional Constant	Integral Constant	Derivative Constant	Interval
Heating Valve Output	1000	20	10	2 sec.

3.6 SYSTEMS FUNCTIONAL PERFORMANCE TESTING

A. This paragraph applies to Systems Functional Performance Testing of systems for all referenced specification Divisions.

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B. Objectives and Scope: The objective of Systems Functional Performance Testing is to demonstrate that each system is operating according to the Contract Documents. Systems Functional Performance Testing facilitates bringing the systems from a state of substantial completion to full dynamic operation. Additionally, during the testing process, areas of noncompliant performance are identified and corrected, thereby improving the operation and functioning of the systems. ~~In general, each system shall be operated through all modes of operation (seasonal, occupied, unoccupied, warm-up, cool-down, part and full-load, fire alarm and emergency power) where there is a specified system response. The Contractor shall verify each sequence in the sequences of operation. Proper responses to such modes and conditions as power failure, freeze condition, low oil pressure, no flow, equipment failure, etc. shall also be tested.~~

C. Development of Systems Functional Performance Test Procedures: Before Systems Functional Performance Test procedures are written, the Contractor shall submit all requested documentation and a current list of change orders affecting equipment or systems, including an updated points list, program code, control sequences and parameters. Using the testing parameters and requirements found in the Contract Documents and approved submittals and shop drawings, the Commissioning Agent will develop specific Systems Functional Test Procedures to verify and document proper operation of each piece of equipment and system to be commissioned. The Contractor shall assist the Commissioning Agent in developing the Systems Functional Performance Test procedures as requested by the Commissioning Agent i.e. by answering questions about equipment, operation, sequences, etc. Prior to execution, the Commissioning Agent will provide a copy of the Systems Functional Performance Test procedures to the VA, the Architect/Engineer, and the Contractor, who shall review the tests for feasibility, safety, equipment and warranty protection.

D. Purpose of Test Procedures: The purpose of each specific Systems Functional Performance Test is to verify and document compliance with the stated criteria of acceptance given on the test form. Representative test formats and examples are found in the Commissioning Plan for this project. (The Commissioning Plan is issued as a separate document and is available for review.) The test procedure forms

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developed by the Commissioning Agent will include, but not be limited to, the following information:

1. System and equipment or component name(s)
2. Equipment location and ID number
3. Unique test ID number, and reference to unique Pre-Functional Checklists and startup documentation, and ID numbers for the piece of equipment
4. Date
5. Project name
6. Participating parties
7. A copy of the specification section describing the test requirements
8. A copy of the specific sequence of operations or other specified parameters being verified
9. Formulas used in any calculations
10. Required pretest field measurements
11. Instructions for setting up the test.
12. Special cautions, alarm limits, etc.
13. Specific step-by-step procedures to execute the test, in a clear, sequential and repeatable format
14. Acceptance criteria of proper performance with a Yes / No check box to allow for clearly marking whether or not proper performance of each part of the test was achieved.
15. A section for comments.
16. Signatures and date block for the Commissioning Agent. A place for the Contractor to initial to signify attendance at the test.

~~E. Test Methods: Systems Functional Performance Testing shall be achieved by manual testing (i.e. persons manipulate the equipment and observe performance) and/or by monitoring the performance and analyzing the results using the control system's trend log capabilities or by standalone data loggers. The Contractor and Commissioning Agent shall determine which method is most appropriate for tests that do not have a method specified.~~

- ~~1. Simulated Conditions: Simulating conditions (not by an overwritten value) shall be allowed, although timing the testing to experience actual conditions is encouraged wherever practical.~~
- ~~2. Overwritten Values: Overwriting sensor values to simulate a condition, such as overwriting the outside air temperature reading~~

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- ~~in a control system to be something other than it really is, shall be allowed, but shall be used with caution and avoided when possible. Such testing methods often can only test a part of a system, as the interactions and responses of other systems will be erroneous or not applicable. Simulating a condition is preferable. e.g., for the above case, by heating the outside air sensor with a hair blower rather than overwriting the value or by altering the appropriate setpoint to see the desired response. Before simulating conditions or overwriting values, sensors, transducers and devices shall have been calibrated.~~
- ~~3. Simulated Signals: Using a signal generator which creates a simulated signal to test and calibrate transducers and DDC constants is generally recommended over using the sensor to act as the signal generator via simulated conditions or overwritten values.~~
- ~~4. Altering Setpoints: Rather than overwriting sensor values, and when simulating conditions is difficult, altering setpoints to test a sequence is acceptable. For example, to see the Air Conditioning compressor lockout initiate at an outside air temperature below 12 C (54 F), when the outside air temperature is above 12 C (54 F), temporarily change the lockout setpoint to be 2 C (4 F) above the current outside air temperature.~~
- ~~5. Indirect Indicators: Relying on indirect indicators for responses or performance shall be allowed only after visually and directly verifying and documenting, over the range of the tested parameters, that the indirect readings through the control system represent actual conditions and responses. Much of this verification shall be completed during systems startup and initial checkout.~~
- ~~F. Setup: Each function and test shall be performed under conditions that simulate actual conditions as closely as is practically possible. The Contractor shall provide all necessary materials, system modifications, etc. to produce the necessary flows, pressures, temperatures, etc. necessary to execute the test according to the specified conditions. At completion of the test, the Contractor shall return all affected building equipment and systems, due to these temporary modifications, to their pretest condition.~~
- ~~G. Sampling: No sampling is allowed in completing Pre-Functional Checklists. Sampling is allowed for Systems Functional Performance~~

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~~Test Procedures execution. The Commissioning Agent will determine the sampling rate. If at any point, frequent failures are occurring and testing is becoming more troubleshooting than verification, the Commissioning Agent may stop the testing and require the Contractor to perform and document a checkout of the remaining units, prior to continuing with Systems Functional Performance Testing of the remaining units.~~

~~Spec Writer Note: Verify that the following paragraph regarding cost of expanded sample testing is allowed for the specific project. Retain or delete the paragraph as necessary.~~

- H. Cost of Retesting: The cost associated with expanded sample System Functional Performance Tests shall be solely the responsibility of the Contractor. Any required retesting by the Contractor shall not be considered a justified reason for a claim of delay or for a time extension by the Contractor.
- I. Coordination and Scheduling: The Contractor shall provide a minimum of 7 days' notice to the Commissioning Agent and the VA regarding the completion schedule for the Pre-Functional Checklists and startup of all equipment and systems. The Commissioning Agent will schedule Systems Functional Performance Tests with the Contractor and VA. The Commissioning Agent will witness and document the Systems Functional Performance Testing of systems. The Contractor shall execute the tests in accordance with the Systems Functional Performance Test Procedure.
- J. Testing Prerequisites: In general, Systems Functional Performance Testing will be conducted only after Pre-Functional Checklists have been satisfactorily completed. The control system shall be sufficiently tested and approved by the Commissioning Agent and the VA before it is used to verify performance of other components or systems. The air balancing and water balancing shall be completed before Systems Functional Performance Testing of air-related or water-related equipment or systems are scheduled. Systems Functional Performance Testing will proceed from components to subsystems to systems. When the proper performance of all interacting individual systems has been

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achieved, the interface or coordinated responses between systems will be checked.

- K. Problem Solving: The Commissioning Agent will recommend solutions to problems found, however the burden of responsibility to solve, correct and retest problems is with the Contractor.

3.7 DOCUMENTATION, NONCONFORMANCE AND APPROVAL OF TESTS

- A. Documentation: The Commissioning Agent will witness, and document the results of all Systems Functional Performance Tests using the specific procedural forms developed by the Commissioning Agent for that purpose. Prior to testing, the Commissioning Agent will provide these forms to the VA and the Contractor for review and approval. The Contractor shall include the filled out forms with the O&M manual data.
- B. Nonconformance: The Commissioning Agent will record the results of the Systems Functional Performance Tests on the procedure or test form. All items of nonconformance issues will be noted and reported to the VA on Commissioning Field Reports and/or the Commissioning Master Issues Log.
1. Corrections of minor items of noncompliance identified may be made during the tests. In such cases, the item of noncompliance and resolution shall be documented on the Systems Functional Test Procedure.
 2. Every effort shall be made to expedite the systems functional Performance Testing process and minimize unnecessary delays, while not compromising the integrity of the procedures. However, the Commissioning Agent shall not be pressured into overlooking noncompliant work or loosening acceptance criteria to satisfy scheduling or cost issues, unless there is an overriding reason to do so by direction from the VA.
 3. As the Systems Functional Performance Tests progresses and an item of noncompliance is identified, the Commissioning Agent shall discuss the issue with the Contractor and the VA.
 4. When there is no dispute on an item of noncompliance, and the Contractor accepts responsibility to correct it:
 - a. The Commissioning Agent will document the item of noncompliance and the Contractor's response and/or intentions. The Systems Functional Performance Test then continues or proceeds to another test or sequence. After the day's work is complete, the Commissioning Agent will submit a Commissioning Field Report to

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the VA. The Commissioning Agent will also note items of noncompliance and the Contractor's response in the Master Commissioning Issues Log. The Contractor shall correct the item of noncompliance and report completion to the VA and the Commissioning Agent.

- b. The need for retesting will be determined by the Commissioning Agent. If retesting is required, the Commissioning Agent and the Contractor shall reschedule the test and the test shall be repeated.
- 5. If there is a dispute about item of noncompliance, regarding whether it is an item of noncompliance, or who is responsible:
 - a. The item of noncompliance shall be documented on the test form with the Contractor's response. The item of noncompliance with the Contractor's response shall also be reported on a Commissioning Field Report and on the Master Commissioning Issues Log.
 - b. Resolutions shall be made at the lowest management level possible. Other parties are brought into the discussions as needed. Final interpretive and acceptance authority is with the Department of Veterans Affairs.
 - c. The Commissioning Agent will document the resolution process.
 - d. Once the interpretation and resolution have been decided, the Contractor shall correct the item of noncompliance, report it to the Commissioning Agent. The requirement for retesting will be determined by the Commissioning Agent. If retesting is required, the Commissioning Agent and the Contractor shall reschedule the test. Retesting shall be repeated until satisfactory performance is achieved.

~~Spec Writer Note: Verify that the following paragraph regarding cost of retesting is allowed for the specific project. Retain or delete the paragraph as necessary.~~

- C. Cost of Retesting: The cost to retest a System Functional Performance Test shall be solely the responsibility of the Contractor. Any required retesting by the Contractor shall not be considered a

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justified reason for a claim of delay or for a time extension by the Contractor.

~~Spec Writer's Note: Verify that the paragraph and subparagraphs below do not conflict with other general or specific contract documents regarding manufacturer's defects. Retain, delete, or modify the paragraphs accordingly.~~

- D. Failure Due to Manufacturer Defect: If 10%, or three, whichever is greater, of identical pieces (size alone does not constitute a difference) of equipment fail to perform in compliance with the Contract Documents (mechanically or substantively) due to manufacturing defect, not allowing it to meet its submitted performance specifications, all identical units may be considered unacceptable by the VA. In such case, the Contractor shall provide the VA with the following:
1. Within one week of notification from the VA, the Contractor shall examine all other identical units making a record of the findings. The findings shall be provided to the VA within two weeks of the original notice.
 2. Within two weeks of the original notification, the Contractor shall provide a signed and dated, written explanation of the problem, cause of failures, etc. and all proposed solutions which shall include full equipment submittals. The proposed solutions shall not significantly exceed the specification requirements of the original installation.
 3. The VA shall determine whether a replacement of all identical units or a repair is acceptable.
 4. Two examples of the proposed solution shall be installed by the Contractor and the VA shall be allowed to test the installations for up to one week, upon which the VA will decide whether to accept the solution.
 5. Upon acceptance, the Contractor shall replace or repair all identical items, at their expense and extend the warranty accordingly, if the original equipment warranty had begun. The replacement/repair work shall proceed with reasonable speed beginning within one week from when parts can be obtained.

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E. Approval: The Commissioning Agent will note each satisfactorily demonstrated function on the test form. Formal approval of the Systems Functional Performance Test shall be made later after review by the Commissioning Agent and by the VA. The Commissioning Agent will evaluate each test and report to the VA using a standard form. The VA will give final approval on each test using the same form, and provide signed copies to the Commissioning Agent and the Contractor.

3.8 DEFERRED TESTING

- A. Unforeseen Deferred Systems Functional Performance Tests: If any Systems Functional Performance Test cannot be completed due to the building structure, required occupancy condition or other conditions, execution of the Systems Functional Performance Testing may be delayed upon approval of the VA. These Systems Functional Performance Tests shall be conducted in the same manner as the seasonal tests as soon as possible. Services of the Contractor to conduct these unforeseen Deferred Systems Functional Performance Tests shall be negotiated between the VA and the Contractor.
- B. Deferred Seasonal Testing: Deferred Seasonal Systems Functional Performance Tests are those that must be deferred until weather conditions are closer to the systems design parameters. The Commissioning Agent will review systems parameters and recommend which Systems Functional Performance Tests should be deferred until weather conditions more closely match systems parameters. The Contractor shall review and comment on the proposed schedule for Deferred Seasonal Testing. The VA will review and approve the schedule for Deferred Seasonal Testing. Deferred Seasonal Systems Functional Performances Tests shall be witnessed and documented by the Commissioning Agent. Deferred Seasonal Systems Functional Performance Tests shall be executed by the Contractor in accordance with these specifications.

3.9 OPERATION AND MAINTENANCE TRAINING REQUIREMENTS

- A. Training Preparation Conference: Before operation and maintenance training, the Commissioning Agent will convene a training preparation conference to include VA's Resident Engineer, VA's Operations and Maintenance personnel, and the Contractor. The purpose of this conference will be to discuss and plan for Training and Demonstration of VA Operations and Maintenance personnel.

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- ~~B. The Contractor shall provide training and demonstration as required by other Division 21, Division 22, Division 23, Division 26, Division 27, Division 28, and Division 31 sections. The Training and Demonstration shall include, but is not limited to, the following:~~
- ~~1. Review the Contract Documents.~~
 - ~~2. Review installed systems, subsystems, and equipment.~~
 - ~~3. Review instructor qualifications.~~
 - ~~4. Review instructional methods and procedures.~~
 - ~~5. Review training module outlines and contents.~~
 - ~~6. Review course materials (including operation and maintenance manuals).~~
 - ~~7. Review and discuss locations and other facilities required for instruction.~~
 - ~~8. Review and finalize training schedule and verify availability of educational materials, instructors, audiovisual equipment, and facilities needed to avoid delays.~~
 - ~~9. For instruction that must occur outside, review weather and forecasted weather conditions and procedures to follow if conditions are unfavorable.~~
- ~~C. Training Module Submittals: The Contractor shall submit the following information to the VA and the Commissioning Agent:~~
- ~~1. Instruction Program: Submit two copies of outline of instructional program for demonstration and training, including a schedule of proposed dates, times, length of instruction time, and instructors' names for each training module. Include learning objective and outline for each training module. At completion of training, submit two complete training manuals for VA's use.~~
 - ~~2. Qualification Data: Submit qualifications for facilitator and/or instructor.~~
 - ~~3. Attendance Record: For each training module, submit list of participants and length of instruction time.~~
 - ~~4. Evaluations: For each participant and for each training module, submit results and documentation of performance-based test.~~
 - ~~5. Demonstration and Training Recording:~~
 - ~~a. General: Engage a qualified commercial photographer to record demonstration and training. Record each training module separately. Include classroom instructions and demonstrations,~~

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~~board diagrams, and other visual aids, but not student practice.
At beginning of each training module, record each chart
containing learning objective and lesson outline.~~

- ~~b. Video Format: Provide high quality color DVD color on standard size DVD disks.~~
 - ~~c. Recording: Mount camera on tripod before starting recording, unless otherwise necessary to show area of demonstration and training. Display continuous running time.~~
 - ~~d. Narration: Describe scenes on video recording by audio narration by microphone while demonstration and training is recorded. Include description of items being viewed. Describe vantage point, indicating location, direction (by compass point), and elevation or story of construction.~~
 - ~~e. Submit two copies within seven days of end of each training module.~~
- ~~6. Transcript: Prepared on 8 1/2 by 11 inch paper, punched and bound in heavy duty, 3 ring, vinyl covered binders. Mark appropriate identification on front and spine of each binder. Include a cover sheet with same label information as the corresponding videotape. Include name of Project and date of videotape on each page.~~

~~D. Quality Assurance:~~

- ~~1. Facilitator Qualifications: A firm or individual experienced in training or educating maintenance personnel in a training program similar in content and extent to that indicated for this Project, and whose work has resulted in training or education with a record of successful learning performance.~~
- ~~2. Instructor Qualifications: A factory authorized service representative, complying with requirements in Division 01 Section "Quality Requirements," experienced in operation and maintenance procedures and training.~~
- ~~3. Photographer Qualifications: A professional photographer who is experienced photographing construction projects.~~

~~E. Training Coordination:~~

- ~~1. Coordinate instruction schedule with VA's operations. Adjust schedule as required to minimize disrupting VA's operations.~~
- ~~2. Coordinate instructors, including providing notification of dates, times, length of instruction time, and course content.~~

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- ~~3. Coordinate content of training modules with content of approved emergency, operation, and maintenance manuals. Do not submit instruction program until operation and maintenance data has been reviewed and approved by the VA.~~

~~F. Instruction Program:~~

- ~~1. Program Structure: Develop an instruction program that includes individual training modules for each system and equipment not part of a system, as required by individual Specification Sections, and as follows:~~
- ~~a. Fire protection systems, including fire alarm, fire pumps, and fire suppression systems.~~
 - ~~b. Intrusion detection systems.~~
 - ~~c. Conveying systems, including elevators, wheelchair lifts, escalators, and automated materials handling systems.~~
 - ~~d. Medical equipment, including medical gas equipment and piping.~~
 - ~~e. Laboratory equipment, including laboratory air and vacuum equipment and piping.~~
 - ~~f. Heat generation, including boilers, feedwater equipment, pumps, steam distribution piping, condensate return systems, heating hot water heat exchangers, and heating hot water distribution piping.~~
 - ~~g. Refrigeration systems, including chillers, cooling towers, condensers, pumps, and distribution piping.~~
 - ~~h. HVAC systems, including air handling equipment, air distribution systems, and terminal equipment and devices.~~
 - ~~i. HVAC instrumentation and controls.~~
 - ~~j. Electrical service and distribution, including switchgear, transformers, switchboards, panelboards, uninterruptible power supplies, and motor controls.~~
 - ~~k. Packaged engine generators, including synchronizing switchgear/switchboards, and transfer switches.~~
 - ~~l. Lighting equipment and controls.~~
 - ~~m. Communication systems, including intercommunication, surveillance, nurse call systems, public address, mass evacuation, voice and data, and entertainment television equipment.~~
 - ~~n. Site utilities including lift stations, condensate pumping and return systems, and storm water pumping systems.~~

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~~G. Training Modules: Develop a learning objective and teaching outline for each module. Include a description of specific skills and knowledge that participants are expected to master. For each module, include instruction for the following:~~

~~1. Basis of System Design, Operational Requirements, and Criteria:~~

~~Include the following:~~

- ~~a. System, subsystem, and equipment descriptions.~~
- ~~b. Performance and design criteria if Contractor is delegated design responsibility.~~
- ~~c. Operating standards.~~
- ~~d. Regulatory requirements.~~
- ~~e. Equipment function.~~
- ~~f. Operating characteristics.~~
- ~~g. Limiting conditions.~~
- ~~h. Performance curves.~~

~~2. Documentation: Review the following items in detail:~~

- ~~a. Emergency manuals.~~
- ~~b. Operations manuals.~~
- ~~c. Maintenance manuals.~~
- ~~d. Project Record Documents.~~
- ~~e. Identification systems.~~
- ~~f. Warranties and bonds.~~
- ~~g. Maintenance service agreements and similar continuing commitments.~~

~~3. Emergencies: Include the following, as applicable:~~

- ~~a. Instructions on meaning of warnings, trouble indications, and error messages.~~
- ~~b. Instructions on stopping.~~
- ~~c. Shutdown instructions for each type of emergency.~~
- ~~d. Operating instructions for conditions outside of normal operating limits.~~
- ~~e. Sequences for electric or electronic systems.~~
- ~~f. Special operating instructions and procedures.~~

~~4. Operations: Include the following, as applicable:~~

- ~~a. Startup procedures.~~
- ~~b. Equipment or system break-in procedures.~~
- ~~c. Routine and normal operating instructions.~~

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- ~~d. Regulation and control procedures.~~
- ~~e. Control sequences.~~
- ~~f. Safety procedures.~~
- ~~g. Instructions on stopping.~~
- ~~h. Normal shutdown instructions.~~
- ~~i. Operating procedures for emergencies.~~
- ~~j. Operating procedures for system, subsystem, or equipment failure.~~
- ~~k. Seasonal and weekend operating instructions.~~
- ~~l. Required sequences for electric or electronic systems.~~
- ~~m. Special operating instructions and procedures.~~
- ~~5. Adjustments: Include the following:~~
 - ~~a. Alignments.~~
 - ~~b. Checking adjustments.~~
 - ~~c. Noise and vibration adjustments.~~
 - ~~d. Economy and efficiency adjustments.~~
- ~~6. Troubleshooting: Include the following:~~
 - ~~a. Diagnostic instructions.~~
 - ~~b. Test and inspection procedures.~~
- ~~7. Maintenance: Include the following:~~
 - ~~a. Inspection procedures.~~
 - ~~b. Types of cleaning agents to be used and methods of cleaning.~~
 - ~~c. List of cleaning agents and methods of cleaning detrimental to product.~~
 - ~~d. Procedures for routine cleaning~~
 - ~~e. Procedures for preventive maintenance.~~
 - ~~f. Procedures for routine maintenance.~~
 - ~~g. Instruction on use of special tools.~~
- ~~8. Repairs: Include the following:~~
 - ~~a. Diagnosis instructions.~~
 - ~~b. Repair instructions.~~
 - ~~c. Disassembly; component removal, repair, and replacement; and reassembly instructions.~~
 - ~~d. Instructions for identifying parts and components.~~
 - ~~e. Review of spare parts needed for operation and maintenance.~~
- ~~H. Training Execution:~~
 - ~~1. Preparation: Assemble educational materials necessary for instruction, including documentation and training module. Assemble~~

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- ~~training modules into a combined training manual. Set up instructional equipment at instruction location.~~
- ~~2. Instruction:~~
- ~~a. Facilitator: Engage a qualified facilitator to prepare instruction program and training modules, to coordinate instructors, and to coordinate between Contractor and Department of Veterans Affairs for number of participants, instruction times, and location.~~
 - ~~b. Instructor: Engage qualified instructors to instruct VA's personnel to adjust, operate, and maintain systems, subsystems, and equipment not part of a system.~~
 - ~~1) The Commissioning Agent will furnish an instructor to describe basis of system design, operational requirements, criteria, and regulatory requirements.~~
 - ~~2) The VA will furnish an instructor to describe VA's operational philosophy.~~
 - ~~3) The VA will furnish the Contractor with names and positions of participants.~~
- ~~3. Scheduling: Provide instruction at mutually agreed times. For equipment that requires seasonal operation, provide similar instruction at start of each season. Schedule training with the VA and the Commissioning Agent with at least seven days' advance notice.~~
- ~~4. Evaluation: At conclusion of each training module, assess and document each participant's mastery of module by use of an oral, or a written, performance-based test.~~
- ~~5. Cleanup: Collect used and leftover educational materials and remove from Project site. Remove instructional equipment. Restore systems and equipment to condition existing before initial training use.~~
- ~~I. Demonstration and Training Recording:~~
- ~~1. General: Engage a qualified commercial photographer to record demonstration and training. Record each training module separately. Include classroom instructions and demonstrations, board diagrams, and other visual aids, but not student practice. At beginning of each training module, record each chart containing learning objective and lesson outline.~~

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- ~~2. Video Format: Provide high quality color DVD color on standard size DVD disks.~~
- ~~3. Recording: Mount camera on tripod before starting recording, unless otherwise necessary to show area of demonstration and training. Display continuous running time.~~
- ~~4. Narration: Describe scenes on videotape by audio narration by microphone while demonstration and training is recorded. Include description of items being viewed. Describe vantage point, indicating location, direction (by compass point), and elevation or story of construction.~~

----- END -----

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SECTION 02 41 00
DEMOLITION

~~SPEC WRITER NOTES:~~

- ~~1. Delete between // ---- // if not applicable to project. Also delete any other item or paragraph not applicable in the section and renumber the paragraphs.~~
- ~~2. Use this Section for projects involving total or large scale demolition. Omit this Section on projects involving minor demolition.~~
- ~~3. Buildings, structures, utilities, etc., required to be removed must be clearly shown.~~
- ~~4. Debris or trash dumps should be shown to the fullest extent. If quantities of materials to be removed cannot be accurately estimated, do not include estimates of quantities. If site clearing is included in project, removal of debris from onsite trash dumps should be included in that specification section, then removal of materials from onsite trash dumps should be included in this specification section.~~
- ~~5. Modify the following paragraphs to reflect specific conditions for the project.~~

PART 1 - GENERAL

1.1 DESCRIPTION:

This section specifies demolition and removal of buildings, portions of buildings, utilities, other structures and debris from trash dumps shown.

1.2 RELATED WORK:

~~A. Demolition and removal of roads, walks, curbs, and on-grade slabs outside buildings to be demolished: // Section 31 20 00, EARTH MOVING // Section 31 20 11, EARTH MOVING (SHORT FORM) //.~~

~~B.A.~~ Safety Requirements: Section 01 35 26 Safety Requirements Article, ACCIDENT PREVENTION PLAN (APP).

~~C.B.~~ Disconnecting utility services prior to demolition: Section 01 00 00, GENERAL REQUIREMENTS.

~~D.C.~~ Reserved items that are to remain the property of the Government: Section 01 00 00, GENERAL REQUIREMENTS.

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~~E.D.~~ Asbestos Removal: Section 02 82 ~~13.2111~~, ~~TRADITIONAL ASBESTOS ABATEMENT~~ ASBESTOS ROOFING ABATEMENT.

~~F.E.~~ Lead Paint: Section 02 83 33.13, ~~LEAD-BASED PAINT REMOVAL AND DISPOSAL~~.

~~G.~~ Environmental Protection: Section 01 57 19, ~~TEMPORARY ENVIRONMENTAL CONTROLS~~.

~~H.F.~~ Construction Waste Management: Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT.

~~I.G.~~ Infectious Control: Section 01 35 26, SAFETY REQUIREMENTS.

1.3 PROTECTION:

- A. Perform demolition in such manner as to eliminate hazards to persons and property; to minimize interference with use of adjacent areas, utilities and structures or interruption of use of such utilities; and to provide free passage to and from such adjacent areas of structures. Comply with requirements of GENERAL CONDITIONS Article, ACCIDENT PREVENTION.
- B. Provide safeguards, including warning signs, barricades, temporary fences, warning lights, and other similar items that are required for protection of all personnel during demolition and removal operations. Comply with requirements of Section 01 00 00, GENERAL REQUIREMENTS, Article PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES AND IMPROVEMENTS.
- C. Maintain fences, barricades, lights, and other similar items around exposed excavations until such excavations have been completely filled.
- D. Provide enclosed dust chutes with control gates from each floor to carry debris to truck beds and govern flow of material into truck. Provide overhead bridges of tight board or prefabricated metal construction at dust chutes to protect persons and property from falling debris.
- E. Prevent spread of flying particles and dust. Sprinkle rubbish and debris with water to keep dust to a minimum. Do not use water if it results in hazardous or objectionable condition such as, but not limited to; ice, flooding, or pollution. Vacuum and dust the work area daily.

~~SPEC WRITER NOTE: Unless the building is to be demolished story by story paragraph F2 should not be used.~~

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- F. In addition to previously listed fire and safety rules to be observed in performance of work, include following:
1. No wall or part of wall shall be permitted to fall outwardly from structures.
 2. Maintain at least one stairway in each structure in usable condition to highest remaining floor. Keep stairway free of obstructions and debris until that level of structure has been removed.
 3. Wherever a cutting torch or other equipment that might cause a fire is used, provide and maintain fire extinguishers nearby ready for immediate use. Instruct all possible users in use of fire extinguishers.
 4. Keep hydrants clear and accessible at all times. Prohibit debris from accumulating within a radius of 4500 mm (15 feet) of fire hydrants.
- G. Before beginning any demolition work, the Contractor shall survey the site and examine the drawings and specifications to determine the extent of the work. The contractor shall take necessary precautions to avoid damages to existing items to remain in place, to be reused, or to remain the property of the ~~// Medical Center //~~ ~~Cemetery Property //~~; any damaged items shall be repaired or replaced as approved by the Contracting Officer's Representative (COR). The Contractor shall coordinate the work of this section with all other work and shall construct and maintain shoring, bracing, and supports as required. The Contractor shall ensure that structural elements are not overloaded and shall be responsible for increasing structural supports or adding new supports as may be required as a result of any cutting, removal, or demolition work performed under this contract. Do not overload structural elements. Provide new supports and reinforcement for existing construction weakened by demolition or removal works. Repairs, reinforcement, or structural replacement must have the COR's approval.
- H. The work shall comply with the requirements of Section 01 57 19, TEMPORARY ENVIRONMENTAL CONTROLS.
- I. The work shall comply with the requirements of Section 01 00 00, GENERAL REQUIREMENTS and Section 01 35 26, SAFETY REQUIREMENTS.

1.4 UTILITY SERVICES:

- A. Demolish and remove outside utility service lines shown to be removed.

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- B. Remove abandoned outside utility lines that would interfere with installation of new utility lines and new construction.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 DEMOLITION:

- A. Completely demolish and remove portions of buildings and structures, including all appurtenances related or connected thereto, as indicated in the Construction Documents., ~~as noted below:~~
- ~~1. As required for installation of new utility service lines.~~
 - ~~2. To full depth within an area defined by hypothetical lines located 1500 mm (5 feet) outside building lines of new structures..~~
- B. Debris, including brick, concrete, stone, metals and similar materials shall become property of Contractor and shall be disposed of by him daily, off the ~~//Medical Center //Cemetery Property //~~ to avoid accumulation at the demolition site. Materials that cannot be removed daily shall be stored in areas specified by the COR. ~~Break-up concrete slabs below grade that do not require removal from present location into pieces not exceeding 600 mm (24 inches) square to permit drainage.~~ Contractor shall dispose debris in compliance with applicable federal, state or local permits, rules and/or regulations.
- ~~C. In removing buildings and structures of more than two stories, demolish work story by story starting at highest level and progressing down to third floor level. Demolition of first and second stories may proceed simultaneously.~~
- D. Remove and legally dispose of all materials, other than earth to remain as part of project work, from any trash dumps shown. Materials removed shall ~~//~~become property of contractor and shall be disposed of in compliance with applicable federal, state or local permits, rules and/or regulations, and ~~//~~be hauled to VA specified disposal site~~//~~. All materials in the indicated trash dump areas, including above surrounding grade and extending to a depth of 1500mm (5feet) below surrounding grade, shall be included as part of the lump sum compensation for the work of this section. Materials that are located beneath the surface of the surrounding ground more than 1500 mm (5 feet), or materials that are discovered to be hazardous, shall be handled as unforeseen. The removal of hazardous material shall be referred to Hazardous Materials specifications.

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E. Remove existing utilities as indicated or uncovered by work and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the COR. When Utility lines are encountered that are not indicated on the drawings, the COR shall be notified prior to further work in that area.

3.2 CLEAN-UP:

On completion of work of this section and after removal of all debris, leave site in clean condition satisfactory to the COR. Clean-up shall include off the ~~//Medical Center //Cemetery Property //~~ disposal of all items and materials not required to remain property of the Government as well as all debris and rubbish resulting from demolition operations.

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**SECTION 02 82 13.21
ASBESTOS ROOFING ABATEMENT**

SPEC WRITER NOTES:

1. These specifications provide general guidance to personnel given the task of designing and executing a OSHA Class II and EPA Category I Non-Friable roofing abatement project. Each abatement is a unique situation and therefore must be tailored for that project. This specification incorporates current regulatory requirements and current best abatement practices, procedures and technology. The Architect/Engineer and/or the Industrial Hygiene consultants may provide additional specification additions or deletions to this specification that, in their professional judgment, will ensure a safe and effective approach to a specific abatement project while maintaining compliance with applicable regulations and VA policy. Any changes must be clearly marked on/attached to this document prior to finalization of the specification so that the changes will be adequately considered in the review process by the VA.
2. These specifications are to be used in conjunction with Contractor selection criteria; special instructions package; and general construction provisions.
3. Paragraphs that are not preceded by a number code are indented as instructions to the specifications writer and identified by the notation "Spec Writer Notes". These paragraphs must be deleted from the final document.
4. Within the text of the specifications, there may be optional procedures which the specification writer could include in the final specification. Procedures which are not chosen must be deleted by the specification writer. Optional text is shown by the notation (//text//).
5. The specification writer, VPIH/CPIH, CPIH/CIH, and A/E must be aware of and read the AEQA 10-95 since it details common errors in specification and

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contract documents for asbestos project. This would be especially helpful if a survey is being conducted prior to an abatement project.

6. A full AHERA survey of the facility would be needed prior to renovation activities, however, if demolition of the facility is planned, a NESHAP survey of the facility would need to be performed.

PART 1 - GENERAL

1.1 SUMMARY OF THE WORK

A. Drawings, general provisions of the contract, including general and supplementary conditions and other Division 01 specifications, shall apply to the work of this section. The contract documents show the work to be done under the contract and related requirements and conditions impacting the project. Related requirements and conditions include applicable codes and regulations, notices and permits, existing site conditions and restrictions on use of the site, requirements for partial owner occupancy during the work, coordination with other work and the phasing of the work. In the event the Asbestos Abatement Contractor discovers a conflict in the contract documents and/or requirements or codes, the conflict must be brought to the immediate attention of the Contracting Officer for resolution. Whenever there is a conflict or overlap in the requirements, the most stringent shall apply. Any actions taken by the Contractor without obtaining guidance from the Contracting Officer shall become the sole risk and responsibility of the Asbestos Abatement Contractor. All costs incurred due to such action are also the responsibility of the Asbestos Abatement Contractor.

B. Extent of Work:

1. Below is a brief description of the estimated quantities of asbestos roofing materials to be abated. These quantities are for informational purposes only and are based on the best information available at the time of the specification preparation. The Contractor shall satisfy himself as to the actual quantities to be abated. Nothing in this section may be interpreted as limiting the extent of work otherwise required by this contract and related documents.

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2. Removal, clean-up and disposal of ACM roofing in an appropriate regulated area in the following approximate quantities;
3. () linear meters (feet) of <50 mm (2 inches) diameter pipe insulation
() square feet of roofing
() square feet of roofing material containing multiple built-up roofs
() square feet of roofing flashing

SPEC WRITER NOTE: Provide the approximate quantities for removal, if roofing system conceals old roof systems identify here. Identify where flashing, sealant, etc., are located. Make sure that provisions are made if asbestos fireproofing is located on roof deck since it may be significantly disturbed during roofing material removal and replacement. Roofing abatement projects may cause roofing debris to fall on top of plaster or suspended ceiling system below, due primarily to holes placed in roof deck. This should be addressed in this specification, if it may occur.

C. Related Work:

1. Section 07 84 00, FIRESTOPPING.
2. Section 02 41 00, DEMOLITION.
3. Division 09, FINISHES

D. Tasks:

1. The work tasks are summarized briefly as follows:
 - a. Pre-abatement activities including pre-abatement meeting(s), inspection(s), notifications, permits, submittal approvals, work-site preparations, emergency procedures arrangements, and standard operating procedures for OSHA Class II and EPA Category I asbestos abatement work.
 - b. Abatement activities including removal, clean-up and disposal of ACM waste, recordkeeping, security, monitoring, and inspections.
 - c. Cleaning and decontamination activities including final visual inspection, air monitoring and certification of decontamination.

E. Abatement Contractors Use of Premises:

1. The Contractor and Contractor's personnel shall cooperate fully with the VA Representative/consultant to facilitate efficient use of

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buildings and areas within buildings. The Contractor shall perform the work in accordance with the VA specifications, drawings, phasing plan and in compliance with any/all applicable Federal, State and Local regulations and requirements.

2. The Contractor shall use the existing facilities in the building strictly within the limits indicated in contract documents as well as the approved VA Design and Construction Procedure. VA Design and Construction Procedure drawings of partially occupied buildings will show the limits of regulated areas; the placement of decontamination facilities; the temporary location of bagged waste ACM; the path of transport to outside the building; and the temporary waste storage area for each building/regulated area. Any variation from the arrangements shown on drawings shall be secured in writing from the VA Representative through the pre-abatement plan of action. The following limitations of use shall apply to existing facilities shown on drawings:

SPEC WRITER NOTE: Provide specific limitations on the use of facility elements such as corridors, stairs, elevators, loading platforms, etc., which are not dedicated for the use of the abatement contractor during his work. Consult with the VA on this and secure verification in writing on the conditions of use.

1.2 VARIATIONS IN QUANTITY

- A. The quantities and locations of ACM as indicated on the drawings and the extent of work included in this section are estimates which are limited by the physical constraints imposed by occupancy of the buildings. Accordingly, minor variations (+/- 10 percent) in quantities of ACM within the regulated area are considered as having no impact on contract price and time requirements of this contract. Where additional work is required beyond the above variation, the Contractor shall provide unit prices for additional work that is newly discovered materials and those prices will be used for additional work under the contract.

SPEC WRITER NOTE: The contract time and price will be adjusted under the provisions of "Differing Site Conditions"; (FAR 52.236-2). The Contractor

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shall have submitted unit prices before
letting the contract.

1.3 STOP ASBESTOS REMOVAL

A. If the Contracting Officer; their field representative; the facility Safety Officer/Manager or their designee, or the VA Professional Industrial Hygienist/Certified Industrial Hygienist (VPIH/CIH) presents a verbal **Stop Asbestos Removal Order**, the Contractor/Personnel shall immediately stop all asbestos removal and adequately wet any exposed ACM. If a verbal Stop Asbestos Removal Order is issued, the VA shall follow-up with a written order to the Contractor as soon as practicable. The Contractor shall not resume any asbestos removal activity until authorized to do so in writing by the VA Contracting Officer. A stop asbestos removal order may be issued at any time the VA Contracting Officer determines abatement conditions/activities are not within VA specification, regulatory requirements or that an imminent hazard exists to human health or the environment. Work stoppage will continue until conditions have been corrected to the satisfaction of the VA. Standby time and costs for corrective actions will be borne by the Contractor, including the VPIH/CIH time. The occurrence of any of the following events shall be reported immediately by the Contractor's competent person to the VA Contracting Office or field representative using the most expeditious means (e.g., verbal or telephonic), followed up with written notification to the Contracting Officer as soon as it is practical. The Contractor shall immediately stop asbestos removal/disturbance activities and initiate fiber reduction activities if:

1. Airborne PCM analysis results equal to or greater than 0.01 f/cc above background levels outside the regulated area;
2. breach or break in regulated area containment barrier(s);
3. serious injury/death at the site;
4. fire/safety emergency at the site;
5. respiratory protection system failure;
6. power failure or loss of wetting agent; or
7. any visible emissions observed outside the regulated area.
8. failure to follow project specification requirements.

1.4 DEFINITIONS

A. General: Definitions and explanations here are neither complete nor exclusive of all terms used in the contract documents, but are general for the work to the extent they are not stated more explicitly in another element of the contract documents. Drawings must be recognized as diagrammatic in nature and not completely descriptive of the requirements indicated therein.

B. Glossary:

Abatement - Procedures to control fiber release from asbestos-containing materials. Includes removal, encapsulation, enclosure, demolition, and renovation activities related to asbestos containing materials (ACM).

Aerosol - Solid or liquid particulate suspended in air.

Adequately wet - Sufficiently mixed or penetrated with liquid to prevent the release of particulates. If visible emissions are observed coming from the ACM, then that material has not been adequately wetted.

Aggressive method - Removal or disturbance of building material by sanding, abrading, grinding, or other method that breaks, crumbles, or disintegrates intact ACM.

Aggressive air sampling - EPA AHERA defined clearance sampling method using air moving equipment such as fans and leaf blowers to aggressively disturb and maintain in the air residual fibers after abatement.

AHERA - Asbestos Hazard Emergency Response Act. Asbestos regulations for schools issued in 1987.

Aircell - Pipe or duct insulation made of corrugated cardboard which contains asbestos.

Air monitoring - The process of measuring the fiber content of a known volume of air collected over a specified period of time. The NIOSH 7400 Method, Issue 3, Fifth Edition is used to determine the fiber levels in air. For personal samples, area air samples and clearance air testing using Phase Contrast Microscopy (PCM) analysis, the NIOSH Method 7402 (Issue 2, Fourth Edition) can be used when it is necessary to confirm fibers counted by PCM as being asbestos. The AHERA TEM analysis may be used for background, area samples and clearance samples when required by this specification, or at the discretion of the VPIH/CIH as appropriate.

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Air sample filter - The filter used to collect fibers which are then counted. The filter is made of mixed cellulose ester (MCE) membrane for PCM (Phase Contrast Microscopy, 25 mm, 3-piece with 2 inches Static Extension Cowl, 0.8 micron pore size) and MCE for TEM (Transmission Electron Microscopy, 25 mm, 3-piece with 2 inches Static Extension Cowl, 0.45 micron pore size).

Amended water - Water to which a surfactant (wetting agent) has been added to increase the penetrating ability of the liquid.

Asbestos - Includes chrysotile, amosite, crocidolite, tremolite asbestos, anthophyllite asbestos, actinolite asbestos, and any of these minerals that have been chemically treated or altered. Asbestos also includes PACM, as defined below.

Asbestos Hazard Abatement Plan (AHAP) - Asbestos work procedures required to be submitted by the contractor before work begins.

Asbestos-containing material (ACM) - Any material containing more than one percent of asbestos.

Asbestos contaminated elements (ACE) - Building elements such as ceilings, walls, lights, or ductwork that are contaminated with asbestos.

Asbestos-contaminated soil (ACS) - Soil found in the work area or in adjacent areas such as crawlspaces or pipe tunnels which is contaminated with asbestos-containing material debris and cannot be easily separated from the material.

Asbestos-containing waste (ACW) material - Asbestos-containing material or asbestos contaminated objects requiring disposal.

Asbestos Project Monitor - Some states require that any person conducting asbestos abatement air sampling, clearance inspections and clearance air sampling be licensed as an asbestos project monitor.

Asbestos waste decontamination facility - A system consisting of drum/bag washing facilities and a temporary storage area for cleaned containers of asbestos waste. Used as the exit for waste and equipment leaving the regulated area. In an emergency, it may be used to evacuate personnel.

Authorized person - Any person authorized by the VA, the Contractor, or government agency and required by work duties to be present in regulated areas.

Authorized visitor - Any person approved by the VA; the contractor; or any government agency representative having jurisdiction over the regulated area (e.g., OSHA, Federal and State EPA).

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Barrier - Any surface that isolates the regulated area and inhibits fiber migration from the regulated area.

Containment Barrier - An airtight barrier consisting of walls, floors, and/or ceilings of sealed plastic sheeting which surrounds and seals the outer perimeter of the regulated area.

Critical Barrier - The barrier responsible for isolating the regulated area from adjacent spaces, typically constructed of 2-layers of 6-mil independently installed plastic sheeting (Polyethylene) secured in place at openings such as doors, windows, penetrations or any other opening into the regulated area.

Primary Barrier - Plastic barriers placed over critical barriers and exposed directly to abatement work or to secondary barrier.

Secondary Barrier - Any additional plastic barriers used to isolate and provide protection from debris during abatement work.

Breathing zone - The hemisphere forward of the shoulders with a radius of about 150 - 225 mm (6 - 9 inches) from the worker's nose.

Bridging encapsulant - An encapsulant that forms a layer on the surface of the ACM.

Building/facility owner - The legal entity, including a lessee, which exercises control over management and recordkeeping functions relating to a building and/or facility in which asbestos activities take place.

Bulk testing - The collection and analysis of suspect asbestos containing materials.

Certified Industrial Hygienist (CIH) - A person certified in the comprehensive practice of industrial hygiene by the American Board of Industrial Hygiene.

Class I asbestos work - Activities involving the removal of Thermal System Insulation (TSI) and surfacing ACM and Presumed Asbestos Containing Material (PACM).

Class II asbestos work - Activities involving the removal of ACM which is not thermal system insulation or surfacing material. This includes, but is not limited to, the removal of asbestos-containing wallboard, floor tile and sheeting, roofing and siding shingles, and construction mastic.

Clean room/Changing room - An uncontaminated room having facilities for the storage of employee's street clothing and uncontaminated materials and equipment.

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Clearance sample - The final air sample taken after all asbestos work has been done and visually inspected. Performed by the VA's Professional Industrial Hygiene Consultant/Certified Industrial Hygienist (VPIH/CIH).

Closely resemble - The major workplace conditions which have contributed to the levels of historic asbestos exposure, are no more protective than conditions of the current workplace.

Competent person - In addition to the definition in 29 CFR 1926.32(f), one who is capable of identifying existing asbestos hazards in the workplace and selecting the appropriate control strategy for asbestos exposure, who has the authority to take prompt corrective measures to eliminate them, as specified in 29 CFR 1926.32(f); in addition, for Class I and II work who is specially trained in a training course which meets the criteria of EPA's Model Accreditation Plan (40 CFR 763) for supervisor.

Contractor's Professional Industrial Hygienist (CPIH/CIH) - The asbestos abatement contractor's industrial hygienist. The industrial hygienist must meet the qualification requirements of a PIH and may report to a certified industrial hygienist (CIH).

Count - Refers to the fiber count or the average number of fibers greater than five microns in length with a length-to-width (aspect) ratio of at least 3 to 1, per cubic centimeter of air.

Crawl space - An area which can be found either in or adjacent to the work area. This area has limited access and egress and may contain asbestos materials and/or asbestos contaminated soil.

Decontamination area/unit - An enclosed area adjacent to and connected to the regulated area and consisting of an equipment room, shower room, and clean room, which is used for the decontamination of workers, materials, and equipment that are contaminated with asbestos.

Demolition - The wrecking or taking out of any load-supporting structural member and any related razing, removing, or stripping of asbestos products.

Disposal bag - Typically 6-mil thick sift-proof, dustproof, leak-tight container used to package and transport asbestos waste from regulated areas to the approved landfill. Each bag/container must be labeled/marked in accordance with EPA, OSHA and DOT requirements.

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Disturbance - Asbestos Operations and Maintenance Activities (OSHA Class III) that disrupt the matrix of ACM or PACM, crumble or pulverize ACM or PACM, or generate visible debris from ACM or PACM. Disturbance includes cutting away small amounts of ACM or PACM, no greater than the amount that can be contained in one standard sized glove bag or waste bag in order to access a building component. In no event shall the amount of ACM or PACM so disturbed exceed that which can be contained in one glove bag or disposal bag, which shall not exceed 60 inches in length or width.

Drum - A rigid, impermeable container made of cardboard fiber, plastic, or metal which can be sealed in order to be sift-proof, dustproof, and leak-tight.

Employee exposure - The exposure to airborne asbestos that would occur if the employee were not wearing respiratory protection equipment.

Encapsulant - A material that surrounds or embeds asbestos fibers in an adhesive matrix and prevents the release of fibers.

Encapsulation - Treating ACM with an encapsulant.

Enclosure - The construction of an air tight, impermeable, permanent barrier around ACM to control the release of asbestos fibers from the material and also eliminate access to the material.

Equipment room - A contaminated room located within the decontamination area that is supplied with impermeable bags or containers for the disposal of contaminated protective clothing and equipment.

Fiber - A particulate form of asbestos, 5 microns or longer, with a length to width (aspect) ratio of at least 3 to 1.

Fibers per cubic centimeter (f/cc) - Abbreviation for fibers per cubic centimeter, used to describe the level of asbestos fibers in air.

Filter - Media used in respirators, vacuums, or other machines to remove particulate from air.

Firestopping - Material used to close the open parts of a structure in order to prevent a fire from spreading.

Friable asbestos containing material - Any material containing more than one (1) percent asbestos as determined using the method specified in 40 CFR 763, Polarized Light Microscopy, that, when dry, can be crumbled, pulverized, or reduced to powder by hand pressure.

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Glovebag - Not more than a 60 x 60 inch impervious plastic bag-like enclosure affixed around an asbestos-containing material, with glove-like appendages through which materials and tools may be handled.

High efficiency particulate air (HEPA) filter - An ASHRAE MERV 17 filter capable of trapping and retaining at least 99.97 percent of all mono-dispersed particles of 0.3 micrometers in diameter.

HEPA vacuum - Vacuum collection equipment equipped with a HEPA filter system capable of collecting and retaining asbestos fibers.

Homogeneous area - An area of surfacing, thermal system insulation or miscellaneous ACM that is uniform in color, texture and date of application.

HVAC - Heating, Ventilation and Air Conditioning

Industrial hygienist (IH) - A professional qualified by education, training, and experience to anticipate, recognize, evaluate and develop controls for occupational health hazards. Meets definition requirements of the American Industrial Hygiene Association (AIHA).

Industrial hygienist technician (IH Technician) - A person working under the direction of an IH or CIH who has special training, experience, certifications and licenses required for the industrial hygiene work assigned. Some states require that an industrial hygienist technician conducting asbestos abatement air sampling, clearance inspection and clearance air sampling be licensed as an asbestos project monitor.

Intact - The ACM has not crumbled, been pulverized, or otherwise deteriorated so that the asbestos is no longer likely to be bound with its matrix.

Lockdown - Applying encapsulant, after a final visual inspection, on all abated surfaces at the conclusion of ACM removal prior to removal of critical barriers.

National Emission Standards for Hazardous Air Pollutants (NESHAP) - EPA's rule to control emissions of asbestos to the environment (40 CFR part 61, Subpart M).

Negative initial exposure assessment - A demonstration by the employer which complies with the criteria in 29 CFR 1926.1101 (f) (2) (iii), that employee exposure during an operation is expected to be consistently below the PEL or Excursion Limit (EL).

Negative pressure - Air pressure which is lower than the surrounding area, created by exhausting air from a sealed regulated area through

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HEPA equipped filtration units. OSHA requires maintaining -0.02 inch water column gauge inside the negative pressure enclosure.

Negative pressure respirator - A respirator in which the air pressure inside the facepiece is negative during inhalation relative to the air pressure outside the respirator facepiece.

Non-friable ACM - Material that contains more than 1 percent asbestos but cannot be crumbled, pulverized, or reduced to powder by hand pressure.

Organic vapor cartridge - The type of cartridge used on air purifying respirators to remove organic vapor hazardous air contaminants.

Outside air - The air outside buildings and structures, including, but not limited to, the air under a bridge or in an open ferry dock.

Owner/operator - Any person who owns, leases, operates, controls, or supervises the facility being demolished or renovated or any person who owns, leases, operates, controls, or supervises the demolition or renovation operation, or both.

Penetrating encapsulant - Encapsulant that is absorbed into the ACM matrix without leaving a surface layer.

Permissible exposure limit (PEL) - The level of exposure OSHA allows for as an eight (8) hour time-weighted average (TWA). For asbestos fibers, the eight (8) hour time-weighted average PEL is 0.1 fibers per cubic centimeter (0.1 f/cc) of air and the 30-minute Excursion Limit (EL) is 1.0 fibers per cubic centimeter (1 f/cc).

Personal protective equipment (PPE) - equipment designed to protect user from injury and/or specific job hazard. Such equipment may include protective clothing, hard hats, safety glasses, fall protection, and respirators.

Personal sampling/monitoring - Representative air samples obtained in the breathing zone for one or more workers within the regulated area using a filter cassette and a calibrated air sampling pump to determine asbestos exposure.

Pipe tunnel - An area, typically located adjacent to mechanical spaces or boiler rooms in which the pipes servicing the heating system in the building are routed to allow the pipes to access heating elements. These areas may contain asbestos pipe insulation, asbestos fittings, debris or asbestos-contaminated soil.

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Polarized light microscopy (PLM) - Light microscopy using dispersion staining techniques and refractive indices to identify and quantify the type of asbestos present in a bulk sample.

Polyethylene sheeting - Strong plastic barrier material 4 to 6-mils thick, semi-transparent, flame retardant per NFPA 241.

Positive/negative fit check - A method of verifying the seal of a facepiece respirator by temporarily occluding the filters and breathing in (inhaling) and then temporarily occluding the exhalation valve and breathing out (exhaling) while checking for inward or outward leakage of the respirator, respectively.

Presumed ACM (PACM) - Thermal system insulation, surfacing, and flooring material installed in buildings prior to 1981. If the building owner has actual knowledge, or should have known through the exercise of due diligence that other materials are ACM, they too must be treated as PACM. The designation of PACM may be rebutted pursuant to 29 CFR 1926.1101 (k) (5).

Professional IH - An IH who meets the definition requirements of AIHA; meets the definition requirements of OSHA as a "Competent Person" at 29 CFR 1926.1101 (b); has completed two specialized EPA approved courses on management and supervision of asbestos abatement projects; has formal training in respiratory protection and waste disposal; and has a minimum of four projects of similar complexity with this project of which at least three projects serving as the supervisory IH. The PIH may be either the VA's PIH (VPIH/CIH) or Contractor's PIH (CPIH/CIH).

Project designer - A person who has successfully completed the training requirements for an asbestos abatement project designer as required by 40 CFR 763 Subpart E, Appendix C, Part I; (B) (5).

Assigned protection factor - A value assigned by OSHA/NIOSH to indicate the expected protection provided by each respirator class, when the respirator is properly selected and worn correctly. The number indicates the reduction of exposure level from outside to inside the respirator facepiece.

Qualitative fit test (QLFT) - A fit test using a challenge material that can be sensed by the wearer if leakage in the respirator occurs.

Quantitative fit test (QNFT) - A fit test using a challenge material which is quantified outside and inside the respirator thus allowing the determination of the actual fit factor.

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Regulated area - An area established by the employer to demarcate where Class I, II, III asbestos work is conducted, and any adjoining area where debris and waste from such asbestos work may accumulate; and a work area within which airborne concentrations of asbestos exceed, or there is a reasonable possibility they may exceed the PEL.

Regulated ACM (RACM) - Friable ACM; Category I non-friable ACM that has become friable; Category I non-friable ACM that will be or has been subjected to sanding, grinding, cutting, or abrading or; Category II non-friable ACM that has a high probability of becoming or has become crumbled, pulverized, or reduced to powder by the forces expected to act on the material in the course of the demolition or renovation operation.

Removal - All operations where ACM, PACM and/or RACM is taken out or stripped from structures or substrates, including demolition operations.

Renovation - Altering a facility or one or more facility components in any way, including the stripping or removal of asbestos from a facility component which does not involve demolition activity.

Repair - Overhauling, rebuilding, reconstructing, or reconditioning of structures or substrates, including encapsulation or other repair of ACM or PACM attached to structures or substrates.

Shower room - The portion of the PDF where personnel shower before leaving the regulated area.

Supplied air respirator (SAR) - A respiratory protection system that supplies minimum Grade D respirable air per ANSI/Compressed Gas Association Commodity Specification for Air, G-7.1-2018.

Surfacing ACM - A material containing more than 1 percent asbestos that is sprayed, troweled on or otherwise applied to surfaces for acoustical, decorative, fireproofing and other purposes.

Surfactant - A chemical added to water to decrease water's surface tension thus making it more penetrating into ACM.

Thermal system ACM - A material containing more than 1 percent asbestos applied to pipes, fittings, boilers, breeching, tanks, ducts, or other structural components to prevent heat loss or gain.

Transmission electron microscopy (TEM) - A microscopy method that can identify and count asbestos fibers.

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VA Professional Industrial Hygienist (VPIH/CIH) - The Department of Veterans Affairs Professional Industrial Hygienist must meet the qualifications of a PIH, and may report to a Certified Industrial Hygienist (CIH).

VA Representative - The VA official responsible for on-going project work.

VA Total - means a building or substantial part of the building is completely removed, torn or knocked down, bulldozed, flattened, or razed, including removal of building debris.

Visible emissions - Any emissions, which are visually detectable without the aid of instruments, coming from ACM/PACM/RACM/ACS or ACM waste material.

Waste/Equipment decontamination facility (W/EDF) - The area in which equipment is decontaminated before removal from the regulated area.

Waste generator - Any owner or operator whose act or process produces asbestos-containing waste material.

Waste shipment record - The shipping document, required to be originated and signed by the waste generator, used to track and substantiate the disposition of asbestos-containing waste material.

Wet cleaning - The process of thoroughly eliminating, by wet methods, any asbestos contamination from surfaces or objects.

C. Referenced Standards Organizations: See Section 01 42 19 REFERENCED STANDARDS.

1.5 APPLICABLE CODES AND REGULATIONS

A. General Applicability of Codes, Regulations, and Standards:

1. All work under this contract shall be done in strict accordance with all applicable Federal, State, and Local regulations, standards and codes governing asbestos abatement, and any other trade work done in conjunction with the abatement. All applicable codes, regulations and standards are adopted into this specification and will have the same force and effect as this specification.
2. The most recent edition of any relevant regulation, standard, document or code shall be in effect. Where conflict among the requirements or with these specifications exists, the most stringent requirement(s) shall be utilized.
3. Copies of all standards, regulations, codes and other applicable documents, including this specification and those listed in Section

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1.5 shall be available at the worksite in the clean change area of the worker decontamination system and/or the Contractor's on-site Field Office. These standards, regulations, codes and other applicable documents, including this specification and those listed in Section 1.5 may be made available electronically.

B. Contractor Responsibility: The Asbestos Abatement Contractor (Contractor) shall assume full responsibility and liability for compliance with all applicable Federal, State and Local regulations related to any and all aspects of the asbestos abatement project. The Contractor is responsible for providing and maintaining training, accreditations, medical exams, medical records, personal protective equipment (PPE), including respiratory protection, and respirator fit testing, as required by applicable Federal, State and Local regulations. The Contractor shall hold the VA and VPIH/CIH consultants harmless for any Contractor's failure to comply with any applicable work, packaging, transporting, disposal, safety, health, or environmental requirement on the part of himself, his employees, or his subcontractors. The Contractor will incur all costs of the CPIH/CIH, including all sampling/analytical costs to assure compliance with OSHA/EPA/State/Local requirements related to failure to comply with the regulations applicable to the work.

C. Federal Requirements:

1. Federal requirements which govern some aspect of asbestos abatement include, but are not limited to, the following regulations:

a. Occupational Safety and Health Administration (OSHA)

- 1) Title 29 CFR 1926.1101 - Construction Standard for Asbestos
- 2) Title 29 CFR 1926 Subpart E - Personal Protective Equipment and Life Saving Equipment
- 3) Title 29 CFR 1910.134 - Respiratory Protection
- 4) Title 29 CFR 1926 - Construction Industry Standards
- 5) Title 29 CFR 1926.33 - Access to Employee Exposure and Medical Records
- 6) Title 29 CFR 1926.59 same as 1910.1200 - Hazard Communication
- 7) Title 29 CFR 1926 Subpart C - General Safety and Health Provisions and Subpart D - Occupational Health and Environmental Controls

b. Environmental Protection Agency (EPA):

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1) 40 CFR 61 Subpart M - National Emission Standard for Hazardous Air Pollutants - Asbestos.

2) 40 CFR 763.80 - Asbestos Hazard Emergency Response Act (AHERA) and Asbestos Hazard Abatement Reauthorization Act (ASHARA)

c. Department of Transportation (**DOT**)

1) Title 49 CFR 171 - 180 - Transportation

D. //State Requirements:

1. State requirements that apply to the asbestos abatement work, disposal, clearance, etc., include, but are not limited to, the following://

SPEC WRITER NOTE: Provide pertinent information on applicable state regulations, statutes, and guidance documents. Acquire copies of the applicable documents and include the appropriate citations in the specifications. Contact state agencies for standards and requirements.

E. //Local Requirements:

1. If local requirements are more stringent than federal or state standards, the local standards are to be followed.//

SPEC WRITER NOTE: Include local requirements here if they apply.

F. Standards:

1. Standards which govern asbestos abatement activities include, but are not limited to, the following:

a. American National Standards Institute (ANSI)/ASSP Z9.2-2018 - Fundamentals Governing the Design and Operation of Local Exhaust Systems and ANSI/ASSE Z88.2-2015 - Practices for Respiratory Protection.

b. Underwriters Laboratories (UL) 586-2009 - UL Standard for Safety of HEPA Filter Units, 9th Edition; ANSI Approval 2017-12-19.

2. Standards which govern encapsulation work include, but are not limited to the following:

a. American Society for Testing and Materials International (ASTM)

3. Standards which govern the fire and safety concerns in abatement work include, but are not limited to, the following:

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- a. National Fire Protection Association (NFPA) 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations.
- b. NFPA 701 - Standard Methods for Fire Tests for Flame Resistant Textiles and Film.
- c. NFPA 101 - Life Safety Code

G. EPA Guidance Documents:

- 1. EPA guidance documents which discuss asbestos abatement work activities are listed below. These documents are made part of this section by reference.
- 2. Guidance for Controlling ACM in Buildings (Purple Book) EPA 560/5-85-024
- 3. Asbestos Waste Management Guidance EPA 530-SW-85-007
- 4. A Guide to Respiratory Protection for the Asbestos Abatement Industry EPA-560-OPTS-86-001
- 5. Guide to Managing Asbestos in Place (Green Book) TS 799 20T July 1990

H. Notices:

- 1. State and Local agencies: Send written notification as required by State and Local regulations including the local fire department prior to beginning any work on ACM as follows:

SPEC WRITER NOTE: Provide pertinent notification requirements for EPA, State and Local Notification. Note: Many states do not use the federal EPA notification quantity requirements. Additionally, note that demolition requires notification whether asbestos is present or not present. If courtesy notification is available, consider requiring this to minimize calls to the regulatory offices. Most times, regulatory entities can respond that they are aware of asbestos abatement activities.

- 2. Copies of notifications shall be submitted to the VA for the facility's records at the same time frame notification is given to EPA, State, and Local authorities prior to beginning any work on ACM as follows.

I. Permits/Licenses: The contractor shall apply for and have all required permits and licenses to perform asbestos abatement work as required by Federal, State, and Local regulations.

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SPEC WRITER NOTE: Permits and/or licenses may be required by authorities to perform asbestos abatement work. Indicate here any such requirements.

J. Posting and Filing of Regulations: Maintain two (2) copies of applicable federal, state, and local regulations. Post one copy of each at the regulated area where workers will have daily access to the regulations and keep another copy in the Contractor's office.

K. VA Responsibilities prior to commencement of work:

1. Notify occupants adjacent to regulated areas of project dates and requirements for relocation, if needed. **Note: Notification of adjacent personnel is required by OSHA in 29 CFR 1926.1101 (k) to prevent unnecessary or unauthorized access to the regulated area.**
2. Submit to the Contractor results of background air sampling; including location of samples, person who collected the samples, equipment utilized, calibration data and method of analysis. During abatement, submit to the Contractor, results of bulk material analysis and air sampling data collected during the course of the abatement. This information shall not release the Contractor from any responsibility for OSHA compliance.

L. Site Security:

1. Regulated area access is to be restricted only to authorized, trained/accredited and protected personnel. These may include the Contractor's employees, employees of Subcontractors, VA employees and Representatives, State and Local inspectors, and any other designated individuals. A list of authorized personnel shall be established prior to commencing the project and shall be located immediately outside the boundary of the regulated work area.
2. Entry into the regulated area by unauthorized individuals shall be reported immediately to the Competent Person by anyone observing the entry. The Competent person shall immediately notify the VA Representative.
3. A log book shall be maintained outside the boundary of the regulated work area. Anyone who enters the regulated area must record their name, affiliation, time in, and time out for each entry.
4. Access to the regulated area shall be via scaffolding, interior stair systems, or approved equivalent. All other access (doors,

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windows, hallways, etc.) shall be sealed or locked to prevent entry to or exit from the regulated area. The only exceptions for this requirement are emergency exits. Emergency exits shall not be locked from the inside or outside.

5. The Contractor's Competent Person shall control site security during abatement operations in order to isolate work in progress and protect adjacent personnel. A 24-hour security system shall be provided at the entrance to the regulated area to assure that all entrants are logged in/out and that only authorized personnel are allowed entrance.
6. The Contractor will have the VA's assistance in notifying adjacent personnel of the presence, location and quantity of ACM in the regulated area and enforcement of restricted access by the VA's employees.
7. The regulated area shall be locked during non-working hours and secured by VA Representative or Competent Person. The VA Police should be informed of asbestos abatement regulated areas to provide security checks during facility rounds and emergency response.

M. Emergency Action Plan and Arrangements:

1. An Emergency Action Plan shall be developed prior to commencing abatement activities and shall be agreed to by the Contractor and the VA. The Plan shall meet the requirements of 29 CFR 1926, Subpart C, Standard 1926.35 Employee Emergency Action Plans.
2. Emergency procedures shall be in written form and prominently posted in the clean room and equipment room of the decontamination unit. Everyone, prior to entering the regulated area, must read and sign these procedures to acknowledge understanding of the regulated area layout, location of emergency exits and emergency procedures.
3. Emergency planning shall include written notification of police, fire, and emergency medical personnel of planned abatement activities; work schedule; layout of regulated area; and access to the regulated area, particularly barriers that may affect response capabilities.
4. Emergency planning shall include consideration of fire, explosion, hazardous atmospheres, electrical hazards, slips/trips and falls, confined spaces, and heat stress illness. Written procedures for

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response to emergency situations shall be developed and employee training in procedures shall be provided.

5. Employees shall be trained in regulated area/site evacuation procedures in the event of workplace emergencies.
 - a. For non-life-threatening situations - employees injured or otherwise incapacitated shall be decontaminated following normal procedures with assistance from fellow workers, if necessary, before exiting the regulated area to obtain proper medical treatment.
 - b. For life-threatening injury or illness, worker decontamination shall take least priority after measures to stabilize the injured worker, medical personnel shall remove them from the regulated area if back or neck injury is present, and secure proper medical treatment.
6. Telephone numbers of any/all emergency response personnel shall be prominently posted in the clean room, along with the location of the nearest telephone.
7. The Contractor shall provide verification of first aid/CPR training for personnel responsible for providing first aid/CPR. OSHA requires medical assistance within 3-4 minutes of a life-threatening injury/illness. Bloodborne Pathogen training shall also be verified for those personnel required to provide first aid/CPR.
8. The Emergency Action Plan shall provide for a Contingency Plan in the event that an incident occurs that may require the modification of the standard operating procedures during abatement. Such incidents include, but are not limited to, fire; accident; power failure; extreme heat; inclement weather; high wind; and lightning. The Contractor shall detail procedures to be followed in the event of an incident assuring that asbestos abatement work is stopped and wetting is continued until correction of the problem.

N. Pre-Construction Meeting:

1. Prior to commencing the work, the Contractor shall meet with the VPIH/CIH to present and review, as appropriate, the items following this paragraph. The Contractor's Competent Person(s) who will be on-site shall participate in the pre-start meeting. The pre-start meeting is to discuss and determine procedures to be used during the project. At this meeting, the Contractor shall provide:

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SPEC WRITER NOTE: At this place, the specification writer should assure all State and Local requirements are listed and submitted.

- a. Proof of Contractor licensing.
- b. Proof the Competent Person(s) is trained and accredited and approved for working in this State. Verification of the experience of the Competent Person(s) shall also be presented.
- c. A list of all workers who will participate in the project, including experience and verification of training and accreditation.
- d. A list of and verification of training for all personnel who have current first-aid/CPR training. A minimum of one person per shift must have adequate training.
- e. Current medical written opinions for all personnel working on-site meeting the requirements of 29 CFR 1926.1101(m).
- f. Current fit-tests for all personnel wearing respirators on-site meeting the requirements of 29 CFR 1926. 1101(h) and Appendix C.
- g. A copy of the Contractor's Asbestos Hazard Abatement Plan for Class II Asbestos Abatement. In these procedures, the following information must be detailed, specific for this project.
 - 1) Regulated area preparation procedures;
 - 2) Notification requirements procedure of Contractor as required in 29 CFR 1926.1101(d) Multi-Employer Worksites;
 - 3) If required, decontamination area set-up/layout and decontamination procedures for employees; extent of 6-mil polyethylene sheeting placed //25'// out from the outer perimeter of the work area to aid in cleaning roofing material debris that may fall off building; method of demarcating the regulated work area with Asbestos Danger Tape and Asbestos Danger Signs; method for fall protection; method for accessing roof; and method for bringing material to lined roll-off or approved equivalent.
 - 4) Abatement methods/procedures and equipment to be used;
 - 5) Personal protective equipment to be used.
- h. At this meeting the Contractor shall provide all submittals as required.

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- i. Procedures for handling, packaging and disposal of asbestos waste.
- j. Emergency Action Plan and Contingency Plan Procedures.

1.6 PROJECT COORDINATION

A. The following are the minimum administrative and supervisory personnel necessary for coordination of the work.

1. Personnel:

- a. Administrative and supervisory personnel shall consist of a qualified Competent Person(s) as defined by OSHA in the Construction Standards and the Asbestos Construction Standard; Contractor Professional Industrial Hygienist and Industrial Hygiene Technicians. These employees are the Contractor's Representatives responsible for compliance with these specifications and all other applicable requirements.
- b. Non-supervisory personnel shall consist of an adequate number of qualified personnel to meet the schedule requirements of the project. Personnel shall meet required qualifications. Personnel utilized on-site shall be pre-approved by the VA Representative. A request for approval shall be submitted for any person to be employed during the project giving the person's name; last four digits of social security number; qualifications; accreditation card with color picture, if required by state; Certificate of Worker's Acknowledgment; and Affidavit of Medical Surveillance and Respiratory Protection and current Respirator Fit Test.
- c. Minimum qualifications for Contractor and assigned personnel are:
 - 1) The Contractor has conducted within the last three (3) years, three (3) projects of similar complexity and dollar value as this project; has not been cited and penalized for serious violations of Federal (and State as applicable) EPA and OSHA asbestos regulations in the past three (3) years; has adequate liability/occurrence insurance for asbestos work as required by the State; is licensed in applicable States; has adequate and qualified personnel available to complete the work; has comprehensive standard operating procedures for asbestos work; and has adequate materials, equipment and supplies to perform the work.

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- 2) The Competent Person has four (4) years of abatement experience of which two (2) years were as the Competent Person on the project; meets the OSHA definition of a Competent Person; has been the Competent Person on two (2) projects of similar size and complexity as this project within the past three (3) years; has completed EPA AHERA/OSHA/State/Local training requirements/accreditation(s) and refreshers; and has all required OSHA documentation related to medical and respiratory protection.
- 3) The Contractor Professional Industrial Hygienist/CIH (CPIH/CIH) shall have five (5) years of monitoring experience and supervision of asbestos abatement projects; has participated as senior IH on five (5) abatement projects, three (3) of which are similar in size and complexity as this project; has specialized EPA AHERA/OSHA training in asbestos abatement management, respiratory protection, waste disposal and asbestos inspection; has completed the NIOSH 582 Course or equivalent, Contractor/Supervisor course; and has appropriate medical/respiratory protection records/documentation.
- 4) The Abatement Personnel shall have completed the EPA AHERA/OSHA abatement worker course; have training on the standard operating procedures of the Contractor; has one year of asbestos abatement experience within the past three (3) years of similar size and complexity; has applicable medical and respiratory protection documentation; has certificate of training/current refresher and State accreditation/license.

d. All personnel shall be in compliance with OSHA construction safety training as applicable and submit certification.

1.7 RESPIRATORY PROTECTION

- A. General - Respiratory Protection Program: The Contractor shall develop and implement a written Respiratory Protection Program (RPP) which is in compliance with OSHA requirements found at 29 CFR 1926.1101 and 29 CFR 1910.134. ANSI Standard Z88.2-2015 provides excellent guidance for developing a respiratory protection program. All respirators used must be NIOSH approved for asbestos abatement activities. The written RPP shall, at a minimum, contain the basic requirements found at 29 CFR 1910.134 (c) - Respiratory Protection Program.

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- B. Respiratory Protection Program Coordinator: The Respiratory Protection Program Coordinator (RPPC) must be identified and shall have two (2) years of experience coordinating RPP of similar size and complexity. The RPPC must submit a signed statement attesting to the fact that the program meets the above requirements.
- C. Selection and Use of Respirators: The procedure for the selection and use of respirators must be submitted to the VA as part of the Contractor's qualifications. The procedure must be written clearly enough for workers to understand. A copy of the Respiratory Protection Program must be available in the clean room of the decontamination unit/area or in the onsite Contractor's office, for reference by employees or authorized visitors.
- D. Minimum Respiratory Protection: Shall be a ½-mask negative pressure air purifying respirator equipped with Combination P100 filters //and Organic Vapor Cartridge//, provided personal air samples in the workplace remain at or below 0.1 f/cc, //and the applicable PEL for solvents, // both// determined as an 8-hour TWA. Full face powered air purifying respirator equipped with P100 filters //and Organic Vapor Cartridge// shall be required until Contractor demonstrates that personal air samples are at or below 0.1 f/cc, //and the applicable PEL for solvents, // //both// determined as an 8-hour TWA. A higher level of respiratory protection shall be required, if fiber levels exceed 1 f/cc as an 8-hour TWA, inside the regulated work area. Respirator selection shall meet the requirements of 29 CFR 1926.1101 (h) and 29 CFR 1910.134 (d) (3) (i) (A) Table 1, except as indicated in this paragraph. Abatement personnel must have a respirator for their exclusive use.
- E. Medical Written Opinion: No employee shall be allowed to wear a respirator unless a physician or other licensed health care professional has provided a written determination they are medically qualified to wear the class of respirator to be used on the project while wearing whole body impermeable garments and subjected to heat or cold stress.
- F. Respirator Fit Test: All personnel wearing respirators shall have a current qualitative/quantitative fit test which was conducted in accordance with 29 CFR 1910.134 (f) and Appendix A. Quantitative fit tests shall be done for PAPRs which have been put into a failure mode.

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- G. Respirator Fit Check: The Competent Person shall assure that the positive/negative pressure user seal check is done each time the respirator is donned by an employee. Head coverings must cover respirator head straps. Any situation that prevents an effective facepiece to face seal as evidenced by failure of a user seal check shall preclude that person from entering the regulated area until resolution of the problem.
- H. Maintenance and Care of Respirators: The Respiratory Protection Program Coordinator shall submit evidence and documentation showing compliance with 29 CFR 1910.134 (h) Maintenance and Care of Respirators.

1.8 WORKER PROTECTION

- A. Training of Abatement Personnel: Prior to beginning any abatement activity, all personnel shall be trained in accordance with OSHA 29 CFR 1926.1101 (k) (9) and any additional State/Local requirements. Training must include, at a minimum, the elements listed at 29 CFR 1926.1101 (k) (9) (viii). Training shall have been conducted by a third party, EPA/State approved trainer meeting the requirements of EPA 40 CFR 763 Appendix C (AHERA MAP). Initial training certificates and current refresher and accreditation proof must be submitted for each person working at the site.
- B. Medical Examinations: Medical examinations meeting the requirements of 29 CFR 1926.1101 (m) shall be provided for all personnel working in the regulated area, regardless of exposure levels. A current physician's written opinion as required by 29 CFR 1926.1101 (m) (4) shall be provided for each person and shall include in the medical opinion that the person has been evaluated for working in a heat and cold stress environment while wearing personal protective equipment (PPE) and is able to perform the work without risk of material health impairment.
- C. Personal Protective Equipment: Provide whole body clothing, head coverings, foot coverings and any other personal protective equipment as determined by conducting the hazard assessment required by OSHA at 29 CFR 1910.132 (d). The Competent Person shall ensure the integrity of personal protective equipment worn for the duration of the project. Duct tape shall be used to secure all suit sleeves to wrists and to secure foot coverings at the ankle.
- D. Regulated Area Entry Procedure: The Competent Person shall ensure that each time workers enter the regulated area, they //remove// //place

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protective disposable coverall over// ALL street clothes in the clean room of the decontamination unit/area established by the competent person and put on //new disposable coveralls, // head coverings, fall protection, PPE, a clean respirator, and then proceed to the regulated work area.

- F. Decontamination Procedure: The Competent Person shall require all personnel to adhere to following decontamination procedures whenever they leave the regulated area.
1. When exiting the regulated area, //HEPA vacuum and Wet Wipe outer disposable coveralls and respirator//, remove all disposable PPE and dispose of in a disposal bag provided at the outer boundary of the regulated area.
 2. Carefully decontaminate and clean the respirator. Put in a clean container/bag.
 3. Where containment is not required for ACM removal, an adequate washing station will be provided for the employees for removal of disposable PPE and to clean the respirator.
- F. Regulated Area Requirements: The Competent Person shall meet all requirements of 29 CFR 1926.1101 (o) and assure that all requirements for OSHA Class II regulated areas at 29 CFR 1926.1101 (e) and this specification are met applicable to OSHA Class II work. All personnel in the regulated area shall not be allowed to eat, drink, smoke, chew tobacco or gum, apply cosmetics, or in any way interfere with the fit of their respirator.

1.9 DECONTAMINATION FACILITIES

- A. Description: Provide each regulated area with a fiber drum with a disposal bag in it for personnel waste materials. The Competent Person shall provide a specific area at the outer boundary of the regulated area and designate it specifically for the decontamination of personnel and respirators.

SPEC. WRITER NOTE: OSHA does not require
a decontamination unit for Class II work.

- B. Waste/Equipment Decontamination Area (W/Eda): The Competent Person shall provide a specific area in the regulated area designated for removal of all waste, equipment and contaminated material from the regulated area.

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- C. Waste/Equipment Decontamination Procedures: Contain all waste in 6-mil poly bags or approved equivalent. Clean/decontaminate bags and pass through a double 6-mil flap doorway into another bag or fiber drum. Remove to disposal dumpster/gondola/enclosed vehicle.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. General Requirements (All Abatement Projects): Prior to the start of work, the contractor shall provide and maintain a sufficient quantity of materials and equipment to assure continuous and efficient work throughout the duration of the project. Work shall not start unless the following items have been delivered to the site and the CPIH/CIH has submitted verification to the VA's Representative.
1. All materials shall be delivered in their original package, container or bundle bearing the name of the manufacturer and the brand name (where applicable).
 2. Store all materials subject to damage off the ground, away from wet or damp surfaces and under cover sufficient enough to prevent damage or contamination. Flammable and combustible materials cannot be stored inside buildings. Replacement materials shall be stored inside buildings. Replacement materials shall be stored outside of the regulated area until abatement is completed.
 3. The Contractor shall not block or hinder use of buildings by patients, staff, and visitors to the VA in partially occupied buildings by placing materials/equipment in any unauthorized location.
 4. The Competent Person shall inspect for damaged, deteriorating or previously used materials. Such materials shall not be used and shall be removed from the worksite and disposed of properly.
 5. Critical penetration points into the building shall be sealed with 2-layers of 6-mil independently installed plastic sheeting (Polyethylene) secured in place at openings such as ducts, windows, louvers, penetrations or any other opening into the regulated area. A perimeter drop cloth of 1-layer of 6-mil plastic sheeting will be placed around the outside perimeter of the roof to capture falling debris and to facilitate clean-up at ground level, if necessary. Fire retardant poly shall be used throughout.

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6. The method of attaching polyethylene sheeting shall be agreed upon in advance by the Contractor and the VA and selected to minimize damage to equipment and surfaces. Method of attachment may include any combination of moisture resistant duct tape, poly tape, furring strips, spray glue, staples, nails, screws, lumber and plywood for enclosures other effective procedures capable of sealing polyethylene to dissimilar finished or unfinished surfaces under both wet and dry conditions.
7. Polyethylene sheeting utilized for the PDF, if used, shall be opaque white or black in color, 6-mil fire retardant poly.
8. Installation and plumbing hardware, showers, hoses, drain pans, sump pumps and waste water filtration system shall be provided by the Contractor.
9. An adequate number of HEPA vacuums, scrapers, sprayers, nylon brushes, brooms, disposable mops, rags, sponges, staple guns, shovels, ladders and scaffolding of suitable height and length as well as meeting OSHA requirements, fall protection devices, water hose to reach all areas in the regulated area, airless spray equipment, and any other tools, materials or equipment required to conduct the abatement project. All electrically operated hand tools, equipment, electric cords shall be connected to GFCI protection.
10. Special protection for objects in the regulated area shall be detailed to prevent damage from scaffolds, water and falling material).
11. Disposal bags - 2-layers of 6-mil poly or approved equivalent for asbestos waste shall be pre-printed with labels, markings and address as required by OSHA, EPA and DOT regulations.
12. The VA shall be provided an advance copy of the Safety Data Sheets (SDS) as required for all hazardous chemicals under OSHA 29 CFR 1910.1200 - Hazard Communication in the pre-project submittal. Chlorinated compounds shall not be used with any spray adhesive, mastic remover or other product. Appropriate encapsulant(s) shall be provided.
13. OSHA DANGER demarcation signs, as many and as required by OSHA 29 CFR 1926.1101(k) (7) shall be provided and placed by the Competent Person. All other posters and notices required by Federal, State and

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Local regulations shall be posted at the outer boundary of the regulated area or approved equivalent.

14. Adequate and appropriate PPE for the project and number of personnel/shifts shall be provided. All personal protective equipment issued must be based on a written hazard assessment conducted under 29 CFR 1910.132(d).

2.2 CONTAINMENT BARRIERS AND COVERINGS IN THE REGULATED AREA

- A. General: Using critical barriers, seal off the regulated area to completely isolate the regulated area from adjacent spaces. All penetrations to the regulated area must be covered with 2-layers of 6-mil fire retardant poly to prevent contamination and to facilitate clean-up. Should adjacent areas become contaminated, immediately stop work and clean up the contamination at no additional cost to the Government. Provide firestopping and identify all fire barrier penetrations due to abatement work as specified in Section 2.2.7; FIRESTOPPING.
- B. Preparation Prior to Sealing the Regulated Area: Place all tools, scaffolding, materials and equipment needed for working in the regulated area prior to erecting any plastic sheeting. Lock out and tag out any HVAC systems in the regulated area.
- C. Controlling Access to the Regulated Area: Access to the regulated area is allowed only at the Competent Person's designated location and shall serve as a personnel decontamination facility (PDF)/Area, if required. All other means of access shall be eliminated and OSHA Danger demarcation signs posted as required by OSHA.
- D. Critical Barriers: Critical penetration points into the building shall be sealed with 2-layers of 6-mil independently installed plastic sheeting (Polyethylene) secured in place at openings such as ducts, windows, louvers, penetrations or any other opening into the regulated area.
- E. Secondary Barriers: A perimeter drop cloth of 1-layer of 6-mil plastic sheeting will be placed around the outside perimeter of the roof to capture falling debris and to facilitate clean-up at ground level, if necessary.
- F. Extension of the Regulated Area: If the enclosure of the regulated area is breached in any way that could allow contamination to occur, the affected area shall be included in the regulated area and constructed

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as per this section. If the affected area cannot be added to the regulated area, decontamination measures must be started immediately and continue until air monitoring indicates background levels are met.

G. Firestopping:

1. Through penetrations caused by cables, cable trays, pipes, ducts, sleeves must be firestopped with a fire-rated firestop system providing an air tight seal.
2. Firestop materials that are not equal to the wall or ceiling penetrated shall be brought to the attention of the VA Representative. The Contractor shall list all areas of penetration, the type of sealant used, and whether or not the location is fire rated. Any discovery of penetrations during abatement shall be brought to the attention of the VA Representative immediately. All walls, floors and ceilings are considered fire rated unless otherwise determined by the VA Representative or Fire Marshall.
3. Any visible openings whether or not caused by a penetration shall be reported by the Contractor to the VA Representative for a sealant system determination. Firestops shall meet ASTM E814 and UL 1479 requirements for the opening size, penetrant, and fire rating needed.

H. Scope of Services of the VPIH/CIH Consultant:

1. The purpose of the work of the VPIH/CIH is to: assure quality; adherence to the specification; resolve problems; prevent the spread of contamination beyond the regulated area; and assure clearance at the end of the project. In addition, their work includes performing the final inspection and testing to determine whether the regulated area or building has been adequately decontaminated. All air monitoring is to be done utilizing PCM/TEM. The VPIH/CIH will perform the following tasks:
 - a. Task 1: Establish background levels before abatement begins by collecting background samples. Retain samples for possible TEM analysis.
 - b. Task 2: Perform representative air monitoring, inspection, and testing outside the regulated area during actual abatement work to detect any faults in the regulated area isolation and any adverse impact on the surroundings from regulated area activities.

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- c. Task 3: Perform unannounced visits to spot check overall compliance of work with contract/specifications. These visits may include any inspection, monitoring, and testing inside and outside the regulated area and all aspects of the operation except personnel monitoring.
- d. Task 4: Provide support to the VA Representative such as evaluation of submittals from the Contractor, resolution of conflicts, interpret data, etc.
- e. Task 5: Perform, in the presence of the VA Representative, final inspection and testing of a decontaminated regulated area at the conclusion of the abatement to certify compliance with all regulations and VA requirements/specifications.
- f. Task 6: Issue certificate of decontamination for each regulated area and project report.

SPEC WRITER NOTE: Buildings which will be totally demolished must have abatement plans approved by the VPIH/CIH consultant and be periodically inspected for compliance. See Section 02 82 13.41, ASBESTOS ABATEMENT FOR TOTAL DEMOLITION PROJECTS.

- 2. All documentation, inspection results and testing results generated by the VPIH/CIH will be available to the Contractor for information and consideration. The Contractor shall cooperate with and support the VPIH/CIH for efficient and smooth performance of their work.
 - 3. The monitoring and inspection results of the VPIH/CIH will be used by the VA to issue any Stop Removal orders to the Contractor during abatement work and to accept or reject a regulated area or building as decontaminated.
- I. Monitoring, Inspection and Testing by Contractor CPIH/CIH: The Contractor's CPIH/CIH is responsible for managing all monitoring, inspections, and testing required by these specifications, as well as any and all regulatory requirements adopted by these specifications. The CPIH/CIH is responsible for the continuous monitoring of all subsystems and procedures which could affect the health and safety of the Contractor's personnel. Safety and health conditions and the provision of those conditions inside the regulated area for all persons entering the regulated area is the exclusive responsibility of the

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Contractor/Competent Person. The person performing the personnel and area air monitoring inside the regulated area shall be an IH Technician, who shall be trained and shall have specialized field experience in sampling and analysis. The IH Technician shall have successfully completed a NIOSH 582 Course or equivalent and provide documentation. The IH Technician shall participate in the AIHA Asbestos Analysis Registry or participate in the Proficiency Analytical Testing program of AIHA for fiber counting quality control assurance. The IH Technician shall also be an accredited EPA AHERA/State Contractor/Supervisor and Building Inspector. The IH Technician shall have participated in five abatement projects collecting personal and area samples and have experience in substantially similar projects in size and scope. The analytical laboratory used by the Contractor to analyze the samples shall be AIHA accredited for asbestos PAT and approved by the VA prior to start of the project. A daily log shall be maintained by the CPIH/CIH or IH Technician, documenting all OSHA requirements for personal and area air monitoring for asbestos in 29 CFR 1926.1101(f), (g) and Appendix A. This log shall be made available to the VA Representative and the VPIH/CIH upon request. The log will contain, at a minimum, information on personnel or area samples, other persons represented by the sample, the date of sample collection, start and stop times for sampling, sample volume, flow rate, and fibers/cc. The CPIH/CIH shall collect and analyze samples for each representative job being done in the regulated area, i.e., removal, wetting, clean-up, and load-out. No fewer than two (2) personal air samples or 25 percent of representative workforce per shift shall be collected, whichever is greater, in the regulated area; a minimum of three (3) area air samples at locations inside the building but immediately outside the regulated work area; one (1) area air sample shall be collected daily in the Clean Room of the PDF; and one (1) area air sample shall be collected daily at the approximate location of HEPA exhaust discharge. In addition to the continuous monitoring required, the CPIH/CIH will perform inspection and testing at the final stages of abatement for each regulated area as specified in the CPIH/CIH responsibilities. Additionally, the CPIH/CIH will monitor and record pressure readings within the containment daily with a minimum of two readings at the

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beginning and at the end of a shift, and submit the data in the daily report.

2.3 MONITORING, INSPECTION AND TESTING

A. General:

1. Perform throughout abatement work monitoring, inspection and testing inside and around the regulated area in accordance with the OSHA requirements and these specifications. OSHA requires that the employee exposure to asbestos must not exceed 0.1 fiber per cubic centimeter (f/cc) of air, averaged over an 8-hour work shift. The CPIH/CIH is responsible for and shall inspect and oversee the performance of the Contractor IH Technician. The IH Technician shall continuously inspect and monitor conditions inside the regulated area to ensure compliance with these specifications. In addition, the CPIH/CIH shall personally manage air sample collection, analysis, and evaluation for personnel, regulated area, and //insert number of samples, if any// daily area air samples inside the building, but outside the regulated area. At a minimum three (3) perimeter area air samples shall be collected daily at the ground level around the regulated area perimeter, preferably downwind of the regulated work area. Additional inspection and testing requirements are also indicated in other parts of this specification.
2. The VA will employ an independent industrial hygienist (VPIH/CIH) consultant and/or use its own IH to perform various services on behalf of the VA. The VPIH/CIH will perform representative monitoring, inspection, testing, and other support services to ensure that VA patients, employees, and visitors will not be adversely affected by the abatement work, and that the abatement work proceeds in accordance with these specifications, that the abated areas or abated buildings have been successfully decontaminated. The work of the VPIH/CIH consultant in no way relieves the Contractor from their responsibility to perform the work in accordance with contract/specification requirements, to perform continuous inspection, monitoring and testing for the safety of their employees, and to perform other such services as specified. The cost of the VPIH/CIH and their services will be borne by the VA except for any repeat of final inspection and testing that may be

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required due to unsatisfactory initial results. Any repeated final inspections and/or testing, if required, will be paid for by the Contractor.

3. If fibers counted by the VPIH/CIH during abatement work, either inside or outside the regulated area, utilizing the NIOSH 7400 air monitoring method, exceed the specified respective limits, the Contractor shall stop work. The Contractor may request confirmation of the results by analysis of the samples by TEM. Request must be in writing and submitted to the VA's Representative. Cost for the confirmation of results will be borne by the Contractor for both the collection and analysis of samples and for the time delay that may/does result for this confirmation. Confirmation sampling and analysis will be the responsibility of the CPIH/CIH with review and approval of the VPIH/CIH. An agreement between the CPIH/CIH and the VPIH/CIH shall be reached on the exact details of the confirmation effort, in writing, including such things as the number of samples, location, collection, quality control on-site, analytical laboratory, interpretation of results and any follow-up actions. This written agreement shall be co-signed by the IH's and delivered to the VA's Representative.

B. Scope Of Services of the VPIH/CIH Consultant

1. The purpose of the work of the VPIH/CIH is to: assure quality; resolve problems; and prevent the spread of contamination beyond the regulated area. In addition, their work includes performing the final inspection and testing to determine whether the regulated area or building has been adequately decontaminated. All air monitoring is to be done utilizing PCM/TEM. The VPIH/CIH will perform the following tasks:
 - a. Task 1: Establish background levels before abatement begins by collecting background samples. Retain samples for possible TEM analysis.//
 - b. Task 2: Perform representative air monitoring, inspection, and testing outside the regulated area during actual abatement work to detect any faults in the regulated area isolation and any adverse impact on the surroundings from regulated area activities.//

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- c. Task 3: Perform unannounced visits to spot check overall compliance of work with contract/specifications. These visits may include any inspection, monitoring, and testing inside and outside the regulated area and all aspects of the operation except personnel monitoring.
- d. Task 4: Provide support to the VA Representative such as evaluation of submittals from the Contractor, resolution of unforeseen developments, etc.
- e. Task 5: Perform, in the presence of the VA Representative, final inspection and testing of a decontaminated regulated area or building at the conclusion of the abatement and clean-up work to certify compliance with all regulations and the VA requirements/specifications.
- f. Task 6: Issue certificate of decontamination for each regulated area or building and project report.

SPEC WRITER NOTE: Buildings which will be totally demolished must have abatement plans approved by the VPIH/CIH consultant and be periodically inspected for compliance. See Section 02 82 13.41, ASBESTOS ABATEMENT FOR TOTAL DEMOLITION PROJECTS.

- 2. All documentation, inspection results and testing results generated by the VPIH/CIH will be available to the Contractor for information and consideration. The Contractor shall cooperate with and support the VPIH/CIH for efficient and smooth performance of their work.
 - 3. The monitoring and inspection results of the VPIH/CIH will be used by the VA to issue any Stop Removal orders to the Contractor during abatement work and to accept or reject a regulated area or building as decontaminated.
- C. Monitoring, Inspection and Testing by Contractor CPIH: The Contractor's CPIH/CIH is responsible for managing all monitoring, inspections, and testing required by these specifications, as well as any and all regulatory requirements adopted by these specifications. The CPIH/CIH is responsible for the continuous monitoring of all subsystems and procedures which could affect the health and safety of the Contractor's personnel. Safety and health conditions and the provision of those conditions inside the regulated area for all persons entering the

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regulated area is the exclusive responsibility of the Contractor/Competent Person. The person performing the personnel and area air monitoring inside the regulated area shall be an IH Technician, who shall be trained and shall have specialized field experience in sampling and analysis. The IH Technician shall have successfully completed a NIOSH 582 Course or equivalent and provide documentation. The IH Technician shall participate in the AIHA Asbestos Analysis Registry or participate in the Proficiency Analytical Testing program of AIHA for fiber counting quality control assurance. The IH Technician shall also be an accredited EPA AHERA/State Contractor/Supervisor and Building Inspector. The IH Technician shall have participated in five abatement projects collecting personal and area samples and have experience in substantially similar projects in size and scope. The analytical laboratory used by the Contractor to analyze the samples shall be AIHA accredited for asbestos PAT and approved by the VA prior to start of the project. A daily log shall be maintained by the CPIH/CIH or IH Technician, documenting all OSHA requirements for personal and area air monitoring for asbestos in 29 CFR 1926.1101 (f), (g) and Appendix A. This log shall be made available to the VA Representative and the VPIH/CIH upon request. The log will contain, at a minimum, information on personnel or area samples, other persons represented by the sample, the date of sample collection, start and stop times for sampling, sample volume, flow rate, and fibers/cc. The CPIH/CIH shall collect and analyze samples for each Representative job being done in the regulated area, i.e., removal, wetting, clean-up, and load-out. The CPIH/CIH shall personally manage air sample collection, analysis, and evaluation for personnel, regulated area, and //insert number of samples, if any// daily area air samples inside the building, but outside the regulated area. At a minimum three (3) perimeter area air samples shall be collected daily at the ground level around the regulated area perimeter, preferably downwind of the regulated work area. In addition to the continuous monitoring required, the CPIH/CIH will perform inspection and testing at the final stages of abatement for each regulated area as specified in the CPIH/CIH responsibilities.

2.4 ASBESTOS HAZARD ABATEMENT PLAN

A. The Contractor shall have an established Asbestos Hazard Abatement Plan (AHAP) in printed form and loose leaf folder consisting of simplified text, diagrams, sketches, and pictures that establish and explain clearly the ways and procedures to be followed during all phases of the work by the Contractor's personnel. The AHAP must be modified as needed to address specific requirements of the project. The AHAP shall be submitted for review and approval prior to the start of any abatement work. The minimum topics and areas to be covered by the AHAP(s) are:

1. Minimum Personnel Qualifications
2. Contingency Plans and Arrangements
3. Security and Safety Procedures
4. Respiratory Protection/Personal Protective Equipment Program and Training
5. Medical Surveillance Program and Recordkeeping
6. Regulated Area Requirements for Abatement
7. Decontamination Entry/Exit Procedures
8. Monitoring, Inspections, and Testing
9. Disposal of ACM waste
10. Regulated Area Decontamination/Clean-up
11. Regulated Area Visual and Air Clearance
12. Project Completion/Closeout

2.5 SUBMITTALS

A. Pre-Start Meeting Submittals:

1. Submit to the VA a minimum of 14 days prior to the pre-start meeting the following for review and approval. Meeting this requirement is a prerequisite for the pre-start meeting for this project:
 - a. Submit a detailed work schedule for the entire project reflecting contract documents and the phasing/schedule requirements from the CPM chart.
 - b. Submit a staff organization chart showing all personnel who will be working on the project and their capacity/function. Provide their qualifications, training, accreditations, and licenses, as appropriate. Provide a copy of the "Certificate of Worker's Acknowledgment" and the "Affidavit of Medical Surveillance and Respiratory Protection" for each person.

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- c. Submit Asbestos Hazard Abatement Plan developed specifically for this project, incorporating the requirements of the specifications, prepared, signed and dated by the CPIH/CIH.
- d. Submit the specifics of the materials and equipment to be used for this project with manufacturer names, model numbers, performance characteristics, pictures/diagrams, and number available for the following:
 - 1) HEPA vacuums, air monitoring pumps, calibration devices, and emergency power generating system.
 - 2) Waste water filtration system and containment barriers.
 - 3) Encapsulants, surfactants, hand held sprayers, airless sprayers, and fire extinguishers.
 - 4) Respirators, protective clothing, fall protection and other required personal protective equipment.
 - 5) Fire safety equipment to be used in the regulated area.
- e. Submit the name, location, and phone number of the approved landfill; proof/verification the landfill is approved for ACM disposal; the landfill's requirements for ACM waste; the type of vehicle to be used for transportation; and name, address, and phone number of subcontractor, if used. Proof of asbestos training for transportation personnel shall be provided.
- f. Submit required notifications and arrangements made with regulatory agencies having regulatory jurisdiction and the specific contingency/emergency arrangements made with local health, fire, ambulance, hospital authorities and any other notifications/arrangements.
- g. Submit the name, location and verification of the laboratory and/or personnel to be used for analysis of air and/or bulk samples. Personal air monitoring must be done in accordance with OSHA 29 CFR 1926.1101 (f) and Appendix A. Area or clearance air monitoring shall be conducted in accordance with EPA AHERA protocols.
- h. Submit qualifications verification: Submit the following evidence of qualifications. Make sure that all references are current and verifiable by providing current phone numbers and documentation.
 - 1) Asbestos Abatement Company: Project experience within the past 3 years; listing projects first most similar to this project:

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- Project Name; Type of Abatement; Duration; Cost; Reference Name/Phone Number; Final Clearance; and Completion Date
- 2) List of project(s) halted by owner, A/E, IH, regulatory agency in the last 3 years: Project Name; Reason; Date; Reference Name/Number; Resolution
 - 3) List asbestos regulatory citations (e.g., OSHA), notices of violations (e.g., Federal, State and Local EPA), penalties, and legal actions taken against the company including the company's officers (including damages paid) in the last 3 years. Provide copies and all information needed for verification.
- i. Submit information on personnel: Provide a resume; address each item completely; copies of certificates, accreditations, and licenses. Submit an affidavit signed by the CPIH/CIH stating that all personnel submitted below have medical records in accordance with OSHA 29 CFR 1926.1101(m) and that the company has implemented a medical surveillance program and written respiratory protection program, and maintains recordkeeping in accordance with the above regulations. Submit the phone number and doctor/clinic/hospital used for medical evaluations.
- 1) CPIH/CIH and IH Technician: Name; years of abatement experience; list of projects similar to this one; certificates, licenses, accreditations for proof of AHERA/OSHA specialized asbestos training; professional affiliations; medical opinion; and current respirator fit test.
 - 2) Competent Person(s)/Supervisor(s): Number; names; last four digits of social security numbers; years of abatement experience as Competent Person/Supervisor; list of similar projects in size/complexity as Competent Person/Supervisor; as a worker; certificates, licenses, accreditations; proof of AHERA/OSHA specialized asbestos training; maximum number of personnel supervised on a project; medical opinion (asbestos surveillance and respirator use); and current respirator fit test.
 - 3) Workers: Numbers; names; last four digits of social security numbers; years of abatement experience; certificates, licenses, accreditations; training courses in asbestos

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abatement and respiratory protection; medical opinion (asbestos surveillance and respirator use); and current respirator fit test.

- j. Submit copies of State license for asbestos abatement; copy of insurance policy, including exclusions with a letter from agent stating in plain language the coverage provided and the fact that asbestos abatement activities are covered by the policy; copy of AHAP(s) incorporating the requirements of this specification; information on who provides your training, how often; who provides medical surveillance, how often; who performs and how is personal air monitoring of abatement workers conducted; a list of references of independent laboratories/IH's familiar with your air monitoring and standard operating procedures; and copies of monitoring results of the five referenced projects listed and analytical methods used.
- k. Rented equipment must be decontaminated prior to returning to the rental agency.
- l. Submit, before the start of work, the manufacturer's technical data for all types of encapsulants, all SDS and application instructions.

B. Submittals During Abatement:

- 1. The Competent Person shall maintain and submit a daily log at the regulated area documenting the dates and times of the following: purpose, attendees and summary of meetings; all personnel entering/exiting the regulated area; document and discuss the resolution of unusual events such as barrier breaching, equipment failures, emergencies, and any cause for stopping work; Representative air monitoring and results/TWAs/ELs. Submit this information daily to the VA's Representative.
- 2. The CPIH/CIH shall document and maintain the inspection and approval of the regulated area preparation prior to start of work and daily during work.
 - a. Removal of any poly barriers.
 - b. Visual inspection/testing by the CPIH/CIH or IH Technician prior to application of lockdown encapsulant, if used.
 - c. Packaging and removal of ACM waste from regulated area.

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- d. Disposal of ACM waste materials; copies of Waste Shipment Records/landfill receipts to the VA's Representative on a weekly basis.
- C. Submittals at Completion of Abatement: The CPIH/CIH shall submit a project report consisting of the daily log book requirements and documentation of events during the abatement project including Waste Shipment Records signed by the landfill's agent. It will also include information on the containment and transportation of waste from the containment with applicable Chain of Custody forms. The report shall include a certificate of completion, signed and dated by the CPIH/CIH, in accordance with Attachment #1. All clearance and perimeter area samples must be submitted. The VA Representative will retain the abatement report after completion of the project and provide copies of the abatement report to VAMC Office of Engineer and the Safety Office.

PART 3 - EXECUTION

3.1 REGULATED AREA PREPARATIONS

- A. Pre-Abatement Activities: The VA Representative, upon receipt, review, and substantial approval of all pre-abatement submittals and verification by the CPIH/CIH that all materials and equipment required for the project are on the site, will arrange for a pre-abatement meeting between the Contractor, the CPIH/CIH, Competent Person, the VA Representative, and the VPIH/CIH. The purpose of the meeting is to discuss any aspect of the submittals needing clarification or amplification and to discuss any aspect of the project execution and the sequence of the operation. The Contractor shall be prepared to provide any supplemental information/documentation to the VA's Representative regarding any submittals, documentation, materials or equipment. Upon satisfactory resolution of any outstanding issues, the VA's Representative will issue a written order to proceed to the Contractor. No abatement work of any kind described in the following provisions shall be initiated prior to the VA written order to proceed.
- B. Pre-Abatement Inspections and Preparations:
 - 1. Before any work begins on the construction of the regulated area, the Contractor will:
 - a. Conduct a space-by-space inspection with an authorized VA Representative and prepare a written inventory of all existing damage in those spaces where asbestos abatement will occur. Still

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or video photography may be used to supplement the written damage inventory. Document will be signed and certified as accurate by both parties.

- b. The VA Representative, the Contractor, and the VPIH/CIH must be aware of VA AEQA 10-95 indicating the failure to identify asbestos in the areas listed as well as common issues when preparing specifications and contract documents. This is especially critical when demolition is planned, because AHERA surveys are non-destructive, and ACM may remain undetected. A NESHAP (destructive) ACM inspection shall be conducted on all building structures that will be demolished. Ensure the following areas are inspected on the project: Lay-in ceilings concealing ACM; ACM behind walls/windows from previous renovations; inside utility chases/walls; transite piping/ductwork/sheets; behind radiators; lab fume hoods; transite lab countertops; roofing materials; //fireproofing on deck below roofing materials requires significant consideration// below window sills; water/sewer lines; electrical conduit coverings; crawl spaces (previous abatement contamination); flooring/mastic covered by carpeting/new flooring; exterior insulated wall panels; on underground fuel tanks; steam line trench coverings.
 - c. Inspect existing firestopping in the regulated area. Correct as needed.
- C. Pre-Abatement Construction and Operations:
- 1. Perform all preparatory work for the first regulated area in accordance with the approved work schedule and with this specification.
 - 2. Upon completion of all preparatory work, the CPIH/CIH will inspect the work and systems and will notify the VA's Representative when the work is completed in accordance with this specification. The VA's Representative may inspect the regulated area and the systems with the VPIH/CIH and may require that upon satisfactory inspection, the Contractor's employees perform all major aspects of the approved SOP's, especially worker protection, fall protection, respiratory systems, contingency plans, decontamination procedures, and monitoring to demonstrate satisfactory operation.

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3. The CPIH/CIH shall document the pre-abatement activities described above and deliver a copy to the VA's Representative.
4. Upon satisfactory inspection of the installation of and operation of systems the VA's Representative will notify the Contractor in writing to proceed with the asbestos abatement work in accordance with this specification.

3.2 REGULATED AREA PREPARATIONS

- A. OSHA Danger Signs: Post OSHA DANGER signs meeting the specifications of OSHA 29 CFR 1926.1101 at any location and approaches to the regulated area where airborne concentrations of asbestos may exceed ambient background levels. Signs shall be posted at a distance sufficiently far enough away from the regulated area to permit any personnel to read the sign and take the necessary measures to avoid exposure. Additional signs will be posted following construction of the regulated area enclosure.
- B. Shut Down - Lock Out Electrical: Shut down and lock out/tag out electric power to the regulated area. Provide temporary power and lighting (if working at night). Insure safe installation including GFCI of temporary power sources and equipment by compliance with all applicable electrical code requirements and OSHA requirements for temporary electrical systems. Electricity shall be provided by the VA.
- C. Shut Down - Lock Out HVAC:
 1. Shut down and lock out/tag out heating, cooling, and air conditioning system (HVAC) components that are in, supply or pass through the regulated area.
 2. Investigate the regulated area and agree on pre-abatement condition with the VA's Representative. Seal all intake and exhaust vents in the regulated area with duct tape and 2-layers of independently installed 6-mil poly. Also, seal any seams in system components that pass through the regulated area. Remove all contaminated HVAC system filters and place in labeled 6-mil poly disposal bags for disposal as asbestos waste.
- D. Sanitary Facilities: The Contractor shall provide sanitary facilities for abatement personnel and maintain them in a clean and sanitary condition throughout the abatement project.

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E. Water for Abatement: The VA will provide water for abatement purposes. The Contractor shall connect to the existing VA system. The service to the shower, if used, shall be supplied with backflow prevention.

SPEC WRITER NOTE: Typically roofing
abatement projects will OMIT this Section
F.

F. Pre-Cleaning Movable Objects:

1. Pre-cleaning of ACM contaminated items shall be performed after the enclosure has been erected and negative pressure has been established in the work area. After items have been pre-cleaned and decontaminated, they may be removed from the work area for storage until the completion of abatement in the work area.
2. Pre-clean all movable objects within the regulated area using a HEPA filtered vacuum and wet cleaning methods as appropriate. After cleaning, these objects shall be removed from the regulated area and carefully stored in an uncontaminated location.

SPEC WRITER NOTE: Typically roofing
abatement projects will OMIT this Section
G.

G. Pre-Cleaning Fixed Objects:

1. Pre-cleaning of ACM contaminated items shall be performed after the regulated area has been established. PPE must be donned during all pre-cleaning activities.
2. Pre-clean all surfaces in the regulated area using HEPA filtered vacuums and wet cleaning methods as appropriate. Do not use any methods that would raise dust such as dry sweeping or vacuuming with equipment not equipped with HEPA filters. Do not disturb asbestos-containing materials during this pre-cleaning phase.

SPEC WRITER NOTE: Typically roofing
abatement projects will OMIT this Section
H.

H. PRE-CLEANING SURFACES IN THE REGULATED AREA

1. Pre-cleaning of ACM contaminated items shall be performed after the regulated area has been established. PPE must be donned during all pre-cleaning activities.
2. Pre-clean all surfaces in the regulated area using HEPA filtered vacuums and wet cleaning methods as appropriate. Do not use any

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methods that would raise dust such as dry sweeping or vacuuming with equipment not equipped with HEPA filters. Do not disturb asbestos-containing materials during this pre-cleaning phase.

3.3 CONTAINMENT BARRIERS AND COVERINGS IN THE REGULATED AREA:

- A. General: Using critical barriers, seal off the perimeter to the regulated area to completely isolate the regulated area from adjacent spaces. All penetrations in the regulated area must be covered with 2-layers of independently installed 6-mil fire retardant poly to prevent contamination and to facilitate clean-up. Should adjacent areas become contaminated, immediately stop work and clean up the contamination at no additional cost to the Government
- B. Preparation Prior To Sealing Off: Place all tools, scaffolding, materials and equipment needed for working in the regulated area prior to erecting any plastic sheeting. Lock out and tag out any HVAC systems in the regulated area.
- C. Controlling Access to the Regulated Area: Access to the regulated area is allowed only at the Competent Person's designated location and shall serve as a personnel decontamination facility (PDF)/Area, if required. All other means of access shall be eliminated and OSHA Danger demarcation signs posted as required by OSHA.
- D. Critical Barriers: Critical penetration points into the building shall be sealed with 2-layers of 6-mil independently installed plastic sheeting (Polyethylene) secured in place at openings such as ducts, windows, louvers, penetrations or any other opening into the regulated area.
- E. Extension of the Regulated Area: If the enclosure of the regulated area is breached in any way that could allow contamination to occur, the affected area shall be included in the regulated area and constructed as per this section. If the affected area cannot be added to the regulated area, decontamination measures must be started immediately and continue until air monitoring indicates background levels are met

3.4 REMOVAL OF OSHA CLASS II AND EPA CATEGORY I ROOFING

- A. General: The VA must be notified at least 24 hours in advance of any waste removed from the containment. All applicable requirements of OSHA, EPA, State, Local and DOT shall be followed during Class II work. Keep materials intact; do not disturb; wet while working with it; wrap

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as soon as possible with 2-layers of 6-mil plastic for disposal or approved equivalent

- B. Outdoor Work Areas: On some projects, work must be performed on exterior areas of the building. If outdoor work is to be performed, all applicable OSHA, State and Local regulations must be followed to ensure that outdoor work areas are in compliance so that workers, the general public and the environment are protected.
- C. Scaffold Fall Protection: Each employee more than 10 feet above a lower level shall be protected from falls by guardrails or a fall arrest system. Fall arrest system includes harnesses, components of the harness/belt such as Dee-rings, and snap hooks, lifelines, and anchorage points. Lifelines must be independent of supports lines and suspension ropes and not attached to the same anchorage point as the support or suspension rope. OSHA's scaffolding standard defines a competent person as "one who is capable of identifying existing and predictable hazards in the surroundings or working conditions, which are unsanitary, hazardous to employees, and who has authorization to take prompt corrective measures to eliminate them." The competent person will determine if it is safe for employees to work on or from a scaffold or roof during storms or high winds and to ensure that a personal fall arrest system will protect the employees. The competent person will also inspect the scaffold and scaffold components for visible defects before each work shift and after any occurrence which could affect the structural integrity and to authorize prompt corrective measures.
- D. Roof Fall Protection: The competent person shall determine if the walking/working surfaces on which the employees are to work have the strength and structural integrity to support the employees safely. Each employee on a walking/working surface (horizontal and vertical surface) with an unprotected side or edge which is 6 feet or more above a lower level shall be protected from falling by the use of guardrail systems, safety net systems, or personal fall arrest system.
- E. Removal of Roofing:
 - 1. Roofing material shall be removed in an intact state to the extent that it is feasible.

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2. Wet methods shall be used to remove roofing materials that are not intact, or that will be rendered not intact during removal, unless such wet methods are not feasible or will create safety hazards.
3. Cutting machines shall be continuously misted during use, unless a competent person determines that misting substantially decreases worker safety.
4. When removing built-up roofs with asbestos-containing roofing felts and an aggregate surface using a power roof cutter, all dust resulting from the cutting operation shall be collected by a HEPA dust collector, or shall be HEPA vacuumed by vacuuming along the cut line. When removing built-up roofs with asbestos-containing roofing felts and a smooth surface using a power roof cutter, the dust resulting from the cutting operation shall be collected either by a HEPA dust collector or HEPA vacuuming along the cut line, or by gently sweeping and then carefully and completely wiping up the still-wet dust and debris left along the cut line.
5. Asbestos-containing material that has been removed from a roof shall not be dropped or thrown to the ground. Unless the material is carried or passed to the ground by hand, it shall be lowered to the ground via covered, dust-tight chute, crane or hoist.
6. Any ACM that is not intact shall be lowered to the ground as soon as is practicable, but in any event no later than the end of the work shift. While the material remains on the roof it shall either be kept wet, placed in an impermeable waste bag, or wrapped in plastic sheeting.
7. Intact ACM shall be lowered to the ground as soon as is practicable, but in any event no later than the end of the work shift.
8. Upon being lowered, unwrapped material shall be transferred to a closed receptacle in such manner so as to preclude the dispersion of dust.
9. Roof level heating and ventilation air intake sources shall be isolated or the ventilation system shall be shut down //or both//. The sealing of air intake sources shall be coordinated with VA Facility Engineering Personnel and occupant location to ensure acceptable IAQ is maintained within the facility as per ASHRAE Standard 1955.

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10. Notwithstanding any other provision of this section, removal or repair of sections of intact roofing less than 25 square feet in area does not require use of wet methods or HEPA vacuuming as long as manual methods which do not render the material non-intact are used to remove the material and no visible dust is created by the removal method used. In determining whether a job involves less than 25 square feet, the employer shall include all removal and repair work performed on the same roof on the same day.

F. All waste must be wrapped in two layers of 6-mil poly or approved equivalent and lowered carefully to the ground. Roofing may be lowered by way of a dust-tight chute.

3.5 DISPOSAL OF OSHA CLASS II AND EPA CATEGORY I WASTE MATERIAL

A. General: Dispose of waste ACM and debris which is packaged in accordance with these specifications, OSHA, EPA and DOT. The landfill requirements for packaging must also be met. Transport will be in compliance with 49 CFR 171-180 regulations. Disposal shall be done at an approved landfill in accordance with State and Local requirements, which are generally more restrictive than Federal EPA. Disposal of non-friable ACM shall be done in accordance with applicable State and Local regulations.

3.6 PROJECT DECONTAMINATION

A. General:

1. The entire work related to project decontamination shall be performed under the close supervision and monitoring of the CPIH/CIH.
2. If the asbestos abatement work is in an area which was contaminated prior to the start of abatement, the decontamination will be done by cleaning the primary barrier poly prior to its removal and cleanings of the surfaces of the regulated area after the primary barrier removal.
3. If the asbestos abatement work is in an area which was uncontaminated prior to the start of abatement, the decontamination will be done by cleaning the primary barrier poly prior to its removal, thus preventing contamination of the building when the regulated area critical barriers are removed.

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- B. Regulated Area Clearance: Air testing and other requirements which must be met before release of the Contractor and re-occupancy of the regulated area space are specified in Final Testing Procedures.
- C. Work Description: Decontamination includes the cleaning and clearance of the air in the regulated area and the decontamination and removal of the enclosures/facilities installed prior to the abatement work including primary/critical barriers, PDF and W/EDF facilities, if used.
- D. Pre-Decontamination Conditions:
 - 1. Before decontamination starts, all ACM waste from the regulated area shall be removed, all waste collected and removed, and the secondary barrier of poly removal and disposed of along with any gross debris generated by the work.
 - 2. At the start of decontamination, the following shall be in place:
 - a. Critical barriers at openings such as ducts, windows, louvers, penetrations or any other opening into the regulated area.
 - b. Decontamination facilities, if required for personnel and equipment in operating condition.
- E. Cleaning: Carry out a first cleaning of all surfaces of the regulated area including items of remaining poly sheeting, tools, scaffolding, ladders/staging by wet methods and HEPA vacuuming. Do not use dry dusting/sweeping/air blowing methods. Use each surface of a wetted cleaning cloth one time only and then dispose of as contaminated waste. Continue this cleaning until there is no visible residue from abated surfaces or poly or other surfaces. Remove all filters in the air handling system and dispose of as ACM waste in accordance with these specifications.

3.7 VISUAL INSPECTION AND AIR CLEARANCE TESTING

- A. General: Notify the VA Representative 24 hours in advance for the performance of the final visual inspection and testing. The final visual inspection and testing will be performed by the VPIH/CIH after the cleaning.
- B. Visual Inspection: Final visual inspection will include the entire regulated area, all poly sheeting, seals over ducts, windows, louvers, penetrations or any other opening into the regulated area. If any debris, residue, dust or any other suspect material is detected, the cleaning shall be repeated at no additional cost to the VA. Dust/material samples may be collected and analyzed at no additional cost to

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the VA at the discretion of the VPIH/CIH to confirm visual findings. When the regulated area is visually clean the final testing can be done.

C. Air Clearance Testing:

1. After an acceptable final visual inspection by the VPIH/CIH and VA Representative, the VPIH/CIH will perform the final clearance testing. Air samples will be collected and analyzed in accordance with procedures for AHERA, for this project, five (5) PCM samples shall be collected for clearance and a minimum of two field blank
2. If the results of the PCM are acceptable, remove the critical barriers. Any small quantities of residue material found upon removal of the poly shall be removed with a HEPA vacuum and localized isolation. If significant quantities are found as determined by the VPIH/CIH, then the entire area affected shall be cleaned as specified in the final cleaning.
3. If release criteria are met, proceed to perform the abatement closeout and to issue the certificate of completion in accordance with these specifications.

D. Final Air Clearance Procedures:

1. Contractor's Release Criteria: Work in a regulated area is complete when the regulated area is visually clean and airborne fiber levels have been reduced to or below 0.01 f/cc, as measured by PCM methods.
2. Air Monitoring and Final Clearance Sampling: To determine if the elevated airborne fiber counts encountered during abatement operations have been reduced to the specified level, the VPIH/CIH will secure samples and analyze them according to the following procedures:
 - a. Fibers Counted: "Fibers" referred to in this section shall be either all fibers regardless of composition as counted in the NIOSH 7400 PCM method.
 - b. All clearance air testing samples shall be collected on 0.8µ MCE filters for PCM analysis. Air samples will be collected in areas subject to normal air circulation. A minimum of 5 PCM samples will be collected with at least 3850 Liters of air sampled. All results must be less than 0.01 f/cc for clearance.

3.8 ABATEMENT CLOSEOUT AND CERTIFICATE OF COMPLIANCE

A. Completion of Abatement Work:

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1. After thorough decontamination, complete asbestos abatement work upon meeting the regulated area clearance criteria and fulfilling the following:
 - a. Remove all equipment, materials, and debris from the project area.
 - b. Package and dispose of all asbestos waste as required.
 - c. Fulfill other project closeout requirements as specified elsewhere in this specification.
- B. Certificate of Completion by Contractor: The CPIH shall complete and sign the "Certificate of Completion" in accordance with Attachment 1 at the completion of the abatement and decontamination of the regulated area.
- C. Work Shifts: All work shall generally be done during administrative hours (8:00 AM to 4:30 PM) Monday - Friday excluding Federal Holidays. Any change in the work schedule must be approved in writing by the VA Representative.

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**ATTACHMENT #1
CERTIFICATE OF COMPLETION**

DATE: _____ VA Project #: _____

PROJECT NAME: _____ Abatement Contractor: _____

VAMC/ADDRESS: _____

1. I certify that I have personally inspected, monitored and supervised the abatement work of (specify regulated area or Building):
which took place from / / to / /
2. That throughout the work all applicable requirements/regulations and the VA's specifications were met.
3. That any person who entered the regulated area was protected with the appropriate personal protective equipment and respirator and that they followed the proper entry and exit procedures and the proper operating procedures for the duration of the work.
4. That all employees of the Abatement Contractor engaged in this work were trained in respiratory protection, were experienced with abatement work, had proper medical surveillance documentation, were fit-tested for their respirator, and were not exposed at any time during the work to asbestos without the benefit of appropriate respiratory protection.
5. That I performed and supervised all inspection and testing specified and required by applicable regulations and VA specifications.
6. That the conditions inside the regulated area were always maintained in a safe and healthy condition and the maximum fiber count never exceeded 0.5 f/cc, except as described below.
7. That all abatement work was done in accordance with OSHA requirements and the manufacturer's recommendations.

CPIH/CIH Signature/Date: _____

CPIH/CIH Print Name: _____

Abatement Contractor Signature/Date: _____

Abatement Contractor Print Name: _____

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**ATTACHMENT #2
CERTIFICATE OF WORKER'S ACKNOWLEDGMENT**

PROJECT NAME: _____ DATE: _____

PROJECT ADDRESS: _____

ABATEMENT CONTRACTOR'S NAME: _____

WORKING WITH ASBESTOS CAN BE HAZARDOUS TO YOUR HEALTH. INHALING ASBESTOS HAS BEEN LINKED WITH VARIOUS TYPES OF CANCERS. IF YOU SMOKE AND INHALE ASBESTOS FIBERS, YOUR CHANCES OF DEVELOPING LUNG CANCER IS GREATER THAN THAT OF THE NON-SMOKING PUBLIC.

Your employer's contract with the owner for the above project requires that: You must be supplied with the proper personal protective equipment including an adequate respirator and be trained in its use. You must be trained in safe and healthy work practices and in the use of the equipment found at an asbestos abatement project. You must receive/have a current medical examination for working with asbestos. These things shall be provided at no cost to you. By signing this certificate of worker's acknowledgement you are indicating to the owner that your employer has met these obligations.

RESPIRATORY PROTECTION: I have been trained in the proper use of respirators and have been informed of the type of respirator to be used on the above indicated project. I have a copy of the written Respiratory Protection Program issued by my employer. I have been provided for my exclusive use, at no cost, with a respirator to be used on the above indicated project.

TRAINING COURSE: I have been trained by a third party, State/EPA accredited trainer in the requirements for an AHERA/OSHA Asbestos Abatement Worker training course, 32-hours minimum duration. I currently have a valid State accreditation certificate. The topics covered in the course include, as a minimum, the following:

- Physical Characteristics and Background Information on Asbestos
- Potential Health Effects Related to Exposure to Asbestos
- Employee Personal Protective Equipment
- Establishment of a Respiratory Protection Program
- State of the Art Work Practices
- Personal Hygiene
- Additional Safety Hazards
- Medical Monitoring
- Air Monitoring
- Relevant Federal, State and Local Regulatory Requirements, Procedures, and Standards
- Asbestos Waste Disposal

MEDICAL EXAMINATION: I have had a medical examination within the past 12 months which was paid for by my employer. This examination included: health history, occupational history, pulmonary function test, and may have included a chest x-ray evaluation. The physician issued a positive written opinion after the examination.

Signature: _____

Printed Name: _____

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Social Security Number: _____

Witness: _____

ATTACHMENT #3

**AFFIDAVIT OF MEDICAL SURVEILLANCE, RESPIRATORY PROTECTION AND
TRAINING/ACCREDITATION**

VA PROJECT NAME AND NUMBER: _____

VA MEDICAL FACILITY: _____

ABATEMENT CONTRACTOR'S NAME AND ADDRESS: _____

1. I verify that the following individual

Name: _____ Social Security Number: _____

who is proposed to be employed in asbestos abatement work associated with the above project by the named Abatement Contractor, is included in a medical surveillance program in accordance with 29 CFR 1926.1101(m), and that complete records of the medical surveillance program as required by 29 CFR 1926.1101(m)(n) and 29 CFR 1910.20 are kept at the offices of the Abatement Contractor at the following address.

Address: _____

2. I verify that this individual has been trained, fit-tested and instructed in the use of all appropriate respiratory protection systems and that the person is capable of working in safe and healthy manner as expected and required in the expected work environment of this project.

3. I verify that this individual has been trained as required by 29 CFR 1926.1101(k). This individual has also obtained a valid State accreditation certificate. Documentation will be kept on-site.

4. I verify that I meet the minimum qualifications criteria of the VA specifications for a CPIH.

Signature of CPIH/CIH: _____ Date: _____

Printed Name of CPIH/CIH: _____

Signature of Contractor: _____ Date: _____

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Printed Name of Contractor: _____

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ATTACHMENT #4
ABATEMENT CONTRACTOR/COMPETENT PERSON(S) REVIEW AND ACCEPTANCE OF THE
VA'S ASBESTOS SPECIFICATIONS

VA Project Location: _____

VA Project #: _____

VA Project Description: _____

This form shall be signed by the Asbestos Abatement Contractor Owner and the Asbestos Abatement Contractor's Competent Person(s) prior to any start of work at the VA related to this Specification. If the Asbestos Abatement Contractor's/Competent Person(s) has not signed this form, they shall not be allowed to work on-site.

I, the undersigned, have read VA's Asbestos Specification regarding the asbestos abatement requirements. I understand the requirements of the VA's Asbestos Specification and agree to follow these requirements as well as all required rules and regulations of OSHA/EPA/DOT and State/Local requirements. I have been given ample opportunity to read the VA's Asbestos Specification and have been given an opportunity to ask any questions regarding the content and have received a response related to those questions. I do not have any further questions regarding the content, intent and requirements of the VA's Asbestos Specification.

At the conclusion of the asbestos abatement, I will certify that all asbestos abatement work was done in accordance with the VA's Asbestos Specification and all ACM was removed properly and no fibrous residue remains on any abated surfaces.

Abatement Contractor Owner's Signature _____ Date _____

- - - E N D - - -

SECTION 03 39 35
CONCRETE SEALER

PART 1 - GENERAL

1.1 DESCRIPTION

A. This section includes.

1. Concrete sealer/~~densifier/hardener~~.

~~**1.2 RELATED SECTIONS**~~

~~A. Section 03 30 00 - CAST IN PLACE CONCRETE: Cast in Place Concrete.~~

1.3 REFERENCES

~~A. Corps of Engineers Spec: CECS 03300 4 79.~~

~~B.A. _____~~ USDA approved.

~~C.B. _____~~ Department of Navy, GSA, VA approved.

1.4 SYSTEM

A. Description: A sealer/~~densifier/hardener~~ that will react with concrete surfaces to produce a dense, hydrophobic, insoluble, moisture barrier to seal out contaminants, while hardening and densifying surface.

B. Design Criteria:

1. ACI 302 Class 1 through 4 concrete floors.
2. May be used on Class 5 and 6 floors when used with mineral or metallic aggregate hardeners and toppings, and Class 9, Super-flat floors.
3. Complies with all Federal and State VOC requirements.
4. Independent Test Data, ASTM C 779, Procedure A, reduction of surface abrasion by 50% or more at the 30 minute time interval.

1.5 SUBMITTALS

A. Submit in accordance with Section 01 33 23 SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

B. Product Data: Manufacturer's data sheets on each product to be used, including:

1. Preparation instructions and recommendations.
2. Storage and handling requirements and recommendations.
3. Installation methods.

C. Samples: Furnish representative samples ~~if requested by the Architect~~.

D. Project Specific Information:

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1. Test reports.
2. Certificate of rate of application.
3. Signed warranty.

E. Application Rate: From Installer, based on project requirements.

1.6 QUALITY ASSURANCE

A. Manufacturer Qualifications: Minimum 5 year experience manufacturing similar products.

B. Installer Qualifications: Minimum 2 year experience installing similar products.

1. Approved in writing by manufacturer.

~~** NOTE TO SPECIFIER ** Include a mock up if the project size and/or quality warrant taking such a precaution. The following is one example of how a mock up on a large project might be specified. When deciding on the extent of the mock up, consider all the major different types of work on the project.~~

~~C. Installer Qualifications: Minimum 2 year experience installing similar products.~~

~~D. Mock Up: Provide a mock up for evaluation of surface preparation techniques and application workmanship.~~

- ~~1. Finish areas designated by Architect.~~
- ~~2. Do not proceed with remaining work until workmanship is approved by Architect.~~
- ~~3. Refinish mock up area as required to produce acceptable work.~~

1.7 PRE-INSTALLATION MEETINGS

A. Convene minimum two weeks prior to starting work of this section.

B. At the ~~Architect's~~ Contracting Officer's Representative (COR)'s request, technical personnel shall be available for a pre-job conference to review installation procedures.

1.8 DELIVERY, STORAGE AND HANDLING

A. Deliver to jobsite in sealed, labeled containers.

B. Store and handle to prevent damage to product and environment.

1.9 PROJECT CONDITIONS

A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's recommended limits.

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B. Environmental requirements: Comply with applicable VOC and EPA requirements.

1.10 SEQUENCING

A. Ensure that products of this section are supplied to affected trades in time to prevent interruption of construction progress.

1.11 WARRANTY

A. Warranty: Provide written warranty signed jointly by applicator, manufacturer and contractor. Warrant installation for a period of 10 years from date of substantial completion against dusting from abrasion.

PART 2 PRODUCTS

~~2.1 MANUFACTURERS~~

~~A. Acceptable Manufacturer: L&M(TM) Construction Chemicals, which is located at: 1 LATICRETE Park N.; Bethany, CT 06524-3423; Toll Free Tel: 800-362-3331; Tel: 402-453-6600; Email: request info (info@lmcc.com); Web: www.laticrete.com/lmcc~~

~~** NOTE TO SPECIFIER ** Delete one of the following two paragraphs; coordinate with requirements of Division 1 section on product options and substitutions.~~

~~B. Substitutions: Not permitted.~~

~~C. Requests for substitutions will be considered in accordance with provisions of Section 01 60 00 - Product Requirements.~~

2.2 CONCRETE SEALER/~~DENSIFIER/HARDENER~~

A. Product: ~~SEAL HARD by L&M Construction Chemicals;~~ a solution of 100% active ingredient chemicals which penetrate concrete to seal, densify, dustproof and harden to resist water and oil penetration, and contamination.

1. Active ingredients: 100 percent.
2. Type: Alkali Silicate.
3. Flash Point: None.
4. Specific Gravity: 1.155.
5. VOC: gm/L: 0.
6. Solids minimum: 30 percent.

PART 3 EXECUTION

3.1 EXAMINATION

- A. Do not begin installation until substrates have been properly prepared.
- B. If substrate preparation is the responsibility of another installer, notify Architect the COR of unsatisfactory preparation before proceeding.
- C. Assure surfaces are clean and free of all contaminants, and any film forming curing compounds or sealers.
- D. Existing conditions, prior to installation:
 - 1. Assure concrete has been cured a minimum of 3 days.
 - 2. Assure concrete is clean and free of membrane forming curing compounds and/or other sealers.
 - 3. Concrete is free of laitance, grease, oil and contaminants.
- E. Protect concrete from construction activity staining.

3.2 PREPARATION

- A. Clean surfaces thoroughly prior to installation.
- B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.
- C. Protection: Protect adjacent surfaces/areas from damage due to over spray; especially glass and painted surfaces.

3.3 INSTALLATION

- A. Apply in accord with manufacturer's instructions:
 - 1. Apply directly from sealer container onto prepared surfaces, undiluted.
 - 2. Application equipment: Mechanical "walk-behind" or riding scrubber.
 - 3. Apply at minimum rate of 1 gallon per 150-200 sq. ft.
 - 4. Allow surfaces to remain wet with sealer for 30-60 minutes.
 - 5. Remove excess sealer at end of application procedure by water flushing and then squeegee dry.
 - 6. Apply in 1 coat typical.
 - 7. Apply 2 coats to porous or rough concrete surfaces.
 - 8. Apply 2 coats to floor areas that will be exposed to frequent oil spills.

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B. Leave area broom clean.

3.4 PROTECTION

A. Protect installed products until completion of project.

B. Touch-up, repair or replace damaged products ~~before Substantial~~
Completion.

- - - E N D - - -

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SECTION 05 12 00
STRUCTURAL STEEL FRAMING

SPEC WRITER NOTE: Delete text between
// _____ // not applicable to project.
Edit remaining text to suit project.

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
1. Structural steel shapes, plates, and bars.
 2. Structural pipe.
 3. Bolts, nuts, and washers.

1.2 RELATED REQUIREMENTS

SPEC WRITER NOTE: Update and retain
references only when specified elsewhere
in this section.

- A. Materials Testing And Inspection During Construction: Section 01 45 29, TESTING LABORATORY SERVICES.
- B. Roof Decks Framing: Section 03 51 16, GYPSUM CONCRETE ROOF DECKS.
- C. Floor Decks Framing: Section 03 55 11, GYPSUM CONCRETE FLOOR DECKS.
- D. Steel Joist: Section 05 21 00, STEEL JOIST FRAMING.
- E. Steel Decking: Section 05 31 00, STEEL DECKING.
- F. Composite Steel Deck: Section 05 36 00, COMPOSITE METAL DECKING.
- G. Fireproofing: Section 07 81 00, APPLIED FIREPROOFING.
- H. Steel Framing: // Section 08 44 13, GLAZED ALUMINUM CURTAIN WALLS //
Section 13 34 19, METAL BUILDING SYSTEMS //.
- I. Steel Finishes: Section 09 06 00, SCHEDULE FOR FINISHES.
- J. Steel Support: Section 10 13 00, DIRECTORIES.
- K. Painting: Section 09 91 00, PAINTING.
- L. Steel Piles: Section 31 62 00, DRIVEN PILES.

SPEC WRITER NOTE: AISC has a
certification program confirming that
certified structural steel fabricating
plant has procedures and commitment to
produce fabricated steel of required
quality for given category of structural
steel framing. Consider deleting category
certification if there is minimal amount

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of steel required. Category is STD:
Standard for Building Structures.

1.3 APPLICABLE PUBLICATIONS

- A. Comply with references to extent specified in this section.
- B. American Institute of Steel Construction (AISC):
 - 1. AISC Manual - Steel Construction Manual, 14th Ed.
 - 2. 303-10 - Code of Structural Steel Buildings and Bridges.
 - 3. 360-10: Specification for Structural Steel Buildings.
- C. The American Society of Mechanical Engineers (ASME):
 - 1. B18.22.1-09 - Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers.
- D. American Welding Society (AWS):
 - 1. D1.1/D1.1M-15 - Structural Welding Code - Steel.
- E. ASTM International (ASTM):
 - 1. A6/A6M-14 - General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling.
 - 2. A36/A36M-14 - Carbon Structural Steel.
 - 3. A53/A53M-12 - Pipe, Steel, Black and Hot-Dip, Zinc-Coated, Welded and Seamless.
 - 4. A123/A123M-15 - Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
 - 5. A242/A242M-13 - High-Strength Low-Alloy Structural Steel.
 - 6. A283/A283M-13 - Low and Intermediate Tensile Strength Carbon Steel Plates.
 - 7. A307-14 - Carbon Steel Bolts, Studs, and Threaded Rod 60,000 PSI Tensile Strength.
 - 8. A500/A500M-13 - Cold-Formed Welded and Seamless Carbon Steel Structural Tubing and Rounds and Shapes.
 - 9. A501/A501M-14 - Hot-Formed Welded and Seamless Carbon Steel Structural Tubing and Rounds and Shapes.
 - 10. A572/A572M-15 - High-Strength Low-Alloy Columbium-Vanadium Structural Steel.
 - 11. A992/A992M-15 - Structural Shapes.
 - 12. F2329/F2329M-15 - Zinc Coating, Hot-Dip, Requirements for Application to Carbon and Alloy steel Bolts, Screws, washers, Nuts, and Special Threaded Fasteners.

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13. F3125/F3125M-15 - Standard Specification for High Strength Structural Bolts, Steel and Alloy Steel, Heat Treated, 120 ksi (830 MPa) and 150 ksi (1040 MPa) Minimum Tensile Strength, Inch and Metric Dimensions

F. Master Painters Institute (MPI):

1. No. 18 - Primer, Zinc Rich, Organic.

G. Military Specifications (Mil. Spec.):

1. MIL-P-21035 - Paint, High Zinc Dust Content, Galvanizing, Repair.

H. Occupational Safety and Health Administration (OSHA):

1. 29 CFR 1926.752(e) - Guidelines For Establishing The Components Of A Site-Specific Erection Plan.

2. 29 CFR 1926-2001 - Safety Standards for Steel Erection.

I. Research Council on Structural Connections (RCSC) of The Engineering Foundation:

1. Specification for Structural Joints Using ASTM F3125 Bolts.

1.4 SUBMITTALS

A. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

B. Submittal Drawings:

1. Show size, configuration, and fabrication and installation details.

C. Sustainable Construction Submittals:

SPEC WRITER NOTE: Retain sustainable construction submittals appropriate to product.

1. Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

D. Test Reports: Certify products comply with specifications.

1. Welders' qualifying tests.

E. Certificates: Certify each product complies with specifications.

1. Structural steel.

2. Steel connections.

3. Welding materials.

4. Shop coat primer paint.

F. Qualifications: Substantiate qualifications comply with specifications.

1. Fabricator // with project experience list //.

2. Installer // with project experience list //.

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- 3. Welders and welding procedures.
- G. Delegated Design Drawings and Calculations: Signed and sealed by responsible Architect/Engineer.
 - 1. Connection calculations.

SPEC WRITER NOTE: Include cooling tower supports, only when required and not detailed on structural drawings.

- 2. // Cooling Tower Supports. //
- H. Record Surveys: Signed and sealed by responsible surveyor or engineer.

1.5 QUALITY ASSURANCE

- A. Fabricator Qualifications: AISC Quality Certification participant designated as AISC Certified Plant, Category STD.
 - 1. Regularly fabricates specified products.
 - 2. Fabricated specified products with satisfactory service on five similar installations for minimum five years.
 - a. // Project Experience List: Provide contact names and addresses for completed projects. //
- B. Installer Qualifications: AISC Quality Certification Program participant designated as AISC-Certified Erector, Category ACSE.
 - 1. Regularly installs specified products.
 - 2. Installed specified products with satisfactory service on five similar installations for minimum five years.
 - a. // Project Experience List: Provide contact names and addresses for completed projects. //
- C. Before commencement of Work, ensure steel erector provides written notification required by OSHA 29 CFR 1926.752(e). Submit a copy of the notification to Contracting Officer's Representative.
- D. Welders and Welding Procedures Qualifications: AWS D1.1/D1.1M.

1.6 WARRANTY

SPEC WRITER NOTE: Always retain construction warranty. FAR includes Contractor's one year labor and material warranty.

- A. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

PART 2 - PRODUCTS

2.1 SYSTEM PERFORMANCE

- A. Delegated Design: Prepare submittal documents including design calculations and drawings signed and sealed by registered design professional, licensed in state where project is located.
- B. Design structural steel framing connections complying with specified performance:
 - 1. Load Capacity: // Resist loads indicated on drawings // Resist full capacity of supported framing member //. Account for connection and member loads and eccentricities.
 - a. Request additional design criteria when necessary to complete connection design.
 - 2. Configuration: Design and detail all connections for each member size, steel grade and connection type to resist the loads and reactions indicated on the drawings or specified herein. Use details consistent with details shown on drawings, supplementing where necessary. The details shown on drawings are conceptual and do not indicate the required weld sizes or number of bolts unless specifically noted. Use rational engineering design and standard practice in detailing, accounting for all loads and eccentricities in both the connection and the members. Promptly notify the Contracting Officer Representative of any location where the connection design criteria is not clearly indicated. The design of all connections is subject to the review and acceptance of the Contracting Officer's Representative. Submit structural calculations prepared and sealed by a qualified engineer registered in the state where the project is located. Submit calculations for review before preparation of detail drawings.

SPEC WRITER NOTE: Include cooling tower supports when required for project. Add other specific components requiring delegated design.

- C. Design cooling tower supports meeting specified performance:
 - 1. Coordinate design criteria with cooling tower specified in Section 23 65 00, COOLING TOWERS.

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2. Design Loads: Support operating cooling tower maximum dead load and minimum dead load when combined with lateral loads. Resist wind, snow, // seismic // loads indicated on drawings and supplied by Cooling Tower manufacturer.
3. Configuration: Design framing member sizes, dimensions, and locations to suit cooling tower load, size, and configuration. Submit detailed drawings and design calculations, prepared by a registered Professional Engineer, for approval before members are fabricated.

2.2 MATERIALS

SPEC WRITER NOTE: For each structural steel shape, select reference standard. First two subparagraphs are typical; last subparagraph is considered specialty steel.

A. W-Shapes:

1. ASTM A992/A992M.
2. ASTM A572/A572M; Grade // 50 // 60 // 65 //.
3. ASTM A529; Grade // 50 // 55 //.

B. // M // S // -Shapes:

1. ASTM A36/A36M.
2. ASTM A572/A572M; Grade // 50 // 60 // 65 //.
3. ASTM A529; Grade // 50 // 55 //.

C. Channel and Angles:

1. ASTM A36/A36M.
2. ASTM A572/A572M; Grade // 50 // 60 // 65 //.
3. ASTM A529; Grade // 50 // 55 //.

D. Plates and Bars:

1. ASTM A36/A36M.
2. ASTM A572/A572M; Grade // 50 // 60 // 65 //.
3. ASTM A529; Grade // 50 // 55 //.

E. Hollow Structural Sections:

SPEC WRITER NOTE: Select first subparagraph for cold formed and second subparagraph for hot formed HSS framing.

1. ASTM A500/A500M.

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- 2. ASTM A501/A501M.
- F. Structural Pipe: ASTM A53/A53M, Grade B.
- G. Bolts, Nuts and Washers: // Galvanized for galvanized framing // and // plain finish for other framing //.
 - 1. High-strength bolts, including nuts and washers: ASTM F3125.
 - 2. Bolts and nuts, other than high-strength: ASTM A307, Grade A.
 - 3. Plain washers, other than those in contact with high-strength bolt heads and nuts: ASME B18.22.1.
- H. Welding Materials: AWS D1.1, type to suit application.

2.3 PRODUCTS - GENERAL

- A. Basis of Design: Section 09 06 00, SCHEDULE FOR FINISHES.
- B. Sustainable Construction Requirements:

SPEC WRITER NOTE:

- 1 Specify products containing greatest recycled content practicable to maximize material recovery. See EPA Comprehensive Procurement Guidelines (CPG) for guidance about individual products and available recycled content. Section 01 81 13 sets overall project recycled content requirements.
- 2. Steel recycled content depends upon furnace type. AISC reports industry wide 32 percent for basic oxygen furnace and 93 percent for electric arc furnace.

- 1. Steel Recycled Content: 30 percent total recycled content, minimum.

SPEC WRITER NOTE:

- 1. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS includes comprehensive product list setting VOC limits for low-emitting materials.
- 2. Retain subparagraphs applicable to products specified in this section.

- 2. Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:
 - a. Paints and coatings.

2.4 FABRICATION

- A. Fabricate structural steel according to Chapter M, AISC 360.
- B. Shop and Field Connections:

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1. Weld connections according to AWS D1.1/D1.1M. Welds shall be made only by welders and welding operators who have been previously qualified by tests as prescribed in AWS D1.1 to perform type of work required.
2. High-Strength Bolts: High-strength bolts tightened to a bolt tension minimum 70 percent of their minimum tensile strength. Tightening done with properly calibrated wrenches, by turn-of-nut method or by use of direct tension indicators (bolts or washers). Tighten bolts in connections identified as slip-critical using Direct Tension Indicators. Twist-off torque bolts are not an acceptable alternate fastener for slip critical connections.

2.5 FINISHES

SPEC WRITER NOTE: Retain field painting only when steel is shop primed or galvanized.

- A. Shop Priming:
 1. Prime paint structural steel according to AISC 303, Section 6.
 - a. Interstitial Space Structural Steel: Prime paint, unless indicated to receive sprayed on fireproofing.
- B. Shop Finish Painting: Apply primer and finish paint as specified in Section 09 91 00, PAINTING.
- C. Do not paint:
 1. Surfaces within 50 mm (2 inches) of field welded joints.
 2. Surfaces indicated to be encased in concrete.
 3. Surfaces receiving sprayed on fireproofing.
 4. Beam top flanges receiving shear connector studs applied.

SPEC WRITER NOTE: Retain galvanizing for structural steel exterior applications.

- D. Structural Steel Galvanizing: ASTM A123/A123M, hot dipped, after fabrication. Touch-up after erection: Clean and wire brush any abraded and other spots worn through zinc coating, including threaded portions of bolts and welds and touch-up with galvanizing repair paint.
 1. Galvanize structural steel framing installed at exterior locations.
- E. Bolts, Nuts, and Washers Galvanizing: ASTM F2329, hot-dipped.

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2.6 ACCESSORIES

- A. General: Shop paint steel according to AISC 303, Section 6.
- B. Finish Paint System: Primer and finish as specified in Section 09 91 00, PAINTING.
- C. Galvanizing Repair Paint: MPI No. 18.

PART 3 - EXECUTION

3.1 ERECTION

- A. Erect structural steel according to AISC 303 and AISC 360.
- B. Set structural steel accurately at locations and elevations indicated on drawings.
- C. Maintain erection tolerances of structural steel within AISC 303 requirements.
 - 1. Pour Stop Elevation Tolerance: 6 mm (1/4 inch), maximum, before concrete placement.
- D. Weld and bolt connections as specified for shop connections.

SPEC WRITER NOTE: Retain field painting only when steel is shop primed or galvanized.

3.2 FIELD PAINTING

- A. After welding, clean and prime weld areas to match adjacent finish.
- B. Touch-up primer damaged by construction operations.
- C. Apply galvanizing repair paint to galvanized coatings damaged by construction operations.
- D. Finish Painting: As specified in Section 09 91 00, PAINTING.

3.3 FIELD QUALITY CONTROL

- A. Record Survey:
 - 1. Engage registered land surveyor or registered civil engineer as specified in Section 01 00 00, GENERAL REQUIREMENTS to perform survey.
 - 2. Measure and record structural steel framing plumbness, level, and alignment after completing bolting and welding and before installation of work supported by structural steel.
 - 3. Identify deviations from allowable tolerances specified in AISC Manual.

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SECTION 05 31 00
STEEL DECKING

SPEC WRITER NOTE:

1. Delete text between // // not applicable to project. Edit remaining text to suit project.
2. Refer to Section 05 36 00, COMPOSITE METAL DECKING for floor deck requirements.

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Single pan fluted metal form deck supporting concrete fill as roof substrate.
2. Corrugated metal form deck supporting concrete fill as roof substrate.
3. Single pan fluted metal roof deck as roof substrate.
4. Acoustic metal roof deck as roof substrate.

1.2 RELATED WORK

SPEC WRITER NOTE: Update and retain references only when specified elsewhere in this section.

- A. Section 01 81 13. SUSTAINABLE CONSTRUCTION REQUIREMENTS.
- B. Section 05 21 00, STRUCTURAL STEEL FRAMING: Structural Steel Shapes.
- C. Section 09 06 00, SCHEDULE FOR FINISHES: Color.
- D. Section 09 91 00, PAINTING: Finish Painting.

1.3 APPLICABLE PUBLICATIONS

- A. Comply with references to extent specified in this section.
- B. AISI - American Iron and Steel Institute.
S100-16.....Specification for the Design of Cold-formed
Steel Structural Members.
- C. American Welding Society (AWS):
D1.1/D1.1M-20.....Structural Welding Code - Steel.
1.3/D1.3M-18..... Structural Welding Code - Sheet Steel.
- D. ASTM International (ASTM):
A36/A36M-19.....Standard Specification for Carbon Structural
Steel.

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- A653/A653M-20.....Standard Specification for Steel Sheet,
Zinc-Coated (Galvanized) or Zinc-Iron
Alloy-Coated (Galvannealed) by the Hot-Dip
Process.
- A1008/A1008M-20.....Standard Specification for Steel, Sheet,
Cold-Rolled, Carbon, Structural, High-Strength
Low-Alloy, High-Strength Low-Alloy with
Improved Formability, Solution Hardened, and
Baked Hardenable.
- C423-17.....Standard Test Method for Sound Absorption and
Sound Absorption Coefficients by the
Reverberation Room Method.
- E119-20.....Standard Test Methods for Fire Tests of
Building Construction and Materials.
- E. FM Global (FM):
1-28-15.....Wind Design.
Factory Mutual Research Approval Guide.
- F. Master Painters Institute (MPI):
No. 18.....Primer, Zinc Rich, Organic.
- G. Military Specifications (Mil. Spec.):
MIL-P-21035B..... Paint, High Zinc Dust Content, Galvanizing
Repair.
- H. Steel Deck Institute (SDI):
No. 31-07.....Design Manual for Composite Deck, Form Decks,
and Roof Decks.
- I. UL LLC (UL):
Listed Online Certifications Directory.
580.....Tests for Uplift Resistance of Roof Assemblies.

1.4 SUBMITTALS

- A. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA,
AND SAMPLES. All items indicated below are required submittals
requiring Contracting Officer's Representative (COR) review and
approval.
- B. Submittal Drawings:
1. Show layout, connections to supporting members, anchorage, sump
pans, accessories, deck openings and reinforcements.

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2. Show similar information necessary for completing installation as shown and specified, including supplementary framing, ridge and valley plates, cant strips, cut openings, special jointing or other accessories.
3. Show welding, side lap, closure, deck reinforcing and closure reinforcing details.
4. Show openings required for work of other trades, including openings not shown on structural drawings. Indicate where temporary shoring is required to satisfy design criteria.

C. Manufacturer's Literature and Data:

1. Description of each product.
2. Show steel decking section properties and structural characteristics.

D. Sustainable Construction Submittals:

SPEC WRITER NOTE: Retain sustainable construction submittals appropriate to product.

1. Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

E. Certificates: Certify each product complies with specifications.

1. Fire Resistance Product Listing: For each metal deck type and thickness supporting concrete slab or fill.

SPEC WRITER NOTE: Retain paragraph below when UL uplift resistance and fire resistance requirements are specified.

2. Show steel decking is UL Listed for specified application.

SPEC WRITER NOTE: Retain NRC certification for acoustic roof deck.

3. Show noise reduction coefficient test results.

F. Qualifications: Substantiate qualifications comply with specifications.

1. Welders and welding procedures.

SPEC WRITER NOTE: Retain paragraph below for Factory Mutual Insurance requirement or similar form by others.

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- G. Insurance Certification: Assist the Government in preparation and submittal of roof installation acceptance certification as may be necessary in connection with fire and extended coverage insurance.

1.5 QUALITY ASSURANCE

SPEC WRITER NOTE: Delete paragraph below if FM compliance not required for roof deck. Coordinate with "Insurance Certification" above.

- A. FM Listing: Provide metal roof deck units which have been evaluated by Factory Mutual Global and are listed in "Factory Mutual Research Approval Guide" for "Class 1" fire rated construction.
- B. Welders and Welding Procedures Qualifications: AWS D1.3/D1.3M.

1.6 WARRANTY

SPEC WRITER NOTE: Always retain construction warranty. FAR includes Contractor's one year labor and material warranty.

- A. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

PART 2 - PRODUCTS

2.1 SYSTEM PERFORMANCE

- A. Design steel decking and accessories according to AISI S100.

SPEC WRITER NOTE: Retain one paragraph for wind uplift resistance. Class 90 is maximum for UL Listing. Greater uplift resistance ratings are available from FM.

1. Wind Uplift Resistance and Corner Conditions:
 - a. Eave Overhang: 2.1 kPa (45 per square foot), minimum.
 - b. Other Roof Areas: 1.4 kPa (30 per square foot), minimum.
2. Wind Uplift Resistance and Corner Conditions: UL 580, Class // 60 // 90 // or higher UL Class required by wind loading in the location of the project.
3. Wind Uplift Resistance and Corner Conditions: FM 1-28; Class // 1-60 // 1-90 // or higher UL Class required by wind loading in the location of the project.
4. Fire Resistance: ASTM E119; as component of // 1 // 2 // hour rated roof assembly.

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SPEC WRITER NOTE: Retain NRC requirement
for acoustic roof deck.

- 5. Noise Reduction Coefficient (NRC): Minimum 0.90 when tested according to ASTM C423.
- 6. Design side and end closures and attachment to supporting steel to safely support wet weight of concrete and construction loads.
- 7. Cantilever Closure Deflection: 3 mm (1/8 inch), maximum.

2.2 MATERIALS

SPEC WRITER NOTE: Select galvanized
coating class G60 for normal conditions
and G90 for corrosive environments such
as salt air.

- A. Galvanized Steel Sheet: ASTM A653/A653M; // G60 // G90 // coating.

SPEC WRITER NOTE: Include painted steel
sheet when finish painting is required.

- B. Painted Steel Sheet: ASTM A1008/A1008M, Grade C or D, shop primed.
- C. Primer for Shop Painted Sheets: Manufacturer's standard primer (2 coats). When finish painting of steel decking is specified in Section 09 91 00, PAINTING primer coating shall be compatible with specified finish painting.
- D. Steel Shapes: ASTM A36/A36M.
- E. Acoustic Deck: Cellular deck profile, SDI Publication No. 31.

SPEC WRITER NOTE: Delete paragraph below
if acoustical deck is not required.

- F. Acoustic Insulation: Manufacturer's standard mineral fiber type, profile matching deck flute profile.

2.3 PRODUCTS - GENERAL

- A. Basis of Design: Section 09 06 00, SCHEDULE FOR FINISHES.
- B. Sustainable Construction Requirements:

SPEC WRITER NOTE:

- 1. Specify products containing greatest recycled content practicable to maximize material recovery. See EPA Comprehensive Procurement Guidelines (CPG) for guidance about individual products and available recycled content. Section 01 81 13 sets overall project recycled content requirements.

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2. Steel recycled content depends upon furnace type. AISC reports industry wide 32 percent for basic oxygen furnace and 93 percent for electric arc furnace.

1. Steel Recycled Content: 30 percent total recycled content, minimum.

2.4 METAL ROOF DECK

- A. Metal Roof Deck: // UL Listed // FM Global approved // as metal roof deck panels.

1. Steel decking of the type, depth, thickness, and section properties as shown.

SPEC WRITER NOTE: Two types of form decks: Type 1 form deck is 50 or 75 mm (2 or 3 inches) deep deck similar to composite metal decking but with no shear deformations, and is used for reinforced and thick slabs where unshored composite is not enough. Type 2 form deck is shallow 14 or 25 mm (9/16 or 1 inch) corrugated type (centering) used with closely spaced bar joists.

- B. Metal Form Deck - Type 1: Single pan fluted units as permanent form for reinforced concrete slabs.

1. Depth and Thickness: As indicated on drawings.
2. Material: // Galvanized sheet steel. // Painted sheet steel. //

- C. Metal Form Deck - Type 2: Corrugated deck units as permanent form for reinforced concrete slabs.

1. Depth and Thickness: As indicated on drawings.
2. Material: // Galvanized sheet steel. // Painted sheet steel. //

- D. Metal Roof Deck: Single pan fluted units with flat horizontal top surfaces as permanent support for superimposed loads.

SPEC WRITER NOTE: Retain required deck styles. Wide Rib (Type B) is typical.

1. Deck Style:

- a. Wide Rib (Type B) deck.
- b. Intermediate Rib (Type F) deck.
- c. Narrow Rib (Type A) deck.
- d. Deep Rib (Type N) deck.

2. Depth and Thickness: As indicated on drawings.

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SPEC WRITER NOTE: Select material and finish. Use painted roof deck only for dry climate with short construction exposure.

3. Material: // Galvanized sheet steel. // Painted sheet steel. //

SPEC WRITER NOTE: Retain paragraph below for acoustic roof deck. Verify NRC value.

E. Acoustic Metal Roof Deck Units: Single-pan fluted units with perforated vertical webs.

1. Depth and Thickness: As indicated on drawings.

2. Material: // Galvanized sheet steel. // Painted sheet steel. //

3. Provide acoustical insulation to fill roof deck flutes.

F. Do not use steel deck for hanging supports of building components including suspended ceilings, electrical light fixtures, plumbing, heating, or air conditioning pipes or ducts or electrical conduits.

G. Include integral system for steel decking units used for interstitial levels.

1. Provide system suitable for simple point of attachment for light duty hanger devices.

2. Provide system suitable to allow for flexibility for attaching hangers for support of suspended ceilings, electrical, plumbing, heating, or air conditioning items, weight not to exceed 50 kg/m² (10 psf).

3. Provide a minimum spacing pattern of 300 mm (12 inches) on centers longitudinally and 600 mm (24 inches) on centers transversely.

4. Maximum allowable load suspended from any hanger: 23 kg (50 pounds).

5. System consisting of fold-down type hanger tabs or lip hanger is acceptable.

2.5 FABRICATION

A. Fabricate steel decking in sufficient lengths to extend over 3 or more supports, except for interstitial levels.

1. Cut metal deck units to proper length in shop.

B. Fabricate accessories required to complete installation of steel decking.

1. Exposed to View: Fabricate from sheet steel matching metal decking.

2. Concealed from View: Fabricate from galvanized sheet steel.

C. Sheet Metal Accessories:

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1. Metal Cover Plates: For end-abutting decking, to close gaps at changes in deck direction, columns, walls and openings.
 - a. Sheet Steel: Minimum 1.0 mm (0.04 inch) thick.
2. Continuous Sheet Metal Edging: At openings, concrete slab edges and roof deck edges.
 - a. Sheet Steel: Minimum 1.0 mm (0.04 inch) thick.
3. Metal Closure Strips: For openings between decking and other construction. Form to configurations required to provide tight-fitting closures at open ends of flutes and sides of decking.
 - a. Sheet Steel: Minimum 1.0 mm (0.04 inch) thick.
4. Ridge and Valley Plates: Minimum 100 mm (4 inch) wide ridge and valley plates where roof slope exceeds 1/24 (1/2 inch per foot).
 - a. Sheet Steel: Minimum 1.0 mm (0.04 inch) thick.
5. Cant Strips: Provide bent metal 45 degree leg cant strips where indicated on the drawings. Fabricate cant strips with minimum 125 mm (5 inch) face width.
 - a. Sheet Steel: Minimum 0.8 mm (0.03 inch) thick.
6. Seat Angles for Deck: Provide where beam does not frame into column.
7. Sump Pans for Roof Drains: Fabricated from single piece galvanized sheet steel with level bottoms and sloping sides to direct water flow to drain. Provide sump pans of adequate size to receive roof drains and with bearing flanges minimum 75 mm (3 inches) wide. Recess pans minimum 38 mm (1-1/2 inches) below roof deck surface, unless otherwise shown or required by deck configuration. Drain holes will be field cut.
 - a. Sheet Steel: Minimum 1.7 mm (0.06 inch) thick.

2.6 FINISHES

- A. Shop prime painted sheet steel with two coats of primer.

2.7 ACCESSORIES

- A. Primer: Manufacturer's standard primer compatible with finish painting specified in Section 09 91 00, PAINTING.
- B. Welding Materials: AWS D1.1, type to suit application.
- C. Galvanizing Repair Paint: MPI No. 18.
- D. Touch-Up Paint: Match shop finish.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Examine and verify substrate suitability for product installation.

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- B. Protect existing construction and completed work from damage.
- C. Remove contaminants from structural steel surfaces where steel decking will be welded.
- D. Verify structural steel framing installation is completed, plumbed, and aligned with temporary bracing installed where required.
- E. Coordinate with structural steel erector to prevent overloading of structural members when placing steel decking for installation.

3.2 ERECTION

- A. Do not use floor deck units for storage or working platforms until permanently secured. Do not overload deck units once placed. Replace deck units that become damaged after erection and before casting concrete at no cost additional to the Government.
- B. Place steel decking at right angles to supporting members with ends located over supports.
- C. Lap end joints 50 mm (2 inches), minimum.

SPEC WRITER NOTE: FM Global requires screwed side laps (not welded) for roof deck thinner than 1 mm (0.04 inches). Welded side laps may be mandatory in seismic areas.

D. Fluted Form Deck Fastening:

- 1. Fasten form deck to steel supporting members by welding.
 - a. Welds: 16 mm (5/8 inch) diameter puddle welds or elongated welds of equal strength.
 - b. Weld Spacing: Maximum 300 mm (12 inches) on center with minimum two welds per unit at each support.
 - c. Where two units abut, fasten each unit individually to supporting steel framework.
- 2. End Closure Fastening: Tack weld or self-tapping No. 8 or larger machine screws at 900 mm (3 feet) on center.
 - a. Longitudinal End Closure Fastening: Tack weld only.
- 3. Weld side laps of adjacent decking units.
 - a. Fastener Locations: Mid-span and maximum 900 mm (3 feet) on center.

E. Corrugated Form Deck Fastening:

SPEC WRITER NOTE: Check weld pattern and side lap fastener spacing to develop diaphragm shear strength required.

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1. Weld end laps of corrugated form deck units in valley of side lap and at middle of sheet.
 - a. Weld Spacing: Maximum 380 mm (15 inches) on center.
2. Weld corrugated deck to intermediate supports in X-pattern. Weld in valley of side laps on every other support and in valley of center corrugation on remaining support.
 - a. Weld Spacing: Maximum 760 mm (30 inches) on center.

SPEC WRITER NOTE: Check weld pattern and side lap fastener spacing to develop diaphragm shear strength required.

F. Roof Deck Fastening:

1. Fasten decking to steel supporting members by welding.
 - a. Welds: 16 mm (5/8 inch) diameter puddlewelds or elongated welds of equal strength.
 - b. Weld Spacing: Maximum 300 mm (12 inches) on center at every support. Use closer spacing where required for lateral force resistance by diaphragm action.
2. Fasten split or partial decking panels to structure in every valley.
3. Fasten decking to each supporting member at ribs where side laps occur.
 - a. Power driven fasteners is acceptable in lieu of welding if strength equivalent to welding specified above is provided. Submit test data and design calculations verifying equivalent design strength.
4. Mechanically fasten decking side laps with self-tapping No. 8 or larger machine screws.
 - a. Fastener Locations: Mid-span and maximum 900 mm (3 feet) on center.
5. Provide additional fastening necessary to comply with // UL Listing // FM Approval // for specified performance.

SPEC WRITER NOTE: Delete next 2 paragraphs below if corrugated type form deck is not required.

G. Cutting and Fitting:

1. Field cut steel decking to accommodate columns and other penetrating items.

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2. Cut openings located and dimensioned on Structural Drawings.
 3. Coordinate openings for other penetrations shown on approved submittal drawings but not shown on Structural Drawings.
 - a. Cut and reinforce required opening.
 4. Make cuts neat and trim using metal saw, drill or punch-out device. Cutting with torches is prohibited.
 5. Do not make cuts in the metal deck that are not shown on the approved metal decking submittal drawings.
 - a. When additional openings are required, submit scaled drawing, locating required opening and other openings and supports in immediate area.
 - b. Do not cut the opening until drawing is approved by Contracting Officer's Representative.
 - c. Provide additional reinforcing and framing required for opening.
 - d. Failure to comply with these requirements is cause for rejection of the work and removal and replacement of the affected steel decking.
 6. Opening Reinforcement: Provide additional metal reinforcement and closure pieces as required for strength, continuity of decking, and support of other work.
- H. Touch up damaged factory finishes.
1. Apply galvanizing repair paint to damaged galvanized surfaces.
 2. Apply touch up paint to damaged shop painted surfaces.

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SECTION 05 50 00
METAL FABRICATIONS

~~SPEC WRITER NOTE:~~

- ~~1. Delete between //____// if not applicable to project. Also delete any other item or paragraph not applicable in the section and renumber the paragraphs.~~
- ~~2. Use terminology as listed and coordinate drawings to use the same term.~~
- ~~3. This section includes custom-fabricated items.~~
- ~~4. List additional items required by the project.~~
- ~~5. VA standard details numbers are identified in list in parenthesis following each item; delete numbers from project specification.~~

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This section specifies items and assemblies fabricated from structural steel shapes and other materials as shown and specified.
- B. Items specified.

- ~~1. Support for Wall and Ceiling Mounted Items: (SD055000-01, SD055000-02, SD102113-01, SD102600-01, SD123100-01 & SD123100-02)~~
- ~~2. Frames:~~
- ~~3. Guards~~
- ~~4. Covers and Frames for Pits and Trenches.~~
- ~~5. Gratings~~
- ~~6. Loose Lintels~~
- ~~7. Shelf Angles~~
- ~~8. Gas Racks~~
- ~~9. Plate Door Sill~~
- ~~10. Safety Nosings~~
11. Ladders
12. Railings+
- ~~13. Catwalks and Platforms~~
- ~~14. Trap Doors with Ceiling Hatch~~
- ~~15. Sidewalk Access Doors~~

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~~16. Screened Access Doors~~

~~17. Steel Counter or Bench Top Frame and Leg~~

~~18. Steel Pipe Bollards~~

1.2 RELATED WORK

~~A. Railings attached to steel stairs: Section 05 51 00, METAL STAIRS.~~

~~B. Colors, finishes, and textures: Section 09 06 00, SCHEDULE FOR FINISHES.~~

C. Prime and finish painting: Section 09 91 00, PAINTING.

~~D. Stainless steel corner guards: Section 10 26 00, WALL AND DOOR PROTECTION.~~

1.3 SUBMITTALS

A. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

B. Manufacturer's Literature and Data. +

Grating, each type	Floor plate
Trap door	Wheel guards
Ceiling hatch	Sidewalk Access door
Manhole Covers	Safety nosing

C. Shop Drawings:

1. Each item specified, showing complete detail, location in the project, material and size of components, method of joining various components and assemblies, finish, and location, size and type of anchors.

2. Mark items requiring field assembly for erection identification and furnish erection drawings and instructions.

3. Provide templates and rough-in measurements as required.

D. Manufacturer's Certificates:

1. Anodized finish as specified.
2. Live load designs as specified.

E. Design Calculations for specified live loads including dead loads.

F. Furnish setting drawings and instructions for installation of anchors to be preset into concrete and masonry work, and for the positioning of items having anchors to be built into concrete or masonry construction.

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1.4 QUALITY ASSURANCE

- A. Each manufactured product shall meet, as a minimum, the requirements specified, and shall be a standard commercial product of a manufacturer regularly presently manufacturing items of type specified.
- B. Each product type shall be the same and be made by the same manufacturer.
- C. Assembled product to the greatest extent possible before delivery to the site.
- D. Include additional features, which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product.

1.5 APPLICABLE PUBLICATIONS

- A. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.
- B. American Society of Mechanical Engineers (ASME):
 - B18.6.1-97.....Wood Screws
 - B18.2.2-87(R2010).....Square and Hex Nuts
- C. American Society for Testing and Materials (ASTM):
 - A36/A36M-14.....Structural Steel
 - A47-99(R2014).....Malleable Iron Castings
 - A48-03(R2012).....Gray Iron Castings
 - A53-12.....Pipe, Steel, Black and Hot-Dipped, Zinc-Coated
Welded and Seamless
 - A123-15.....Zinc (Hot-Dip Galvanized) Coatings on Iron and
Steel Products
 - A240/A240M-15.....Standard Specification for Chromium and
Chromium-Nickel Stainless Steel Plate, Sheet
and Strip for Pressure Vessels and for General
Applications.
 - A269-15.....Seamless and Welded Austenitic Stainless Steel
Tubing for General Service
 - A307-14.....Carbon Steel Bolts and Studs, 60,000 PSI
Tensile Strength
 - A391/A391M-07(R2015)....Grade 80 Alloy Steel Chain
 - A786/A786M-15.....Rolled Steel Floor Plate

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- B221-14.....Aluminum and Aluminum-Alloy Extruded Bars,
Rods, Wire, Shapes, and Tubes
- B456-11.....Electrodeposited Coatings of Copper Plus Nickel
Plus Chromium and Nickel Plus Chromium
- B632-08.....Aluminum-Alloy Rolled Tread Plate
- C1107-13.....Packaged Dry, Hydraulic-Cement Grout
(Nonshrink)
- D3656-13.....Insect Screening and Louver Cloth Woven from
Vinyl-Coated Glass Yarns
- F436-16.....Hardened Steel Washers
- F468-06(R2015).....Nonferrous Bolts, Hex Cap Screws, Socket Head
Cap Screws and Studs for General Use
- F593-13.....Stainless Steel Bolts, Hex Cap Screws, and
Studs
- F1667-15.....Driven Fasteners: Nails, Spikes and Staples
- D. American Welding Society (AWS):
- D1.1-15.....Structural Welding Code Steel
- D1.2-14.....Structural Welding Code Aluminum
- D1.3-18.....Structural Welding Code Sheet Steel
- E. National Association of Architectural Metal Manufacturers (NAAMM)
- AMP 521-01(R2012).....Pipe Railing Manual
- AMP 500-06.....Metal Finishes Manual
- MBG 531-09(R2017).....Metal Bar Grating Manual
- MBG 532-09.....Heavy Duty Metal Bar Grating Manual
- F. Structural Steel Painting Council (SSPC)/Society of Protective
Coatings:
- SP 1-15.....No. 1, Solvent Cleaning
- SP 2-04.....No. 2, Hand Tool Cleaning
- SP 3-04.....No. 3, Power Tool Cleaning
- G. Federal Specifications (Fed. Spec):
- RR-T-650E.....Treads, Metallic and Nonmetallic, Nonskid

PART 2 - PRODUCTS

~~SPEC WRITER NOTE:-~~

- ~~1. Update materials requirements to agree with applicable requirements (types, grades, classes,) specified in the referenced Applicable Publications.~~
- ~~2. Coordinate with structural design criteria and specify live loads where~~

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~~fork lift and other vehicles will
subject plates, gratings and trap
doors in floors or pavements to
concentrated loads.~~

2.1 DESIGN CRITERIA

- A. In addition to the dead loads, design fabrications to support the following live loads unless otherwise specified.
- B. Ladders and Rungs: 120 kg (250 pounds) at any point.
- C. Railings and Handrails: 900 N (200 pounds) in any direction at any point.
- ~~D. Floor Plates, Gratings, Covers, Trap Doors, Catwalks, and Platforms: 500 kg/m² (100 pounds per square foot). Use _____ kg (pounds) for concentrated loads. Use _____ kg/m² (pounds per square foot) for vehicle loads in the following areas : _____.~~
- ~~E. Manhole Covers: 1200 kg/m² (250 pounds per square foot).~~

2.2 MATERIALS

- A. Structural Steel: ASTM A36.
- B. Stainless Steel: ASTM A240, Type 302 or 304.
- C. Aluminum, Extruded: ASTM B221, Alloy 6063-T5 unless otherwise specified. For structural shapes use alloy 6061-T6 and alloy 6061-T4511.
- D. Floor Plate:
 - 1. Steel ASTM A786.
 - 2. Aluminum: ASTM B632.
- E. Steel Pipe (Bollard): ASTM A53.
 - 1. Galvanized for exterior locations.
 - 2. Type S, Grade A unless specified otherwise.
 - 3. NPS (inside diameter) as shown.
- F. Cast-Iron: ASTM A48, Class 30, commercial pattern.
- G. Malleable Iron Castings: A47.
- H. Primer Paint: As specified in Section 09 91 00, PAINTING.
- I. Stainless Steel Tubing: ASTM A269, type 302 or 304.
- J. Modular Channel Units:
 - 1. Factory fabricated, channel shaped, cold formed sheet steel shapes, complete with fittings bolts and nuts required for assembly.
 - 2. Form channel within turned pyramid shaped clamping ridges on each side.

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3. Provide case hardened steel nuts with serrated grooves in the top edges designed to be inserted in the channel at any point and be given a quarter turn so as to engage the channel clamping ridges. Provide each nut with a spring designed to hold the nut in place.
4. Factory finish channels and parts with oven baked primer when exposed to view. Channels fabricated of ASTM A525, G90 galvanized steel may have primer omitted in concealed locations. Finish screws and nuts with zinc coating.
5. Fabricate snap-in closure plates to fit and close exposed channel openings of not more than 0.3 mm (0.0125 inch) thick stainless steel.

K. Grout: ASTM C1107, pourable type.

L. Insect Screening: ASTM D3656.

2.3 HARDWARE

A. Rough Hardware:

1. Furnish rough hardware with a standard plating, applied after punching, forming and assembly of parts; galvanized, cadmium plated, or zinc-coated by electro-galvanizing process. Galvanized G-90 where specified.
2. Use G90 galvanized coating on ferrous metal for exterior work unless non-ferrous metal or stainless is used.

B. Fasteners:

1. Bolts with Nuts:
 - a. ASME B18.2.2.
 - b. ASTM A307 for 415 MPa (60,000 psi) tensile strength bolts.
 - c. ASTM F468 for nonferrous bolts.
 - d. ASTM F593 for stainless steel.
2. Screws: ASME B18.6.1.
3. Washers: ASTM F436, type to suit material and anchorage.
4. Nails: ASTM F1667, Type I, style 6 or 14 for finish work.

2.4 FABRICATION GENERAL

A. Material

1. Use material as specified. Use material of commercial quality and suitable for intended purpose for material that is not named or its standard of quality not specified.
2. Use material free of defects which could affect the appearance or service ability of the finished product.

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B. Size:

1. Size and thickness of members as shown.
2. When size and thickness is not specified or shown for an individual part, use size and thickness not less than that used for the same component on similar standard commercial items or in accordance with established shop methods.

C. Connections

1. Except as otherwise specified, connections may be made by welding, riveting or bolting.
2. Field riveting will not be approved.
3. Design size, number and placement of fasteners, to develop a joint strength of not less than the design value.
4. Holes, for rivets and bolts: Accurately punched or drilled and burrs removed.
5. Size and shape welds to develop the full design strength of the parts connected by welds and to transmit imposed stresses without permanent deformation or failure when subject to service loadings.
6. Use Rivets and bolts of material selected to prevent corrosion (electrolysis) at bimetallic contacts. Plated or coated material will not be approved.
7. Use stainless steel connectors for removable members machine screws or bolts.

D. Fasteners and Anchors

1. Use methods for fastening or anchoring metal fabrications to building construction as shown or specified.
2. Where fasteners and anchors are not shown, design the type, size, location and spacing to resist the loads imposed without deformation of the members or causing failure of the anchor or fastener, and suit the sequence of installation.
3. Use material and finish of the fasteners compatible with the kinds of materials which are fastened together and their location in the finished work.
4. Fasteners for securing metal fabrications to new construction only, may be by use of threaded or wedge type inserts or by anchors for welding to the metal fabrication for installation before the concrete is placed or as masonry is laid.

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5. Fasteners for securing metal fabrication to existing construction or new construction may be expansion bolts, toggle bolts, power actuated drive pins, welding, self drilling and tapping screws or bolts.

E. Workmanship

1. General:

- a. Fabricate items to design shown.
- b. Furnish members in longest lengths commercially available within the limits shown and specified.
- c. Fabricate straight, true, free from warp and twist, and where applicable square and in same plane.
- d. Provide holes, sinkages and reinforcement shown and required for fasteners and anchorage items.
- e. Provide openings, cut-outs, and tapped holes for attachment and clearances required for work of other trades.
- f. Prepare members for the installation and fitting of hardware.
- g. Cut openings in gratings and floor plates for the passage of ducts, sumps, pipes, conduits and similar items. Provide reinforcement to support cut edges.
- h. Fabricate surfaces and edges free from sharp edges, burrs and projections which may cause injury.

2. Welding:

- a. Weld in accordance with AWS.
- b. Welds shall show good fusion, be free from cracks and porosity and accomplish secure and rigid joints in proper alignment.
- c. Where exposed in the finished work, continuous weld for the full length of the members joined and have depressed areas filled and protruding welds finished smooth and flush with adjacent surfaces.
- d. Finish welded joints to match finish of adjacent surface.

3. Joining:

- a. Miter or butt members at corners.
- b. Where frames members are butted at corners, cut leg of frame member perpendicular to surface, as required for clearance.

4. Anchors:

- a. Where metal fabrications are shown to be preset in concrete, weld 32 x 3 mm (1-1/4 by 1/8 inch) steel strap anchors, 150 mm (6

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inches) long with 25 mm (one inch) hooked end, to back of member at 600 mm (2 feet) on center, unless otherwise shown.

- b. Where metal fabrications are shown to be built into masonry use 32 x 3 mm (1-1/4 by 1/8 inch) steel strap anchors, 250 mm (10 inches) long with 50 mm (2 inch) hooked end, welded to back of member at 600 mm (2 feet) on center, unless otherwise shown.
5. Cutting and Fitting:
- a. Accurately cut, machine and fit joints, corners, copes, and miters.
 - b. Fit removable members to be easily removed.
 - c. Design and construct field connections in the most practical place for appearance and ease of installation.
 - d. Fit pieces together as required.
 - e. Fabricate connections for ease of assembly and disassembly without use of special tools.
 - f. Joints firm when assembled.
 - g. Conceal joining, fitting and welding on exposed work as far as practical.
 - h. Do not show rivets and screws prominently on the exposed face.
 - i. The fit of components and the alignment of holes shall eliminate the need to modify component or to use exceptional force in the assembly of item and eliminate the need to use other than common tools.

~~SPEC WRITER NOTE: If more than one finish is used on project, specify applicable finish under the item. Coordinate paragraphs to delete finishes not used.~~

F. Finish:

- 1. Finish exposed surfaces in accordance with NAAMM AMP 500 Metal Finishes Manual.
- 2. Aluminum: NAAMM AMP 501.
 - a. Mill finish, AA-M10, as fabricated, use unless specified otherwise.
 - b. Clear anodic coating, AA-C22A41, chemically etched medium matte, with Architectural Class 1, 0.7 mils or thicker.
 - c. Colored anodic coating, AA-C22A42, chemically etched medium matte with Architectural Class 1, 0.7 mils or thicker.

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- d. Painted: AA-C22R10.
- 3. Steel and Iron: NAAMM AMP 504.
 - a. Zinc coated (Galvanized): ASTM A123, G90 unless noted otherwise.
 - b. Surfaces exposed in the finished work:
 - 1) Finish smooth rough surfaces and remove projections.
 - 2) Fill holes, dents and similar voids and depressions with epoxy type patching compound.
 - c. Shop Prime Painting:
 - 1) Surfaces of Ferrous metal:
 - a) Items not specified to have other coatings.
 - b) Galvanized surfaces specified to have prime paint.
 - c) Remove all loose mill scale, rust, and paint, by hand or power tool cleaning as defined in SSPC-SP2 and SP3.
 - d) Clean of oil, grease, soil and other detrimental matter by use of solvents or cleaning compounds as defined in SSPC-SP1.
 - e) After cleaning and finishing apply one coat of primer as specified in Section 09 91 00, PAINTING.
 - 2) Non ferrous metals: Comply with MAAMM-500 series.
- 4. Stainless Steel: NAAMM AMP-504 Finish No. 4.

SPEC WRITER NOTE: Specify items to
receive chromium plating.

- 5. Chromium Plating: ASTM B456, satin or bright as specified, Service Condition No. SC2.

G. Protection:

- 1. Insulate aluminum surfaces that will come in contact with concrete, masonry, plaster, or metals other than stainless steel, zinc or white bronze by giving a coat of heavy-bodied alkali resisting bituminous paint or other approved paint in shop.
- 2. Spot prime all abraded and damaged areas of zinc coating which expose the bare metal, using zinc rich paint on hot-dip zinc coat items and zinc dust primer on all other zinc coated items.

2.5 SUPPORTS

A. General:

- 1. Fabricate ASTM A36 structural steel shapes as shown.
- 2. Use clip angles or make provisions for welding hangers and braces to overhead construction.

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3. Field connections may be welded or bolted.

~~B. For Ceiling Hung Toilet Stall:~~

- ~~1. Use a continuous steel channel above pilasters with hangers centered over pilasters.~~
- ~~2. Make provision for installation of stud bolts in lower flange of channel.~~
- ~~3. Provide a continuous steel angle at wall and channel braces spaced as shown.~~
- ~~4. Use threaded rod hangers.~~
- ~~5. Provide diagonal angle brace where the suspended ceiling over toilet stalls does not extend to side wall of room.~~

~~//6. Provide supports for ceiling hung pilasters at dressing booths and entrance screen to toilet room similar to support for toilet stall pilasters. //~~

~~C. For Wall Mounted Items:~~

- ~~1. For items supported by metal stud partitions.~~
- ~~2. Steel strip or hat channel minimum of 1.5 mm (0.0598 inch) thick.~~
- ~~3. Steel strip minimum of 150 mm (6 inches) wide, length extending one stud space beyond end of item supported.~~
- ~~4. Steel hat channels where shown. Flange cut and flattened for anchorage to stud.~~
- ~~5. Structural steel tube or channel for grab bar at water closets floor to structure above with clip angles or end plates formed for anchors.~~
- ~~6. Use steel angles for thru wall counters. Drill angle for fasteners at ends and not over 100 mm (4 inches) on center between ends.~~

~~D. For Trapeze Bars:~~

- ~~1. Construct assembly above ceilings as shown and design to support not less than a 340 kg (750 pound) working load at any point.~~
- ~~2. Fabricate trapeze supports as shown, with all exposed members, including screws, nuts, bolts and washers, fabricated of stainless steel.~~
- ~~3. Fabricate concealed components of structural steel shapes unless shown otherwise.~~
- ~~4. Stainless steel ceiling plate drilled for eye bolt.~~
- ~~5. Continuously weld connections where welds shown.~~

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- ~~6. Use modular channel where shown with manufacturers bolts and fittings.~~
 - ~~a. Weld ends of steel angle braces to steel plates and secure to modular channel units as shown. Drill plates for anchor bolts.~~
 - ~~b. Fabricate eye bolt, special clamp bolt, and plate closure full length of modular channel at ceiling line and secure to modular channel unit with manufacturers standard fittings.~~
- ~~E. For Intravenous Track and Cubical Curtain Track:~~
 - ~~1. Fabricate assembly of steel angle as shown.~~
 - ~~2. Drill angle bent ends for anchor screws to acoustical suspension system and angle for hanger wires.~~
 - ~~3. Provide pipe sleeve welded to angle.~~
- ~~F. Supports at Ceiling for Radiographic (x-ray) Equipment:~~
 - ~~1. Fabricate hangers braces, and track of modular channel units assembly as shown.~~
 - ~~2. Fabricate steel plates for anchor to structure above.~~
 - ~~3. Drill bent plates for bolting at mid height at concrete beams.~~
- ~~G. For Operating Room Light:~~
 - ~~1. Fabricate as shown to suit equipment furnished.~~
 - ~~2. Drill leveling plate for light fixture bolts.~~
- ~~H. Supports in Orthopedic Brace Shop:~~
 - ~~1. Fabricate from 25 mm (one inch) steel pipe, fasten to steel angles above and extend to a point 150 mm (6 inches) below finished ceiling.~~
 - ~~2. Lower end of the pipe shall have a standard pipe thread.~~
 - ~~3. Provide an escutcheon plate at ceiling.~~
- ~~I. Supports for Accordion Partition Tracks, Exercise Equipment, and Items at Various Conditions at Suspended Ceilings:~~
 - ~~1. Fabricate of structural steel shapes as shown.~~
 - ~~2. Drill for anchor bolts of suspended item.~~
- ~~J. Supports for Communion Rail Posts in Chapel:~~
 - ~~1. Fabricate one steel plate support for each post as shown.~~
 - ~~2. Drill for fasteners.~~

~~2.6 FRAMES~~

- ~~A. Elevator Entrance Wall Opening.~~
 - ~~1. Fabricate of channel shapes, plates, and angles as shown.~~
 - ~~2. Weld or bolt head to jamb as shown.~~

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~~3. Weld clip angles to bottom of frame and top of jamb members extended to structure above for framed construction.~~

~~a. Provide holes for anchors.~~

~~b. Weld head to jamb members.~~

~~B. Channel Door Frames:~~

~~1. Fabricate of structural steel channels of size shown.~~

~~2. Miter and weld frames at corners.~~

~~3. Where anchored to masonry or embedded in concrete, weld to back of frame at each jamb, 5 mm (3/16 inch) thick by 44 mm (1-3/4 inch) wide steel strap anchors with ends turned 50 mm (2 inches), and of sufficient length to extend at least 300 mm (12 inches) into wall. Space anchors 600 mm (24 inches) above bottom of frame and 600 mm (24 inches) o.c. to top of jamb. Weld clip angles to bottom of jambs and provide holes for expansion bolts.~~

~~4. Where anchored to concrete or masonry in prepared openings, drill holes at jambs for anchoring with expansion bolts. Weld clip angles to bottom of frame and provide holes for expansion bolt anchors as shown. Drill holes starting 600 mm (24 inches) above bottom of frame and 600 mm (24 inches) o.c. to top of jamb and at top of jamb. Provide pipe spacers at holes welded to channel.~~

~~5. Where closure plates are shown, continuously weld them to the channel flanges.~~

~~6. Weld continuous 19 x 19 x 3 mm (3/4 x 3/4 x 1/8 inch) thick steel angles to the interior side of each channel leg at the head and jambs to form a caulking groove.~~

~~7. Prepare frame for installation of hardware specified in Section 08-71 00, DOOR HARDWARE.~~

~~a. Cut a slot in the lock jamb to receive the lock bolt.~~

~~b. Where shown use continuous solid steel bar stops at perimeter of frame, weld or secure with countersunk machine screws at not more than 450 mm (18 inches) on center.~~

~~C. Frames for Breech Opening:~~

~~1. Fabricate from steel channels, or combination of steel plates and angles to size and contour shown.~~

~~2. Weld strap anchors on back of frame at not over 600 mm (2 feet) on centers for concrete or masonry openings.~~

~~D. Frames for Lead Lined Doors:~~

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- ~~1. Obtain accurate dimensions and templates from suppliers of lead lined doors, finish hardware, and hollow steel door frames.~~
- ~~2. Fabricate as shown for use in connection with lead lined doors.~~
- ~~3. Deliver assembled frames with removable shipping spreaders at top and bottom.~~
- ~~4. Extend angles at jambs from floor to structural slab above. At floors of interstitial spaces, terminate jamb sections and provide anchors as shown.~~
- ~~5. Continuously weld plates and reinforcements to frame members and head members of angle frames between jambs.~~
- ~~6. Weld strap anchors, not over 600 mm (24 inches) on centers, to the back of angles for embedment in masonry or concrete unless shown otherwise.~~
- ~~7. Type 15 Door Frames:~~
 - ~~a. Structural steel angle frames with plate or bar full height to heads. Extend reinforcing at hinge cutouts two inches beyond cutout.~~
 - ~~b. Fabricate top anchorage to beam side at mid height.~~
 - ~~c. Weld clip angles to both legs of angle at top and bottom.~~
 - ~~d. Drill clips and plates, at top and bottom for anchoring jamb angles with two 9 mm (3/8 inch) expansion bolts at each location.~~
 - ~~e. Cut rabbet for pivot hinges and lock strike.~~

~~SPEC WRITER NOTE: See Section 10 26 00, WALL AND DOOR PROTECTION for interior guards used in finished spaces. Use structural steel items for service and shop areas. Use paragraph B where door frame is not used.~~

~~2.7 GUARDS~~

~~A. Wall Corner Guards:~~

- ~~1. Fabricate from steel angles and furnish with anchors as shown.~~
- ~~2. Continuously weld anchor to angle.~~

~~B. Guard Angles for Overhead Doors:~~

- ~~1. Cut away top portion of outstanding leg of angle and extend remaining portion of angle up wall.~~
- ~~2. Weld filler piece across head of opening to jamb angles.~~
- ~~3. Make provisions for fasteners and anchorage.~~

~~C. Channel Guard at Loading Platform:~~

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~~1. Fabricate from steel channel of size shown.~~

~~2. Weld anchors to channels as shown.~~

~~3. Drill channel for bumper anchor bolts.~~

~~D. Edge Guard Angles for Openings in slabs.~~

~~1. Fabricate from steel angles of sizes and with anchorage shown.~~

~~2. Where size of angle is not shown, provide 50 x 50 x 6 mm (2 x 2 x 1/4 inch) steel angle with 32 x 5 mm (1-1/4 x 3/16 inch) strap anchors, welded to back.~~

~~3. Miter or butt angles at corners and weld.~~

~~4. Use one anchor near end and three feet on centers between end anchors.~~

~~E. Wheel Guards:~~

~~1. Construct wheel guards of not less than 16 mm (5/8 inch) thick cast iron.~~

~~2. Provide corner type, with flanges for bolting to walls.~~

2.8 COVERS AND FRAMES FOR PITS AND TRENCHES

~~A. Fabricate covers to support live loads specified.~~

~~B. Galvanized steel members after fabrication in accordance with ASTM A123, G-90 coating.~~

~~C. Steel Covers:~~

~~1. Use 6 mm (1/4 inch) thick floor plate for covers unless otherwise shown. Use gratings where shown as specified in paragraph GRATINGS. Use smooth floor plate unless noted otherwise.~~

~~2. Provide clearance at all sides to permit easy removal of covers.~~

~~3. Make cutouts within 6 mm (1/4 inch) of penetration for passage of pipes and ducts.~~

~~4. Drill covers for flat head countersunk screws.~~

~~5. Make cover sections not to exceed 2.3 m² (25 square feet) in area and 90 kg (200 pounds) in weight.~~

~~6. Fabricate trench cover sections not be over 900 mm (3 feet) long and if width of trench is more than 900 mm (3 feet) or over, equip one end of each section with an angle or "T" bar stiffener to support adjoining plate.~~

~~7. Use two, 13 mm (1/2 inch) diameter steel bar flush drop handles for each cover section.~~

~~SPEC WRITER NOTE: Use cast iron for manhole covers and frames not specified~~

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~~under mechanical and electrical sections.
Use circular manhole covers only. Use
cast iron for exterior work.~~

~~D. Cast Iron Covers~~

- ~~1. Fabricate covers to support live loads specified.~~
- ~~2. Fabricate from ASTM A48, cast iron, 13 mm (1/2 inch) minimum metal thickness, cast with stiffeners as required.~~
- ~~3. Fabricate as flush type with frame, reasonably watertight and be equipped with flush type lifting rings. Provide seals where watertight covers noted.~~
- ~~4. Make covers in sections not over 90 kg (200 pounds) except round covers.~~

~~E. Steel Frames:~~

- ~~1. Form frame from structural steel angles as shown. Where not shown use 63 x 63 x 6 mm (2-1/2 x 2-1/2 x 1/4 inch) angles for frame openings over 1200 mm (4 feet) long and 50 x 50 x 6 mm (2 ix 2 x 1/4 inch) for frame openings less than 1200 mm (4 feet).~~
- ~~2. Fabricate intermediate supporting members from steel "T's" or angles; located to support cover section edges.~~
- ~~3. Where covers are required use steel border bars at frames so that top of cover will be flush with frame and finish floor.~~
- ~~4. Weld steel strap anchors to frame. Space straps not over 600 mm (24 inches) o.c., not shown otherwise between end anchors. Use 6 x 25 x 200 mm (1/4 x 1 x 8 inches) with 50 mm (2 inch) bent ends strap anchors unless shown otherwise.~~
- ~~5. Drill and tap frames for screw anchors where plate covers occur.~~

~~SPEC WRITER NOTE: Verify frames are
detailed to show shapes and anchors.~~

~~F. Cast Iron Frames:~~

- ~~1. Fabricate from ASTM A48 cast iron to shape shown.~~
- ~~2. Provide anchors for embedding in concrete, spaced near ends and not over 600 mm (24 inches) apart.~~

~~SPEC WRITER NOTE:~~

- ~~1. Design bar gratings in accordance with specified ANSI/NAAMM MBC-531 or MBC-532.~~
- ~~2. Use manufacture's standard load data for selection where applicable.~~

~~2.9 GRATINGS~~

- ~~A. Fabricate gratings to support live loads specified and a concentrated load as specified.~~
- ~~B. Provide clearance at all sides to permit easy removal of grating.~~
- ~~C. Make cutouts in gratings with 6 mm (1/4 inch) minimum to 25 mm (one inch) maximum clearance for penetrations or passage of pipes and ducts. Edge band cutouts.~~
- ~~D. Fabricate in sections not to exceed 2.3 m² (25 square feet) in area and 90 kg (200 pounds) in weight.~~
- ~~E. Fabricate sections of grating with end banding bars.~~
- ~~F. Fabricate angle frames and supports, including anchorage as shown.~~
 - ~~1. Fabricate intermediate supporting members from "T's" or angles.~~
 - ~~2. Locate intermediate supports to support grating section edges.~~
 - ~~3. Fabricate frame to finish flush with top of grating.~~
 - ~~4. Locate anchors at ends and not over 600 mm (24 inches) o.c.~~
 - ~~5. Butt or miter, and weld angle frame at corners.~~

~~SPEC WRITER NOTE:~~

- ~~1. Riveted grating performs better where subject to vehicle loads. Specify locations where required.~~
- ~~2. Considered serrated bars where hazards are constant, specify locations of serrated gratings.~~

~~G. Steel Bar Gratings:~~

- ~~1. Fabricate grating using steel bars, frames, supports and other members shown in accordance with Metal Bar Grating Manual.~~
- ~~2. Galvanize steel members after fabrication in accordance with ASTM A123, G-90 for exterior gratings, gratings in concrete floors, and interior grating where specified.~~
- ~~3. Interior gratings: Prime paint unless specified galvanized.~~
- ~~//4. Use serrated bars for exterior gratings and interior gratings in the following areas: //~~
- ~~//5. Use riveted grating in the following areas: //~~

~~H. Aluminum Bar Gratings:~~

- ~~1. Fabricate grating and frame assembly from aluminum as shown in accordance with Metal Bar Grating Manual.~~
- ~~2. Use 25 x 5 mm (1 x 3/16 inch) minimum size bearing bars.~~
- ~~3. Mill finish unless specified otherwise.~~

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~~//4. Use serrated bars for exterior gratings and interior gratings in the following areas: //~~

~~SPEC WRITER NOTE:~~

- ~~1. Identify interior surfaces and finish.~~
- ~~2. Clearly detail railing connection to grating assembly for specified live loads.~~

~~I. Plank Gratings:~~

- ~~1. Conform to Fed. Spec. RR-G-1602.~~
- ~~2. Manufacturers standard widths, lengths and side channels to meet live load requirements.~~
- ~~3. Galvanize exterior steel gratings ASTM A123, G-90 after fabrication.~~
- ~~4. Fabricate interior steel gratings from galvanized steel sheet, ASTM A525, where bearing on concrete or masonry.~~
- ~~5. Fabricate other interior grating from steel sheet and finish with shop prime paint. Prime painted galvanized sheet may be used.~~

~~J. Cast Iron Gratings:~~

- ~~1. Fabricate gratings to support a live load of 23940 Pa (500 pounds per square foot).~~
- ~~2. Fabricate gratings and frames for gutter type drains from cast iron conforming to ASTM A48.~~
- ~~3. Fabricate gratings in section not longer than 1200 mm (4 feet) or over 90 kg (200 pounds) and fit so as to be readily removable.~~

~~SPEC WRITER NOTE: Coordinate with structural section and drawings for specifying sizes.~~

~~2.10 LOOSE LINTELS~~

- ~~A. Furnish lintels of sizes shown. Where size of lintels is not shown, provide the sizes specified.~~
- ~~B. Fabricate lintels with not less than 150 mm (6 inch) bearing at each end for nonbearing masonry walls, and 200 mm (8 inch) bearing at each end for bearing walls.~~
- ~~C. Provide one angle lintel for each 100 mm (4 inches) of masonry thickness as follows except as otherwise specified or shown.~~
 - ~~1. Openings 750 mm to 1800 mm (2 1/2 feet to 6 feet) — 100 x 90 x 8 mm (4 x 3 1/2 x 5/16 inch).~~
 - ~~2. Openings 1800 mm to 3000 mm (6 feet to 10 feet) — 150 x 90 x 9 mm (6 x 3 1/2 x 3/8 inch).~~

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- ~~D. For 150 mm (6 inch) thick masonry openings 750 mm to 3000 mm (2-1/2 feet to 10 feet) use one angle 150 x 90 x 9 mm (6 x 3-1/2 x 3/8 inch).~~
- ~~E. Provide bearing plates for lintels where shown.~~
- ~~F. Weld or bolt upstanding legs of double angle lintels together with 19 mm (3/4 inch bolts) spaced at 300 mm (12 inches) on centers.~~
- ~~G. Insert spreaders at bolt points to separate the angles for insertion of metal windows, louver, and other anchorage.~~
- ~~H. Where shown or specified, punch upstanding legs of single lintels to suit size and spacing of anchor bolts.~~
- ~~I. Elevator Entrance:~~
 - ~~1. Fabricate lintel from plate bent to channel shape, and provide a minimum of 100 mm (4 inch) bearing each end.~~
 - ~~2. Cut away the front leg of the channel at each end to allow for concealment behind elevator hoistway entrance frame.~~

~~SPEC WRITER NOTE Coordinate with structural section for shelf angles part of steel framing this paragraph is for angles to concrete framing.~~

~~2.11 SHELF ANGLES~~

- ~~A. Fabricate from steel angles of size shown.~~
- ~~B. Fabricate angles with horizontal slotted holes for 19 mm (3/4 inch) bolts spaced at not over 900 mm (3 feet) on centers and within 300 mm (12 inches) of ends.~~
- ~~C. Provide adjustable malleable iron inserts for embedded in concrete framing.~~

~~SPEC WRITER NOTE:~~

- ~~1. Where doors on a curb open onto a roof area over base flashing provide a plate door sill to protect base flashing and counter flashing.~~
- ~~2. Use checkered plate formed to channel shape.~~
- ~~3. Verify details to show channel projecting beyond face of wall 100 mm (4 inches), full width of door opening, and at least 225 mm (9 inches) above top of roof surface; allowing for 200 mm (8 inch) minimum height of base flashing and counter flashing.~~

~~2.12 PLATE DOOR SILL~~

- ~~A. Fabricate of checkered plate as detailed.~~

~~1. Aluminum Plate: ASTM B632, 3 mm (0.125 inch) thick.~~

~~2. Steel Plate: ASTM A786, 3 mm (0.125 inch thick), galvanized G90.~~

~~B. Fabricate for anchorage with flat head countersunk bolts at each end and not over 300 mm (12 inches), o.c.~~

~~2.13 SAFETY NOSINGS~~

~~A. Fed. Spec. RR-T-650, Type C.~~

~~1. Aluminum: Class 2, Style 2.~~

~~2. Cast iron: Class 4.~~

~~B. Fabricate nosings for exterior use from cast aluminum, and nosings for interior use from either cast aluminum or cast iron. Use one Class throughout.~~

~~C. Fabricate nosings approximately 100 mm (4 inches) wide with not more than 9 mm (3/8 inch) nose.~~

~~D. Provide nosings with integral type anchors spaced not more than 100 mm (4 inches) from each end and intermediate anchors spaced approximately 375 mm (15 inches) on center.~~

~~E. Fabricate nosings to extend within 100 mm (4 inches) of ends of concrete stair treads except where shown to extend full width.~~

~~F. Fabricate nosings to extend full width between stringers of metal stairs and full width of door openings.~~

~~G. On curved steps fabricate to terminate at point of curvature of steps having short radius curved ends.~~

~~SPEC WRITER NOTE:~~

~~1. Do not use ladders or ships ladders where stairs are required by OSHA for access by maintenance personnel to equipment except for elevator pits.~~

~~2. Roofs require stairs to equipment.~~

2.14 LADDERS

A. Steel Ladders:

1. Fixed-rail type with steel rungs shouldered and headed into and welded to rails.

2. Fabricate angle brackets of 50 mm (2 inch) wide by 13 mm (1/2 inch) thick steel; brackets spaced maximum of 1200 mm (4 feet) apart and of length to hold ladder 175 mm (7 inches) from wall to center of rungs. Provide turned ends or clips for anchoring.

3. Provide holes for anchoring with expansion bolts through turned ends and brackets.

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4. Where shown, fabricate side rails curved, twisted and formed into a gooseneck.
5. Galvanize exterior ladders after fabrication, ASTM A123, G-90.

~~SPEC WRITER NOTE: Verify details, show size and dimensions of components, or specify.~~

~~B. Aluminum Ladders:~~

- ~~1. Fixed rail type, constructed of structural aluminum, with mill finish.~~
- ~~2. Fabricate side rails and rungs of size and design shown, with the rungs shouldered and headed into and welded to the rails.~~
- ~~3. Where shown fabrication side rails curved, twisted and formed into gooseneck.~~
- ~~4. Fabricate angle brackets at top and bottom and intermediate brackets where shown. Drill for bolting.~~

C. Ladder Rungs:

1. Fabricate from 25 mm (one inch) diameter steel bars.
2. Fabricate so that rungs will extend at least 100 mm (4 inches) into wall with ends turned 50 mm (2 inches), project out from wall 175 mm (7 inches), be 400 mm (16 inches) wide and be designed so that foot cannot slide off end.
3. Galvanized after fabrication, ASTM A123, G-90 rungs for exterior use and for access to pits.

~~SPEC WRITER NOTE:~~

- ~~1. Specify railings attached to steel stairs in Section 05 51 00, METAL STAIRS.~~
- ~~2. Specify loose handrail and other railings in this section, verify railings are detailed for compliance with NFPA 101 in public areas.~~

2.15 RAILINGS

- A. In addition to the dead load design railing assembly to support live load specified.
- B. Fabrication General:
 1. Provide continuous welded joints, dressed smooth and flush.
 2. Standard flush fittings, designed to be welded, may be used.
 3. Exposed threads will not be approved.
 4. Form handrail brackets to size and design shown.

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5. Exterior Post Anchors.

- a. Fabricate tube or pipe sleeves with closed ends or plates as shown.
- b. Where inserts interfere with reinforcing bars, provide flanged fittings welded or threaded to posts for securing to concrete with expansion bolts.
- c. Provide heavy pattern sliding flange base plate with set screws at base of pipe or tube posts. ~~// Base plates are not required on pipe sleeves where ornamental railings occur. //~~

~~6. Interior Post Anchors:~~

- ~~a. Provide flanged fittings for securing fixed posts to floor with expansion bolts, unless shown otherwise.~~
- ~~b. Weld or thread flanged fitting to posts at base.~~
- ~~c. For securing removable posts to floor, provide close fitting sleeve insert or inverted flange base plate with stud bolts or rivets concrete anchor welded to the base plate.~~
- ~~d. Provide sliding flange base plate on posts secured with set screws.~~
- ~~e. Weld flange base plate to removable posts set in sleeves.~~

C. Handrails/Guardrails:

1. Close free ends of rail with flush metal caps welded in place except where flanges for securing to walls with bolts are shown.
2. Make provisions for attaching handrail brackets to wall, posts, and handrail as shown.

D. Steel Pipe Railings:

1. Fabricate of steel pipe with welded joints.
2. Number and space of rails as shown.
3. Space posts for railings not over 1800 mm (6 feet) on centers between end posts.
4. Form handrail brackets from malleable iron.
5. Fabricate removable sections with posts at end of section.
6. Removable Rails:
 - a. Provide "U" shape brackets at each end to hold removable rail as shown. Use for top and bottom horizontal rail when rails are joined together with vertical members.

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- b. Secure rail to brackets with 9 mm (3/8 inch) stainless steel through bolts and nuts at top rail only when rails joined with vertical members.
 - c. Continuously weld brackets to post.
 - d. Provide slotted bolt holes in rail bracket.
 - e. Weld bolt heads flush with top of rail.
 - f. Weld flanged fitting to post where posts are installed in sleeves.
7. Opening Guard Rails:
- a. Fabricate rails with flanged fitting at each end to fit between wall opening jambs.
 - b. Design flange fittings for fastening with machine screws to steel plate anchored to jambs.
 - c. Fabricate rails for floor openings for anchorage in sleeves.

~~8. Gates:~~

- ~~a. Fabricate from steel pipe as specified for railings.~~
- ~~b. Fabricate gate fittings from either malleable iron or wrought steel.~~
- ~~c. Hang each gate on suitable spring hinges of clamp on or through bolted type. Use bronze hinges for exterior gates.~~
- ~~d. Provide suitable stops, so that gate will swing as shown.~~
- ~~//e. Provide padlock eyes where shown. //~~

~~9. Chains:~~

- ~~a. Chains: ASTM A391, Grade 63, straight link style, normal size chain bar 8 mm (5/16 inch) diameter, eight links per 25 mm (foot) and with boat type snap hook on one end, and through type eye bolt on other end.~~
- ~~b. Fabricate eye bolt for attaching chain to pipe posts, size not less than 9 mm (3/8 inch) diameter.~~
- ~~c. Fabricate anchor at walls, for engagement of snap hook of either a 9 mm (3/8 inch) diameter eye bolt or punched angle.~~
- ~~d. Galvanize chain and bolts after fabrication.~~

~~E. Aluminum Railings:~~

- ~~1. Fabricate from extruded aluminum.~~
- ~~2. Use tubular posts not less than 3 mm (0.125 inch) wall thickness for exterior railings.~~

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- ~~3. Punch intermediate rails and bottom of top rails for passage of posts and machine to a close fit.~~
- ~~4. Where shown use extruded channel sections for top rail with 13 mm (1/2 inch) thick top cover plates and closed ends.~~
- ~~5. Fabricate brackets of extruded or wrought aluminum as shown.~~
- ~~6. Fabricate stainless pipe sleeves with closed bottom at least six inches deep having internal dimensions at least 13 mm (1/2 inch) greater than external dimensions of posts where set in concrete.~~

F. Stainless Steel Railings:

1. Fabricate from 38 mm (1-1/2 inches) outside diameter stainless steel tubing, ASTM A269, having a wall thickness of 1.6 mm (0.065 inch).
2. Join sections by an internal connector to form hairline joints where field assembled.
3. Fabricate with continuous welded connections.
4. Fabricate brackets of stainless steel to design shown.
5. Fabricate stainless steel sleeves at least 150 mm (6 inches) deep having internal dimensions at least 13 mm (1/2 inch) greater than external dimensions of post.

~~SPEC WRITER NOTE: If custom designed ornamental railings are used on the project write paragraph describing the requirements not shown and insert here.~~

~~//G. Ornamental Railings: //~~

~~SPEC WRITER NOTE:~~

- ~~1. Include catwalks, platforms, railings, ladders, supports and hangers only when not specified in structural steel specification.~~
- ~~2. Coordinate to insure steel stairs for catwalks are specified in Section 05 51 00, METAL STAIRS.~~
- ~~3. Design catwalks to support live load specified.~~
- ~~4. Design catwalks to conform to OSHA safety requirements.~~

~~2.16 CATWALKS~~

- ~~A. Fabricate catwalks including platforms, railings, ladders, supports and hangers, and arrangement of members as shown on drawings.~~
- ~~B. Fabricate stairs as specified in Section 05 51 00, METAL STAIRS.~~
- ~~C. Fabricate steel ladders as specified under paragraph LADDERS unless shown otherwise.~~

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~~D. Fabricate steel pipe railings as specified under paragraph RAILINGS.~~

~~E. Catwalk and platforms floor surfaces as shown.~~

- ~~1. Steel gratings as specified under paragraph gratings, either bar or plank type.~~
- ~~2. Steel floor plate.~~
- ~~3. Aluminum floor plate.~~

~~F. Prime paint catwalk system.~~

~~2.17 TRAP DOOR AND FRAMES WITH CEILING HATCH~~

~~A. Design to support a live load as specified.~~

~~B. Frames:~~

- ~~1. Fabricate steel angle frame to set in concrete slabs and design to set flush with finished concrete slab or curb. If not shown use 63 x 63 x 6 mm (2-1/2 x 2-1/2 x 1/4 inch) angles.~~
- ~~2. Miter steel angles at corners and weld together.~~
- ~~3. Weld steel bar stops to vertical leg of frame, to support doors flush with the top of the frame.~~
- ~~4. Weld steel strap anchors on each side not over 600 mm (24 inches) on center to the backs of the frames. If not shown use 6 x 50 x 200 mm (1/4 x 2 x 8 inch) long straps with 50 mm (2 inch bent) ends.~~
- ~~5. Form frames from steel angles with welded corners for reinforcing and bracing of well lining and support of ceiling hatch.~~

~~C. Covers:~~

- ~~1. Use 6 mm (1/4 inch) thick steel floor plate.~~
- ~~2. Where double leaf covers are shown, reinforce at meeting edges.~~
- ~~3. Use wrought steel hinges with fixed brass pins.~~
 - ~~a. Weld to cover.~~
 - ~~b. Secure to frame with machine screws.~~
- ~~4. Where ladders occur, install hinges on the side opposite the ladder.~~
- ~~5. Provide two bar type drop handles, flush with cover when closed for each leaf.~~

~~SPEC WRITER NOTE: Use well linings where ceiling hatch occurs or as closure of interstitial space when no ceiling hatch occurs.~~

~~D. Well Lining:~~

- ~~1. Fabricate well linings, for access through concrete floor slabs and suspended ceilings, from hatch to ceiling hatch or ceiling openings.~~

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- ~~2. Use steel sheet and shapes of size and thickness as shown. If not shown use 1.5 mm (0.0598 inch) thick steel sheet.~~
- ~~3. If not shown use 50 x 50 x 6 mm (2 x 2 x 1/4 inch) angle braces from ceiling level on each side angled at 45 degrees to structure above.~~
- ~~4. Use 25 x 25 x 3 mm (1 x 1 x 1/8 inch) angle bottom flange trim welded to well lining where no ceiling hatch occurs.~~

~~E. Ceiling Hatch:~~

- ~~1. Construct hatch with "T" or angle frame designed to support edge of ceiling and hatch, weld to well lining.~~
- ~~2. Form hatch panels of 3 mm (1/8 inch) steel, 5 mm (3/16 inch) aluminum or 1 mm (0.0359 inch) thick steel of pan type construction with 25 mm (one inch) of mineral fiber insulation between.~~
- ~~3. Use counter balance device, hinges, latch, hangers and other accessories required for installation and operation of hatch with not over 90 N (20 pounds) of force.~~
- ~~4. Fabricate panels flush and reinforced to remain flat.~~
- ~~5. Locate hatch panel flush with frame.~~

~~F. Finish with baked on prime coat.~~

~~SPEC WRITER NOTE:~~

- ~~1. Use sidewalk doors for fuel tanks specified under Mechanical Specifications. When manhole cover is not used.~~
- ~~2. Coordinate to insure details and specification agree.~~
- ~~3. Coordinate with site specifications. Do not duplicate in this section.~~

~~2.18 SIDEWALK DOOR~~

- ~~A. Use flush, watertight, gutter type design.~~
- ~~B. Cover fabricate of 6 mm (1/4 inch) thick, diamond pattern floor plate.~~
- ~~C. Use automatic lock hold open feature and be hung on two flush type heavy bronze hinges capable of 90 degree swing on each door leaf.~~
- ~~D. Equip with locking and latching device and lifting devices; operable and accessible from both sides of doors.~~
- ~~E. Doors removable without disturbing frame.~~
- ~~F. Provide gutters at all joints for drainage of water.~~

~~SPEC WRITER NOTE:~~

- ~~1. Use screened access doors at emergency exit from pipe basements, between service rooms and pipe basements, and~~

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~~where shown, to vent spaces. Do not vent conditioned spaces.~~

~~2. Verify doors are detailed; coordinate to insure details and specifications agree.~~

~~3. Use not less than 900 x 900 mm (3 foot by 3 foot) door.~~

~~2.19 SCREENED ACCESS DOORS AND FRAMES~~

~~A. Galvanized ASTM A123, G-90 after fabrication.~~

~~B. Wall frame:~~

~~1. Fabricate frame from steel angles or channels as shown.~~

~~2. Continuously weld 38 x 13 mm (1-1/2 x 1/2 inch) steel channel door stop to angle frame. Cut out lock strike opening in channel.~~

~~3. Miter and weld channel frame at corners. Reinforce corner with 3 mm (1/8 inch) plate angle.~~

~~4. Reinforce channel frame with 3 x 150 mm (1/8 x 6 inch) long steel plate at channel back to cutout for latch. Cutout lock strike opening in channel face. Drill and tap for hinge anchorage.~~

~~5. Drill jambs for 6 mm (1/4 inch) bolt anchors at top and bottom and not over 450 mm (18 inches) between top and bottom.~~

~~6. Fabricate frame for door to sit flush with face of frame.~~

~~C. Doors~~

~~1. Fabricate door using steel channel frame with 3 mm (1/8 inch) angle plate reinforcing at corners.~~

~~2. Miter and weld corners.~~

~~3. Fabricate lock box of 1.6 mm (1/16 inch) plate and weld to channel surround.~~

~~4. Provide wire mesh constructed of 3.5 mm (0.135 inch) diameter galvanized steel wire crimped and woven into 38 mm (1-1/2 inch) diamond mesh pattern. Fasten the wire mesh to door frames by bending the ends of each strand of wire over through channel clinched and welded to channel door frame.~~

~~5. Weld steel plate back bands to channel door frame at hinge stiles only.~~

~~6. Screen on doors in exterior walls.~~

~~a. Fabricate rewirable frame for screen from either extruded or tubular aluminum.~~

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- ~~b. Design to allow for removing or replacement frame and screening or adjoining items without damage.~~
 - ~~c. Use aluminum insect screening specified.~~
 - ~~d. Use stainless steel fasteners for securing screen to door.~~
 - ~~D. Hardware:~~
 - ~~1. Install hinged door to fixed frame with two 63 mm (2-1/2 inch) brass or bronze hinges.~~
 - ~~2. Install lock or latch specified in Section 08 71 00, DOOR HARDWARE in lockbox.~~
- ~~SPEC WRITER NOTE: Coordinate with Section 06 20 00 for Countertops or Benches requiring steel frame and legs.~~

~~2.20 STEEL COUNTER OR BENCH TOP FRAME AND LEGS~~

- ~~A. Fabricate channel or angle frame with mitered and welded corners as shown.~~
- ~~B. Drill top of frame with 6 mm (1/4inch) holes spaced 200 mm (8 inches) on center for securing countertop.~~
- ~~C. Fabricate legs of angle or pipe shapes and continuously weld to frame.~~
- ~~D. Finish frame with backed on enamel prime coat.~~

~~2.21 STEEL PIPE BOLLARD~~

~~Provide bollard in accordance with ASTM A53 with dimensions as shown in standard detail SD320523-04. Anchor posts in concrete and fill solidly with concrete with a minimum compressive strength of 17 MPa 2500psi.~~

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Set work accurately, in alignment and where shown, plumb, level, free of rack and twist, and set parallel or perpendicular as required to line and plane of surface.
- B. Items set into concrete or masonry.
 - 1. Provide temporary bracing for such items until concrete or masonry is set.
 - 2. Place in accordance with setting drawings and instructions.
 - 3. Build strap anchors, into masonry as work progresses.
- C. Set frames of gratings, covers, corner guards, trap doors and similar items flush with finish floor or wall surface and, where applicable, flush with side of opening.
- D. Field weld in accordance with AWS.

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1. Design and finish as specified for shop welding.
2. Use continuous weld unless specified otherwise.
- E. Install anchoring devices and fasteners as shown and as necessary for securing metal fabrications to building construction as specified.
Power actuated drive pins may be used except for removable items and where members would be deformed or substrate damaged by their use.
- F. Spot prime all abraded and damaged areas of zinc coating as specified and all abraded and damaged areas of shop prime coat with same kind of paint used for shop priming.
- G. Isolate aluminum from dissimilar metals and from contact with concrete and masonry materials as required to prevent electrolysis and corrosion.
- H. Secure escutcheon plate with set screw.

3.2 INSTALLATION OF SUPPORTS

- A. Anchorage to structure.
 1. Secure angles or channels and clips to overhead structural steel by continuous welding unless bolting is shown.
 2. Secure supports to concrete inserts by bolting or continuous welding as shown.
 3. Secure supports to mid height of concrete beams when inserts do not exist with expansion bolts and to slabs, with expansion bolts.
unless shown otherwise.
 4. Secure steel plate or hat channels to studs as detailed.

~~B. Ceiling Hung Toilet Stalls:~~

- ~~1. Securely anchor hangers of continuous steel channel above pilasters to structure above.~~
- ~~2. Bolt continuous steel angle at wall to masonry or weld to face of each metal stud.~~
- ~~3. Secure brace for steel channels over toilet stall pilasters to wall angle supports with bolts at each end spaced as shown.~~
- ~~4. Install diagonal angle brace where the suspended ceiling over toilet stalls does not extend to side wall of room.~~
- ~~5. Install stud bolts in lower flange of channel before installing furred down ceiling over toilet stalls.~~
- ~~6. Install support for ceiling hung pilasters at entrance screen to toilet room similar to toilet stall pilasters.~~

~~C. Supports for Wall Mounted items:~~

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- ~~1. Locate center of support at anchorage point of supported item.~~
- ~~2. Locate support at top and bottom of wall hung cabinets.~~
- ~~3. Locate support at top of floor cabinets and shelving installed against walls.~~
- ~~4. Locate supports where required for items shown.~~

~~D. Support at Ceiling for X-ray Tube Stand and Radiographic Equipment:~~

- ~~1. Bolt modular steel channel frames to hangers as shown, anchored to structure above.~~
- ~~2. Fasten frames with modular channel manufacturers fittings, bolts, and nuts. Space modular channel supports and hangers as shown and as required to suit equipment furnished.~~
- ~~3. Install closure plates in channels at ceiling where channel opening is visible. Coordinate and cut plates to fit tight against equipment anchors after equipment anchors are installed.~~

~~E. Ceiling Support for Operating Light:~~

- ~~1. Anchor support to structure above as shown.~~
- ~~2. Set leveling plate as shown level with ceiling.~~
- ~~3. Secure operating light to leveling plate in accordance with light manufacturer's requirements.~~

~~F. Supports for intravenous (IV) Track and Cubicle Curtain Track:~~

- ~~1. Install assembly where shown after ceiling suspension grid is installed.~~
- ~~2. Drill angle for bolt and weld nut to angle prior to installation of tile.~~

~~G. Support for cantilever grab bars:~~

- ~~1. Locate channels or tube in partition for support as shown, and extend full height from floor to underside of structural slab above.~~
- ~~2. Anchor at top and bottom with angle clips bolted to channels or tube with two, 9 mm (3/8 inch) diameter bolts.~~
- ~~3. Anchor to floors and overhead construction with two 9 mm (3/8 inch) diameter bolts.~~
- ~~4. Fasten clips to concrete with expansion bolts, and to steel with machine bolts or welds.~~

~~H. Supports for Trapeze Bars:~~

- ~~1. Secure plates to overhead construction with fasteners as shown.~~
- ~~2. Secure angle brace assembly to overhead construction with fasteners as shown and bolt plate to braces.~~

~~3. Fit modular channel unit flush with finish ceiling, and secure to plate with modular channel unit manufacturer's standard fittings through steel shims or spreaders as shown.~~

~~a. Install closure plates in channel between eye bolts.~~

~~b. Install eyebolts in channel.~~

~~I. Support for Communion Rail Posts:~~

~~1. Anchor steel plate supports for posts as shown.~~

~~2. Use four bolts per plate, locate two at top and two at bottom.~~

~~3. Use lag bolts.~~

~~3.3 COVERS AND FRAMES FOR PITS AND TRENCHES~~

~~A. Set frame and cover flush with finish floor.~~

~~B. Secure plates to frame with flat head countersunk screws.~~

~~C. Set gratings loose in drainage trenches or over pits unless shown anchored.~~

~~3.4 FRAMES FOR LEAD LINED DOORS~~

~~A. Secure jamb angle clips and plates, at top and bottom with two, 9 mm (3/8 inch) expansion bolts to concrete.~~

~~B. Secure 150 x 90 x 13 mm (6 x 3-1/2 x 1/2 inch) angle to steel framing for anchorage when expansion bolts to concrete is not possible.~~

~~C. Secure clips by welding to steel.~~

~~D. At interstitial spaces, anchor jamb angles as shown.~~

~~3.5 DOOR FRAMES~~

~~A. Secure clip angles at bottom of frames to concrete slab with expansion bolts as shown.~~

~~B. Level and plumb frame; brace in position required.~~

~~C. At masonry, set frames in walls so anchors are built in as the work progresses unless shown otherwise.~~

~~D. Set frames in formwork for frames cast into concrete.~~

~~E. Where frames are set in prepared openings, bolt to wall with spacers and expansion bolts.~~

~~3.6 OTHER FRAMES~~

~~A. Set frame flush with surface unless shown otherwise.~~

~~B. Anchor frames at ends and not over 450 mm (18 inches) on centers unless shown otherwise.~~

~~C. Set in formwork before concrete is placed.~~

~~3.7 GUARDS~~

~~A. Steel Angle Corner Guards:~~

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- ~~1. Build into masonry as the work progress.~~
- ~~2. Set into formwork before concrete is placed.~~
- ~~3. Set angles flush with edge of opening and finish floor or wall or as shown.~~
- ~~4. At existing construction fasten angle and filler piece to adjoining construction with 16 mm (5/8 inch) diameter by 75 mm (3 inch) long expansion bolts 450 mm (18 inches) on center.~~
- ~~5. Install Guard Angles at Edges of // Trench // Stairwell // Openings in Slab // Dock Leveler // Overhead Doors where shown.~~

~~B. Channel Guard at Top Edge of Concrete Platforms:~~

- ~~1. Install in formwork before concrete is placed.~~
- ~~2. Set channel flush with top of the platform.~~

~~C. Wheel Guards:~~

- ~~1. Set flanges of wheel guard at least 50 mm (2 inches) into pavement.~~
- ~~2. Anchor to walls as shown, expansion bolt if not shown.~~

~~3.8 GRATINGS~~

- ~~A. Set grating flush with finish floor, top of curb, or areaway wall. Set frame so that horizontal leg of angle frame is flush with face of wall except when frame is installed on face of wall.~~
- ~~B. Set frame in formwork before concrete is placed.~~
- ~~C. Where grating terminates at a wall bolt frame to concrete or masonry with expansion bolts unless shown otherwise.~~
- ~~D. Secure removable supporting members in place with stainless steel bolts.~~
- ~~E. Bolt gratings to supports.~~

~~3.9 STEEL LINTELS~~

- ~~A. Use lintel sizes and combinations shown or specified.~~
- ~~B. Install lintels with longest leg upstanding, except for openings in 150 mm (6 inch) masonry walls install lintels with longest leg horizontal.~~
- ~~C. Install lintels to have not less than 150 mm (6 inch) bearing at each end for nonbearing walls, and 200 mm (8 inch) bearing at each end for bearing walls.~~

~~3.10 SHELF ANGLES~~

- ~~A. Anchor shelf angles with 19 mm (3/4 inch) bolts unless shown otherwise in adjustable malleable iron inserts, set level at elevation shown.~~
- ~~B. Provide expansion space at end of members.~~

~~3.11 PLATE DOOR SILL~~

- ~~A. Install after roofing base flashing and counter flashing work is completed.~~
- ~~B. Set in sealant and bolt to curb.~~

~~3.12 SAFETY NOSINGS~~

- ~~A. Except as specified and where preformed rubber treads are shown or specified install safety nosings at the following:~~
 - ~~1. Exterior concrete steps.~~
 - ~~2. Door sills of areaway entrances curbs.~~
 - ~~3. Exposed edges of curbs of door sills at transformer and service rooms.~~
 - ~~4. Interior concrete steps, including concrete filled treads of metal stairs of service stairs.~~
- ~~B. Install flush with horizontal and vertical surfaces.~~
- ~~C. Install nosing to within 100 mm (4 inches) of ends of concrete stair treads, except where shown to extend full width.~~
- ~~D. Extend nosings full width of door openings.~~
- ~~E. Extend nosings, full width between stringers of metal stairs, and terminate at point of curvature of steps having short radius curved ends.~~

3.13 LADDERS

- A. Anchor ladders to walls and floors with expansion bolts through turned lugs or angle clips or brackets.
- ~~B. In elevator pits, set ladders to clear all elevator equipment where shown on the drawings.~~
 - ~~1. Where ladders are interrupted by division beams, anchor ladders to beams by welding, and to floors with expansion bolts.~~
 - ~~2. Where ladders are adjacent to division beams, anchor ladders to beams with bent steel plates, and to floor with expansion bolts.~~
- C. Ladder Rungs:
 - 1. Set ladder rungs into formwork before concrete is placed. // Build ladder rungs into masonry as the work progresses. //
 - 2. Set step portion of rung 150 mm (6 inches) from wall.
 - 3. Space rungs approximately 300 mm (12 inches) on centers.
 - 4. Where only one rung is required, locate it 400 mm (16 inches) above the floor.

3.14 RAILINGS

A. Steel Posts:

1. Secure fixed posts to concrete with expansion bolts through flanged fittings except where sleeves are shown with pourable grout.
2. Install sleeves in concrete formwork.
3. Set post in sleeve and pour grout to surface. Apply beveled bead of urethane sealant at perimeter of post or under flange fitting as specified in Section 07 92 00, JOINT SEALANTS—on exterior posts.
4. Secure removable posts to concrete with either machine screws through flanged fittings which are secured to inverted flanges embedded in and set flush with finished floor, or set posts in close fitting pipe sleeves without grout.
5. Secure sliding flanged fittings to posts at base with set screws.
6. Secure fixed flanged fittings to concrete with expansion bolts.
7. Secure posts to steel with welds.

~~SPEC WRITER NOTE: Clearly identify location of aluminum and stainless railings if not identified on drawings.~~

B. Aluminum Railing, Stainless Steel Railing, and Ornamental Railing Posts:

1. Install pipe sleeves in concrete formwork.
2. Set posts in sleeve and pour grout to surface on exterior locations and to within 6 mm (1/4 inch) of surface for interior locations except to where posts are required to be removable.
3. Apply beveled bead of urethane sealant over sleeve at post perimeter for exterior posts and flush with surface for interior posts as specified in Section 07 92 00, JOINT SEALANTS.

C. Anchor to Walls:

1. Anchor rails to concrete or solid masonry with machine screws through flanged fitting to steel plate.
 - a. Anchor steel plate to concrete or solid masonry with expansion bolts.
 - b. Anchor steel plate to hollow masonry with toggle bolts.
2. Anchor flanged fitting with toggle bolt to steel support in frame walls.

D. Removable Rails:

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1. Rest rails in brackets at each end and secure to bracket with stainless steel bolts and nuts where part of a continuous railing.
2. Rest rail posts in sleeves where not part of a continuous railing.
Do not grout posts.

~~E. Gates:~~

- ~~1. Hang gate to swing as shown.~~
- ~~2. Bolt gate hinges to jamb post with clamp on or through bolts.~~

~~F. Chains:~~

- ~~1. Eye bolt chains to pipe posts.~~
- ~~2. Eye bolt anchoring at walls.~~
 - ~~a. Expansion bolt to concrete or solid masonry.~~
 - ~~b. Toggle bolt to hollow masonry of frame wall installed support.~~

G. Handrails/Guardrails:

1. Anchor brackets for metal handrails as detailed.
2. Install brackets within 300 mm (12 inches) of return of walls, and at evenly spaced intermediate points not exceeding 1200 mm (4 feet) on centers unless shown otherwise.
3. Expansion bolt to concrete or solid masonry.
4. Toggle bolt to installed supporting frame wall and to hollow masonry unless shown otherwise.

~~3.15 CATWALK AND PLATFORMS~~

- ~~A. Expansion bolt members to concrete unless shown otherwise.~~
- ~~B. Bolt or weld structural components together including ladders and stairs to support system.~~
- ~~C. Weld railings to structural framing.~~
- ~~D. Bolt or weld walk surface to structural framing.~~
- ~~E. Smooth field welds and spot prime damaged prime paint surface.~~
- ~~F. Fasten removable members with stainless steel fasteners.~~

~~3.16 SIDEWALK DOOR, TRAP DOORS, AND FRAMES~~

- ~~A. Set frame flush with finished concrete slab or curb.~~
- ~~B. Secure well linings to structure with expansion bolts unless shown otherwise.~~
- ~~C. Bolt ceiling hatch to well lining angle brace and to angle iron frames near corners and 300 mm (12 inches) on centers with not less than 9 mm (3/8 inch) roundhead machine screws.~~
- ~~D. Coordinate sidewalk door drain connections with plumbing work.~~

~~3.17 SCREENED ACCESS DOOR~~

- ~~A. Set frame in opening so that clearance at jambs is equal and secure with expansion bolts.~~
- ~~B. Use shims at bolts to prevent deformation of frame members in prepared openings.~~
- ~~C. Set frame in mortar bed and build in anchors as the masonry work progresses.~~
- ~~D. Grout jambs solid with mortar.~~
- ~~E. Secure insect screen to inside of door with stainless steel fasteners on doors in exterior walls.~~

~~3.18 STEEL COMPONENTS FOR MILLWORK ITEMS~~

~~Coordinate and deliver to Millwork fabricator for assembly where millwork items are secured to metal fabrications.~~

~~3.19 INSTALLATION OF STEEL PIPE BOLLARD~~

~~Set bollards vertically in concrete piers. Compressive strength of concrete piers shall be 21MPa 3000psi. For dimensions of concrete piers See standard detail SD320523-04.~~

3.20 CLEAN AND ADJUSTING

- A. Adjust movable parts including hardware to operate as designed without binding or deformation of the members centered in the opening or frame and, where applicable, contact surfaces fit tight and even without forcing or warping the components.
- B. Clean after installation exposed prefinished and plated items and items fabricated from stainless steel, aluminum and copper alloys, as recommended by the metal manufacture and protected from damage until completion of the project.

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SECTION 07 01 50.19
PREPARATION FOR RE-ROOFING

~~SPEC WRITER NOTES:~~

- ~~1. Delete text between // // not applicable to project. Edit remaining text to suit project.~~
- ~~2. Edit Summary article below after establishing re-roofing scope and editing section to correspond to Project requirements.~~

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Complete roof removal for new roof system installation.
2. Partial roof removal for new roof system installation.
3. Roofing membrane and selective roofing system component removal for new roof membrane installation.
- ~~4. Existing roofing membrane preparation for new roofing // membrane // system // installation.~~

~~SPEC WRITER NOTE: Modify paragraph below to describe existing roofing system.~~

- B. Existing Roofing System: ~~// Built-up asphalt // Built-up coal tar // EPDM, // CSPE // PVC // TPO, and standing seam metal roof // Modified bituminous //~~. System components include:

~~SPEC WRITER NOTE: System components are listed in order from top down. Edit to suit existing roofing system.~~

- ~~1. Vegetation with growing media // water retention mat // and root barrier.~~
- ~~2. Pavers and paver supports.~~
- ~~3. Aggregate ballast.~~
- ~~4. Roof insulation // and drainage board //.~~
- ~~5. Aggregate surfacing.~~
- ~~6-1.~~ _____ Roofing membrane.
- ~~7-2.~~ _____ Cover board.
- ~~8-3.~~ _____ Roof insulation.
- ~~9-4.~~ _____ Vapor retarder.
- ~~10-5.~~ _____ Substrate board.

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1.2 RELATED WORK

~~SPEC WRITER NOTE: Update and retain references only when specified elsewhere in this section.~~

- A. SECTION 02 82 13.21, ASBESTOS ROOFING ABATEMENT: Asbestos Removal.
- ~~B. Section 05 31 00, STEEL DECKING: Replacement Roof Deck.~~
- ~~C. Section 06 10 00, ROUGH CARPENTRY: Replacement Roof Deck and Parapet Sheathing.~~
- ~~D. //Section 07 51 00, BUILT-UP BITUMINOUS ROOFING: New Roofing System. //~~
- ~~E. //Section 07 51 13.11, HOT APPLIED BUILT-UP ASPHALT ROOFING: New Roofing System. //~~
- ~~F. //Section 07 51 13.13, COLD APPLIED BUILT-UP ASPHALT ROOFING: New Roofing System. //~~
- ~~G. //Section 07 52 16.12, HOT MOPPED STYRENE BUTADIENE STYRENE MODIFIED BITUMINOUS MEMBRANE ROOFING: New Roofing System. //~~
- ~~H. //Section 07 52 16.11, COLD ADHESIVE STYRENE BUTADIENE STYRENE MODIFIED BITUMINOUS MEMBRANE, ROOFING: New Roofing System. //~~
- ~~I. //Section 07 53 23, ETHYLENE PROPYLENE DIENE MONOMER (EPDM) ROOFING: New Roofing System. //~~
- ~~J. //Section 07 54 16, ETHYLENE INTERPOLYMER (KEE) ROOFING: New Roofing System. //~~
- ~~K.B.~~ //Section 07 54 19, POLYVINYL-CHLORIDE (PVC) ROOFING: New Roofing System. //
- ~~L.C.~~ //Section 07 54 23, THERMOPLASTIC POLYOLEFIN (TPO) ROOFING: New Roofing System. //
- ~~M. //Section 07 56 00, FLUID APPLIED ROOFING: New Roofing System. //~~
- ~~N. //Section 07 57 13, SPRAYED POLYURETHANE FOAM ROOFING: New Roofing System. //~~
- ~~O.D.~~ Section 07 60 00, FLASHING AND SHEET METAL: Sheet Metal Counterflashing.

1.3 APPLICABLE PUBLICATIONS

- A. Comply with references to extent specified in this section.
- B. American National Standards Institute/Single-Ply Roofing Institute (ANSI/SPRI):
- FX-1 (R2016).....Standard Field Test Procedure for Determining the Withdrawal Resistance of Roofing Fasteners.
- C. American Society for Nondestructive Testing (ASNT):

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SNT-TC-1A (2019).....Personnel Qualification and Certification for
Nondestructive Testing.

D. ASTM International (ASTM):

C208-12(2017)e2.....Cellulosic Fiber Insulating Board.

C578-19.....Rigid, Cellular Polystyrene Thermal Insulation.

C728-17a.....Perlite Thermal Insulation Board.

C1177/C1177M-17.....Glass Mat Gypsum Substrate for Use as
Sheathing.

C1153-10(2015).....Location of Wet Insulation in Roofing Systems
Using Infrared Imaging.

C1278/C1278M-17.....Standard Specification Fiber-Reinforced Gypsum
Panel.

D4263-83(2018).....Indicating Moisture in Concrete by the Plastic
Sheet Method.

E. U.S. Department of Commerce National Institute of Standards and
Technology (NIST):

DOC PS 1-19.....Structural Plywood.

DOC PS 2-18.....Performance Standard for Wood-Based
Structural-Use Panels.

1.4 PREINSTALLATION MEETINGS

A. Conduct preinstallation meeting minimum 30 days before beginning Work
of this section.

~~SPEC WRITER NOTE: Edit participant list
to ensure entities influencing outcome
attend.~~

1. Required Participants:

a. Contracting Officer's Representative (COR).

b. ~~//~~Architect/Engineer (as requested by COR).~~//~~

c. ~~//~~Inspection and Testing Agency.~~//~~

d. Contractor.

e. Installer.

~~SPEC WRITER NOTE: Include manufacturer's
representative when work affects existing
warranted roof.~~

f. ~~//~~Manufacturer's field representative.~~//~~

g. Other installers responsible for adjacent and intersecting work,
including mechanical and electrical equipment installers.

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SPEC WRITER NOTE: Edit meeting agenda to incorporate project specific topics.

2. Meeting Agenda: Distribute agenda to participants minimum 3 days before meeting.
 - a. Removal and installation schedule.
 - b. Removal and installation sequence.
 - c. Preparatory work.
 - d. Protection before, during, and after installation.
 - e. Removal and installation.
 - f. Temporary roofing including daily terminations.
 - g. Transitions and connections to other work.
 - h. Inspecting and testing.
 - i. Other items affecting successful completion.
3. Document and distribute meeting minutes to participants to record decisions affecting installation.

1.5 SUBMITTALS

- A. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.
- B. Submittal Drawings:
 1. Show size, configuration, and installation details.
- C. Manufacturer's Literature and Data:
 1. Description of each product.
 2. Description of temporary roof system and components.
 3. List of patching materials.
 4. Recover board fastening requirements.
 5. Temporary roofing installation instructions and ~~//~~removal instructions; and .//preparation instructions to receive new roofing.~~//~~

~~SPEC WRITER NOTE: Retain paragraph below when existing roof is under warranty.~~

6. Existing roofing warrantor's instructions.
- D. Photographs: Document existing conditions potentially affected by roofing operations before work begins.

~~SPEC WRITER NOTE: Include field inspection report only when existing roof is under warranty.~~

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E. Field Inspection Reports:

1. Certify warrantor inspected completed roofing and existing warranty remains in effect.

~~SPEC WRITER NOTE: Retain paragraph below if Infrared Roof Moisture Survey is conducted as specified in Part 3 for partial roof replacement.~~

F. Infrared Roof Moisture Survey Report.

1.6 QUALITY ASSURANCE

A. Installer Qualifications:

1. Same installer as ~~// Section 07 51 00, BUILT UP BITUMINOUS ROOFING // Section 07 51 13.11, HOT-APPLIED BUILT-UP ASPHALT ROOFING // Section 07 51 13.13, COLD-APPLIED BUILT-UP ASPHALT ROOFING // Section 07 52 16.11, COLD-APPLIED STYRENE BUTADIENE STYRENE MODIFIED BITUMINOUS MEMBRANE ROOFING // Section 07 52 16.12, HOT-MOPPED STYRENE BUTADIENE STYRENE MODIFIED BITUMINOUS MEMBRANE, ROOFING, // Section 07 53 23, ETHYLENE PROPYLENE DIENE MONOMER (EPDM) ROOFING // Section 07 54 16, ETHYLENE INTERPOLYMER (KEE) ROOFING // Section 07 54 19, POLYVINYL-CHLORIDE (PVC) ROOFING, // Section 07 54 23, THERMOPLASTIC POLYOLEFIN (TPO) ROOFING and Section 07 40 00, ROOFING AND SIDING PANELS // Section 07 56 00, FLUID-APPLIED ROOFING // 07 57 13, SPRAYED POLYURETHANE FOAM ROOFING //~~
2. Licensed to perform asbestos abatement in Project jurisdiction when removal of asbestos-containing material is required.
3. Approved by existing roofing system warrantor when work affects existing roofing system under warranty.

1.7 FIELD CONDITIONS

- A. Building Occupancy: Perform work to minimize disruption to normal building operations.

~~SPEC WRITER NOTE: Retain subparagraph below only when impaired deck areas exist and building will be occupied during work.~~

1. Verify occupants are evacuated from affected building areas when working on structurally impaired roof decking above occupied areas.
2. Provide notice minimum 72 hours before beginning activities affecting normal building operations.

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B. Existing Roofing Available Information:

1. The following are available for Contractor reference:

~~SPEC WRITER NOTES:~~

- ~~1. Retain three subparagraphs below if applicable.~~
- ~~2. If roof moisture survey results are not available, include testing specified in Part 3.~~

- a. ~~Roof moisture survey~~As-Built Drawings.
 - b. ~~Test cores analysis~~Hazardous Materials Testing Report.
 - c. Construction drawings and project manual.
2. Examine available information before beginning work of this section.

~~SPEC WRITER NOTE: Insert section number and title of Division 07 roofing section used for roof replacement.~~

C. Weather Limitations: Proceed with reroofing preparation only during dry

weather conditions as specified for new roofing installation in ~~Section~~

~~// Section 07 51 00, BUILT UP BITUMINOUS ROOFING // Section~~
~~07 51 13.11, HOT-APPLIED BUILT-UP ASPHALT ROOFING // Section~~
~~07 51 13.13, COLD-APPLIED BUILT-UP ASPHALT ROOFING // Section~~
~~07 52 16.11, COLD ADHESIVE STYRENE-BUTADIENE-STYRENE MODIFIED~~
~~BITUMINOUS MEMBRANE ROOFING // Section 07 52 16.12, HOT-MOPPED~~
~~STYRENE-BUTADIENE-STYRENE MODIFIED BITUMINOUS MEMBRANE, ROOFING//~~
~~Section 07 53 23, ETHYLENE-PROPYLENE-DIENE MONOMER (EPDM) ROOFING //~~
~~Section 07 54 16, ETHYLENE INTERPOLYMER (KEE) ROOFING // Section~~
~~07 54 19, POLYVINYL-CHLORIDE (PVC) ROOFING, //~~ Section 07 54 23,
THERMOPLASTIC POLYOLEFIN (TPO) ROOFING ~~// Section 07 56 00,~~
~~FLUID-APPLIED ROOFING // Section 07 57 13, SPRAYED POLYURETHANE FOAM~~
~~ROOFING //and Section 07 40 00, ROOFING AND SIDING PANELS.~~

1. Remove only as much roofing in one day as can be made watertight in same day.

~~SPEC WRITER NOTE: Retain paragraph below when no known hazardous materials exist.~~

D. Hazardous materials are not expected in existing roofing system.

1. Known hazardous materials were removed before start of work.
2. Do not disturb suspected hazardous materials. When discovered, notify Contracting Officer's Representative.

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3. Hazardous materials discovered during execution of the work will be removed by Government as work of a separate contract.

~~SPEC WRITER NOTE: Retain paragraph below when hazardous materials exist.~~

E. Hazardous Materials are known to exist in existing roofing system.

1. See Section 02 82 13.21, ASBESTOS ROOFING ABATEMENT for hazardous material removal and disposal.
2. Coordinate reroofing preparation with asbestos roofing abatement to prevent water from entering existing roofing system and building.

~~SPEC WRITER NOTE: Ensure report of hazardous material is available or attached.~~

3. Hazardous material report is ~~// attached at the end of this section~~
~~// available in _____~~ //from the VA for review and use.

1.8 WARRANTY

SPEC WRITER NOTE: Always retain construction warranty. FAR includes Contractor's one year labor and material warranty.

A. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

~~SPEC WRITER NOTE: Retain this article if portions of existing roof still under warranty are to remain or if an adjoining existing roof is under warranty.~~

B. Existing Warranties: Perform work to maintain existing roofing warranty in effect.

1. Notify warrantor before beginning, and upon completion of reroofing.
2. Obtain warrantor's instructions for maintaining existing warranty.

PART 2 - PRODUCTS

2.1 MATERIALS

~~SPEC WRITER NOTE: Retain first paragraph below when patching and retaining portions of existing roofing system.~~

A. Patching Materials: Match existing roofing system materials.

B. Plywood Sheathing: ~~See Section 06 10 00, ROUGH CARPENTRY..~~

C. Metal Flashing: See Section 07 60 00, FLASHING AND SHEET METAL.

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D. Temporary Protection Materials:

1. Expanded Polystyrene (EPS) Insulation: ASTM C578-19.
2. Plywood: NIST DOC PS 1-19, Grade CD Exposure 1-18.
3. Oriented Strand Board (OSB): NIST DOC PS 2-18, Exposure 1.

E. Temporary Roofing System Materials: Contractor's option.

~~F. Recover Board: One of the following:~~

~~SPEC WRITER NOTE: Retain one of paragraphs below when new roofing is installed over existing roofing.~~

- ~~1. Insulation: See Section 07 22 00, ROOF AND DECK INSULATION.~~
- ~~2. Fiber Board: ASTM C208 12(2017)e2, Type II, fiber board; 13 mm (1/2 inch) thick.~~
- ~~3. Glass Mat Gypsum Board: ASTM C1177/C1177M-17, water resistant; // 6 mm (1/4 inch) // 13 mm (1/2 inch) // Type X, 16 mm (5/8 inch) // thick.~~
- ~~4. Fiber Reinforced Gypsum Board: ASTM C1278/C1278M-17, water resistant; // 6 mm (1/4 inch) // 10 mm (3/8 inch) // 13 mm (1/2 inch) // 16 mm (5/8 inch) // thick.~~
- ~~5. Perlite: ASTM C728-17a; // 3 mm (1/2 inch) // 19 mm (3/4 inch) // 25 mm (1 inch) // thick.~~

~~G.F.~~ Fasteners: Type and size required by roof membrane manufacturer to resist wind uplift.

PART 3 - EXECUTION

3.1 EXAMINATION

~~SPEC WRITER NOTE: Where partial removals are indicated, it is advisable to perform moisture testing to confirm extent.~~

- A. Infrared Roof Moisture Survey: Ground-based, walk-over type performed according to ASTM C1153-10(2015).
1. Record the entire survey on DVD and provide one copy to Contracting Officer's Representative with report.
 2. Include in report thermograms of suspect areas and corresponding daytime photos of same locations.
 3. Conduct inspection by NDT test technician certified to at least Level 2 in Thermal/Infrared test method according to ASNT SNT-TC-1A.
 4. Mark out roof areas determined to be wet to indicate ~~//minimum//~~ areas to be removed.

3.2 PREPARATION

A. Examine and verify substrate suitability for product installation.

~~SPEC WRITER NOTE: Retain first paragraph below when portions of existing roof will remain.~~

B. Protect existing roofing system indicated to remain.

1. Cover roof membrane with temporary protection materials without impeding drainage.
2. Limit traffic and material storage to protected areas.
3. Maintain temporary protection until replacement roofing is completed.

C. Protect existing construction and completed work from damage.

D. Protect landscaping from damage.

E. Maintain access to existing walkways and adjacent occupied facilities.

F. Coordinate use of rooftop fresh air intakes with Contracting Officer's Representative to minimize effect on indoor air quality.

G. Ensure temporary protection materials are available for immediate use in case of unexpected rain.

H. Ensure roof drainage remains functional.

1. Keep drainage systems clear of debris.
2. Prevent water from entering building and existing roofing system.

I. Coordinate rooftop utilities remaining active during roofing work with Contracting Officer's Representative.

3.3 RE-ROOFING PREPARATION - GENERAL

A. Notify Contracting Officer's Representative of planned operations, daily.

1. Identify location and extent of roofing removal.
2. Request authorization to proceed.

~~3.4 OVERBURDEN REMOVAL~~

~~SPEC WRITER NOTE: Retain first paragraph below for loose laid roofing membrane and decorative ballast.~~

~~A. Remove aggregate ballast.~~

- ~~1. Store aggregate ballast for reuse.~~

~~B. Remove loose aggregate from bituminous membrane surface.~~

~~C. Remove pavers and paver support.~~

- ~~1. Store undamaged pavers and paver supports for reuse.~~

~~2. Dispose of damaged pavers.~~

~~D. Remove plants, planting medium, // water retention mat, // and root barrier from vegetated roof assembly.~~

~~1. Store materials and plants for reuse.~~

~~2. Protect plants from root exposure and drying.~~

~~SPEC WRITER NOTE: Retain first paragraph below for protected roofing membrane.~~

~~E. Remove insulation // and drainage board // from protected roofing membrane.~~

~~1. Store insulation and drainage board for reuse.~~

3.53.4 COMPLETE ROOFING SYSTEM REMOVAL

A. Remove existing roofing system completely, exposing structural roof deck.

~~SPEC WRITER NOTE: Retain subparagraph below when roofing system components are known. Edit to suit existing conditions.~~

1. Remove ~~//~~ cover board, ~~//~~ roof insulation, ~~//~~ vapor retarder, ~~//~~ and ~~//~~ substrate board~~//~~.

2. Remove or cut-off roofing system fasteners.

3.63.5 PARTIAL ROOFING SYSTEM REMOVAL

A. Remove existing roofing completely, exposing structural roof deck at locations and to extent indicated on drawings.

1. Remove ~~//~~ cover board, ~~//~~ roof insulation, ~~//~~ vapor retarder, ~~//~~ and ~~//~~ substrate board~~//~~.

2. Remove or cut-off roofing system fasteners.

~~SPEC WRITER NOTE: Retain article below if removal is limited to existing roofing membrane and selected components within roofing system identified for removal.~~

3.7 ROOFING MEMBRANE AND SELECTIVE ROOFING SYSTEM COMPONENT REMOVAL

~~A. Remove existing roofing membrane, only, in locations and to extent indicated on drawings.~~

~~B. Visually inspect // cover board, // roof insulation, // vapor retarder, // and // substrate board // for moisture immediately after roof membrane removal.~~

~~1. Coordinate with Contracting Officer's Representative to observe inspections.~~

~~2. Identify wet roofing system components required to be removed.~~

~~3. Mark roofing system removal locations and extents.~~

~~C. Remove wet roofing system components.~~

~~1. Remove or cut off roofing system fasteners when removals expose structural roof deck.~~

~~D. Patch selective roofing system removals immediately after inspection and repair.~~

~~E. Install patching materials to match existing roofing system.~~

~~F. Patch roofing membrane to maintain building watertight, unless new roofing membrane is installed same day as removal and repair.~~

3.83.6 DECK PREPARATION

A. Inspect structural roof deck after roofing system removal.

~~SPEC WRITER NOTE: Retain first paragraph below for concrete deck.~~

B. Concrete Roof Decks:

1. Visually confirm concrete roof deck is dry.

2. Perform moisture test according to ASTM D4263-83(2018) each day for each separate roof area.

a. Proceed with roofing work only when moisture is not observed.

~~SPEC WRITER NOTE: Retain first paragraph below for steel roof deck.~~

C. Steel Roof Decks:

1. Visually inspect structural roof deck installation and fasteners.

a. Notify Contracting Officer's Representative of unsuitable conditions and inadequate fastenings potentially affecting roof system performance.

~~SPEC WRITER NOTE: Retain first paragraph below and revise to add fastening requirements.~~

2. Secure roof deck with additional fastenings as ~~// indicated on drawings. //~~ determined by Contracting Officer's Representative. ~~//~~

~~SPEC WRITER NOTE: Retain paragraph below and revise to include replacement criteria. Coordinate specification of type of deck with "Related Requirements" Paragraph in Part 1 "Summary" Article.~~

3. Replace roof deck as ~~// indicated on drawings. //~~ determined by Contracting Officer's Representative. ~~//~~

- a. Replacement Roof Deck: See ~~//~~Section 05 31 00, STEEL DECKING.~~//~~
~~Section 06 10 00, ROUGH CARPENTRY. //~~

3.93.7 TEMPORARY ROOFING

- A. Install temporary roofing to maintain building watertight.

~~SPEC WRITER NOTE: Retain one of two paragraphs below.~~

- B. Remove temporary roofing before installing new roofing.

- C. Prepare temporary roofing to receive new roofing.

3.103.8 EXISTING MEMBRANE PREPARATION FOR NEW ROOFING

~~SPEC WRITER NOTE: Retain this article when new roof is installed over existing roofing membrane.~~

- A. Remove existing roofing surface projections and irregularities. Produce smooth surface to receive recover boards.

1. Broom clean existing surface.

3.113.9 BASE FLASHING REMOVAL

- A. Expose base flashings to permit removal.

1. Two-Piece Counterflashings: Remove cap flashing and store for reuse.
2. Single Piece Counterflashings: Carefully bend counterflashing.
3. Metal Copings: Remove decorative cap and store for reuse.

- B. Remove existing base flashings.

1. Clean substrates to receive new flashings.

- C. Replace counterflashings damaged during removal.

1. Counterflashings: See Section 07 60 00 FLASHING AND SHEET METAL.

~~SPEC WRITER NOTES:~~

- ~~1. Retain paragraph below if replacing parapet sheathing.~~
- ~~2. Parapet sheathing is specified in Section 06 10 00, ROUGH CARPENTRY.~~

- ~~D. Remove existing parapet sheathing and inspect parapet framing.~~

- ~~1. Notify Contracting Officer's Representative of damaged framing.~~

- ~~E. Install // pressure-preservative treated // exterior~~

~~fire-retardant treated // plywood sheathing, // 15 mm (19/32 inch) // thick.~~

3.12 RECOVER BOARD INSTALLATION

- ~~A. Install recover boards over existing // roof insulation // roofing membrane // with butted joints. Stagger end joints in adjacent rows.~~

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~~B. Fasten recover boards to resist wind uplift.~~

~~SPEC WRITER NOTES: Insert Division 07
roofing section number and title for new
roofing membrane installation.~~

- ~~1. Fastening Requirements: See // Section 07 51 00, BUILT UP BITUMINOUS
ROOFING // Section 07 51 13.11, HOT APPLIED BUILT UP ASPHALT ROOFING
// Section 07 51 13.13, COLD ADHESIVE BUILT UP ASPHALT ROOFING //
Section 07 52 16.11, COLD APPLIED STYRENE BUTADIENE STYRENE MODIFIED
BITUMINOUS MEMBRANE ROOFING // Section 07 52 16.12, HOT MOPPED
STYRENE BUTADIENE STYRENE MODIFIED BITUMINOUS MEMBRANE, ROOFING //
Section 07 53 23, ETHYLENE PROPYLENE DIENE MONOMER (EPDM) ROOFING //
Section 07 54 16, ETHYLENE INTERPOLYMER (KEE) ROOFING // Section
07 54 19, POLYVINYL CHLORIDE (PVC) ROOFING // Section 07 54 23,
THERMOPLASTIC POLYOLEFIN (TPO) ROOFING // Section 07 56 00,
FLUID APPLIED ROOFING // Section 07 57 13, SPRAYED POLYURETHANE FOAM
ROOFING //.~~
- ~~2. Uplift Resistance: Base on pull out resistance determined by
specified field testing.~~

3.133.10 FIELD QUALITY CONTROL

~~SPEC WRITER NOTE: Section 01 45 29,
TESTING LABORATORY SERVICES includes VA
provided testing for large projects and
contractor provided testing for small
projects. Coordinate testing
responsibility.~~

A. Field Tests: Performed by contractor.~~testing laboratory specified in
Section 01 45 29, TESTING LABORATORY SERVICES.~~

~~SPEC WRITER NOTE: ANSI/SPRI FX-1 sets
testing frequency as 10 tests for first
4,650 square meter (50,000 square feet)
and five tests for each additional
4,650 square meter (50,000 square feet).
Specify frequency to suit project
conditions.~~

1. Fastener Pull Out Tests: ANSI/SPRI FX-1(2016).
- B. Existing Roofing System Warrantor Services:
 1. Inspect reroofing preparation and roofing installation to verify compliance with existing warranty conditions.

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2. Submit reports of field inspections, and supplemental instructions issued during inspections.

3.143.11 DISPOSAL

- A. Collect waste materials in containers.
- B. Remove waste materials from project site, regularly, to prevent accumulation.
- C. Legally dispose of waste materials.

- - E N D - -

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SECTION 07 08 00
FACILITY EXTERIOR CLOSURE COMMISSIONING

~~SPEC WRITER NOTES:~~

- ~~1. Delete between // if not applicable to project. Also delete any other item or paragraph not applicable in the section and renumber the paragraphs.~~
- ~~2. The spec writer shall review the Physical Security Design Manual for VA Facilities to determine and include in this section any Mission Critical or Life Safety requirements called out.~~

PART 1 - GENERAL

1.1 DESCRIPTION

- A. The requirements of this Section apply to all sections of Division 07 ~~and Division 08.~~
- B. This project will have selected building systems commissioned. The complete list of equipment and systems to be commissioned is specified in Section 01 91 00 GENERAL COMMISSIONING REQUIREMENTS. The commissioning process, which the Contractor is responsible to execute, is defined in Section 01 91 00 GENERAL COMMISSIONING REQUIRMENTS. A Commissioning Agent (CxA) appointed by the ~~//VA//~~ ~~//Contractor//~~ will manage the commissioning process.

1.2 RELATED WORK

- A. Section 01 00 00 GENERAL REQUIREMENTS.
- B. Section 01 91 00 GENERAL COMMISSIONING REQUIREMENTS.
- C. Section 01 33 23 SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

1.3 SUMMARY

- A. This Section includes requirements for commissioning the Facility exterior closure, related subsystems and related equipment. This Section supplements the general requirements specified in Section 01 91 00 General Commissioning Requirements.
- B. Refer to Section 01 91 00 GENERAL COMMISSIONING REQUIREMENTS for more details regarding commissioning processes and procedures, as well as roles and responsibilities for all Commissioning Team members.

1.4 DEFINITIONS

- A. Refer to Section 01 91 00 GENERAL COMMISSIONING REQUIREMENTS for definitions

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1.5 COMMISSIONED SYSTEMS

- A. Commissioning of a system or systems specified in Division 07 and Division 08 is part of the construction process. Documentation and testing of these systems, as well as training of the VA's Operation and Maintenance personnel in accordance with the requirements of Section 01 91 00 and of Division 07 ~~and 08~~, is required in cooperation with the VA and the Commissioning Agent.
- B. The Facility exterior closure systems commissioning will include the systems listed in Section 01 91 00 General Commissioning Requirements:

1.6 SUBMITTALS

- A. The commissioning process requires review of selected Submittals that pertain to the systems to be commissioned. The Commissioning Agent will provide a list of submittals that will be reviewed by the Commissioning Agent. This list will be reviewed and approved by the VA prior to forwarding to the Contractor. Refer to Section 01 33 23 SHOP DRAWINGS, PRODUCT DATA, and SAMPLES for further details.
- B. The commissioning process requires Submittal review simultaneously with engineering review. Specific submittal requirements related to the commissioning process are specified in Section 01 91 00 GENERAL COMMISSIONING REQUIREMENTS.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 CONSTRUCTION INSPECTIONS

- A. Commissioning of the building envelope systems will require inspection of individual elements of the envelope construction throughout the construction period. The Contractor shall coordinate with the Commissioning Agent in accordance with Section 01 91 00 and the Commissioning plan to schedule envelope inspections as required to support the Commissioning Process.

3.2 PRE-FUNCTIONAL CHECKLISTS

- A. The Contractor shall complete Pre-Functional Checklists to verify systems, subsystems, and equipment installation is complete and systems are ready for Systems Functional Performance Testing. The Commissioning Agent will prepare Pre-Functional Checklists to be used to document equipment installation. The Contractor shall complete the checklists. Completed checklists shall be submitted to the VA and to the Commissioning Agent for review. The Commissioning Agent may spot check

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a sample of completed checklists. If the Commissioning Agent determines that the information provided on the checklist is not accurate, the Commissioning Agent will return the marked-up checklist to the Contractor for correction and resubmission. If the Commissioning Agent determines that a significant number of completed checklists for similar equipment are not accurate, the Commissioning Agent will select a broader sample of checklists for review. If the Commissioning Agent determines that a significant number of the broader sample of checklists is also inaccurate, all the checklists for the type of equipment will be returned to the Contractor for correction and resubmission. Refer to SECTION 01 91 00 GENERAL COMMISSIONING REQUIREMENTS for submittal requirements for Pre-Functional Checklists, Equipment Startup Reports, and other commissioning documents.

3.3 CONTRACTORS TESTS

- A. Contractor tests as required by other sections of Division 07 or Division 08 shall be scheduled and documented in accordance with Section 01 00 00 GENERAL REQUIREMENTS. All testing shall be incorporated into the project schedule. Contractor shall provide no less than 7 calendar days' notice of testing. The Commissioning Agent will witness selected Contractor tests at the sole discretion of the Commissioning Agent. Contractor tests shall be completed prior to scheduling Systems Functional Performance Testing.

3.4 SYSTEMS FUNCTIONAL PERFORMANCE TESTING

- A. The Commissioning Process includes Systems Functional Performance Testing that is intended to test systems functional performance under steady state conditions, to test system reaction to changes in operating conditions, and system performance under emergency conditions. The Commissioning Agent will prepare detailed Systems Functional Performance Test procedures for review and approval by the COR. The Contractor shall review and comment on the tests prior to approval. The Contractor shall provide the required labor, materials, and test equipment identified in the test procedure to perform the tests. The Commissioning Agent will witness and document the testing. The Contractor shall sign the test reports to verify tests were performed. See Section 01 91 00 GENERAL COMMISSIONING REQUIREMENTS, for additional details.

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3.5 TRAINING OF VA PERSONNEL

A. Training of the VA operation and maintenance personnel is required in cooperation with the COR and Commissioning Agent. Provide competent, factory authorized personnel to provide instruction to operation and maintenance personnel concerning the location, operation, and troubleshooting of the installed systems. Contractor shall submit training agendas and trainer resumes in accordance with the requirements of Section 01 91 00. The instruction shall be scheduled in coordination with the VA COR after submission and approval of formal training plans. Refer to Section 01 91 00 GENERAL COMMISSIONING REQUIREMENTS and Division 07 ~~and 08 Sections~~ for additional Contractor training requirements.

----- END -----

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SECTION 07 22 00
ROOF AND DECK INSULATION

~~SPEC WRITER NOTES:~~

- ~~1. Delete text between // and // not applicable to project. Edit remaining text to suit project.~~
- ~~2. Follow recommendations of National Roofing Contractors Association Roofing Manual for design criteria.~~
- ~~3. Comply with ASHRAE Standard 90.1.~~
- ~~4. Ensure positive slope to drains occurs; minimum of 1/48 (1/4 inch per foot) without any "gutters" (no slopes) between drains.~~
- ~~5. Do not use on slopes over 1/12 (1 inch per foot) unless text is edited for this.~~
- ~~6. Ensure drains are located at points of maximum deflection; such as at mid-spans and bottom of slopes.~~
- ~~7. This section is intended for use where insulation is under the roofing or waterproofing membrane.~~
- ~~8. Do not use polystyrene insulation under bituminous roofing or waterproofing membrane.~~
- ~~9. Anchor insulation to deck. Do not specify loosely laid insulation.~~
- ~~10. Coordinate with work of Division 26 ELECTRICAL. Do not install electrical conduit in roof insulation.~~
- ~~11. Confirm applicable version of sustainability requirements.~~

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
- B. Roof and deck insulation, ~~//~~substrate board, ~~//~~vapor retarder, ~~//~~and ~~//~~cover board ~~//~~on ~~//new//~~existing~~//~~ concrete ~~//~~and~~//~~ metal deck ~~//~~substrates ready to receive roofing or waterproofing membrane.
- C. Repairs and alteration work to existing roof insulation.

1.2 RELATED WORK

~~SPEC WRITER NOTE: Update and retain references only when specified elsewhere in this section.~~

- A. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Non-Flooring Adhesives and Sealants VOC Limits.

~~B. Section 06 10 00, ROUGH CARPENTRY: Wood Cants, Blocking, and Edge Strips.~~

1.3 APPLICABLE PUBLICATIONS

- A. Comply with references to extent specified in this section.
- B. American Society of Civil Engineers
ASCE 7-16.....Minimum Design Loads and Associated Criteria
for Buildings and Other Structures
- C. American Society of Heating, Refrigeration and Air Conditioning
(ASHRAE):
Standard 90.1-13.....Energy Standard for Buildings Except Low-Rise
Residential Buildings.
- D. ASTM International (ASTM):
C208-12(2017)e2.....Cellulosic Fiber Insulating Board.
C552-17e1.....Cellular Glass Thermal Insulation.
C726-17.....Mineral Fiber Roof Insulation Board.
C728-17a.....Perlite Thermal Insulation Board.
C1177/C1177M-17.....Glass Mat Gypsum Substrate for Use as
Sheathing.
C1278/C1278M-17.....Fiber-Reinforced Gypsum Panel.
C1289-19.....Faced Rigid Cellular Polyisocyanurate Thermal
Insulation Board.
C1396/C1396M-17.....Gypsum Board.
D41/D41M-11 (2016).....Asphalt Primer Used in Roofing, Dampproofing,
and Waterproofing.
D312/D312M-16a.....Asphalt Used in Roofing.
D1970/D1970M-20.....Self-Adhering Polymer Modified Bituminous Sheet
Materials Used as Steep Roofing Underlayment
for Ice Dam Protection.
D2178/D2178M-15a.....Asphalt Glass Felt Used in Roofing and
Waterproofing.
D2822/D2822M-05(2011)e1.Asphalt Roof Cement, Asbestos Containing.
D4586/D4586M-07(2018)...Asphalt Roof Cement, Asbestos-Free.
E84-20.....Surface Burning Characteristics of Building
Materials.
F1667-18a.....Driven Fasteners: Nails, Spikes, and Staples.
- E. National Roofing Contractors Association (NRCA):

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Manual-15.....The NRCA Roofing Manual: Membrane Roof Systems-
2019.

F. UL LLC (UL):

Listed Online Certifications Directory.

G. U.S. Department of Agriculture (USDA):

USDA BioPreferred Program Catalog.

H. U.S. Department of Commerce National Institute of Standards and
Technology (NIST):

DOC PS 1-19.....Structural Plywood.

DOC PS 2-18.....Performance Standard for Wood-Based
Structural-Use Panels.

1.4 SUBMITTALS

A. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA,
AND SAMPLES.

B. Submittal Drawings:

1. Show size, configuration, and installation details.

a. Nailers, cants, and terminations.

b. Layout of insulation showing slopes, tapers, penetrations, and
edge conditions.

C. Manufacturer's Literature and Data:

1. Description of each product.

D. Samples:

1. Roof insulation, each type.

2. Fasteners, each type.

E. Sustainable Construction Submittals:

~~SPEC WRITER NOTE: Retain sustainable
construction submittals appropriate to
product.~~

1. Recycled Content: Identify post-consumer and pre-consumer recycled
content percentage by weight.

2. Biobased Content:

a. Show type and quantity for each product.

3. Low Pollutant-Emitting Materials:

a. Show volatile organic compound types and quantities.

b. Certify each ~~//~~composite wood ~~//~~and agrifiber ~~//~~product
contain no added urea formaldehyde.

F. Qualifications: Substantiate qualifications meet specifications.

1. Installer.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Same installer as Division 07 roofing section installer.

1.6 DELIVERY

- A. Comply with recommendations of NRCA Manual.
B. Deliver products in manufacturer's original sealed packaging.
C. Mark packaging, legibly. Indicate manufacturer's name or brand, type, and manufacture date.
D. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.7 STORAGE AND HANDLING

- A. Comply with recommendations of NRCA Manual.
B. Store products indoors in dry, weathertight facility.
C. Protect products from damage during handling and construction operations.

1.8 FIELD CONDITIONS

- A. Environment: Install products when existing and forecasted weather permit installation according to manufacturer's instructions.

1.9 WARRANTY

~~SPEC WRITER NOTE: Always retain construction warranty. FAR includes Contractor's one year labor and material warranty.~~

- A. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."
B. Manufacturer's Warranty: Warrant substrate board, vapor retarder, insulation, and cover board against material and manufacturing defects as part of Division 07 roofing system warranty.

PART 2 - PRODUCTS

2.1 SYSTEM PERFORMANCE

~~A. SPEC WRITER NOTE: On existing roofs confirm available insulation thickness and modify as required~~

~~B-A.~~ _____ Insulation Thermal Performance:

1. Overall Average R-Value: RSI-57 (R-33), minimum.
2. Any Location R-Value: RSI-17 (R-10), minimum.

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~~C.B.~~ _____ Fire and Wind Uplift Resistance: Provide roof insulation complying with requirements specified in Division 07 roofing section.

~~SPEC WRITER NOTE: Retain surface burning characteristics when insulation is installed over combustible and steel roof decks.~~

~~D.C.~~ _____ Insulation on ~~// Combustible //~~ Metal ~~//~~ Decking: UL labeled indicating compliance with one of the following:

1. UL Listed.
2. Insulation Surface Burning Characteristics: When tested according to ASTM E84.
 - a. Flame Spread Rating: 75 maximum.
 - b. Smoke Developed Rating: 150 maximum.

2.2 PRODUCTS - GENERAL

- A. Provide each product from one manufacturer.
- B. Sustainable Construction Requirements:

~~SPEC WRITER NOTE: Specify products containing greatest recycled content practicable to maximize material recovery. See EPA Comprehensive Procurement Guidelines (CPG) for guidance about individual products and available recycled content. Section 01 81 13 sets overall project recycled content requirements.~~

1. Insulation Recycled Content:

~~SPEC WRITER NOTE: Retain one or more of insulation types below based on project requirements.~~

- a. Mineral Fiber: 75 percent total recycled content, minimum.
- b. Fiberglass: 20 percent total recycled content, minimum.
- c. Cellulose: 75 percent post-consumer recycled content, minimum.
- d. Perlite Composite Board: 23 percent post-consumer recycled content, minimum.
- e. Rigid Foam: 9 percent total recycled content, minimum.
- f. Glass Fiber Reinforced Rigid Foam: 6 percent total recycled content, minimum.

~~SPEC WRITER NOTE:~~

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- ~~1. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS includes comprehensive product list setting VOC limits for low-emitting materials.~~
- ~~2. Retain subparagraphs applicable to products specified in this section.~~

2. Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:

- a. Non-flooring adhesives and sealants.
- b. Composite wood and agrifiber.

~~SPEC WRITER NOTES:~~

- ~~1. Retain paragraph below when it is appropriate to incorporate Federal Guiding Principles IV from Guiding Principles for Federal Leadership in High Performance and Sustainable Buildings incorporated in Executive Order 13423 "Strengthening Federal Environmental, Energy, and Transportation Management, dated January 24, 2007, Enhance Indoor Environmental Quality: Biobased Content Requirement, and and Executive Order 13693 Planning for Federal Sustainability in the Next Decade, 2015,. Requirement differs from related requirements in LEED definitions.~~
- ~~2. Project's General Requirements should indicate goals for percentages of bio-based, rapidly renewable, and certified sustainable wood products.~~

3. Bio-Based Materials: Where applicable, provide products designated by USDA and meeting or exceeding USDA recommendations for bio-based content, and products meeting Rapidly Renewable Materials and certified sustainable wood content definitions; refer to www.biopreferred.gov.

2.3 ADHESIVES

- A. Primer: ASTM D41/D41M.
- B. Asphalt: ASTM D312, Type III or IV for vapor retarders and insulation.
- C. Modified Asphaltic Insulation Adhesive: Insulation manufacturer's recommended modified asphaltic, asbestos-free, cold-applied adhesive

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formulated to adhere roof insulation to substrate or to another insulation layer.

- D. Bead-Applied Urethane Insulation Adhesive: Insulation manufacturer's recommended bead-applied, low-rise, one- or multicomponent urethane adhesive formulated to adhere roof insulation to substrate or to another insulation layer.
- E. Full-Spread Applied Urethane Insulation Adhesive: Insulation manufacturer's recommended spray-applied, low-rise, two-component urethane adhesive formulated to adhere roof insulation to substrate or to another insulation layer.
- F. Roof Cement: Asbestos free, ASTM D2822/D2822M, Type I or Type II; or, ASTM D4586/D4586M, Type I or Type II.

2.4 ROOF AND DECK INSULATION

- A. Roof and Deck Insulation, General: Preformed roof insulation boards approved by roofing manufacturer.

~~SPEC WRITER NOTE: Retain one or more of insulation types below based on project requirements. If retaining more than one type, indicate locations of different types on drawings.~~

- B. Polyisocyanurate Board Insulation: ASTM C1289, Type II, Class 1, Grade 2, faced with glass fiber reinforced cellulosic felt facers on both major surfaces of the core foam.

~~C. Cellular Glass Board Insulation: ASTM C552, Type IV, kraft paper sheet faced.~~

~~D. Perlite Board Insulation: ASTM C728, expanded perlite particles, selected binders, and cellulosic fibers with surface treated to reduce bitumen absorption.~~

~~E. Cellulosic Fiber Board Insulation: ASTM C208, Type II, // Grade 1 for built-up asphalt or modified bitumen roofing // Grade 2 for single-ply roofing //.~~

~~F.C.~~ _____ Tapered Roof Insulation System:

1. Fabricate of mineral fiberboard, polyisocyanurate, perlite board, or cellular glass. Use only one insulation material for tapered sections. Use only factory-tapered insulation.
2. Cut to provide high and low points with crickets and slopes as shown.

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3. Minimum thickness of tapered sections; 38 mm (1-1/2 inch).
4. Minimum slope 1/48 (1/4 inch per 12 inches).

~~SPEC WRITER NOTE: Specify Composite Nail Base Insulated Roof Sheathing for steep slope roof installations such as shingle, slate, tile, or metal.~~

~~G. Composite Nail Base Insulated Roof Sheathing:~~

- ~~1. Oriented Strand Board Surfaced, Polyisocyanurate Foam Sheathing: Polyisocyanurate thermal insulation ASTM C1289, Type V, insulation thickness as shown, with oriented strand board laminated to top surface.~~
- ~~2. Oriented Strand Board: NIST DOC PS 1, Exposure 1, // 11 mm (7/16 inch) // 16 mm (5/8 inch) // thick.~~
- ~~3. Bottom surface faced with felt facers.~~

2.5 INSULATION ACCESSORIES

- A. Glass (Felt): ASTM D2178/D2178M, Type VI, heavy duty ply sheet.
- B. Cants and Tapered Edge Strips:

~~SPEC WRITER NOTE: Retain first paragraph below for roofing systems requiring wood cant strips at roof perimeter and major openings.~~

- ~~1. Wood Cant Strips: Refer to Section 06 10 00, ROUGH CARPENTRY.~~
2. Insulation Cant Strips: ASTM C208, Type II, Grade 1, cellulosic-fiber insulation board.
3. Tapered Edge Strips: 1/12 (1 inch per 12 inches), from 0 mm (0 inches), 300 mm to 450 mm (12 inches to 18 inches) wide.
 - a. Cellulosic Fiberboard: ASTM C208.
 - b. Mineral Fiberboard: ASTM C726.
 - c. Perlite Board: ASTM C728.
- C. Vapor Retarder:
 1. Glass-Fiber Felts: ASTM D2178/D2178M, Type IV, asphalt impregnated.
 2. Self-Adhering Sheet Vapor Retarder: ASTM D1970/D1970M, minimum 1.0 mm (40 mils) thick membrane of HDPE film fully coated with asphalt adhesive, or 0.76 to 1.0 mm (30 to 40 mils) thick membrane of butyl rubber based adhesive backed by a layer of high density cross-laminated polyethylene; maximum permeance rating of 6 ng/Pa/s/sq. m (0.1 perms).

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D. Substrate Board:

~~SPEC WRITER NOTE: Select one or more paragraphs below for roofing systems requiring support for vapor retarder over steel deck, or where specific substrate board is required by specified fire-resistance-rated roof assembly design.~~

1. Gypsum Board: ASTM C1396/C1396M, 16 mm (5/8 inch) thick, Type X.
2. Glass-Mat, Water-Resistant Gypsum Roof Board: ASTM C1177/C1177M, ~~//~~13 mm (1/2 inch) or~~//~~Type X, 16 mm (5/8 inch) ~~//~~thick, factory primed.
- ~~3. Cellulosic Fiber Reinforced, Water-Resistant Gypsum Roof Board: ASTM C1278/C1278M, // 6 mm (1/4 inch) // 10 mm (3/8 inch) // 13 mm (1/2 inch) // 16 mm (5/8 inch) // thick.~~
- ~~4. Perlite Board Insulation: ASTM C728, // 19 mm (3/4 inch) // 25 mm (1 inch) //.~~

E. Cover Board:

~~SPEC WRITER NOTE: Typically retain one paragraph from four below for VA roofing systems. Cover board is required under single-ply membranes installed over plastic foam insulation board based upon NRCA recommendations.~~

1. Glass-Mat, Water-Resistant Gypsum Roof Board: ASTM C1177/C1177M, ~~//~~6 mm (1/4 inch), ~~//~~13 mm (1/2 inch), or~~//~~16 mm (5/8 inch) ~~//~~thick, factory primed.
- ~~2. Cellulosic Fiber Reinforced, Water-Resistant Gypsum Roof Board: ASTM C1278/C1278M, // 6 mm (1/4 inch) // 10 mm (3/8 inch) // 13 mm (1/2 inch) // 16 mm (5/8 inch) // thick.~~
- ~~3. Cellulosic Fiber Insulation Board: ASTM C208, Type II, Grade 2, 13 mm (1/2 inch) thick.~~
- ~~4. Oriented Strand Board: DOC PS 2, Exposure 1, 11 mm (7/16 inch) thick.~~

2.6 ACCESSORIES

- A. Fasteners: Corrosion-resistant carbon steel fasteners and galvalume-coated steel or plastic round plates for fastening substrate board and insulation to roof deck.
- B. Nails: ASTM F1667; type to suit application.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Comply with requirements of Division 07 roofing section.

3.2 PREPARATION

- A. Examine and verify substrate suitability for product installation.
- B. Protect existing construction and completed work from damage.

3.3 INSTALLATION - GENERAL

- A. Install products according to manufacturer's instructions.
 - 1. When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's Representative consideration.
- B. Comply with requirements of UL for insulated steel roof deck.
- C. Attach substrate board and other products to meet requirements of Division 07 roofing section.

3.4 SUBSTRATE BOARD INSTALLATION

~~SPEC WRITER NOTES:~~

- ~~1. The National Roofing Contractors Association recommends substrate board be installed over steel deck. Use substrate board over steel deck on all VA new construction.~~
- ~~2. Use substrate board over steel deck to provide continuous support for vapor retarder where required.~~

- A. Fasten substrate board to top flanges of steel decking to resist uplift pressures according requirements for specified roofing system.
 - 1. Locate the long dimension edge joints solidly bearing on top of decking ribs.

3.5 VAPOR RETARDER INSTALLATION

~~SPEC WRITER NOTES:~~

- ~~1. Review requirement for vapor retarders for use over decks to receive insulation for heated buildings where the January average means temperature is below 4.5 degrees C (40 degrees F), the relative humidity is 45 percent or greater. Vapor retarders should typically be considered for heated portions of buildings where a high humidity condition is expected such as a kitchen, indoor swimming pool, or laundry.~~

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~~2. Follow NRCA procedures for determining if vapor retarders are required, including venting, and ASHRAE for calculations; however, under conditions of extreme humidity and cold weather conditions, a vapor retarder is required and water vapor transmission analysis is required. Ensure that the vapor retarder is shown at all locations where required on the Drawings.~~

A. Vapor Retarder Installation, General:

1. Install continuous vapor retarder on roof decks where indicated.
2. At vertical surfaces, turn up vapor retarder to top of insulation or base flashing.
3. Seal penetrations through vapor retarder with roof cement to prevent moisture entry from below.

B. Cast in Place Concrete Decks, Except Insulating Concrete:

1. Prime deck as specified.
2. Apply two plies of asphalt saturated felt mopped down to deck.

C. Precast Concrete Unit Decks Without Concrete Topping:

1. Prime deck as specified.
2. Apply two plies of asphalt saturated felt.
3. Mop to deck, keeping bitumen 100 mm (4 inches) away from joints of precast units. Bridge joints with felt. Mop between plies as specified.

3.6 INSULATION INSTALLATION

~~SPEC WRITER NOTES:~~

- ~~1. Calculate dew point location to determine amount of insulation required over vapor retarder, if any. Ensure dew point occurs within insulation thickness to prevent condensation occurring on interior surface of vapor retarder.~~
- ~~2. Ensure that the total insulation thickness will span flutes in metal deck.~~
- ~~3. Specify R value for each layer of insulation to avoid ambiguities.~~
- ~~4. Indicate types of insulation required when more than one type is needed due to roofing system designs. Edit Part 2 PRODUCTS to correspond to acceptable insulation types.~~

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- ~~5. Use two layers, except on steel decks used for canopies or unheated structures.~~
- ~~6. Mechanical fasteners shall not extend through both layers of insulation over heated spaces.~~
- ~~7. Comply with International Building Code requirements for use of plastic insulation.~~
- ~~8. Use cellular glass or mineral fiber board for tapered edge strips at gravel stop in facilities with raised edges.~~
- ~~9. For tapered insulation systems, use minimum thickness at roof drains of 38 mm (1-1/2 inch).~~
- ~~10. Use crickets to eliminate "gutters" in roof slopes. Do not allow "flat areas" where ponding will occur. Provide dishes at drains, 13 mm (1/2 inch) deep. Coordinate with drawing details.~~
- ~~11. Use cellular glass board for plaza at promenade decks.~~

A. Insulation Installation, General:

~~SPEC WRITER NOTE: Base sheet is not required for all insulation. Confirm the requirement to include it according to the roofing system.~~

1. Base Sheet: Where required by roofing system, install one lapped base sheet specified in Division 07 roofing section by mechanically fastening to roofing substrate before installation of insulation.

~~SPEC WRITER NOTE: Coordinate cant strip requirement with roof systems referencing this section.~~

2. Cant Strips: Install ~~// preformed insulation cant strips // wood cant strips specified in Section 06 10 00 ROUGH CARPENTRY //~~ at junctures of roofing system with vertical construction.

~~SPEC WRITER NOTE: Retain paragraph below for re-roofing projects.~~

3. Use same insulation as existing for roof repair and alterations unless specified otherwise.

B. Insulation Thickness:

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1. Thickness of roof insulation shown on drawings is nominal. Provide thickness required to comply with specified thermal performance.

~~SPEC WRITER NOTE: Verify sufficient R-Value occurs over vapor retarder to prevent condensation, especially over insulating concrete decks.~~

2. Insulation on Metal Decks: Provide insulation in minimum thickness recommended by insulation manufacturer to span deck flutes. Support edges of insulation on metal deck ribs.
3. When actual insulation thickness differs from drawings, coordinate alignment and location of roof drains, flashing, gravel stops, fascias and similar items.

~~SPEC WRITER NOTE: Ensure that roof slopes high and low points are shown on the roof plan, including crickets. Do not have "gutters" or level areas between drains. Ensure that drains are located at low points. Ensure that correct geometry is shown for slope.~~

4. Where tapered insulation is used, maintain insulation thickness at high points and roof edges shown on drawings.
 - a. Low Point Thickness: Minimum 38 mm (1-1/2 inches).
5. Use minimum two layers of insulation when required thickness is 68 mm (2.7 inch) or greater.
- C. Lay insulating units with close joints, in regular courses and with end joints staggered.
 1. Stagger joints between layers minimum 150 mm (6 inches).
- D. Lay units with long dimension perpendicular to the rolled (longitudinal) direction of the roofing felt.
- E. Seal cut edges at penetrations and at edges against blocking with bitumen or roof cement.

~~SPEC WRITER NOTE: Insulation is required to be mechanically anchored to steel decks. Delete non applicable text. Refer to Division 07 roofing section requirements for resistance to wind storm uplift. Design for wind conditions of area where building is constructed.~~

- F. Cut to fit tightly against blocking or penetrations.

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G. Cover all insulation installed on the same day; comply with temporary protection requirements of Division 07 roofing section.

H. Installation Method:

1. Adhered Insulation:

- a. Prime substrate as required.
- b. Set each layer of insulation firmly in solid mopping of hot asphalt.
- c. Set each layer of insulation firmly in ribbons of bead-applied insulation adhesive.
- d. Set each layer of insulation firmly in uniform application of full-spread insulation adhesive.

~~2. Mechanically Fastened Insulation:~~

- ~~a. Fasten insulation according to requirements in Division 07 roofing section.~~
- ~~b. Fasten insulation to resist uplift pressures specified in Division 07 roofing section and ASCE-7.~~

~~3. Mechanically Fastened and Adhered Insulation:~~

- ~~a. Fasten first layer of insulation according to "Mechanically Fastened Insulation" requirements.~~
- ~~b. Fasten each subsequent layer of insulation according to "Adhered Insulation" requirements.~~

3.7 COVER BOARD INSTALLATION

- A. Install cover boards over insulation with long joints in continuous straight lines with staggered end joints.
- B. Offset cover board joints from insulation joints 150 mm (6 inches), minimum.

~~SPEC WRITER NOTE: Do not mechanically fasten cover board under singly ply membranes to avoid potential excessive wear and puncture of membrane at exposed cover board fasteners.~~

- C. Secure cover boards according to ~~//~~"Adhered Insulation" ~~//~~
~~"Mechanically Fastened Insulation"~~ ~~//~~ requirements.

- - E N D - -

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SECTION 07 40 00
ROOFING AND SIDING PANELS

~~SPEC WRITER NOTE:—~~

~~Delete between // // if not applicable to project. Also delete any other item or paragraph not applicable in the section and renumber the paragraphs.~~

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This section specifies ~~// insulated // //~~uninsulated ~~// metal // wall // and roof panels // // and fire rated composite metal wall and roof systems //~~as shown on contract documents.

1.2 RELATED WORK

- A. ~~//~~Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.~~//~~
- B. Section 07 92 00, JOINT SEALANTS: Sealant.
- ~~C. Section 09 06 00, SCHEDULE FOR FINISHES: Color and texture of finish.~~

1.3 QUALITY ASSURANCE

- A. Manufacturer's Qualifications: Provide metal ~~wall and roof panels //~~ ~~and composite metal wall and roof systems //~~ products of a manufacturer regularly engaged for not less than five (5) years in the fabrication of metal ~~panels //~~ ~~and composite metal wall and roof systems //~~ of the type and design indicated.
- B. Installer: A firm with three (3) years of successful experience with installation of roofing and siding panels of type and scope equivalent to Work of this Section. Submit installer qualifications.

1.4 FIRE RATING

- A. ~~Composite metal wall // and roof Roof//~~ systems to have a fire rating ~~of // // hours when tested in accordance with ASTM E119.~~to match adjacent existing roof.

1.5 SUBMITTALS

- A. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.
- B. ~~//~~Sustainable Design Submittals, as described below:
- // Postconsumer recycled content as specified in PART 2 - PRODUCTS.~~// //~~

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- C. Samples: Metal panel, 152 mm (6 inch) square, showing finish, each color and texture.
- D. Shop Drawings: ~~// Wall // // and rRoof //~~ panels, showing details of construction and installation. ~~// Collateral steel framing, // // U value, // thickness and kind of material, closures, flashing, fastenings and related components and accessories. Show interfaces and relationships to work at other trades and continuity with adjacent thermal, weather, air and vapor barriers.~~
- E. Manufacturer's Literature and Data: ~~// Wall // // and rRoof //~~ panels
- F. Fire Test Report: Report of fire test by recognized testing laboratory for fire rating specified, showing details of construction.
- G. Manufacturer's Certificates: Indicating manufacturer's qualifications specified.
- H. Installer qualifications.
- I. Manufacturer warranty.

1.6 QUALITY ASSURANCE

- A. Approval by Contracting Officer Representative (COR) is required of products of proposed manufacturer.
- B. Certify manufacturer has five (5) years continuous documented experience in fabrication of metal roofing and siding panels.
- C. Source: For each material type required for work of this section, provide primary materials, which are products of one manufacturer. Provide secondary or accessory materials, which are acceptable to manufacturers of primary materials.
- D. Installer: A firm with a minimum of three (3) years' experience in type of work required by this section and which is acceptable to manufacturers of primary materials.

1.7 WARRANTY

- A. Construction Warranty: Comply with FAR clause 52.246-21 "Warranty of Construction".
- B. Manufacturer Warranty: Manufacturer shall warranty their metal ~~// roofing // // and wall panels //~~ for a minimum of ten (10) years ~~// //~~ from the date of installation and final acceptance by the Government. Submit manufacturer warranty.
- C. Warranty on Panel Finishes: Manufacturer's shall warrant their ~~// roofing // // and wall panel //~~ finish and provide standard agreement to repair finish or replace metal panels that show evidence

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of deterioration of factory-applied finishes within specified warranty period.

1. Exposed Panel Finish: Deterioration includes, but is not limited to, the following:
 - a. Color fading more than 5 Hunter units when tested according to ASTM D2244.
 - b. Chalking in excess of a No. 8 rating when testing according to ASTM D4214.
 - c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.
2. Finish Warranty Period: ~~// 20 //~~ ~~// 10 //~~ years from date of installation and final acceptance by the COR.

1.8 APPLICABLE PUBLICATIONS

- A. The publications listed below form a part of this specification to the extend referenced. The publications are referenced in the text by the basic designation only.
- B. American Architecture Manufacturers Association (AAMA):
 - 611-14..... Anodized Architectural Aluminum
 - 621-02..... Voluntary Specifications for High Performance Organic Coatings on Coil Coated Architectural Hot Dipped Galvanized (HDG) and Zinc-Aluminum Coated Steel Substrates
 - 2605-13..... Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels
- C. American Iron and Steel Institute (AISI):
 - SG03-02..... Cold-Formed Steel Design Manual
- D. ASTM International (ASTM):
 - A463/A463M-15..... Steel Sheet, Cold-Rolled, Aluminum-Coated, by the Hot-Dip Process
 - A653/A653M-20..... Steel Sheet, Zinc-Coated (Galvanized), or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
 - A924/A924M-19..... Steel Sheet, Metallic Coated by the Hot-Dip Process

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A1008/A1008M-18	Steel, Sheet, Cold-Rolled, Carbon, Structural, High Strength Low Alloy
B209-14	Aluminum and Aluminum Alloy Sheet and Plate
B209M-14	Aluminum and Aluminum Alloy Sheet and Plate (Metric)
C553-19	Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications
C591-20	Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation
C612-14 (2019)	Mineral Fiber Block and Board Thermal Insulation
C1396/C1396M-17	Gypsum Board
D2244-16	Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates
D4214-07 (2015)	Test Methods for Evaluating the Degree of Chalking of Exterior Paint Films
E119-20	Fire Test of Building Construction and Materials
E283-19	Test Method for Determining Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen
E331-00 (2016)	Test Method for Water Penetration of Exterior Windows, Skylight, Doors, and Curtain Walls by Uniform Static Air Pressure Difference
E1592-05 (2017)	Standard Test Method for Structural Performance of Sheet Metal Roof and Siding Systems by Uniform Static Air Pressure Method
E1646-95 (2018)	Test Method for Water Penetration of Exterior Metal Roof Panel Systems by Uniform Static Air Pressure Difference
E1680-16	Test Method for Rate of Air Leakage Through Exterior Metal Roof Panel Systems
E1980-11 (2019)	Calculating Solar Reflectance Index of Horizontal and Low-Sloped Opaque Surfaces

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E2140-01(2017) Test Method for Water Penetration of Metal Roof
Panel Systems by Static Water Pressure Head

E. Cool Roof Rating Council (CRRC):
Standard-14

F. FM Global:
4471-10 Class 1 Panel Roofs

G. Underwriters Laboratories (UL):
580-05(R2018) Tests for Uplift Resistance of Roof Assemblies
Fire Resistance Directory

PART 2 - PRODUCTS

~~SPEC WRITER NOTE: Delete this entire
paragraph if roof panels are not used.~~

2.1 PERFORMANCE REQUIREMENTS ROOF PANELS

- A. Energy Performance: Provide roof panels according to one of the following when tested according to CRRC-1:
1. Three-year, aged solar reflectance of not less than 0.55 and emissivity of not less than 0.75.
 2. Three-year, aged Solar Reflectance Index (SRI) of not less than 64
~~// — //~~ when calculated according to ASTM E1980.

~~SPEC WRITER NOTE:—
Coordinate wind and other design with
that identified on drawings or required
by code for location of project.~~

- B. Structural Performance: Provide metal panel systems capable of withstanding the effects of the following loads, based on testing according to ASTM E1592.

- ~~1. Wind Loads: // — // //As indicated//.~~
- ~~2. Other Design Loads: // — // //As indicated//.~~
- ~~3. Deflection Limits: For wind loads, no greater than // 1/180 //
// 1/240 // of the span.~~
- ~~4. Minimum Panel rib depth: // — //.~~
- ~~5. Panel width: //12 inches// //24 inches// //36 inches//.~~

- C. Air Infiltration: Air leakage of not more than 0.3 liter/second per square meter (0.06 cfm/square foot) when tested according to ASTM E1680 or ASTM E283. ~~at the following test pressure difference:~~

- ~~1. Test Pressure Difference: // 75 Pa (1.57 pound force/square foot) //
// 300 Pa (6.24 pound force/square foot) //.~~

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~~D. Water Penetration under Static Pressure: No water penetration when tested according to ASTM E1646 or ASTM E331. at the following test pressure difference:~~

~~1.D. Test Pressure Difference: // 137 Pa (2.86 pound force/square foot) // // 300 Pa (6.24 pound force/square foot) //.~~

E. Hydrostatic-Head Resistance: No water penetration when tested according to ASTM E2140.

F. Wind-Uplift Resistance: Provide metal roof panel assemblies that comply with UL 580 for wind-uplift-resistance class indicated.

~~1. Uplift Rating: // UL 30 // // UL 60 // // UL 90 //.~~

G. FM Global Listing: Provide metal roof panels and composite component materials that comply with requirements FM Global 4771 as part of a panel roofing system and that are listed in FM Global's "Approval Guide" for Class 1 or noncombustible construction, as applicable. Identify materials with FM Global markings.

~~1. Fire/Windstorm Classification: Class 1A // 60 // // 75 // // 90 // // 120 // // 105 //.~~

~~2. Hail Resistance: // MH // // SH //.~~

H. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes by preventing buckling, opening of joints, overstressing of components, failure of joints sealants, failure of connections, and other detrimental effects. Base calculations on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

~~1. Temperature Change (Range): 67 degrees C (120 degrees F), ambient, 100 degrees C (180 degrees F), material surfaces.~~

~~SPEC WRITER NOTE: Delete this entire paragraph if wall panels are not used.~~

~~2.2 PERFORMANCE REQUIREMENTS FOR WALL PANELS~~

~~SPEC WRITER NOTE:~~

~~Coordinate wind load and other design load requirements with that identified on drawings or required by code for location of project.~~

~~A. Structural Performance: Provide metal panel systems capable of withstanding the effects of the following loads, based on testing according to ASTM E1592.~~

~~1. Wind Loads: // // //As indicated//.~~

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- ~~2. Other Design Loads: // // //As indicated//.~~
- ~~3. Deflection Limits: For wind loads, no greater than // 1/180 //~~
~~// 1/240 // of the span.~~
- ~~4. Minimum Panel rib depth: // //.~~
- ~~5. Panel width: //12 inches// //24 inches// //36 inches//.~~
- ~~B. Air Infiltration: Air leakage of not more than 0.3 liter/second per~~
~~square meter (0.06 cfm/square foot) when tested according to ASTM E283~~
~~at the following test pressure difference:~~
- ~~1. Test Pressure Difference: // 75 Pa (1.57 pound/square foot) // //~~
~~300 Pa (6.24 pound force/square foot) //.~~
- ~~C. Water Penetration under Static Pressure: No water penetration when~~
~~tested according to ASTM E331 at the following test pressure~~
~~difference:~~
- ~~1. Test Pressure Difference: // 137 Pa (2.86 pound force/square foot)~~
~~// // 300 Pa (6.24 pound force/square foot) //.~~
- ~~D. Thermal Movements: Allow for thermal movements from ambient and surface~~
~~temperature changes by preventing buckling, opening of joints,~~
~~overstressing of components, failure of joints sealants, failure of~~
~~connections, and other detrimental effects. Base calculations on~~
~~surface temperatures of materials due to both solar heat gain and~~
~~nighttime sky heat loss.~~
- ~~1. Temperature Change (Range): 67 degrees C (120 degrees F), ambient;~~
~~100 degrees C (180 degrees F), material surfaces.~~
- ~~E. // Fire Resistance Ratings: Comply with ASTM E119, testing by a~~
~~qualified testing agency. Identify products with appropriate markings~~
~~of applicable testing agency.~~
- ~~1. Indicate design designations from UL's Fire Directory or from the~~
~~listings of another qualified testing agency. //~~

~~SPEC WRITER NOTE:—~~

~~Make material requirements agree with~~
~~applicable requirements specified in the~~
~~referenced Applicable Publications.~~
~~Update and specify only that which~~
~~applies to the project.~~

2.32.2 SHEET STEEL

- A. Minimum 0.8 mm (0.31 inch) thick for wall and roof panels.
- B. Steel, Sheet, Galvanized: ASTM A653/A653M and AISI SG03-3, Structural.

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1. Grade 40, galvanized coating conforming to ASTM A924/A924M, Class Z 275 G-90.
- C. Steel, Sheet, Commercial: ASTM A1008, Type C.
- D. Steel, Sheet, Aluminized: ASTM A463/A463M and AISI SG03-3. Steel to be coated on both sides with 0.15 Kg/square meter (0.5 ounce of aluminum per square foot).
- E. ~~//~~Recycled Content of Steel Products: Postconsumer recycled content not less than ~~//~~30 ~~////~~~~//~~ percent.~~//~~

2.42.3 ALUMINUM PLATE AND SHEET

- A. ASTM B209M (B209).

2.52.4 FASTENERS

- A. ~~//~~Fasteners for Steel Panels: Galvanized or cadmium plated steel.~~//~~
- B. ~~//~~Fasteners for Aluminum Panels to be aluminum or stainless steel.~~//~~
- C. ~~//~~Fasteners of size, type and holding strength as recommended by panel manufacturer.~~//~~

2.62.5 GYPSUM BACKING BOARD

- A. ASTM C1396/C1396M, Type X, Plain face, Square edge.

2.72.6 THERMAL INSULATING MATERIALS

- A. Urethane or Isocyanurate Board: ASTM C591, Type I.
- B. Mineral Fiber Blankets: ASTM C553, Type I.
- C. Mineral Fiber Board: ASTM C612, Class I.

2.82.7 FABRICATION

- A. General:
 1. Furnish panels in one continuous length, ~~//~~ for full height, ~~//~~ or at least one story height for wall panels ~~//~~ with no horizontal joints, except at cut-outs or openings as required for the passage of pipes, conduits, vents and the like.
 2. Construct panels by pressing members together to form a structural unit with closed ends.
 3. Overall thickness of panels is shown of the contract documents.
 4. Provide connection between panels by ~~//~~ interlocking male and female joints ~~//~~ ~~//~~ interlocking joints filled with sealant ~~//~~. Seal joints between related components as required to make the work water-tight. Refer to Section 07 92 00, JOINT SEALANTS for sealing compounds.
 5. Provide ~~//~~ collateral steel framing, ~~//~~ metal and bituminous closures, fastenings, flashing, clip, caulking, ~~//~~ panel reinforcements for support of mechanical and electrical work as

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shown on the contract documents, ~~///~~ and related components and accessories.

- a. Sub-girts: 1.0 mm (0.0396 inches) thick galvanized steel hat channels deigned to receive panel fasteners or clips.
- b. Accessories, fastenings, and flashings to be the same material and finish as the panels. Thickness and installation of accessories and flashing to be as recommended by the panel manufacturer.

~~B. Insulated Metal Panels:~~

~~1. Panels to consist of a structurally reinforced insulated core, fastened between an exterior face sheet and an interior liner sheet.~~

~~2. Exterior Face Sheets:~~

- a. ~~/// 0.8 /// 1.0 /// 1.25 /// mm (/// 0.032 /// 0.040 /// 0.050 /// inch) thick aluminum.~~
- b. ~~/// 1.2 /// 0.9 /// 0.6 /// 0.5 /// mm (/// 0.0478 /// 0.0359 /// 0.0229 /// 0.0239 /// inch) thick uncoated steel.~~
- c. ~~/// 1.3 /// 1.0 /// 0.85 /// 0.7 /// 0.6 /// mm (/// 0.0516 /// 0.0396 /// 0.0336 /// 0.0276 /// 0.0247 /// inch) thick galvanized steel.~~
- d. ~~/// 0.8 /// 1.25 /// mm (/// 0.032 /// 0.050 /// inch) thick aluminized steel.~~

~~3. Interior Liner Face Sheet:~~

- a. ~~/// 0.8 /// 1.0 /// 1.25 /// mm (/// 0.032 /// 0.040 /// 0.050 /// inch) thick aluminum.~~
- b. ~~/// 1.2 /// 0.9 /// 0.6 /// 0.5 /// mm (/// 0.0478 /// 0.0359 /// 0.0229 /// 0.0239 /// inch) thick uncoated steel.~~
- c. ~~/// 1.3 /// 1.0 /// 0.85 /// 0.7 /// 0.6 /// mm (/// 0.0516 /// 0.0396 /// 0.0336 /// 0.0276 /// 0.0247 /// inch) thick galvanized steel.~~

~~SPEC WRITER NOTE:~~

~~Verify use of urethane or isocyanurate insulation in VA hospital patient and other public areas.~~

~~4. Insulation:~~

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~~a. Provide // urethane board // // isocyanurate // // mineral fiber blanket // // mineral fiber board // having a "U" value of // 0.85 // // 1.2 // // // W/ (square meter x K) (// 0.15 // // 0.21 // // // Btu/ [h x square foot x degrees F]).~~

~~C.B.~~ Uninsulated Metal Panels:

1. Panels to consist of a structurally reinforced air-space fastened between two metal face sheets.

2. Panels:

- ~~a. // 0.8 // // 1.0 // // 1.25 // // mm (// 0.032 // // 0.040 // // 0.050 // inch) thick aluminum. Match existing adjacent roof panels.~~
- ~~b. // 1.2 // // 0.9 // // 0.6 // // 0.5 // mm (// 0.0478 // // 0.0359 // // 0.0229 // // 0.0239 // inch) thick uncoated steel.~~
- ~~c. // 1.3 // // 1.0 // // 0.85 // // 0.7 // // 0.6 // mm (// 0.0516 // // 0.0396 // // 0.0336 // // 0.0276 // // 0.0247 // inch) thick galvanized steel.~~
- ~~d. // 0.8 // // 1.25 // mm (// 0.032 // // 0.050 // inch) thick aluminized steel.~~

~~D. Composite Metal Wall Systems:~~

~~1. Panels consisting of an exterior face sheet, sub-girts, gypsum backing board panels, insulation, and interior liner sheet.~~

~~2. Panel Composition:~~

- ~~a. Exterior face sheet of 0.9 mm (0.0359 inch) thick sheet steel of indicated configuration and pattern.~~
- ~~b. Gypsum backing board used for wall panels to be of the same type for each layer.~~
- ~~c. Interior liner sheet of 0.7 mm (0.0276 inch) thick galvanized sheet steel of flush pattern.~~
- ~~d. Insulation to be mineral fiber blankets installed on interior face of liner sheet.~~

~~E. Fabricate wall louvers and frames used in conjunction with walls panels to be of same material, thickness and finish as exterior face sheets of wall system. Louver assembly to be designed and installed to prevent infiltration of water into structure.~~

2.92.8-FINISH

- A. For ~~insulated and~~ uninsulated ~~wall and~~ roof panels ~~// and composite wall and roof panels //~~, provide finishes as follows for face sheets. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
- B. ~~//Energy Performance: Provide roof panels with solar reflectance index not less than // when calculated according to per ASTM E1980 based on testing identical products by a qualified testing agency.//.~~
- C. Provide aluminum alloy for color coating as required to produce specified color. Provide color ~~to match existing adjacent roofs specified in Section 09 06 00, SCHEDULE FOR FINISHES~~. Color for sheet aluminum to not deviate more than the colors of extrusion samples.
1. ~~//Two-Coat Fluoropolymer: AAMA 2605. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat.//~~
 2. ~~//Three-Coat Fluoropolymer: AAMA 2605. Fluoropolymer finish coating not less than 70 percent PVDF resin by weight in both color coat and clear topcoat.//~~
 3. ~~//Mica Fluoropolymer: AAMA 2605. Two-coat fluoropolymer finish with suspended mica flakes containing not less than 70 percent PVDF resin by weight in color coat.//~~
 4. ~~//Metallic Fluoropolymer: AAMA 2605. Three-coat fluoropolymer finish with suspended metallic flakes containing not less than 70 percent PVDF resin by weight in both color coat and clear topcoat.//~~
 5. ~~//FEVE Fluoropolymer: AAMA 2605. Two-coat fluoropolymer finish containing 100 percent fluorinated ethylene vinyl ether resin in color coat.//~~
 6. ~~//Siliconized Polyester: Epoxy primer and silicone-modified, polyester-enamel topcoat; with a dry film thickness of not less than 0.005 mm (0.2 mil) for primer and 0.02 mm 0.8 mil for topcoat.//~~
 7. ~~//Exposed Anodized Finish://~~
 - a. Clear Anodic Finish: AAMA 611, ~~//AA-M12C22A41, Class I, 0.018 mm (0.7 mil) (min.); //M12C22A31, Class II, 0.010 mm (0.4 mil) (min.).//~~
 - b. Color Anodic Finish: AAMA 611, ~~//AA-M12C22A42/A44, Class I, 0.018 mm (0.7 mil) (min.); //AA-M12C22A32/A34, Class II, 0.010 mm (0.4 mil) (min.).//~~

- D. Provide finishes for steel face sheets as follows. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
1. ~~Two-Coat Fluoropolymer~~: AAMA 2605. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat.~~Two-Coat Fluoropolymer~~
 2. ~~Three-Coat Fluoropolymer~~: AAMA 2605. Fluoropolymer finish coating not less than 70 percent PVDF resin by weight in both color coat and clear topcoat.~~Three-Coat Fluoropolymer~~
 3. ~~Mica Fluoropolymer~~: AAMA 2605. Two-coat fluoropolymer finish with suspended mica flakes containing not less than 70 percent PVDF resin by weight in color coat.~~Mica Fluoropolymer~~
 4. ~~Metallic Fluoropolymer~~: AAMA 2605. Three-coat fluoropolymer finish with suspended metallic flakes containing not less than 70 percent PVDF resin by weight in both color coat and clear topcoat.~~Metallic Fluoropolymer~~
 5. ~~FEVE Fluoropolymer~~: AAMA 2605. Two-coat fluoropolymer finish containing 100 percent fluorinated ethylene vinyl ether resin in color coat.~~FEVE Fluoropolymer~~
 6. ~~Siliconized Polyester~~: Epoxy primer and silicone-modified, polyester-enamel topcoat; with a dry film thickness of not less than 0.005 mm (0.2 mil) for primer and 0.02 mm (0.8 mil) for topcoat.~~Siliconized Polyester~~
 7. ~~Concealed Finish~~: Apply pretreatment and manufacturer's standard white or light-colored acrylic or polyester backer finish consisting of prime coat and wash coat with a minimum total dry film thickness of 3 mm (0.5 mil).

PART 3 - EXECUTION

3.1 INSTALLATION

- A. General: Install panels in accordance with the manufacturer's approved erection instructions and diagrams, except as specified otherwise.
- B. Install panels in full and firm contact with supports and with each other at side and end laps.
- C. Where panels are cut in the field, or where factory applied coverings or coatings are abraded or damaged in handling or installation, make finish repairs with material of the same type and color as the weather coating, before being installed.
- D. Seal cut ends and edges, including those at openings through the sheets.

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- E. Correct defects or errors in the materials in a manner approved by the COR.
- F. Replace defective materials which cannot be corrected with nondefective material.
- G. Provide molded closure strips where indicated and whenever sheets terminate with open ends after installation.

~~H. Wall Panels:-~~

- ~~1. Apply panels with the configuration in a vertical position.~~
- ~~2. Provide panels in // the longest obtainable lengths, with end laps occurring only at structural members // full heights from base to eave with no horizontal joints except at the junctions of door frames, window frames, louver panels, and similar locations.//~~
- ~~3. Seal side and end laps with joint sealing material.~~
- ~~4. Flash and seal walls at the base, at the top, around windows, door frames, framed louvers, and other similar openings. Install closure strips, flashings, and sealing material in an approved manner that will assure complete weather tightness.~~
- ~~5. Flashing is not required where approved "self-flashing" panels are used.//~~

~~I.H.~~ //Roof Panels:

- 1. Apply roofing panels with the configurations parallel to the slope of the roof. Provide roofing panels in //the longest lengths obtainable, with end laps occurring only at structural members, // full lengths from ridge (or ridge panel) to eaves, //top to eaves on shed roofs, //with no transverse joints except at the junction of ventilators, curbs, skylights, chimneys and similar openings//.
- 2. Lay side laps away from the prevailing wind, and seal side and end laps with joint sealing material.
- 3. Flash and seal the roof at the ridge, at eaves and rakes, at projections through the roof, and elsewhere as necessary.
- 4. Install closure strips, flashing, and sealing material in a manner that will assure complete weather tightness.//

~~J.I.~~ Flashing:

- 1. Provide flashing and related closures and accessories in connection with the preformed metal panels as indicated and as necessary to provide a watertight installation.

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2. Install details of installation, which are not indicated, in accordance with the panel manufacturer's printed instruction and details, or the approved shop drawings.
3. Allow for expansion and contraction of flashing.

K-J. Fasteners:

1. Space fasteners in accordance with the manufacturer's recommendations, and as necessary to withstand the design loads indicated.
2. Install fasteners in valleys or crowns as recommended by the manufacturer of the panel being used.
3. Install fasteners in straight lines within a tolerance of 13 mm (1/2-inch) in the length of a bay.
4. Drive exposed penetrating type fasteners normal to the surface, and to a uniform depth to seat gasketed washers properly, and drive so as not to damage factory applied coating.
5. Exercise care in drilling pilot holes for fastenings to keep drills perpendicular and centered in valleys, or crowns, as applicable. After drilling, remove metal filings and burrs from holes prior to installing fasteners and washers. Do not torque fasteners to exceed values recommended by the manufacturer.
6. Remove panels deformed or otherwise damaged by over-torqued fastenings, and provide new panels.
7. Remove metal shavings and filings from roofs on completion to prevent rusting and discoloration of the panels.

3.2 ISOLATION OF ALUMINUM

- A. Isolate aluminum in contact with or fastened to dissimilar metals other than stainless steel, white bronze, or other metal compatible with aluminum by one of the following:
 1. Painting the dissimilar metal with a prime coat of Zinc-Molybdate followed by two coats of aluminum paint.
 2. Placing a non-abrasive tape or gasket between the aluminum and the dissimilar metal.
- B. Paint aluminum in contact with, or built into mortar, concrete, plaster, or other masonry materials with a coat of alkali-resistant bituminous paint.

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- C. Paint aluminum in contact with wood or other absorptive materials that may become repeatedly wet, with two coats of bituminous paint, or two coats of aluminum paint. Seal joints with caulking material.

3.3 PROTECTION AND CLEANING

- A. Protect panels and other components from damage during and after erection, and until project is accepted by the COR.
- B. After completion of work, all exposed finished surfaces of panels are to be cleaned of soil, discoloration and disfiguration. Touch-up abraded surfaces of panels.

- - - E N D - - -

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SECTION 07 54 19
POLYVINYL-CHLORIDE (PVC) ROOFING

~~SPEC WRITER NOTES:~~

- ~~1. Delete text between // _____ // not applicable to project. Edit remaining text to suit project.~~
- ~~2. Follow National Roofing Contractors Association "The NRCA Roofing Manual" recommendations for design criteria.~~
- ~~3. Slopes: Do not use on slopes over 1/10 (1 inch per 12 inches). Provide 1/50 (1/4 inch per 12 inches) minimum slope to drains without any "Gutters" (no slopes between drains). NO EXCEPTIONS TO MINIMUM SLOPE. Slope crickets 1/50 (1/4 inch per 12 inches).~~
- ~~4. Coordinate with plumbing requirements for roof drains and drain locations at low points and mid span where maximum deflection occurs. Do not put drains at columns or on slopes. Coordinate with insulation to provide sumps at drains.~~
- ~~5. Coordinate details and systems used to provide code required fire rated roofing system.~~
- ~~6. Adhered system is preferred over mechanically fastened system.~~
- ~~7. Coordinate with Section 07 22 00, ROOF AND DECK INSULATION. Decrease R-value 5 percent when mechanical fasteners are used through insulation to compensate for parallel heat flow.~~
- ~~9. Do not use over polystyrene, urethane, or wood fiberboard insulation under membrane.~~
- ~~10. Do not use over bituminous materials where direct contact occurs, including grease, oil, or other substances not compatible with PVC. Use thin layer of insulation, slip sheet or separator sheet depending upon method of attachment or felt back sheet when minimum amount of asphalt occurs.~~
- ~~11. Terminate base flashings minimum 200 mm (8 inches) above roof surface~~

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~~including curb for building expansion joints.~~
~~12. Do not put expansion joints at roof surface level.~~
~~13. Do not use "pitch pocket" or "sealant pockets" in lieu of base flashings and cap flashings.~~
~~14. This specification is for use over concrete, cellular insulating concrete decks, or insulation. Insert additional text when installed directly to other decks or insulation systems not specified in Section 07 22 00, ROOF AND DECK INSULATION.~~
~~15. Do not use pipe boots that provide less than 200 mm (8 inch) height above roof.~~

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Polyvinyl chloride (PVC) sheet roofing ~~//adhered // mechanically fastened //~~ to insulated ~~//concrete //~~ metal ~~//~~ roof deck.

1.2 RELATED REQUIREMENTS

~~SPEC WRITER NOTE: Update and retain references only when specified elsewhere in this section.~~

~~A. Non-Flooring Adhesives and Sealants VOC Limits: Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.~~

B. Preparation of Existing Membrane Roofs and Repair Areas: Section 07 01 50.19, PREPARATION FOR REROOFING.

C. Substrate Board, Vapor Retarder, Roof Insulation, and Cover Board: Section 07 22 00, ROOF AND DECK INSULATION.

D. Roof Membrane Color: Section 09 06 00, SCHEDULE FOR FINISHES.

1.3 APPLICABLE PUBLICATIONS

A. Comply with references to extent specified in this section.

B. American National Standards Institute/Single-Ply Roofing Institute (ANSI/SPRI):

1. FX-1-01(R2006) - Standard Field Test Procedure for Determining the Withdrawal Resistance of Roofing Fasteners.

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2. RP-4 2013 - Wind Design Standard for Ballasted Single-ply Roofing Systems.
- C. American Society of Civil Engineers/Structural Engineering Institute (ASCE/SEI):
 1. 7-10 - Minimum Design Loads for Buildings and Other Structures.
- D. American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc. (ASHRAE):
 1. 90.1-13 - Energy Standard for Buildings Except Low-Rise Residential Buildings.
- E. ASTM International (ASTM):
 1. C67-14 - Sampling and Testing Brick and Structural Clay Tile.
 2. C140/C140M-15 - Sampling and Testing Concrete Masonry Units and Related Units.
 3. C578-15b - Specification for Rigid, Cellular Polystyrene Thermal Insulation.
 4. C936/C936M-15 - Solid Concrete Interlocking Paving Units.
 5. C1371-15 - Determination of Emittance of Materials Near Room Temperature Using Portable Emissometers.
 6. C1549-09(2014) - Determination of Solar Reflectance Near Ambient Temperature Using a Portable Solar Reflectometer.
 7. D751-06(2011) - Test Methods for Coated Fabrics.
 8. D4263-83(2012) - Indicating Moisture in Concrete by the Plastic Sheet Method.
 9. D4434/D4434M-12 - Poly (Vinyl Chloride) Sheet Roofing.
 10. E96/E96M-15 - Water Vapor Transmission of Materials.
 11. E408-13 - Total Normal Emittance of Surfaces Using Inspection-Meter Techniques.
 12. E1918-06(2015) - Measuring Solar Reflectance of Horizontal and Low-Sloped Surfaces in the Field.
 13. E1980-11 - Measuring Solar Reflectance of Horizontal and Low-Sloped Surfaces in the Field.
- F. Cool Roof Rating Council (CRRC):
 1. 1-15 - Product Rating Program.
- G. Florida Department of Business and Professional Regulation (FL):
 1. Approved - Product Approval.
- H. National Roofing Contractors Association (NRCA):

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1. Manual-15 - The NRCA Roofing Manual: Membrane Roofing Systems.
- I. U.S. Department of Agriculture (USDA): USDA BioPreferred Catalog.
- J. UL LLC (UL):
 1. 580-06 - Tests for Uplift Resistance of Roof Assemblies.
 2. 1897-15 - Uplift Tests for Roof Covering Systems.
- K. U.S. Department of Commerce National Institute of Standards and Technology (NIST):
 1. DOC PS 1-09 - Structural Plywood.
 2. DOC PS 2-04 - Performance Standard for Wood-Based Structural-Use Panels.
- L. U.S. Environmental Protection Agency (EPA):

~~SPEC WRITER NOTE: Energy Star Version 2.3 is in effect until June 30, 2017. Version 3.0 becomes effective July 1, 2017.~~

1. Energy Star - ENERGY STAR Program Requirements for Roof Products
~~// Version 2.3. //~~Version 3.0. ~~//~~

1.4 PREINSTALLATION MEETINGS

- A. Conduct preinstallation meeting at the Project site minimum 30 days before beginning Work of this section.

~~SPEC WRITER NOTE: Edit participant list to ensure entities influencing outcome attend.~~

1. Required Participants:
 - a. Contracting Officer's Representative.
 - b. ~~//~~Architect/Engineer. ~~//~~
 - c. ~~//~~Inspection and Testing Agency. ~~//~~
 - d. Contractor.
 - e. Installer.
 - f. ~~//~~Manufacturer's field representative. ~~//~~
 - g. Other installers responsible for adjacent and intersecting work, including roof deck, flashings, roof specialties, roof accessories, utility penetrations, rooftop curbs and equipment, and lightning protection. , and // _____ //, and // _____ /.

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~~SPEC WRITER NOTE: Edit meeting agenda to
incorporate project specific topics.~~

2. Meeting Agenda: Distribute agenda to participants minimum 3 days before meeting.
 - a. Installation schedule.
 - b. Installation sequence.
 - c. Preparatory work.
 - d. Protection before, during, and after installation.
 - e. Installation.
 - f. Terminations.
 - g. Transitions and connections to other work.
 - h. Inspecting and testing.
 - i. Other items affecting successful completion.
 - j. Pull out test of fasteners.
 - k. Material storage, including roof deck load limitations.
3. Document and distribute meeting minutes to participants to record decisions affecting installation.

1.5 SUBMITTALS

- A. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.
- B. Submittal Drawings:
 1. Roofing membrane layout.

~~SPEC WRITER NOTE: Retain paragraph below
for mechanically fastened membrane.~~

- ~~2. Roofing membrane fastener pattern and spacing.~~
- ~~3. Roofing membrane seaming and joint details.~~
- ~~4. Roof membrane penetration details.~~
- ~~5. Base flashing and termination details.~~
- ~~6. Paver layout.~~
- ~~7. Paver anchoring locations and details.~~
- C. Manufacturer's Literature and Data:
 1. Description of each product.
 2. Minimum fastener pull out resistance.
 3. Installation instructions.
 4. Warranty.

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~~SPEC WRITER NOTE: Retain paragraph below
when retaining requirement for use of
Federally mandated products under Quality
Control Article above.~~

5. Product Data for Federally-Mandated Bio-Based Materials: For roof materials, indicating USDA designation and compliance with definitions for bio-based products, Rapidly Renewable Materials, and certified sustainable wood content.

D. Sustainable Construction Submittals:

~~SPEC WRITER NOTE: Retain sustainable
construction submittals appropriate to
product.~~

1. Solar Reflectance Index (SRI) for roofing membrane.
2. Low Pollutant-Emitting Materials:
 - a. Show volatile organic compound types and quantities.
3. Energy Star label for roofing membrane.

E. Samples:

1. Roofing Membrane: 150 mm (6 inch) square.
2. Base Flashing: 150 mm (6 inch) square.
3. Fasteners: Each type.
4. Roofing Membrane Seam: 300 mm (12 inches) square.

F. Certificates: Certify products comply with specifications.

1. Fire and windstorm classification.

~~SPEC WRITER NOTE: Retain paragraph below
for Florida and Gulf coast projects.~~

- ~~2. High wind zone design requirements.~~
3. Energy performance requirements.

G. Qualifications: Substantiate that qualifications comply with specifications.

1. Installer, including supervisors ~~//~~with project experience list ~~//~~.
2. Manufacturer's field representative ~~//~~with project experience list ~~//~~.

H. Field quality control reports.

- ~~H-1.~~ 1. Submit within seven (&) days of tests and observations.

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~~SPEC WRITER NOTE: Retain paragraph below
for reroofing projects.~~

- I. Temporary protection plan. Include list of proposed temporary materials, and procedures for protecting respective facilities.
- J. Operation and Maintenance Data:
 - 1. Submit Operation and Maintenance manuals at Project Close-out.

1.6 QUALITY ASSURANCE

- A. Installer Qualifications:
 - 1. Approved by roofing system manufacturer as installer for roofing system with specified warranty.
 - 2. Regularly installs specified products.
 - 3. Installed specified products with satisfactory service on five similar installations for minimum five years.
 - a. ~~//~~Project Experience List: Provide contact names and addresses for completed projects. ~~//~~
 - 4. Employs full-time supervisors experienced installing specified system and able to communicate with Contracting Officer's Representative and installer's personnel.
- B. Manufacturer's Field Representative:
 - 1. Manufacturer's full-time technical employee or independent roofing inspector.
 - 2. Individual certified by Roof Consultants Institute as Registered Roof Observer.

1.7 DELIVERY

- A. Deliver products in manufacturer's original sealed packaging.
- B. Mark packaging, legibly. Indicate manufacturer's name or brand, type, production run number, and manufacture date.
- C. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

- A. Comply with NRCA Manual storage and handling requirements.
- B. Store products indoors in dry, weathertight facility.
- C. Store adhesives according to manufacturer's instructions.
- D. Protect products from damage during handling and construction operations.

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- E. Products stored on the roof deck must not cause permanent deck deflection.

1.9 FIELD CONDITIONS

- A. Environment:

~~SPEC WRITER NOTE: Coordinate installation temperature with available adhesives. Solvent based adhesives can be used at lower temperatures.~~

1. Product Temperature: Minimum 4 degrees C (40 degrees F) and rising before installation.
2. Weather Limitations: Install roofing only during dry current and forecasted weather conditions.

1.10 WARRANTY

~~SPEC WRITER NOTE: Always retain construction warranty. FAR includes Contractor's one year labor and material warranty.~~

- A. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

~~SPEC WRITER NOTE:
1. Specify extended manufacturer's warranties for materials only.
2. Contracting Officer's Representative must approve specification of a manufacturer's warranty.~~

- B. Manufacturer's Warranty: Warrant roofing system against material and manufacturing defects and agree to repair any leak caused by a defect in the roofing system materials or workmanship of the installer.

~~SPEC WRITER NOTE: Specify customarily available warranty period for specified products.~~

1. Warranty Period: ~~10~~ 20 years.

PART 2 - PRODUCTS

2.1 SYSTEM DESCRIPTION

- A. Roofing System: ~~//Adhered //Mechanically fastened~~ roofing membrane, base flashing, roof insulation, fasteners, cover boards, ~~substrate boards~~ // vapor retarders, ~~// pavers~~ // copings, ~~//edge metal,~~ //and ~~//walkway pads~~.

2.2 SYSTEM PERFORMANCE

- A. Design roofing system meeting specified performance:
1. Load Resistance: ASCE/SEI 7; Design criteria as indicated on Drawings.

~~SPEC WRITER NOTE: Specify actual loads for project if not indicated on the Drawings.~~

- a. Uplift Pressures:

1) Corner Uplift Pressure: // _____ // kPa/sq. m
(// _____ // psf).

2) Perimeter Uplift Pressure: // _____ // kPa/sq. m
(// _____ // psf).

3) Field-of-Roof Uplift Pressure: // _____ // kPa/sq. m
(// _____ // psf).

~~SPEC WRITER NOTE: Retain one or more paragraphs below for compliance with:~~

- ~~a. Guiding Principles for Federal Leadership in High Performance and Sustainable Buildings incorporated in Executive Order 13423 "Strengthening Federal Environmental, Energy, and Transportation Management, dated January 24, 2007.~~
- ~~b. Energy Policy Act of 2005 (EPA 2005).~~
- ~~c. Energy Independence and Security Act of 2007 (EISA 2007).~~
- ~~d. LEED mandate.~~
- ~~e. Conformance with locally applicable requirements.~~

2. Energy Performance:

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~~SPEC WRITER NOTE: Retain paragraph below
for Energy Star compliance.~~

a. EPA Energy Star Listed for low-slope roof products.

~~SPEC WRITER NOTE: Retain paragraph below
for LEED heat island effect (roof) credit
compliance.~~

b. ~~ASTM E1980; Minimum 78 Solar Reflectance Index (SRI).~~

~~SPEC WRITER NOTE: Retain paragraph below
for California Energy Commission (CEC)-
Title 24 compliance.~~

c. ~~CRRC 1; Minimum 0.70 initial solar reflectance and minimum 0.75
emissivity.~~

~~SPEC WRITER NOTE: Typically retain below
for VA new construction and reroofing
projects in ASHRAE Climate Zones 1
through 3 and elsewhere where cool roof
technology is cost effective.~~

d. Three-Year Aged Performance: Minimum 0.55 solar reflectance
tested in according to ASTM C1549 or ASTM E1918, and minimum
0.75 thermal emittance tested in according to ASTM C1371 or
ASTM E408.

- 1) Where tested aged values are not available:
 - a) Calculate compliance adjusting initial solar reflectance
according to ASHRAE 90.1.
 - b) Provide roofing system with minimum 64 three-year aged
Solar Reflectance Index calculated according to
ASTM E1980 with 12 W/sq. m/degree K (2.1 BTU/h/sq. ft.)
convection coefficient.

2.3 PRODUCTS - GENERAL

- A. Basis of Design: Section 09 06 00, SCHEDULE FOR FINISHES.
- B. Provide roof system components from one manufacturer.
- C. Sustainable Construction Requirements:

~~SPEC WRITER NOTE:
1. Retain paragraph below when complying
with Federal Guiding Principles IV from
Guiding Principles for Federal Leadership~~

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~~in High Performance and Sustainable Buildings incorporated in Executive Order 13423 "Strengthening Federal Environmental, Energy, and Transportation Management, dated January 24, 2007, Enhance Indoor Environmental Quality Biobased Content Requirement. Requirement differs from related requirements in LEED definitions.~~

~~2. Indicate project goals for percentages of bio-based, rapidly renewable, and certified sustainable wood products in Section 01 00 00, GENERAL REQUIREMENTS.~~

1. Bio-Based Materials: Where applicable, provide products designated by USDA and meeting or exceeding USDA recommendations for bio-based content, and products meeting Rapidly Renewable Materials and certified sustainable wood content definitions; refer to www.biopreferred.gov.

~~SPEC WRITER NOTE:~~

- ~~1. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS includes comprehensive product list setting VOC limits for low-emitting materials.~~
- ~~2. Retain subparagraphs applicable to products specified in this section.~~

2. Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:
 - a. Non-flooring adhesives and sealants.

2.4 PVC ROOFING MEMBRANE

SPEC WRITER NOTE:

1. Type III and Type IV are available with or without fabric backing. Type II is available without fabric backing, only.
2. Use fabric-backed sheet for adhered systems to cellular insulating concrete, structural concrete, or when re-roofing over other incompatible substrates where

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the manufacturer recommends a fabric backing for separation.

3. Use sheets without fabric backing when adhering to rigid insulation boards and cover boards.

4. Do not use PVC sheet in ballasted applications. Use Type II for adhesive applications only.

5. Use adhered systems over cover boards on new VA construction unless other application is specifically approved by VA.

- A. PVC Sheet: ASTM D4434/D4434M, ~~//~~Type II ~~--~~ reinforced, ~~//~~Type III - fabric reinforced, ~~//~~ or ~~//~~Type IV - fabric reinforced ~~//~~.

1. Backing: ~~//~~Without fabric backing ~~//~~ With fabric backing ~~//~~.

2. Thickness: ~~//~~1.5 mm (60 mils) ~~//~~ 2.0 mm (80 mils) ~~//~~.

~~SPEC WRITER NOTE: PVC is available in white, gray, tan, and other colors. Confirm color selection meets energy performance requirements.~~

3. Color: See Section 09 06 00, SCHEDULE OF FINISHES.

- B. Additional Properties:

1. Water Vapor Permeance, ASTM E96/E96M: Minimum 8 ng/Pa/s/sq. m (0.14 perms) (Water Method).

2.5 MEMBRANE ACCESSORY MATERIALS

- A. Flashing Sheet: Manufacturer's standard; same material, type, reinforcement, thickness, and color as roofing membrane.

~~SPEC WRITER NOTE: Solvent based adhesives may be required for roofing installation in cold climates.~~

- B. Factory Formed Flashings: Inside and outside corners, pipe boots, and other special flashing shapes to minimize field fabrication.

- C. Splice Lap Sealant: Manufacturer's standard for exposed lap edge, matching roof membrane color.

- D. Bonding Adhesive: Manufacturer's standard, ~~//~~water or ~~//~~solvent ~~//~~ based, to suit substrates.

- E. Termination Bars: Manufacturer's standard, stainless steel or aluminum, 25 mm wide by 3 mm thick (1 inch wide by 1/8 inch thick) factory drilled for fasteners.

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- ~~F. Battens: Manufacturer's standard, galvanized or galvanized steel, 25 mm wide by 1.3 mm thick (1 inch wide by 0.05 inch thick) factory punched for fasteners.~~
- G. Pipe Compression Clamp:
1. Stainless steel drawband.
 2. Worm drive clamp device.
- H. Fasteners: Manufacturer's standard coated steel with metal or plastic plates to suit application.
- ~~I. Protection Sheet: UV-resistant fabric and weight recommended by roofing manufacturer for installation under pavers.~~
- J. Miscellaneous Accessories: Provide other accessories required by manufacturer for complete, watertight installation.

2.6 ROOF PAVERS

~~SPEC WRITER NOTE:~~

- ~~1. Ensure pavers are detailed showing size and shape.~~
- ~~2. Do not exceed 600 mm square (24 inches square) for non-interlocking units with approximate weight of 23 kg (50 lbs.) each.~~
- ~~3. Pavers require 73 kg/sq. m (15 lbs./sq. ft.) minimum for fire rating.~~
- ~~4. Interlocking pavers are preferred over non-interlocking pavers.~~
- ~~5. Use interlocking type that have been tested in a wind tunnel for wind uplift.~~
- ~~6. Do not use light weight aggregate pavers.~~
- ~~7. Retain ribbed bottom surface when pavers are installed directly on roof membrane.~~

- ~~A. Roof Pavers: Precast, normal weight, // interlocking // non-interlocking // concrete units // with ribbed bottom surface for drainage //.~~
- ~~1. Weight: Minimum 73 kg/sq. m (15 lbs./sq. ft.).~~
 - ~~2. Size: As indicated on drawings.~~
 - ~~3. Compressive Strength: ASTM C140; Minimum (8,000 psi).~~
 - ~~4. Water Absorption: ASTM C936; Maximum 5 percent.~~
 - ~~5. Freeze Thaw: ASTM C67; Maximum 1 percent mass loss.~~

2.7 ACCESSORIES

- A. Temporary Protection Materials:
1. Expanded Polystyrene (EPS) Insulation: ASTM C578.
 2. Plywood: NIST DOC PS 1, Grade CD Exposure 1.
 3. Oriented Strand Board (OSB): NIST DOC PS 2, Exposure 1.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine and verify substrate suitability for product installation with roofing installer and roofing inspector present.

~~SPEC WRITER NOTE: Require firestopping verification for fire rated roof assemblies.~~

1. Verify roof penetrations are complete, secured against movement, ~~//~~and firestopped ~~//~~.
 2. Verify roof deck is adequately secured to resist wind uplift.
 3. Verify roof deck is clean, dry, and in-plane ready to receive roofing system.
- B. Correct unsatisfactory conditions before beginning roofing work.

3.2 PREPARATION

- A. Complete roof deck construction before beginning roofing work:
1. Install curbs, blocking, edge strips, nailers, and other components to which roofing membrane and base flashing are attached.
 2. Coordinate roofing membrane installation with flashing work and roof insulation work so insulation and flashing are installed concurrently to permit continuous roofing operations.
 3. Document installation of related materials to be concealed before installing roofing work.
- B. Dry out wet substrate surfaces ~~// including roof deck flutes //~~. Apply materials to dry substrates, only.
- C. Broom clean roof decks. Remove dust, dirt and debris.
- D. Remove projections capable of damaging roofing materials.
- E. Concrete Decks, except Insulating Concrete:
1. Test concrete decks for moisture according to ASTM D4263 before installing roofing materials.

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2. Prime concrete decks. Keep primer back 100 mm (four inches) from precast concrete deck joints.
 3. Allow primer to dry before application of bitumen.
- F. Insulating Concrete Decks:
1. Allow deck to dry out minimum five days after installation before installing roofing materials.
 2. Allow additional drying time when precipitation occurs before installing roofing materials.
- G. Existing Membrane Roofs and Repair Areas:
1. Comply with Section 07 01 50.19 PREPARATION FOR REROOFING.

3.3 TEMPORARY PROTECTION

- A. Install temporary protection at end of each day's work, when work is halted indefinitely, and when precipitation is imminent. Comply with approved temporary protection plan.
- B. Install temporary cap flashing over top of base flashings where permanent flashings are not in place to protect against water intrusion into roofing system. Securely anchor in place to prevent blow off and damage by construction activities.
- C. Temporarily seal exposed insulation surfaces within roofing membrane.
1. Apply temporary seal and water cut off by extending roofing membrane beyond insulation and securely embedding edge of the roofing membrane in 6 mm (1/4 inch) thick by 50 mm (2 inches) wide strip of temporary closure sealant. Weight roofing membrane edge with sandbags, to prevent displacement; space sandbags maximum 2400 mm (8 feet) on center.
 2. Direct water away from work. Provide drainage, preventing water accumulation.
 3. Check daily to ensure temporary seal remains watertight. Reseal open areas and weight down.
- D. Before the work resumes, cut off and discard portions of roof membrane in contact with temporary seal.
1. Cut minimum 150 mm (6 inches) back from sealed edges and surfaces.
- E. Remove sandbags and store for reuse.

3.4 INSTALLATION, GENERAL

- A. Install products according to manufacturer's instructions ~~//~~and approved submittal drawings ~~//~~.

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1. When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's Representative consideration.
- B. Comply with NRCA Manual installation requirements.

~~SPEC WRITER NOTE: The code requires roof coverings on fully adhered or mechanically fastened non-ballasted roofs to be tested according to UL 580 or UL 1897.~~

- C. Comply with ~~//~~UL 580 or ~~//~~UL 1897 ~~//~~for uplift resistance.
- D. Do not allow membrane and flashing to contact surfaces contaminated with asphalt, coal tar, oil, grease, or other substances incompatible with PVC.

3.5 ROOFING INSTALLATION

- A. Install membrane perpendicular to long dimension of insulation boards.
- B. Begin membrane installation at roof low point and work towards high point. Lap membrane shingled in water flow direction.
- C. Position membrane free of buckles and wrinkles.
- D. Roll membrane out; inspect for defects as membrane is unrolled. Remove defective areas.
 1. Allow 30 minutes for membrane to relax before proceeding.
 2. Lap edges and ends minimum 50 mm (2 inches).
 3. Heat weld or solvent weld laps. Apply pressure to develop full adhesion with minimum seam strength according to ASTM D4434/D4434M.
 4. Check seams to ensure continuous adhesion and correct defects.
 5. Finish seam edges beveled bead of sealant.
 6. Finish seams same day as membrane is installed.
 7. Anchor membrane perimeter to roof deck and parapet wall as indicated on drawings.

~~E. Membrane Perimeter Anchorage:~~

- ~~1. Install batten with fasteners at perimeter of each roof area, curb flashing, expansion joints and similar penetrations on top of roof membrane as indicated on drawings.~~
- ~~2. Space fasteners maximum 300 mm (12 inches) on center, starting 25 mm (1 inch) from ends.~~
- ~~3. When battens are cut round corners and eliminate sharp corners.~~

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- ~~a. Stop batten where batten interferes with drainage. Space ends of batten 150 mm (6 inch) apart.~~
- ~~4. Cover batten with 150 mm (6 inch) wide roof membrane strip; heat or solvent weld to roofing membrane and seal edges.~~
- ~~5. At // gravel stops, // fascia eaves, // turn roofing membrane down over front edge of blocking or nailer. Secure roofing membrane to vertical portion of nailer with fasteners spaced maximum 150 mm (6 inches) on center.~~
- ~~6. At parapet walls, intersecting building walls and curbs, secure roofing membrane to structural deck fasteners 150 mm (6 inches) on center or as shown in NRCA Manual.~~

F. Adhered System Installation:

1. Apply bonding adhesive in quantities required by roofing membrane manufacturer.
2. Fold sheet back on itself after rolling out and coat bottom side of roofing membrane and top substrate with adhesive. Do not coat the lap joint area.
3. After adhesive has set according to adhesive manufacturer's instructions, roll roofing membrane into adhesive minimizing voids and wrinkles.
4. Repeat for other half of sheet.
5. Cut voids and wrinkles to lay flat. Clean and patch cut area.

~~G. Mechanically Fastened System Installation:~~

- ~~1. Secure roofing membrane to structural deck with fasteners through battens to achieve specified wind uplift performance.~~
 - ~~a. Drill pilot holes for fasteners installed into cast in place concrete. Drill hole minimum 10 mm (3/8 inch) deeper than fastener penetration.~~
- ~~2. When fasteners are installed within membrane laps, locate battens minimum 13 mm (1/2 inch) from edge of sheets.~~
- ~~3. Where fasteners are installed over roofing membrane after seams are welded, cover fasteners with minimum 175 mm (7 inch) diameter PVC membrane cap centered over fasteners. Where battens are used cover batten with minimum 175 mm (7 inch) wide PVC strip centered over batten. Weld cap to roofing membrane and finish edges with lap sealant.~~

3.6 FLASHING INSTALLATION

- A. Install flashings on same day as roofing membrane is installed. When flashing cannot be completely installed in one day, complete installation until flashing is watertight and provide temporary covers or seals.

~~SPEC WRITER NOTE:~~

- ~~1. Ensure roof drain flashing details are shown on drawings with sump to depress notched clamping ring below roof surface.~~
- ~~2. Offset drains in sump to side of steel beams so drain is not above low point when roof slope terminates on top of beam.~~

B. Flashing Roof Drains:

1. Install roof drain flashing according to roofing membrane manufacturer's instruction.
 - a. Install metal drain flashing in asphalt roof cement, holding cement back from edge of metal flange.
 - b. Do not allow roof cement to contact PVC roofing membrane.
 - c. Adhere roofing membrane to metal flashing with bonding adhesive.
2. Turn metal drain flashing and roofing membrane down into drain body. Install clamping ring and strainer.

SPEC WRITER NOTES:

1. See NRCA Manual for base flashing details.
2. Use with metal cap flashing.
3. Do not use "pitch pocket" or "sealant pocket" construction detail.
4. Coordinate with sheet metal work to provide metal cap flashing for base flashing on curbs and walls and penetrations. Use surface mounted reglets on existing walls.
5. Do not terminate base flashing or membrane edge exposed on top of parapet walls or in reglets on horizontal or sloped wash surface.
6. Terminate only under cap flashings or coping covers except gravel stops and for draw bands on pipe boots.
7. Use 200 mm (8 inch) minimum height for base flashing.

C. Installing Base Flashing and Pipe Flashing:

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1. Install flashing sheet to pipes, walls and curbs to minimum 200 mm (8 inches) height above roof surfaces and extend roofing manufacturer's standard lap dimension onto roofing membranes.
 - a. Adhere flashing with bonding adhesive.
 - b. Form inside and outside corners of flashing sheet according to NRCA Manual.
 - c. Form pipe flashing according to NRCA Manual.
 - d. Lap ends roofing manufacturer's standard dimension.
 - e. Weld flashing sheets together, and weld flashing sheets to roofing membranes. Finish exposed edges with lap sealant.
 2. Anchor top of flashing to walls and curbs with fasteners spaced maximum 150 mm (6 inches) on center. Use surface mounted fastening strip on ducts. Use pipe clamps on pipes or other round penetrations.
 3. Apply sealant to top edge of flashing.
- D. Installing Building Expansion Joints:

~~SPEC WRITER NOTES:~~

- ~~1. Do not put expansion joints at roof membrane level.~~
- ~~2. Design joints to be installed on curbs minimum 200 mm (8 inches) high especially at walls.~~
- ~~3. Detail expansion joint.~~

1. Install base flashing on curbs as specified.
 2. Coordinate installation with ~~// metal expansion joint cover // roof expansion joint system //~~.
 3. Install flexible tubing 1-1/2 times width of joint centered over joint. Cover tubing with flashing sheet adhered to base flashing and lapping base flashing roofing manufacturer's standard dimension. Finish edges of laps with lap sealant.
- E. Repairs to Membrane and Flashings:
1. Remove sections of roofing membrane and flashing sheets that are creased, wrinkled, or fishmouthed.
 2. Cover removed areas, cuts and damaged areas with patch extending 100 mm (4 inches) beyond damaged, cut, or removed area. Weld patch to roofing membrane or flashing sheet. Finish edge of lap with lap sealant.

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3.7 FLEXIBLE WALKWAYS

~~SPEC WRITER NOTES:~~

- ~~1. Use manufactured walkway pad for walkways over roof areas which do not have paver walkways between equipment.~~
- ~~2. Clearly indicate on roof plan PVC walkways.~~

- A. Walkway Pad: PVC walkway pad with slip resistant surface and molded channels on the underside for water drainage.

~~3.8 PAVER INSTALLATION~~

~~SPEC WRITER NOTES:~~

- ~~1. Use pavers in the following locations as a minimum:~~
 - ~~a. At working and access areas of equipment requiring servicing.~~
 - ~~b. At equipment having discharges detrimental to roof membrane, under gooseneck discharges from kitchens and chemical exhausts.~~
 - ~~c. At landing points for hatches, ladders, and doors entering roof level.~~
- ~~2. Confirm that walkways and pavers are shown on drawings.~~
- ~~3. Specify pavers and anchorage for pavers when weight of pavers does not meet requirements for wind velocities according to ANSI/SPRI RP-4.~~
- ~~4. Pavers without interlocking connectors require strapping together and edge clamps when pavers do not provide minimum weight for wind uplift resistance.~~
- ~~5. Use mechanical strapping to create perimeter anchor, at penetrations, cuts at valleys, over drains, and where partial or cut units occur.~~
- ~~6. Detail strapping, perimeter restraints, edge clamps and location of strapping. Do not anchor through base flashing or into cants.~~
- ~~7. Interlocking connectors:~~
 - ~~a. Use 400 mm (16 inches) on center minimum spacing of connectors.~~
 - ~~b. Decrease spacing to 300, 200, or 100 mm (12, 8, or 4 inches) on center for greater wind velocities.~~

- ~~A. Saw cut or core drill pavers for cut units.~~

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- ~~B. Apply protection sheet over roof membrane. Overlap and install additional layers as recommended by roofing manufacturer.~~
- ~~C. Install pavers with butt joints in running bond with minimum one half-length units at ends.~~
- ~~1. Stagger end joints; generally locate joints near midpoint of adjacent rows, except where end joints occur in valleys. Miter end joints to fit in valleys.~~
 - ~~2. Cut to fit within 13 mm (1/2 inch) of penetrations.~~
- ~~D. Install interlocking connectors in channel units for complete tie in of units, including cut units. Use corner spacings for distance of 1200 mm (4 feet) or more around roof drains, penetrations, and other vertical surfaces in field of roof area.~~
- ~~1. Space connectors at // _____ // mm (// _____ // inches) on center at corners for 3 m (10 foot) square area.~~
 - ~~2. Space connectors at // _____ // mm (// _____ // inches) on center at perimeter for 1800 mm (6 foot) wide strip.~~
 - ~~3. Space connectors at // _____ // mm (// _____ // inches) on center in field.~~
 - ~~4. Install pavers under the perimeter retainer as shown.~~
- ~~E. Install strapping where shown.~~
- ~~1. Limit strap lengths to maximum of 9 m (30 feet).~~
 - ~~2. Install straps at corner connection to perimeter retainer at approximate 45 degree angle at approximate 3 to 3.6 m (10 to 12 feet) from corner.~~
 - ~~3. Install straps on both sides of valleys, hips, and ridges, with cross straps spaced maximum 1200 mm (4 feet) on center between the end straps.~~
 - ~~4. Install straps at perimeter of penetrations more than two paves in width or length.~~
 - ~~5. Anchor straps to each paver with two fasteners per unit.~~
 - ~~6. Pre-drill holes for fasteners in pavers.~~

3.9 FIELD QUALITY CONTROL

~~SPEC WRITER NOTES: Section 01 45 29, TESTING LABORATORY SERVICES includes VA provided testing for large projects and contractor provided testing for small~~

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~~projects. Coordinate testing-
responsibility.~~

A. Field Tests: Performed by testing laboratory specified in Section
01 45 29, TESTING LABORATORY SERVICES.

~~SPEC WRITER NOTE: ANSI/SPRI FX-1 sets
testing frequency as 10 tests for first
4,650 sq. m (50,000 sq. ft.) and five
tests for each additional 4,650 sq. m
(50,000 sq. ft.). Specify frequency to
suit project conditions.~~

1. Fastener Pull Out Tests: ANSI/SPRI FX-1; one test for every 230 sq. m (2,500 sq. ft.) of deck. Perform tests for each combination of fastener type and roof deck type before installing roof insulation.
 - a. Test at locations selected by Contracting Officer's Representative.
 - b. Do not proceed with roofing work when pull out resistance is less than manufacturer's required resistance.
 - c. Test Results:
 - 1) Repeat tests using different fastener type or use additional fasteners achieve pull out resistance required to meet specified wind uplift performance.
 - 2) Patch cementitious deck to repair areas of fastener tests holes.

SPEC WRITER NOTE: Select one or both of following two paragraphs based upon project requirements. VA may elect to perform or hire roofing inspector. VA may also elect to require contractor to retain roofing inspector, either as qualified representative of manufacturer or independent third party inspector.

2. Examine and probe roofing membrane and flashing seams in presence of Contracting Officer's Representative and Manufacturer's field representative.
3. Probe seams to detect marginal welds, voids, skips, and fishmouths.
4. Cut 100 mm (4 inch) wide by 300 mm (12 inch) long samples through seams where directed by Contracting Officer's Representative.

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5. Cut one sample for every 450 m (1500 feet) of seams.
6. Cut samples perpendicular to seams.
7. Failure of samples to pass ASTM D751 test will be cause for rejection of work.
8. Repair areas where samples are taken and where marginal bond, voids, and skips occur.
9. Repair fishmouths and wrinkles by cutting to lay flat. Install patch over cut area extending 100 mm (4 inches) beyond cut.

B. Manufacturer Services:

1. Inspect initial installation, installation in progress, and completed work.
2. Issue supplemental installation instructions necessitated by field conditions.
3. Prepare and submit inspection reports.
4. Certify completed installation complies with manufacturer's instructions and warranty requirements.

3.10 CLEANING

- A. Remove excess adhesive before adhesive sets.
- B. Clean exposed roofing surfaces. Remove contaminants and stains // to comply with specified solar reflectance performance //.

3.11 PROTECTION

- A. Protect roofing system from ~~//~~traffic and ~~//~~construction operations.
 1. Protect roofing system when used for subsequent work platform, materials storage, or staging.
 2. Distribute scaffolding loads to exert maximum 50 percent roofing system materials compressive strength.
- B. Loose lay temporary insulation board overlaid with plywood or OSB.
 1. Weight boards to secure against wind uplift.
- C. Remove protection when ~~// no longer required //~~ when directed by Contacting Officer's Representative ~~//~~.
- D. Repair damage.

- - E N D - -

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SECTION 07 54 23
THERMOPLASTIC POLYOLEFIN (TPO) ROOFING

~~SPEC WRITER NOTES:~~

- ~~1. Delete text between // _____ // not applicable to project. Edit remaining text to suit project.~~
- ~~2. Slopes: Do not use on slopes over 1: 12 (one inch per 12 inches). Provide 1: 50 (1/4 inch per 12 inches) minimum to drains without any "Cutters" (no slopes between drains). NO EXCEPTION TO MINIMUM SLOPE. Slope crickets 1: 50 (1/4 inch per foot).~~
- ~~3. Coordinate with plumbing requirements for roof drains and drain locations at low points and mid span where maximum deflection occurs. Do not put drains at columns or on slopes. Coordinate with insulation to provide sumps at drains.~~
- ~~4. Coordinate details and systems used to provide code required fire rated roofing system.~~
- ~~5. Adhered system is preferred over mechanically anchored system.~~
- ~~6. Coordinate with Section 07 22 00, ROOF AND DECK INSULATION.~~
- ~~7. Coordinate with Section 07 22 00, ROOF AND DECK INSULATION for roof insulation under the membrane. Decrease R value 5 percent when mechanical fasteners are used through insulation to compensate for parallel heat flow.~~
- ~~8. Do not use polystyrene, urethane, or wood fiberboard insulation under membrane.~~
- ~~9. Do not use over bituminous materials where direct contact occurs, including grease, oil, or other substances not compatible with TPO. Use a thin layer of insulation, slip sheet or separator sheet depending upon method of attachment.~~
- ~~10. Terminate base flashings minimum 200 mm (8 inches) above roof surface including curb for building expansion joints.~~
- ~~11. Do not put expansion joints at roof surface level.~~
- ~~12. Do not use "pitch pockets" or "sealant pockets" in lieu of base flashings and cap flashings.~~

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~~13. This specification is for use over concrete, cellular insulating concrete decks, or insulation. Insert additional text when installed directly to other decks or insulation systems not specified in Section 07 22 00, ROOF AND DECK INSULATION.~~
~~14. Do not use pipe boots that provide less than 200 mm (8 inch) height above roof.~~

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Thermoplastic Polyolefin (TPO) sheet roofing ~~//adhered //~~
~~mechanically fastened //~~ to roof deck.

1.2 RELATED WORK

~~SPEC WRITER NOTE: Update and retain references only when specified elsewhere in this section.~~

- A. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Non-Flooring Adhesives and Sealants VOC Limits.
- B. Section 07 01 50.19, PREPARATION FOR REROOFING: Preparation of Existing Membrane Roofs and Repair Areas.
- C. Section 07 22 00, ROOF AND DECK INSULATION: Roof Insulation.

1.3 APPLICABLE PUBLICATIONS

- A. Comply with references to extent specified in this section.
- B. American National Standards Institute/Single-Ply Roofing Institute (ANSI/SPRI):
FX-1-16.....Standard Field Test Procedure for Determining the Withdrawal Resistance of Roofing Fasteners.
- C. American Society of Civil Engineers/Structural Engineering Institute (ASCE/SEI):
7-16.....Minimum Design Loads for Buildings and Other Structures.
- D. American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc. (ASHRAE):
90.1-13.....Energy Standard for Buildings Except Low-Rise Residential Buildings.
- E. ASTM International (ASTM):

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- C67-20.....Sampling and Testing Brick and Structural Clay
Tile.
- C140/C140M-20a.....Sampling and Testing Concrete Masonry Units and
Related Units.
- C1371-15.....Determination of Emittance of Materials Near
Room Temperature Using Portable Emissometers.
- C1549-16.....Determination of Solar Reflectance Near Ambient
Temperature Using a Portable Solar
Reflectometer.
- D1876-08(2015)e1.....Peel Resistance of Adhesives (T-Peel Test).
- D4263-83(2018).....Indicating Moisture in Concrete by the Plastic
Sheet Method.
- D4434/D4434M-15.....Poly(Vinyl Chloride) Sheet Roofing.
- D6878/D6878M-13.....Thermoplastic Polyolefin Based Sheet Roofing.
- E408-13.....Total Normal Emittance of Surfaces Using
Inspection-Meter Techniques.
- E1918-16.....Measuring Solar Reflectance of Horizontal and
Low-Sloped Surfaces in the Field.
- E1980-11(2019).....Calculating Solar Reflectance Index of
Horizontal and Low-Sloped Opaque Surfaces.
- F. Cool Roof Rating Council (CRRC):
- 1-20.....Product Rating Program.
- G. National Roofing Contractors Association (NRCA):
- Manual-19.....The NRCA Roofing Manual: Membrane Roofing
Systems.
- H. U.S. Department of Agriculture (USDA):
- BioPreferred® Program Catalog.
- I. UL LLC (UL):
- 580-06.....Tests for Uplift Resistance of Roof Assemblies.
- 1897-20.....Uplift Tests for Roof Covering Systems.
- J. U.S. Department of Commerce National Institute of Standards and
Technology (NIST):
- DOC PS 1-19.....Structural Plywood.
- DOC PS 2-18.....Performance Standard for Wood-Based
Structural-Use Panels.
- K. U.S. Environmental Protection Agency (EPA):

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Energy Star.....ENERGY STAR Program Requirements for Roof
Products Version 3.0.

1.4 PREINSTALLATION MEETINGS

- A. Conduct pre-installation meeting at project site minimum 30 days before beginning Work of this section.

~~SPEC WRITER NOTE: Edit participant list to ensure entities influencing outcome attend.~~

1. Required Participants:

- a. Contracting Officer's Representative (COR).
- b. ~~//~~Architect/Engineer (as requested by COR).~~//~~
- c. ~~//~~Inspection and Testing Agency.~~//~~
- d. Contractor.
- e. Installer.
- f. ~~//~~Manufacturer's field representative.~~//~~
- g. Other installers responsible for adjacent and intersecting work, including roof deck, flashings, roof penetrations, roof accessories, utility penetrations, rooftop curbs and equipment. and // _____ //.

~~SPEC WRITER NOTE: Edit meeting agenda to incorporate project specific topics.~~

2. Meeting Agenda: Distribute agenda to participants minimum 3 days before meeting.
 - a. Installation schedule.
 - b. Installation sequence.
 - c. Preparatory work.
 - d. Protection before, during, and after installation.
 - e. Installation.
 - f. Terminations.
 - g. Transitions and connections to other work.
 - h. Inspecting and testing.
 - i. Other items affecting successful completion.
 - j. Pullout test of fasteners.
 - k. Material storage, including roof deck load limitations.
3. Document and distribute meeting minutes to participants to record decisions affecting installation.

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1.5 SUBMITTALS

A. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

B. Submittal Drawings:

1. Roof membrane layout.

~~SPEC WRITER NOTE: Retain paragraph below
for mechanically fastened membrane.~~

~~2. //Roofing membrane fastener pattern and spacing.//~~

~~3.2.~~ _____ Roofing membrane seaming and joint details.

~~4.3.~~ _____ Roof membrane penetration details.

~~5.4.~~ _____ Base flashing and termination details.

~~6. Paver layout.~~

~~7. Paver anchoring locations and details.~~

C. Manufacturer's Literature and Data:

1. Description of each product.
2. Minimum fastener pullout resistance.
3. Installation instructions.
4. Warranty.

D. Samples:

1. Roofing Membrane: 150 mm (6 inch) square.
2. Base Flashing: 150 mm (6 inch) square.
3. Fasteners: Each type.
4. Roofing Membrane Seam: 300 mm (12 inches) square.

E. Sustainable Construction Submittals:

~~SPEC WRITER NOTE: Retain sustainable
construction submittals appropriate to
product.~~

1. Solar Reflectance Index (SRI) for roofing membrane.

2. Biobased Content:

a. Show type and quantity for each product.

3. Low Pollutant-Emitting Materials:

a. Show volatile organic compound types and quantities.

4. Energy Star label for roofing membrane.

F. Certificates: Certify products comply with specifications.

1. Fire and windstorm classification.

~~SPEC WRITER NOTE: Retain paragraph below
for Florida and Gulf coast projects.~~

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~~High wind zone design requirements.~~

~~2. Energy performance requirements.~~

- G. Qualifications: Substantiate qualifications comply with specifications.
 - 1. Installer, including supervisors with project experience list.
 - 2. Manufacturer's field representative with project experience list.
- H. Field quality control reports.

~~SPEC WRITER NOTE: Retain paragraph below
for reroofing projects.~~

- I. Temporary protection plan. Include list of proposed temporary materials, and procedures to protect facilities from damage.
- J. Operation and Maintenance Data:
 - 1. Operation and Maintenance instructions.

1.6 QUALITY ASSURANCE

- A. Installer Qualifications:
 - 1. Approved by roofing system manufacturer as installer for roofing system with specified warranty.
 - 2. Regularly installs specified products.
 - 3. Installed specified products with satisfactory service on five similar installations for minimum five years.
 - a. Project Experience List: Provide contact names and addresses for completed projects.
 - 4. Employs full-time supervisors experienced installing specified system and able to communicate with Contracting Officer's Representative and installer's personnel.
- B. Manufacturer's Field Representative:
 - 1. Manufacturer's full-time technical employee or independent roofing inspector.
 - 2. Individual certified by Roof Consultants Institute as Registered Roof Observer.

1.7 DELIVERY

- A. Deliver products in manufacturer's original sealed packaging.
- B. Mark packaging, legibly. Indicate manufacturer's name or brand, type, and manufacture date.
- C. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

- A. Comply with NRCA Manual storage and handling requirements.
- B. Store products indoors in dry, weathertight facility.
- C. Store adhesives according to manufacturer's instructions.
- D. Protect products from damage during handling and construction operations.
- E. Products stored on the roof deck must not cause permanent deck deflection.

1.9 FIELD CONDITIONS

- A. Environment:

~~SPEC WRITER NOTE: Coordinate installation temperature with available adhesives. Solvent based adhesives can be used at lower temperatures.~~

- 1. Product Temperature: Minimum 4 degrees C (40 degrees F) for minimum 48 hours before installation.
- 2. Weather Limitations: Install roofing only during dry current and forecasted weather conditions.

1.10 WARRANTY

~~SPEC WRITER NOTE: Always retain construction warranty. FAR includes Contractor's one year labor and material warranty.~~

- A. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

~~SPEC WRITER NOTE:
1. Specify extended manufacturer's warranties for materials only.
2. Contracting Officer's Representative must approve specification of a manufacturer's warranty.~~

- B. Manufacturer's Warranty: Warrant roofing system against material and manufacturing defects and agree to repair any leak caused by a defect in the roofing system materials or workmanship of the installer.

~~SPEC WRITER NOTE: Specify customarily available warranty period for specified products.~~

- 1. Warranty Period: 10 years.

PART 2 - PRODUCTS

2.1 SYSTEM DESCRIPTION

- A. Roofing System: Thermoplastic Polyolefin (TPO) sheet roofing ~~//adhered~~
~~// mechanically fastened //~~ to roof deck.

2.2 SYSTEM PERFORMANCE

- A. Design roofing system complying with specified performance:

1. Load Resistance: ASCE/SEI 7; Design criteria: ~~//as indicated on~~
Drawings~~//~~.

~~SPEC WRITER NOTE: Specify actual loads~~
~~when known for project.~~

~~a. //Uplift Pressures:~~

- ~~1) Corner Uplift Pressure: // _____ // kPa/square meter~~
~~(// _____ // per square foot).~~
- ~~2) Perimeter Uplift Pressure: // _____ // kPa/square meter~~
~~(// _____ // per square foot).~~
- ~~3) Field of Roof Uplift Pressure: // _____ // kPa/square meter~~
~~(// _____ // per square foot). //~~

~~SPEC WRITER NOTE:~~

- ~~1. Energy performance requirements apply~~
~~only when white roofing membrane is~~
~~specified.~~
- ~~2. Retain one or more paragraphs below~~
~~for compliance with:~~
- ~~a. Guiding Principles for Federal~~
~~Leadership in High Performance and~~
~~Sustainable Buildings incorporated~~
~~in Executive Order 13423~~
~~"Strengthening Federal~~
~~Environmental, Energy, and~~
~~Transportation Management, dated~~
~~January 24, 2007.~~
- ~~b. Energy Policy Act of 2005 (EPAAct).~~
- ~~c. Energy Independence and Security~~
~~Act of 2007 (EISA).~~
- ~~d. LEED mandate.~~
- ~~e. Conformance with locally applicable~~
~~requirements.~~

2. Energy Performance:

~~SPEC WRITER NOTE: Retain paragraph below~~
~~for Energy Star compliance.~~

- a. EPA Energy Star Listed for low-slope roof products.

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~~SPEC WRITER NOTE: Retain paragraph below for LEED heat island effect (roof) credit compliance.~~

- b. ASTM E1980; Minimum 78 Solar Reflectance Index (SRI).

SPEC WRITER NOTE: Retain paragraph below for California Energy Commission (CEC) Title 24 compliance.

- c. CRRC-1; Minimum 0.70 initial solar reflectance and minimum 0.75 emissivity.

~~SPEC WRITER NOTE: Typically retain below for VA new construction and reroofing projects in ASHRAE Climate Zones 1 through 3 and elsewhere where cool roof technology is cost effective.~~

- d. Three-Year Aged Performance: Minimum 0.55 solar reflectance tested in according to ASTM C1549 or ASTM E1918, and minimum 0.75 thermal emittance tested in according to ASTM C1371 or ASTM E408.

- 1) Where tested aged values are not available:

- a) Calculate compliance adjusting initial solar reflectance according to ASHRAE 90.1.
- b) Provide roofing system with minimum 64 three-year aged Solar Reflectance Index calculated according to ASTM E1980 with 12 W/square meter/degree K (2.1 BTU/hour/square foot) convection coefficient.

2.3 PRODUCTS - GENERAL

A. Provide roof system components from one manufacturer.

B. Sustainable Construction Requirements:

1. ~~//~~Solar Reflectance Index: 78 minimum.~~//~~

~~SPEC WRITER NOTE:~~

~~1. Retain paragraph below when complying with Federal Guiding Principles IV from Guiding Principles for Federal Leadership in High Performance and Sustainable Buildings incorporated in Executive Order 13423 "Strengthening Federal Environmental, Energy, and Transportation Management, dated January 24, 2007, Enhance Indoor Environmental Quality - Biobased Content Requirement. Requirement differs from related requirements in LEED definitions.~~

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~~2. Indicate project goals for percentages of bio-based, rapidly renewable, and certified sustainable wood products in Section 01 00 00, GENERAL REQUIREMENTS.~~

2. ~~//~~Biobased Content: Where applicable, provide products designated by USDA and meeting or exceeding USDA recommendations for bio-based content, and products meeting Rapidly Renewable Materials and certified sustainable wood content definitions; refer to www.biopreferred.gov.~~//~~

~~SPEC WRITER NOTE:~~

- ~~1. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS includes comprehensive product list setting VOC limits for low emitting materials.~~
- ~~2. Retain subparagraphs applicable to products specified in this section.~~

3. Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:
- a. Non-flooring adhesives and sealants.

2.4 TPO ROOFING MEMBRANE

~~SPEC WRITER NOTES:~~

- ~~1. Use fabric-backed sheet for adhered systems to cellular insulating concrete, structural concrete, or when re-roofing over other incompatible substrates where the manufacturer recommends a fabric backing for separation.~~
- ~~2. Use sheets without fabric backing when adhering to rigid insulation board cover boards.~~
- ~~3. Do not use TPO sheet in ballasted applications.~~
- ~~4. Use adhered systems over cover boards on new VA construction unless other application is specifically approved by VA.~~
- ~~5. TPO membrane sheet is available in gray and several other colors that will not meet the DoE cool roof initiative standards.~~

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1. TPO Sheet: ASTM D6878/D6878M, internally fabric or scrim reinforced, 1.5 mm (60 mils) thick, ~~//with no backing~~ ~~//with fabric backing~~.

2.5 MEMBRANE ACCESSORY MATERIALS

- A. Sheet Flashing: Manufacturer's standard sheet flashing of same material, type, reinforcement, thickness, and color as TPO sheet membrane.
- B. Factory Formed Flashings: Inside and outside corners, pipe boots, and other special flashing shapes to minimize field fabrication.
- C. Bonding Adhesive: Manufacturer's standard, water based.
- D. Metal Termination Bars: Manufacturer's standard, stainless-steel or aluminum, 25 mm wide by 3 mm thick (1-inch wide by 1/8 inch thick) factory drilled for fasteners.
- E. Battens: Manufacturer's standard, galvanized or galvanized steel sheet, 25 mm wide by 1.3 mm thick (1-inch wide by 0.05 inch thick), factory punched for fasteners.
- ~~F. Fasteners: Manufacturer's standard coated steel with metal or plastic plates, to suit application.~~
- ~~G.F.~~ _____ Primers, Sealers, T-Joint Covers, Lap Sealants, and Termination Reglets: As specified by roof membrane manufacturer.
- ~~H.G.~~ _____ Adhesive and sealant materials recommended by roofing system manufacturer for intended use, identical to materials utilized in approved listed roofing system, and compatible with roofing membrane.

2.6 WALKWAY PADS

- A. Manufacturer's standard, slip-resistant rolls, minimum 900 mm (3 feet) wide by 5 mm (3/16 inch) thick.

2.7 ROOF PAVERS

~~SPEC WRITER NOTES:~~

- ~~1. Ensure pavers are detailed showing size and shape.~~
- ~~2. Do not exceed 600 mm square (24 inches square) for non interlocking units with approximate weight of 23 kg (50 pounds) each.~~
- ~~3. Pavers require 73 kg/square meter (15 pounds/square feet.) minimum for fire rating.~~
- ~~4. Interlocking pavers are preferred over non interlocking pavers.~~

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- ~~5. Use interlocking type that has been tested in a wind tunnel for wind uplift meeting project requirements.~~
- ~~6. Do not use light weight aggregate pavers.~~
- ~~7. Extensive use of pavers is not appropriate for roof systems with solar reflective design requirements.~~

- ~~A. Roof Pavers: Precast, normal weight, // interlocking // non-interlocking // concrete units // with ribbed bottom surface for drainage //.~~
- ~~1. Weight: Minimum 73 kg/square meter (15 pounds/square foot).~~
- ~~2. Compressive Strength: ASTM C140/C140M; minimum 55 MPa (8,000 psi).~~
- ~~3. Freeze Thaw: ASTM C67; maximum 1 percent mass loss.~~
- ~~4. Units of size, shape, and thickness as shown on drawings.~~

2.82.7 ACCESSORIES

- A. Temporary Protection Materials:
 - 1. Expanded Polystyrene (EPS) Insulation: ASTM C578.
 - 2. Plywood: NIST DOC PS 1, Grade CD Exposure 1.
 - 3. Oriented Strand Board (OSB): NIST DOC PS 2, Exposure 1.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine and verify substrate suitability with roofing Installer and roofing inspector present.

~~SPEC WRITER NOTE: Require firestopping verification for fire rated roof assemblies.~~

- 1. Verify roof penetrations are complete, secured against movement, ~~//~~and firestopped~~//~~.
- 2. Verify roof deck is adequately secured to resist wind uplift.
- 3. Verify roof deck is clean, dry, and in-plane ready to receive roofing system.

- B. Correct unsatisfactory conditions before beginning roofing work.

3.2 PREPARATION

- A. Complete roof deck construction before beginning roofing work:
 - 1. Curbs, blocking, edge strips, nailers, cants, and other components to which insulation, roofing, and base flashing is attached in place ready to receive insulation and roofing.

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2. Coordinate roofing membrane installation with flashing work and roof insulation work so insulation and flashing are installed concurrently to permit continuous roofing operations.
 3. Complete installation of flashing, insulation, and roofing in same day except for the area where temporary protection is required when work is stopped for inclement weather or end of work day.
- B. Dry out surfaces ~~//~~including roof deck flutes, ~~//~~that become wet from any cause during progress of the work before roofing work is resumed. Apply materials to dry substrates, only.
- C. Broom clean roof decks. Remove dust, dirt and debris.
- D. Remove projections capable of damaging roofing materials.
- E. Concrete Decks, ~~except Insulating Concrete:~~
1. Test concrete decks for moisture according to ASTM D4263 before installing roofing materials.
 2. Prime concrete decks. Keep primer back 100 mm (4 inches) from precast concrete deck joints.
 3. Allow primer to dry before application of bitumen.
- ~~F. Insulating Concrete Decks:~~
- ~~1. Allow to dry out minimum five days after installation before installing roofing materials.~~
 - ~~2. Allow additional drying time when precipitation occurs before installing roofing materials.~~
- ~~G. Poured Gypsum Decks: Dry out poured gypsum according to manufacturer's instructions before installing roofing materials.~~
- ~~H.F.~~ _____//Existing Membrane Roofs and Repair Areas:
1. Comply with requirements in Section 07 01 50.19 PREPARATION FOR REROOFING.//

3.3 TEMPORARY PROTECTION

- A. Install temporary protection consisting of a temporary seal and water cut-offs at the end of each day's work and when work is halted for an indefinite period or work is stopped when precipitation is imminent.
- B. Install temporary cap flashing over top of base flashings where permanent flashings are not in place to protect against water intrusion into roofing system. Securely anchor in place to prevent blow off and damage by construction activities.
- C. Temporarily seal exposed insulation surfaces within roofing membrane.

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1. Apply temporary seal and water cut off by extending roofing membrane beyond insulation and securely embedding edge of the roofing membrane in 6 mm (1/4 inch) thick by 50 mm (2 inches) wide strip of temporary closure sealant. Weight roofing membrane edge with sandbags, to prevent displacement; space sandbags maximum 2400 mm (8 feet) on center.
 2. Direct water away from work. Provide drainage, preventing water accumulation.
 3. Check daily to ensure temporary seal remains watertight. Reseal open areas and weight down.
- D. Before the work resumes, cut off and discard portions of roof membrane in contact with temporary seal.
1. Cut minimum 150 mm (6 inches) back from sealed edges and surfaces.
- E. Remove sandbags and store for reuse.

~~SPEC WRITER NOTE: Use pull out tests for decks other than wood and wood blocking.~~

3.4 INSTALLATION - GENERAL

- A. Install products according to manufacturer's instructions and approved submittal drawings.
1. When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's Representative consideration.
- B. Comply with NRCA Manual installation requirements.

~~SPEC WRITER NOTE: The code requires roof coverings on fully adhered or mechanically fastened non-ballasted roofs to be tested according to UL 580 or UL 1897.~~

- C. Comply with ~~//~~UL 580 or~~//~~UL 1897 ~~//~~for uplift resistance.
- D. Do not allow membrane and flashing to contact surfaces contaminated with asphalt, coal tar, oil, grease, or other substances incompatible with TPO.

3.5 ROOFING INSTALLATION

- A. Install the membrane so the sheets run perpendicular to the long dimension of the insulation boards.
- B. Begin installation at the low point of the roof and work towards the high point. Lap membrane shingled in water flow direction.

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- C. Position the membrane free of buckles and wrinkles.
- D. Roll membrane out; inspect for defects as membrane is unrolled. Remove defective areas:
1. Lap edges and ends of sheets 50 mm (2 inches) or more as recommended by the manufacturer.
 2. Heat weld laps. Apply pressure as required. Seam strength of laps as required by ASTM D4434/D4434M.
 3. Check seams to ensure continuous adhesion and correct defects.
 4. Finish seam edges with beveled bead of lap sealant.
 5. Finish seams same day as membrane is installed.
 6. Anchor membrane perimeter to roof deck or parapet wall as indicated on drawings.
 7. Repair areas of welded seams where samples have been taken or marginal welds, bond voids, or skips occurs.
 8. Repair fishmouths and wrinkles by cutting to lay flat and installing patch over cut area extending 100 mm (4 inches) beyond cut.
- E. Membrane Perimeter Anchorage:
1. Install batten at perimeter of each roof area, curb flashing, expansion joints and similar penetrations on top of roof membrane as indicated on drawings.
 2. Mechanically Fastening:
 - a. Space fasteners maximum 300 mm (12 inches) on center, starting 25 mm (1 inch) from ends.
 - b. When battens are cut, round edges and corners before installing.
 - c. After mechanically fastening strip cover and seal strip with a 150 mm (6 inch) wide roof membrane strip; heat weld to roof membrane and seal edges.
 - ~~d. At // gravel stops // fascia eants // turn roofing membrane down over front edge of the blocking, cant, or nailer. Secure roofing membrane to vertical portion of nailer; or, if required by the membrane manufacturer, with fasteners spaced maximum 150 mm (6 inches) on centers.~~
 - e.d. At parapet walls intersecting building walls and curbs, secure roofing membrane to structural deck with fasteners 150 mm (6 inches) on centers or as shown in NRCA manual.
- F. Adhered System:

1. Apply bonding adhesive in quantities required by roof membrane manufacturer.
2. Fold sheet back on itself, clean and coat the bottom side of the membrane and the top of substrate with adhesive. Do not coat the lap joint area.
3. After adhesive has set according to adhesive manufacturer's instruction, roll roofing membrane into adhesive minimizing voids and wrinkles.
4. Repeat for other half of sheet.

~~G. Mechanically Fastened System Installation:~~

- ~~1. Secure roofing membrane to structural deck with fasteners through battens to achieve specified wind uplift performance.~~
 - ~~a. Drill pilot holes for fasteners installed into cast-in-place concrete. Drill hole minimum 10 mm (3/8 inch) deeper than fastener penetration.~~
- ~~2. When fasteners are installed within membrane laps, locate battens minimum 13 mm (1/2 inch) from the edge of sheets.~~
- ~~3. Apply lap sealant under battens and anchor to deck while lap sealant is still fluid. Cover fastener head with fastener sealer.~~
- ~~4. Where fasteners are installed over roofing membrane after seams are welded, cover fasteners with minimum 200 mm (8 inch) diameter TPO membrane cap centered over fasteners. Where battens are used cover battens with minimum 200 mm (8 inch) wide TPO strip cap centered over batten. Splice caps to roofing membrane and finish edges with lap sealant.~~

3.6 FLASHING INSTALLATION

- A. Install flashings same day as roofing membrane is installed. When flashing cannot be completely installed in one day, complete installation until flashing is watertight and provide temporary covers or seals.

~~SPEC WRITER NOTES:~~

- ~~1. Ensure roof drain flashing details are shown on drawings with sump to depress notched clamping ring below roof surface.~~
- ~~2. Offset drains in sump to side of steel beams so drain is not above low point when roof slope terminates on top of beam.~~

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B. Flashing Roof Drains:

1. Install roof drain flashing as recommended by roofing membrane manufacturer.
 - a. Coordinate to set the metal drain flashing in asphalt roof cement, holding cement back from the edge of the metal flange.
 - b. Do not allow the roof cement to come in contact with TPO roofing membrane.
 - c. Adhere roofing membrane to metal flashing with bonding adhesive.
2. Turn down the metal drain flashing and roofing membrane into drain body. Install clamping ring and strainer.

~~SPEC WRITER NOTES:~~

- ~~1. See NRCA manual for base flashing details.~~
- ~~2. Use with metal cap flashing.~~
- ~~3. Do not use "pitch pocket" or "sealant pocket" construction detail.~~
- ~~4. Coordinate with sheet metal work to provide metal cap flashing for base flashing on curbs and walls and penetrations.~~
- ~~5. Do not terminate base flashing or membrane edge exposed on top of parapet walls or in reglets on horizontal or sloped wash surface.~~
- ~~6. Terminate only under cap flashings or coping covers except for draw bands on pipe boots and gravel stops.~~
- ~~8. Use 200 mm (8 inch) minimum height for base flashing.~~

C. Installing Base Flashing and Pipe Flashing:

1. Install flashing sheet to pipes, wall or curbs to minimum 200 mm (8 inches) above roof surfaces and extending roofing manufacturer's standard lap dimension onto roofing membranes.
 - a. Adhere flashing with bonding adhesive.
 - b. Form inside and outside corners of flashing sheet according to NRCA manual. Form pipe flashing according to NRCA manual.
 - c. Lap ends roofing manufacturer's standard dimension.
 - d. Heat weld flashing membranes together and flashing membranes to roofing membranes. Finish exposed edges with lap sealant.
 - e. Install flashing membranes according to NRCA manual.
2. Anchor top of flashing to walls and curbs with fasteners spaced maximum 150 mm (6 inches) on center. Use surface mounted fastening

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strip with sealant on ducts. Use pipe clamps on pipes or other round penetrations.

3. Apply sealant to top edge of flashing.

D. Installing Building Expansion Joints:

~~SPEC WRITER NOTES:~~

- ~~1. Do not put expansion joints at roof membrane level.~~
- ~~2. Design joints to be installed on curbs minimum 200 mm (8 inches) high.~~
- ~~3. Detail expansion joint.~~

1. Install base flashing on curbs as specified.

2. Coordinate installation with ~~// metal expansion joint cover. // roof expansion joint system //~~.

3. Install flexible tubing 1-1/2 times the width of joint centered over joint. Cover tubing with flashing sheet adhered to base flashing and lapping base flashing roofing manufacturer's standard dimension. Finish edges of laps with sealant.

E. Repairs to Membrane and Flashings:

1. Remove sections of roofing membrane or flashing that are creased, wrinkled, or fishmouthed.
2. Cover removed areas, cuts and damaged areas with a patch extending 100 mm (4 inches) beyond damaged, cut, or removed area. Heat weld to roofing membrane or flashing sheet. Finish edge of lap with lap sealant.

3.7 WALKWAY PAD INSTALLATION

A. Heat weld walkway sheet to roofing membrane at edges. Weld area 50 mm (2 inches) wide by the entire length of the walkway sheet.

B. Finish edges of laps with lap sealant.

~~C.~~

~~3.8 PAVER INSTALLATION~~

~~SPEC WRITER NOTES:~~

- ~~1. Use pavers in the following locations as a minimum:~~
 - ~~a. At working and access areas of equipment requiring servicing.~~
 - ~~b. At equipment having discharges detrimental to roof membrane, under gooseneck discharges from kitchens and chemical exhausts.~~

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- ~~e. At landing points for hatches, ladders, and doors entering roof level.~~
 - ~~2. Confirm walkways and pavers are shown on drawings.~~
 - ~~3. Pavers:~~
 - ~~a. Pavers without interlocking connectors require strapping together and edge clamps when pavers do not provide minimum weight for wind uplift resistance.~~
 - ~~b. Use mechanical strapping to create perimeter anchor, at penetrations, cuts at valleys, over drains, and where partial or cut units occur.~~
 - ~~c. Detail strapping, perimeter restraints, edge clamps and location of strapping. Do not anchor through base flashing or into eants.~~
 - ~~4. Interlocking connectors:~~
 - ~~a. Use 400 mm (16 inches) on center minimum spacing of connectors.~~
 - ~~b. Decrease spacing to 300, 200, or 100 mm (12, 8, or 4 inches) on center for greater wind velocities.~~
- ~~A. Install pavers as soon as roofing membrane is installed.~~
 - ~~1. Saw cut or core drill pavers for cut units.~~
 - ~~2. Install pavers with butt joints in running bond with minimum one half-length units at ends.~~
 - ~~a. Stagger end joints; generally locate joints near midpoint of adjacent rows, except where end joints occur in valleys. Miter end joints to fit in valleys.~~
 - ~~b. Cut to fit within 13 mm (1/2 inch) of penetrations.~~
 - ~~3. Install interlocking connectors in channel units for complete tie in of units, including cut units. Use corner spacings for a distance of 1200 mm (4 feet) or more around roof drains, penetrations, and other vertical surfaces in the field of the roof area.~~
 - ~~a. Space connectors at // _____ // mm (// _____ // inches) on center at the corners for 3 m (10 foot) square area.~~
 - ~~b. Space connectors at // _____ // mm (// _____ // inches) on center at the perimeter for 1800 mm (6 foot) wide strip.~~
 - ~~c. Space connectors at // _____ // mm (// _____ // inches) on center in the field.~~

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- ~~d. Install pavers under the perimeter retainer as shown on drawings.~~
- ~~4. Install strapping where shown.~~
 - ~~a. Limit strap lengths to a maximum of 9 m (30 feet).~~
 - ~~b. Install straps at corner connection to the perimeter retainer at approximate 45 degree angle at approximate 3 to 3.6 m (10 to 12 feet) from corner.~~
 - ~~c. Install straps on both sides of valleys, hips, and ridges, with cross straps spaced maximum 1200 mm (4 feet) on center between end straps.~~
 - ~~d. Install straps at the perimeter of penetrations more than two pavers in width or length.~~
 - ~~e. Anchor straps to each paver with two fasteners per unit.~~
 - ~~f. Pre-drill holes for fasteners in pavers.~~

3-93.8 FIELD QUALITY CONTROL

~~SPEC WRITER NOTE: Section 01 45 29, TESTING LABORATORY SERVICES includes VA provided testing for large projects and contractor provided testing for small projects. Coordinate testing responsibility.~~

- A. Field Tests: Performed by ~~testing laboratory specified in Section 01 45 29, TESTING LABORATORY SERVICES.~~ Contractor.

~~SPEC WRITER NOTE: ANSI/SPRI FX-1 sets testing frequency as 10 tests for first 4,650 square meter (50,000 square feet) and five tests for each additional 4,650 square meter (50,000 square feet). Specify frequency to suit project conditions.~~

1. Fastener Pull Out Tests: ANSI/SPRI FX-1; one test for every 230 square meter (2,500 square feet) of deck. Perform tests for each combination of fastener type and roof deck type before installing roof insulation.
 - a. Test at locations selected by Contracting Officer's Representative.
 - b. Do not proceed with roofing work when pull out resistance is less than manufacturer's required resistance.
 - c. Test Results:

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- 1) Repeat tests using different fastener type or use additional fasteners achieve pull out resistance required to meet specified wind uplift performance.
 - 2) Patch cementitious deck to repair areas of fastener tests holes.
2. Examine and probe roofing membrane and flashing seams in presence of Contracting Officer's Representative and Manufacturer's field representative.
 3. Probe seams to detect marginal bonds, voids, skips, and fishmouths.
 4. Cut 100 mm (4 inch) wide by 300 mm (12 inch) long samples through seams where directed by Contracting Officer's Representative.
 5. Cut one sample for every 450 m (1500 feet) of seams.
 6. Cut samples perpendicular to seams.
 7. Failure of samples to pass ASTM D1876 test will be cause for rejection of work.
 8. Repair areas where samples are taken and where marginal bond, voids, and skips occur.
 9. Repair fishmouths and wrinkles by cutting to lay flat. Install patch over cut area extending 100 mm (4 inches) beyond cut.
- B. Manufacturer Services:
1. Inspect initial installation, installation in progress, and completed work.
 2. Issue supplemental installation instructions necessitated by field conditions.
 3. Prepare and submit inspection reports.
 4. Certify completed installation complies with manufacturer's instructions and warranty requirements.

3-103.9 CLEANING

- A. Remove excess adhesive before adhesive sets.
- B. Clean exposed roofing surfaces. Remove contaminants and stains ~~++~~to comply with specified solar reflectance performance ~~++~~.

3-113.10 PROTECTION

- A. Protect roofing system from traffic and construction operations.
 1. Protect roofing system when used for subsequent work platform, materials storage, or staging.
 2. Distribute scaffolding loads to exert maximum 50 percent roofing system materials compressive strength.

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- B. Loose lay temporary insulation board overlaid with plywood or OSB.
 - 1. Weight boards to secure against wind uplift.
- C. Remove protective materials immediately before acceptance.
- D. Repair damage.

- - - E N D - - -

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SECTION 07 60 00
FLASHING AND SHEET METAL

~~SPEC WRITER NOTES:-~~

- ~~1. Delete or edit text in // // if not applicable to project. Also delete any other paragraph not applicable in the section.~~
- ~~2. This section requires close coordination with roofing and masonry sections for flashing installed by other trades and for flashings integral with manufactured metal roofing/wall systems where matching finish is required of the same material.~~
- ~~3. Coordinate with drawing terminology for correct and uniform nomenclature.~~
- ~~4. Follow the recommendations of the National Roofing Contractors Association "Roofing and Waterproofing Manual" for design criteria.~~
- ~~5. Modify this section to address reroofing of existing roofing system. Coordinate with Section 07 01 50.19 PREPARATION FOR REROOFING for reroofing of existing facilities.~~

PART 1 - GENERAL

1.1 DESCRIPTION

A. Formed sheet metal work for wall and roof flashing, copings, roof edge metal, fasciae, drainage specialties, and formed expansion joint covers are specified in this section.

1.2 RELATED WORK

~~SPEC WRITER NOTE: Edit Related Work to reflect other sections relating directly to this section or referenced in this section.~~

- ~~A. //Section 07 51 00, BUILT-UP BITUMINOUS ROOFING: Membrane base flashings and stripping.//~~
- ~~B. //Section 07 51 13.11 HOT-APPLIED BUILT-UP ASPHALT ROOFING: Membrane base flashings and stripping.//~~
- ~~C. //07 51 13.13 COLD-APPLIED BUILT-UP ASPHALT ROOFING Membrane base flashings and stripping.//~~
- ~~D. //Section 07 52 16.11 COLD ADHESIVE STYRENE-BUTADIENE-STYRENE MODIFIED BITUMINOUS MEMBRANE ROOFING Membrane base flashings and stripping.//~~

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~~E. //07 52 16.12 HOT-MOPPED STYRENE-BUTADIENE-STYRENE MODIFIED BITUMINOUS
MEMBRANE ROOFING Membrane base flashings and stripping.//~~

~~F. //Section 07 53 23 ETHYLENE-PROPYLENE-DIENE MONOMER ROOFING Membrane
base flashings and stripping.//~~

~~G. //Section 07 54 16 ETHYLENE INTERPOLYMER (KEE) ROOFING Membrane base
flashings and stripping.//~~

~~H.A. //Section 07 54 19 POLYVINYL-CHLORIDE (PVC) ROOFING Membrane base
flashings and stripping.//~~

~~I.B. //Section 07 54 23 THERMOPLASTIC POLYOLEFIN (TPO) ROOFING:
Membrane base flashings and stripping.//~~

~~J.C. Section 07 71 00 ROOF SPECIALTIES: Manufactured flashing,
copings, roof edge metal, and fasciae.~~

~~K. //Section 07 71 00, ROOF SPECIALTIES: Integral flashing components of
manufactured roof specialties and accessories or equipment.//~~

~~L.D. //Section 07 72 00, ROOF ACCESSORIES: Integral flashing
components of manufactured roof specialties and accessories or
equipment.//~~

~~M. Division 07 ROOFING AND WALL SYSTEM: Flashing components of factory
finished roofing and wall systems.~~

~~N.E. Section 07 92 00, JOINT SEALANTS: Joint Sealants.~~

~~O. Section 09 06 00, SCHEDULE FOR FINISHES: Color of factory coated
exterior architectural metal and anodized aluminum items.~~

~~P.F. Section 09 91 00, PAINTING: Paint materials and application.~~

~~Q. Section 13 34 19, METAL BUILDING SYSTEMS: Flashing and sheet metal in
connection with prefabricated metal buildings.~~

~~R. : sections and sections.~~

~~S. //Division 22, PLUMBING: Integral flashing components of manufactured
roof specialties and accessories or equipment.//~~

~~T. Section 22 14 00, FACILITY STORM DRAINAGE: Flashing of Roof Drains.~~

~~U. //Division 23 HVAC: Integral flashing components of manufactured roof
specialties and accessories or equipment.~~

1.3 -APPLICABLE PUBLICATIONS

~~SPEC-WRITER NOTES:~~

- ~~1. Update applicable publications to
current issue at time of project
specification preparation.~~
- ~~2. Update material requirements to agree
with applicable requirements (types,~~

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~~grades, classes,) specified in the
referenced Applicable Publications.~~

- A. Publications listed below form a part of this specification to the extent referenced. Publications are referenced in the text by the basic designation only. Editions of applicable publications current on date of issue of bidding documents apply unless otherwise indicated.
- B. Aluminum Association (AA):
- AA-C22A41.....Aluminum Chemically etched medium matte, with clear anodic coating, Class I Architectural, 0.7-mil thick
- AA-C22A42.....Chemically etched medium matte, with integrally colored anodic coating, Class I Architectural, 0.7 mils thick
- AA-C22A44.....Chemically etched medium matte with electrolytically deposited metallic compound, integrally colored coating Class I Architectural, 0.7-mil thick finish
- C. American National Standards Institute/Single-Ply Roofing Institute/Factory Mutual (ANSI/SPRI/FM):
- 4435/ES-1-11.....Wind Design Standard for Edge Systems Used with Low Slope Roofing Systems
- D. American Architectural Manufacturers Association (AAMA):
- AAMA 620-02.....Voluntary Specification for High Performance Organic Coatings on Coil Coated Architectural Aluminum
- AAMA 621-02.....Voluntary Specification for High Performance Organic Coatings on Coil Coated Architectural Hot Dipped Galvanized (HDG) and Zinc-Aluminum Coated Steel Substrates
- E. ASTM International (ASTM):
- A240/A240M-20.....Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet and Strip for Pressure Vessels and for General Applications.

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- A653/A653M-20.....Steel Sheet Zinc-Coated (Galvanized) or Zinc Alloy Coated (Galvanized) by the Hot- Dip Process
- B32-08 (2014)Solder Metal
- B209-14.....Aluminum and Aluminum-Alloy Sheet and Plate
- B370-12 (2019)Copper Sheet and Strip for Building Construction
- D173/D173M-03 (2018)Bitumen-Saturated Cotton Fabrics Used in Roofing and Waterproofing
- D412-16.....Vulcanized Rubber and Thermoplastic Elastomers-Tension
- D1187/D1187M-97 (2018) ...Asphalt Base Emulsions for Use as Protective Coatings for Metal
- D1784-20.....Rigid Poly (Vinyl Chloride) (PVC) Compounds and Chlorinated Poly (Vinyl Chloride) (CPVC) Compounds
- D3656/D3656M-13.....Insect Screening and Louver Cloth Woven from Vinyl-Coated Glass Yarns
- D4586/D4586M-07 (2018) ...Asphalt Roof Cement, Asbestos Free
- F. Sheet Metal and Air Conditioning Contractors National Association (SMACNA): Architectural Sheet Metal Manual.
- G. National Association of Architectural Metal Manufacturers (NAAMM): AMP 500-06.....Metal Finishes Manual
- H. Federal Specification (Fed. Spec):
- A-A-1925A.....Shield, Expansion; (Nail Anchors)
- UU-B-790A.....Building Paper, Vegetable Fiber
- I. International Code Commission (ICC): International Building Code, Current Edition

1.4 PERFORMANCE REQUIREMENTS

~~SPEC WRITER NOTE: Select applicable subparagraph below based upon FM Approvals 1-28, Table AC 1.4. Coordinate with uplift requirements of Division 07 roofing section.~~

- A. Wind Uplift Forces: Resist the following forces per FM Approvals 1-49:
1. Wind Zone 1: 0.48 to 0.96 kPa (10 to 20 pound force/square foot):
- 1.92-kPa (40 pound force/square foot) perimeter uplift force, 2.87-

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kPa (60 pound force/square foot pound force/square foot) corner uplift force, and 0.96-kPa (20- pound force/square foot) outward force.

2. Wind Zone 1: 1.00 to 1.44 kPa (21 to 30 pound force/square foot): 2.87-kPa (60 pound force/square foot) perimeter uplift force, 4.31-kPa (90 pound force/square foot) corner uplift force, and 1.44-kPa (30 pound force/square foot) outward force.

~~3. Wind Zone 2: 1.48 to 2.15 kPa (31 to 45 pound force/square foot): 4.31-kPa (90 pound force/square foot) perimeter uplift force, 5.74-kPa (120 pound force/square foot) corner uplift force, and 2.15-kPa (45 pound force/square foot) outward force.~~

~~4. Wind Zone 3: 2.20 to 4.98 kPa (46 to 104 pound force/square foot): 9.96-kPa (208 pound force/square foot) perimeter uplift force, 14.94 kPa (312 pound force/square foot) corner uplift force, and 4.98 kPa (104 pound force/square foot) outward force.~~

~~SPEC WRITER NOTE: Standard below applies to shop-fabricated sheet metal copings and roof edge metal items tested under cited standard; it is a requirement of the current International Building Code. NRCA member firms are able to meet below utilizing NRCA-approved and tested details.~~

- B. Wind Design Standard: Fabricate and install ~~//copings~~ and ~~//~~ ~~//roof-edge flashings~~ ~~//~~ tested per ANSI/SPRI/FM ES-1 to resist design pressure ~~//insert design pressure//~~ ~~//~~ indicated on Drawings~~//~~.

1.5 SUBMITTALS

- A. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

~~SPEC WRITER NOTE: Add to shop drawing list special items not specified in this section that are added to this section.~~

- B. Shop Drawings: For all specified items, including:

1. Flashings
2. Copings
3. Gravel Stop-Fascia
4. Gutter and Conductors
5. Expansion joints

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6. Fascia-cant
- C. Manufacturer's Literature and Data: For all specified items, including:
1. Two-piece counterflashing
 2. Thru wall flashing
 3. Expansion joint cover, each type
 4. Nonreinforced, elastomeric sheeting
 5. Copper clad stainless steel
 6. Polyethylene coated copper
 7. Bituminous coated copper
 8. Copper covered paper
 9. Fascia-cant
- D. Certificates: Indicating compliance with specified finishing requirements, from applicator and contractor.

PART 2 - PRODUCTS

2.1 FLASHING AND SHEET METAL MATERIALS

~~SPEC WRITER NOTE: Retain paragraphs below that correspond to project requirements.~~

- A. Stainless Steel: ASTM A240, Type 302B, dead soft temper.
- ~~B. Copper ASTM B370, cold-rolled temper.~~
- ~~C. Bituminous Coated Copper: Minimum copper ASTM B370, weight not less than 1 kg/m² (3 oz/sf). Bituminous coating shall weigh not less than 2 kg/m² (6 oz/sf); or, copper sheets may be bonded between two layers of coarsely woven bitumen-saturated cotton fabric ASTM D173. Exposed fabric surface shall be crimped.~~
- ~~D. Copper Covered Paper: Fabricated of electro-deposit pure copper sheets ASTM B 370, bonded with special asphalt compound to both sides of creped, reinforced building paper, UU-B-790, Type I, style 5, or to a three ply sheet of asphalt impregnated crepe paper. Grooves running along the width of sheet.~~
- ~~E. Polyethylene Coated Copper: Copper sheet ASTM B370, weighing 1 Kg/m² (3 oz/sf) bonded between two layers of (two mil) thick polyethylene sheet.~~
- ~~F.B.~~ Aluminum Sheet: ASTM B209, alloy 3003-H14 ~~//~~except alloy used for color anodized aluminum shall be as required to produce specified color. Alloy required to produce specified color shall have the same structural properties as alloy 3003-H14~~//~~.
- ~~G.C.~~ Galvanized Sheet: ASTM, A653.

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~~H-D.~~ Nonreinforced, Elastomeric Sheeting: Elastomeric substances reduced to thermoplastic state and extruded into continuous homogenous sheet (0.056 inch) thick. Sheeting shall have not less than 7 MPa (1,000 psi) tensile strength and not more than seven percent tension-set at 50 percent elongation when tested in accordance with ASTM D412. Sheeting shall show no cracking or flaking when bent through 180 degrees over a 1 mm (1/32 inch) diameter mandrel and then bent at same point over same size mandrel in opposite direction through 360 degrees at temperature of -30°C (-20 °F).

2.2 FLASHING ACCESSORIES

- A. Solder: ASTM B32; flux type and alloy composition as required for use with metals to be soldered.
- B. Rosin Paper: Fed-Spec. UU-B-790, Type I, Grade D, Style 1b, Rosin-sized sheathing paper, weighing approximately 3 Kg/10 m² (6 pounds/100 square feet).
- C. Bituminous Paint: ASTM D1187, Type I.
- D. Fasteners:
 - 1. Use ~~copper, copper alloy, bronze, brass, or stainless steel for copper and copper clad stainless steel, and~~ stainless steel for stainless steel and aluminum alloy. Use galvanized steel or stainless steel for galvanized steel.
 - 2. Nails:
 - a. Minimum diameter for copper nails: 3 mm (0.109 inch).
 - b. Minimum diameter for aluminum nails 3 mm (0.105 inch).
 - c. Minimum diameter for stainless steel nails: 2 mm (0.095 inch) and annular threaded.
 - d. Length to provide not less than 22 mm (7/8 inch) penetration into anchorage.
 - 3. Rivets: Not less than 3 mm (1/8 inch) diameter.
 - 4. Expansion Shields: Fed Spec A-A-1925A.
- E. Sealant: As specified in Section 07 92 00, JOINT SEALANTS for exterior locations.
- F. Insect Screening: ASTM D3656, 18 by 18 regular mesh.
- G. Roof Cement: ASTM D4586.

2.3 SHEET METAL THICKNESS

- A. Except as otherwise shown or specified use thickness or weight of sheet metal as follows:
- B. Concealed Locations (Built into Construction):
- ~~1. Copper: 30g (10 oz) minimum 0.33 mm (0.013 inch thick).~~
 - ~~2-1.~~ Stainless steel: 0.25 mm (0.010 inch) thick.
 - ~~3. Copper clad stainless steel: 0.25 mm (0.010 inch) thick.~~
 - ~~4-2.~~ Galvanized steel: 0.5 mm (0.021 inch) thick.
- C. Exposed Locations:
- ~~1. Copper: 0.4 Kg (16 oz).~~
 - ~~2-1.~~ Stainless steel: 0.4 mm (0.015 inch).
 - ~~3. Copper clad stainless steel: 0.4 mm (0.015 inch).~~
- D. Thickness of aluminum or galvanized steel is specified with each item.

2.4 FABRICATION, GENERAL

- A. Jointing:
- 1. In general, ~~copper,~~ stainless steel ~~and copper clad stainless steel~~ joints, except expansion and contraction joints, shall be locked and soldered.
 - 2. Jointing of ~~copper over 0.5 Kg (20 oz) weight or~~ stainless steel over 0.45 mm (0.018 inch) thick shall be done by lapping, riveting and soldering.
 - 3. Joints shall conform to following requirements:
 - a. Flat-lock joints shall finish not less than 19 mm (3/4 inch) wide.
 - b. Lap joints subject to stress shall finish not less than 25 mm (one inch) wide and shall be soldered and riveted.
 - c. Unsoldered lap joints shall finish not less than 100 mm (4 inches) wide.
 - 4. Flat and lap joints shall be made in direction of flow.
 - ~~5. Edges of bituminous coated copper, copper covered paper, nonreinforced elastomeric sheeting and polyethylene coated copper shall be jointed by lapping not less than 100 mm (4 inches) in the direction of flow and cementing with asphalt roof cement or sealant as required by the manufacturer's printed instructions.~~
 - ~~6-5.~~ Soldering:

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a. Pre tin both mating surfaces with solder for a width not less than 38 mm (1 1/2 inches) of uncoated copper, stainless steel, and copper clad stainless steel.

~~b. Wire brush to produce a bright surface before soldering lead coated copper.~~

~~c.~~ b. Treat in accordance with metal producers recommendations other sheet metal required to be soldered.

~~d.~~ c. Completely remove acid and flux after soldering is completed.

B. Expansion and Contraction Joints:

1. Fabricate in accordance with the Architectural Sheet Metal Manual recommendations for expansion and contraction of sheet metal work in continuous runs.
2. Space joints as shown or as specified.
3. Space expansion and contraction joints for ~~copper,~~ stainless steel, ~~and copper clad stainless steel~~ at intervals not exceeding 7200 mm (24 feet).
4. Space expansion and contraction joints for aluminum at intervals not exceeding 5400 mm (18 feet), except do not exceed 3000 mm (10 feet) for gravel stops and fascia-cant systems.
5. Fabricate slip-type or loose locked joints and fill with sealant unless otherwise specified.
6. Fabricate joint covers of same thickness material as sheet metal served.

C. Cleats:

1. Fabricate cleats to secure flashings and sheet metal work over 300 mm (12 inches) wide and where specified.
2. Provide cleats for maximum spacing of 300 mm (12 inch) centers unless specified otherwise.
3. Form cleats of same metal and weights or thickness as the sheet metal being installed unless specified otherwise.
4. Fabricate cleats from 50 mm (2 inch) wide strip. Form end with not less than 19 mm (3/4 inch) wide loose lock to item for anchorage. Form other end of length to receive nails free of item to be anchored and end edge to be folded over and cover nail heads.

D. Edge Strips or Continuous Cleats:

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1. Fabricate continuous edge strips where shown and specified to secure loose edges of the sheet metal work.
2. Except as otherwise specified, fabricate edge strips or minimum ~~//0.6 Kg (24 ounce) copper//~~ //0.6 mm (0.024 inch) thick stainless steel// or 1.25 mm (0.050 inch) thick aluminum//.
3. Use material compatible with sheet metal to be secured by the edge strip.
4. Fabricate in 3000 mm (10 feet) maximum lengths with not less than 19 mm (3/4 inch) loose lock into metal secured by edge strip.
5. Fabricate Strips for fascia anchorage to extend below the supporting wood construction to form a drip and to allow the flashing to be hooked over the lower edge at least 19 mm (3/4-inch).
6. Fabricate anchor edge maximum width of 75 mm (3 inches) or of sufficient width to provide adequate bearing area to insure a rigid installation using ~~//1 Kg (32 oz) copper//~~ //0.8 mm (0.031 inch) thick stainless steel// or 1.6 mm (0.0625 inch) thick aluminum//.

E. Drips:

1. Form drips at lower edge of sheet metal counter-flashings (cap flashings), fascias, ~~gravel stops, and~~ wall copings, by folding edge back 13 mm (1/2 inch) and bending out 45 degrees from vertical to carry water away from the wall.
2. Form drip to provide hook to engage cleat or edge strip for fastening for not less than 19 mm (3/4 inch) loose lock where shown.

F. Edges:

1. Edges of flashings concealed in masonry joints opposite drain side shall be turned up 6 mm (1/4 inch) to form dam, unless otherwise specified or shown otherwise.
2. Finish exposed edges of flashing with a 6 mm (1/4 inch) hem formed by folding edge of flashing back on itself when not hooked to edge strip or cleat. Use 6 mm (1/4 inch) minimum penetration beyond wall face with drip for through-wall flashing exposed edge.
3. All metal roof edges shall meet requirements of IBC, current edition.

~~SPEC WRITER NOTE: For alteration work and small projects, minimize number of specified optional flashing materials. Match existing metals.~~

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G. Metal Options:

1. Where options are permitted for different metals use only one metal throughout.
2. Stainless steel may be used in concealed locations for fasteners of other metals exposed to view.
- ~~3. Where copper gravel stops, copings and flashings will carry water onto cast stone, stone, or architectural concrete, or stainless steel.~~

2.5 FINISHES

~~SPEC WRITER NOTES:~~

- ~~1. If more than one finish is used on project, precede finish spec with "Finish for (list items): "Following items are of aluminum: Coping, Gravel Stop and Fascia Cant, Exp. Jt. Cover, Gutter and Downspout.~~
- ~~2. Coordinate with Section 09 06 00, SCHEDULE FOR FINISHES and drawings clearly identify locations of different colors or finish on the same item.~~

- A. Use same finish on adjacent metal or components and exposed metal surfaces unless specified or shown otherwise.
- B. In accordance with NAAMM Metal Finishes Manual AMP 500, unless otherwise specified.
- C. Finish exposed metal surfaces as follows, unless specified otherwise:

~~1. Copper: Mill finish.~~

~~2.1.~~ Stainless Steel: Finish No. 2B or 2D.

~~3.2.~~ Aluminum:

- a. Clear Finish: AA-C22A41 medium matte, clear anodic coating, Class 1 Architectural, 18 mm (0.7 mils) thick.
- b. Colored Finish: AA-C22A42 (anodized) or AA-C22A44 (electrolytically deposited metallic compound) medium matte, integrally colored coating, Class 1 Architectural, 18 mm (0.7 mils) thick. Dyes will not be accepted.
- c. Fluorocarbon Finish: AAMA 620, high performance organic coating.
- d. Mill finish.

~~4.3.~~ Steel and Galvanized Steel:

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- a. Finish painted under Section 09 91 00, PAINTING unless specified as prefinished item.
- b. Manufacturer's finish:
 - 1) Baked on prime coat over a phosphate coating.
 - 2) Baked-on prime and finish coat over a phosphate coating.
 - 3) Fluorocarbon Finish: AAMA 621, high performance organic coating.

~~2.6 THROUGH-WALL FLASHINGS~~

~~SPEC WRITER NOTES:~~

- ~~1. See Sheet Metal Manual for wall flashings. Provide drips on drainage sides.~~
- ~~2. Clearly detail flashings at copings, building set-backs, sills, spandrels, lintels, and grade.~~
- ~~3. Coordinate with waterproofing or dampproofing to show interface.~~

~~A. Form through-wall flashing to provide a mechanical bond or key against lateral movement in all directions. Install a sheet having 2 mm (1/16 inch) deep transverse channels spaced four to every 25 mm (one inch), or ribbed diagonal pattern, or having other deformation unless specified otherwise.~~

- ~~1. Fabricate in not less than 2400 mm (8 feet) lengths; 3000 mm (10 feet) maximum lengths.~~
- ~~2. Fabricate so keying nests at overlaps.~~

~~B. For Masonry Work When Concealed Except for Drip:~~

- ~~1. Either copper, stainless steel, or copper clad stainless steel.~~
- ~~2. Form an integral dam at least 5 mm (3/16 inch) high at back edge.~~
- ~~3. Form exposed portions of flashing with drip, approximately 6 mm (1/4 inch) projection beyond wall face.~~

~~C. For Masonry Work When Exposed Edge Forms a Receiver for Counter Flashing:~~

- ~~1. Use same metal and thickness as counter flashing.~~
- ~~2. Form an integral dam at least 5 mm (3/16 inch) high at back edge.~~
- ~~3. Form exposed portion as snap lock receiver for counter flashing upper edge.~~

~~D. For Flashing at Architectural Precast Concrete Panels or Stone Panels.~~

- ~~1. Use plan flat sheet of stainless steel.~~
- ~~2. Form exposed portions with drip as specified or receiver.~~

~~E. Window Sill Flashing and Lintel Flashing:~~

- ~~1. Use either copper, stainless steel, copper clad stainless steel plane flat sheet, or nonreinforced elastomeric sheeting, bituminous coated copper, copper covered paper, or polyethylene coated copper.~~
- ~~2. Fabricate flashing at ends with folded corners to turn up 5 mm (3/16 inch) in first vertical masonry joint beyond masonry opening.~~
- ~~3. Turn up back edge as shown.~~
- ~~4. Form exposed portion with drip as specified or receiver.~~

~~F. Door Sill Flashing:~~

- ~~1. Where concealed, use either 0.5 Kg (20 ounce) copper, 0.5 mm (0.018 inch) thick stainless steel, or 0.5 mm (0.018 inch) thick copper clad stainless steel.~~
- ~~2. Where shown on drawings as combined counter flashing under threshold, sill plate, door sill, or where subject to foot traffic, use either 0.6 Kg (24 ounce) copper, 0.6 mm (0.024 inch) stainless steel, or 0.6 mm (0.024 inch) thick stainless steel.~~
- ~~3. Fabricate flashing at ends to turn up 5 mm (3/16 inch) in first vertical masonry joint beyond masonry opening with folded corners.~~

~~SPEC WRITER NOTE: Do not use galvanized steel for base flashing.~~

2.72.6 BASE FLASHING

- A. Use metal base flashing at vertical surfaces intersecting built-up roofing without cant strips or where shown.
 1. Use ~~either copper, or~~ stainless steel, thickness specified unless specified otherwise.
 2. When flashing is over 250 mm (10 inches) in vertical height or horizontal width use ~~either 0.5 Kg (20 oz) copper or~~ 0.5 mm (0.018 inch) stainless steel.
 3. Use stainless steel at aluminum roof curbs where flashing contacts the aluminum.
 4. Use ~~either copper, or~~ stainless steel at pipe flashings.
- B. Fabricate metal base flashing up vertical surfaces not less than 200 mm (8 inch) nor more than 400 mm (16 inch).
- C. Fabricate roof flange not less than 100 mm (4 inches) wide unless shown otherwise. When base flashing length exceeds 2400 mm (8 feet) form flange edge with 13 mm (1/2 inch) hem to receive cleats.

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- D. Form base flashing bent from strip except pipe flashing. Fabricate ends for riveted soldered lap seam joints. Fabricate expansion joint ends as specified.
- E. Pipe Flashing: (Other than engine exhaust or flue stack)
1. Fabricate roof flange not less than 100 mm (4 inches) beyond sleeve on all sides.
 2. Extend sleeve up and around pipe and flange out at bottom not less than 13 mm (1/2 inch) and solder to flange and sleeve seam to make watertight.
 3. At low pipes 200 mm (8 inch) to 450 mm (18 inch) above roof:
 - a. Form top of sleeve to turn down into the pipe at least 25 mm (one inch).
 - b. Allow for loose fit around and into the pipe.
 4. At high pipes and pipes with goosenecks or other obstructions which would prevent turning the flashing down into the pipe:
 - a. Extend sleeve up not less than 300 mm (12 inch) above roofing.
 - b. Allow for loose fit around pipe.

2.82.7 COUNTERFLASHING (CAP FLASHING OR HOODS)

~~SPEC WRITER NOTES:-~~

- ~~1. Galvanized steel is not desired due to maintenance, especially painting, if used specify 0.6 mm (0.023 inch) thick material for counter flashing.~~
- ~~2. Use 1.25 mm (0.050 inch) thick aluminum for counter flashing.~~
- ~~3. Identify clearly locations of different metals if used for same item.~~

- A. ~~Either copper or s~~Stainless steel, unless specified otherwise.
- B. Fabricate to lap base flashing a minimum of 100 mm (4 inches) with drip:
1. Form lock seams for outside corners. Allow for lap joints at ends and inside corners.
 2. In general, form flashing in lengths not less than 2400 mm (8 feet) and not more than 3000 mm (10 feet).
 3. Two-piece, lock in type flashing may be used in-lieu-of one piece counter-flashing.
 4. Manufactured assemblies may be used.

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5. Where counterflashing is installed at new work use an integral flange at the top designed to be extended into ~~the masonry joint or~~ reglet in concrete.
 6. Where counterflashing is installed at existing work use surface applied type, formed to provide a space for the application of sealant at the top edge.
- C. One-piece Counterflashing:
1. Back edge turned up and fabricate to lock into reglet in concrete.
 - ~~2. Upper edge formed to extend full depth of masonry unit in mortar joint with back edge turned up 6 mm (1/4 inch).~~
- D. Two-Piece Counterflashing:
- ~~1. Receiver to extend into masonry wall depth of masonry unit with back edge turned up 6 mm (1/4 inch) and exposed edge designed to receive and lock counterflashing upper edge when inserted.~~
 - 2-1. Counterflashing upper edge designed to snap lock into receiver.
- E. Surface Mounted Counterflashing; one or two piece:
1. Use at existing or new surfaces where flashing cannot be inserted in vertical surface.
 2. One piece fabricate upper edge folded double for 65 mm (2 1/2 inches) with top 19 mm (3/4 inch) bent out to form "V" joint sealant pocket with vertical surface. Perforate flat double area against vertical surface with horizontally slotted fastener holes at 400 mm (16 inch) centers between end holes. Option: ~~One piece surface mounted counter-flashing (cap flashing) may be used. Fabricate as detailed on Plate 51 of SMACNA Architectural Sheet Metal Manual.~~
 3. Two pieces: Fabricate upper edge to lock into surface mounted receiver. Fabricate receiver joint sealant pocket on upper edge and lower edge to receive counterflashing, with slotted fastener holes at 400 mm (16 inch) centers between upper and lower edge.
- F. Pipe Counterflashing:
1. Form flashing for water-tight umbrella with upper portion against pipe to receive a draw band and upper edge to form a "V" joint sealant receiver approximately 19 mm (3/4 inch) deep.
 2. Fabricate 100 mm (4 inch) over lap at end.
 3. Fabricate draw band of same metal as counter flashing. Use ~~0.6 Kg (24 oz) copper or~~ 0.33 mm (0.013 inch) thick stainless steel ~~or copper coated stainless steel.~~

4. Use stainless steel bolt on draw band tightening assembly.
5. Vent pipe counter flashing may be fabricated to omit draw band and turn down 25 mm (one inch) inside vent pipe.
- G. Where vented edge decks intersect vertical surfaces, form in one piece, shape to slope down to a point level with and in front of edge-set notched plank; then, down vertically, overlapping base flashing.

~~2.9 GRAVEL STOPS~~

~~A. General:~~

- ~~1. Fabricate in lengths not less than 2400 mm (8 feet) long and maximum of 3000 mm (10 feet).~~
- ~~2. Fabricate internal and external corners as one piece with legs not less than 600 mm (2 feet) or more than 1200 mm (4 feet) long.~~
- ~~3. Fabricate roof flange not less than 100 mm (4 inches) wide.~~
- ~~4. Fabricate top edge to extend above roof not less than 25 mm (one inch) for embedded gravel aggregate and not less than 100 mm (4 inches) for loose laid ballast.~~
- ~~5. Fabricate lower edge outward at an angle of 45 degrees to form drip and as fascia or as counter flashing as shown:~~
 - ~~a. Fabricate of one piece material of suitable width for fascia height of 250 mm (10 inch) maximum or counterflashing lap of not less than 100 mm (4 inch) over base flashing.~~
 - ~~b. Fabricate bottom edge of formed fascia to receive edge strip.~~
 - ~~c. When fascia bottom edge forms counter flashing over roofing lap roofing not less than 150 mm (6 inches).~~

~~B. Formed Flat Sheet Metal Gravel Stops and Fascia:~~

- ~~1. Fabricate as shown of // .05 mm (0.018 inch) thick stainless steel// //0.5 Kg (20 ounce) copper// //1.25 mm (0.050 inch) thick aluminum//.~~
- ~~2. When fascia exceeds 150 mm (6 inches) in depth, form one or more horizontal stops not less than 13 mm (1/2 inch) high in the fascia.~~
- ~~3. Fabricate as two piece fascia when fascia depth exceeds 250 mm (10 inches).~~
- ~~4. At joint between ends of sheets, provide a concealed clip soldered or welded near one end of each sheet to hold the adjoining sheet in lapped position. The clip shall be approximately 100 mm (4 inches) wide and shall be the full depth of the fascia less 25 mm (one inch)~~

~~at top and bottom. Clip shall be of the same thickness as the fascia.~~

- ~~5. Provide edge strip as specified with lower hooked edge bent outward at an angle of 45 degrees.~~

~~C. Formed (Corrugated Sheet) Sheet Metal Gravel Stops and Fascia:~~

- ~~1. Fabricate as shown of //0.4 mm (0.015 inch) thick stainless steel//
//0.5 Kg (16 ounce) copper// //0.8 mm (0.032 inch) thick aluminum//.~~
- ~~2. Sheets shall have 2 mm (1/16 inch) deep corrugations either transversely or diagonally rolled into the sheet. Crimped sheets are not acceptable.~~
- ~~3. Factory fabricate prepackaged system, complete with fastenings.~~
- ~~4. Provide concealed flashing splice plate at joints not less than 150 mm (6 inches) long and continuous edge strip at lower edge of fascia made from same metal.~~
- ~~5. Fabricate as two piece fascia when fascia depth exceeds 175 mm (7 inches).~~

2.10 BITUMEN STOPS

- ~~A. Fabricate bitumen stops for bituminous roofing edges for use with formed sheet metal gravel stops, pipe penetrations, and other penetrations through roof deck without a curb.~~
- ~~B. Fabricate with 19 mm (3/4 inch) vertical legs and 75 mm (3 inch) horizontal legs.~~
- ~~C. When used with gravel stop or metal base flashing use same metal for bitumen stop in thickness specified for concealed locations.~~

2.11 HANGING GUTTERS

~~SPEC WRITER NOTE: See Architectural Sheet Metal Manual for minimum thickness of sheet metal and specify. Detail gutter and show size on drawings.~~

- ~~A. Fabricate gutters of not less than the following:~~

- ~~1. //0.4// //0.5// //0.6//Kg (//16// //20// //24// oz) copper.~~
- ~~2. //0.5// //0.6// //0.8//mm (//0.018// //0.025// //0.031// inch) thick stainless steel.~~
- ~~3. //0.6// //0.8// //1.3//mm (//0.025// //0.032// //0.051//inch) thick aluminum.~~

- ~~B. Fabricate hanging gutters in sections not less than 2400 mm (8 feet) long, except at ends of runs where shorter lengths are required.~~

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~~SPEC WRITER NOTE: When gutter back is
against fascia specify higher back edge.~~

~~C. Building side of gutter shall be //not less than 38 mm (1 1/2 inches)
higher than exterior side// //same height as exterior side//.~~

~~D. Gutter Bead: Stiffen outer edge of gutter by folding edge over
approximately 19 mm (3/4 inch) toward roof and down approximately 19 mm
(3/4 inch) unless shown otherwise.~~

~~E. Gutter Spacers:~~

- ~~1. Fabricate of same material and thickness as gutter.~~
- ~~2. Fabricate 25 mm (one inch) wide strap and fasten to gutters not over
900 mm (36 inches) on center.~~
- ~~3. Turn back edge up 25 mm (one inch) and lap front edge over gutter
bead.~~
- ~~4. Rivet and solder to gutter except rivet and seal to aluminum.~~

~~F. Outlet Tubes:~~

- ~~1. Form outlet tubes to connect gutters to conductors of same metal and
thickness as gutters extend into the conductor 75 mm (3 inch).
Flange upper end of outlet tube 13 mm (1/2 inch).~~
- ~~2. Lock and solder longitudinal seam //except use sealant in lieu of
solder with aluminum//.~~
- ~~3. //Solder tube to gutter. //Seal aluminum tube to gutter and rivet to
gutter.~~
- ~~4. Fabricate basket strainers of same material as gutters.~~

~~G. Gutter Brackets:~~

~~SPEC WRITER NOTE: Do not use spike and
ferrule anchor systems. Use strap hangers
or gutter brackets. Design brackets for
gutter girth.~~

- ~~1. Fabricate of same metal as gutter. Use the following:~~
 - ~~a. //3 by 25 mm (1/8 by 1 inch)// //6 by 25 mm (1/4 by 1 inch)//
copper.~~
 - ~~b. //3 by 25 mm (1/8 by 1 inch)// //3 by 40 mm (1/8 by 1 1/2 inch)//
stainless steel.~~
 - ~~c. //5 by 25 mm (3/16 by 1 inch)// //6 by 25 mm (1/4 by 1 inch)//
aluminum.~~
- ~~2. Fabricate to gutter profile.~~

- ~~3. Drill two 5 mm (3/16 inch) diameter holes in anchor leg for countersunk flat head screws.~~

~~2.12 CONDUCTORS (DOWNSPOUTS)~~

- ~~A. Fabricate conductors of same metal and thickness as gutters in sections approximately 3000 mm (10 feet) long [with 19 mm (3/4 inch) wide flat locked seams].~~

~~SPEC WRITER NOTE: Use open faced downspouts in icing areas.~~

- ~~1. Fabricate open face channel shape with hemmed longitudinal edges.~~
- ~~B. Fabricate elbows by mitering, riveting, and soldering except seal aluminum in lieu of solder. Lap upper section to the inside of the lower piece.~~
- ~~C. Fabricate conductor brackets or hangers of same material as conductor, 2 mm (1/16 inch) thick by 25 mm (one inch) minimum width. Form to support conductors 25 mm (one inch) from wall surface in accordance with Architectural Sheet Metal Manual for rectangular and round shapes.~~
- ~~D. Conductor Heads:~~
 - ~~1. Fabricate of same material as conductor.~~
 - ~~2. Fabricate conductor heads to not less than 250 mm (10 inch) wide by 200 mm (8 inch) deep by 200 mm (8 inches) from front to back.~~
 - ~~3. Form front and side edges channel shape not less than 13 mm (1/2 inch) wide flanges with edge hemmed.~~
 - ~~4. Slope bottom to sleeve to conductor or downspout at not less than 60 degree angle.~~
 - ~~5. Extend wall edge not less than 25 mm (one inch) above front edge.~~
 - ~~6. Solder joints for water tight assembly.~~
 - ~~7. Fabricate outlet tube or sleeve at bottom not less than 50 mm (2 inches) long to insert into conductor.~~

~~2.13 SPLASHPANS~~

~~SPEC WRITER NOTE: Use splash pans where conductors discharge on built-up bituminous roofs.~~

- ~~A. Fabricate splashpans from the following:~~
 - ~~1. 0.4 Kg (16 oz) copper.~~
 - ~~2. 0.4 mm (0.015 inch) thick stainless steel.~~
 - ~~3. 1.25 mm (0.050 inch) thick aluminum.~~

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~~B. Fabricate in accordance with Architectural Sheet Metal Manual Plate 35 with not less than two ribs as shown in alternate section.~~

2-142.8 -REGLETS

~~SPEC WRITER NOTE: Verify reglets are detailed and location clearly shown.~~

A. Fabricate reglets of one of the following materials:

~~1. 0.4 Kg (16 ounce) copper.~~

~~2-1. Stainless steel, not less than 0.3 mm (0.012 inch) thick.~~

~~3-2. Plastic coated extruded aluminum, not less than 1.4 mm (0.055 inch) thick prefilled with butyl rubber sealer and complete with plastic wedges inserted at 1000 mm (40 inches) on centers.~~

~~4-3. Plastic, ASTM D1784, Type II, not less than 2 mm (0.075 inch) thick.~~

B. Fill open-type reglets with fiberboard or other suitable separator, to prevent crushing of the slot during installation.

C. Bend edges of reglets for setting into concrete to an angle of not less than 45 degrees, and make wide enough to provide firm anchorage in the concrete.

D. Fabricate reglets for building into horizontal masonry mortar joints not less than 19 mm (3/4 inch) deep, nor more than 25 mm (one inch) deep.

E. Fabricate mitered corners, fittings, and special shapes as may be required by details.

F. Reglets for concrete may be formed to receive flashing and have a 10 mm (3/8 inch), 45 degree snap lock.

2-152.9 INSULATED EXPANSION JOINT COVERS

A. ~~Either type optional, use only one type throughout. Refer to Section 07 95 13, EXPANSION JOINT COVER ASSEMBLIES.~~

~~B. Types:~~

~~1. Construct of two preformed, stainless steel strips, not less than 0.4 mm (0.015 inch) thick, mechanically and adhesively bonded to both sides of a 2 mm (1/16 inch) thick neoprene or butyl sheet, or to a 0.4 mm (32 mil) thick reinforced chlorinated polyethylene sheet. Adhesively attach a 10 mm (3/8 inch) thick sheet of closed cell, neoprene foam insulation, to the underside of the neoprene, butyl, or chlorinated polyethylene sheet.~~

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~~2. Constructed of a 2 mm (1/16 inch) thick vinyl sheet, flanged at both sides with stainless steel strips not less than 0.4 mm (0.015 inch) thick. Vinyl sheet locked and encased by the stainless steel strip and prepunched for nailing. A 10 mm (3/8 inch) thick closed cell polyvinyl chloride foam insulating strip shall be heat laminated to the underside of the vinyl sheet between the stainless steel strips.~~

~~C. Expansion joint covers shall have factory fabricated mitered corners, crossing tees, and other necessary accessories. Furnish in the longest available lengths.~~

~~D. Metal flange of sufficient width to extend over the top of the curb and down curb sides 50 mm (2 inches) with hemmed edge for lock to edge strip.~~

~~2.162.10~~ -ENGINE EXHAUST PIPE OR FLUE OR STACK FLASHING

A. Flashing at penetrations through roofing shall consist of a metal collar, sheet metal flashing sleeve and hood.

~~SPEC WRITER NOTE: Verify flashing and collar systems is detailed on drawings to comply with specification. No exceptions acceptable. Do not allow wood in contact with collar.~~

B. Fabricate collar with roof flange of 1.2 mm (0.047 inch) minimum thick black iron or galvanized steel sheet.

1. Fabricate inside diameter of collar 100 mm (4 inches) larger than the outside diameter of the item penetration the roofing.

2. Extend collar height from structural roof deck to not less than 350 mm (14 inches) above roof surface.

3. Fabricate collar roof flange not less than 100 mm (4 inches) wide.

4. Option: Collar may be of steel tubing 3 mm (0.125 inch) minimum wall thickness, with not less than four, 50 mm x 100 mm x 3 mm (2 inch by 4 inch by 0.125 inch) thick tabs bottom edge evenly spaced around tube in lieu of continuous roof flange. Full butt weld joints of collar.

C. Fabricate sleeve base flashing with roof flange of ~~either copper,~~ stainless steel, ~~or copper clad stainless steel.~~

1. Fabricate sleeve roof flange not less than 100 mm (4 inches) wide.

2. Extend sleeve around collar up to top of collar.

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3. Flange bottom of sleeve out not less than 13 mm (1/24 inch) and soldered to 100 mm (4 inch) wide flange to make watertight.
4. Fabricate interior diameter 50 mm (2 inch) greater than collar.
- D. Fabricate hood counter flashing from same material and thickness as sleeve.
 1. Fabricate the same as pipe counter flashing except allow not less than 100 mm (4 inch) lap below top of sleeve and to form vent space minimum of 100 mm (4 inch) wide.
 2. Hem bottom edge of hood 13 mm (1/2 inch).
 3. Provide a 50 mm (2 inch) deep drawband.
- E. Fabricate insect screen closure between sleeve and hood. Secure screen to sleeve with sheet metal screws.

2.172.11 SCUPPERS

~~SPEC WRITER NOTE: See SMACNA plates 26 thru 29 for scupper, Detail required on drawings.~~

- A. Fabricate scuppers with minimum of 100 mm (4 inch) wide flange.
- B. Provide flange at top on through wall scupper to extend to top of base flashing.
- C. Fabricate exterior wall side to project not less than 13 mm (1/2 inch) beyond face of wall with drip at bottom outlet edge.
- ~~D. Fabricate not less than 100 mm (4 inch) wide flange to lap behind gravel stop fascia.~~
- ~~E.D. _____~~ Fabricate exterior wall flange for through wall scupper not less than 25 mm (one inch) wide on top and sides with edges hemmed.
- ~~F. Fabricate gravel stop bar of 25 mm x 25 mm (one by one inch) angle strip soldered to bottom of scupper.~~
- ~~G.E. _____~~ Fabricate scupper not less than 200 mm (8 inch) wide and not less than 125 mm (5 inch) high for through wall scupper.
- ~~H.F. _____~~ Solder joints watertight.

2.182.12 GOOSENECK ROOF VENTILATORS

- A. Form of 1.3 mm (0.0508 inch) thick sheet aluminum, reinforce as necessary for rigidity, stiffness, and connection to curb, and to be watertight.
 1. Form lower-edge to sleeve to curb.
 2. Curb:
 - a. Form for 100 mm (4 inch) high sleeve to ventilator.

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- b. Form for concealed anchorage to structural curb and to bear on structural curb.
- c. Form bottom edge of curb as counterflashing to lap base flashing.
- B. Provide open end with 1.6 mm (16 gage), stainless steel wire guard of 13 mm (1/2 inch) square mesh.
 - 1. Construct suitable aluminum angle frame to retain wire guard.
 - 2. Rivet angle frame to end of gooseneck.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. General:
 - 1. Install flashing and sheet metal items as shown in Sheet Metal and Air Conditioning Contractors National Association, Inc., publication, ARCHITECTURAL SHEET METAL MANUAL, except as otherwise shown or specified.
 - 2. Apply Sealant as specified in Section 07 92 00, JOINT SEALANTS.
 - 3. Apply sheet metal and other flashing material to surfaces which are smooth, sound, clean, dry and free from defects that might affect the application.
 - 4. Remove projections which would puncture the materials and fill holes and depressions with material compatible with the substrate. Cover holes or cracks in wood wider than 6 mm (1/4 inch) with sheet metal compatible with the roofing and flashing material used.
 - 5. Coordinate with masonry work for the application of a skim coat of mortar to surfaces of unit masonry to receive flashing material before the application of flashing.
 - 6. Apply a layer of 7 Kg (15 pound) saturated felt followed by a layer of rosin paper to wood surfaces to be covered with copper. Lap each ply 50 mm (2 inch) with the slope and nail with large headed copper nails.
 - 7. Confine direct nailing of sheet metal to strips 300 mm (12 inch) or less wide. Nail flashing along one edge only. Space nail not over 100 mm (4 inches) on center unless specified otherwise.
 - 8. Install bolts, rivets, and screws where indicated, specified, or required in accordance with the SMACNA Sheet Metal Manual. Space rivets at 75 mm (3 inch) on centers in two rows in a staggered position. Use neoprene washers under fastener heads when fastener head is exposed.

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9. Coordinate with roofing work for the installation of metal base flashings and other metal items having roof flanges for anchorage and watertight installation.
10. Nail continuous cleats on 75 mm (3 inch) on centers in two rows in a staggered position.
11. Nail individual cleats with two nails and bend end tab over nail heads. Lock other end of cleat into hemmed edge.
12. Install flashings in conjunction with other trades so that flashings are inserted in other materials and joined together to provide a water tight installation.
13. Where required to prevent galvanic action between dissimilar metal isolate the contact areas of dissimilar metal with sheet lead, waterproof building paper, or a coat of bituminous paint.
14. Isolate aluminum in contact with dissimilar metals others than stainless steel, white bronze or other metal compatible with aluminum by:
 - a. Paint dissimilar metal with a prime coat of zinc-chromate or other suitable primer, followed by two coats of aluminum paint.
 - b. Paint dissimilar metal with a coat of bituminous paint.
 - c. Apply an approved caulking material between aluminum and dissimilar metal.
15. Paint aluminum in contact with or built into mortar, concrete, plaster, or other masonry materials with a coat of bituminous paint.
16. Paint aluminum in contact with absorptive materials that may become repeatedly wet with two coats of bituminous paint or two coats of aluminum paint.
17. Bitumen Stops:
 - a. Install bitumen stops for built-up roof opening penetrations through deck and at formed sheet metal gravel stops.
 - b. Nail leg of bitumen stop at 300 mm (12 inch) intervals to nailing strip at roof edge before roofing material is installed.

~~3.2 THROUGH-WALL FLASHING~~

~~A. General:~~

- ~~1. Install continuous through-wall flashing between top of concrete foundation walls and bottom of masonry building walls; at top of concrete floors; under masonry, concrete, or stone copings and elsewhere as shown.~~

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- ~~2. Where exposed portions are used as a counterflashings, lap base flashings at least 100 mm (4 inches) and use thickness of metal as specified for exposed locations.~~
- ~~3. Exposed edge of flashing may be formed as a receiver for two piece counter flashing as specified.~~
- ~~4. Terminate exterior edge beyond face of wall approximately 6 mm (1/4 inch) with drip edge where not part of counter flashing.~~
- ~~5. Turn back edge up 6 mm (1/4 inch) unless noted otherwise where flashing terminates in mortar joint or hollow masonry unit joint.~~
- ~~6. Terminate interior raised edge in masonry backup unit approximately 38 mm (1 1/2 inch) into unit unless shown otherwise.~~
- ~~7. Under copings terminate both edges beyond face of wall approximately 6 mm (1/4 inch) with drip edge.~~
- ~~8. Lap end joints at least two corrugations, but not less than 100 mm (4 inches). Seal laps with sealant.~~
- ~~9. Where dowels, reinforcing bars and fastening devices penetrate flashing, seal penetration with sealing compound. Sealing compound is specified in Section 07 92 00, JOINT SEALANTS.~~
- ~~10. Coordinate with other work to set in a bed of mortar above and below flashing so that total thickness of the two layers of mortar and flashing are same as regular mortar joint.~~
- ~~11. Where ends of flashing terminate turn ends up 25 mm (1 inch) and fold corners to form dam extending to wall face in vertical mortar or veneer joint.~~
- ~~12. Turn flashing up not less than 200 mm (8 inch) between masonry or behind exterior veneer.~~
- ~~13. When flashing terminates in reglet extend flashing full depth into reglet and secure with lead or plastic wedges spaced 150 mm (6 inch) on center.~~
- ~~14. Continue flashing around columns:~~
 - ~~a. Where flashing cannot be inserted in column reglet hold flashing vertical leg against column.~~
 - ~~b. Counterflash top edge with 75 mm (3 inch) wide strip of saturated cotton unless shown otherwise. Secure cotton strip with roof cement to column. Lap base flashing with cotton strip 38 mm (1 1/2 inch).~~

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~~SPEC WRITER NOTES: Details required of flashing.~~

- ~~1. Verify details show flashing at masonry faced concrete walls and termination of back edge.~~
- ~~2. Coordinate with waterproofing to define interface with metal flashing at joint.~~
- ~~3. Show metal flashing to have not less than 200 mm (8 inch) high vertical portion and termination against or in concrete backup or into masonry backup mortar joint.~~

~~B. Flashing at Top of Concrete Foundation Walls Where concrete is exposed. Turn up not less than 200 mm (8 inch) high and into masonry backup mortar joint or reglet in concrete backup as specified.~~

~~C. Flashing at Top of Concrete Floors (except where shelf angles occur): Place flashing in horizontal masonry joint not less than 200 mm (8 inch) below floor slab and extend into backup masonry joint at floor slab 38 mm (1 1/2 inch).~~

~~D. Flashing at Cavity Wall Construction: Where flashing occurs in cavity walls turn vertical portion up against backup under waterproofing, if any, into mortar joint. Turn up over insulation, if any, and horizontally through insulation into mortar joint.~~

~~E. Flashing at Veneer Walls:~~

- ~~1. Install near line of finish floors over shelf angles or where shown.~~
- ~~2. Turn up against sheathing.~~
- ~~3. At stud framing, hem top edge 19 mm (3/4 inch) and secure to each stud with stainless steel fasteners through sheathing.~~
- ~~4. At concrete backing, extend flashing into reglet as specified.~~
- ~~5. Coordinate with installation of waterproofing or asphalt felt for lap over top of flashing.~~

~~F. Lintel Flashing when not part of shelf angle flashing:~~

- ~~1. Install flashing full length of lintel to nearest vertical joint in masonry over veneer.~~
- ~~2. Turn ends up 25 mm (one inch) and fold corners to form dam and extend end to face of wall.~~
- ~~3. Turn back edge up to top of lintel; terminate back edge as specified for back up wall.~~

~~G. Window Sill Flashing:~~

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- ~~1. Install flashing to extend not less than 100 mm (4 inch) beyond ends of sill into vertical joint of masonry or veneer.~~
- ~~2. Turn back edge up to terminate under window frame.~~
- ~~3. Turn ends up 25 mm (one inch) and fold corners to form dam and extend to face of wall.~~

~~H. Door Sill Flashing:~~

- ~~1. Install flashing under bottom of plate sills of doors over curbs opening onto roofs. Extend flashing out to form counter flashing or receiver for counter flashing over base flashing. Set in sealant.~~
- ~~2. Extend sill flashing 200 mm (8 inch) beyond jamb opening. Turn ends up one inch in vertical masonry joint, extend end to face of wall. Join to counter flashing for water tight joint.~~
- ~~3. Where doors thresholds cover over waterproof membranes install sill flashing over water proof membrane under thresholds. Extend beyond opening to cover exposed portion of waterproof membrane and not less than 150 mm (6 inch) beyond door jamb opening at ends. Turn up approximately 6 mm (1/4 inch) under threshold.~~

~~I. Flashing at Masonry, Stone, or Precast Concrete Copings:~~

- ~~1. Install flashing with drips on both wall faces unless shown otherwise.~~
- ~~2. Form penetration openings to fit tight against dowel or other item with edge turned up. Seal penetrations with sealant.~~

3.33.2 -BASE FLASHING

~~SPEC WRITER NOTE: Limit use of metal base flashing to small penetrations, less than 3000 mm (10 feet) in length and pipes, where possible.~~

A. Install where roof membrane type base flashing is not used and where shown.

1. Install flashing at intersections of roofs with vertical surfaces or at penetrations through roofs, to provide watertight construction.
- ~~2. Install metal flashings and accessories having flanges extending out on top of the built-up roofing before final bituminous coat and roof aggregate is applied.~~
- ~~3. Set flanges in heavy trowel coat of roof cement and nail through flanges into wood nailers over bituminous roofing.~~

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4.2. Secure flange by nailing through roofing into wood blocking with nails spaced 75 mm (3 inch) on centers or, when flange over 100 mm (4 inch) wide terminate in a 13 mm (1/2 inch) folded edge anchored with cleats spaced 200 mm (8 inch) on center. Secure one end of cleat over nail heads. Lock other end into the seam.

- B. For long runs of base flashings install in lengths of not less than 2400 mm (8 feet) nor more than 3000 mm (ten feet). Install a 75 mm (3 inch) wide slip type, loose lock expansion joint filled with sealant in joints of base flashing sections over 2400 mm (8 feet) in length. Lock and solder corner joints at corners.
- C. Extend base flashing up under counter flashing of roof specialties and accessories or equipment not less than 75 mm (3 inch).

3.43.3 COUNTERFLASHING (CAP FLASHING OR HOODS)

~~SPEC WRITER NOTE: At steel pipes a standard screwed iron fitting for clamping counterflashing may be used in lieu of turning metal down into pipe. Check with PLUMBING specification.~~

A. General:

1. Install counterflashing over and in conjunction with installation of base flashings, except as otherwise specified or shown.
2. Install counterflashing to lap base flashings not less than 100 mm (4 inch).
3. Install upper edge or top of counterflashing not less than 225 mm (9 inch) above top of the roofing.
4. Lap joints not less than 100 mm (4 inch). Stagger joints with relation to metal base flashing joints.
5. Use surface applied counterflashing on existing surfaces and new work where not possible to integrate into item.
6. When fastening to concrete ~~or masonry~~, use screws driven in expansion shields set in concrete ~~or masonry~~. Use screws to ~~wood and~~ sheet metal. ~~Set fasteners in mortar joints of masonry work.~~

B. One Piece Counterflashing:

- ~~1. Where flashing is installed at new masonry, coordinate to insure proper height, embed in mortar, and end lap.~~

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2.1. Where flashing is installed in reglet in concrete insert upper edge into reglet. Hold flashing in place with lead wedges spaced not more than 200 mm (8 inch) apart. Fill joint with sealant.

3.2. Where flashing is surface mounted on flat surfaces.

a. When top edge is double folded anchor flat portion below sealant "V" joint with fasteners spaced not over 400 mm (16 inch) on center:

- 1) Locate fasteners in masonry mortar joints.
- 2) Use screws to sheet metal or wood.

b. Fill joint at top with sealant.

4.3. Where flashing or hood is mounted on pipe.

- a. Secure with draw band tight against pipe.
- b. Set hood and secure to pipe with a one by 25 mm x 3 mm (1 x 1/8 inch) bolt on stainless steel draw band type clamp, or a stainless worm gear type clamp.
- c. Completely fill joint at top with sealant.

C. Two-Piece Counterflashing:

~~1. Where receiver is installed at new masonry coordinate to insure proper height, embed in mortar, and lap.~~

2.1. Surface applied type receiver:

- a. Secure to face construction in accordance, with manufacturers' instructions.
- b. Completely fill space at the top edge of receiver with sealant.

3.2. Insert counter flashing in receiver in accordance with fabricator or manufacturer's instructions and to fit tight against base flashing.

D. Where vented edge occur install so lower edge of counterflashing is against base flashing.

E. When counter flashing is a component of other flashing install as shown.

3.53.4 REGLETS

A. Install reglets in a manner to provide a watertight installation.

B. Locate reglets not less than 225 mm (9 inch) nor more than 400 mm (16 inch) above roofing, and not less than 125 mm (5 inch) nor more than 325 mm (13 inch) above cant strip.

C. Butt and align end joints or each section of reglet and securely hold in position until concrete or mortar are hardened:

1. Coordinate reglets for anchorage into concrete with formwork construction.
- ~~2. Coordinate reglets for masonry to locate horizontally into mortar joints.~~

~~3.6 GRAVEL STOPS~~

~~SPEC WRITER NOTE: Install gravel stops on raised curb with tapered edge strip not less than 38 mm (1 1/2 inches) above roof line. Verify detail conforms using blocking.~~

~~A. General:~~

- ~~1. Install gravel stops and fascias with allowance for expansion at each joint; minimum of 6 mm (1/4 inch).~~
- ~~2. Extend roof flange of gravel stop and splice plates not less than four inches out over roofing and nail or screw to wood nailers. Space fasteners on 75 mm (3 inch) centers in staggered pattern.~~
- ~~3. Install continuous cleat for fascia drip edge. Secure with fasteners as close to lower edge as possible on 75 mm (3 inch) centers.~~
- ~~4. Where ends of gravel stops and fascias abut a vertical wall, provide a watertight, flashed and sealant filled joint.~~
- ~~5. Set flange in roof cement when installed over built-up roofing.~~
- ~~6. Edge securement for low-slope roofs: Low-slope membrane roof systems metal edge securement, except gutters, shall be designed in accordance with ANSI/SPRI/FM ES-1, except the basic wind speed shall be determined from Figure 1609, of IBC 2003.~~

~~B. Sheet metal gravel stops and fascia:~~

- ~~1. Install with end joints of splice plates sheets lapped three inches.~~
- ~~2. Hook the lower edge of fascia into a continuous edge strip.~~
- ~~3. Lock top section to bottom section for two-piece fascia.~~

~~C. Corrugated sheet gravel stops and fascia:~~

- ~~1. Install 300 mm (12 inch) wide sheet flashing centered under joint. A combination bottom and cover plate, extending above and beneath the joint, may be used.~~
- ~~2. Hook lower edge of fascia into a continuous edge strip.~~

~~D. Scuppers:~~

- ~~1. Install scupper with flange behind gravel stops; leave 6 mm (1/4 inch) joint to gravel stop.~~

- ~~2. Set scupper at roof water line and fasten to wood blocking.~~
- ~~3. Use sealant to seal joint with fascia gravel stops at ends.~~
- ~~4. Coordinate to lap over conductor head and to discharge water into conductor head.~~

~~3.73.5~~ COPINGS

A. General:

- ~~1. On walls topped with a wood plank, install a continuous edge strip on the front // and rear // edge of the plank. Lock the coping to the edge strip with a 19 mm (3/4 inch) loose lock seam.~~

~~2.1.~~ Where shown turn down roof side of coping and extend down over base flashing as specified for counter-flashing. Secure counter-flashing to lock strip in coping at continuous cleat.

~~3.2.~~ Install ends adjoining existing construction so as to form space for installation of sealants. Sealant is specified in Section 07 92 00, JOINT SEALANTS.

B. Aluminum Coping:

1. Install with 6 mm (1/4 inch) joint between ends of coping sections.
2. Install joint covers, centered at each joint, and securely lock in place.

~~SPEC WRITER NOTE: If prefinished galvanized steel copings are used change par C title and specify as if aluminum with sealant in lieu of solder.~~

C. ~~//Stainless Steel// //Copper//~~ Copings:

1. Join ends of sheets by a 19 mm (3/4 inch) locked and soldered seam, except at intervals of 9600 mm (32 feet), provide a 38 mm (1 1/2 inch) loose locked expansion joint filled with sealant or mastic.
2. At straight runs between 7200 mm (24 feet) and 19200 mm (64 feet) locate expansion joint at center.
3. At straight runs that exceed 9600 mm (32 feet) and form the leg of a corner locate the expansion joint not more than 4800 mm (16 feet) from the corner.

~~3.83.6~~ EXPANSION JOINT COVERS, INSULATED

- A. ~~Install insulated expansion joint covers at locations shown on curbs not less than 200 mm (8 inch) high above roof surface. Refer to Section 07 95 13, EXPANSION JOINT COVER ASSEMBLIES.~~

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- ~~B. Install continuous edge strips of same metal as expansion joint flange, nailed at not less than 75 mm (3 inch) centers.~~
- ~~C. Install insulated expansion joint covers in accordance with manufacturer's directions locking edges to edge strips.~~

~~3.93.7~~-ENGINE EXHAUST PIPE OR STACK FLASHING

- A. Set collar where shown and secure roof tabs or flange of collar to structural deck with 13 mm (1/2 inch) diameter bolts.
- B. Set flange of sleeve base flashing not less than 100 mm (4 inch) beyond collar on all sides as specified for base flashing.
- C. Install hood to above the top of the sleeve 50 mm (2 inch) and to extend from sleeve same distance as space between collar and sleeve beyond edge not sleeve:
 - 1. Install insect screen to fit between bottom edge of hood and side of sleeve.
 - 2. Set collar of hood in high temperature sealant and secure with one by 3 mm (1/8 inch) bolt on stainless steel draw band type, or stainless steel worm gear type clamp. Install sealant at top of head.

~~3.10 HANGING GUTTERS~~

- ~~A. Hang gutters with high points equidistant from downspouts. Slope at not less than 1:200 (1/16 inch per foot).~~
- ~~B. Lap joints, except for expansion joints, at least 25 mm (one inch) in the direction of flow. Rivet and seal or solder lapped joints.~~
- ~~C. Support gutters in brackets spaced not more than 600 mm (24 inch) on centers, brackets attached to facial or wood nailer by at least two screws or nails.~~
 - ~~1. For copper or copper clad stainless steel gutters use brass or bronze brackets.~~
 - ~~2. For stainless steel gutters use stainless steel brackets.~~
 - ~~3. For aluminum gutters use aluminum brackets or stainless steel brackets.~~
 - ~~4. Use brass or stainless steel screws.~~
- ~~D. Secure brackets to gutters in such a manner as to allow free movement of gutter due to expansion and contraction.~~

~~SPEC WRITER NOTE: Insure roof plan shows location of gutters, outlets or conductors, and expansion joints.~~

~~E. Gutter Expansion Joint:~~

- ~~1. Locate expansion joints midway between outlet tubes.~~
- ~~2. Provide at least a 25 mm (one inch) expansion joint space between end baffles of gutters.~~
- ~~3. Install a cover plate over the space at expansion joint.~~
- ~~4. Fasten cover plates to gutter section on one side of expansion joint only.~~
- ~~5. Secure loose end of cover plate to gutter section on other side of expansion joint by a loose-locked slip joint.~~

~~F. Outlet Tubes: Set bracket strainers loosely into gutter outlet tubes.~~

~~3.11 CONDUCTORS (DOWNSPOUTS)~~

- ~~A. Where scuppers discharge into downspouts install conductor head to receive discharge with back edge up behind drip edge of scupper. Fasten and seal joint. Sleeve conductors to gutter outlet tubes and fasten joint and joints between sections.~~
- ~~B. Set conductors plumb and clear of wall, and anchor to wall with two anchor straps, located near top and bottom of each section of conductor. Strap at top shall be fixed to downspout, intermediate straps and strap at bottom shall be slotted to allow not less than 13 mm (1/2 inch) movement for each 3000 mm (10 feet) of downspout.~~
- ~~C. Install elbows, offsets and shoes where shown and required. Slope not less than 45 degrees.~~

~~3.12 SPLASH PANS~~

- ~~A. Install where downspouts discharge on low slope roofs unless shown otherwise.~~
- ~~B. Set in roof cement prior to pour coat installation or sealant compatible with single ply roofing membrane.~~

~~3.13.8 GOOSENECK ROOF VENTILATORS~~

- ~~A. Install on structural curb not less than 200 mm (8 inch) high above roof surface.~~
- ~~B. Securely anchor ventilator curb to structural curb with fasteners spaced not over 300 mm (12 inch) on center.~~
- ~~C. Anchor gooseneck to curb with screws having neoprene washers at 150 mm (6 inch) on center.~~

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**SECTION 07 71 00
ROOF SPECIALTIES**

~~SPEC WRITER NOTES:~~

- ~~1. Delete between // // if not applicable to project. Also delete any other item or paragraph not applicable in the section and renumber the paragraphs.~~
- ~~2. Include standard manufactured components installed on and in roofing other than mechanical, electrical, and structural items.~~

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This section specifies copings, gravel stops, fascias, and expansion joints.

1.2 RELATED WORK

- A. ~~//~~Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.~~//~~
- ~~B. Section 07 21 13, THERMAL INSULATION: General Insulation.~~
- ~~C.~~B. Section 07 22 00, ROOF AND DECK INSULATION: Rigid Insulations for Roofing.
- ~~D.~~C. Section 07 92 00, JOINT SEALANTS: Sealant Material and Installation.
- ~~E. Section 09 06 00, SCHEDULE FOR FINISHES: Color and Texture of Finish~~

1.3 -QUALITY CONTROL

- A. Provide roof accessories that products of manufacturers regularly engaged in producing the kinds of products specified.
- B. For each accessory type provide products made by the same manufacturer.
- C. Assemble each accessory to the greatest extent possible before delivery to the site.
- D. Provide each accessory with FM approval listing for class specified.

1.4 PERFORMANCE REQUIREMENTS

- A. Provide roof accessories that withstand exposure to weather and resist thermal movement without failure, rattling, leaking, or fastener disengagement due to defective manufacture, fabrication, or installation.

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- B. Provide roof accessories listed in FM Approvals "RoofNav" and approved for windstorm classification Class 1-75~~1-75~~~~1-75~~. Identify materials with FM Approval markings.
- C. Manufacture and install roof accessories to allow for thermal movements from ambient and surface temperature changes to prevent buckling, opening of joints, hole elongation, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects.
 - 1. Provide clips that resist rotation and avoid shear stress as a result of thermal movements.
 - 2. For design purposes, base provisions for thermal movement on assumed ambient temperature (range) from minus 18 degrees C (0 degrees F), ambient to 82 degrees C (180 degrees F).

1.5 SUBMITTALS

- A. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.
- B. ~~1. Sustainable Design Submittals~~, as described below:
 - 1. ~~Postconsumer and preconsumer recycled content as specified in PART 2 - PRODUCTS.~~
- C. Samples: Representative sample panel of color-anodized aluminum not less than 101 x 101 mm (4 x 4 inches), except extrusions are to be of a width not less than section to be used. Submit sample that shows coating with integral color and texture. Include manufacturer's identifying label.
- D. Shop Drawings: Each item specified showing design, details of construction, installation and fastenings.
- E. Manufacturer's Literature and Data: Each item specified.
- F. Certificates: Stating that aluminum has been given specified thickness of anodizing.

1.6 APPLICABLE PUBLICATIONS:

- A. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.
- B. ASTM International (ASTM):
 - A240/A240M-20.....Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

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A653/A653M-20.....Steel Sheet Zinc-Coated (Galvanized) or
Zinc-Iron Alloy Coated (Galvannealed) by the
Hot Dip Process

A666-15.....Annealed or Cold-Worked Austenitic Stainless
Steel Sheet, Strip, Plate, and Flat Bar

B209-14.....Aluminum and Aluminum Alloy-Sheet and Plate

B209M-14.....Aluminum and Aluminum Alloy-Sheet and Plate
(Metric)

B221-14.....Aluminum-Alloy Extruded Bars, Rods, Wire,
Shapes, and Tubes

B221M-13.....Aluminum-Alloy Extruded Bars, Rods, Wire,
Shapes, and Tubes (Metric)

B32-08 (2014).....Solder Metal

B370-12 (2019).....Copper Sheet and Strip for Building
Construction

B882-10 (2018).....Pre-Patinated Copper for Architectural
Applications

C612-14 (2019).....Mineral Fiber Block and Board Thermal
Insulation

D1187/D1187M-97 (2018)...Asphalt-Base Emulsions for Use as Protective
Coatings for Metal

D1970/D1970M-20.....Self-Adhering Polymer Modified Bituminous Sheet
Materials Used as Steep Roofing Underlayment
for Ice Dam Protection

D226/D226M-17.....Asphalt-Saturated Organic Felt Used in Roofing
and Waterproofing

D4869/D4969M-16a.....Asphalt-Saturated Organic Felt Underlayment
Used In Steep Slope Roofing

C. National Association of Architectural Metal Manufacturers (NAAMM):
AMP 500-06.....Metal Finishes Manual

D. American Architectural Manufacturers Association (AAMA):
2605-11.....High Performance Organic Coatings on
Architectural Extrusions and Panels.
611-14.....Anodized Architectural Aluminum

E. FM Global (FM):
RoofNav.....Approved Roofing Assemblies and Products

PART 2 - PRODUCTS

~~SPEC WRITER NOTE: Update materials requirements to agree with applicable requirements (types, grades, classes) and balance of roofing system materials specified in the referenced Applicable Publications.~~

2.1 MATERIALS

- A. Aluminum, Extruded: ASTM B221M (B221).
- B. Aluminum Sheet: ASTM B209M (B209).
- C. Galvanized Sheet Steel: ASTM A653/A653M; G-90 coating.
- D. Stainless-Steel Sheet: ASTM A240/A240M or ASTM A666, Type 304.
- ~~E. Copper Sheet: ASTM B370, cold-rolled copper sheet, H00 or H01 temper.~~
- ~~F.E.~~ //Recycled Content of Metal Products: Postconsumer recycled content plus one-half of preconsumer recycled content not less than ~~//30~~ ~~///~~ ~~percent.~~ ~~///~~
- ~~G.F.~~ Insulation: ASTM C612, Class 1 or 2.
- ~~H.G.~~ Asphalt Coating: ASTM D1187, Type I, quick setting.

2.2 UNDERLAYMENT

- A. Self-Adhering Modified Bitumen Underlayment:
 - 1. Provide self-adhering modified bitumen membrane underlayment material in compliance with ASTM D1970/D1970M, suitable for use as underlayment for metal copings and fascias.
 - 2. Provide membrane resistant to cyclical elevated temperatures for extended period of time in high heat service conditions (stable after testing at 116 degrees C (240 degrees F)).
 - 3. Provide membrane with integral non-tacking top surface of polyethylene film or other surface material to serve as separator between bituminous material and metal products to be applied above.
 - 4. Provide primer.
- ~~B. Felt Underlayment: Provide No. 30 asphalt saturated organic, non-perforated felt underlayment in compliance with ASTM D226/D226M, Type II, or ASTM D4869/D4869M.~~
- ~~G.B.~~ Slip Sheet: Provide 0.24 kg per square meter (5 pounds per 100 sf) rosin sized unsaturated building paper for slip sheet.

~~2.3~~ ~~//SOLDER~~

- ~~A. Copper Solder conforming to ASTM B32, //lead-free solder//
//Grade Sn50, 50 percent tin and 50 percent lead//.~~

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~~SPEC WRITER NOTE:~~

~~1. Detail coping system; select thickness of metal for wall thickness span. Select anchoring system to conform to details.~~

2.42.3 COPINGS

- A. Fabricate of aluminum sheet not less than ~~// 1.6 // // 2 //~~
~~// 3.2 // mm (// 0.063 // // 0.08 // // 0.125 // inch) thick; 16 oz.~~
~~copper // 0.5 mm (0.018 inch) 0.812 mm (0.032 inch) thick.; stainless~~
~~steel//~~
- B. Turn outer edges down each face of wall as shown on construction documents.
- C. Maximum lengths of 3.05 M (10 feet).
- D. Shop fabricate external and internal corners as one-piece assemblies with not less than 305 mm (12 inch) leg lengths.
- E. Provide 101 mm (4 inch) wide 0.81 mm (0.032 inch) thick watertight joint covers.
- F. Provide anchor gutter bar of 0.81 mm (0.032 inch) thick with anchor holes formed for underside of joint.
- G. Provide concealed guttered splice plate of 0.81 mm (0.032 inch) thick with butyl or other resilient seal strips anchored to splice plate for underside of joint. Use galvanized steel anchor plate providing compression spring anchoring of coping cover.
- H. Finish: ~~// Two-coat fluoropolymer // // Three-coat fluoropolymer; //~~
~~// Two-coat mica fluoropolymer // // Three-coat metallic~~
~~fluoropolymer // // Clear anodic // // Color anodic; //~~ ~~Color //~~ ~~//~~
as specified.

~~SPEC WRITER NOTE: Detail gravel stop and fascias. If roof is ballasted extend vertical curb to be above paver/gravel surface. Where protective membrane insulation or 1-90 design occurs use system that clamps insulation or pavers down. Provide scuppers at water line for overflow protection when stop exceeds 25 mm (1 inch) in height above roof line.~~

2.5 EXTRUDED ALUMINUM GRAVEL STOPS AND FASCIAS

- ~~A. Fabricate of aluminum not less than 2 mm (0.078 inch) thick.~~

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- ~~B. Turn fascia down face of wall and up above roof as shown in construction documents.~~
- ~~C. Maximum lengths of 3.05 M (10 feet).~~
- ~~D. Shop fabricate external and internal corners as one (1) piece assemblies with not less than 305 mm (12 inch) leg lengths.~~
- ~~E. Provide 101 mm (4 inch) wide 2 mm (0.078 inch) thick watertight joint covers with 152 mm (6 inch) wide 0.8 mm (0.030 inch) thick underside joint flashing.~~
- ~~F. Finish: // Two-coat fluoropolymer // // Three-coat fluoropolymer // // Two-coat mica fluoropolymer // // Three-coat metallic fluoropolymer // // Clear anodic // // Color anodic // Color // // as specified.~~

~~SPEC WRITER NOTE: Use not less than 101 mm (4 inch) high cant above roof.~~

~~2.6 EXTRUDED ALUMINUM FASCIA-CANT SYSTEM~~

- ~~A. The fascia cant system consists of three (3) pieces, an extruded aluminum fascia, a galvanized steel cant, and an aluminum compression clamp.~~
- ~~B. Furnish in stock lengths of not more than 3.05 M (10 feet) long.~~
- ~~C. Form fascia from not less than 2 mm (0.070 inch) thick aluminum. Provide 101 mm (4 inch) wide 0.81 mm (0.032 inch) thick concealed sheet aluminum joint cover plates in back of fascia.~~
- ~~D. Form cant strip from galvanized steel not less than 0.75 mm (0.0299 inch) thick, to profile shown and design to hold lower edge of the fascia.~~
- ~~E. Form compression clamp of not less than 0.81 mm (0.032 inch) thick aluminum designed to hold the top edge of the fascia and the built-up flashing.~~
- ~~F. Internal and external corners:~~
 - ~~1. Factory fabricate and fully weld mitered joints.~~
 - ~~2. Furnish corner sections in // manufacturers standard sizes // // sizes shown // with not less than 305 mm (12 inch) leg lengths.~~

~~SPEC WRITER NOTE: Use sump assembly when downspouts are used to drain roof not having roof drains.~~

- ~~G. Factory fabricated fascia sump assemblies.~~

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- ~~1. Fabricate sump assemblies with stainless steel cores and extruded aluminum cover to match fascia cant.~~
- ~~2. Provide stainless steel outlet, tube sized to suit downspout and solder to core to make watertight.~~
- ~~3. Furnish sump assembly in 508 mm (20 inch) minimum lengths.~~

~~SPEC WRITER NOTE: Use scuppers for overflow safety if only one roof drain occurs.~~

~~H. Factory fabricated scupper assemblies:~~

- ~~1. Fabricate scupper assembly with extended plates to match fascia cant in 508 mm (20 inch) minimum lengths.~~
- ~~2. Extend outlet opening not less than 50 mm (2 inches) with drip edge.~~
- ~~3. Fabricate with stainless steel core or sleeve to drain water from toe of cant and flash in to built up roofing with 101 mm (4 inch) wide flange.~~

~~I. Finish on aluminum: // Two coat fluoropolymer // // Three coat fluoropolymer // // Two coat mica fluoropolymer // // Three coat metallic fluoropolymer // // Clear anodic // // Color anodic // // Color // // as specified.~~

~~SPEC WRITER NOTE: Insure expansion joints are installed on curbs providing not less than 203 mm (8 inch) high base flashing. Detail and show system; allow option for use of both system.~~

2.72.4 EXTRUDED ALUMINUM ROOF EXPANSION JOINT COVERS

- ~~A. Fabricate in 3.0 M (10 foot) lengths with fastener openings slotting for expansion not over 610 mm (24 inch) centers. Refer to Section 07 95 13 EXPANSION JOINT COVER ASSEMBLIES.~~
- ~~B. Provide four way expansion, for joint widths shown on construction documents.~~
- ~~C. Mill finish.~~
- ~~D. Form waterstop or moisture seals of continuous sheets of neoprene, not less than 0.81 mm (0.032 inch) thick.~~
- ~~E. Fabricate corners as one (1) piece assembly with mitered and welded joint and least dimension legs not less than 300 mm (12 inches) long.~~
- ~~F. Factory fabricate end caps and transitions to insure waterproof assembly.~~

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~~G. Five (5) piece assembly:~~

- ~~1. Roof expansion joint cover system consists of an extruded aluminum cover, extruded frame or curb vertical section, galvanized steel cant, and aluminum compression clamp counter flashing, complete with moisture seals. Form cover and vertical section from extruded aluminum, 2 mm (0.080 inch) minimum thickness with spring stainless steel tension or pivot bar.~~
- ~~2. Form cant from galvanized steel not less than 0.8 mm (0.029 inch) thick formed to profile shown on construction documents.~~
- ~~3. Form splice plates of not less than 0.81 mm (0.032 inch) thick aluminum sheet.~~
- ~~4. Form counter flashing member of 1.3 mm (0.050 inch) thick sheet aluminum, secured with screws to the top edge of the vertical section and providing compression clamp over base flashing.~~
- ~~5. Provide compression gasket separating cover from curb bearing.~~

~~H. Two (2) piece assembly:~~

- ~~1. Roof expansion joint system consists of an extruded aluminum cover combination extruded aluminum frame or curb with integral adjustable counter flashing flange, and moisture seals.~~
- ~~2. Form cover from extruded aluminum 2 mm (0.078 inch) minimum thickness.~~
- ~~3. Form cover anchor system of stainless steel pivot bar.~~
- ~~4. Form frame assembly of not less than 2 mm (0.076 inch) aluminum except for flashing portion.~~
- ~~5. Provide compression gasket separating cover from curb at bearing.~~

~~SPEC WRITER NOTE: If more than one finish is used on project, proceed finish spec with "Finish for (list items);" Use mill finish for aluminum unless specified otherwise under item.~~

2.92.5 FINISH:

- A. In accordance with NAAMM AMP 500-505.
- B. ~~//~~Aluminum, Mill Finish: AA-MIX, as fabricated.~~//~~
- C. ~~//~~Aluminum, Clear Anodic Finish AAMA 611: ~~//~~AA-M12C22A41, Class I, 0.017 mm (0.7 mil) thick (min.). ~~//~~~~//~~AA12C22A31 Class II, Architectural, 0.010 mm (0.4 mil) thick (min.).~~//~~

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D. ~~//~~Aluminum Color Anodic Finish AAMA 611: AA-C22A42 (anodized or AA0C22A44 (electrolytically deposited metallic compound), ~~//~~Class 1, Architectural, 0.017 mm (0.7 mil) thick (min.)~~//~~. ~~//~~Class II, Architectural, 0.010 mm (0.4 mil) thick (min.). ~~//~~Dyes will not be accepted.~~//~~

~~E. //Copper Sheet Finishes: // Non-Patinated Finish: Mill finish // // Pre-Patinated Finish: Chemically treated according to ASTM B882 // //~~

~~F.E.~~ ~~//~~Fluoropolymer Finishes: High performance organic coating.

Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

1. ~~//~~Two-Coat Fluoropolymer: AAMA 2605. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat.~~//~~
2. ~~//~~Three-Coat Fluoropolymer: AAMA 2605. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in both color coat and top color coat.~~//~~
3. ~~//3.~~Two-Coat Mica Fluoropolymer: AAMA 2605. Fluoropolymer finish with suspended mica flakes containing not less than 70 percent PVDF resin by weight in color coat.~~//~~
4. ~~//4.~~Three-Coat Metallic Fluoropolymer: AAMA 2605. Fluoropolymer finish with suspended metallic flakes containing not less than 70 percent PVDF resin by weight in both color coat and clear topcoat.~~//~~
5. ~~//5.~~Concealed Surface Finish: Apply pretreatment and manufacturer's standard acrylic or polyester backer finish consisting of prime coat and wash coat with a minimum total dry film thickness of 0.013 mm (0.5 mil).~~// //~~

~~G.F.~~ ~~//~~Stainless-Steel Finish: ~~//~~No. 2B (bright, cold rolled, unpolished;~~// //~~No. 3 (coarse, polished directional satin;~~// //~~No. 4 (bright, polished directional satin~~//~~.~~//~~

~~SPEC WRITER NOTE: Show anchorage location for items specified on drawings.~~

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Examine substrates, areas, and conditions, to verify actual locations, dimensions, and other conditions affecting performance of the Work.
- B. Verify that substrate is sound, dry, smooth, clean, sloped for drainage where applicable, and securely anchored.

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SPEC WRITER NOTE: Edit underlayment types
required for project/details.

C. Underlayment Installation:

1. ~~//~~Self-Adhering Sheet Underlayment:

- a. Apply primer as required by manufacturer.
- b. Comply with temperature restrictions of underlayment manufacturer for installation.
- c. Apply wrinkle free, in shingle fashion to shed water, and with end laps of not less than 152 mm (6 inches) staggered 610 mm (24 inches) between courses.
- d. Overlap side edges not less than 89 mm (3-1/2 inches). Roll laps with roller.
- e. Cover underlayment within 14 days.
- f. Apply continuously under copings and roof-edge fascias and gravel stops.
- g. Coordinate application of self-adhering sheet underlayment under roof specialties with requirements for continuity with adjacent air barrier materials.~~//~~

~~2. //Felt Underlayment:-~~

- ~~a. Install with adhesive for temporary anchorage to minimize use of mechanical fasteners under roof specialties.~~
- ~~b. Apply in shingle fashion to shed water, with lapped joints of not less than 50 mm (2 inches). //~~

~~3.2.~~ Slip Sheet:

- a. Install with tape or adhesive for temporary anchorage to minimize use of mechanical fasteners under roof specialties.
- b. Apply in shingle fashion to shed water, with lapped joints of not less than 50 mm (2 inches).

D. Install roof accessories where indicated in construction documents.

E. Secure with fasteners in accordance with manufacture's printed installation instructions and approved shop drawings unless shown otherwise. Provide fasteners suitable for application, for metal types being secured and designed to meet performance requirements.

F. Where soldered joints are required, clean surfaces to be soldered, removing oils and foreign matter.

1. Pre-tin edges of sheets to be soldered to a width of 38 mm (1-1/2 inches).

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2. Reduce pre-tinning where pre-tinned surface would show in completed work.

~~3. //Tin edges of uncoated copper sheets using solder for copper.//~~

~~4.3.~~ Do not use torches for soldering.

~~5.4.~~ Heat surfaces to receive solder and flow solder into joint.

~~6.5.~~ Fill joint completely.

~~7.6.~~ Completely remove flux and spatter from exposed surfaces.

G. Coordinate to install insulation where shown; see ~~Section 07 21 13,~~
~~THERMAL INSULATION~~ and Section 07 22 00, ROOF AND DECK INSULATION.

H. Comply with section 07 92 00, JOINT SEALANTS to install sealants where required by manufactures installation instructions.

I. Coordinate with roofing work for installation of items in sequence to prevent water infiltration.

~~J. Gravel Stops and Fascias:~~

~~1. Install gravel stops and fascia with butt joints with approximately 6 mm (1/4 inch) space for expansion.~~

~~2. Over each joint provide cover plates of sheet aluminum, complete with concealed sheet aluminum flashing, centered under each joint.~~

~~3. Provide lap cover plates and concealed flashing over the gravel stop and fascia not less than 101 mm (4 inches).~~

~~4. Extend concealed flashing over built-up roofing, embed in roof cement and turn down over face of blocking at roof edge.~~

~~K.J.~~ Aluminum Coping:

1. Install sections of coping with approximately 6 mm (1/4-inch) space between ends of sections.

2. Center joint gutter bar and covers at joints and lock in place.

3. When snap-on system is installed ensure front and back edges are locked in place.

~~L. Fascia Cant System:~~

~~1. Install galvanized steel cant; coordinate with roofing work and after completion of roofing work install extruded aluminum fascia, concealed joint cover plate, and aluminum compression clamp, where shown in construction documents.~~

~~2. Install system to allow for expansion and contraction with 6 mm (1/4 inch) space between extruded aluminum members and galvanized steel cant as required by manufacturer of system.~~

~~3. Offset joints in extruded aluminum members from galvanized steel cant joints.~~

M-K. Expansion Joint Covers:

1. Install to terminate base flashing 203 mm (8 inches) above roof.
2. Install moisture seals to drain water to outlets that do not permit water to enter building.
3. Provide stainless steel screws when exposed.
4. Three piece assembly:
 - a. Install curb section with screws to wood blocking, allowing 6 mm (1/4 inch) at butt joints between sections with splice plate at joint.
 - b. Install cant to wood blocking by nailing along horizontal flange every 152 mm (6 inches), with galvanized roofing nails 25 mm (1 inch) long.
 - c. After completion of base flashing install cap flashing and compression clamp and fasten to the curb or metal cant with stainless steel self-tapping screws with neoprene washers under head spaced approximately 457 mm (18 inches) on center.
 - d. Install expansion joint cover with a 6 mm (1/4 inch) wide end joints.
 - e. Install over end joint a cover plate complete with concealed aluminum flashing, centered under each joint. Fabricate flashing to lap cover not less than 101 mm (4 inches).
5. Two piece assembly:
 - a. Install curb section with screws allowing 6 mm (1/4 inch) space at end joints with splice plate at joint.
 - b. After completion of base flashing bend down cap flashing flange and secure to blocking with screws.
 - c. Install expansion joint cover with 6 mm (1/4 inch) wide space at end joints and tension bars at 610 mm (24 inches) on center.
 - d. Install cover plates with formed aluminum flashing concealed and centered on joint. Flashing to lap cover not less than 101 mm (4 inches).

3.2 PROTECTION OF ALUMINUM

- A. Provide protection for aluminum against galvanic action wherever dissimilar materials are in contact, by painting the contact surfaces of the dissimilar material with two (2) coats of asphalt coating

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(complete coverage), or by separating the contact surfaces with a preformed neoprene tape having pressure sensitive adhesive coating on one (1) side.

- B. Paint aluminum in contact with wood, concrete and masonry, or other absorptive materials, that may become repeatedly wet, with two (2) coats of asphalt coating.

3.3 ADJUSTING

- A. Adjust expansion joints to close tightly and be watertight; insuring maximum allowance for building movement.

3.4 PROTECTION

- A. Protect roof accessories from damage during installation and after completion of the work from subsequent construction.

- - - E N D - - -

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**SECTION 07 72 00
ROOF ACCESSORIES**

~~SPEC WRITER NOTES:~~

~~Delete between // // if not applicable to project. Also delete any other item or paragraph not applicable in the section and renumber the paragraphs.~~
~~Include standard manufactured components installed on and in roofing other than mechanical, electrical, and structural items.~~

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This section specifies ~~roof hatches;~~ equipment supports; ~~gravity ventilators;~~ and metal grating roof walkway system.

1.2 RELATED WORK

- A. ~~//~~Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.~~//~~
- ~~B. Section 07 21 13, THERMAL INSULATION: General insulation.~~
- ~~C.~~B. Section 07 22 00, ROOF AND DECK INSULATION: Rigid insulations for roofing.
- ~~D.~~C. Section 07 92 00, JOINT SEALANTS: Sealant material and installation.
- ~~E. Section 09 06 00, SCHEDULE FOR FINISHES: Color and texture of finish.~~

1.3 -QUALITY ASSURANCE

- A. Provide roof accessories that are the products of manufacturers regularly engaged in producing the kinds of products specified.
- B. For each accessory type provide the same product made by the same manufacturer.
- C. Assemble each accessory to the greatest extent possible before delivery to the site.

1.4 SUBMITTALS

- A. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.
- B. ~~//~~Sustainable Design Submittals, as described below:
1. ~~//~~Postconsumer and preconsumer recycled content as specified in PART 2 - PRODUCTS.~~//~~
- C. Samples: Submit representative sample panel of color anodized aluminum not less than 101 x 101 mm (4 x 4 inches). For extrusions, submit width

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not less than section to be installed. Show coating with integral color and texture and include manufacturer's identifying label.

- D. Shop Drawings: Each item specified showing design, details of construction, installation and fastenings.
- E. Manufacturer's Literature and Data: Each item specified.
- F. Certificates: Stating that aluminum has been given specified thickness of anodizing.

1.5 APPLICABLE PUBLICATIONS

- A. The publications listed below form a part of this specification to the extend referenced. The publications are referenced in the text by the basic designation only.
- B. ASTM International (ASTM):
 - A653/A653M-20.....Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) By the Hot-Dip Process
 - B209-14.....Aluminum and Aluminum-Alloy Sheet and Plate
 - B209M-14.....Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
 - B221-14.....Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes, and Tubes
 - B221M-13.....Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes, and Tubes (Metric)
 - C726-17.....Mineral Wool Roof Insulation Board
 - C1289-19.....Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board
 - D1187/D1187M-97(2018)...Asphalt-Base Emulsions for Use as Protective Coatings for Metal
- C. National Association of Architectural Metal Manufacturers (NAAMM):
 - AMP 500-06 Series.....Metal Finishes Manual
- D. American Architectural Manufacturers Association (AAMA):
 - 2603-20.....Performance Requirements and Test Procedures for Pigmented Organic Coatings on Aluminum Extrusions and Panels (with Coil Coating Appendix).
 - 2605-20.....Performance Requirements and Test Procedures for Superior Performing Organic Coatings on

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Architectural Extrusions and Panels (with Coil
Coating Appendix).

611-14.....Anodized Architectural Aluminum

621-02.....High Performance Organic Coatings on Coil

Coated Architectural Hot Dipped Galvanized
(HDG) and Zinc-Aluminum Coated Steel Substrates

E. American Society of Civil Engineers (ASCE):

ASCE/SEI 7-16.....Minimum Design Loads and Associated Criteria
for Buildings and Other Structures

F. U.S. Occupational Safety and Health Standards (OSHA):

29 CFR 1910 Subpart D... Walking-Working Surfaces (1910.21-1910.30)

PART 2 - PRODUCTS

~~SPEC WRITER NOTE: Update materials
requirements to agree with applicable
requirements (types, grades, classes)
specified in the referenced Applicable
Publications.~~

2.1 MATERIALS

A. Aluminum, Extruded: ASTM B221M (B221).

B. Aluminum Sheet: ASTM B209M (B209).

C. Galvanized Sheet Steel: ASTM A653/A653M; G-90 coating.

D. ~~//~~ Recycled Content of Metal Products: Postconsumer recycled content
plus one-half of preconsumer recycled content not less than ~~//~~ 30 ~~//~~
~~//~~ percent. ~~//~~

E. Asphalt Coating: ASTM D1187/D1187M, Type I, quick setting.

~~2.2 ROOF HATCH (SCUTTLE)~~

~~A. Performance Characteristics:~~

- ~~1. Cover to be reinforced to support a minimum live load of
195 kilogram per square meter (40 pounds per square foot) with a
maximum deflection of 1/150th of the span or 97 kilogram per square
meter (20 pounds per square foot) wind uplift.~~
- ~~2. Operation of the Cover: Smooth and easy with controlled operation
throughout the entire arc of opening and closing.~~
- ~~3. Operation of the Cover: Not affected by temperature.~~
- ~~4. Entire Hatch: Weathertight with fully welded corner joints on cover
and curb.~~

~~B. Shop fabricate from aluminum with mill finish.~~

~~C. Curb and Cover:~~

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- ~~1. Exterior facing: Minimum 2.3 mm (0.09 inch) thick sheet aluminum with mill finish.~~
- ~~2. Interior facing: Minimum 1 mm (0.04 inch) thick sheet aluminum.~~
- ~~3. Minimum of 50 mm (2 inch) thick polyisocyanurate insulation (ASTM C1289) with a U value = 0.47 W/mK (R value = 12) between facings of cover and over exterior face of curb.~~
- ~~4. Form exterior curb facing with an integral 76 mm (3 inch) wide roof flange and cap flashing minimum 2.3 mm (0.09 inch) thick sheet aluminum.~~

~~SPEC WRITER NOTE: Do not use less than 305 mm (12 inch high) curb above roof surface. Where access is to service roof equipment comply with OSHA for stair size, roof opening 750 mm x 2400 mm (2 feet, 6 inches x 8 feet). Show opening size on the drawings. Coordinate/insert height requirements and finish.~~

- ~~5. Make curb // 305 mm (12 inches) // // // above finish roof surface.~~
- ~~6. Form cover to lap curb and cap flashing.~~
- ~~7. Size opening as shown on construction documents.~~
- ~~8. Finish: // //, color // //.~~

~~D. Hardware:~~

- ~~1. Provide spring snap latch with inside and outside operating handles and padlock hasp on inside. Provide two snap latches when hinge side is over 2100 mm (7 feet) long. Bolt hardware into heavy gauge channel reinforcement welded to the underside of the cover and concealed within the insulation space.~~
- ~~2. Provide heavy duty pintle hinges.~~
- ~~3. Provide automatic hold open and operating arm with enclosed torsion or compression spring lifting mechanism.~~
- ~~4. Latch Strike: Stamped component bolted or welded to the curb assembly.~~
- ~~5. Automatically lock in the open position at not less than 70 degrees.~~
- ~~6. Provide weather stripping at cover closure.~~
- ~~7. Galvanize all hardware items.~~

~~E. Assembly:~~

- ~~1. Shop assemble roof scuttle.~~

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~~2. Weld joints exposed to the weather and built into the roofing.~~

~~3. Finish weld smooth where exposed.~~

~~F. Safety Accessories:~~

~~1. Ladder Assist Post: Provide a telescoping tubular section that locks automatically when fully extended. Control upward and downward movement by a stainless steel spring balancing mechanism. Provide unit completely assembled with fasteners for securing to the ladder rungs in accordance with the manufacturer's instructions.~~

~~2. Safety Railing: Provide a fixed, attached to the roof hatch railing assembly including rails, clamps, fasteners, safety barrier at railing opening, and accessories required for a complete installation; complying with 29 CFR 1910.23 requirements.~~

~~SPEC WRITER NOTE: Use following article for roof mounted equipment items other than mechanical equipment items. Prefabricated roof curbs for fans, ventilators and other roof mounted mechanical items are specified in Mechanical Specifications. Coordinate/insert load requirements and finish/colors.~~

2.32.2 EQUIPMENT SUPPORTS

A. Supported Load Capacity: existing equipment, verify in field~~//~~~~//~~.

B. Fabricate equipment supports from 1.3 mm (0.0516 inch) thick galvanized ASTM A653/A653M steel fabricate with welded corners and with seams joined by continuous water and air tight welds.

C. Equipment supports to be internally reinforced with angles 1.22 m (48 inches) on center.

D. Form exterior curb with integral base, ~~//~~ and deck closures for curbs installed on steel decking.~~//~~

E. Use galvanized steel liners for curbs having inside dimension over 305 mm (12 inches).

F. Internally insulate with 38 mm (1-1/2 inch) glass-fiber board insulation (ASTM C726).

G. Fabricate curb with a minimum height of 203 mm (8 inches) above roof surface.

H. Attach preservative treated wood nailers to top of curb. Provide 50 mm (2 inch) by 50 mm (2 inch) minimum nominal size on curb with openings

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and 50 mm (2 inch) thick, width of curb up to 305 mm (12 inches) on equipment support curbs.

- I. Make size of supports suit size of equipment furnished, with height as shown on construction documents, but not less than 203 mm (8 inches) above roof surface.
- J. Top of Equipment Supports: Level with pitch built into curb when deck slopes. Equip supports with water diverter or cricket on side that obstructs water flow.

~~K. Finish: // //; color // //.~~

~~SPEC WRITER NOTE: Use following article for gravity type ventilators not connected to ducts. All ventilators, mechanical and gravity, that are connected to ducts are specified in Mechanical Specifications. Coordinate finish/color.~~

~~2.4 LOW SILHOUETTE GRAVITY VENTILATORS~~

~~A. Fabricate base of 1 mm (0.04 inch) thick aluminum, and vent of 0.8 mm (0.032 inch) thick aluminum.~~

- ~~1. Height not to exceed 305 mm (12 inches) above top of roof curb.~~
- ~~2. Design ventilators to withstand 137 Km (85 miles) per hour wind velocity.~~
- ~~3. Provide ventilators with a removable 18 by 18 mesh by 0.28 mm (0.11 inch) diameter aluminum wire cloth insect screen.~~
- ~~4. //Provide security grille where indicated on construction documents. //~~

~~B. Construct damper of the same material as the ventilator and design to completely close opening or remain wide open. Hold damper in closed position by a brass chain and catch. Extend chains 305 mm (12 inches) below and engage catch when damper is closed.~~

~~C. Finish: // //; color // //.~~

~~2.5 METAL GRATING ROOF WALKWAY SYSTEM~~

~~A. Provide metal grating roof walkway system consisting of prefabricated pans, of 14 gauge, galvanized (G-90 Coating) steel grating with slip resistant surface.~~

~~B. Grating units to be in 610 mm (2 foot) widths and in 3048 to 3658 mm (10 to 12 foot long) sections as required.~~

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- ~~C. Provide complete with support framing, brackets, connectors, nosings and other accessories as required for complete roof walkway system.~~
- ~~1. Include support stands at minimum 1524 mm (5 feet) on center to hold planks a minimum of 228 mm (9 inches) above roof surface.~~
 - ~~2. Provide wind restraint attachment to roof structure of size and spacing required to meet wind uplift requirements.~~
- ~~D. Include step units, nosings framing and connectors to provide changes in elevation as required. Comply with ASCE 7 and 29 CFR 1910.23.~~
- ~~E. Equip walkways with safety railings where required by 29 CFR 1910.23.~~
- ~~F. Provide neoprene rubber pads having a shore A hardness of 80 to 90 Durometer under each support, or bearing surface.~~

~~G.K. Finish: // //; color // //.~~

~~SPEC WRITER NOTE: Edit/select finishes used. Use AAMA 621 for zinc coated steel sheet use AAMA 2605 for aluminum sheet.~~

~~2.6 FINISH:~~

- ~~A. In accordance with NAAMM AMP 500 Series.~~
- ~~B. //Aluminum, Mill Finish: AA-MIX, as fabricated.//~~
- ~~C. //Aluminum, Clear Finish AAMA 611: AA-M12C22A41 medium matte, clear anodic coating, // Class I, Architectural, 0.018 mm (0.7 mils) thick (min.) //. // AA-M12C22A31 Class II, Architectural, 0.010 mm (0.4 mils) thick (min.). // //~~
- ~~D. //Aluminum Colored Finish AAMA 611: AA-C22A42 (anodized or AA-M12C22A44 (electrolytically deposited metallic compound) medium matte, integrally colored coating, // Class 1, Architectural, 0.018 mm (0.7 mils) thick (min.)//. // AA-M12C22A32/A33 Class II, Architectural, 0.010 mm (0.4 mils) thick (min.). // Dyes will not be accepted.//~~
- ~~E. //Baked Enamel or Powder Coat Finish: AAMA 2603 except with a minimum dry film thickness of 0.04 mm (1.5 mils). Comply with coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish. //~~
- ~~F. //Fluoropolymer Finish: High performance organic coating. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturer's written instructions.~~
- ~~1. //Two-Coat Fluoropolymer Finish: AAMA 2605. System consisting of primer and fluoropolymer color topcoat containing not less than 70 percent PVDF resin by weight. //~~

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~~2. //Two Coat Fluoropolymer Finish: AAMA 621. System consisting of primer and fluoropolymer color topecoat containing not less than 70 percent PVDF resin by weight.// //~~

~~SPEC WRITER NOTE: Show anchorage location for items specified on drawings.~~

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install roof specialties where indicated on construction documents.
- B. Secure with fasteners in accordance with manufacture's printed installation instructions and approved shop drawings unless shown otherwise.
- C. Coordinate to install insulation where shown; see ~~Section 07 21 13, THERMAL INSULATION~~ and Section 07 22 00, ROOF AND DECK INSULATION.
- D. Comply with section 07 92 00, JOINT SEALANTS to install sealants where required by manufactures installation instructions require sealant.
- E. Coordinate with roofing work for installation of items in sequence to prevent water infiltration.
 - 1. After completion of base flashing bend down cap flashing flange and secure to blocking with screws.
 - 2. Install expansion joint cover with 6 mm (1/4 inch) wide space at end joints and tension bars at 610 mm (24 inches) on center.
 - 3. Install cover plates with formed aluminum flashing concealed and centered on joint. Flashing to lap cover not less than 101 mm (4 inches).
- F. Equipment Supports: Do not anchor to insulating concrete or metal deck. Anchor only to building structure as per manufacturers recommendations.

3.2 PROTECTION OF ALUMINUM

- A. Provide protection for aluminum against galvanic action wherever dissimilar materials are in contact, by painting the contact surfaces of the dissimilar material with two (2) coats of asphalt coating (complete coverage), or by separating the contact surfaces with a preformed neoprene tape having pressure sensitive adhesive coating on side.
- B. Paint aluminum in contact with wood, concrete and masonry, or other absorptive materials, that may become repeatedly wet, with two coats of asphalt coating.

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~~3.3 ADJUSTING~~

- ~~A. Adjust roof hatch hardware to operate freely and so that cover will operate without binding, close tightly at perimeter, and latch securely.~~

~~3.43.3 PROTECTION~~

- A. Protect roof accessories from damage during installation and after completion of the work from subsequent construction.

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SECTION 07 84 00
FIRESTOPPING

~~SPEC WRITER NOTE: Delete between //---//
if not applicable to project. Also delete
any other item or paragraph not
applicable in the section and renumber
the paragraphs.~~

PART 1 - GENERAL

1.1 DESCRIPTION

A. Provide UL or equivalent approved firestopping system for the closures of openings in walls, floors, and roof decks against penetration of flame, heat, and smoke or gases in fire resistant rated construction.

~~B. Provide UL or equivalent approved firestopping system for the closure of openings in walls against penetration of gases or smoke in smoke partitions.~~

1.2 RELATED WORK

A. ~~//~~Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.~~//~~

B. Section 07 95 13, EXPANSION JOINT COVER ASSEMBLIES: Expansion and seismic joint firestopping.

~~C. Section 07 81 00, APPLIED FIREPROOFING: Spray applied fireproofing.~~

~~D-C.~~ Section 07 92 00, JOINT SEALANTS: Sealants and application.

~~E. Section 23 31 00, HVAC DUCTS AND CASINGS: Fire and smoke damper assemblies in ductwork.~~

~~F. Section 23 37 00, AIR OUTLETS AND INLETS: Fire and smoke damper assemblies in ductwork.~~

1.3 SUBMITTALS

A. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

B. ~~//~~Sustainable Design Submittals, as described below:

1. ~~//~~Volatile organic compounds per volume as specified in

PART 2 - PRODUCTS.~~//~~

C. Installer qualifications.

D. Inspector qualifications.

E. Manufacturers literature, data, and installation instructions for types of firestopping and smoke stopping used.

F. List of FM, UL, or WH classification number of systems installed.

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- G. Certified laboratory test reports for ASTM E814 tests for systems not listed by FM, UL, or WH proposed for use.
- H. Submit certificates from manufacturer attesting that firestopping materials comply with the specified requirements.

1.4 DELIVERY AND STORAGE

- A. Deliver materials in their original unopened containers with manufacturer's name and product identification.
- B. Store in a location providing protection from damage and exposure to the elements.

1.5 QUALITY ASSURANCE

- A. FM, UL, or WH or other approved laboratory tested products will be acceptable.
- B. Installer Qualifications: A firm that has been approved by FM Global according to FM Global 4991 or been evaluated by UL and found to comply with UL's "Qualified Firestop Contractor Program Requirements." Submit qualification data.
- C. Inspector Qualifications: Contractor to engage a qualified inspector to perform inspections and final reports. The inspector to meet the criteria contained in ASTM E699 for agencies involved in quality assurance and to have a minimum of two years' experience in construction field inspections of firestopping systems, products, and assemblies. The inspector to be completely independent of, and divested from, the Contractor, the installer, the manufacturer, and the supplier of material or item being inspected. Submit inspector qualifications.

1.6 APPLICABLE PUBLICATIONS

- A. Publications listed below form a part of this specification to the extent referenced. Publications are referenced in the text by the basic designation only.
- B. ASTM International (ASTM):
 - E84-20.....Surface Burning Characteristics of Building Materials
 - E699-16.....Standard Specification for Agencies Involved in Testing, Quality Assurance, and Evaluating of Manufactured Building Components
 - E814-13a(2017).....Fire Tests of Penetration Firestop Systems
 - E2174-20a.....Standard Practice for On-Site Inspection of Installed Firestop Systems

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E2393-20.....Standard Practice for On-Site Inspection of
Installed Fire Resistive Joint Systems and
Perimeter Fire Barriers

C. FM Global (FM):

Annual Issue Approval Guide Building Materials

4991-13.....Approval of Firestop Contractors

D. Underwriters Laboratories, Inc. (UL):

Annual Issue Building Materials Directory

E. Annual Issue Fire Resistance Directory

723-Edition 11(2018)....Standard for Test for Surface Burning
Characteristics of Building Materials

1479-04(2015).....Fire Tests of Penetration Firestops

F. Intertek Testing Services - Warnock Hersey (ITS-WH):

Annual Issue Certification Listings

G. Environmental Protection Agency (EPA):

40 CFR 59(2014).....National Volatile Organic Compound Emission
Standards for Consumer and Commercial Products

PART 2 - PRODUCTS

2.1 FIRESTOP SYSTEMS

- A. Provide either factory built (Firestop Devices) or field erected (through-Penetration Firestop Systems) to form a specific building system maintaining required integrity of the fire barrier and stop the passage of gases or smoke. Firestop systems to accommodate building movements without impairing their integrity.
- B. Through-penetration firestop systems and firestop devices tested in accordance with ASTM E814 or UL 1479 using the "F" or "T" rating to maintain the same rating and integrity as the fire barrier being sealed. "T" ratings are not required for penetrations smaller than or equal to 101 mm (4 inches) nominal pipe or 0.01 square meter (16 square inches) in overall cross sectional area.
- C. Firestop sealants used for firestopping or smoke sealing to have the following properties:
 - 1. Contain no flammable or toxic solvents.
 - 2. Release no dangerous or flammable out gassing during the drying or curing of products.
 - 3. Water-resistant after drying or curing and unaffected by high humidity, condensation or transient water exposure.

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4. When installed in exposed areas, capable of being sanded and finished with similar surface treatments as used on the surrounding wall or floor surface.
5. ~~++~~VOC Content: Firestopping sealants and sealant primers to comply with the following limits for VOC content when calculated according to 40 CFR 59, (EPA Method 24):
 - a. Sealants: 250 g/L.
 - b. Sealant Primers for Nonporous Substrates: 250 g/L.
 - c. Sealant Primers for Porous Substrates: 775 g/L. ~~++~~
- D. Firestopping system or devices used for penetrations by glass pipe, plastic pipe or conduits, unenclosed cables, or other non-metallic materials to have following properties:
 1. Classified for use with the particular type of penetrating material used.
 2. Penetrations containing loose electrical cables, computer data cables, and communications cables protected using firestopping systems that allow unrestricted cable changes without damage to the seal.
- E. Maximum flame spread of 25 and smoke development of 50 when tested in accordance with ASTM E84 or UL 723. Material to be an approved firestopping material as listed in UL Fire Resistance Directory or by a nationally recognized testing laboratory.
- F. FM, UL, or WH rated or tested by an approved laboratory in accordance with ASTM E814.
- G. Materials to be nontoxic and noncarcinogen at all stages of application or during fire conditions and to not contain hazardous chemicals. Provide firestop material that is free from Ethylene Glycol, PCB, MEK, and asbestos.
- H. For firestopping exposed to view, traffic, moisture, and physical damage, provide products that do not deteriorate when exposed to these conditions.
 1. For piping penetrations for plumbing and wet-pipe sprinkler systems, provide moisture-resistant through-penetration firestop systems.
 2. For floor penetrations with annular spaces exceeding 101 mm (4 inches) or more in width and exposed to possible loading and traffic, provide firestop systems capable of supporting the floor

loads involved either by installing floor plates or by other means acceptable to the firestop manufacturer.

3. For penetrations involving insulated piping, provide through-penetration firestop systems not requiring removal of insulation.

~~2.2 SMOKE STOPPING IN SMOKE PARTITIONS~~

- ~~A. Provide silicone sealant in smoke partitions as specified in Section 07 92 00, JOINT SEALANTS.~~
- ~~B. Provide mineral fiber filler and bond breaker behind sealant.~~
- ~~C. Sealants to have a maximum flame spread of 25 and smoke developed of 50 when tested in accordance with ASTM E84.~~
- ~~D. When used in exposed areas capable of being sanded and finished with similar surface treatments as used on the surrounding wall or floor surface.~~

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Submit product data and installation instructions, as required by article, submittals, after an on-site examination of areas to receive firestopping.
- B. Examine substrates and conditions with installer present for compliance with requirements for opening configuration, penetrating items, substrates, and other conditions affecting performance of firestopping. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Remove dirt, grease, oil, laitance and form-release agents from concrete, loose materials, or other substances that prevent adherence and bonding or application of the firestopping or smoke stopping materials.
- B. Remove insulation on insulated pipe for a distance of 150 mm (6 inches) on each side of the fire rated assembly prior to applying the firestopping materials unless the firestopping materials are tested and approved for use on insulated pipes.
- C. Prime substrates where required by joint firestopping system manufacturer using that manufacturer's recommended products and methods. Confine primers to areas of bond; do not allow spillage and migration onto exposed surfaces.

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- D. Masking Tape: Apply masking tape to prevent firestopping from contacting adjoining surfaces that will remain exposed upon completion of work and that would otherwise be permanently stained or damaged by such contact or by cleaning methods used to remove smears from firestopping materials. Remove tape as soon as it is possible to do so without disturbing seal of firestopping with substrates.

3.3 INSTALLATION

- A. Do not begin firestopping work until the specified material data and installation instructions of the proposed firestopping systems have been submitted and approved.
- B. Install firestopping systems with smoke stopping in accordance with FM, UL, WH, or other approved system details and installation instructions.

~~C. Install smoke stopping seals in smoke partitions.~~

3.4 -CLEAN-UP

- A. As work on each floor is completed, remove materials, litter, and debris.
- B. Clean up spills of liquid type materials.
- C. Clean off excess fill materials and sealants adjacent to openings and joints as work progresses by methods and with cleaning materials approved by manufacturers of firestopping products and of products in which opening and joints occur.
- D. Protect firestopping during and after curing period from contact with contaminating substances or from damage resulting from construction operations or other causes so that they are without deterioration or damage at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, cut out and remove damaged or deteriorated firestopping immediately and install new materials to provide firestopping complying with specified requirements.

3.5 INSPECTIONS AND ACCEPTANCE OF WORK

- A. Do not conceal or enclose firestop assemblies until inspection is complete and approved by the Contracting Officer Representative (COR).
- B. Furnish service of approved inspector to inspect firestopping in accordance with ASTM E2393 and ASTM E2174 for firestop inspection, and document inspection results. Submit written reports indicating locations of and types of penetrations and type of firestopping used at each location; type is to be recorded by UL listed printed numbers.

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SECTION 07 92 00
JOINT SEALANTS

~~SPEC WRITER NOTE: Delete between //---//
if not applicable to project. Also delete
any other item or paragraph not
applicable in the section and renumber
the paragraphs.~~

PART 1 - GENERAL

1.1 DESCRIPTION:

- A. This section covers interior and exterior sealant and their application, wherever required for complete installation of building materials or systems.

1.2 RELATED WORK (INCLUDING BUT NOT LIMITED TO THE FOLLOWING):

- ~~//A. Sustainable Design Requirements: Section 01 81 13, SUSTAINABLE
CONSTRUCTION REQUIREMENTS.//~~
- ~~//B. Sealing of Site Work Concrete Paving: Section 32 05 23, CEMENT AND
CONCRETE FOR EXTERIOR IMPROVEMENTS. //~~
- ~~//C. Masonry Control and Expansion Joint: Section 04 20 00, UNIT
MASONRY. //~~
- ~~//D. Firestopping Penetrations: Section 07 84 00, FIRESTOPPING. //~~
- ~~//E. Glazing: Section 08 80 00, GLAZING. //~~
- ~~//F. Glazed Aluminum Curtain Wall: Section 08 44 13, GLAZED ALUMINUM
CURTAIN WALLS. //~~
- ~~//G. Sound Rated Gypsum Partitions/Sound Sealants: Section 09 29 00,
GYPSUM BOARD. //~~
- ~~//H. Mechanical Work: // Section 22 05 11 COMMON WORK RESULTS FOR
PLUMBING//Section 23 05 10 Common Work Results for Boiler Plant and
Steam Generation// Section 23 05 11, COMMON WORK RESULTS FOR HVAC//.~~

1.3 QUALITY ASSURANCE:

- A. Installer Qualifications: An experienced installer with a minimum of three (3) years' experience and who has specialized in installing joint sealants similar in material, design, and extent to those indicated for this Project and whose work has resulted in joint-sealant installations with a record of successful in-service performance. Submit qualification.
- B. Source Limitations: Obtain each type of joint sealant through one (1) source from a single manufacturer.

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~~SPEC WRITER NOTE: Review and edit testing requirements to suit complexity of project.~~

C. Product Testing: Obtain test results from a qualified testing agency based on testing current sealant formulations within a 12-month period.

1. Testing Agency Qualifications: An independent testing agency qualified according to ASTM C1021.
2. Test elastomeric joint sealants for compliance with requirements specified by reference to ASTM C920, and where applicable, to other standard test methods.

3. Test elastomeric joint sealants according to SWRI's Sealant Validation Program for compliance with requirements specified by reference to ASTM C920 for adhesion and cohesion under cyclic movement, adhesion-in peel, and indentation hardness.

4. Test other joint sealants for compliance with requirements indicated by referencing standard specifications and test methods.

D. Lab Tests: Submit samples of materials that will be in contact or affect joint sealants to joint sealant manufacturers for tests as follows:

1. Adhesion Testing: Before installing elastomeric sealants, test their adhesion to protect joint substrates according to the method in ASTM C794 to determine if primer or other specific joint preparation techniques are required.
2. Compatibility Testing: Before installing elastomeric sealants, determine compatibility when in contact with glazing and gasket materials.
3. Stain Testing: Perform testing per ASTM C1248 on interior and exterior sealants to determine if sealants or primers will stain adjacent surfaces. No sealant work is to start until results of these tests have been submitted to the Contracting Officer Representative (COR) and the COR has given written approval to proceed with the work.

~~//E. Preconstruction Field Adhesion Testing: Before installing elastomeric sealants, field test their adhesion to joint substrates according to Method A, Field Applied Sealant Joint Hand Pull Tab, in Appendix X1.1 in ASTM C1193 or Method A, Tail Procedure, in ASTM C1521.//~~

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~~//1. Locate test joints where indicated in construction documents or, if not indicated, as directed by COR. //~~

~~2. Conduct field tests for each application indicated below:~~

~~a. Each type of elastomeric sealant and joint substrate indicated.~~

~~b. Each type of non elastomeric sealant and joint substrate indicated.~~

~~3. Notify COR seven (7) days in advance of dates and times when test joints will be erected. //~~

~~//4. Arrange for tests to take place with joint sealant manufacturer's technical representative present. //~~

~~//F. Mockups: Before installing joint sealants, apply elastomeric sealants as follows to verify selections and to demonstrate aesthetic effects and qualities of materials and execution: //~~

~~//1. Joints in mockups of assemblies that are indicated to receive elastomeric joint sealants. //~~

1.4 CERTIFICATION:

A. Contractor is to submit to the COR written certification that joints are of the proper size and design, that the materials supplied are compatible with adjacent materials and backing, that the materials will properly perform to provide permanent watertight, airtight or vapor tight seals (as applicable), and that materials supplied meet specified performance requirements.

1.5 SUBMITTALS:

A. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

~~//B. Sustainable Design Submittals, as described below:~~

~~//1. Volatile organic compounds per volume as specified in PART 2 - PRODUCTS.//~~

C. Installer qualifications.

D. Contractor certification.

E. Manufacturer's installation instructions for each product used.

F. Cured samples of exposed sealants for each color.

G. Manufacturer's Literature and Data:

1. Primers

2. Sealing compound, each type, including compatibility when different sealants are in contact with each other.

H. Manufacturer warranty.

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1.6 PROJECT CONDITIONS:

A. Environmental Limitations:

1. Do not proceed with installation of joint sealants under following conditions:
 - a. When ambient and substrate temperature conditions are outside limits permitted by joint sealant manufacturer or are below 4.4 degrees C (40 degrees F).
 - b. When joint substrates are wet.

B. Joint-Width Conditions:

1. Do not proceed with installation of joint sealants where joint widths are less than those allowed by joint sealant manufacturer for applications indicated.

C. Joint-Substrate Conditions:

1. Do not proceed with installation of joint sealants until contaminants capable of interfering with adhesion are removed from joint substrates.

1.7 DELIVERY, HANDLING, AND STORAGE:

- A. Deliver materials in manufacturers' original unopened containers, with brand names, date of manufacture, shelf life, and material designation clearly marked thereon.
- B. Carefully handle and store to prevent inclusion of foreign materials.
- C. Do not subject to sustained temperatures exceeding 32 degrees C (90 degrees F) or less than 5 degrees C (40 degrees F).

1.8 DEFINITIONS:

- A. Definitions of terms in accordance with ASTM C717 and as specified.
- B. Backing Rod: A type of sealant backing.
- C. Bond Breakers: A type of sealant backing.
- D. Filler: A sealant backing used behind a back-up rod.

1.9 WARRANTY:

- A. Construction Warranty: Comply with FAR clause 52.246-21 "Warranty of Construction".
- B. Manufacturer Warranty: Manufacturer shall warranty their sealant for a minimum of five (5) ~~++~~ years from the date of installation and final acceptance by the Government. Submit manufacturer warranty.

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1.10 APPLICABLE PUBLICATIONS:

- A. Publications listed below form a part of this specification to extent referenced. Publications are referenced in text by basic designation only.
- B. ASTM International (ASTM):
- C509-06.....Elastomeric Cellular Preformed Gasket and
Sealing Material
 - C612-14.....Mineral Fiber Block and Board Thermal
Insulation
 - C717-14a.....Standard Terminology of Building Seals and
Sealants
 - C734-06 (R2012).....Test Method for Low-Temperature Flexibility of
Latex Sealants after Artificial Weathering
 - C794-10.....Test Method for Adhesion-in-Peel of Elastomeric
Joint Sealants
 - C919-12.....Use of Sealants in Acoustical Applications.
 - C920-14a.....Elastomeric Joint Sealants.
 - C1021-08 (R2014).....Laboratories Engaged in Testing of Building
Sealants
 - C1193-13.....Standard Guide for Use of Joint Sealants.
 - C1248-08 (R2012).....Test Method for Staining of Porous Substrate by
Joint Sealants
 - C1330-02 (R2013).....Cylindrical Sealant Backing for Use with Cold
Liquid Applied Sealants
 - C1521-13.....Standard Practice for Evaluating Adhesion of
Installed Weatherproofing Sealant Joints
 - D217-10.....Test Methods for Cone Penetration of
Lubricating Grease
 - D1056-14.....Specification for Flexible Cellular Materials—
Sponge or Expanded Rubber
 - E84-09.....Surface Burning Characteristics of Building
Materials
- C. Sealant, Waterproofing and Restoration Institute (SWRI).
The Professionals' Guide
- D. Environmental Protection Agency (EPA):
- 40 CFR 59 (2014).....National Volatile Organic Compound Emission
Standards for Consumer and Commercial Products

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PART 2 - PRODUCTS

~~SPEC WRITER NOTES:~~

- ~~1. Make material requirements agree with applicable requirements specified in referenced Applicable Publications.~~
- ~~2. Update and specify only that which applies to project. Coordinate sealant selection and sealant material edited for local conditions and where sealant is to be applied.~~
- ~~3. For joints with anticipated movement in excess of Class 25 rating input sealant class required.~~
- ~~4. Coordinate sealant types and designations (S-#'s) with drawings.~~
- ~~5. Coordinate color requirements of sealants.~~

2.1 SEALANTS:

A. Exterior Sealants:

- ~~//1. //S-#//~~ Vertical surfaces, provide non-staining ASTM C920, Type S or M, Grade NS, Class 25, ~~//---, //~~ Use NT. ~~//~~
- ~~//2. //S-#//~~ Horizontal surfaces, provide ASTM C920, Type S or M, Grade P, Class 25, Use T. ~~//~~
3. Provide location(s) of exterior sealant as follows:
 - a. Joints formed where frames and subsills of windows, doors, louvers, and vents adjoin masonry, concrete, or metal frames. Provide sealant at exterior surfaces of exterior wall penetrations.
 - b. Metal to metal.
 - c. Masonry to masonry or stone.
 - d. Stone to stone.
 - e. Cast stone to cast stone.
 - f. Masonry expansion and control joints.
 - g. Wood to masonry.
 - h. Masonry joints where shelf angles occur.
 - i. Voids where items penetrate exterior walls.
 - j. Metal reglets, where flashing is inserted into masonry joints, and where flashing is penetrated by coping dowels.

B. Floor Joint Sealant:

1. ASTM C920, Type S or M, Grade P, Class 25, ~~//---, //~~ Use T. ~~//S-#//~~
2. ~~//S-#//~~ Provide location(s) of floor joint sealant as follows.

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- a. Seats of metal thresholds exterior doors.
- b. Control and expansion joints in floors, slabs, ceramic tile, and walkways.

~~C. Interior Sealants:~~

- ~~//1. VOC Content of Interior Sealants: Sealants and sealant primers used inside the weatherproofing system are to comply with the following limits for VOC content when calculated according to 40 CFR 59, (EPA Method 24):~~
 - ~~a. Architectural Sealants: 250 g/L.~~
 - ~~b. Sealant Primers for Nonporous Substrates: 250 g/L.~~
 - ~~c. Sealant Primers for Porous Substrates: 775 g/L.//~~
- ~~2. //S-#// Vertical and Horizontal Surfaces: ASTM C920, Type S or M, Grade NS, Class 25, //—, // Use NT.~~
- ~~3. //S-#// Food Service: Use a Vinyl Acetate Homopolymer, or other low VOC, non-toxic sealant approved for use in food preparation areas.~~
- ~~4. Provide location(s) of interior sealant as follows:~~
 - ~~a. Typical narrow joint 6 mm, (1/4 inch) or less at walls and adjacent components.~~
 - ~~b. Perimeter of doors, windows, access panels which adjoin concrete or masonry surfaces.~~
 - ~~c. Interior surfaces of exterior wall penetrations.~~
 - ~~d. Joints at masonry walls and columns, piers, concrete walls or exterior walls.~~
 - ~~e. Perimeter of lead faced control windows and plaster or gypsum wallboard walls.~~
 - ~~f. Exposed isolation joints at top of full height walls.~~
 - ~~g. Joints between bathtubs and ceramic tile; joints between shower receptors and ceramic tile; joints formed where nonplanar tile surfaces meet.~~
 - ~~h. Joints formed between tile floors and tile base cove; joints between tile and dissimilar materials; joints occurring where substrates change.~~
 - ~~i. Behind escutcheon plates at valve pipe penetrations and showerheads in showers.~~

~~D. Acoustical Sealant:~~

- ~~1. Conforming to ASTM C919; flame spread of 25 or less; and a smoke developed rating of 50 or less when tested in accordance with~~

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~~ASTM E84. Acoustical sealant have a consistency of 250 to 310 when tested in accordance with ASTM D217; remain flexible and adhesive after 500 hours of accelerated weathering as specified in ASTM C734; and be non-staining.~~

~~2. Provide location(s) of acoustical sealant as follows:~~

- ~~a. Exposed acoustical joint at sound rated partitions.~~
- ~~b. Concealed acoustic joints at sound rated partitions.~~
- ~~c. Joints where item pass through sound rated partitions.~~

2.2 COLOR:

- A. Sealants used with exposed masonry are to match color of mortar joints.
- B. Sealants used with unpainted concrete are to match color of adjacent concrete.
- C. Color of sealants for other locations to be light gray or aluminum, unless otherwise indicated in construction documents.

2.3 JOINT SEALANT BACKING:

- A. General: Provide sealant backings of material and type that are nonstaining; are compatible with joint substrates, sealants, primers, and other joint fillers; and are approved for applications indicated by sealant manufacturer based on field experience and laboratory testing.
- B. Cylindrical Sealant Backings: ASTM C1330, of type indicated below and of size and density to control sealant depth and otherwise contribute to producing optimum sealant performance:
 - 1. Type C: Closed-cell material with a surface skin.
- C. Elastomeric Tubing Sealant Backings: Neoprene, butyl, EPDM, or silicone tubing complying with ASTM D1056 or synthetic rubber (ASTM C509), nonabsorbent to water and gas, and capable of remaining resilient at temperatures down to minus 32 degrees C (minus 26 degrees F). Provide products with low compression set and of size and shape to provide a secondary seal, to control sealant depth, and otherwise contribute to optimum sealant performance.
- D. Bond-Breaker Tape: Polyethylene tape or other plastic tape recommended by sealant manufacturer for preventing sealant from adhering to rigid, inflexible joint-filler materials or joint surfaces at back of joint where such adhesion would result in sealant failure. Provide self-adhesive tape where applicable.

~~//2.4 WEEPS:~~

~~A. Weep/Vent Products: Provide the following unless otherwise indicated or approved.~~

~~1. Round Plastic Tubing: Medium density polyethylene, 10 mm (3/8 inch) OD by thickness of stone or masonry veneer. //~~

2.5 FILLER:

- A. Mineral fiberboard: ASTM C612, Class 1.
- B. Thickness same as joint width.
- C. Depth to fill void completely behind back-up rod.

2.6 PRIMER:

- A. As recommended by manufacturer of caulking or sealant material.
- B. Stain free type.

2.7 CLEANERS-NON POROUS SURFACES:

- A. Chemical cleaners compatible with sealant and acceptable to manufacturer of sealants and sealant backing material. Cleaners to be free of oily residues and other substances capable of staining or harming joint substrates and adjacent non-porous surfaces and formulated to promote adhesion of sealant and substrates.

PART 3 - EXECUTION

3.1 INSPECTION:

- A. Inspect substrate surface for bond breaker contamination and unsound materials at adherent faces of sealant.
- B. Coordinate for repair and resolution of unsound substrate materials.
- C. Inspect for uniform joint widths and that dimensions are within tolerance established by sealant manufacturer.

3.2 PREPARATIONS:

- A. Prepare joints in accordance with manufacturer's instructions and SWRI (The Professionals' Guide).
- B. Clean surfaces of joint to receive caulking or sealants leaving joint dry to the touch, free from frost, moisture, grease, oil, wax, lacquer paint, or other foreign matter that would tend to destroy or impair adhesion.
 - 1. Clean porous joint substrate surfaces by brushing, grinding, blast cleaning, mechanical abrading, or a combination of these methods to produce a clean, sound substrate capable of developing optimum bond with joint sealants.

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2. Remove loose particles remaining from above cleaning operations by vacuuming or blowing out joints with oil-free compressed air. Porous joint surfaces include but are not limited to the following:
 - a. Concrete.
 - b. Masonry.
 - c. Unglazed surfaces of ceramic tile.
3. Remove laitance and form-release agents from concrete.
4. Clean nonporous surfaces with chemical cleaners or other means that do not stain, harm substrates, or leave residues capable of interfering with adhesion of joint sealants. Nonporous surfaces include but are not limited to the following:
 - a. Metal.
 - b. Glass.
 - c. Porcelain enamel.
 - d. Glazed surfaces of ceramic tile.
- C. Do not cut or damage joint edges.
- D. Apply non-staining masking tape to face of surfaces adjacent to joints before applying primers, caulking, or sealing compounds.
 1. Do not leave gaps between ends of sealant backings.
 2. Do not stretch, twist, puncture, or tear sealant backings.
 3. Remove absorbent sealant backings that have become wet before sealant application and replace them with dry materials.
- E. Apply primer to sides of joints wherever required by compound manufacturer's printed instructions or as indicated by pre-construction joint sealant substrate test.
 1. Apply primer prior to installation of back-up rod or bond breaker tape.
 2. Use brush or other approved means that will reach all parts of joints. Avoid application to or spillage onto adjacent substrate surfaces.

3.3 BACKING INSTALLATION:

- A. Install backing material, to form joints enclosed on three sides as required for specified depth of sealant.
- B. Where deep joints occur, install filler to fill space behind the backing rod and position the rod at proper depth.
- C. Cut fillers installed by others to proper depth for installation of backing rod and sealants.

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- D. Install backing rod, without puncturing the material, to a uniform depth, within plus or minus 3 mm (1/8 inch) for sealant depths specified.
- E. Where space for backing rod does not exist, install bond breaker tape strip at bottom (or back) of joint so sealant bonds only to two opposing surfaces.

~~SPEC WRITER NOTE: Detail joints correctly
for symmetry of sealant installation. See
ASTM C1193.~~

3.4 SEALANT DEPTHS AND GEOMETRY:

- A. At widths up to 6 mm (1/4 inch), sealant depth equal to width.
- B. At widths over 6 mm (1/4 inch), sealant depth 1/2 of width up to 13 mm (1/2 inch) maximum depth at center of joint with sealant thickness at center of joint approximately 1/2 of depth at adhesion surface.

3.5 INSTALLATION:

- A. General:
 - 1. Apply sealants and caulking only when ambient temperature is between 5 degrees C and 38 degrees C (40 degrees and 100 degrees F).
 - 2. Do not install polysulfide base sealants where sealant may be exposed to fumes from bituminous materials, or where water vapor in continuous contact with cementitious materials may be present.
 - 3. Do not install sealant type listed by manufacture as not suitable for use in locations specified.
 - 4. Apply caulking and sealing compound in accordance with manufacturer's printed instructions.
 - 5. Avoid dropping or smearing compound on adjacent surfaces.
 - 6. Fill joints solidly with compound and finish compound smooth.
 - 7. Tool exposed joints to form smooth and uniform beds, with slightly concave surface conforming to joint configuration per Figure 5A in ASTM C1193 unless shown or specified otherwise in construction documents. Remove masking tape immediately after tooling of sealant and before sealant face starts to "skin" over. Remove any excess sealant from adjacent surfaces of joint, leaving the working in a clean finished condition.
 - 8. Finish paving or floor joints flush unless joint is otherwise detailed.
 - 9. Apply compounds with nozzle size to fit joint width.

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10. Test sealants for compatibility with each other and substrate. Use only compatible sealant. Submit test reports.

11. Replace sealant which is damaged during construction process.

~~//B. Weeps: Place weep holes and vents in joints where moisture may accumulate, including at base of cavity walls, above shelf angles, at all flashing, and as indicated on construction documents.~~

~~1. Use round plastic tubing to form weep holes.~~

~~2. Space weep holes formed from plastic tubing not more than 406 mm (16 inches) o.c.~~

~~3. Trim tubing material used in weep holes flush with exterior wall face after sealant has set.//~~

C. For application of sealants, follow requirements of ASTM C1193 unless specified otherwise. Take all necessary steps to prevent three-sided adhesion of sealants.

~~D. Interior Sealants: Where gypsum board partitions are of sound rated, fire rated, or smoke barrier construction, follow requirements of ASTM C919 only to seal all cut outs and intersections with the adjoining construction unless specified otherwise.~~

~~1. Apply a 6 mm (1/4 inch) minimum bead of sealant each side of runners (tracks), including those used at partition intersections with dissimilar wall construction.~~

~~2. Coordinate with application of gypsum board to install sealant immediately prior to application of gypsum board.~~

~~3. Partition intersections: Seal edges of face layer of gypsum board abutting intersecting partitions, before taping and finishing or application of veneer plaster joint reinforcing.~~

~~4. Openings: Apply a 6 mm (1/4 inch) bead of sealant around all cutouts to seal openings of electrical boxes, ducts, pipes and similar penetrations. To seal electrical boxes, seal sides and backs.~~

~~5. Control Joints: Before control joints are installed, apply sealant in back of control joint to reduce flanking path for sound through control joint.~~

3.6 FIELD QUALITY CONTROL:

~~SPEC WRITER NOTE: Revise and edit this article to suit sealant requirements of project.~~

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//A. Field-Adhesion Testing: Field-test joint-sealant adhesion to joint substrates according to Method A, Field-Applied Sealant Joint Hand Pull Tab, in Appendix X1 in ASTM C1193 or Method A, Tail Procedure, in ASTM C1521.

1. Extent of Testing: Test completed elastomeric sealant joints as follows:
 - a. Perform 10 tests for first 305 m (1000 feet) of joint length for each type of elastomeric sealant and joint substrate.
 - b. Perform one test for each 305 m (1000 feet) of joint length thereafter or one test per each floor per elevation. //

//B. Inspect joints for complete fill, for absence of voids, and for joint configuration complying with specified requirements. Record results in a field adhesion test log. //

//C. Inspect tested joints and report on following:

1. Whether sealants in joints connected to pulled-out portion failed to adhere to joint substrates or tore cohesively. Include data on pull distance used to test each type of product and joint substrate.
2. Compare these results to determine if adhesion passes sealant manufacturer's field-adhesion hand-pull test criteria.
3. Whether sealants filled joint cavities and are free from voids.
4. Whether sealant dimensions and configurations comply with specified requirements. //

//D. Record test results in a field adhesion test log. Include dates when sealants were installed, names of persons who installed sealants, test dates, test locations, whether joints were primed, adhesion results and percent elongations, sealant fill, sealant configuration, and sealant dimensions. //

//E. Repair sealants pulled from test area by applying new sealants following same procedures used to originally seal joints. Ensure that original sealant surfaces are clean and new sealant contacts original sealant. //

//F. Evaluation of Field-Test Results: Sealants not evidencing adhesive failure from testing or noncompliance with other indicated requirements, will be considered satisfactory. Remove sealants that fail to adhere to joint substrates during testing or to comply with other requirements. Retest failed applications until test results prove sealants comply with indicated requirements. //

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3.7 CLEANING:

- A. Fresh compound accidentally smeared on adjoining surfaces: Scrape off immediately and rub clean with a solvent as recommended by manufacturer of the adjacent material or if not otherwise indicated by the caulking or sealant manufacturer.
- B. Leave adjacent surfaces in a clean and unstained condition.

- - - E N D - - -

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SECTION 07 95 13
EXPANSION JOINT COVER ASSEMBLIES

~~SPEC WRITER NOTE:~~

- ~~1. Delete text between // // not applicable to project. Edit remaining text to suit project.~~
- ~~2. Do not use aluminum, brass, or bronze for joints exceeding 100 mm (4 inches) wide, when wheel loads exceed 136 kg (300 lbs.), and for seismic joints.~~
- ~~3. Use stainless steel for seismic joints and joints exceeding 100 mm (4 inches) wide.~~
- ~~4. Use steel for joints in garages, service corridors, and other areas where wheel loads exceed 136 kg (300 lbs.) and appearance is not critical.~~

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Prefabricated ~~floor, wall, and ceiling~~ // seismic and // building expansion joint assemblies.
 - a. Metal plate covers at ~~// floor // wall // and ceiling~~ // roof joints.
 - b. Elastomeric joint covers at ~~// wall // and ceiling~~ // roof joints.

~~SPEC WRITER NOTE: Preformed joint is for expansion and contraction joints, not building expansion joints.~~

~~e. Preformed elastomeric sealant joint at interior // floor // and wall // control joints.~~

~~d.c.~~ Exterior wall joints.

1.2 RELATED WORK

~~SPEC WRITER NOTE: Update and retain references only when specified elsewhere in this Section.~~

- A. Section 05 50 00, METAL FABRICATIONS: Steel Plate Expansion Joint Covers.
- B. Section 07 60 00, FLASHING AND SHEET METAL: Sheet Metal Expansion Joint Seals.

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C. Section 07 72 00, ROOF ACCESSORIES: Roof Expansion Joint Cover Assemblies.

~~D. Section 09 06 00, SCHEDULE FOR FINISHES: Color of Elastomer Inserts, Filler Strips, Exterior Wall Seals and Metal Finishes.~~

~~SPEC WRITER NOTE: Update and retain references only when specified elsewhere in this Section.~~

1.3 APPLICABLE PUBLICATIONS

A. Comply with references to extent specified in this Section.

B. American Society of Civil Engineers (ASCE):

ASCE/SEI 7-10 - Minimum Design Loads For Buildings and Other Structures.

C. ASTM International (ASTM):

A36/A36M-19 - Structural Steel.

A240/A240M-20 - Chromium and Chromium-Nickel Stainless Steel Plate, Sheet and Strip for Pressure Vessels and for General Applications.

A283/A283M-18 - Low and Intermediate Tensile Strength Carbon Steel Plates.

A786/A786M-15 - Hot-Rolled Carbon, Low-Alloy, High-Strength Low-Alloy, and Alloy Steel Floor Plates.

B36/B36M-18 - Brass, Plate, Sheet, Strip, and Rolled Bar.

B121/B121M-16 - Lead Brass Plate, Sheet, Strip and Rolled Bar.

B209-14 - Aluminum and Aluminum-Alloy Sheet and Plate.

B209M-14 - Aluminum and Aluminum-Alloy Sheet and Plate (Metric).

B221-14 - Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes.

B221M-13 - Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric).

B455/B455M-20 - Copper-Zinc-Lead Alloy (Leaded-Brass) Extruded Shapes.

C864-05(2019) - Dense Elastomeric Compression Seal Gaskets, Setting Blocks, and Spacers.

D1187/D1187M-97(2018) - Asphalt-Base Emulsions for Use as Protective Coatings for Metal.

E1399/E1399M-97(2017) - Standard Test Method for Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems.

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E1966-15(2019) - Standard Test Method for Fire-Resistive Joint Systems.

D. National Association of Architectural Metal Manufacturers (NAAMM):

AMP 500-06 - Metal Finishes Manual.

E. UL LLC (UL):

2079-15 - Standard for Tests for Fire Resistance of Building Joint
Systems.

1.4 PREINSTALLATION MEETINGS

A. Conduct preinstallation meeting ~~//~~at project site ~~//~~minimum 30 days
before beginning Work of this Section.

~~SPEC WRITER NOTE: Edit participant list
to ensure entities influencing outcome
attend.~~

1. Required Participants:

a. Contracting Officer's Representative (COR).

b. ~~//~~Architect/Engineer (as requested by COR).~~//~~

c. ~~//~~Inspection and Testing Agency.~~//~~

d. Contractor.

e. Installer.

f. ~~//~~Manufacturer's field representative.~~//~~

g. Other installers responsible for adjacent and intersecting work,
~~including // _____ //~~.

~~SPEC WRITER NOTE: Edit meeting agenda to
incorporate project specific topics.~~

2. Meeting Agenda: Distribute agenda to participants minimum 3 days
before meeting.

a. Installation schedule.

b. Installation sequence.

c. Preparatory work.

d. Protection before, during, and after installation.

e. Installation.

f. Terminations.

g. Transitions and connections to other work.

h. Other items affecting successful completion.

3. Document and distribute meeting minutes to participants to record
decisions affecting installation.

1.5 SUBMITTALS

- A. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.
- B. Submittal Drawings:
1. Include large-scale details indicating profiles of each type of expansion joint cover, splice joints between joint sections, transitions to other assemblies, terminations, anchorages, fasteners, and relationship to adjoining work and finishes.
 2. Show size, configuration, and fabrication and installation details.
 3. Include composite drawings showing work specified in other Sections coordinated with expansion joints.
- C. Manufacturer's Literature and Data:
1. Description of each product specified.
 2. Show movement capability of each cover assembly ~~//~~ and suitability of material used in exterior seals for ultraviolet exposure ~~//~~.
 3. Description of materials and finishes.
 4. Installation instructions.
- D. Samples: Submit 300 mm (12 inch) long samples.
1. Each type and color of metal finish for each required thickness and alloy.
 2. Each type and color of flexible seal.
- E. Sustainable Construction Submittals:
- ~~SPEC WRITER NOTE: Retain sustainable construction submittals appropriate to product.~~
1. Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.
 2. Low Pollutant-Emitting Materials:
 - a. Identify volatile organic compound types and quantities.
- F. Qualifications: Substantiate qualifications comply with specifications.
1. Installer ~~//~~ with project experience list ~~//~~.
- G. Certificates: Indicate products comply with specifications.
1. Fire rated expansion joint cover assemblies.
- H. Operation and Maintenance Data:
1. Care instructions for each exposed finish product.

1.6 QUALITY ASSURANCE

- A. Installer Qualifications:

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1. Regularly installs specified products.
2. Installed specified products with satisfactory service on five similar installations for minimum five years.
 - a. ~~//~~Project Experience List: Provide contact names and addresses for completed projects.~~//~~

1.7 DELIVERY

- A. Deliver products in manufacturer's original sealed packaging.
- B. Mark packaging, legibly. Indicate manufacturer's name or brand, type, ~~//~~color, ~~//~~and manufacture date.
- C. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

- A. Store products indoors in dry, weathertight facility.
- B. Protect products from damage during handling and construction operations.

1.9 FIELD CONDITIONS

- A. Field Measurements: Verify field conditions affecting expansion joint cover assembly fabrication and installation. Show field measurements on Submittal Drawings.
 1. Coordinate field measurement and fabrication schedule to avoid delay.

1.10 WARRANTY

~~SPEC WRITER NOTE: Always retain construction warranty. FAR includes Contractor's one year labor and material warranty.~~

- A. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

PART 2 - PRODUCTS

2.1 SYSTEM DESCRIPTION

- A. Provide joint cover assemblies that permit unrestrained movement of joint without disengagement of cover, and, where applicable, maintain moisture, watertight and fire-rated protection.
- B. Provide templates to related trades for location of support and anchorage items.

2.2 SYSTEM PERFORMANCE

- A. Design expansion joint cover assemblies complying with specified performance.

SPEC WRITER NOTE: Specify movement here when same criteria apply to each joint. Specify movement with individual joints when criteria differ between joints. Coordinate with structural requirements.

- B. Joint Movement: ASTM E1399.

1. Nominal Joint Width: ~~//~~ ~~mm~~ (~~//~~ ~~2~~ inches).
2. Minimum Movement Capability: ~~//~~ 25 ~~//~~ 50 ~~//~~ percent.
3. Movement Type: Thermal, ~~and~~ wind ~~//~~ and seismic ~~//~~.

- C. Floor Joints: Live loads, including rolling loads.

1. Load Resistance: ASCE/SEI 7; Design criteria as indicated on Drawings.
2. Maximum Deflection: ~~//~~ 1/360 ~~//~~ of span, maximum.

- D. Fire Rated Joints: ASTM E1399, ASTM E1966, or UL 2079, including hose stream test at full-rated period.

1. Fire rating: Match adjacent floor, wall, and ceiling construction.
2. System: Capable of anticipated movement while maintaining fire rating.
3. Coverless Applications: Maintain fire rating without joint cover system.

~~SPEC WRITER NOTE:~~

- ~~1. Make material requirements agree with applicable requirements specified in applicable publications.~~
- ~~2. Update and specify only what applies to the Project.~~

2.3 MATERIALS

- A. Stainless Steel: ASTM A240/A240M, Type 302 or 304.
B. Structural Steel Shapes: ASTM A36/A36M.
C. Steel Plate: ASTM A283/A283M, Grade C.
D. Rolled Steel Floor Plate: ASTM A786/A786M.
E. Aluminum:

1. Extruded: ASTM B221M (ASTM B221), alloy 6063-T5, 6063-T6, or 6061-T6.
2. Plate and Sheet: ASTM B209M (ASTM B209), alloy 6061-T6.

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- F. Bronze: Manufacturer's standard alloy.
1. Extruded: ASTM B455.
 2. Plate: ASTM B121.
- G. Brass: ASTM B36/B36M.
- H. Elastomeric Sealant: As specified in Section 07 92 00, JOINT SEALANTS.
- I. Elastomeric Seals:
1. Flexible extruded polyvinyl chloride, meeting a Shore A hardness of 75 with UV stabilizer. Manufacturer's standard colors.
- J. Thermoplastic Rubber:
1. ASTM C864.
 2. Dense Neoprene or other material standard with expansion joint manufacturers having the same physical properties.
- K. Compression Seals: Pre-compressed secondary sealant using preformed expanding foam sealant; open-cell polyurethane foam impregnated with polymer-modified acrylic adhesive.
- L. Water Barrier Sheets: Neoprene or EPDM flexible sheet materials minimum 45 mils thick.
1. Provide with drain tubes for horizontal applications.
- M. Vinyl Invertor Sealant Waterstops: Manufacturer's standard shapes and grade.
- N. Moisture Barrier: Fabric reinforced clear vinyl sheet material sized to accommodate opening.
- O. Flexible Membrane: 1.5 mm (60 mil) EPDM sheet, with manufacturer's standard support foam.
- P. Fire Barrier: Labeled by an approved independent testing laboratory for fire resistance ~~//rating //ratings //~~ indicated for maximum joint width.
- a. Thermal Insulation: Manufacturer's standard with factory cut miters and transitions.
 - b. Fire Barrier Lengths:
 - 1) Joint widths up to and including 150 mm (6 inches): Maximum 15 m (50 feet) to minimize field splicing.
 - 2) Other Joint widths: 3 m (10 foot) with overlapping ends for field splicing.
- Q. Ceramic Blanket: Manufacturer's standard joint filler to achieve fire rating indicated.

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R. Butyl Caulk Tape: Self adhering double sided butyl rubber sealant tape with easy-release silicone coated paper.

2.4 PRODUCTS - GENERAL

~~A. Basis of Design: Section 09 06 00, SCHEDULE FOR FINISHES.~~

~~B.A.~~ Provide each product from one manufacturer.

1. Provide ceiling and wall expansion joint cover assemblies design matching floor to wall and floor to floor expansion joint cover design.
2. Provide expansion joint cover assembly designs, profiles, materials and configuration indicated, as required to accommodate joint size variations in adjacent surfaces, and anticipated movement.

~~C.B.~~ Sustainable Construction Requirements:

~~SPEC WRITER NOTE: Specify product recycled content when known to be achievable for Project location. Section 01 81 13 sets overall project recycled content requirements.~~

1. Steel Recycled Content: 30 percent total recycled content, minimum.

~~SPEC WRITER NOTE: Specialty Steel Industry of North America (SSINA) reports average 75 - 85 percent recycled content for stainless steel.~~

2. Stainless Steel Recycled Content: 70 percent total recycled content, minimum.

~~SPEC WRITER NOTE: Aluminum Association (AA) reports 2008 industry average 85 percent recycled content for aluminum in building construction industry.~~

3. Aluminum Recycled Content: 80 percent total recycled content, minimum.

~~SPEC WRITER NOTE: Edit list showing VOC content to retain only those products required for Project.~~

4. Low Pollutant-Emitting Materials: Maximum VOC content by weight.
 - a. Non-Flooring Adhesives and Sealants.

2.5 FABRICATION

A. Fabricate Expansion Joint Cover Assemblies:

1. As complete assembly ready for installation.

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2. In longest practicable lengths to minimize number of end joints.
3. With factory mitered corners where joint changes directions or abuts other materials.
 - a. With closure materials and transition pieces, tee-joints, corners, curbs, cross-connections and other assemblies.

~~SPEC WRITER NOTE: Larger sized joints and seismic joints may require sheet metal and EPDM for fire rated conditions.~~

4. Joints within enclosed spaces such as chase walls, include 1 mm (0.04 inch) thick galvanized steel cover where conventional expansion joint cover is not used.
5. Where floor slab is fire rated provide ceramic blanket at joints.
6. Seal Strip: Factory-formed and bonded to metal frames and anchor members.
7. Compression Seals: Fabricate from expanding foam as secondary seal and elastomeric sealant to sizes and profiles shown.

~~SPEC WRITER NOTE:~~

- ~~1. Verify details show various type of assemblies.~~
- ~~2. Do not use sole source details.~~
- ~~3. Use joints designed for three-way movement where possible.~~
- ~~4. Use metal cover plate over seal on joints that are accessible from walks or other pedestrian traffic.~~

B. Floor-to-Floor Metal Plate Joints:

1. Frames: Metal, continuous on both sides of joint designed to support cover plate.
 - a. Flush Design: Seating surface and raised floor rim to accommodate adjacent flooring.
 - b. Anchorage: Concealed bolt and steel anchors for embedment in concrete.
2. Cover Plate: Metal, matching frames where exposed.
 - a. Supported Load: 19.2 MPa (400 psf), minimum.
 - b. Rattle-free due to traffic.
3. Fillers: Resilient material between raised rim of frame and edge of cover plate, where shown.
 - a. No gaps or bulges over full design range joint movement.
4. Fire Barrier: As required for fire resistance rating.

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5. Water Stop: Manufacturer's standard, continuous, full length of joint.
6. Seismic: As required by Code.
7. Finishes: As specified in ~~Section 09 06 00, SCHEDULE FOR FINISHES~~the drawings.

C. Floor-to-Wall Metal Plate Joints:

1. Frames: Metal, continuous on floor side of joint only.
 - a. Provide wall side frame where required by manufacturer's design.
2. Cover Plates: Angle cover plates with countersunk flat-head exposed fasteners for securing cover plate to wall unless shown otherwise.
 - a. Fastener Spacing: As recommended by manufacturer.
3. Joint Design: Match adjacent floor to floor design.
4. Fire Barrier: As required for fire resistance rating.
5. Water Stop: Manufacturer's standard, continuous, full length of joint.
6. Seismic: As required by Code.
7. Finishes: As specified in ~~Section 09 06 00, SCHEDULE FOR FINISHES~~the drawings.

~~SPEC WRITER NOTE: Use cover plates at wall to wall joints to protect seals behind plates.~~

~~D. Interior Wall Joint Cover Assemblies:~~

- ~~1. Frame: Metal, surface mounted, concealed fastening to wall on one sides of joint.~~
- ~~2. Cover Plate: Metal, smooth surface, lap both sides of joint and permitting free movement on one side.~~
 - ~~a. Fabricate with concealed attachment of cover to frame when cover is in close contact with adjacent wall surface finish.~~
 - ~~b. Use angle cover plates at intersecting walls.~~
- ~~3. Joint Design: Match adjacent floor to floor design.~~
- ~~4. Fire Barrier: As required for fire resistance rating.~~
- ~~5. Seismic: As required by Code.~~
- ~~6. Finishes: As specified in Section 09 06 00, SCHEDULE FOR FINISHES.~~

~~E.D.~~ Exterior Wall Joint Assemblies:

1. Design seal for variable movement and prevention of water and air infiltration.
2. Frame: Metal, concealed, for fastening to wall on one side of joint.

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3. Cover Plate: Metal, surface mounted, lap both sides of joint, permitting free movement on one side.
 - a. Fabricate with concealed attachment of cover to frame for cover with cover in close contact with adjacent finish surfaces.
 - b. Use angle cover plate at intersecting walls.
4. Water Seal: Vinyl seal strip as secondary seal behind primary seal.
5. Seismic: As required by Code.
6. Finish: As specified in ~~Section 09-06-00, SCHEDULE FOR FINISHES~~the drawings.

F.E. Extruded Thermoplastic Rubber Joint Assemblies:

1. Frames: Aluminum, both sides of joint.
2. Primary Seal: Flexible rubber on exposed face after frame installation with factory welded watertight miters and transitions.
 - a. Anchor spaced at ends and not over 600 mm (24 inches).
 - 1) Variable movement extruded rubber primary seal designed to remain in aluminum frame, throughout movement of joint.
 - b. ~~//~~ Flush mounted seal minimum 3 mm (0.12 inch) thick with dual movement grooves designed for plus or minus 50 percent, movement of joint width. ~~//~~
 - c. ~~//~~ Seismic seal minimum 3 mm (0.12 inch) thick with multi-movement grooves designed for plus or minus 100 percent movement of joint width. ~~//~~
 - d. ~~//~~ Recessed front face seal minimum 3 mm (0.12 inch) thick with no movement grooves, designed for plus or minus 50 percent movement of joint width. ~~//~~
 - e. Provide pantographic wind load supports, maximum 2400 mm (8 feet) on center to support seal systems of 300 mm (12 inches) wide and greater.
3. Secondary Seal: Continuous vinyl sheet seal.
4. Finishes: As specified in ~~Section 09-06-00, SCHEDULE FOR FINISHES~~the drawings.

~~SPEC WRITER NOTE:~~

- ~~1. Considering insert type with acoustical ceilings, not fire rated.~~
- ~~2. Do not use for walls.~~

~~G. Ceiling and Soffit Assemblies:~~

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- ~~1. Frames: Metal, continuous on both sides of joint, flush mounted with no exposed fasteners.~~
- ~~2. Flexible Insert: Variable movement semi-rigid vinyl locked into frame.~~
 - ~~a. Face Style: Flush or accordion, as shown, to span joint width without sagging.~~
- ~~3. Seismic: As required by Code.~~
- ~~4. Finishes: As specified in Section 09 06 00, SCHEDULE FOR FINISHES.~~

~~SPEC WRITER NOTE:~~

- ~~1. Design plate for span of joint and for wheel loads of minimum 1360 kg (3000 pounds), for garage floor covers.~~
- ~~2. Coordinate with Section 05 50 00, METAL FABRICATIONS if steel expansion joints are specified there for garage floors having joints over 50 mm (2 inches) wide.~~

~~H. Garage Floor Joint Cover Plate:~~

- ~~1. Frame: Angle edge frame on both sides of joint, size as shown.~~
 - ~~a. Anchors: Stud bolts minimum 100 mm (4 inches) long and 10 mm (3/8 inch) diameter welded to angle spaced maximum 600 mm (24 inches) on center.~~
 - ~~b. Drill and tap one frame for cover plate fasteners.~~
- ~~2. Cover Plate: Aluminum or steel cover plate minimum 10 mm (3/8 inch) thick with edges beveled, smooth finish, drilled for countersunk fasteners at ends and maximum 600 mm (24 inches) on center.~~
- ~~3. Seismic: As required by Code.~~
- ~~4. Finishes: As specified in Section 09 06 00, SCHEDULE FOR FINISHES.~~

~~SPEC WRITER NOTE: Joint application is for expansion and contraction joints of interior floor and wall substrates. This is not a building expansion joint. Typical size is 10 mm (3/8 inch) with maximum joint width of 38 mm (1 1/2 inches).~~

~~I. Preformed Sealant Joint: Factory installed elastomeric sealant between extruded aluminum angle frame both sides.~~

- ~~1. Frames: Extruded aluminum angle on both sides of joint.~~
- ~~2. Filler: Elastomeric sealant.~~

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~~3. Anticipated movement: 25 percent maximum.~~

~~4. Finishes: As specified in Section 09 06 00, SCHEDULE FOR FINISHES.~~

2.6 FINISHES

A. Carbon Steel: NAAMM AMP 500, Galvanized G90.

B. Stainless Steel: NAAMM AMP 500, No. 2B bright finish.

C. Aluminum Anodized Finish: NAAMM AMP 500.

1. Clear Anodized Finish: AA-C22A41; Class I Architectural, 0.018 mm (0.7 mil) thick.

~~SPEC WRITER NOTE:~~

~~1. Use of colored coatings on floor covers is not recommended.~~

~~2. Choose the type of color anodized finish or leave both for the contractor to decide.~~

2. Color Anodized Finish: AA-C22A42 or AA-C22A44; Class I Architectural, 0.018 mm (0.7 mil) thick.

D. Aluminum Paint Finish:

~~SPEC WRITER NOTE: Use first finish for extrusions and second for coil coated sheet.~~

1. Fluorocarbon Finish: AAMA 2605; 70 percent fluoropolymer resin, 2-coat system.

2. Fluorocarbon Finish: AAMA 605; 70 percent fluoropolymer resin, 2-coat system.

E. Bronze Finish: NAAMM-AMP 500, M32 mechanical finish, directional textured, natural medium satin.

2.7 ACCESSORIES

A. General: Manufacturer's standard anchors, fasteners, set screws, spaces, protective coating, and filler materials, adhesive and other accessories required for installation.

~~SPEC WRITER NOTE: Retain barrier coating to separate dissimilar metals and to separate metals from cementitious materials.~~

B. Barrier Coating: ASTM D1187/D1187M.

C. Adhesives: Low pollutant-emitting, water based type recommended by adhered product manufacturer for each application.

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- D. Fasteners: Type and size recommended by expansion joint cover assembly manufacturer.
1. Exterior Applications: Stainless steel.
 2. Fasteners for Aluminum: Stainless steel.
 3. Other Applications: Galvanized steel or stainless steel.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Examine and verify substrate suitability for product installation.
1. Provide items embedded in concrete and masonry in time for building into work without delaying work.

~~SPEC WRITER NOTE: Edit barrier coating for metals specific for project; and whether to separate dissimilar metals and or to separate metals from cementitious materials.~~

- B. Protect existing construction and completed work from damage.
- C. Apply barrier coating to ~~//aluminum_//brass//bronze or//steel //~~ surfaces in contact with ~~//dissimilar metals //~~and cementitious materials ~~//to minimum 0.7 mm (30 mils) dry film thickness.~~

3.2 INSTALLATION

- A. Install products according to manufacturer's instructions ~~//and approved submittal drawings//~~.
1. When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's Representative consideration.
- B. Install anchorage devices and fasteners for securing expansion joint assemblies to in-place construction where anchors are not embedded in concrete and masonry.
1. Secure with metal fasteners, type and size to suit application.
- C. Perform cutting, drilling and fitting required for installation of expansion joint cover assemblies.
- D. Install joint cover assemblies aligned and positioned in correct relationship to expansion joint opening and adjoining finished surfaces measured from established lines and levels.
1. Allow for thermal expansion and contraction of metal to avoid buckling.
 2. Accommodate joint opening size at time of installation.

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- E. Set floor covers at elevations flush with adjacent finished flooring, unless shown otherwise.
- F. Grout floor frames set in prepared recesses.
- G. Locate wall, ceiling and soffit covers in continuous contact with adjacent surfaces. Secure with required accessories.
- H. Locate anchors at interval recommended by manufacturer, but minimum 75 mm (3 inches) from each end, and, maximum 600 mm (24 inches) on centers.
- I. Maintain continuity of expansion joint cover assemblies with end joints held to a minimum and metal members aligned mechanically using splice joints.
- J. Cut and fit ends to accommodate thermal expansion and contraction of metal to avoid buckling of frames and cover plates.
- K. Flush Metal Cover Plates:
 - 1. Secure flexible filler between frames to allow compression and expansion.
 - 2. Adhere flexible filler materials to frames with adhesive or pressure-sensitive tape as recommended by manufacturer.
- L. Waterstops:
 - 1. Install in conjunction with floor joints, and where shown.
 - 2. Install continuously to prevent water damage to finish spaces.
 - 3. Seal waterstop to frames to prevent water leakage.
 - 4. Install drainage tubes from waterstops to discharge collected water in nearest plumbing air gap drain.
- M. Fire Barriers:
 - 1. Install in compliance with tested assembly.
 - 2. Install at joints in floors and in fire rated walls.
 - 3. Use fire barrier sealant furnished with expansion joint assembly.
- N. Apply sealant where required to prevent water and air infiltration.
- O. Vertical Exterior Extruded Thermoplastic Rubber.
 - 1. Install side frames mounted on sealant or butyl caulk tape with appropriate anchors 600 mm (24 inches) on center complete with secondary seal.
 - 2. Install primary seals retained in extruded aluminum side frames.
- P. Extruded Thermoplastic Rubber or Seals:
 - 1. For straight sections, install preformed seals in continuous lengths.

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2. Vulcanize or heat-seal field spliced joints to provide watertight joints as recommended by manufacturer.

Q. Preformed Elastomeric Sealant Joint:

1. Locate joint directly over joints in wall and floor substrates.
2. Fasten full length to substrate using construction adhesive.
3. Install flush or slightly below finish material.

3.3 CLEANING

- A. Remove excess adhesive before adhesive sets.
- B. Clean exposed metal surfaces. Remove contaminants and stains.

3.4 PROTECTION

- A. Cover floor joints with plywood where wheel traffic occurs before Substantial completion.
- B. Remove protective covering when adjacent work areas are completed. Clean exposed surfaces in compliance with manufacture's printed instructions.

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SECTION 09 91 00
PAINTING

~~SPEC WRITER NOTE: Delete between // // if not applicable to project. Delete, modify and add to text as required to suit information shown on the construction documents and specified in Section 09 06 00, SCHEDULE FOR FINISHES.~~

PART 1 - GENERAL

1.1 DESCRIPTION

A. Work of this Section includes all labor, materials, equipment, and services necessary to complete the painting and finishing as shown on the construction documents and/or specified herein, including, but not limited to, the following:

1. Prime coats which may be applied in shop under other sections.
2. Prime painting unprimed surfaces to be painted under this Section.
3. Painting items furnished with a prime coat of paint, including touching up of or repairing of abraded, damaged or rusted prime coats applied by others.

4. Painting ferrous metal (except stainless steel) exposed to view.

5. Painting galvanized ferrous metals exposed to view.

~~6. Painting interior concrete block exposed to view.~~

~~7. Painting gypsum drywall exposed to view.~~

~~8. Painting of wood exposed to view, except items which are specified to be painted or finished under other Sections of these specifications. Back painting of all wood in contact with concrete, masonry or other moisture areas.~~

~~9.~~6. Painting pipes, pipe coverings, conduit, ducts, insulation, hangers, supports and other mechanical and electrical items and equipment exposed to view.

~~10. Painting surfaces above, behind or below grilles, gratings, diffusers, louvers lighting fixtures, and the like, which are exposed to view through these items.~~

~~11.~~7. Painting includes shellacs, stains, varnishes, coatings specified, and striping or markers and identity markings.

~~12.~~8. Incidental painting and touching up as required to produce proper finish for painted surfaces, including touching up of factory finished items.

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~~13-9.~~ Painting of any surface not specifically mentioned to be painted herein or on construction documents, but for which painting is obviously necessary to complete the job, or work which comes within the intent of these specifications, is to be included as though specified.

1.2 RELATED WORK

A. Section 01 35 26, SAFETY REQUIREMENTS: Activity Hazard Analysis.

B. ~~//~~Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.~~//~~

C. Section 02 83 33.13, LEAD-BASED PAINT REMOVAL AND DISPOSAL: Lead Paint Removal.

D. ~~//~~Section 04 05 13, MASONRY MORTARING: Masonry Repairs. ~~//~~

E. ~~//~~Section 04 05 16, MASONRY GROUTING: Masonry Repairs. ~~//~~

F-D. Division 05 METALS: Shop prime painting of steel and ferrous metals.

G. Division 08 OPENINGS: Shop prime painting of steel and ferrous metals.

H. Section 08 14 00, INTERIOR WOOD DOORS: Prefinished flush doors with transparent finishes.

I. Section 09 06 00, SCHEDULE FOR FINISHES: Type of Finish, Color, and Gloss Level of Finish Coat.

J. Section 09 94 19, MULTICOLOR INTERIOR FINISHING: Multi-color Textured Wall Finish.

K. Section 09 96 59, RESINOUS SPECIALTY GLAZED COATING SYSTEMS FOR WALLS, CEILINGS, WALLBOARD, AND BLOCK CMU (RES-W1, RES-W2): Glazed wall surfacing or tile like coatings.

L. Section 09 96 59, RESINOUS SPECIALTY GLAZED COATING SYSTEMS FOR WALLS, CEILINGS, WALLBOARD, AND BLOCK CMU (RES-W1, RES-W2): Glazed wall surfacing or tile like coatings.

M. Division 10 SPECIALTIES: Shop prime painting of steel and ferrous metals.

N. Division 11 EQUIPMENT: Shop prime painting of steel and ferrous metals.

O. Division 12 FURNISHINGS: Shop prime painting of steel and ferrous metals.

P. Division 13 SPECIAL CONSTRUCTION: Shop prime painting of steel and ferrous metals.

Q. Division 14 CONVEYING EQUIPMENT: Shop prime painting of steel and ferrous metals.

R. Division 21 FIRE SUPPRESSION: Shop prime painting of steel and ferrous metals.

S. Division 22 PLUMBING: Shop prime painting of steel and ferrous metals.

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~~T. Division 23 HEATING; VENTILATION AND AIR-CONDITIONING: Shop prime painting of steel and ferrous metals.~~

~~U. Division 26 ELECTRICAL: Shop prime painting of steel and ferrous metals.~~

~~V. Division 27 COMMUNICATIONS: Shop prime painting of steel and ferrous metals.~~

~~W. Division 28 ELECTRONIC SAFETY AND SECURITY: Shop prime painting of steel and ferrous metals.~~

~~X. Division 32 EXTERIOR IMPROVEMENTS: Shop prime painting of steel and ferrous metals.~~

~~Y. Section 32 17 23, PAVEMENT MARKINGS: Asphalt and concrete pavement marking.~~

1.3 ~~SUBMITTALS~~

A. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

B. ~~//~~ Sustainable Design Submittals as described below:

1. ~~//~~ Volatile organic compounds per volume as specified in
PART 2 - PRODUCTS. ~~// //~~

C. Painter qualifications.

D. Manufacturer's Literature and Data:

1. Before work is started, or sample panels are prepared, submit manufacturer's literature and technical data, the current Master Painters Institute (MPI) "Approved Product List" indicating brand label, product name and product code as of the date of contract award, will be used to determine compliance with the submittal requirements of this specification. The Contractor may choose to use subsequent MPI "Approved Product List", however, only one (1) list may be used for the entire contract and each coating system is to be from a single manufacturer. All coats on a particular substrate must be from a single manufacturer. No variation from the MPI "Approved Product List" where applicable is acceptable.

E. Sample Panels:

1. After painters' materials have been approved and before work is started, submit sample panels showing each type of finish and color specified.

2. Panels to Show Color: Composition board, 100 x 250 mm (4 x 10 inch).

~~3. Panel to Show Transparent Finishes: Wood of same species and grain pattern as wood approved for use, 100 x 250 mm (4 x 10 inch face)~~

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~~minimum, and where both flat and edge grain will be exposed, 250 mm (10 inches) long by sufficient size, 50 x 50 mm (2 x 2 inch) minimum or actual wood member to show complete finish.~~

~~4.3.~~ Attach labels to panel stating the following:

a. Federal Specification Number or manufacturers name and product number of paints used.

~~b. Specification code number specified in Section 09 06 00, SCHEDULE FOR FINISHES.~~

~~e.b.~~ Product type and color.

~~d.c.~~ Name of project.

~~5.4.~~ Strips showing not less than 50 mm (2 inch) wide strips of undercoats and 100 mm (4 inch) wide strip of finish coat.

F. Sample of identity markers if used.

G. Manufacturers' Certificates indicating compliance with specified requirements:

1. Manufacturer's paint substituted for Federal Specification paints meets or exceeds performance of paint specified.

2. High temperature aluminum paint.

3. Epoxy coating.

~~4. Intumescent clear coating or fire retardant paint.~~

~~5. Plastic floor coating.~~

1.4 DELIVERY AND STORAGE

A. Deliver materials to site in manufacturer's sealed container marked to show following:

1. Name of manufacturer.

2. Product type.

3. Batch number.

4. Instructions for use.

5. Safety precautions.

B. In addition to manufacturer's label, provide a label legibly printed as following:

1. Federal Specification Number, where applicable, and name of material.

2. Surface upon which material is to be applied.

3. Specify Coat Types: Prime; body; finish; etc.

C. Maintain space for storage, and handling of painting materials and equipment in a ventilated, neat and orderly condition to prevent spontaneous combustion from occurring or igniting adjacent items.

- D. Store materials at site at least 24 hours before using, at a temperature between 7 and 30 degrees C (45 and 85 degrees F).

1.5 QUALITY ASSURANCE

- A. Qualification of Painters: Use only qualified journeyman painters for the mixing and application of paint on exposed surfaces. Submit evidence that key personnel have successfully performed surface preparation and application of coating on a minimum of three (3) similar projects within the past three (3) years.
- B. Paint Coordination: Provide finish coats which are compatible with the prime paints used. Review other Sections of these specifications in which prime paints are to be provided to ensure compatibility of the total coatings system for the various substrates. Upon request from other subcontractors, furnish information on the characteristics of the finish materials proposed to be used, to ensure that compatible prime coats are used. Provide barrier coats over incompatible primers or remove and re-prime as required. Notify the Contracting Officer Representative (COR) in writing of any anticipated problems using the coating systems as specified with substrates primed by others.

1.6 ~~//~~ MOCK-UP PANEL

~~SPEC WRITER NOTES:-~~

- ~~1. Define spaces and other finishes where sample panel is required on construction documents.~~
- ~~2. Mock-up must be approved by COR in the project's design phase before including requirement in specification.~~

- A. In addition to the samples specified herein to be submitted for approval, apply in the field, at their final location, each type and color of approved paint materials, ~~applied 3.05 m (10 feet) wide, floor to ceiling of wall surfaces,~~ before proceeding with the remainder of the work, for approval by the COR. ~~Paint mock-ups to include one (1) door and frame assembly.~~
- B. Finish and texture approved by COR will be used as a standard of quality and workmanship for remainder of work.
- C. Repaint individual areas which are not approved, as determined by the COR, until approval is received. ~~//~~

1.7 ~~REGULATORY REQUIREMENTS~~

- A. Paint materials are to conform to the restrictions of the local Environmental and Toxic Control jurisdiction.
1. Volatile Organic Compounds (VOC) Emissions Requirements: Field-applied paints and coatings that are inside the waterproofing system to not exceed limits of authorities having jurisdiction.

2. Lead-Base Paint:

- a. Comply with Section 410 of the Lead-Based Paint Poisoning Prevention Act, as amended, and with implementing regulations promulgated by Secretary of Housing and Urban Development.
- b. Regulations concerning prohibition against use of lead-based paint in federal and federally assisted construction, or rehabilitation of residential structures are set forth in Subpart F, Title 24, Code of Federal Regulations, Department of Housing and Urban Development.
- c. Do not use coatings having a lead content over 0.06 percent by weight of non-volatile content.
- d. For lead-paint removal, see Section 02 83 33.13, LEAD-BASED PAINT REMOVAL AND DISPOSAL.
3. Asbestos: Provide materials that do not contain asbestos.
4. Chromate, Cadmium, Mercury, and Silica: Provide materials that do not contain zinc-chromate, strontium-chromate, Cadmium, mercury or mercury compounds or free crystalline silica.
5. Human Carcinogens: Provide materials that do not contain any of the ACGIH-BKLT and ACGHI-DOC confirmed or suspected human carcinogens.
6. Use high performance acrylic paints in place of alkyd paints.

1.8 SAFETY AND HEALTH

- A. Apply paint materials using safety methods and equipment in accordance with the following:
1. Comply with applicable Federal, State, and local laws and regulations, and with the ACCIDENT PREVENTION PLAN, including the Activity Hazard Analysis (AHA) as specified in Section 01 35 26, SAFETY REQUIREMENTS. The AHA is to include analyses of the potential impact of painting operations on painting personnel and on others involved in and adjacent to the work zone.
- B. Safety Methods Used During Paint Application: Comply with the requirements of SSPC PA Guide 10.

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C. Toxic Materials: To protect personnel from overexposure to toxic materials, conform to the most stringent guidance of:

1. The applicable manufacturer's Material Safety Data Sheets (MSDS) or local regulation.
2. 29 CFR 1910.1000.
3. ACHIH-BKLT and ACGIH-DOC, threshold limit values.

1.9 APPLICABLE PUBLICATIONS

A. Publications listed below form a part of this specification to the extent referenced. Publications are referenced in the text by basic designation only.

B. American Conference of Governmental Industrial Hygienists (ACGIH):
ACGIH TLV-BKLT-2012.....Threshold Limit Values (TLV) for Chemical Substances and Physical Agents and Biological Exposure Indices (BEIs)
ACGIH TLV-DOC-2012.....Documentation of Threshold Limit Values and Biological Exposure Indices, (Seventh Edition)

C. ASME International (ASME):
A13.1-07(R2013).....Scheme for the Identification of Piping Systems

D. Code of Federal Regulation (CFR):
40 CFR 59.....Determination of Volatile Matter Content, Water Content, Density Volume Solids, and Weight Solids of Surface Coating

E. Commercial Item Description (CID):
A-A-1272A.....Plaster Gypsum (Spackling Compound)

F. Federal Specifications (Fed Spec):
TT-P-1411A.....Paint, Copolymer-Resin, Cementitious (For Waterproofing Concrete and Masonry Walls) (CEP)

G. Master Painters Institute (MPI):
1.....Aluminum Paint
3.....Primer, Alkali Resistant, Water Based
4.....Interior/ Exterior Latex Block Filler
5.....Exterior Alkyd Wood Primer
6.....Exterior, Latex for Exterior Wood Primer
7.....Exterior Oil Wood Primer
8.....Exterior Alkyd, Flat MPI Gloss Level 1
9.....Exterior Alkyd Enamel MPI Gloss Level 6
10.....Exterior Latex, Flat

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- 11.....Exterior Latex, Semi-Gloss
- 15.....Exterior Latex, Low Sheen (MPI Gloss Level 3-4)
- 17.....Primer, Bonding, Waterbased
- 18.....Organic Zinc Rich Primer
- 22.....Aluminum Paint, High Heat (up to 590° - 1100F)
- 23.....Primer, Metal, Surface Tolerant

- 27.....Exterior / Interior Alkyd Floor Enamel, Gloss
- 31.....Polyurethane, Moisture Cured, Clear Gloss
- 36.....Knot Sealer
- 39.....Primer, Latex, for Interior Wood
- 40.....Exterior, Latex High Build
- 42.....Textured Coating, Latex, Flat
- 43.....Interior Satin Latex, MPI Gloss Level 4
- 44.....Interior Low Sheen Latex, MPI Gloss Level 2
- 45.....Interior Primer Sealer
- 46.....Interior Enamel Undercoat
- 47.....Interior Alkyd, Semi-Gloss, MPI Gloss Level 5
- 48.....Interior Alkyd, Gloss, MPI Gloss Level 6
- 50.....Interior Latex Primer Sealer
- 51.....Interior Alkyd, Eggshell, MPI Gloss Level 3
- 52.....Interior Latex, MPI Gloss Level 3
- 53.....Interior Latex, Flat, MPI Gloss Level 1
- 54.....Interior Latex, Semi-Gloss, MPI Gloss Level 5
- 59.....Interior/Exterior Alkyd Porch & Floor Enamel, Low
Gloss
- 60.....Interior/Exterior Latex Porch & Floor Paint, Low
Gloss
- 66.....Interior Alkyd Fire Retardant, Clear Top-Coat (ULC
Approved)
- 67.....Interior Latex Fire Retardant, Top-Coat (ULC
Approved)
- 68.....Interior/ Exterior Latex Porch & Floor Paint,
Gloss
- 71.....Polyurethane, Moisture Cured, Clear, Flat
- 77.....Epoxy Cold Cured, Gloss
- 79.....Marine Alkyd Metal Primer

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90.....Interior Wood Stain, Semi-Transparent
91.....Wood Filler Paste
94.....Exterior Alkyd, Semi-Gloss
95.....Fast Drying Metal Primer
98.....High Build Epoxy Coating
99.....Sealer, Water-based, for Concrete Floors
101.....Epoxy Anti-Corrosive Metal Primer
107.....Primer, Rust-Inhibitive, Water-based

108.....High Build Epoxy Coating, Low Gloss
113.....Elastomeric, Pigmented, Exterior, Water-based,
Flat
114.....Interior Latex, Gloss
115.....Epoxy-Modified Latex, Interior Gloss (MPI gloss
level 6)
118.....Dry Fall, Latex Flat
119.....Exterior Latex, High Gloss (acrylic)
134.....Galvanized Water Based Primer
135.....Non-Cementitious Galvanized Primer
138.....Interior High Performance Latex, MPI Gloss Level 2
139.....Interior High Performance Latex, MPI Gloss Level 3
140.....Interior High Performance Latex, MPI Gloss Level 4
141.....Interior High Performance Latex (SG) MPI Gloss
Level 5
144.....Latex, Interior, Institutional Low Odor / VOC,
(MPI Gloss Level 2)
145.....Latex, Interior, Institutional Low Odor / VOC,
(MPI Gloss Level 3)
146.....Latex, Interior, Institutional Low Odor / VOC,
(MPI Gloss Level 4)
151.....Light Industrial Coating, Interior, Water-based,
(MPI Gloss Level 3)
153.....Light Industrial Coating, Interior, Water-based,
(MPI Gloss Level 4)
163.....Exterior Water Based Semi-Gloss Light Industrial
Coating, MPI Gloss Level 5

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164.....Exterior, Water Based, Gloss, Light Industrial
Coating, MPI Gloss Level 6

H. Society for Protective Coatings (SSPC):

SSPC SP 1-82 (R2004).....Solvent Cleaning

SSPC SP 2-82 (R2004).....Hand Tool Cleaning

SSPC SP 3-28 (R2004).....Power Tool Cleaning

SSPC SP 10/NACE No.2.....Near-White Blast Cleaning

SSPC PA Guide 10.....Guide to Safety and Health Requirements

I. Maple Flooring Manufacturer's Association (MFMA):

J. U.S. National Archives and Records Administration (NARA):

29 CFR 1910.1000.....Air Contaminants

K. Underwriter's Laboratory (UL)

PART 2 - PRODUCTS

~~SPEC WRITER NOTES:~~

- ~~1. Coordinate material requirements to agree with applicable requirements specified in the referenced Applicable Publications.~~
- ~~2. Update and specify only that which applies to the project with paint schedule and Section 09 06 00, SCHEDULE FOR FINISHES.~~

2.1 MATERIALS:

- A. Conform to the coating specifications and standards referenced in PART 3. Submit manufacturer's technical data sheets for specified coatings and solvents.

2.2 PAINT PROPERTIES:

- A. Use ready-mixed (including colors), except two component epoxies, polyurethanes, polyesters, paints having metallic powders packaged separately and paints requiring specified additives.
- B. Where no requirements are given in the referenced specifications for primers, use primers with pigment and vehicle, compatible with substrate and finish coats specified.
- C. Provide undercoat paint produced by the same manufacturer as the finish coats. Use only thinners approved by the paint manufacturer and use only to recommended limits.
- D. ~~//~~VOC Content: For field applications that are inside the weatherproofing system, paints and coating to comply with VOC content limits of authorities having jurisdiction and the following VOC content limits:
1. Flat Paints and Coatings: 50 gram/liter.

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2. Non-flat Paints and Coatings: 150 gram/liter.
3. Dry-Fog Coatings: 400 gram/liter.
4. Primers, Sealers, and Undercoaters: 200 gram/liter.
5. Anticorrosive and Antirust Paints applied to Ferrous Metals: 250 gram/liter.
6. Zinc-Rich Industrial Maintenance Primers: 340 gram/liter.
7. Pretreatment Wash Primers: 420 gram/liter.
8. Shellacs, Clear: 730 gram/liter.
9. Shellacs, Pigmented: 550 gram/liter. ~~++~~

E. VOC test method for paints and coatings is to be in accordance with 40 CFR 59 (EPA Method 24). Part 60, Appendix A with the exempt compounds' content determined by Method 303 (Determination of Exempt Compounds) in the South Coast Air Quality Management District's (SCAQMD) "Laboratory Methods of Analysis for Enforcement Samples" manual.

2.3 PLASTIC TAPE:

- A. Pigmented vinyl plastic film in colors as specified ~~in Section 09 06 00,~~
~~SCHEDULE FOR FINISHES or specified.~~
- B. Pressure sensitive adhesive back.
- C. ~~++~~ Snap on coil plastic markers. ~~++~~
- D. Widths as shown on construction documents.

2.4 BIOBASED CONTENT

- A. Paint products shall comply with following bio-based standards for biobased materials:

Material Type	Percent by Weight
Interior Paint	20 percent biobased material
Interior Paint- Oil Based and Solvent Alkyd	67 percent biobased material
Exterior Paint	20 percent biobased material
Wood & Concrete Stain	39 percent biobased content
Polyurethane Coatings	25 percent biobased content
Water Tank Coatings	59 percent biobased content
Wood & Concrete Sealer- Membrane Concrete Sealers	11 percent biobased content
Wood & Concrete Sealer- Penetrating Liquid	79 percent biobased content

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- B. The minimum-content standards are based on the weight (not the volume) of the material.

PART 3 - EXECUTION

3.1 JOB CONDITIONS:

- A. Safety: Observe required safety regulations and manufacturer's warning and instructions for storage, handling and application of painting materials.
1. Take necessary precautions to protect personnel and property from hazards due to falls, injuries, toxic fumes, fire, explosion, or other harm.
 2. Deposit soiled cleaning rags and waste materials in metal containers approved for that purpose. Dispose of such items off the site at end of each day's work.
- B. Atmospheric and Surface Conditions:
1. Do not apply coating when air or substrate conditions are:
 - a. Less than 3 degrees C (5 degrees F) above dew point.
 - b. Below 10 degrees C (50 degrees F) or over 35 degrees C (95 degrees F), unless specifically pre-approved by the COR and the product manufacturer. Under no circumstances are application conditions to exceed manufacturer recommendations.
 - c. When the relative humidity exceeds 85 percent; or to damp or wet surfaces; unless otherwise permitted by the paint manufacturer's printed instructions.
 2. ~~Maintain interior temperatures until paint dries hard.~~
 3. Do no exterior painting when it is windy and dusty.
 4. Do not paint in direct sunlight or on surfaces that the sun will warm.
 5. Apply only on clean, dry and frost-free surfaces except as follows:
 - a. Apply water thinned acrylic and cementitious paints to damp (not wet) surfaces only when allowed by manufacturer's printed instructions.
 - b. Concrete and masonry when permitted by manufacturer's recommendations, dampen surfaces to which water thinned acrylic and cementitious paints are applied with a fine mist of water on hot dry days to prevent excessive suction and to cool surface.
 6. Varnishing:
 - a. Apply in clean areas and in still air.
 - b. Before varnishing vacuum and dust area.
 - c. Immediately before varnishing wipe down surfaces with a tack rag.

3.2 INSPECTION:

- A. Examine the areas and conditions where painting and finishing are to be applied and correct any conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions are corrected to permit proper installation of the work.

3.3 GENERAL WORKMANSHIP REQUIREMENTS:

- A. Application may be by brush or roller. Spray application only upon acceptance from the COR in writing.
- B. Furnish to the COR a painting schedule indicating when the respective coats of paint for the various areas and surfaces will be completed. This schedule is to be kept current as the job progresses.
- C. Protect work at all times. Protect all adjacent work and materials by suitable covering or other method during progress of work. Upon completion of the work, remove all paint and varnish spots from floors, glass and other surfaces. Remove from the premises all rubbish and accumulated materials of whatever nature not caused by others and leave work in a clean condition.
- D. Remove and protect hardware, accessories, device plates, lighting fixtures, and factory finished work, and similar items, or provide in place protection. Upon completion of each space, carefully replace all removed items by workmen skilled in the trades involved.
- E. When indicated to be painted, remove electrical panel box covers and doors before painting walls. Paint separately and re-install after all paint is dry.
- F. Materials are to be applied under adequate illumination, evenly spread and flowed on smoothly to avoid runs, sags, holidays, brush marks, air bubbles and excessive roller stipple.
- G. Apply materials with a coverage to hide substrate completely. When color, stain, dirt or undercoats show through final coat of paint, the surface is to be covered by additional coats until the paint film is of uniform finish, color, appearance and coverage, at no additional cost to the Government.
- H. All coats are to be dry to manufacturer's recommendations before applying succeeding coats.
- I. All suction spots or "hot spots" in plaster after the application of the first coat are to be touched up before applying the second coat.

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~~J. Do not apply paint behind frameless mirrors that use mastic for adhering to wall surface.~~

~~SPEC WRITER NOTES:~~

- ~~1. Insure other technical sections specify acceptable surface conditions to receive paint including patching and repair of new and existing surfaces.~~
- ~~2. Check structural sections specifying ferrous metal; mechanical and electrical sections of the specifications for proper surface condition and compatible prime coats to suit finishes specified. For instance, finish on concrete required to have cementitious coating; type of shop coat on bar joists required to be painted; will any parts of mechanical equipment have to be field painted; what kind of primers are specified, if any.~~

3.4 SURFACE PREPARATION:

A. General:

1. The Contractor shall be held wholly responsible for the finished appearance and satisfactory completion of painting work. Properly prepare all surfaces to receive paint, which includes cleaning, sanding, and touching-up of all prime coats applied under other Sections of the work. Broom clean all spaces before painting is started. All surfaces to be painted or finished are to be completely dry, clean and smooth.
2. See other sections of specifications for specified surface conditions and prime coat.
3. Perform preparation and cleaning procedures in strict accordance with the paint manufacturer's instructions and as herein specified, for each particular substrate condition.
4. Clean surfaces before applying paint or surface treatments with materials and methods compatible with substrate and specified finish. Remove any residue remaining from cleaning agents used. Do not use solvents, acid, or steam on concrete and masonry. Schedule the cleaning and painting so that dust and other contaminants from the cleaning process will not fall in wet, newly painted surfaces.
5. Maximum Moisture Content of Substrates: When measured with an electronic moisture meter as follows:
 - a. Concrete: 12 percent.

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- b. Fiber-Cement Board: 12 percent.
- c. Masonry (Clay and CMU's): 12 percent.
- d. Wood: 15 percent.
- e. Gypsum Board: 12 percent.
- f. Plaster: 12 percent.

~~B. Wood:~~

- ~~1. Sand to a smooth even surface and then dust off.~~
- ~~2. Sand surfaces showing raised grain smooth between each coat.~~
- ~~3. Wipe surface with a tack rag prior to applying finish.~~
- ~~4. Surface painted with an opaque finish:~~
 - ~~a. Coat knots, sap and pitch streaks with MPI 36 (Knot Sealer) before applying paint.~~
 - ~~b. Apply two coats of MPI 36 (Knot Sealer) over large knots.~~
- ~~5. After application of prime or first coat of stain, fill cracks, nail and screw holes, depressions and similar defects with wood filler paste. Sand the surface to make smooth and finish flush with adjacent surface.~~
- ~~6. Before applying finish coat, reapply wood filler paste if required, and sand surface to remove surface blemishes. Finish flush with adjacent surfaces.~~
- ~~7. Fill open grained wood such as oak, walnut, ash and mahogany with MPI 91 (Wood Filler Paste), colored to match wood color.~~
 - ~~a. Thin filler in accordance with manufacturer's instructions for application.~~
 - ~~b. Remove excess filler, wipe as clean as possible, dry, and sand as specified.~~

~~C.B.~~ Ferrous Metals:

- 1. Remove oil, grease, soil, drawing and cutting compounds, flux and other detrimental foreign matter in accordance with SSPC-SP 1 (Solvent Cleaning).
- 2. Remove loose mill scale, rust, and paint, by hand or power tool cleaning, as defined in SSPC-SP 2 (Hand Tool Cleaning) and SSPC-SP 3 (Power Tool Cleaning). ~~//~~Where high temperature aluminum paint is used, prepare surface in accordance with paint manufacturer's instructions.~~//~~
- 3. Fill dents, holes and similar voids and depressions in flat exposed surfaces of hollow steel doors and frames, access panels, roll-up steel

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doors and similar items specified to have semi-gloss or gloss finish with TT-F-322D (Filler, Two-Component Type, For Dents, Small Holes and Blow-Holes). Finish flush with adjacent surfaces.

- a. Fill flat head countersunk screws used for permanent anchors.
 - b. Do not fill screws of item intended for removal such as glazing beads.
4. Spot prime abraded and damaged areas in shop prime coat which expose bare metal with same type of paint used for prime coat. Feather edge of spot prime to produce smooth finish coat.
 5. Spot prime abraded and damaged areas which expose bare metal of factory finished items with paint as recommended by manufacturer of item.

~~D.C. // Zinc-Coated (Galvanized) Metal and ,// // Aluminum, // // Copper~~
~~and Copper Alloys //~~ Surfaces Specified Painted:

1. Clean surfaces to remove grease, oil and other deterrents to paint adhesion in accordance with SSPC-SP 1 (Solvent Cleaning).
2. Spot coat abraded and damaged areas of zinc-coating which expose base metal on hot-dip zinc-coated items with MPI 18 (Organic Zinc Rich Coating). Prime or spot prime with MPI 134 (Waterborne Galvanized Primer) or MPI 135 (Non-Cementitious Galvanized Primer) depending on finish coat compatibility.

~~E. Masonry, Concrete, Cement Board, Cement Plaster and Stucco:~~

- ~~1. Clean and remove dust, dirt, oil, grease efflorescence, form release agents, laitance, and other deterrents to paint adhesion.~~
- ~~2. Use emulsion type cleaning agents to remove oil, grease, paint and similar products. Use of solvents, acid, or steam is not permitted.~~
- ~~3. Remove loose mortar in masonry work.~~
- ~~4. Replace mortar and fill open joints, holes, cracks and depressions with new mortar specified in Section 04 05 13, MASONRY MORTARING // Section 04 05 16, MASONRY GROUTING //. Do not fill weep holes. Finish to match adjacent surfaces.~~
- ~~5. Neutralize Concrete floors to be painted by washing with a solution of 1.4 Kg (3 pounds) of zinc sulfate crystals to 3.8 L (1 gallon) of water, allow to dry three (3) days and brush thoroughly free of crystals.~~
- ~~6. Repair broken and spalled concrete edges with concrete patching compound to match adjacent surfaces as specified in~~

~~Division 03, CONCRETE Sections. Remove projections to level of adjacent surface by grinding or similar methods.~~

~~F. Gypsum Plaster and Gypsum Board:~~

- ~~1. Remove efflorescence, loose and chalking plaster or finishing materials.~~
- ~~2. Remove dust, dirt, and other deterrents to paint adhesion.~~
- ~~3. Fill holes, cracks, and other depressions with CID-A-A-1272A finished flush with adjacent surface, with texture to match texture of adjacent surface. Patch holes over 25 mm (1-inch) in diameter as specified in Section for plaster or gypsum board.~~

3.5 PAINT PREPARATION:

- A. Thoroughly mix painting materials to ensure uniformity of color, complete dispersion of pigment and uniform composition.
- B. Do not thin unless necessary for application and when finish paint is used for body and prime coats. Use materials and quantities for thinning as specified in manufacturer's printed instructions.
- C. Remove paint skins, then strain paint through commercial paint strainer to remove lumps and other particles.
- D. Mix two (2) component and two (2) part paint and those requiring additives in such a manner as to uniformly blend as specified in manufacturer's printed instructions unless specified otherwise.
- E. For tinting required to produce exact shades specified, use color pigment recommended by the paint manufacturer.

3.6 APPLICATION:

- A. Start of surface preparation or painting will be construed as acceptance of the surface as satisfactory for the application of materials.
- B. Unless otherwise specified, apply paint in three (3) coats; prime, body, and finish. When two (2) coats applied to prime coat are the same, first coat applied over primer is body coat and second coat is finish coat.
- C. Apply each coat evenly and cover substrate completely.
- D. Allow not less than 48 hours between application of succeeding coats, except as allowed by manufacturer's printed instructions, and approved by COR.

~~SPEC WRITER NOTE: Do not allow spray painting at existing buildings occupied during the course of the work. Spray painting may be allowed in certain areas of~~

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~~new additions or separate buildings.~~
~~Coordinate restrictions with COR.~~

- E. Apply by brush or roller. Spray application for new or existing occupied spaces only upon approval by acceptance from COR in writing.

~~SPEC WRITER NOTE: Check application requirements with manufacturer of materials specified to determine if below paragraphs are applicable.~~

- ~~1. Apply painting materials specifically required by manufacturer to be applied by spraying.~~
- ~~2. In new construction and in existing occupied spaces, where paint is applied by spray, mask or enclose with polyethylene, or similar air tight material with edges and seams continuously sealed including items specified in "Building and Structural Work Field Painting"; "Work not Painted"; motors, controls, telephone, and electrical equipment, fronts of sterilizes and other recessed equipment and similar prefinished items.~~

- ~~F. Do not paint in closed position operable items such as access doors and panels, window sashes, overhead doors, and similar items except overhead roll-up doors and shutters.~~

3.7 **PRIME PAINTING:**

- A. After surface preparation, prime surfaces before application of body and finish coats, except as otherwise specified.
- B. Spot prime and apply body coat to damaged and abraded painted surfaces before applying succeeding coats.
- C. Additional field applied prime coats over shop or factory applied prime coats are not required except for exterior exposed steel apply an additional prime coat.
- D. Prime rabbets for stop and face glazing of wood, and for face glazing of steel.

~~E. Wood and Wood Particleboard:~~

- ~~1. Use same kind of primer specified for exposed face surface.~~
 - ~~a. Exterior wood: MPI 7 (Exterior Oil Wood Primer) for new construction and MPI 5 (Exterior Alkyd Wood Primer) for repainting bare wood primer except where MPI 90 (Interior Wood Stain, Semi-Transparent) is scheduled.~~

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- ~~b. Interior wood except for transparent finish: MPI 45 (Interior Primer Sealer) or MPI 46 (Interior Enamel Undercoat), thinned if recommended by manufacturer.~~
- ~~c. Transparent finishes as specified under "Transparent Finishes on Wood Except Floors Article" // and "Finish for Wood Floors Article" //.~~
- ~~2. Apply two (2) coats of primer MPI 7 (Exterior Oil Wood Primer) or MPI 5 (Exterior Alkyd Wood Primer) or sealer MPI 45 (Interior Primer Sealer) or MPI 46 (Interior Enamel Undercoat) to surfaces of wood doors, including top and bottom edges, which are cut for fitting or for other reason.~~
- ~~3. Apply one (1) coat of primer MPI 7 (Exterior Oil Wood Primer) or MPI 5 (Exterior Alkyd Wood Primer) or sealer MPI 45 (Interior Primer Sealer) or MPI 46 (Interior Enamel Undercoat) as soon as delivered to site to surfaces of unfinished woodwork, except concealed surfaces of shop-fabricated or assembled millwork and surfaces specified to have varnish, stain or natural finish.~~
- ~~4. Back prime and seal ends of exterior woodwork, and edges of exterior plywood specified to be finished.~~
- ~~5. Apply MPI 67 (Interior Latex Fire Retardant, Top-Coat (UL Approved) to wood for fire retardant finish.~~

F.E. Metals except boilers, incinerator stacks, and engine exhaust pipes:

- 1. Steel and iron: ~~//MPI 79 (Marine Alkyd Metal Primer); //MPI 95 (Fast Drying Metal Primer) //~~. Use MPI 101 (Cold Curing Epoxy Primer) where ~~//MPI 77 (Epoxy Cold Cured, Gloss; //MPI 98 (High Build Epoxy Coating); or //MPI 108 (High Build Epoxy Marine Coating) //~~ finish as is specified.
- 2. Zinc-coated steel and iron: ~~//MPI 134 (Waterborne Galvanized Primer); //MPI 135 (Non-Cementitious Galvanized Primer) //~~.
- 3. Aluminum scheduled to be painted: MPI 95 (Fast Drying Metal Primer).
- 4. Terne Metal: ~~//MPI 79 (Marine Alkyd Metal Primer); //MPI 95 (Fast Drying Metal Primer) //~~.
- 5. Copper and copper alloys scheduled to be painted: MPI 95 (Fast Drying Metal Primer).
- 6. Machinery not factory finished: MPI 9 (Exterior Alkyd Enamel).
- 7. Asphalt coated metal: MPI 1 (Aluminum Paint).

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8. Metal over 94 degrees C (201 degrees F), Boilers, Incinerator Stacks, and Engine Exhaust Pipes: MPI 22 (High Heat Resistant Coating).

~~G. Gypsum Board // and Hardboard //:~~

- ~~1. Surfaces scheduled to have // MPI 10 (Exterior Latex, Flat) // // MPI 11 (Exterior Latex, Semi Gloss) // // MPI 119 (Exterior Latex, High Gloss (acrylic)) // // MPI 53 (Interior Latex, Flat) //, MPI Gloss Level 1 // MPI 52 (Interior Latex, MPI Gloss Level 3) // // MPI 54 (Interior Latex, Semi Gloss, MPI Gloss Level 5) // // MPI 114 (Interior Latex, Gloss) finish: Use // MPI 10 (Exterior Latex, Flat) // // MPI 11 (Exterior Latex, Semi Gloss) // // MPI 119 (Exterior Latex, High Gloss (acrylic)) // // MPI 53 (Interior Latex, MPI Gloss Level 3) // // MPI 52 (Interior Latex, MPI Gloss Level 3) // // MPI 54 (Interior Latex, Semi Gloss, MPI Gloss Level 5) // // MPI 114 (Interior Latex, Gloss) respectively //.~~

~~SPEC WRITER NOTE: List the names of other spaces, where steam will be generated or that have high humidity for pigmented sealer.~~

- ~~2. Primer: // MPI 50 (Interior Latex Primer Sealer) except use MPI 45 (Interior Primer Sealer) // // MPI 46 (Interior Enamel Undercoat) // in shower and bathrooms.~~
- ~~3. Surfaces scheduled to receive vinyl coated fabric wall covering:~~
- ~~a. // Use MPI 45 (Interior Primer Sealer) // // MPI 46 (Interior Enamel Undercoat) //.~~
- ~~b. Use // MPI 101 (Cold Curing Epoxy Primer) for surfaces scheduled to receive MPI 77 (Epoxy Cold Cured, Gloss) // // MPI 98 (High Build Epoxy Coating) // // MPI 108 (High Build Epoxy Marine Coating) finish //.~~

~~H. Gypsum Plaster and Veneer Plaster:~~

- ~~1. Surfaces scheduled to receive vinyl coated fabric wall covering: Use MPI 45 (Interior Primer Sealer).~~
- ~~2. MPI 45 (Interior Primer Sealer), except use MPI 50 (Interior Latex Primer Sealer) when an alkyd flat finish is specified.~~
- ~~3. Surfaces scheduled to have // MPI 10 (Exterior Latex, Flat) // // MPI 11 (Exterior Latex, Semi Gloss) // // MPI 119 (Exterior Latex, High Gloss (acrylic)) // // MPI 53 (Interior Latex, Flat, MPI Gloss Level 1) // // MPI 52 (Interior Latex, MPI Gloss Level 3) // // MPI 54~~

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- ~~(Interior Latex, Semi-Gloss, MPI Gloss Level 5) // // MPI 114 (Interior Latex, Gloss) // finish: Use MPI 10 (Exterior Latex, Flat) // MPI 11 (Exterior Latex, Semi-Gloss) // // MPI 119 (Exterior Latex, High Gloss (acrylic)) // // MPI 53 (Interior Latex, Flat, MPI Gloss Level 1) // // MPI 52 Latex, MPI Gloss Level 3) // // MPI 54 (Interior Latex, Semi-Gloss, MPI Gloss Level 5) // // MPI 114 (Interior Latex, Gloss) // respectively.~~
- ~~4. Use // MPI 101 (Cold Curing Epoxy Primer) for surfaces scheduled to receive // // MPI 77 (Epoxy Cold Cured, Gloss) // // MPI 108 (High Build Epoxy Marine Coating) // finish.~~
- ~~I. Concrete Masonry Units except glazed or integrally colored and decorative units:-~~
- ~~1. MPI 4 (Block Filler) on interior surfaces.~~
- ~~2. Prime exterior surface as specified for exterior finishes.~~
- ~~J. Cement Plaster or stucco // Concrete Masonry, Brick Masonry // and // Cement board // Interior Surfaces of Ceilings and Walls:-~~
- ~~1. // MPI 53 (Interior Latex, Flat, MPI Gloss Level 1) // // MPI 52 (Interior Latex, MPI Gloss Level 3) // // MPI 54 (Interior Latex, Semi-Gloss, MPI Gloss Level 5) // // MPI 114 (Interior Latex, Gloss) // except use two (2) coats where substrate has aged less than six (6) months.~~
- ~~2. Use // MPI 138 (Interior High Performance Latex, MPI Gloss Level 2) // // MPI 139 (Interior High Performance Latex, MPI Gloss level 3) // // MPI 140 (Interior High Performance latex, MPI Gloss Level 4) // // MPI 141 (Interior High Performance Latex, MPI Gloss Level 5) // // MPI 114 (Interior Latex, Gloss) // // TT-P-1411A (Paint, Copolymer Resin, Cementitious) Type II // // MPI 77 (Epoxy Cold Cured, Gloss // // MPI 98 (High Build Epoxy Coating) // MPI 108 (High Build Epoxy Marine Coating) as scheduled.~~
- ~~K. Concrete Floors: // MPI 68 (Interior/ Exterior Latex Porch & Floor Paint, Gloss) // // MPI 60 (Interior/ Exterior Latex Porch & Floor Paint, Low Gloss) //. // MPI 99 (Water-based Acrylic Curing and Sealing Compound).//~~
- 3.8 -EXTERIOR FINISHES:**
- ~~A. Apply following finish coats where specified in Section 09 06 00, SCHEDULE FOR FINISHES.~~

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~~B. Wood:~~

- ~~1. Do not apply finish coats on surfaces concealed after installation, top and bottom edges of wood doors and sash, or on edges of wood framed insect screens.~~
- ~~2. Two (2) coats of // MPI 10 Exterior Latex, Flat) // // MPI 11 (Exterior Latex, Semi-Gloss) // // MPI 119 (Exterior Latex, High Gloss (acrylic)) // on exposed surfaces, except where transparent finish is specified.~~
- ~~3. Two (2) coats of // MPI 31 (Polyurethane, Moisture Cured, Clear Gloss) // // MPI 71 (Polyurethane, Moisture Cured, Clear Flat) // for transparent finish.~~

~~C.A. Steel and Ferrous Metal~~//, Including Tern//:

- ~~1. Two (2) coats of~~// MPI 8 (Exterior Alkyd, Flat); // MPI 9 (Exterior Alkyd Enamel); or ~~// MPI 94 (Exterior Alkyd, Semi-Gloss)~~ // on exposed surfaces, except on surfaces over 94 degrees C (201 degrees F).
- ~~2. One (1) coat of MPI 22 (High Heat Resistant Coating) on surfaces over 94 degrees K (290 degrees F).~~ and on surfaces of // boiler // ~~//, incinerator // //, stacks // engine exhaust pipes.~~

~~D. Machinery without factory finish except for primer: One (1) coat // MPI 8 (Exterior Alkyd, Flat) // // MPI 9 (Exterior Alkyd Enamel) // // MPI 94 (Exterior Alkyd, Semi-Gloss) //.~~

~~SPEC WRITER NOTES:~~

- ~~1. The following finishes are applicable to brick, concrete masonry units, concrete, cement board, cement plaster and stucco.~~
- ~~2. Exterior concrete, brick, stucco or cement plaster and cement boards are normally not painted. Coordinate with Section 09 06 00, SCHEDULE FOR FINISHES and specify additional surfaces scheduled for paint.~~
- ~~3. These paints will fill and seal pores and waterproof the wall but allow transmission of water vapor.~~
- ~~4. Cementitious paint TT-P-1411A (Paint, Copolymer Resin, Cementitious (CEP)) is a factory prepared mix ready to apply by brush.~~

~~E. Concrete Masonry Units // Brick // // Cement Plaster // // Concrete //:~~

~~1. General:~~

- ~~a. Where specified in Section 09 06 00, SCHEDULE FOR FINISHES or shown.~~
- ~~b. Mix as specified in manufacturer's printed directions.~~

- ~~e. Do not mix more paint than can be used within four (4) hours after mixing. Discard paint that has started to set.~~
- ~~d. Dampen warm surfaces above 24 degrees C (75 degrees F) with fine mist of water before application of paint. Do not leave free water on surface.~~
- ~~e. Cure paint with a fine mist of water as specified in manufacturer's printed instructions.~~
- ~~2. Use two (2) coats of TT P 1411 (Paint, Co polymer Resin, Cementitious), unless specified otherwise.~~

~~3.0 INTERIOR FINISHES:~~

- ~~A. Apply following finish coats over prime coats in spaces or on surfaces specified in Section 09 06 00, SCHEDULE FOR FINISHES.~~

~~SPECS WRITER NOTE: List other metals and finish coats required for field painting in Section 09 06 00, SCHEDULE FOR FINISHES.~~

~~B. Metal Work:~~

- ~~1. Apply to exposed surfaces.~~
- ~~2. Omit body and finish coats on surfaces concealed after installation except electrical conduit containing conductors over 600 volts.~~
- ~~3. Ferrous Metal, Galvanized Metal, and Other Metals Scheduled:~~
 - ~~a. Apply two (2) coats of MPI 47 (Interior Alkyd, Semi-Gloss) unless specified otherwise.~~
 - ~~b. Two (2) coats of // MPI 48 (Interior Alkyd Gloss) // // MPI 51 (Interior Alkyd, Eggshell) //.~~
 - ~~c. One (1) coat of MPI 46 (Interior Enamel Undercoat) plus one coat of MPI 47 (Interior Alkyd, Semi-Gloss) on exposed interior surfaces of alkyd-amine enamel prime finished windows.~~
 - ~~d. One (1) coat of MPI 101 primer over two (2) coats of waterborne light industrial coating MPI 163 on exposed surfaces in // battery rooms // // pool area // // chlorinator rooms //. Steel is to be blast cleaned to SSPC 10/NACE No. 2.~~
 - ~~e. Machinery: One (1) coat MPI 9 (Exterior Alkyd Enamel).~~
 - ~~f. Asphalt Coated Metal: One (1) coat MPI 1 (Aluminum Paint).~~
 - ~~g. Ferrous Metal over 94 degrees K (290 degrees F): Boilers, Incinerator Stacks, and Engine Exhaust Pipes: One (1) coat MPI 22 (High Heat Resistant Coating).~~

~~C. Gypsum Board:~~

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1. ~~One (1) coat of // MPI 45 (Interior Primer Sealer) // // MPI 46 (Interior Enamel Undercoat) // plus one (1) coat of MPI 139 (Interior High Performance Latex, MPI Gloss level 3).~~
2. ~~Two (2) coats of MPI 138 (Interior High Performance Latex, MPI Gloss Level 2).~~
3. ~~One (1) coat of // MPI 45 (Interior Primer Sealer) // // MPI 46 (Interior Enamel Undercoat) // plus one (1) coat of MPI 54 (Interior Latex, Semi Gloss, MPI Gloss Level 5) or MPI 114 (Interior Latex, Gloss).~~
4. ~~One (1) coat of // MPI 45 (Interior Primer Sealer) // MPI 46 (Interior Enamel Undercoat) // plus one (1) coat of MPI 48 (Interior Alkyd Gloss).~~

~~D. Plaster:~~

1. ~~One (1) coat of // MPI 45 (Interior Primer Sealer) // MPI 46 (Interior Enamel Undercoat) // MPI 50 (Interior Latex Primer Sealer) // plus one (1) coat of MPI 139 (Interior High Performance Latex, MPI Gloss level 3).~~
2. ~~Two (2) coats of MPI 51 (Interior Alkyd, Eggshell).~~
3. ~~One (1) coat of // MPI 45 (Interior Primer Sealer) // // MPI 46 (Interior Enamel Undercoat) // or MPI 50 (Interior Latex Primer Sealer) plus one (1) coat of 139 (Interior High Performance Latex, MPI Gloss level 3).~~
4. ~~One (1) coat MPI 101 (Cold Curing Epoxy Prime).~~

~~E. Masonry and Concrete Walls:~~

1. ~~Over MPI 4 (Interior/Exterior Latex Block Filler) on CMU surfaces.~~
2. ~~Two (2) coats of // MPI 53 (Interior Latex, Flat, MPI Gloss Level 1) // // MPI 52 (Interior Latex, MPI Gloss Level 3) // // MPI 54 (Interior Latex, Semi Gloss, MPI Gloss Level 5) // // MPI 114 (Interior Latex, Gloss) //.~~
3. ~~Two (2) coats of // MPI 138 (Interior High Performance Latex, MPI Gloss Level 2) // // MPI 139 (Interior High Performance Latex, MPI Gloss Level 3) // // MPI 140 (Interior High Performance Latex MPI Gloss Level 4) // // MPI 141 (Interior High Performance Latex MPI Gloss Level 5) // // MPI 114 (Interior Latex, Gloss) //.~~

~~F. Wood:~~

1. ~~Sanding:~~
 - a. ~~Use 220 grit sandpaper.~~

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- ~~b. Sand sealers and varnish between coats.~~
- ~~c. Sand enough to scarify surface to assure good adhesion of subsequent coats, to level roughly applied sealer and varnish, and to knock off "whiskers" of any raised grain as well as dust particles.~~
- ~~2. Sealers:~~
 - ~~a. MPI 31 (gloss) or MPI 71 (flat) thinned as recommended by manufacturer at rate of one (1) part of thinner to four (4) parts of varnish.~~
 - ~~b. Apply sealers specified except sealer may be omitted where pigmented, penetrating, or wiping stains containing resins are used.~~
 - ~~c. Allow manufacturer's recommended drying time before sanding, but not less than 24 hours or 36 hours in damp or muggy weather.~~
 - ~~d. Sand as specified.~~
- ~~3. Paint Finish:~~
 - ~~a. One (1) coat of // MPI 45 (Interior Primer Sealer) // // MPI 46 (Interior Enamel Undercoat) // plus one (1) coat of MPI 47 (Interior Alkyd, Semi Gloss.~~
 - ~~b. One (1) coat // MPI 66 (Interior Alkyd Fire retardant, Clear Top Coat (UL Approved) // // MPI 67 (Interior Latex Fire Retardant, Top Coat (UL Approved), intumescent type, on exposed wood // in attics with floors used for mechanical equipment // // and above ceilings where shown //.~~
 - ~~c. One (1) coat of // MPI 45 Interior Primer Sealer) // // MPI 46 (Interior Enamel Undercoat) // plus one (1) coat of MPI 48 (Interior Alkyd Gloss).~~
 - ~~d. Two (2) coats of MPI 51 (Interior Alkyd, Eggshell).~~
- ~~4. Transparent Finishes on Wood Except Floors.~~
 - ~~a. Natural Finish:~~
 - ~~One (1) coat of sealer // MPI 31 (gloss) // // MPI 71 (flat) // thinned with thinner recommended by manufacturer at rate of one (1) part of thinner to four (4) parts of varnish.~~
 - ~~Two (2) coats of MPI 71 (Polyurethane, Moisture Cured, Clear Flat // MPI 31 (Polyurethane, Moisture Cured, Clear Gloss.~~

~~SPEC WRITER NOTES:~~

- ~~1. Stain may be used when transparent finishes are specified to change the color of sapwood to match heartwood, and to enhance or even the color of the wood~~

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~~as required to match the finish specified.~~

~~2. Verify requirements for stain with Section 09 06 00, SCHEDULE FOR FINISHES and woods used.~~

~~b. Stain Finish:~~

~~One (1) coat of MPI 90 (Interior Wood Stain, Semi-Transparent).~~

~~Use wood stain of type and color required to achieve finish specified. Do not use varnish type stains.~~

~~One (1) coat of sealer // MPI 31 (gloss) // // MPI 71 (flat) // thinned as recommended by manufacturer at rate of one (1) part of thinner to four (4) parts of varnish.~~

~~Two (2) coats of // MPI 71 (Polyurethane, Moisture Cured, Clear Flat) // // MPI 31 (Polyurethane Moisture Cured, Clear Gloss) //.~~

~~c. Varnish Finish:~~

~~One (1) coat of sealer // MPI 31 (gloss) // // MPI 71 (flat) // thinned as recommended by manufacturer at rate of one (1) part of thinner to four (4) parts of varnish.~~

~~Two (2) coats of // MPI 71 (Polyurethane, Moisture Cured, Clear Flat) // // MPI 31 (Polyurethane Moisture Cured, Clear Gloss) //.~~

~~d. Fire Retardant Intumescent Varnish:~~

~~MPI 66 (Interior Alkyd Fire Retardant, Clear Top Coat (UL Approved)) Intumescent Type, Fire Retardant Coating where scheduled: Two (2) coats.~~

~~5. Finish for Wood Floors:~~

~~a. Hardwood Flooring:~~

~~Apply MPI 91 (Wood Filler Paste) to open grained wood. Remove surplus filler and wipe clean.~~

~~Sand lightly when dry. Remove dust.~~

~~Apply two (2) coats of CID-A-A-2335 (Sealer, Surface).~~

~~Apply two (2) thin coats of P-W-155 (Wax Floor, Water Emulsion) and machine buff to uniform luster.~~

~~b. Stage Floor: Sand only. No filling, sealing, or waxing is required.~~

~~c. // Exercise Area // // Recreation Hall //, // Gymnasium //,~~

~~// Handball Boards in Exercise Area // Floor Finish:~~

~~Floor Sealer Formulation: Pliable, penetrating type, MFMA Group I, Sealers.~~

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~~Finish-Coat Formulation: Formulated for gloss finish and multicoat application.~~

~~Type: MFMA Group 5, Water Based Finishes.~~

~~Allow 48 hours between coats.~~

~~Apply in one (1) continuous operation with squeegee or lamb's wool applicator with application free from streaks in accordance with plastic coating manufacturer's directions.~~

~~SPEC WRITER NOTE: Verify stripe width, layouts and colors are to be shown in construction documents.~~

~~d. Striping:~~

~~Where striping is shown on construction documents for wood floors, apply pressure sensitive adhesive back vinyl plastic tape stripes in widths shown in construction documents.~~

~~Do striping when floor coating is dry.~~

~~Install stripes to straight lines and true curves.~~

~~Provide colors as specified in Section 09 06 00, SCHEDULE FOR FINISHES or indicated in construction documents.~~

~~G. Cement Board: One (1) coat of // MPI 138 (Interior High Performance Latex, MPI Gloss Level 2) // // MPI 139 (Interior High Performance Latex, MPI Gloss Level 3) // // MPI 140 (Interior High Performance Latex MPI Gloss Level 4) // // MPI 141 (Interior High Performance Latex, MPI Gloss Level 5 // // MPI 114 (Interior Latex, Gloss) //.~~

~~H. Concrete Floors: One (1) coat of MPI 68 (Interior/ Exterior Latex Porch & Floor Paint, Gloss).~~

~~I. Miscellaneous:~~

~~1. Apply where specified in Section 09 06 00, SCHEDULE FOR FINISHES.~~

~~2. MPI 1 (Aluminum Paint): Two (2) coats of aluminum paint.~~

~~3. Existing acoustical units scheduled to be repainted except acoustical units with a vinyl finish:~~

~~a. Clean units free of dust, dirt, grease, and other deterrents to paint adhesion.~~

~~b. Mineral fiber units: One (1) coat of // MPI 53 (Interior Latex, Flat, MPI Gloss Level 1) // // MPI 52 (Interior Latex, MPI Gloss Level 3) // // MPI 54 (Interior Latex, Semi-Gloss, MPI Gloss Level 5) // // MPI 114 (Interior Latex, Gloss) //.~~

- ~~e. Units of organic fiber or other material not having a class A rating: One (1) coat of // MPI 66 (Interior Alkyd Fire Retardant, Clear Top Coat (UL Approved)) // // MPI 67 (Interior Latex Fire Retardant, Top Coat (UL Approved)) // fire retardant paint.~~
- ~~4. Interstitial floor markings: One (1) coat // MPI 27 (Exterior/ Interior Alkyd Floor Enamel, Gloss) // // MPI 59 ((Interior/ Exterior Alkyd Porch & Floor Enamel, Low Gloss) // // MPI 68 (Interior/ Exterior Latex Porch & Floor Paint, Gloss) // // MPI 60 (interior/ Exterior Latex Porch & Floor Paint, Low Gloss) //.~~

~~3.103.9~~ **REFINISHING EXISTING PAINTED SURFACES:**

- A. Clean, patch and repair existing surfaces as specified under "Surface Preparation". No "telegraphing" of lines, ridges, flakes, etc., through new surfacing is permitted. Where this occurs, sand smooth and re-finish until surface meets with COR's approval.
- B. Remove and reinstall items as specified under "General Workmanship Requirements".
- C. Remove existing finishes or apply separation coats to prevent non compatible coatings from having contact.
- D. Patched or Replaced Areas in Surfaces and Components: Apply spot prime and body coats as specified for new work to repaired areas or replaced components.
- E. Except where scheduled for complete painting apply finish coat over plane surface to nearest break in plane, such as corner, reveal, or frame.
- F. In existing rooms and areas where alterations occur, clean existing stained and natural finished wood retouch abraded surfaces and then give entire surface one (1) coat of ~~// MPI 31 (Polyurethane, Moisture Cured, Clear Gloss); or // MPI 71 (Polyurethane, Moisture Cured, Clear Flat)~~ ~~//.~~
- G. Refinish areas as specified for new work to match adjoining work unless specified or scheduled otherwise.
- H. Coat knots and pitch streaks showing through old finish with MPI 36 (Knot Sealer) before refinishing.
- I. Sand or dull glossy surfaces prior to painting.
- J. Sand existing coatings to a feather edge so that transition between new and existing finish will not show in finished work.

~~3.113.10~~ **PAINT COLOR:**

A. Color and gloss of finish coats is specified in ~~Section 09 06 00, SCHEDULE FOR FINISHES.~~ the drawings.

B. ~~For additional requirements regarding color see Articles, "REFINISHING EXISTING PAINTED SURFACE" and "MECHANICAL AND ELECTRICAL FIELD PAINTING SCHEDULE".~~

~~C.B.~~ Coat Colors:

1. Color of priming coat: Lighter than body coat.
2. Color of body coat: Lighter than finish coat.
3. Color prime and body coats to not show through the finish coat and to mask surface imperfections or contrasts.

~~D. Painting, Caulking, Closures, and Fillers Adjacent to Casework:~~

1. ~~Paint to match color of casework where casework has a paint finish.~~
2. ~~Paint to match color of wall where casework is stainless steel, plastic laminate, or varnished wood.~~

~~3.12~~ **MECHANICAL AND ELECTRICAL WORK FIELD PAINTING SCHEDULE:**

A. ~~Field painting of mechanical and electrical consists of cleaning, touching up abraded shop prime coats, and applying prime, body and finish coats to materials and equipment if not factory finished in space scheduled to be finished.~~

B. ~~In spaces not scheduled to be finish painted in Section 09 06 00, SCHEDULE FOR FINISHES paint as specified below.~~

C. ~~Paint various systems specified in Division 02 - EXISTING CONDITIONS, Division 21 - FIRE SUPPRESSION, Division 22 - PLUMBING, Division 23 - HEATING, VENTILATION AND AIR CONDITIONING, Division 26 - ELECTRICAL, Division 27 - COMMUNICATIONS, and Division 28 - ELECTRONIC SAFETY AND SECURITY.~~

D. ~~Paint after tests have been completed.~~

E. ~~Omit prime coat from factory prime-coated items.~~

F. ~~Finish painting of mechanical and electrical equipment is not required when located in interstitial spaces, above suspended ceilings, in concealed areas such as pipe and electric closets, pipe basements, pipe tunnels, trenches, attics, roof spaces, shafts and furred spaces except on electrical conduit containing feeders 600 volts or more.~~

G. ~~Omit field painting of items specified in "BUILDING AND STRUCTURAL WORK FIELD PAINTING"; "Building and Structural Work not Painted".~~

H. ~~Color:~~

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~~1. Paint items having no color specified in Section 09 06 00, SCHEDULE FOR FINISHES to match surrounding surfaces.~~

~~2. Paint colors as specified in Section 09 06 00, SCHEDULE FOR FINISHES except for following:~~

~~SPEC WRITER NOTE: Do not change the following color designation.~~

~~a. White: Exterior unfinished surfaces of enameled plumbing fixtures. Insulation coverings on breeching and uptake inside boiler house, drums and drum heads, oil heaters, condensate tanks and condensate piping.~~

~~b. Gray: Heating, ventilating, air conditioning and refrigeration equipment (except as required to match surrounding surfaces), and water and sewage treatment equipment and sewage ejection equipment.~~

~~c. Aluminum Color: Ferrous metal on outside of boilers and in connection with boiler settings including supporting doors and door frames and fuel oil burning equipment, and steam generation system (bare piping, fittings, hangers, supports, valves, traps and miscellaneous iron work in contact with pipe).~~

~~d. Federal Safety Red: Exposed fire protection piping, hydrants, post indicators, electrical conduits containing fire alarm control wiring, and fire alarm equipment.~~

~~e. Federal Safety Orange: Entire lengths of electrical conduits containing feeders 600 volts or more.~~

~~f. Color to match brickwork sheet metal covering on breeching outside of exterior wall of boiler house.~~

~~I. Apply paint systems on properly prepared and primed surface as follows:~~

~~1. Exterior Locations:~~

~~a. Apply two (2) coats of // MPI 8 (Exterior Alkyd, Flat) // // MPI 94 (Exterior Alkyd, Semi-gloss) // // MPI 9 (Exterior Alkyd Enamel) // to the following ferrous metal items:~~

~~Vent and exhaust pipes with temperatures under 94 degrees C (201 degrees F), roof drains, fire hydrants, post indicators, yard hydrants, exposed piping and similar items.~~

~~b. Apply two (2) coats of // MPI 10 (Exterior Latex, Flat) // // MPI 11 (Exterior Latex, Semi-Gloss) // // MPI 119 (Exterior Latex, High-Gloss (acrylic)) // to galvanized and zinc-copper alloy metal.~~

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~~e. Apply one (1) coat of MPI 22 (High Heat Resistant Coating),
650 degrees C (1200 degrees F) to incinerator stacks, boiler stacks,
and engine generator exhaust.~~

~~2. Interior Locations:~~

~~a. Apply two (2) coats of MPI 47 (Interior Alkyd, Semi Gloss) to
following items:~~

~~Metal under 94 degrees C (201 degrees F) of items such as bare
piping, fittings, hangers and supports.~~

~~Equipment and systems such as hinged covers and frames for control
cabinets and boxes, cast iron radiators, electric conduits and
panel boards.~~

~~Heating, ventilating, air conditioning, plumbing equipment, and
machinery having shop prime coat and not factory finished.~~

~~b. Ferrous metal exposed in hydrotherapy equipment room and chlorinator
room of water and sewerage treatment plants: One (1) coat of MPI 101
(Cold Curing Epoxy Primer) and one (1) coat of // MPI 77 (Epoxy Cold
Cured, Gloss // // MPI 98 (High Build Epoxy Coating)) // // MPI 108
(High Build Epoxy Marine coating) //.~~

~~c. Apply one (1) coat of MPI 50 (Interior Latex Primer Sealer) and
one (1) coat of // MPI 53 (Interior Latex, Flat, MPI Gloss Level 1)
// // MPI 44 (Interior Low Sheen Latex) // // MPI 52 (Interior
Latex, MPI Gloss Level 3) // //MPI 43 (Interior Satin Latex) // // MPI 54
(Interior Latex, Semi-Gloss, MPI Gloss Level 5) // // MPI 114
(Interior Latex, Gloss) // on finish of insulation on boiler
breaching and uptakes inside boiler house, drums, drumheads, oil
heaters, feed water heaters, tanks and piping.~~

~~d. Apply two (2) coats of MPI 22 (High Heat Resistant Coating) to
ferrous metal surface over 94 degrees K (290 degrees F) of following
items:~~

~~Garbage and trash incinerator.~~

~~Medical waste incinerator.~~

~~Exterior of boilers and ferrous metal in connection with boiler
settings including supporting members, doors and door frames and
fuel oil burning equipment.~~

~~Steam line flanges, bare pipe, fittings, valves, hangers and
supports over 94 degrees K (290 degrees F).~~

~~Engine generator exhaust piping and muffler.~~

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~~e. Paint electrical conduits containing cables rated 600 volts or more using two (2) coats of // MPI 9 (Exterior Alkyd Enamel) // // MPI 8 (Exterior Alkyd, Flat) // // MPI 94 (Exterior Alkyd, Semi-gloss) // in the Federal Safety Orange color in exposed and concealed spaces full length of conduit.~~

~~3. Other exposed locations:~~

~~a. Metal surfaces, except aluminum, of cooling towers exposed to view, including connected pipes, rails, and ladders: Two (2) coats of MPI 1 (Aluminum Paint).~~

~~b. Cloth jackets of insulation of ducts and pipes in connection with plumbing, air conditioning, ventilating refrigeration and heating systems: One (1) coat of MPI 50 (Interior Latex Primer Sealer) and one (1) coat of // MPI 10 (Exterior Latex, Flat) // // MPI 11 (Exterior Latex Semi Gloss // // MPI 119 (Exterior Latex, High Gloss (acrylic)) //.~~

3.133.11-BUILDING AND STRUCTURAL WORK FIELD PAINTING:

A. Painting and finishing of ~~interior and~~ exterior work except as specified here-in-after.

1. Painting and finishing of new ~~//~~ and existing ~~//~~ work including colors and gloss of finish selected is specified in ~~the drawings~~ Finish Schedule, Section 09 06 00, SCHEDULE FOR FINISHES.

2. Painting of disturbed, damaged and repaired or patched surfaces when entire space is not scheduled for complete repainting or refinishing.

3. Painting of ferrous metal and galvanized metal.

~~4. Painting of wood with fire retardant paint exposed in attics, when used as mechanical equipment space (except shingles).~~

~~5.4.~~ Identity painting and safety painting.

B. Building and Structural Work not Painted:

1. Prefinished items:

~~a. Casework, doors, elevator entrances and cabs, metal panels, wall covering, and similar items specified factory finished under other sections.~~

~~b-a.~~ Factory finished equipment and pre-engineered metal building components such as metal roof and wall panels.

2. Finished surfaces:

a. Hardware except ferrous metal.

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- b. Anodized aluminum, stainless steel, chromium plating, copper, and brass, except as otherwise specified.
- c. Signs, fixtures, and other similar items integrally finished.

~~3. Concealed surfaces:~~

- ~~a. Inside dumbwaiter, elevator and duct shafts, interstitial spaces, pipe basements, crawl spaces, pipe tunnels, above ceilings, attics, except as otherwise specified.~~
- ~~b. Inside walls or other spaces behind access doors or panels.~~
- ~~c. Surfaces concealed behind permanently installed casework and equipment.~~

~~4. Moving and operating parts:~~

- ~~a. Shafts, chains, gears, mechanical and electrical operators, linkages, and sprinkler heads, and sensing devices.~~
- ~~b. Tracks for overhead or coiling doors, shutters, and grilles.~~

~~5.3. Labels:~~

- a. Code required label, such as Underwriters Laboratories Inc., Intertek Testing Service or Factory Mutual Research Corporation.
- b. Identification plates, instruction plates, performance rating, and nomenclature.

~~6. Galvanized metal:~~

- ~~a. Exterior chain link fence and gates, corrugated metal arcways, and gratings.~~
- ~~b. Gas Storage Racks.~~
- ~~c. Except where specifically specified to be painted.~~

~~7. Metal safety treads and nosings.~~

~~8. Gaskets.~~

~~SPEC WRITER NOTES:~~

- ~~1. Edit other exposed concrete surfaces not required to be painted.~~
- ~~2. Coordinate with Section 09 06 00, SCHEDULE FOR FINISHES to schedule exceptions to not painted surfaces.~~

~~9. Concrete curbs, gutters, pavements, retaining walls, exterior exposed foundations walls and interior walls in pipe basements.~~

~~10. Face brick.~~

~~11. Structural steel encased in concrete, masonry, or other enclosure.~~

~~12. Structural steel to receive sprayed-on fire proofing.~~

~~13. Ceilings, walls, columns in interstitial spaces.~~

~~14. Ceilings, walls, and columns in pipe basements.~~

~~15. Wood Shingles.~~

~~3.14 IDENTITY PAINTING SCHEDULE:~~

~~A. Identify designated service in new buildings or projects with extensive remodeling in accordance with ASME A13.1, unless specified otherwise, on exposed piping, piping above removable ceilings, piping in accessible pipe spaces, interstitial spaces, and piping behind access panels. For existing spaces where work is minor match existing.~~

~~1. Legend may be identified using snap on coil plastic markers or by paint stencil applications.~~

~~2. Apply legends adjacent to changes in direction, on branches, where pipes pass through walls or floors, adjacent to operating accessories such as valves, regulators, strainers and cleanouts a minimum of 12.2 M (40 feet) apart on straight runs of piping. Identification next to plumbing fixtures is not required.~~

~~3. Locate Legends clearly visible from operating position.~~

~~4. Use arrow to indicate direction of flow using black stencil paint.~~

~~5. Identify pipe contents with sufficient additional details such as temperature, pressure, and contents to identify possible hazard. Insert working pressure shown on construction documents where asterisk appears for High, Medium, and Low Pressure designations as follows:~~

~~a. High Pressure — 414 kPa (60 psig) and above.~~

~~b. Medium Pressure — 104 to 413 kPa (15 to 59 psig).~~

~~c. Low Pressure — 103 kPa (14 psig) and below.~~

~~d. Add Fuel oil grade numbers.~~

~~6. Legend name in full or in abbreviated form as follows:~~

~~SPEC WRITER NOTES:~~

~~1. Check with mechanical sections to determine legends required, and pressures.~~

~~2. Define Fuel oil grade.~~

PIPING	COLOR OF EXPOSED PIPING	COLOR OF BACKGROUND	COLOR OF LETTERS	LEGEND ABBREVIATIONS
-------------------	------------------------------------	--------------------------------	-----------------------------	---------------------------------

Blow-off	Green	White	Blow-off
---------------------	------------------	------------------	---------------------

Boiler Feedwater	Green	White	Blr Feed
-----------------------------	------------------	------------------	---------------------

~~A/C Condenser Water~~

Supply	Green	White	A/C Cond Wtr Sup
-------------------	------------------	------------------	-----------------------------

~~A/C Condenser Water~~

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Return	Green	White	A/C Cond Wtr Ret
Chilled Water Supply	Green	White	Ch. Wtr Sup
Chilled Water Return	Green	White	Ch. Wtr Ret
Shop Compressed Air	Blue	White	Shop Air
Air Instrument Controls	Green	White	Air Inst Cont
Drain Line	Green	White	Drain
Emergency Shower	Green	White	Emg Shower
High Pressure Steam	Green	White	H.P. _____*
High Pressure Condensate			
Return	Green	White	H.P. Ret _____*
Medium Pressure Steam	Green	White	M. P. Stm _____*
Medium Pressure Condensate			
Return	Green	White	M.P. Ret _____*
Low Pressure Steam	Green	White	L.P. Stm _____*
Low Pressure Condensate			
Return	Green	White	L.P. Ret _____*
High Temperature Water			
Supply	Green	White	H. Temp Wtr Sup
High Temperature Water			
Return	Green	White	H. Temp Wtr Ret
Hot Water Heating Supply	Green	White	H. W. Htg Sup
Hot Water Heating Return	Green	White	H. W. Htg Ret
Gravity Condensate Return	Green	White	Gravity Cond Ret
Pumped Condensate Return	Green	White	Pumped Cond Ret
Vacuum Condensate Return	Green	White	Vac Cond Ret
Fuel Oil - Grade // //	Brown	White	Fuel Oil Grade // //
(Diesel Fuel included under Fuel Oil)			
Boiler Water Sampling	Green	White	Sample
Chemical Feed	Green	White	Chem Feed
Continuous Blow Down	Green	White	Cont. B D
Pumped Condensate	Green	White	Pump Cond
Pump Recirculating	Green	White	Pump Recirc.
Vent Line	Green	White	Vent
Alkali	Orange	Black	Alk
Bleach	Orange	Black	Bleach
Detergent	Yellow	Black	Det
Liquid Supply	Yellow	Black	Liq Sup
Reuse Water	Yellow	Black	Reuse Wtr
Cold Water (Domestic)	White	Green	White
Hot Water (Domestic)			
— Supply	White	Yellow	Black
— Return	White	Yellow	Black
— Tempered Water	White	Yellow	Black

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~~Ice Water~~

~~— Supply ————— White ————— Green ————— White ————— Ice Wtr~~

~~— Return ————— White ————— Green ————— White ————— Ice Wtr Ret~~

~~Reagent Grade Water ————— Green ————— White ————— RG~~

~~Reverse Osmosis ————— Green ————— White ————— RO~~

~~Sanitary Waste ————— Green ————— White ————— San Waste~~

~~Sanitary Vent ————— Green ————— White ————— San Vent~~

~~Storm Drainage ————— Green ————— White ————— St Drain~~

~~Pump Drainage ————— Green ————— White ————— Pump Disch~~

~~Chemical Resistant Pipe~~

~~— Waste ————— Orange ————— Black ————— Acid Waste~~

~~— Vent ————— Orange ————— Black ————— Acid Vent~~

~~Atmospheric Vent ————— Green ————— White ————— ATV~~

~~Silver Recovery ————— Green ————— White ————— Silver Ree~~

~~Oral Evacuation ————— Green ————— White ————— Oral Evac~~

~~Fuel Gas ————— Yellow ————— Black ————— Gas~~

~~Fire Protection Water~~

~~— Sprinkler ————— Red ————— Red ————— White ————— Auto Spr~~

~~— Standpipe ————— Red ————— Red ————— White ————— Stand~~

~~— Sprinkler ————— Red ————— Red ————— White ————— Drain~~

~~SPEC WRITER NOTE: If solar hot water system is on project, include the following.~~

~~// Hot Water Supply Dom./~~

~~Solar Water ————— Green ————— White ————— H.W. Sup Dom/SW~~

~~Hot Water Return Dom./~~

~~Solar Water ————— Green ————— White ————— H.W. Ret Dom/SW //~~

~~SPEC WRITE NOTE: Coordinate with Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS / Section 27 05 33, RACEWAYS AND BOXES FOR COMMUNICATIONS SYSTEMS / Section 28 05 28.33, CONDUIT AND BACKBOXES FOR ELECTRONIC SAFETY AND SECURITY. Electrical conduits carrying high voltage require labels in compliance with Occupational Safety and Health Office. Label is to be listed as class 5000, 15000, and 25000 and not exact voltage.~~

~~7. Electrical Conduits containing feeders over 600 volts, paint legends using 50 mm (2 inch) high black numbers and letters, showing the voltage class rating. Provide legends where conduits pass through walls~~

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- ~~and floors and at maximum 6096 mm (20 foot) intervals in between. Use labels with yellow background with black border and words Danger High Voltage Class, // 5000 // // 15000 // // 25000 //.~~
- ~~8. See Sections for methods of identification, legends, and abbreviations of the following:~~
- ~~a. Regular compressed air lines: Section 22 15 00, GENERAL SERVICE COMPRESSED AIR SYSTEMS.~~
 - ~~b. Dental compressed air lines: Section 22 61 13.74, DENTAL COMPRESSED AIR PIPING / Section 22 61 19.74, DENTAL COMPRESSED AIR EQUIPMENT.~~
 - ~~c. Laboratory gas and vacuum lines: Section 22 62 00, VACUUM SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES / Section 22 63 00, GAS SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES.~~
 - ~~d. Oral evacuation lines: Section 22 62 19.74, DENTAL VACUUM AND EVACUATION EQUIPMENT.~~
 - ~~e. Medical Gases and vacuum lines: Section 22 62 00, VACUUM SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES / Section 22 63 00, GAS SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES.~~
 - ~~f. Conduits containing high voltage feeders over 600 volts: Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS / Section 27 05 33, RACEWAYS AND BOXES FOR COMMUNICATIONS SYSTEMS / Section 28 05 28.33, CONDUITS AND BACKBOXES FOR ELECTRONIC SAFETY AND SECURITY.~~
- ~~B. Fire and Smoke Partitions:~~
- ~~1. Identify partitions above ceilings on both sides of partitions except within shafts in letters not less than 64 mm (2 1/2 inches) high.~~
 - ~~2. Stenciled message: "SMOKE BARRIER" or, "FIRE BARRIER" as applicable.~~
 - ~~3. Locate not more than 6096 mm (20 feet) on center on corridor sides of partitions, and with a least one (1) message per room on room side of partition.~~
 - ~~4. Use semi-gloss paint of color that contrasts with color of substrate.~~
- ~~C. Identify columns in pipe basements and interstitial space:~~
- ~~1. Apply stenciled number and letters to correspond with grid numbering and lettering indicated on construction documents.~~
 - ~~2. Paint numbers and letters 101 mm (4 inches) high, locate 45 mm (18 inches) below overhead structural slab.~~
 - ~~3. Apply on four (4) sides of interior columns and on inside face only of exterior wall columns.~~

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~~4. Color:~~

~~a. Use black on concrete columns.~~

~~b. Use white or contrasting color on steel columns.~~

~~3.153.12~~ **-PROTECTION CLEAN UP, AND TOUCH-UP:**

- A. Protect work from paint droppings and spattering by use of masking, drop cloths, removal of items or by other approved methods.
- B. Upon completion, clean paint from hardware, glass and other surfaces and items not required to be painted of paint drops or smears.
- C. Before final inspection, touch-up or refinished in a manner to produce solid even color and finish texture, free from defects in work which was damaged or discolored.

- - - E N D - - -

SECTION 13 05 41
SEISMIC RESTRAINT REQUIREMENTS FOR NON-STRUCTURAL COMPONENTS

SPEC WRITER NOTE: Delete between //----//
if not applicable to the project. Also,
delete any other item or paragraph not
applicable in the section and renumber
the paragraphs.

PART 1 - GENERAL

1.1 DESCRIPTION:

- A. Provide seismic restraint in accordance with the requirements of the drawings, VA Handbook H18-8: Seismic Design Requirements and this specification in order to maintain the integrity of non-structural components and equipment of the building so that they remain safe and functional in case of seismic event.
- B. The design of seismic restraints of non-structural components to resist seismic load shall be based on Seismic Design parameters indicated below in accordance with VA H-18-8 in conjunction with ASCE 7// and ASCE 41//, as specified in H-18-8 Section 4.0, for existing building retrofit projects. Specific requirements for Critical and Essential facilities are covered in Section 4.0 of H-18-8, including applying $I_p = 1.5$ for all nonstructural components in Critical facilities.
 - 1. International Building Code //2018// Edition
 - 2. // American Society of Civil Engineers Seismic Evaluation and Retrofit of Existing Buildings ASCE 41-17.//
 - 3. American Society of Civil Engineers Minimum Design Loads and Associated Criteria for Buildings and Other Structures (ASCE 7)//7-16//
 - 4. Facility Occupancy Category per VA H-18-8: //Critical//
//Essential// // Ancillary// Facility
 - 5. Site Class: //A// //B// //C// //D// //E// //F//
 - 6. Building Risk Category: //I// //II// //III// //IV//
 - 7. Mapped MCE_R 0.2 s period Spectral Response Acceleration Parameter (S_s): // //
 - 8. Mapped MCE_R 1.0 s period Spectral Response Acceleration Parameter (S_1): // //
 - 9. Short period Spectral Response Acceleration Parameter (S_d): // //
 - 10. Short period Spectral Response Acceleration Parameter (S_{d1}): // //

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11. Building Seismic Design Category: //A// //B// //C// //D// //E//
//F//

12. Component//s// Importance Factor//s// (Ip): //1.0 // //1.5//

13. Component//s// Response Modification Factor//s// (Rp): // //

14. Component//s// Overstrength Factor//s//: // //

C. Definitions: Non-structural building components are components or systems that are not part of the building's structural system whether inside or outside, above or below grade. Non-structural components of buildings include but are not limited to (Refer to VA H-18-8, ASCE 7 and ASCE 41 for additional examples):

1. Architectural Elements: Facades that are not part of the structural system and its shear resistant elements; cornices and other architectural projections and parapets that do not function structurally; glazing; nonbearing partitions; suspended ceilings; stairs isolated from the basic structure; cabinets; bookshelves; medical equipment; and storage racks, etc.
2. Electrical Elements: Power and lighting systems; substations; switchgear and switchboards; auxiliary engine-generator sets; transfer switches; motor control centers; motor generators; selector and controller panels; fire protection and alarm systems; special life support systems; and telephone and communication systems, etc.
3. Mechanical Elements: Heating, ventilating, and air-conditioning systems; medical gas systems; plumbing systems; sprinkler systems; pneumatic systems; boiler/chiller/utility plant/other equipment and components, etc.
4. Transportation Elements: Mechanical, electrical and structural elements for transport systems, i.e., elevators and dumbwaiters, including hoisting equipment and counterweights.

1.2 RELATED WORK:

SPEC WRITER NOTE: Include here all
applicable specification sections.

Related specifications include but are not limited to those shown below.
Coordinate all work with the applicable specification for that work.

- A. Cast-In-Place Concrete: Section 03 30 00, CAST-IN-PLACE CONCRETE
- B. Structural Steel Framing: Section 05 12 00, STRUCTURAL STEEL FRAMING
- C. Metal Fabrication: Section 05 50 00, METAL FABRICATIONS

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D. Accordion Folding Partitions: Section 10 22 26.13, FOLDING ACCORDION PARTITIONS

E. Acoustical Ceilings: Section 09 51 00 ACOUSTICAL CEILINGS

F. Linear Metal Ceilings: Section 09 54 23, LINEAR METAL CEILINGS

G. Interior Lighting: Section 26 51 00, INTERIOR LIGHTING

1.3 QUALITY CONTROL:

A. Shop-Drawing Preparation:

1. Non-structural seismic restraint systems shop drawings and delegated design calculations shall be prepared by a professional structural engineer with a minimum of 5 years' experience in the design and detailing of seismic force restraints. The professional structural engineer shall be registered in the state where the project is located and submit qualifications with list of projects illustrating compliance with the experience requirement of this section.
2. Submit design tables and information used for the design-force levels, stamped and signed by a professional structural engineer registered in the State where project is located.

B. Coordination:

1. Do not install seismic restraints until seismic restraint submittals are approved by the Contracting Officers Representative (COR).
2. Coordinate trapezes or other multi-pipe hanger systems prior to submission of shop drawings for review.

C. Seismic Certification:

In structures assigned to Seismic Design Category C, D, E, or F, permanent equipment and components are to have Special Seismic Certification in accordance with requirements of section 13.2.2 of ASCE 7, including those required in existing buildings within Section 13.7.1.3.3, 13.7.7.3.3 and 13.7.8.3.3 of ASCE 41, except for equipment and components that are considered inherently rugged as listed in Section 4.2.2 of VA H18-8, and shall comply with section 13.2.6 of ASCE 7.

1.4 SUBMITTALS:

A. Submit a complete and coordinated set of bracing and signed and sealed anchorage drawings and calculations for all non-structural elements requiring seismic restraint by the delegated professional structural engineer mentioned in Section 1.3.A.1 for review prior to installation including:

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1. Description, layout, and location of all items to be anchored or braced with anchorage or brace points noted and dimensioned.
 2. Details of all anchorage and bracing at large scale with all members, parts brackets shown, together with all connections, bolts, welds etc. clearly identified and specified. Details shall be coordinated with all project conditions and trades prior to shop drawing submission for review.
 3. Complete calculations including but not limited to seismic design criteria, computer model input and output, seismic design forces and capacities, design tables and information used for all proprietary design elements such as post installed anchors, stamped and signed by a professional structural engineer specified in section 1.3 A.1.
 4. For all post installed anchorages submit the appropriate International Code Council Engineering Service (ICC-ES) evaluation reports, California's Office of Statewide Health Planning and Development (OSHPD) pre-approvals, or lab test reports verifying compliance with OSHPD Interpretation of Regulations 28-6.
 5. Delegated professional structural engineer qualifications.
- B. Submit for review prior to installation, the following for seismic protection of piping in addition to items noted in Section 1.4.A:
1. Single-line piping diagrams on a floor-by-floor basis. Show all suspended piping for a given floor on the same plain.
 2. Type of pipe (Copper, steel, cast iron, insulated, non-insulated, etc.).
 3. Pipe contents.
 4. Structural framing for the seismic and gravity support and the main superstructure for which the bracing and or anchorage is attached.
 5. Location of all gravity load pipe supports and spacing requirements.
 6. Numerical value of gravity load reactions.
 7. Location of all seismic bracing.
 8. Numerical value of applied seismic brace loads.
 9. Type of connection (Vertical support, vertical support with seismic brace etc.).
 10. Seismic brace reaction type (tension or compression): Details illustrating all support and bracing components, methods of connections, and specific anchors to be used.

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- C. Submit for review prior to installation, the following items for seismic protection of suspended ductwork and suspended electrical and communication cables, in addition to items noted in Section 1.4.A:
1. Details illustrating all support and bracing components, methods of connection, and specific anchors to be used.
 2. Numerical value of applied gravity and seismic loads and seismic loads acting on support and bracing components.
 3. Maximum spacing of hangers and bracing.

1.5 APPLICABLE PUBLICATIONS:

- A. The Publications listed below (including amendments, addenda revisions, supplements and errata) form a part of this specification to the extent referenced. The publications are referenced in text by basic designation only.

SPEC WRITER NOTE: Edit applicable
publications to indicate the most recent
edition.

- B. American Concrete Institute (ACI):
355.2-19.....Qualification for Post-Installed Mechanical
Anchors in Concrete and Commentary
- C. American Institute of Steel Construction (AISC):
Load and Resistance Factor Design, Volume 1, Second Edition
- D. ASTM International (ASTM):
A36/A36M-19.....Standard Specification for Carbon Structural
Steel
A53/A53M-18.....Standard Specification for Pipe, Steel, Black
and Hot-Dipped, Zinc-Coated, Welded and
Seamless
A307-14e1.....Standard Specifications for Carbon Steel Bolts,
Studs, and Threaded Rod 60,000 PSI Tensile
Strength
A325-14.....Standard Specification for Structural Bolts,
Steel, Heat Treated, 120/105 ksi Minimum
Tensile Strength
A325M-14.....Standard Specification for High-Strength Bolts
for Structural Steel Joints [Metric]

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- A490-14a.....Standard Specification for Heat-Treated Steel
Structural Bolts, 150 ksi Minimum Tensile
Strength
- A490M-14a.....Standard Specification for High-Strength Steel
Bolts, Classes 10.9 and 10.9.3, for Structural
Steel Joints [Metric]
- A500/A500M-18.....Standard Specification for Cold-Formed Welded
and Seamless Carbon Steel Structural Tubing in
Rounds and Shapes
- A501/A501M-14.....Standard Specification for Hot-Formed Welded
and Seamless Carbon Steel Structural Tubing
- A615/A615M-20.....Standard Specification for Deformed and Plain
Carbon Steel Bars for Concrete Reinforcement
- A992/A992M-11 (2015).....Standard Specification for Steel for Structural
Shapes for Use in Building Framing
- A996/A996M-16.....Standard Specification for Rail Steel and Axle
Steel Deformed Bars for Concrete Reinforcement
- E488/E488M-18.....Standard Test Methods for Strength of Anchors
in Concrete Elements
- E. American Society of Civil Engineers
1. Minimum Design Loads and Associated Criteria for Buildings and Other
Structures (ASCE 7) Edition as indicated in section 1.1 B of this
specification. Associated Criteria for Buildings and Other
Structures (ASCE 7): //7-16//
- F. International Building Code (IBC) Edition as indicated in Section 1.1 B
of this specification.
- G. VA Handbook H18-8 Seismic Design Requirements, VA H-18-8, November
2019 (REVISED MAY 1, 2020)
- H. National Uniform Seismic Installation Guidelines (NUSIG)
- I. Sheet Metal and Air Conditioning Contractors National Association
- J. (SMACNA): Seismic Restraint Manual - Guidelines for Mechanical Systems,
3RD EDITION 2008 and Addendum

1.6 REGULATORY REQUIREMENT:

- A. IBC as shown in Section 1.1 B of this specification.

DESIGNER AND SPEC WRITER NOTES:

1. The design professional's
responsibilities are to identify the
components to be restrained, to

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identify the lateral force coefficient, to provide details for restraints on the construction drawings, and to review and approve seismic restraint shop drawing details prepared by the Contractor/Vendor.

2. The designer (structural engineer) shall specify on the drawings:
 - a) the short-period acceleration (S_{DS});
 - b) the importance factor (I_p) for non-structural elements; and
 - c) the maximum interstory drifts permitted in H-18-8.

B. Exceptions: The omission of seismic restraints shall be allowed only in accordance with VA H18-8, ASCE 7 and //ASCE 41//.

PART 2 - PRODUCTS

2.1 STEEL:

- A. Structural Steel: ASTM A36 // A36M // A992 //.
- B. Structural Tubing: ASTM A500, Grade B.
- C. Structural Tubing: ASTM A501.
- D. Steel Pipe: ASTM A53/A53M, Grade B.
- E. Bolts & Nuts: ASTM // A307 // A325 // A325M // A490 // A490M //.

2.2 CAST-IN-PLACE CONCRETE:

- A. Concrete: 28 day strength, f'_c = // 20.7 MPa (3,000 psi) // 27.5 MPa (4,000 psi) // 34.5 MPa 5000 psi
- B. Reinforcing Steel: ASTM A615/615M or ASTM A996/A996M deformed.

PART 3 - EXECUTION

3.1 CONSTRUCTION, GENERAL:

- A. Provide equipment supports and anchoring devices to withstand the seismic design forces, so that when seismic design forces are applied, the equipment cannot displace, overturn, or become inoperable.
- B. Provide anchorages in conformance with recommendations of the equipment manufacturer and as shown on approved shop drawings and calculations.
- C. Construct seismic restraints and anchorage to allow for thermal expansion.
- D. Testing Before Final Inspection:
 1. Test 10-percent of anchors in masonry and concrete per ASTM E488, and ACI 355.2 to determine that they meet the required load capacity. If any anchor fails to meet the required load, test the

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- next 20 consecutive anchors, which are required to have zero failure, before resuming the 10-percent testing frequency.
2. Before scheduling Final Inspection, submit a report on this testing indicating the number and location of testing, and what anchor-loads were obtained.
 3. Construct seismic restraints and anchorages to not interfere with other trades or damage existing or in-situ elements of the constructed building.

3.2 EQUIPMENT RESTRAINT AND BRACING:

- A. See drawings for equipment to be restrained or braced.

SPEC WRITER NOTE:

SMACNA does not cover all conditions such as, providing details for seismic restraints of equipment or details of flexible joints when crossing seismic or expansion joints, or bracing of in-line equipment, etc. Also, SMACNA provides guidelines and its details must be used with extreme care and verified per current VA H-18-8 requirements and Section 1.4A.

3.3 MECHANICAL DUCTWORK AND PIPING; BOILER PLANT STACKS AND BREACHING; ELECTRICAL BUSWAYS, CONDUITS, AND CABLE TRAYS; AND TELECOMMUNICATION WIRES AND CABLE TRAYS

- A. Support and brace mechanical ductwork and piping; electrical busways, conduits and cable trays; and telecommunication wires and cable trays including boiler plant stacks and breeching to resist directional forces (lateral, longitudinal and vertical).
- B. Brace duct and breeching branches with a minimum of 1 brace per branch.
- C. Provide supports and anchoring so that, upon application of seismic forces, piping remains fully connected as operable systems which will not displace sufficiently to damage adjacent or connecting equipment, or building members.
- D. Piping Connections: Provide flexible connections where pipes connect to equipment. Make the connections capable of accommodating relative differential movements between the pipe and equipment under conditions of earthquake shaking.

3.4 PARTITIONS

SPEC WRITER NOTE:

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1. Masonry walls used as interior partitions or as exterior surfaces of buildings shall be properly anchored to the structure, and shall be designed to carry lateral loads imposed due to earthquake along with their own weight and other lateral dead, live and or wind load forces.
- A. In buildings with flexible structural frames, anchor partitions to only structural element, such as a floor slab, and separate such partition by a physical gap from all other structural elements.
- B. Properly anchor masonry walls to the structure for restraint, so as to carry lateral loads imposed due to earthquake along with their own weight and other lateral forces.

3.5 CEILINGS AND LIGHTING FIXTURES

- A. At intervals required to meet the seismic demand forces, laterally brace suspended ceilings against lateral and vertical movements, and provide with a physical separation at the walls.
- B. Independently support and laterally brace all lighting fixtures. Refer to applicable portion of lighting specification, Section 26 51 00, INTERIOR LIGHTING.

3.6 FACADES AND GLAZING

DESIGNER NOTES:

1. Heavy rigid facades should be used only on rigid structural systems; they should never be attached to relatively flexible building frames.
 2. Review building design to ensure contractor's ability to comply with the following paragraphs.
 3. Design brick veneer anchors and back-up wall for applicable seismic forces at the project location.
 4. Design attachments to structure for all façade materials to accommodate applicable seismic forces at the project location.
- A. Do not install concrete masonry unit filler walls in a manner that can restrain the lateral deflection of the building frame. Provide a gap with adequately sized resilient filler to separate the structural frame from the non-structural filler wall.

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- B. Tie brick veneers to a separate wall that is independent of the steel frame as shown on construction drawings to ensure strength against applicable seismic forces at the project location.
- C. Install attachments to structure for all façade materials as shown on construction drawings to ensure strength against applicable seismic forces at the project location.

3.7 STORAGE RACKS, CABINETS, AND BOOKCASES

- A. Install storage racks to withstand earthquake forces and anchored to the floor or laterally braced from the top to the structural elements.
- B. Anchor medical supply cabinets to the floor or walls and equip them with properly engaged, lockable latches.
- C. Anchor filing cabinets that are more than 2 drawers high to the floor or walls, and equip all drawers with properly engaged, lockable latches.
- D. Anchor bookcases that are more than 30 inches high to the floor or walls, and equip any doors with properly engaged, lockable latches.

- - - E N D - - -